
The RBioc Book

SEAN DAVIS

University of ColoradoAnschutz School of Medicine

2026-06-02

Table of contents

Preface	1
What are the goals?	2
Adult learners	2
AI is part of the solution	4
I. Introduction	6
1. About R	7
1.1. What you'll learn	7
1.2. What is R?	7
1.3. Why use R?	9
1.4. Why not use R?	10
1.5. R license and the open source ideal	10
1.6. Where this is going	11
1.7. Summary	11
1.8. Reflection	11
2. RStudio	13
2.1. What you'll learn	13
2.2. Getting started with RStudio	13
2.3. The RStudio Interface	14
2.4. Alternatives to RStudio	17
3. R mechanics	21
3.1. What you'll learn	21
3.2. Starting R	21
3.3. <i>RStudio</i> : A Quick Tour	21
3.4. Interacting with R	22
3.4.1. Expressions	23
3.4.2. Assignment	23
3.5. Rules for Names in R	25
3.6. About R functions	26
3.7. Resources for Getting Help	26

Table of contents

3.8. Reflection	27
4. Up and Running with R	28
4.1. The R User Interface	28
4.1.1. An exercise	32
4.2. Objects	32
4.3. Functions	40
4.3.1. Sample with Replacement	44
4.4. Writing Your Own Functions	46
4.4.1. The Function Constructor	47
4.5. Arguments	49
4.6. Scripts	52
4.7. Summary	53
5. Packages and more dice	54
5.1. Packages	54
5.1.1. Installing R packages	55
5.1.2. Installing vs loading (library) R packages	55
5.1.3. Finding R packages	56
5.1.4. Creating a package	57
5.2. Are our dice fair?	57
5.3. Bonus exercise	60
6. Reading and writing data files	61
6.1. What you'll learn	61
6.2. Organizing your work with RStudio Projects	61
6.2.1. Creating a project	62
6.2.2. A sensible project layout	62
6.2.3. Why this fixes file paths	63
6.3. CSV files	63
6.3.1. Writing a CSV file	64
6.3.2. Reading a CSV file	65
6.3.3. Reading a CSV straight from the web	66
6.4. Excel files	67
6.4.1. Reading an Excel file	67
6.4.2. Writing an Excel file	69
6.5. Additional formats	69
6.6. Exercises	70
6.7. Summary	71

Table of contents

II. R Data Structures	72
Chapter overview	73
7. Vectors	76
7.1. What you'll learn	76
7.2. A vector is a column of measurements	76
7.3. Creating vectors	78
7.4. Vector operations	80
7.5. Logical vectors	81
7.5.1. Logical operators	82
7.6. Indexing vectors	83
7.7. Named vectors	85
7.8. Character vectors, a.k.a. strings	86
7.9. Missing values, a.k.a. NA	88
7.10. Exercises	89
7.11. Summary	92
8. Matrices	93
8.1. What you'll learn	94
8.2. Creating a matrix	94
8.3. Accessing elements of a matrix	97
8.4. Changing values in a matrix	99
8.5. Calculations on matrix rows and columns	101
8.6. Exercises	103
8.7. Summary	112
9. Lists: a catch-all container	113
9.1. What you'll learn	114
9.2. Creating a list	114
9.3. Inspecting your list: what's inside?	115
9.3.1. <code>str()</code> : the structure function	115
9.3.2. <code>length()</code> , <code>names()</code> , and <code>class()</code>	116
9.4. Accessing list elements: getting things out	117
9.4.1. The mighty <code>[[]]</code> and <code>\$</code> for single items	117
9.4.2. The subsetting <code>[]</code> for new lists	118
9.5. Modifying lists	119
9.5.1. Adding and updating elements	119
9.5.2. Removing elements	120
9.6. A biological example: a self-contained gene record	121
9.7. Exercises	122
9.8. Summary	123

Table of contents

10. Data Frames	125
10.1. What you'll learn	125
10.2. Dataset	125
10.3. Reading in data	126
10.4. Inspecting data.frames	127
10.5. Accessing variables (columns) and subsetting	130
10.6. Aggregating and exploring	132
10.6.1. More advanced indexing and subsetting	133
10.7. Aggregating data	137
10.8. Creating a data.frame from scratch	138
10.9. Saving a data.frame	139
10.10 Exercises	139
10.11 Summary	141
10.12 Reflection	142
11. Factors	143
11.1. What you'll learn	143
11.2. A vector with a built-in dictionary	143
11.3. Counting categories with <code>table()</code>	145
11.4. Converting a factor back	145
11.5. Controlling the levels	146
11.6. Exercises	147
11.7. Summary	149
III. Exploratory data analysis	150
What you'll learn in this part	151
Where to begin	152
12. Introduction to dplyr: mammal sleep dataset	153
12.1. What you'll learn	153
12.2. What is dplyr?	154
12.3. Why is dplyr useful?	154
12.4. Data: mammals sleep	154
12.5. The dplyr verbs	155
12.6. Selecting columns with <code>select()</code>	156
12.7. Selecting rows with <code>filter()</code>	158
12.8. Chaining verbs with the pipe, <code> ></code>	160
12.9. Re-ordering rows with <code>arrange()</code>	161
12.10 Creating new columns with <code>mutate()</code>	163
12.11 Summarizing with <code>summarise()</code>	164

Table of contents

12.12	Grouping with <code>group_by()</code>	165
12.13	Exercises	166
12.14	Summary	169
13.	Case Study: Behavioral Risk Factor Surveillance System	170
13.1.	What you'll learn	170
13.2.	Loading the dataset	171
13.3.	Inspecting the data	171
13.4.	Summary statistics	172
13.5.	Visualizing single variables	173
13.6.	Analyzing relationships between variables	175
13.7.	Comparing two groups: a t-test	177
13.8.	Fitting a model: weight and height	179
13.9.	Exercises	185
13.10	Summary	188
14.	Exploring data with <code>ggplot2</code>	190
14.1.	What you'll learn	190
14.2.	<code>ggplot()</code>	192
14.3.	<code>geom_*()</code>	193
14.4.	Grouping	196
14.5.	Scales	197
14.6.	Facets	198
14.7.	Labels and titles	200
14.8.	Theming	201
14.9.	Exercises	203
14.10	Summary	206
15.	A self-guided tour of data visualization in R	208
15.1.	What you'll learn	208
15.2.	Two principles that make any plot better	208
15.2.1.	Maximize the data-ink ratio	209
15.2.2.	Label everything	212
15.2.3.	Use color deliberately	213
15.3.	The grammar of graphics, layer by layer	217
15.3.1.	Step 1: data and aesthetics	218
15.3.2.	Step 2: add a geom	219
15.3.3.	Step 3: map a third variable to color	220
15.3.4.	Step 4: layer on a second geom	221
15.3.5.	Step 5: control scales	222
15.3.6.	Step 6: facet into small multiples	223

Table of contents

15.3.7. Step 7: labels and theme	224
15.4. Specialized plots for biology	225
15.4.1. Sets and intersections: UpSet plots	225
15.4.2. Heatmaps for data matrices	226
15.4.3. Visualizing the genome (further exploration)	229
15.5. Exercises	229
15.6. Summary	232
 IV. statistics	 234
 16. Working with distribution functions	 235
16.1. What you'll learn	235
16.2. pnorm	236
16.3. dnorm	241
16.4. qnorm	241
16.5. rnorm	245
16.6. Worked example: IQ scores	251
16.7. Exercises	251
16.8. Summary	252
 17. The t-statistic and t-distribution	 254
17.1. What you'll learn	254
17.2. Background	254
17.3. The Z-score and probability	255
17.3.1. Small diversion: a two-sided pnorm	257
17.4. The t-distribution	258
17.4.1. p-values based on Z vs t	261
17.4.2. Experiment: does the choice of distribution actually matter?	262
17.5. Checkpoint: t vs normal	266
17.6. t.test	266
17.6.1. One-sample	267
17.6.2. Two-sample	268
17.6.3. From a data frame	268
17.6.4. Equivalence to a linear model	270
17.7. Power calculations	271
17.8. Exercises	274
17.9. Summary	276
17.10 Resources	277

Table of contents

18. K-means clustering	278
18.1. What you'll learn	278
18.2. What k-means does	278
18.3. The k-means algorithm	279
18.4. Pros and cons of k-means clustering	280
18.5. A worked example: the diauxic shift	281
18.5.1. The data and experimental background	281
18.5.2. Getting the data	281
18.5.3. Preprocessing: keep only the informative genes	282
18.5.4. Clustering	285
18.5.5. Reading the clusters as biology	286
18.6. Exercises	287
18.7. Summary	290
 V. Machine Learning	 291
19. Introduction to machine learning	292
19.1. What you'll learn	292
19.2. The three kinds of machine learning	293
19.3. The machine learning workflow	295
19.4. Overfitting and underfitting	298
19.5. Cross-validation and model selection	301
19.6. Exercises	302
19.7. Summary	303
 20. Supervised Learning	 305
20.1. What you'll learn	305
20.2. A framework for understanding supervised algorithms	306
20.3. Our shared dataset: Palmer penguins	306
20.4. Linear regression	308
20.5. K-nearest neighbours	309
20.6. Penalized regression	310
20.6.1. Ridge regression	310
20.6.2. LASSO regression	311
20.6.3. Elastic net	312
20.7. Classification and regression trees (CART)	313
20.8. Random forests	316
20.9. Summary	317
20.10 Exercises	318

Table of contents

21. The mlr3verse framework	321
21.1. What you'll learn	321
21.2. Why a framework at all?	321
21.3. The mlr3verse ecosystem	322
21.4. Tasks: the problem and the data	323
21.5. Learners: the algorithms	324
21.6. Measures: scoring the result	325
21.7. Resamplings: honest validation	326
21.8. How the four fit together	327
21.9. Summary	327
21.10 Exercises	328
 22. Worked example: classifying cancer types from gene expression	 331
22.1. What you'll learn	332
22.2. A quick recap of the mlr3 workflow	332
22.3. Setup	334
22.4. Understanding the problem	334
22.5. Data preparation	335
22.6. Feature selection and data cleaning	336
22.7. Creating the task	337
22.8. Splitting the data	337
22.9. k-nearest-neighbours	337
22.9.1. Create the learner	338
22.9.2. Train	338
22.9.3. Predict	339
22.9.4. Assess	339
22.10 Classification tree	340
22.10.1. Train	340
22.10.2. Predict	343
22.10.3. Assess	343
22.11 Random forest	344
22.11.1. Train	344
22.11.2. Predict and assess	345
22.12 Summary	347
22.13 Exercises	347
 23. Worked example: predicting age from DNA methylation	 349
23.1. What you'll learn	350
23.2. Setup	350
23.3. The data	350

Table of contents

23.4. Linear regression	352
23.4.1. Train	352
23.4.2. Predict	353
23.4.3. Assess	353
23.5. Regression tree	355
23.5.1. Predict	356
23.5.2. Assess	356
23.6. Random forest	358
23.6.1. Predict	359
23.6.2. Assess	359
23.7. Summary	361
23.8. Exercises	361
24. Worked example: predicting gene expression from histone marks	363
24.1. What you'll learn	364
24.2. Setup	364
24.3. The data	366
24.4. Create the task and split	369
24.5. Linear regression	369
24.5.1. Train and predict	369
24.5.2. Assess	370
24.6. Penalized regression	372
24.6.1. Train	372
24.6.2. Predict and assess	373
24.7. Summary	376
24.8. Exercises	376
VI. Bioconductor	378
25. Accessing public omics data with GEOquery	379
25.1. What you'll learn	379
25.2. Fetching a dataset from GEO	380
25.3. Summarizing the sample annotations	381
25.4. Keeping the most variable genes	382
25.5. Running and interpreting the PCA	382
25.6. Summary	385
25.7. Exercises	385
26. Storing experiments with SummarizedExperiment	387
26.1. What you'll learn	387

Table of contents

26.2. Anatomy of a <code>SummarizedExperiment</code>	388
26.2.1. Assays	390
26.2.2. ‘Row’ (regions-of-interest) data	391
26.2.3. ‘Column’ (sample) data	392
26.2.4. Experiment-wide metadata	393
26.3. Common operations on <code>SummarizedExperiment</code>	394
26.3.1. Subsetting	394
26.3.2. Getters and setters	395
26.3.3. Range-based operations	397
26.4. Constructing a <code>SummarizedExperiment</code>	397
26.4.1. <code>rowData</code> , or feature information	398
26.4.2. <code>colData</code> , or sample information	399
26.4.3. assays, or the data	399
26.4.4. Putting it all together	401
26.4.5. Getting <code>logRatios</code>	401
26.5. Summary	403
26.6. Exercises	403
27. Analyzing microbiome data	405
27.1. What you’ll learn	405
27.2. Getting started	406
27.3. Accessing microbiome datasets	406
27.4. Data containers	407
27.5. Loading a microbiome dataset	408
27.5.1. Experimental measurements: <code>assays</code>	411
27.5.2. Sample information: <code>colData</code>	412
27.5.3. Measured feature information: <code>rowData</code>	415
27.5.4. <code>rowTree</code>	416
27.6. Wrangling and subsetting	418
27.6.1. Subsetting samples	419
27.6.2. Agglomerating data	420
27.7. Community indices	421
27.7.1. Alpha diversity	421
27.7.2. Beta diversity	426
27.8. Microbial composition	428
27.8.1. Abundance	429
27.8.2. Prevalence	431
27.9. Heatmaps	432
27.10 Summary	433
27.11 Exercises	433
27.12 Resources	435

Table of contents

28. Genomic ranges with real ChIP-seq peaks	436
28.1. What you'll learn	436
28.2. The BED file format	437
28.3. Loading the libraries	437
28.4. Importing the CTCF peaks	438
28.5. Looking inside a GRanges	440
28.6. Some quick summaries	443
28.7. Where is each peak?	444
28.8. Reshaping ranges	444
28.8.1. Shifting	445
28.8.2. Resizing	445
28.8.3. Flanking	446
28.9. Coverage	447
28.10 Exercises	450
28.11 Summary	451
29. Genomic ranges and features	453
29.1. What you'll learn	453
29.2. Bioconductor and GenomicRanges	454
29.2.1. Subsetting GRanges objects	463
29.2.2. Interval operations on one GRanges object	464
29.2.3. Set operations for GRanges objects	469
29.3. GRangesList	471
29.3.1. Basic <i>GRangesList</i> accessors	473
29.4. Relationships between region sets	475
29.4.1. Overlaps	475
29.4.2. Nearest feature	479
29.5. Gene models	481
29.6. Summary	485
29.7. Exercises	486
30. Ranges Exercises	489
30.1. What you'll learn	489
30.2. Exercise 1: explore a DNase peak set	490
30.3. Exercise 2: compare two replicates	504
30.4. Exercise 3: DNase sites relative to gene promoters	642
30.5. Exercise 4: enrichment of histone marks at CpG islands	645
30.6. Summary	648
31. Genomic ranges and ATAC-seq	649
<i>R</i> / Bioconductor packages used	649

Table of contents

31.1. What you'll learn	649
31.2. Background	650
31.3. Informatics overview	650
31.4. Working with sequencing data in Bioconductor	654
31.5. Data import and quality control	654
31.6. Coverage	657
31.7. Fragment lengths	659
31.8. Viewing data in IGV	665
31.9. Exercises	666
31.10 Summary	668
31.11 Additional work	668
MACS2	669
32. Single-cell RNA-seq analysis	670
32.1. What you'll learn	670
32.2. Setup and data loading	671
32.2.1. Loading the libraries	671
32.2.2. Exploring available datasets	671
32.3. Dimensionality reduction	676
32.3.1. Feature selection: identifying highly variable genes	676
32.3.2. Non-linear dimensionality reduction with t-SNE	679
32.3.3. Visualizing cell types and clusters	681
32.4. Finding marker genes	684
32.5. Interactive visualization with iSEE	686
32.6. Exercises	687
32.7. Summary	690
32.8. Bonus: working with 10x Genomics data	690
32.8.1. Understanding 10x Genomics output formats	690
32.8.2. Loading 10x data from Matrix Market files	691
32.8.3. Loading from HDF5 files	691
32.8.4. Essential quality control after loading raw data	692
32.8.5. Filtering low-quality cells and genes	693
32.9. Further reading and resources	694
References	695
Appendices	699
A. Git and GitHub	699
A.1. install Git and GitHub CLI	699

Table of contents

A.2. Configure Git	700
A.3. Create a GitHub account	700
A.4. Login to GitHub CLI	701
A.5. Introduction to Version Control with Git	701
A.5.1. Key Git Commands We'll Learn Today:	701
A.6. The Toy Example: An R Script	702
A.7. Let's Get Started with Git!	702
A.7.1. Step 1: Initialize Your Git Repository	702
A.7.2. Step 2: Your First Commit	704
A.7.3. Step 3: Making and Undoing a Change	704
A.7.4. Step 4: Branching Out	704
A.7.5. Step 5: Seeing Branches in Action	705
A.7.6. Step 6: Merging Your Work	706
B. Further reading, cheat sheets, and datasets	707
B.1. Cheat sheets	707
B.2. Free online books	707
B.3. Package and project documentation	708
B.4. Interactive practice: swirl	708
B.5. Datasets used in this book	708
B.6. AI assistants	709
C. AI Tools for Enhanced Learning and Productivity in Bioinformatics	710
C.1. Introduction: AI as Your Bioinformatics Learning Companion	710
C.2. AI for Understanding Statistical Concepts and R Functions	711
C.3. AI for Data Exploration and Visualization Strategies	712
C.4. AI for Boosting Productivity and Troubleshooting	713
C.5. Best Practices for Prompt Engineering	713
C.6. Ethical Considerations and Limitations	714
C.7. De-skilling, never-skilling, and how to stay sharp	716
C.8. Using AI constructively: when to lean in, and when to step back	718
C.9. The Future: AI in Bioinformatics and Data Science	718
C.10. Practical Exercises and Activities	719
D. Matrix Exercises	721
D.1. Data preparation	721
D.2. Exercises	721

List of Figures

1.	Schema theory of learning. While probably not entirely applicable to how adults learn, it does highlight the importance of building on prior knowledge. When new knowledge is introduced, it is more quickly stored into long-term memory if strongly-related concepts are already available for connection. . .	1
2.	Why do adults choose to learn something?	3
3.	How to stay stuck in data science (or anything). The “Read-Do” loop tends to deliver the best results. Too much reading between doing can be somewhat effective. Reading and simply copy-paste is probably the least effective. When working through material, experiment. Try to break things. Incorporate your own experience or applications whenever possible.	4
1.1.	Worldwide search interest in R and Python over time, from Google Trends. Search interest is a rough proxy for general popularity — it does not capture how heavily specific fields lean on each language. Bioinformatics, in particular, remains deeply invested in R.	8
2.1.	The RStudio interface. In this layout, the source pane is in the upper left, the console is in the lower left, the environment panel is in the top right and the viewer/help/files panel is in the bottom right.	15
2.2.	Dealing with limited screen real estate can be a challenge, particularly when you want to open another window to, for example, view a web page. You can resize the panes by sliding the center divider (red arrows) or by clicking on the minimize/maximize buttons (see blue arrow).	16
2.3.	Jupyter Notebook interface. This is an interactive environment for writing and executing R code, along with rich text support for documentation. . . .	18
2.4.	Visual Studio Code (VSCode) with the R extension. This is a lightweight alternative to RStudio that provides syntax highlighting, code completion, and integrated terminal support.	19
2.5.	Positron Workbench interface. This IDE supports R and Python, providing a similar interface to RStudio with additional features for data science workflows.	20

List of Figures

4.1.	Your computer does your bidding when you type R commands at the prompt in the bottom line of the console pane. Don't forget to hit the Enter key. When you first open RStudio, the console appears in the pane on your left, but you can change this with File > Tools > Global Options in the menu bar.	29
4.2.	Assignment creates an object in the environment pane.	35
4.3.	"When R performs element-wise execution, it matches up vectors and then manipulates each pair of elements independently."	38
4.4.	"R will repeat a short vector to do element-wise operations with two vectors of uneven lengths."	39
4.5.	"When you link functions together, R will resolve them from the innermost operation to the outermost. Here R first looks up die, then calculates the mean of one through six, then rounds the mean."	41
4.6.	"Every function in R has the same parts, and you can use function to create these parts. Assign the result to a name, so you can call the function later."	51
4.7.	"When you open an R Script (File > New File > R Script in the menu bar), RStudio creates a fourth pane (or puts a new tab in the existing pane) above the console where you can write and edit your code."	52
5.1.	Installing vs loading R packages.	56
5.2.	In an ideal world, a histogram of the results would look like this	57
5.3.	Histogram of the sums from 100 rolls of our fair dice	59
5.4.	Histogram with 100000 rolls much more closely approximates the pyramidal shape we anticipated	60
6.1.	A pictorial representation of R's most common data structures are vectors, matrices, arrays, lists, and dataframes. Figure from Hands-on Programming with R.	74
7.1.	"Pictorial representation of three vector examples. The first vector is a numeric vector. The second is a 'logical' vector. The third is a character vector. Vectors also have indices and, optionally, names."	77
8.1.	A matrix is a collection of column vectors, all the same length and the same type.	93
13.1.	Distribution of respondent ages in the BRFSS subset.	174
13.2.	Weight distribution compared between the sexes.	175
13.3.	Scatterplot of age versus weight.	176
13.4.	Female respondent weight in 1990 versus 2010.	178
13.5.	Weight distribution for 2010 male respondents.	180
13.6.	Height distribution for 2010 male respondents.	181

List of Figures

13.7. Weight versus height for 2010 male respondents.	182
13.8. The fitted regression line through the weight-versus-height data, with one point marked in red.	184
13.9. Residuals-versus-fitted diagnostic plot for the weight-versus-height model. .	185
14.1. Initialize the plot with <code>ggplot()</code>	192
14.2. Add points to the plot with <code>geom_point()</code>	193
14.3. Modify point colour, size, and transparency.	194
14.4. Add a line of best fit with <code>geom_smooth()</code>	195
14.5. Group points by smoker status.	196
14.6. Modify scales for the x and y axes and the colours.	198
14.7. Create facets based on the obese variable.	199
14.8. Add labels and a title to the plot.	201
14.9. Customize the plot's appearance with a theme.	202
15.1. Edward Tufte's landmark book, <i>The Visual Display of Quantitative Information</i> , popularized the data-ink ratio.	210
15.2. A cluttered plot with a low data-ink ratio. The gray background and heavy gridlines are decoration, not information.	211
15.3. The same data with a high data-ink ratio. The eye goes straight to the points. .	212
15.4. RColorBrewer palettes, grouped into sequential (top), qualitative (middle), and diverging (bottom) families.	214
15.5. The subset of RColorBrewer palettes that remain distinguishable for viewers with common color-vision deficiencies.	216
15.6. Data plus aesthetics, but no geom: <code>ggplot()</code> sets up the axes but draws nothing yet.	218
15.7. Adding <code>geom_point()</code> draws one point per penguin.	219
15.8. Mapping species to the color aesthetic reveals three clusters.	220
15.9. A second geom (a linear trend line) layered over the points, one line per species.	221
15.10A Brewer qualitative palette via <code>scale_color_brewer()</code> and tidy axis breaks. .	222
15.11Faceting by species puts each species in its own panel for side-by-side comparison.	223
15.12The finished plot: data, aesthetics, two geoms, a scale, facets, labels, and a clean theme.	224
15.13An UpSet plot of movie genres. Each bar is the size of one intersection; the dots below say which genres it combines. Read it as a scalable Venn diagram. .	226
15.14A clustered heatmap of scaled car measurements. Rows (cars) and columns (features) are reordered so similar profiles sit together; the color shows how far above (red) or below (blue) average each value is.	228
15.15Bill length vs. depth, colored by species.	230

List of Figures

15.16	Body-mass distribution by species.	231
15.17	The boxplot, now properly labeled.	232
16.1.	The <code>pnorm</code> function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.	237
16.2.	The <code>pnorm</code> function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.	238
16.3.	The <code>pnorm</code> function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.	239
16.4.	The <code>pnorm</code> function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.	240
16.5.	The <code>dnorm</code> function returns the height of the normal distribution at a given point.	242
16.6.	The <code>dnorm</code> function returns the height of the normal distribution at a given point.	243
16.7.	The <code>dnorm</code> function returns the height of the normal distribution at a given point.	244
16.8.	The <code>qnorm</code> function is the inverse of the <code>pnorm</code> function in that it takes a probability and gives the quantile.	245
16.9.	The <code>qnorm</code> function is the inverse of the <code>pnorm</code> function in that it takes a probability and gives the quantile.	246
16.10	The <code>qnorm</code> function is the inverse of the <code>pnorm</code> function in that it takes a probability and gives the quantile.	247
16.11	The <code>qnorm</code> function is the inverse of the <code>pnorm</code> function in that it takes a probability and gives the quantile.	248
16.12	The <code>qnorm</code> function is the inverse of the <code>pnorm</code> function in that it takes a probability and gives the quantile.	249
16.13	The <code>rnorm</code> function takes a number of samples and returns a vector of random numbers from the normal distribution (with mean=0, sd=1 as defaults)	250
17.1.	t-distributions for various degrees of freedom. The tails are fatter for smaller degrees of freedom — the visible cost of estimating the standard deviation from the data.	260
18.1.	K-means clustering takes a dataset and divides it into k clusters.	279
18.2.	Histogram of per-gene standard deviations across the time course. Most genes barely vary (the tall bars on the left); a long right tail marks the genes that change the most.	284

List of Figures

18.3. Expression profiles for the four clusters found by k-means. Each line is one gene; the x-axis runs left to right across the time course, and the y-axis is the (standardized) expression level. Genes within a cluster share a trend, which suggests they may be co-regulated.	286
19.1. A simple view of machine learning according to scikit-learn.	294
19.2. Data splitting and the train/validate/test paradigm. For models without hyperparameters, only the training and testing sets are needed. For models that must learn their <i>structure</i> (such as the <i>k</i> in k-nearest neighbours) as well as their parameters (like the slopes in linear regression), a separate validation set is essential.	297
19.3. Data simulated according to the function $f(x) = \sin(2\pi x) + N(0, 0.25)$ fitted with four different models. A) A simple linear model demonstrates <i>underfitting</i> . B) A linear model with a sin term ($y = \sin(2\pi x)$) and C) a loess model with a wide span (0.5) demonstrate <i>good fits</i> . D) A loess model with a narrow span (0.05) is a good example of <i>overfitting</i>	300
20.1. A classification tree for a breast-cancer screening example. Each internal (blue) node tests one feature; following the branch that matches a sample leads, eventually, to a leaf (green/red) that gives the predicted class.	315
20.2. Graphical representation of a random forest: many decision trees, each grown on a slightly different view of the data, combine their votes into one prediction.	316
22.1. The mlr3 ecosystem. A consistent set of objects — tasks, learners, measures, and resampling strategies — that snap together the same way for every algorithm.	333
23.1. Regression diagnostic plots. Top left: residuals vs. fitted values (look for no pattern). Top right: normal Q-Q plot (points should follow the line). Bottom left: scale-location. Bottom right: residuals vs. leverage.	353
24.1. What is the combined effect of histone marks on gene expression?	365
24.2. Boxplots of the original and the centered-and-scaled features. After scaling, every feature sits on a comparable scale.	367
24.3. Heatmap of 500 randomly sampled genes (rows) across the histone marks (columns). Blocks of similar color reveal groups of genes with similar mark patterns.	368
25.1. Samples projected onto the first two principal components, colored by cancer type and shaped by tumor/normal status.	384

List of Figures

26.1. Summarized Experiment. There are three main components, the <code>colData()</code> , the <code>rowData()</code> and the <code>assays()</code> . The accessors for the various parts of a complete <code>SummarizedExperiment</code> object match the names.	389
27.1. Anatomy of a <code>TreeSummarizedExperiment</code> object. Compared to the <code>SingleCellExperiment</code> objects, <code>TreeSummarizedExperiment</code> has five additional slots. rowTree : the hierarchical structure on the rows of the assays. rowLinks : the link information between rows of the assays and the <code>rowTree</code> . colTree : the hierarchical structure on the columns of the assays. colLinks : the link information between columns of the assays and the <code>colTree</code> . referenceSeq (optional): the reference sequence data per feature (row).	410
28.1. Distribution of CTCF peak widths. The sharp spike reflects the peak caller's fixed-width output.	441
28.2. Distribution of CTCF peak signal values. Most peaks are weak; a long tail of strong sites stretches to the right.	442
28.3. Per-base coverage from randomly placed reads across the first 10,000 bases of a simulated chromosome.	449
28.4. How often each coverage depth occurs across the simulated chromosome. . .	450
29.1. The structure of a <code>GRanges</code> object, which behaves a bit like a vector of ranges, although the analogy is not perfect. A <code>GRanges</code> object is composed of the “Ranges” part the lefthand box, the “metadata” columns (the righthand box), and a “seqinfo” part that describes the names and lengths of associated sequences. Only the “Ranges” part is required. The figure also shows a few of the “accessors” and approaches to subsetting a <code>GRanges</code> object.	459
29.2. The structure of a <code>GRangesList</code> , which is a <code>list</code> of <code>GRanges</code> objects. While the analogy is not perfect, a <code>GRangesList</code> behaves a bit like a list. Each element in the <code>GRangesList</code> is a <code>GRanges</code> object. A common use case for a <code>GRangesList</code> is to store a list of transcripts, each of which have exons as the regions in the <code>GRanges</code>	471
29.3. A graphical representation of range operations demonstrated on a gene model.	482
31.1. Chromatin accessibility methods, compared. Representative DNA fragments generated by each assay are shown, with end locations within chromatin defined by colored arrows. Bar diagrams represent data signal obtained from each assay across the entire region. The footprint created by a transcription factor (TF) is shown for ATAC-seq and DNase-seq experiments.	651

List of Figures

31.2. Multimodal chromatin comparisons. From (Buenrostro et al. 2013), Figure 4. (a) CTCF footprints observed in ATAC-seq and DNase-seq data, at a specific locus on chr1. (b) Aggregate ATAC-seq footprint for CTCF (motif shown) generated over binding sites within the genome. (c) CTCF predicted binding probability inferred from ATAC-seq data, position weight matrix (PWM) scores for the CTCF motif, and evolutionary conservation (PhyloP). The right-most column is the CTCF ChIP-seq data (ENCODE) for this GM12878 cell line, demonstrating high concordance with predicted binding probability.	652
31.3. A BAM file in text form. The output of <code>samtools view</code> is the text representation of a BAM file (called SAM format). Bioconductor and many other tools use BAM files for input. BAM files also often have a companion index <code>.bai</code> file that enables random access into the file, so you can read just one genomic region without reading the whole thing.	653
31.4. Reads per chromosome. In our example data, only chromosomes 21 and 22 are present.	656
31.5. Read counts normalized by chromosome length. Not a critical plot here, but with a full genome it shows each chromosome’s relative contribution given its size.	657
31.6. Distribution of per-base coverage on chromosomes 21 and 22 (log-scaled y axis). Most bases have low coverage; the long tail of high-coverage bases corresponds to the peaks where reads pile up.	659
31.7. Relationship between fragment length and nucleosome number, from Buenrostro et al. (2013). Short fragments come from nucleosome-free DNA; longer fragments span one or more nucleosomes, producing periodic peaks roughly one nucleosome (~147 bp) apart.	660
31.8. Fragment-length (insert-size) density for our ATAC-seq reads.	661
31.9. Fragment-length distribution drawn with <code>ggplot2</code> , zoomed to fragments under 500 bp. The large peak of short fragments is the nucleosome-free signal; the bump near ~200 bp is mononucleosome fragments.	662
31.10 Average signal around the TSS. Nucleosome-free reads peak sharply at position 0 (the TSS); mononucleosome reads stay comparatively flat.	664
31.11 Per-promoter signal at the TSS. Mononucleosome reads on the left, nucleosome-free on the right; both scaled to the same range. Each row is one promoter.	665
32.1. The <code>SingleCellExperiment</code> object.	674

List of Figures

32.2. A plot of the mean vs variance of log-expression for all the genes in the dataset, with the highly variable genes highlighted in red. Here, the x-axis represents the mean log-expression of each gene, and the y-axis represents the variance of log-expression. The blue curve represents the trend of variance as a function of mean expression, while the red points represent the highly variable genes.	678
32.3.	682
32.4.	683
32.5.	684
32.6.	686
32.7.	689
A.1. This is an overview of how git works along with the commands that make it tick. See this video	703

List of Tables

7.1. Atomic (simplest) data types in R.	78
16.1. Functions for the normal distribution	236
29.1. Classes within the GenomicRanges package. Each class has a slightly different use case.	455
29.2. Methods for accessing, manipulating single objects	456
29.3. Methods for comparing and combining multiple GenomicRanges-class objects	457
31.1. Commonly used Bioconductor and their high-level use cases.	654
C.1. Table AI Tools for Bioinformatics Learning	710

Preface

The goal of the material is NOT to teach you R or Bioconductor, but rather to provide enough groundwork to enable you to learn R and Bioconductor on your own. The idea is that we learn faster and better by developing “schemata” that allow us to organize and understand new information. When a new concept is introduced, it is easier to understand if it can be related to something we already know (see Figure 1).

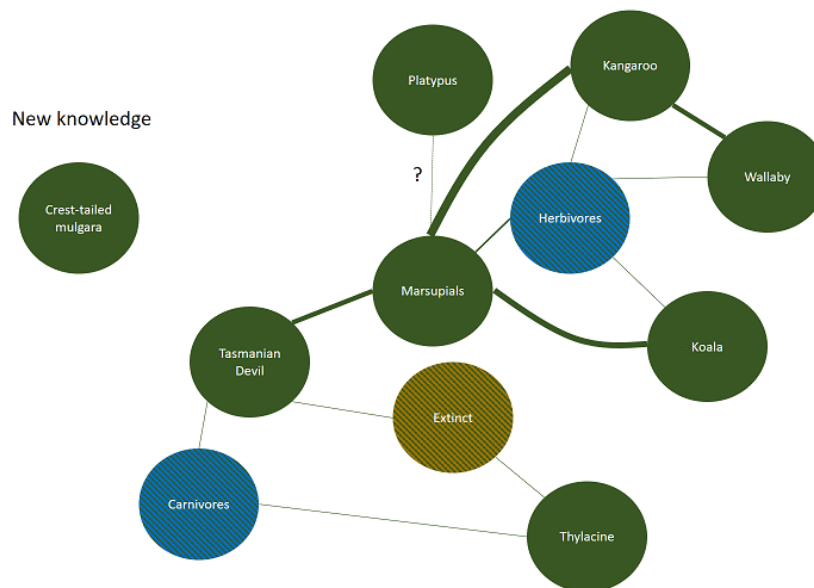


Figure 1.: Schema theory of learning. While probably not entirely applicable to how adults learn, it does highlight the importance of building on prior knowledge. When new knowledge is introduced, it is more quickly stored into long-term memory if strongly-related concepts are already available for connection.

This book is a collection of resources meant to help build your data science, statistical, and computational schemata. It is meant to be largely self-directed, but for those looking to teach data science, it can also be used as a guide for structuring a course. Material is a bit variable in terms of difficulty, prerequisites, and format which is a reflection of the

What are the goals?

organic creation of the material. See below for additional thoughts on adult learning and how it relates to this material.

What are the goals?

We can often get lost in the weeds of technical details of lessons, focusing only on the syntax and semantics of R and Bioconductor. Let's take a step back and consider the big picture. The goals are to:

1. Have a foundation for reading and writing R code to solve problems and understand data.
2. Be able to find and use online resources including AI and tutorials to solve problems and learn new concepts.
3. Be able to effectively communicate with others about R code and data science concepts.
4. Most importantly, to develop the confidence to become a self-directed learner, experimenting with concepts and practice of data science to address real-world problems.

Adult learners

Adult Learning Theory, also known as Andragogy, is the concept and practice of designing, developing, and delivering instructional experiences for adult learners. It is based on the belief that adults learn differently than children, and thus, require distinct approaches to engage, motivate, and retain information (Center 2016). The term was first introduced by Malcolm Knowles, an American educator who is known for his work in adult education (Knowles et al. 2005).

One of the fundamental principles of Adult Learning Theory is that adults are self-directed learners. This means that we prefer to take control of our own learning process and set personal goals for themselves. We are motivated by our desire to solve problems or gain knowledge to improve our lives (see Figure 2). As a result, educational content for adults should be relevant and applicable to real-life situations. Furthermore, adult learners should be given opportunities to actively engage in the learning process by making choices, setting goals, and evaluating their progress.

Another key aspect of Adult Learning Theory is the role of experience. We bring a wealth of experience to the learning process, which serves as a resource for new learning. We often have well-established beliefs, values, and mental models that can influence our willingness to accept new ideas and concepts. Therefore, it is essential to acknowledge and respect our

Adult learners

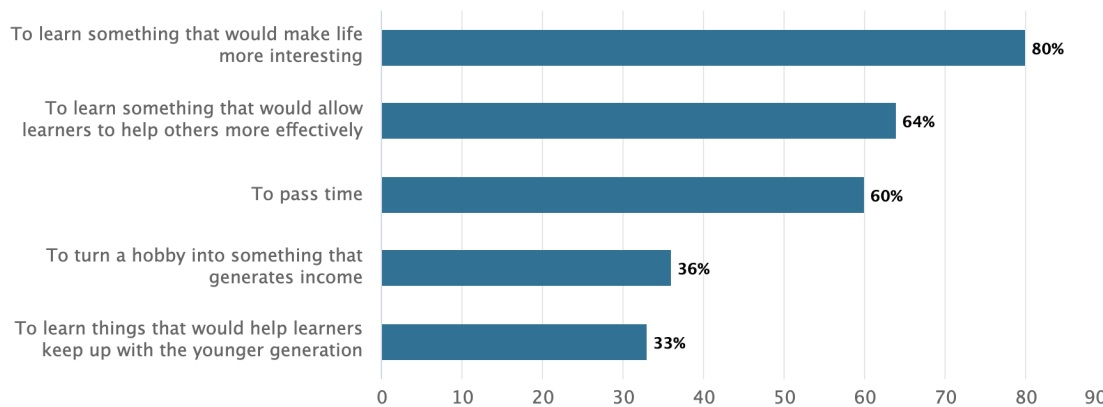


Figure 2.: Why do adults choose to learn something?

shared and unique past experiences and create an environment where we all feel comfortable sharing our perspectives.

To effectively learn as a group of adult learners, it is crucial to establish a collaborative learning environment that promotes open communication and fosters trust among participants. We all appreciate and strive for a respectful and supportive atmosphere where we can express our opinions without fear of judgment. Instructors should help facilitate discussions, encourage peer-to-peer interactions, and incorporate group activities and collaboration to capitalize on the collective knowledge of participants.

Additionally, adult learners often have multiple responsibilities outside of the learning environment, such as work and family commitments. As a result, we require flexible learning opportunities that accommodate busy schedules. Offering a variety of instructional formats, such as online modules, self-paced learning, or evening classes, can help ensure that adult learners have access to education despite any time constraints.

Adult learners benefit from a learner-centered approach that focuses on the individual needs, preferences, and interests of each participant can greatly enhance the overall learning experience. In addition, we tend to be more intrinsically motivated to learn when we have a sense of autonomy and can practice and experiment (see Figure 3) with new concepts in a safe environment.

Understanding Adult Learning Theory and its principles can significantly enhance the effectiveness of teaching and learning as adults. By respecting our autonomy, acknowledging our experiences, creating a supportive learning environment, offering flexible learning opportunities, and utilizing diverse teaching methods, we can better cater to the unique needs and preferences of adult learners.

AI is part of the solution

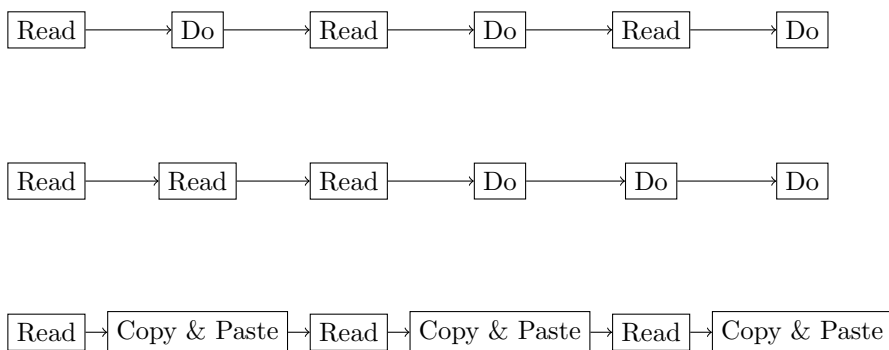


Figure 3.: How to stay stuck in data science (or anything). The “Read-Do” loop tends to deliver the best results. Too much reading between doing can be somewhat effective. Reading and simply copy-paste is probably the least effective. When working through material, experiment. Try to break things. Incorporate your own experience or applications whenever possible.

In practice, that means that we will not be prescriptive in our approach to teaching data science. We will not tell you what to do, but rather we will provide you with a variety of options and you can choose what works best for you. We will also provide you with a variety of resources and you can choose where to focus your time. Given that we cannot possibly cover everything, we will provide you with a framework for learning and you can fill in the gaps as you see fit. A key component of our success as adult learners is to gain the confidence to ask questions and problem-solve on our own.

AI is part of the solution

Artificial Intelligence (AI) is becoming an increasingly important tool in education, and it can be a powerful ally in our quest to learn data science. As it turns out, AI is excellent at helping us learn. It can provide tailored explanations, generate practice problems, and even simulate real-world scenarios for us to work through. AI can also help us find resources, summarize complex topics, and even provide feedback on our work. However, it is important to remember that AI is a tool, not a replacement for our own learning. We still need to engage with the material, ask questions, and seek out additional resources when needed. AI can help us learn more efficiently, but it cannot do the learning for us.

I encourage you to use AI tools to help you learn and to even do work for you when appropriate; AI is an excellent partner in analysis and programming. However, I also encourage you to be critical of the information that AI provides. AI is not perfect and can make mistakes. Throughout this course and book, use AI to dive deeper into the material,

AI is part of the solution

to provide additional explanations, and to generate additional depth and breadth to the material.

Part I.

Introduction

1. About R

Imagine you’ve just gotten back a spreadsheet with expression values for 20,000 genes across 100 samples, and your advisor wants to know which genes differ between treated and untreated cells — by Friday. Clicking through that by hand is hopeless. This is exactly the kind of problem R was built for: load the data, filter and reshape it, run the statistics, and draw a publication-quality figure, all in a handful of lines you can save, rerun, and share.

This chapter is the gentlest possible start. We won’t write much code yet. Instead, we’ll answer two questions that matter before you type a single command: *what is R*, and *why is it worth your time as a biologist?* Once you’re convinced it’s worth learning, the next chapter gets you set up with RStudio and your first commands.

1.1. What you’ll learn

- Describe what R is and where it came from, in plain language.
- Explain why R is so widely used in genomics and bioinformatics.
- Recognize the trade-offs — the things R is *not* good at — so you know when to reach for another tool.
- Understand what “open source” means and why it matters for reproducible science.

1.2. What is R?

R is a programming language and free software environment built for working with data: cleaning it, analyzing it, modeling it, and turning it into graphics. You write short instructions, R carries them out, and you see the results immediately. That’s the whole loop.

R didn’t appear out of nowhere. It’s a dialect of an older language called S, developed in the 1970s at Bell Laboratories. The first version of R was written by Robert Gentleman and Ross Ihaka and released in 1995 (Ihaka’s own [slide deck](#) tells the story well). Since then a volunteer group called the R Core Team — statisticians and computer scientists —

1. About R

has maintained and improved it, releasing a new version roughly every year. So when you use R, you’re standing on three decades of careful work, and you’re not paying a cent for it.

R has become especially popular in genomics and bioinformatics, and that popularity has held up over time even as other languages have risen alongside it (Figure 1.1).

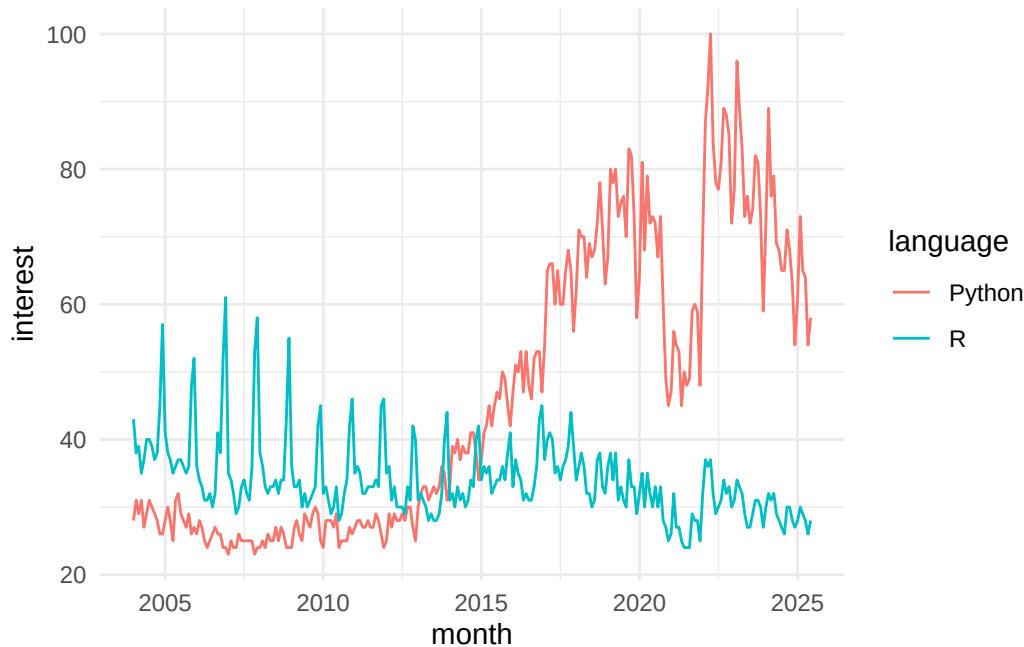


Figure 1.1.: Worldwide search interest in R and Python over time, from [Google Trends](#). Search interest is a rough proxy for general popularity — it does not capture how heavily specific fields lean on each language. Bioinformatics, in particular, remains deeply invested in R.

The figure plots search interest for R and Python over the years. Python is the more general-purpose language and gets more total searches, but don’t read too much into that: a search-popularity chart can’t see *which* communities depend on *which* tool. In bioinformatics specifically — think Bioconductor, the ecosystem this book leans on heavily — R remains a first-class citizen, and that’s a big part of why we teach it here.

1.3. Why use R?

So why has R earned such a devoted following among biologists and data scientists? A few reasons stand out.

It's free and open source. R costs nothing to download, use, modify, or share, and a large worldwide community keeps building it. We'll come back to what "open source" really means in the section on R's license below.

There's a package for almost everything. Thousands of add-on packages extend R's reach, including a deep catalogue of tools written specifically for genomics and bioinformatics. Many have been tested by thousands of users — and, like R itself, they're free.

It's built for wrangling messy data. Real biological data arrives messy. R gives you powerful, well-worn tools to clean, reshape, and summarize even large datasets — not always *easy*, but reliably *possible*.

The graphics are publication quality. R's plotting tools produce figures you can drop straight into a paper or poster, and you can tune nearly every detail to say exactly what you mean.

Your work is reproducible. Because an R analysis is a *script* — a written record of every step — you (or a colleague, or a reviewer) can rerun it and get the same answer. Point-and-click tools rarely leave that kind of trail.

It runs everywhere and scales. R works on Windows, Mac, and Linux, and it can talk to other languages like FORTRAN and C when you need extra speed. The same code serves a quick afternoon analysis or a major project.

That last point is worth a concrete example. I can write and test an analysis on my Mac laptop, then install the *exact same* code on our 20,000-core compute cluster and run it in parallel across a hundred samples, watch the jobs as they go, and have R write the results into a database when it finishes. One language carried me from a back-of-the-envelope idea to a full-scale genomics pipeline without rewriting anything.

i Open source, in one sentence

"Open source" means the program's underlying code is published for anyone to read, change, and redistribute. For science that's not a nicety — it's what lets others inspect exactly how a result was produced and verify it for themselves. R, and the Bioconductor packages you'll meet later, are open source top to bottom.

1.4. Why not use R?

No tool is perfect, and it's healthier to know R's rough edges going in than to be blindsided by them later. R is not always the best tool for every job. It can be slower than languages like C for certain heavy number-crunching, and its documentation can be terse or assume more than a beginner knows. Some community-contributed packages are excellent; others are unmaintained or buggy, so a little judgment helps when you pick one. R also has no single, polished point-and-click interface — that's a feature for reproducibility, but it does mean you commit to writing code. And other languages, notably Python, Julia, and Rust, are increasingly used in biological data science, so R is one strong choice among several rather than the only one.

💡 R will not hold your hand — and that's okay

You'll hear people joke that R doesn't hold your hand; sometimes it *slaps* it instead, with a cryptic error message. Here's the reassuring truth: every R user, including experienced ones, sees errors constantly. An error is not a verdict on you. It's just R saying "I didn't understand that," and the fix is usually a typo, a missing comma, or a quick web search away. We'll return to reading errors calmly in later chapters — for now, just know that getting stuck is normal and temporary.

1.5. R license and the open source ideal

R is free — genuinely, completely free — and distributed under the GNU General Public License. Among other things, that license lets anyone:

- Download the source code.
- Modify it however they like.
- Redistribute the modified code (even for money), as long as it stays under the same GNU license.

That last condition is the clever part: improvements have to flow back to the community on the same open terms. The practical upshot for you is reassuring — R will always be available, will always be open, and can keep growing without being locked behind a paywall or controlled by a single company.

1.6. Where this is going

R is a *programming language*, which means you get things done by writing code rather than clicking buttons. That can feel intimidating at first, but it’s also the source of everything we just praised: scripts you can rerun, functions you can reuse, and analyses you can share. You can work with R two ways — *interactively*, typing one command at a time and seeing each result right away, or as a *script*, saving a whole sequence of commands to run together. You’ll use both.

In the next chapter, on RStudio, we’ll install RStudio — a friendly home base for writing and running R — and take a tour of its panels. The chapter after that gets you up and running with your first real R code, where you’ll build a pair of virtual dice and roll them.

Try it

You don’t need anything installed to look ahead: open the [Google Trends page](#) behind Figure 1.1 and add a third language to the comparison. It’s a gentle reminder that the tools you choose are part of a living, shifting community.

1.7. Summary

You now have the lay of the land. R is a free, open-source language with three decades of history, a vast library of packages, strong graphics, and a deep commitment to reproducibility — which is exactly why it’s a mainstay of genomics and bioinformatics. You also know its limits: it can be slow for some tasks, its error messages can sting, and it’s one good option among several. With the “why” settled, you’re ready to set up the tools and start writing code.

1.8. Reflection

Take a moment to check your understanding before moving on. You don’t need to write anything down — just see whether you can answer each in a sentence or two.

1. In your own words, what does it mean that R is “open source,” and why does that matter for a published scientific result?
2. Name two reasons a biologist might choose R over a point-and-click spreadsheet program for analyzing experimental data.

1. About R

3. The Google Trends figure shows Python with more overall search interest than R. Why is that *not* a good reason to conclude R is unimportant in bioinformatics?
4. What is one situation where R might *not* be the best tool, and what would you consider reaching for instead?

2. RStudio

RStudio is an integrated development environment (IDE) for R. It provides a graphical user interface (GUI) for R, making it easier to write and execute R code. RStudio also provides several other useful features, including a built-in console, syntax-highlighting editor, and tools for plotting, history, debugging, workspace management, and workspace viewing. RStudio is available in both free and commercial editions; the commercial edition provides some additional features, including support for multiple sessions and enhanced debugging.

2.1. What you'll learn

- Install R and RStudio and launch the RStudio IDE.
- Identify the four panes of the RStudio interface and what each is for.
- Recognize that only the **console** talks to R directly — the other panes are a convenience layer on top.
- Customize the layout to suit your screen and preferences.
- Name the main alternatives to RStudio (Jupyter, VS Code, Positron, plain command-line R) and when you might reach for them.

2.2. Getting started with RStudio

To get started with RStudio, you first need to install both R and RStudio on your computer. Follow these steps:

1. Download and install R from the [official R website](#).
2. Download and install RStudio from the [official RStudio website](#).
3. Launch RStudio. You should see the RStudio interface with four panels.

2. RStudio

R versions

RStudio works with all versions of R, but it is recommended to use the latest version of R to take advantage of the latest features and improvements. You can check your R version by running `version` (no parentheses) in the R console. You can check the latest version of R on the [R-project website](#).

2.3. The RStudio Interface

RStudio's interface consists of four panels (see Figure 2.1):

- **Console** This panel displays the R console, where you can enter and execute R commands directly. The console also shows the output of your code, error messages, and other information.
- **Source** This panel is where you write and edit your R scripts. You can create new scripts, open existing ones, and run your code from this panel.
- **Environment** This panel displays your current workspace, including all variables, data objects, and functions that you have created or loaded in your R session.
- **Plots, Packages, Help, and Viewer** These panels display plots, installed packages, help files, and web content, respectively.

Do I need to use RStudio?

No. You can use R without RStudio. However, RStudio makes it easier to write and execute R code, and it provides several useful features that are not available in the basic R console. Note that the only part of RStudio that is actually interacting with R directly is the console. The other panels are simply providing a GUI that enhances the user experience.

Customizing the RStudio Interface

You can customize the layout of RStudio to suit your preferences. To do so, go to **Tools > Global Options > Appearance**. Here, you can change the theme, font size, and panel layout. You can also resize the panels as needed to gain screen real estate (see Figure 2.2).

2. RStudio

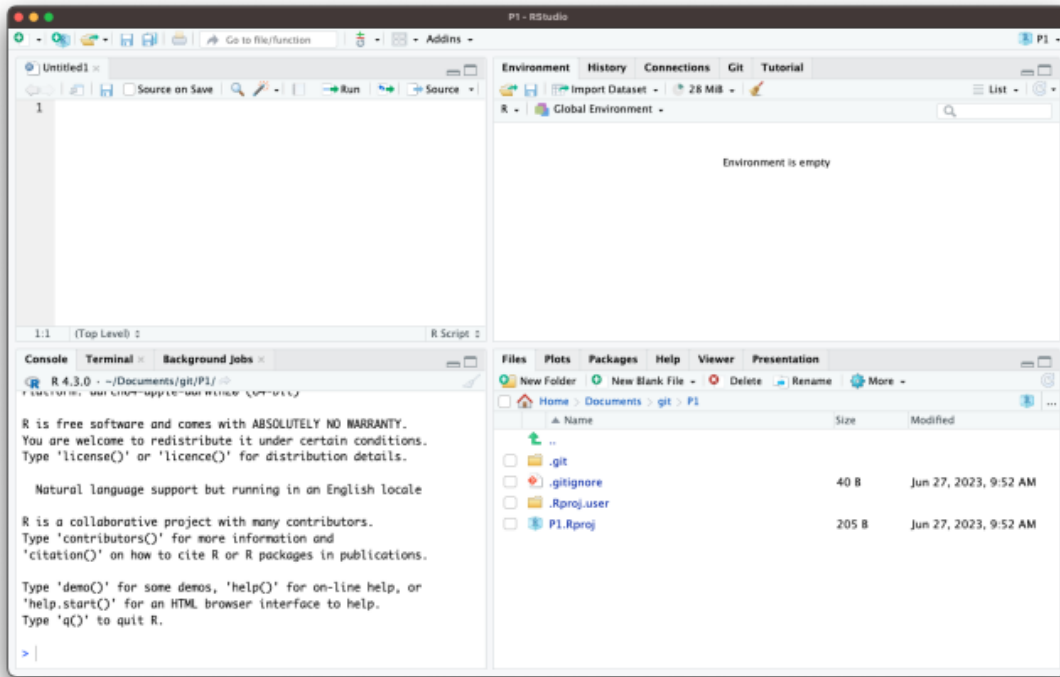


Figure 2.1.: The RStudio interface. In this layout, the **source** pane is in the upper left, the **console** is in the lower left, the **environment** panel is in the top right and the **viewer/help/files** panel is in the bottom right.

2. RStudio

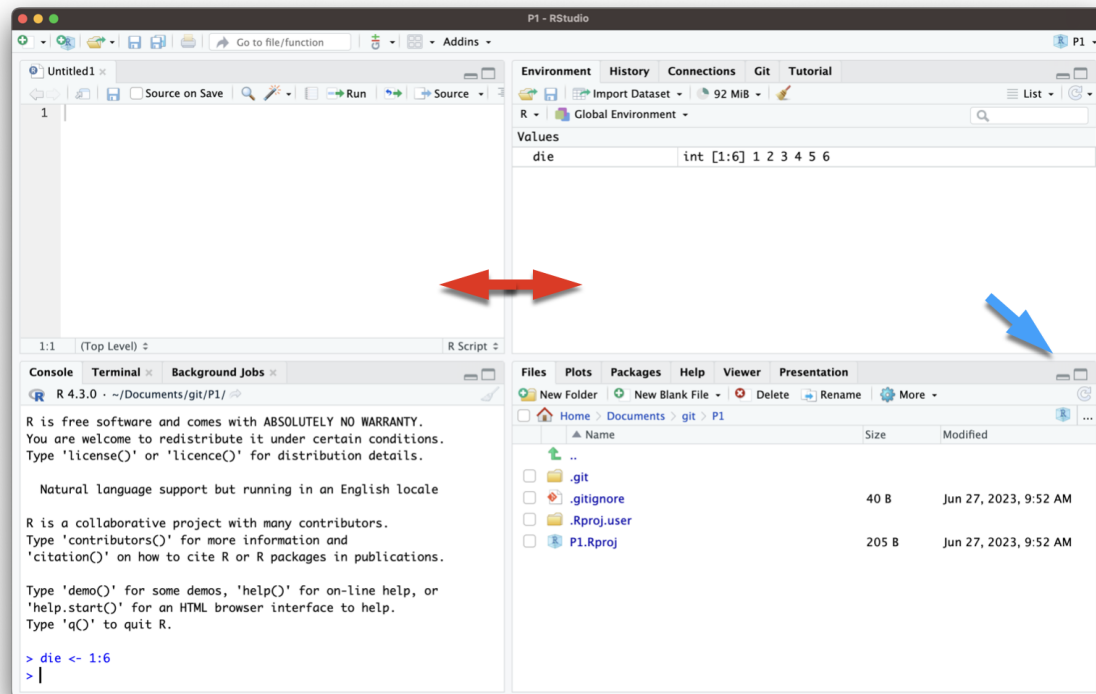


Figure 2.2.: Dealing with limited screen real estate can be a challenge, particularly when you want to open another window to, for example, view a web page. You can resize the panes by sliding the center divider (red arrows) or by clicking on the minimize/maximize buttons (see blue arrow).

2. *RStudio*

In summary, R and RStudio are powerful tools for genomics data analysis. By understanding the advantages of using R and RStudio and familiarizing yourself with the RStudio interface, you can efficiently analyze and visualize your data. In the following chapters, we will delve deeper into the functionality of R, Bioconductor, and various statistical methods to help you gain a comprehensive understanding of genomics data analysis.

2.4. Alternatives to RStudio

While RStudio is a popular choice for R development, there are several alternatives you can consider:

1. **Jupyter Notebooks:** Jupyter Notebooks provide an interactive environment for writing and executing R code, along with rich text support for documentation. You can use the IRKernel to run R code in Jupyter.

2. RStudio

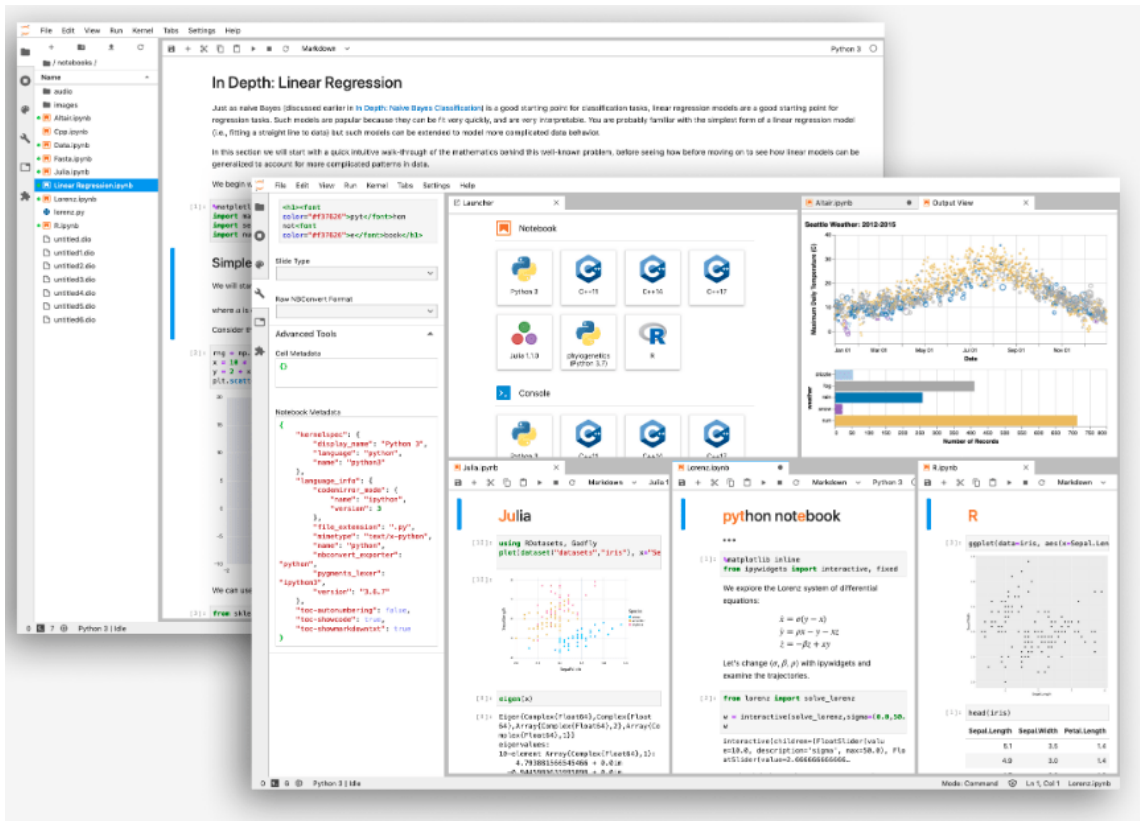


Figure 2.3.: Jupyter Notebook interface. This is an interactive environment for writing and executing R code, along with rich text support for documentation.

2. **Visual Studio Code:** With the R extension for Visual Studio Code, you can write and execute R code in a lightweight editor. This setup provides features like syntax highlighting, code completion, and integrated terminal support.

2. RStudio

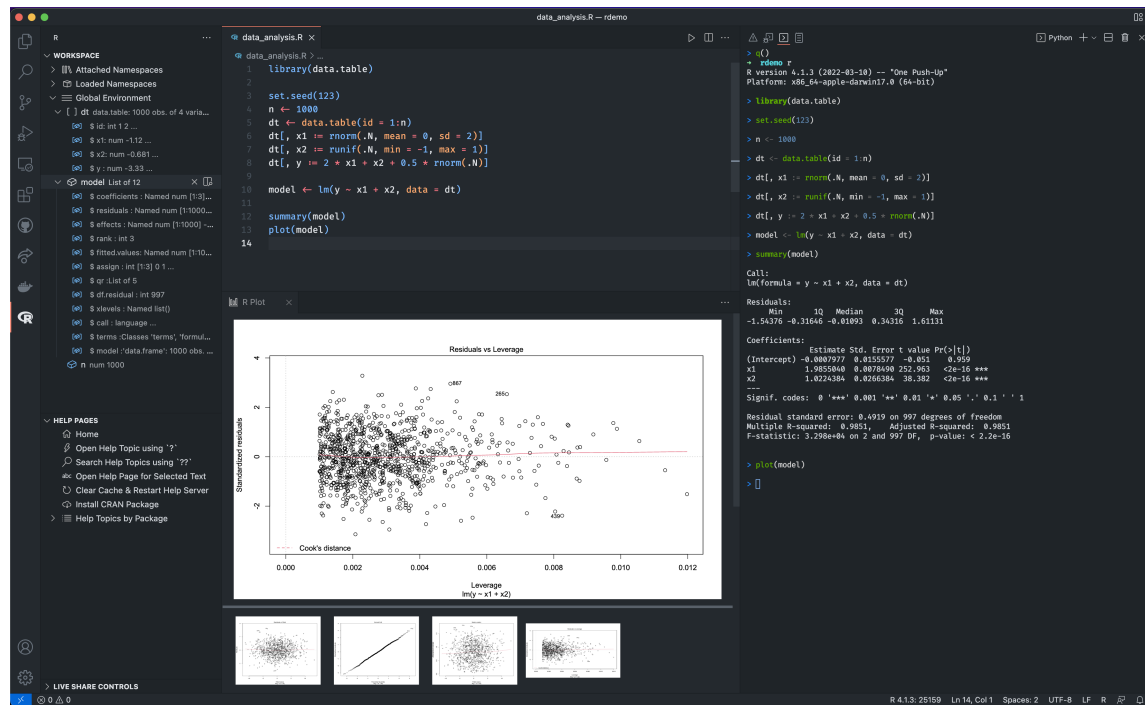


Figure 2.4.: Visual Studio Code (VSCode) with the R extension. This is a lightweight alternative to RStudio that provides syntax highlighting, code completion, and integrated terminal support.

3. **Positron Workbench**: This is a commercial IDE that supports R and Python. It provides a similar interface to RStudio but with additional features for data science workflows, including support for multiple languages and cloud integration.

2. RStudio

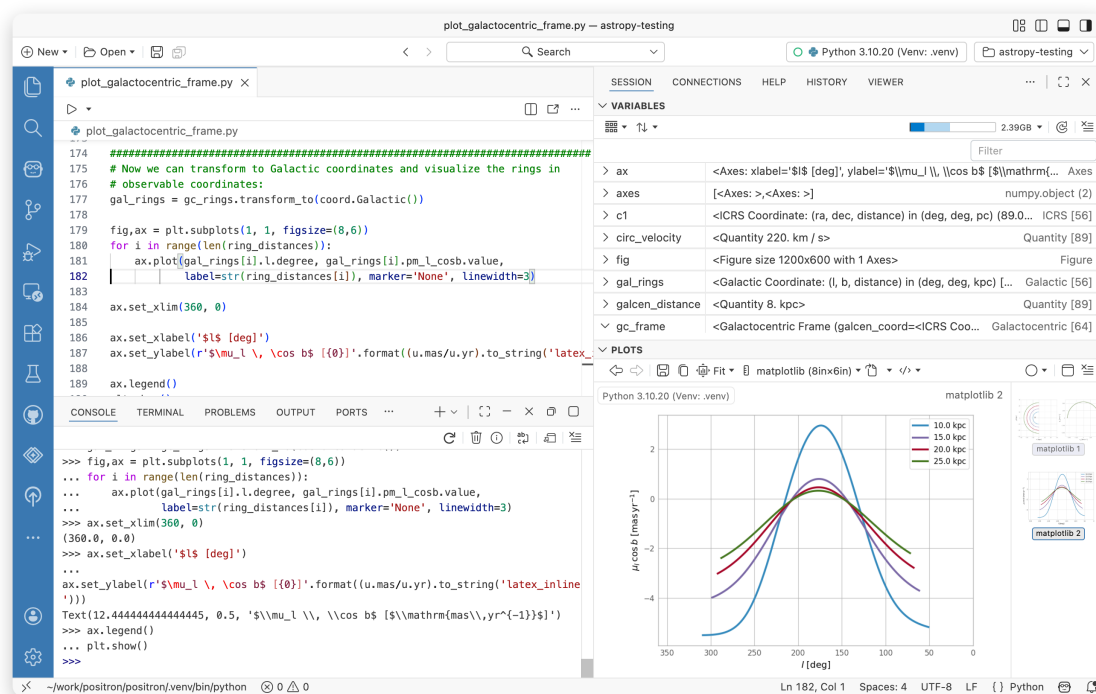


Figure 2.5.: Positron Workbench interface. This IDE supports R and Python, providing a similar interface to RStudio with additional features for data science workflows.

4. **Command Line R:** For those who prefer a minimalistic approach, you can use R directly from the command line. This method lacks the GUI features of RStudio but can be efficient for quick tasks, scripting, automation, or when working on remote servers.

Each of these alternatives has its own strengths and weaknesses, so you may want to try a few to see which one best fits your workflow. All are available for free, and you can install them alongside RStudio if you wish to use multiple environments. Each can be installed in Windows, Mac, and Linux.

3. R mechanics

3.1. What you'll learn

- Start R and interact with it by typing at the console.
- Tell the two basic kinds of input apart: **assignments** (storing a value) and **expressions** (computing a result).
- Use R as a calculator, applying its arithmetic operators and order of operations.
- Assign values to named **objects** with the `<-` operator, and inspect, change, and remove them.
- Apply R's rules for naming objects.
- Look up documentation with `help()` and `?`.

3.2. Starting R

We've installed R and RStudio. Now, let's start R and get going. How to start R depends a bit on the operating system (Mac, Windows, Linux) and interface. In this course, we will largely be using an Integrated Development Environment (IDE) called *RStudio*, but there is nothing to prohibit using R at the command line or in some other interface (and there are a few).

3.3. *RStudio*: A Quick Tour

The RStudio interface has multiple panes. All of these panes are simply for convenience except the “Console” panel, typically in the lower left corner (by default). The console pane contains the running R interface. If you choose to run R outside RStudio, the interaction will be *identical* to working in the console pane. This is useful to keep in mind as some environments, such as a computer cluster, encourage using R without RStudio.

- Panes
- Options
- Help

3. R mechanics

- Environment, History, and Files

3.4. Interacting with R

The only meaningful way of interacting with R is by typing into the R console. At the most basic level, anything that we type at the command line will fall into one of two categories:

1. Assignments

```
x = 1  
y <- 2
```

2. Expressions

```
1 + pi + sin(42)
```

```
[1] 3.225071
```

The assignment type is obvious because either the `<-` or `=` are used. Note that when we type expressions, R will return a result. In this case, the result of R evaluating `1 + pi + sin(42)` is 3.2250711.

The standard R prompt is a “>” sign. When present, R is waiting for the next expression or assignment. If a line is not a complete R command, R will continue the next line with a “+”. For example, typing the following with a “Return” after the second “+” will result in R giving back a “+” on the next line, a prompt to keep typing.

```
1 + pi +  
sin(3.7)
```

```
[1] 3.611757
```

R can be used as a glorified calculator by using R expressions. Mathematical operations include:

- Addition: `+`
- Subtraction: `-`
- Multiplication: `*`
- Division: `/`
- Exponentiation: `^`
- Modulo: `%%`

3. R mechanics

The `^` operator raises the number to its left to the power of the number to its right: for example `3^2` is 9. The modulo returns the remainder of the division of the number to the left by the number on its right, for example 5 modulo 3 or `5 %% 3` is 2.

3.4.1. Expressions

```
5 + 2
28 %% 3
3^2
5 + 4 * 4 + 4 ^ 4 / 10
```

Note that R follows order-of-operations and groupings based on parentheses.

```
5 + 4 / 9
(5 + 4) / 9
```

3.4.2. Assignment

While using R as a calculator is interesting, to do useful and interesting things, we need to assign *values* to *objects*. To create objects, we need to give it a name followed by the assignment operator `<-` (or, entirely equivalently, `=`) and the value we want to give it:

```
weight_kg <- 55
```

`<-` is the assignment operator. Assigns values on the right to objects on the left, it is like an arrow that points from the value to the object. Using an `=` is equivalent (in nearly all cases). Learn to use `<-` as it is good programming practice.

i What about `<-` and `=` for assignment?

The `<-` and `=` both work fine for assignment. You'll see both used and it is up to you to choose a standard for yourself. However, some programming communities, such as Bioconductor, will strongly suggest using the `<-` as it is clearer that it represents an *assignment* operation.

Objects can be given any name such as `x`, `current_temperature`, or `subject_id` (see below). You want your object names to be explicit and not too long. They cannot start with a number (`2x` is not valid but `x2` is). R is case sensitive (e.g., `weight_kg` is different

3. R mechanics

from `Weight_kg`). There are some names that cannot be used because they represent the names of fundamental functions in R (e.g., `if`, `else`, `for`, see [here](#) for a complete list). In general, even if it's allowed, it's best to not use other function names, which we'll get into shortly (e.g., `c`, `T`, `mean`, `data`, `df`, `weights`). When in doubt, check the help to see if the name is already in use. It's also best to avoid dots (`.`) within a variable name as in `my.dataset`. It is also recommended to use nouns for variable names, and verbs for function names.

When assigning a value to an object, R does not print anything. You can force to print the value by typing the name:

```
weight_kg
```

```
[1] 55
```

Now that R has `weight_kg` in memory, which R refers to as the “global environment”, we can do arithmetic with it. For instance, we may want to convert this weight in pounds (weight in pounds is 2.2 times the weight in kg).

```
2.2 * weight_kg
```

```
[1] 121
```

We can also change a variable's value by assigning it a new one:

```
weight_kg <- 57.5  
2.2 * weight_kg
```

```
[1] 126.5
```

This means that assigning a value to one variable does not change the values of other variables. For example, let's store the animal's weight in pounds in a variable.

```
weight_lb <- 2.2 * weight_kg
```

and then change `weight_kg` to 100.

3. R mechanics

```
weight_kg <- 100
```

What do you think is the current content of the object `weight_lb`, 126.5 or 220?

You can see what objects (variables) are stored by viewing the Environment tab in Rstudio. You can also use the `ls()` function. You can remove objects (variables) with the `rm()` function. You can do this one at a time or remove several objects at once. You can also use the little broom button in your environment pane to remove everything from your environment.

```
ls()
rm(weight_lb, weight_kg)
ls()
```

What happens when you type the following, now?

```
weight_lb # oops! you should get an error because weight_lb no longer exists!
```

3.5. Rules for Names in R

R allows users to assign names to objects such as variables, functions, and even dimensions of data. However, these names must follow a few rules.

- Names may contain any combination of letters, numbers, underscore, and “.”
- Names may not start with numbers, underscore.
- R names are case-sensitive.

Examples of valid R names include:

```
pi
x
camelCaps
my_stuff
MY_Stuff
this.is.the.name.of.the.man
ABC123
abc1234asdf
.hi
```

3.6. About R functions

When you see a name followed by parentheses (), you are likely looking a name that represents an R function (or method, but we'll sidestep that distinction for now). Examples of R functions include `print()`, `help()`, and `ls()`. We haven't seen examples yet, but when a name is followed by [], that name represents a variable of some kind and the [] are used for "subsetting" the variable. So:

- Name followed by () is a function.
- Name with [] means a variable that is being subset.

In many cases, when you see a new function used, you may not know what it does. The R `help()` function takes the name of another function and gives back the R help document for that function if there is one. The next section reviews that technique.

3.7. Resources for Getting Help

There is extensive built-in help and documentation within R. A separate page contains a collection of [additional resources](#).

If the name of the function or object on which help is sought is known, the following approaches with the name of the function or object will be helpful. For a concrete example, examine the help for the `print` method.

```
help(print)
help('print')
?print
```

There are also tons of online resources that Google will include in searches if online searching feels more appropriate.

I strongly recommend using `help("newfunction")` for all functions that are new or unfamiliar to you.

There are also many open and free resources and reference guides for R.

- [Quick-R](#): a quick online reference for data input, basic statistics and plots
- R reference card [PDF](#) by Tom Short
- Rstudio [cheatsheets](#)

3.8. Reflection

- Can you recognize the difference between *assignment* and *expressions* when interacting with R?
- Can you demonstrate an assignment to a variable?
- Do you know the rules for “names” in R?
- Are you able to get help using the R `help()` function?
- Do you know that functions are recognizable as names followed by `()`?

4. Up and Running with R

In this chapter, we’re going to get an introduction to the R language, so we can dive right into programming. We’re going to create a pair of virtual dice that can generate random numbers. No need to worry if you’re new to programming. We’ll return to many of the concepts here in more detail later.

To simulate a pair of dice, we need to break down each die into its essential features. A die can only show one of six numbers: 1, 2, 3, 4, 5, and 6. We can capture the die’s essential characteristics by saving these numbers as a group of values in the computer. Let’s save these numbers first and then figure out a way to “roll” our virtual die.

4.1. The R User Interface

The RStudio interface is simple. You type R code into the bottom line of the RStudio console pane and then click Enter to run it. The code you type is called a *command*, because it will command your computer to do something for you. The line you type it into is called the *command line*.

When you type a command at the prompt and hit Enter, your computer executes the command and shows you the results. Then RStudio displays a fresh prompt for your next command. For example, if you type `1 + 1` and hit Enter, RStudio will display:

```
> 1 + 1
[1] 2
>
```

You’ll notice that a `[1]` appears next to your result. R is just letting you know that this line begins with the first value in your result. Some commands return more than one value, and their results may fill up multiple lines. For example, the command `100:130` returns 31 values; it creates a sequence of integers from 100 to 130. Notice that new bracketed numbers appear at the start of the second and third lines of output. These numbers just mean that the second line begins with the 14th value in the result, and the third line begins with the 25th value. You can mostly ignore the numbers that appear in brackets:

4. Up and Running with R

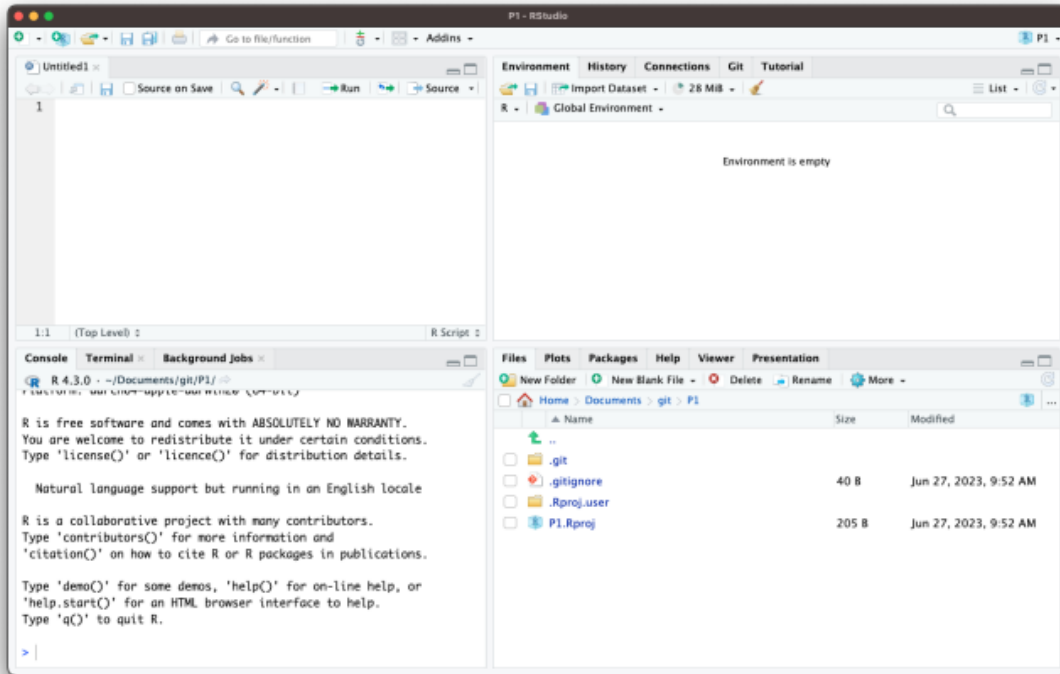


Figure 4.1.: Your computer does your bidding when you type R commands at the prompt in the bottom line of the console pane. Don't forget to hit the Enter key. When you first open RStudio, the console appears in the pane on your left, but you can change this with **File > Tools > Global Options** in the menu bar.

4. Up and Running with R

```
> 100:130
[1] 100 101 102 103 104 105 106 107 108 109 110 111 112
[14] 113 114 115 116 117 118 119 120 121 122 123 124 125
[25] 126 127 128 129 130
```

Tip

The colon operator (:) returns every integer between two integers. It is an easy way to create a sequence of numbers.

When do we compile?

In some languages, like C, Java, and FORTRAN, you have to compile your human-readable code into machine-readable code (often 1s and 0s) before you can run it. If you've programmed in such a language before, you may wonder whether you have to compile your R code before you can use it. The answer is no. R is a dynamic programming language, which means R automatically interprets your code as you run it.

If you type an incomplete command and press Enter, R will display a + prompt, which means R is waiting for you to type the rest of your command. Either finish the command or hit Escape to start over:

```
> 5 -
+
+ 1
[1] 4
```

If you type a command that R doesn't recognize, R will return an error message. If you ever see an error message, don't panic. R is just telling you that your computer couldn't understand or do what you asked it to do. You can then try a different command at the next prompt:

```
> 3 % 5
Error: unexpected input in "3 % 5"
>
```

4. Up and Running with R

Tip

Whenever you get an error message in R, consider googling the error message. You'll often find that someone else has had the same problem and has posted a solution online. Simply cutting-and-pasting the error message into a search engine will often work

Once you get the hang of the command line, you can easily do anything in R that you would do with a calculator. For example, you could do some basic arithmetic:

```
2 * 3
```

```
[1] 6
```

```
4 - 1
```

```
[1] 3
```

```
# this obeys order-of-operations  
6 / (4 - 1)
```

```
[1] 2
```

Tip

R treats the hashtag character, #, in a special way; R will not run anything that follows a hashtag on a line. This makes hashtags very useful for adding comments and annotations to your code. Humans will be able to read the comments, but your computer will pass over them. The hashtag is known as the *commenting symbol* in R.

Cancelling commands

Some R commands may take a long time to run. You can cancel a command once it has begun by pressing ctrl + c or by clicking the “stop sign” if it is available in Rstudio. Note that it may also take R a long time to cancel the command.

4. Up and Running with R

4.1.1. An exercise

That's the basic interface for executing R code in RStudio. Think you have it? If so, try doing these simple tasks. If you execute everything correctly, you should end up with the same number that you started with:

1. Choose any number and add 2 to it.
2. Multiply the result by 3.
3. Subtract 6 from the answer.
4. Divide what you get by 3.

```
10 + 2
```

```
[1] 12
```

```
12 * 3
```

```
[1] 36
```

```
36 - 6
```

```
[1] 30
```

```
30 / 3
```

```
[1] 10
```

4.2. Objects

Now that you know how to use R, let's use it to make a virtual die. The `:` operator from a couple of pages ago gives you a nice way to create a group of numbers from one to six. The `:` operator returns its results as a **vector** (we are going to work with vectors in more detail), a one-dimensional set of numbers:

```
1:6  
## 1 2 3 4 5 6
```

4. Up and Running with R

That’s all there is to how a virtual die looks! But you are not done yet. Running `1:6` generated a vector of numbers for you to see, but it didn’t save that vector anywhere for later use. If we want to use those numbers again, we’ll have to ask your computer to save them somewhere. You can do that by creating an R *object*.

R lets you save data by storing it inside an R object. What is an object? Just a name that you can use to call up stored data. For example, you can save data into an object like `a` or `b`. Wherever R encounters the object, it will replace it with the data saved inside, like so:

```
a <- 1
a
```

```
[1] 1
```

```
a + 2
```

```
[1] 3
```

i What just happened?

1. To create an R object, choose a name and then use the less-than symbol, `<`, followed by a minus sign, `-`, to save data into it. This combination looks like an arrow, `<-`. R will make an object, give it your name, and store in it whatever follows the arrow. So `a <- 1` stores 1 in an object named `a`.
2. When you ask R what’s in `a`, R tells you on the next line.
3. You can use your object in new R commands, too. Since `a` previously stored the value of 1, you’re now adding 1 to 2.

! Assignment vs expressions

Everything that you type into the R console can be assigned to one of two categories:

- Assignments
- Expressions

An expression is a command that tells R to do something. For example, `1 + 2` is an expression that tells R to add 1 and 2. When you type an expression into the R console, R will evaluate the expression and return the result. For example, if you type `1 + 2` into the R console, R will return 3. Expressions can have “side effects”

4. Up and Running with R

but they don't explicitly result in anything being added to R memory.

```
5 + 2
```

```
[1] 7
```

```
28 %% 3
```

```
[1] 1
```

```
3^2
```

```
[1] 9
```

```
5 + 4 * 4 + 4 ^ 4 / 10
```

```
[1] 46.6
```

While using R as a calculator is interesting, to do useful and interesting things, we need to assign values to objects. To create objects, we need to give it a name followed by the assignment operator `<-` (or, entirely equivalently, `=`) and the value we want to give it:

```
weight_kg <- 55
```

So, for another example, the following code would create an object named `die` that contains the numbers one through six. To see what is stored in an object, just type the object's name by itself:

```
die <- 1:6  
die
```

```
[1] 1 2 3 4 5 6
```

When you create an object, the object will appear in the environment pane of RStudio, as shown in Figure 4.2. This pane will show you all of the objects you've created since opening RStudio.

4. Up and Running with R

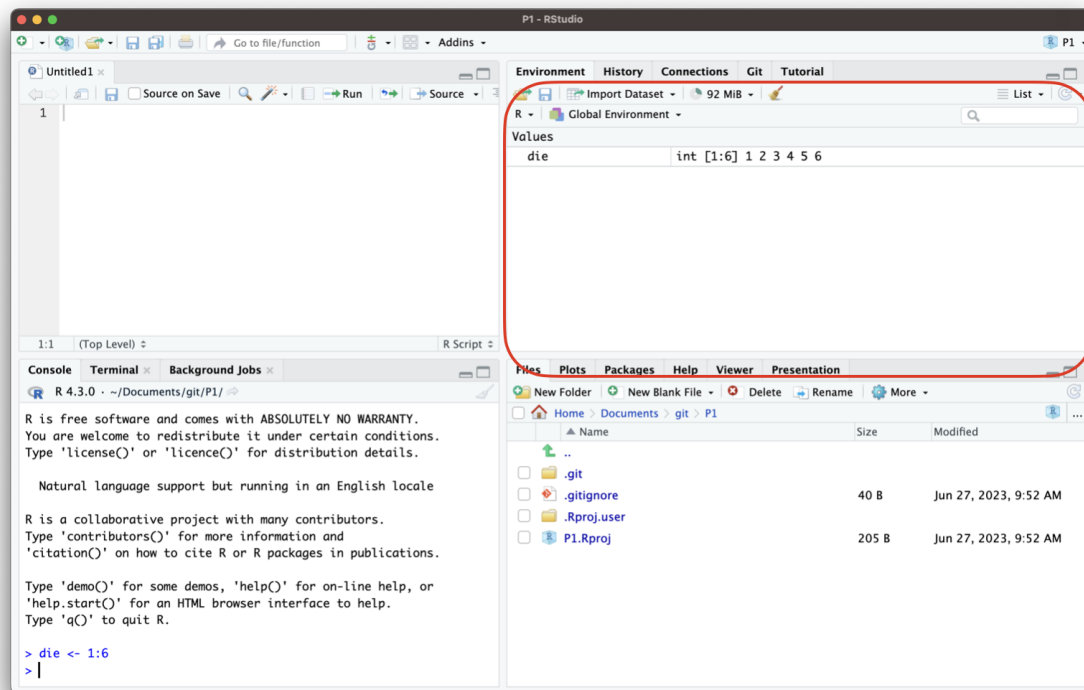


Figure 4.2.: Assignment creates an object in the environment pane.

4. Up and Running with R

You can name an object in R almost anything you want, but there are a few rules. First, a name cannot start with a number. Second, a name cannot use some special symbols, like `^`, `!`, `$`, `@`, `+`, `-`, `/`, or `*`:

Good names	Names that cause errors
a	1trial
b	\$
FOO	^mean
my_var	2nd
.day	!bad

Capitalization matters

R is case-sensitive, so `name` and `Name` will refer to different objects:

```
> Name = 0
> Name + 1
[1] 1
> name + 1
Error: object 'name' not found
```

The error above is a common one!

Finally, R will overwrite any previous information stored in an object without asking you for permission. So, it is a good idea to *not* use names that are already taken:

```
my_number <- 1
my_number
```

```
[1] 1
```

```
my_number <- 999
my_number
```

```
[1] 999
```

You can see which object names you have already used with the function `ls`:

4. Up and Running with R

```
ls()
```

Your environment will contain different names than mine, because you have probably created different objects.

You can also see which names you have used by examining RStudio's environment pane.

We now have a virtual die that is stored in the computer's memory and which has a name that we can use to refer to it. You can access it whenever you like by typing the word *die*.

So what can you do with this die? Quite a lot. R will replace an object with its contents whenever the object's name appears in a command. So, for example, you can do all sorts of math with the die. Math isn't so helpful for rolling dice, but manipulating sets of numbers will be your stock and trade as a data scientist. So let's take a look at how to do that:

```
die - 1
```

```
[1] 0 1 2 3 4 5
```

```
die / 2
```

```
[1] 0.5 1.0 1.5 2.0 2.5 3.0
```

```
die * die
```

```
[1] 1 4 9 16 25 36
```

R uses *element-wise execution* when working with a *vector* like *die*. When you manipulate a set of numbers, R will apply the same operation to each element in the set. So for example, when you run *die - 1*, R subtracts one from each element of *die*.

When you use two or more vectors in an operation, R will line up the vectors and perform a sequence of individual operations. For example, when you run *die * die*, R lines up the two *die* vectors and then multiplies the first element of vector 1 by the first element of vector 2. R then multiplies the second element of vector 1 by the second element of vector 2, and so on, until every element has been multiplied. The result will be a new vector the same length as the first two {Figure 4.3}.

If you give R two vectors of unequal lengths, R will repeat the shorter vector until it is as long as the longer vector, and then do the math, as shown in Figure 4.4. This isn't a

4. Up and Running with R

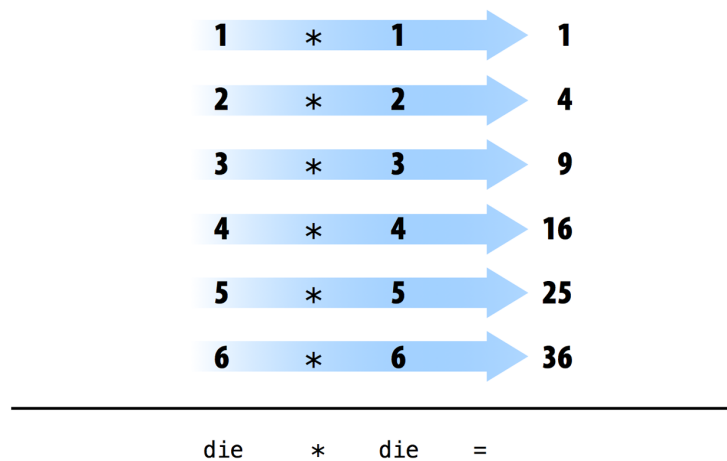


Figure 4.3.: “When R performs element-wise execution, it matches up vectors and then manipulates each pair of elements independently.”

permanent change—the shorter vector will be its original size after R does the math. If the length of the short vector does not divide evenly into the length of the long vector, R will return a warning message. This behavior is known as *vector recycling*, and it helps R do element-wise operations:

```
1:2
```

```
[1] 1 2
```

```
1:4
```

```
[1] 1 2 3 4
```

```
die
```

```
[1] 1 2 3 4 5 6
```

```
die + 1:2
```

```
[1] 2 4 4 6 6 8
```

4. Up and Running with R

```
die + 1:4
```

Warning in `die + 1:4`: longer object length is not a multiple of shorter object length

```
[1] 2 4 6 8 6 8
```

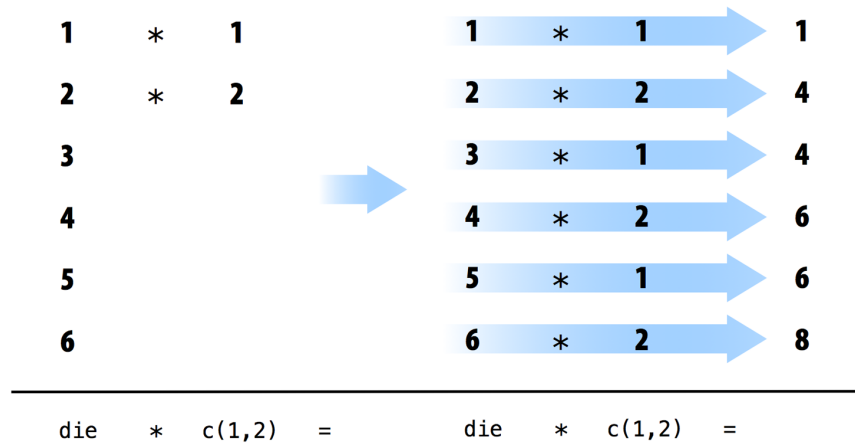


Figure 4.4.: “R will repeat a short vector to do element-wise operations with two vectors of uneven lengths.”

Element-wise operations are a very useful feature in R because they manipulate groups of values in an orderly way. When you start working with data sets, element-wise operations will ensure that values from one observation or case are only paired with values from the same observation or case. Element-wise operations also make it easier to write your own programs and functions in R.

! Element-wise operations are not matrix operations

It is important to know that operations with vectors are not the same that you might expect if you are expecting R to perform “matrix” operations. R can do inner multiplication with the `%*%` operator and outer multiplication with the `%o%` operator:

```
# Inner product (1*1 + 2*2 + 3*3 + 4*4 + 5*5 + 6*6)
die %*% die
# Outer product
die %o% die
```

4. Up and Running with R

Now that you can do math with your `die` object, let's look at how you could “roll” it. Rolling your die will require something more sophisticated than basic arithmetic; you'll need to randomly select one of the die's values. And for that, you will need a *function*.

4.3. Functions

R has many functions and puts them all at our disposal. We can use functions to do simple and sophisticated tasks. For example, we can round a number with the `round` function, or calculate its factorial with the `factorial` function. Using a function is pretty simple. Just write the name of the function and then the data you want the function to operate on in parentheses:

```
round(3.1415)
```

```
[1] 3
```

```
factorial(3)
```

```
[1] 6
```

The data that you pass into the function is called the function's *argument*. The argument can be raw data, an R object, or even the results of another R function. In this last case, R will work from the innermost function to the outermost [Figure 4.5](#).

```
mean(1:6)
```

```
[1] 3.5
```

```
mean(die)
```

```
[1] 3.5
```

```
round(mean(die))
```

```
[1] 4
```

4. Up and Running with R

```
round(mean(die))  
  ↓  
round(mean(1:6))  
  ↓  
round(3.5)  
  ↓  
4
```

Figure 4.5.: “When you link functions together, R will resolve them from the innermost operation to the outermost. Here R first looks up `die`, then calculates the mean of one through six, then rounds the mean.”

Returning to our die, we can use the `sample` function to randomly select one of the die’s values; in other words, the `sample` function can simulate rolling the `die`.

The `sample` function takes *two* arguments: a vector named `x` and a number named `size`. `sample` will return `size` elements from the vector:

```
sample(x = 1:4, size = 2)
```

```
[1] 2 1
```

To roll your die and get a number back, set `x` to `die` and sample one element from it. You’ll get a new (maybe different) number each time you roll it:

```
sample(x = die, size = 1)
```

```
[1] 3
```

```
sample(x = die, size = 1)
```

```
[1] 3
```

```
sample(x = die, size = 1)
```

```
[1] 5
```

4. Up and Running with R

Many R functions take multiple arguments that help them do their job. You can give a function as many arguments as you like as long as you separate each argument with a comma.

You may have noticed that I set `die` and `1` equal to the names of the arguments in `sample`, `x` and `size`. Every argument in every R function has a name. You can specify which data should be assigned to which argument by setting a name equal to data, as in the preceding code. This becomes important as you begin to pass multiple arguments to the same function; names help you avoid passing the wrong data to the wrong argument. However, using names is optional. You will notice that R users do not often use the name of the first argument in a function. So you might see the previous code written as:

```
sample(die, size = 1)
```

```
[1] 3
```

Often, the name of the first argument is not very descriptive, and it is usually obvious what the first piece of data refers to anyways.

But how do you know which argument names to use? If you try to use a name that a function does not expect, you will likely get an error:

```
round(3.1415, corners = 2)
## Error in round(3.1415, corners = 2) : unused argument(s) (corners = 2)
```

If you're not sure which names to use with a function, you can look up the function's arguments with `args`. To do this, place the name of the function in the parentheses behind `args`. For example, you can see that the `round` function takes two arguments, one named `x` and one named `digits`:

```
args(round)
```

```
function (x, digits = 0, ...)
NULL
```

Did you notice that `args` shows that the `digits` argument of `round` is already set to 0? Frequently, an R function will take optional arguments like `digits`. These arguments are considered optional because they come with a default value. You can pass a new value to an optional argument if you want, and R will use the default value if you do not. For example, `round` will round your number to 0 digits past the decimal point by default. To override the default, supply your own value for `digits`:

4. Up and Running with R

```
round(3.1415)
```

```
[1] 3
```

```
round(3.1415, digits = 2)
```

```
[1] 3.14
```

```
# pi happens to be a built-in value in R  
pi
```

```
[1] 3.141593
```

```
round(pi)
```

```
[1] 3
```

You should write out the names of each argument after the first one or two when you call a function with multiple arguments. Why? First, this will help you and others understand your code. It is usually obvious which argument your first input refers to (and sometimes the second input as well). However, you'd need a large memory to remember the third and fourth arguments of every R function. Second, and more importantly, writing out argument names prevents errors.

If you do not write out the names of your arguments, R will match your values to the arguments in your function by order. For example, in the following code, the first value, `die`, will be matched to the first argument of `sample`, which is named `x`. The next value, `1`, will be matched to the next argument, `size`:

```
sample(die, 1)
```

```
[1] 3
```

As you provide more arguments, it becomes more likely that your order and R's order may not align. As a result, values may get passed to the wrong argument. Argument names prevent this. R will always match a value to its argument name, no matter where it appears in the order of arguments:

4. Up and Running with R

```
sample(size = 1, x = die)
```

```
[1] 1
```

4.3.1. Sample with Replacement

If you set `size = 2`, you can *almost* simulate a pair of dice. Before we run that code, think for a minute why that might be the case. `sample` will return two numbers, one for each die:

```
sample(die, size = 2)
```

```
[1] 3 6
```

I said this “almost” works because this method does something funny. If you use it many times, you’ll notice that the second die never has the same value as the first die, which means you’ll never roll something like a pair of threes or snake eyes. What is going on?

By default, `sample` builds a sample *without replacement*. To see what this means, imagine that `sample` places all of the values of `die` in a jar or urn. Then imagine that `sample` reaches into the jar and pulls out values one by one to build its sample. Once a value has been drawn from the jar, `sample` sets it aside. The value doesn’t go back into the jar, so it cannot be drawn again. So if `sample` selects a six on its first draw, it will not be able to select a six on the second draw; six is no longer in the jar to be selected. Although `sample` creates its sample electronically, it follows this seemingly physical behavior.

One side effect of this behavior is that each draw depends on the draws that come before it. In the real world, however, when you roll a pair of dice, each die is independent of the other. If the first die comes up six, it does not prevent the second die from coming up six. In fact, it doesn’t influence the second die in any way whatsoever. You can recreate this behavior in `sample` by adding the argument `replace = TRUE`:

```
sample(die, size = 2, replace = TRUE)
```

```
[1] 1 6
```

4. Up and Running with R

The argument `replace = TRUE` causes `sample` to sample *with replacement*. Our jar example provides a good way to understand the difference between sampling with replacement and without. When `sample` uses replacement, it draws a value from the jar and records the value. Then it puts the value back into the jar. In other words, `sample` *replaces* each value after each draw. As a result, `sample` may select the same value on the second draw. Each value has a chance of being selected each time. It is as if every draw were the first draw.

Sampling with replacement is an easy way to create *independent random samples*. Each value in your sample will be a sample of size one that is independent of the other values. This is the correct way to simulate a pair of dice:

```
sample(die, size = 2, replace = TRUE)
```

```
[1] 5 3
```

Congratulate yourself; you’ve just run your first simulation in R! You now have a method for simulating the result of rolling a pair of dice. If you want to add up the dice, you can feed your result straight into the `sum` function:

```
dice <- sample(die, size = 2, replace = TRUE)
dice
```

```
[1] 5 1
```

```
sum(dice)
```

```
[1] 6
```

What would happen if you call `dice` multiple times? Would R generate a new pair of dice values each time? Let’s give it a try:

```
dice
```

```
[1] 5 1
```


4. Up and Running with R

```
dice
```

```
[1] 5 1
```

```
dice
```

```
[1] 5 1
```

The name `dice` refers to a *vector* of two numbers. Calling more than once does not change the value. Each time you call `dice`, R will show you the result of that one time you called `sample` and saved the output to `dice`. R won't rerun `sample(die, 2, replace = TRUE)` to create a new roll of the dice. Once you save a set of results to an R object, those results do not change.

However, it *would* be convenient to have an object that can re-roll the dice whenever you call it. You can make such an object by writing your own R function.

4.4. Writing Your Own Functions

To recap, you already have working R code that simulates rolling a pair of dice:

```
die <- 1:6
dice <- sample(die, size = 2, replace = TRUE)
sum(dice)
```

```
[1] 8
```

You can retype this code into the console anytime you want to re-roll your dice. However, this is an awkward way to work with the code. It would be easier to use your code if you wrapped it into its own function, which is exactly what we'll do now. We're going to write a function named `roll` that you can use to roll your virtual dice. When you're finished, the function will work like this: each time you call `roll()`, R will return the sum of rolling two dice:

4. Up and Running with R

```
roll()
## 8

roll()
## 3

roll()
## 7
```

Functions may seem mysterious or fancy, but they are *just another type of R object*. Instead of containing data, they contain code. This code is stored in a special format that makes it easy to reuse the code in new situations. You can write your own functions by recreating this format.

4.4.1. The Function Constructor

Every function in R has three basic parts: a name, a body of code, and a set of arguments. To make your own function, you need to replicate these parts and store them in an R object, which you can do with the `function` function. To do this, call `function()` and follow it with a pair of braces, `{}`:

```
my_function <- function() {}
```

This function, as written, doesn't do anything (yet). However, it is a valid function. You can call it by typing its name followed by an open and closed parenthesis:

```
my_function()
```

NULL

`function` will build a function out of whatever R code you place between the braces. For example, you can turn your dice code into a function by calling:

```
roll <- function() {
  die <- 1:6
  dice <- sample(die, size = 2, replace = TRUE)
  sum(dice)
}
```

4. Up and Running with R

i Indentation and readability

Notice each line of code between the braces is indented. This makes the code easier to read but has no impact on how the code runs. R ignores spaces and line breaks and executes one complete expression at a time. Note that in other languages like python, spacing is extremely important and part of the language.

Just hit the Enter key between each line after the first brace, {. R will wait for you to type the last brace, }, before it responds.

Don't forget to save the output of `function` to an R object. This object will become your new function. To use it, write the object's name followed by an open and closed parenthesis:

```
roll()
```

```
[1] 7
```

You can think of the parentheses as the “trigger” that causes R to run the function. If you type in a function's name *without* the parentheses, R will show you the code that is stored inside the function. If you type in the name *with* the parentheses, R will run that code:

```
roll
```

```
function ()
{
  die <- 1:6
  dice <- sample(die, size = 2, replace = TRUE)
  sum(dice)
}
```

```
roll()
```

```
[1] 7
```

The code that you place inside your function is known as the *body* of the function. When you run a function in R, R will execute all of the code in the body and then return the result of the last line of code. If the last line of code doesn't return a value, neither will your function, so you want to ensure that your final line of code returns a value. One way

4. Up and Running with R

to check this is to think about what would happen if you ran the body of code line by line in the command line. Would R display a result after the last line, or would it not?

Here's some code that would display a result:

```
dice
1 + 1
sqrt(2)
```

And here's some code that would not:

```
dice <- sample(die, size = 2, replace = TRUE)
two <- 1 + 1
a <- sqrt(2)
```

Again, this is just showing the distinction between expressions and assignments.

4.5. Arguments

What if we removed one line of code from our function and changed the name `die` to `bones` (just a name—don't think of it as important), like this?

```
roll2 <- function() {
  dice <- sample(bones, size = 2, replace = TRUE)
  sum(dice)
}
```

Now I'll get an error when I run the function. The function **needs** the object `bones` to do its job, but there is no object named `bones` to be found (you can check by typing `ls()` which will show you the names in the environment, or memory).

```
roll2()
## Error in sample(bones, size = 2, replace = TRUE) :
##   object 'bones' not found
```

You can supply `bones` when you call `roll2` if you make `bones` an argument of the function. To do this, put the name `bones` in the parentheses that follow `function` when you define `roll2`:

4. Up and Running with R

```
roll2 <- function(bones) {  
  dice <- sample(bones, size = 2, replace = TRUE)  
  sum(dice)  
}
```

Now `roll2` will work as long as you supply `bones` when you call the function. You can take advantage of this to roll different types of dice each time you call `roll2`.

Remember, we're rolling pairs of dice:

```
roll2(bones = 1:4)
```

```
[1] 5
```

```
roll2(bones = 1:6)
```

```
[1] 8
```

```
roll2(1:20)
```

```
[1] 20
```

Notice that `roll2` will still give an error if you do not supply a value for the `bones` argument when you call `roll2`:

```
roll2()  
## Error in sample(bones, size = 2, replace = TRUE) :  
## argument "bones" is missing, with no default
```

You can prevent this error by giving the `bones` argument a default value. To do this, set `bones` equal to a value when you define `roll2`:

```
roll2 <- function(bones = 1:6) {  
  dice <- sample(bones, size = 2, replace = TRUE)  
  sum(dice)  
}
```

4. Up and Running with R

Now you can supply a new value for `bones` if you like, and `roll2` will use the default if you do not:

```
roll2()
```

```
[1] 7
```

You can give your functions as many arguments as you like. Just list their names, separated by commas, in the parentheses that follow `function`. When the function is run, R will replace each argument name in the function body with the value that the user supplies for the argument. If the user does not supply a value, R will replace the argument name with the argument's default value (if you defined one).

To summarize, `function` helps you construct your own R functions. You create a body of code for your function to run by writing code between the braces that follow `function`. You create arguments for your function to use by supplying their names in the parentheses that follow `function`. Finally, you give your function a name by saving its output to an R object, as shown in Figure 4.6.

Once you've created your function, R will treat it like every other function in R. Think about how useful this is. Have you ever tried to create a new Excel option and add it to Microsoft's menu bar? Or a new slide animation and add it to Powerpoint's options? When you work with a programming language, you can do these types of things. As you learn to program in R, you will be able to create new, customized, reproducible tools for yourself whenever you like.

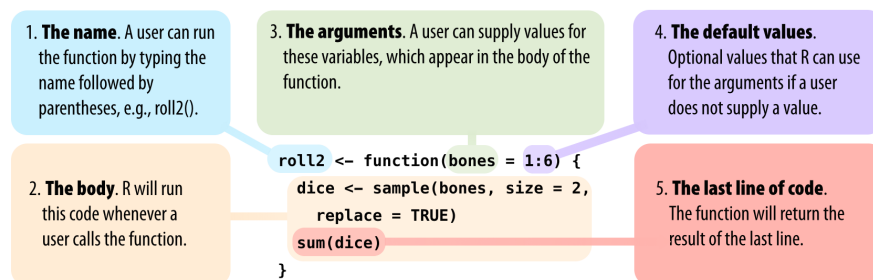


Figure 4.6.: “Every function in R has the same parts, and you can use `function` to create these parts. Assign the result to a name, so you can call the function later.”

4.6. Scripts

Scripts are code that are saved for later reuse or editing. An R script is just a plain text file that you save R code in. You can open an R script in RStudio by going to **File > New File > R script** in the menu bar. RStudio will then open a fresh script above your console pane, as shown in Figure 4.7.

I strongly encourage you to write and edit all of your R code in a script before you run it in the console. Why? This habit creates a reproducible record of your work. When you're finished for the day, you can save your script and then use it to rerun your entire analysis the next day. Scripts are also very handy for editing and proofreading your code, and they make a nice copy of your work to share with others. To save a script, click the scripts pane, and then go to **File > Save As** in the menu bar.

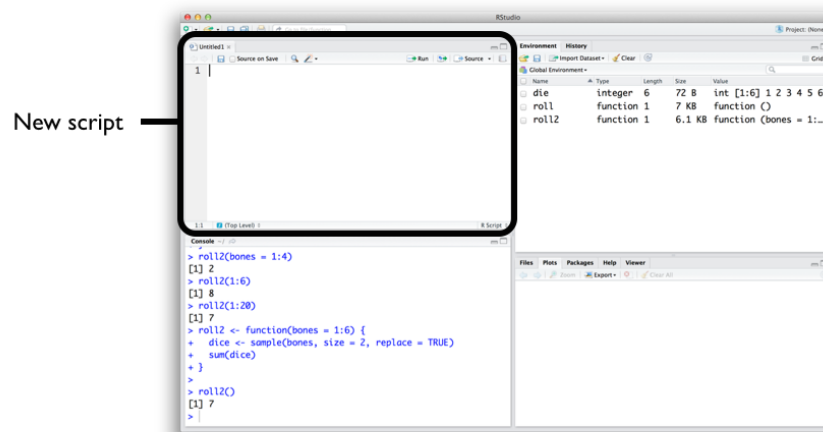


Figure 4.7.: “When you open an R Script (File > New File > R Script in the menu bar), RStudio creates a fourth pane (or puts a new tab in the existing pane) above the console where you can write and edit your code.”

RStudio comes with many built-in features that make it easy to work with scripts. First, you can automatically execute a line of code in a script by clicking the Run button at the top of the editor panel.

R will run whichever line of code your cursor is on. If you have a whole section highlighted, R will run the highlighted code. Alternatively, you can run the entire script by clicking the Source button. Don't like clicking buttons? You can use Control + Return as a shortcut for the Run button. On Macs, that would be Command + Return.

4. Up and Running with R

If you're not convinced about scripts, you soon will be. It becomes a pain to write multi-line code in the console's single-line command line. Let's avoid that headache and open your first script now before we move to the next chapter.

Tip

Extract function

RStudio comes with a tool that can help you build functions. To use it, highlight the lines of code in your R script that you want to turn into a function. Then click **Code > Extract Function** in the menu bar. RStudio will ask you for a function name to use and then wrap your code in a **function** call. It will scan the code for undefined variables and use these as arguments.

You may want to double-check RStudio's work. It assumes that your code is correct, so if it does something surprising, you may have a problem in your code.

4.7. Summary

We've covered a lot of ground already. You now have a virtual die stored in your computer's memory, as well as your own R function that rolls a pair of dice. You've also begun speaking the R language.

The two most important components of the R language are objects, which store data, and functions, which manipulate data. R also uses a host of operators like `+`, `-`, `*`, `/`, and `<-` to do basic tasks. As a data scientist, you will use R objects to store data in your computer's memory, and you will use functions to automate tasks and do complicated calculations.

5. Packages and more dice

We now have code that allows us to roll two dice and add the results together. To keep things interesting, let's aim to weight the dice so that we can fool our friends into thinking we are lucky.

First, though, we should prove to ourselves that our dice are fair. We can investigate the behavior of our dice using two powerful and general tools;

- Simulation (or repetition or repeated sampling)
- Visualization

For the repetition part of things, we will use a built-in R function, `replicate`. For visualization, we are going to use a convenient plotting function, `qplot`. However, `qplot` does not come built into R. We must install a *package* to gain access to it.

5.1. Packages

R is a powerful language for data science and programming, allowing beginners and experts alike to manipulate, analyze, and visualize data effectively. One of the most appealing features of R is its extensive library of packages, which are essential tools for expanding its capabilities and streamlining the coding process.

An R package is a collection of reusable functions, datasets, and compiled code created by other users and developers to extend the functionality of the base R language. These packages cover a wide range of applications, such as data manipulation, statistical analysis, machine learning, and data visualization. By utilizing existing R packages, you can leverage the expertise of others and save time by avoiding the need to create custom functions from scratch.

Using others' R packages is incredibly beneficial as it allows you to take advantage of the collective knowledge of the R community. Developers often create packages to address specific challenges, optimize performance, or implement popular algorithms or methodologies. By incorporating these packages into your projects, you can enhance your productivity, reduce development time, and ensure that you are using well-tested and reliable code.

5.1.1. Installing R packages

To install an R package, you can use the `install.packages()` function in the R console or script. For example, to install the popular data manipulation package “dplyr,” simply type `install.packages("dplyr")`. This command will download the package from the Comprehensive R Archive Network (CRAN) and install it on your local machine. Keep in mind that you only need to install a package once, unless you want to update it to a newer version.

For those who are going to be using R for bioinformatics or biological data science, you will also want to install packages from Bioconductor, which is a repository of R packages specifically designed for bioinformatics and computational biology. To install Bioconductor packages, you can use the `BiocManager::install()` function.

To use this recommended approach, you first need to install the `BiocManager` package, which is the package manager for Bioconductor.

```
install.packages('BiocManager')
```

This is a one-time installation. After that, you can install any R, Bioconductor, rOpenSci, or even GitHub package using the `BiocManager::install()` function. For example, to install the `ggplot2` package, which is widely used for data visualization, you would run:

```
BiocManager::install("ggplot2")
```

5.1.2. Installing vs loading (library) R packages

After installing an R package, you will need to load it into your R session before using its functions. To load a package, use the `library()` function followed by the package name, such as `library(dplyr)`. Loading a package makes its functions and datasets available for use in your current R session. Note that you need to load a package every time you start a new R session.

```
library(ggplot2)
```

Now, the functionality of the *ggplot2* package is available in our R session.

5. Packages and more dice

💡 Installing vs loading packages

The main thing to remember is that you only need to install a package once, but you need to load it with `library` each time you wish to use it in a new R session. R will unload all of its packages each time you close RStudio.



Figure 5.1.: Installing vs loading R packages.

As in {Figure 5.1}, screw in the lightbulb (eg., `BiocManager::install`) only once and then to use it, you need to turn on the switch each time you want to use it (`library`).

5.1.3. Finding R packages

Finding useful R packages can be done in several ways. First, browsing CRAN (<https://cran.r-project.org/>) and Bioconductor (more later, <https://bioconductor.org>) are an excellent starting points, as they host thousands of packages categorized by topic. Additionally, online forums like Stack Overflow and R-bloggers can provide valuable recommendations based on user experiences. Social media platforms such as Twitter, where developers and data scientists often share new packages and updates, can also be a helpful resource. Finally, don't forget to ask your colleagues or fellow R users for their favorite packages, as they may have insights on which ones best suit your specific needs.

5.1.4. Creating a package

While it may seem overwhelming, creating a package can be fairly simple with the assistance of R packages that provide tips and templates. If interested in exploring package creation, some good starting points are:

- [devtools](#)
- [usethis](#)
- [biocthis](#)
- [roxygen2](#)

5.2. Are our dice fair?

Well, let's review our code.

```
roll2 <- function(bones = 1:6) {
  dice = sample(bones, size = 2, replace = TRUE)
  sum(dice)
}
```

If our dice are fair, then each number should show up equally. What does the sum look like with our two dice?

combinations of die 1 and die 2 that add up to sum						(6,1)					
					(5,1)	(5,2)	(6,2)				
				(4,1)	(4,2)	(4,3)	(5,3)	(6,3)			
			(3,1)	(3,2)	(3,3)	(3,4)	(4,4)	(5,4)	(6,4)		
		(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)	
	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)
	2	3	4	5	6	7	8	9	10	11	12
	sum										

Figure 5.2.: In an ideal world, a histogram of the results would look like this

5. Packages and more dice

Read the help page for `replicate` (i.e., `help("replicate")`). In short, it suggests that we can repeat our dice rolling as many times as we like and `replicate` will return a *vector* of the sums for each roll.

```
rolls = replicate(n = 100, roll2())
```

What does `rolls` look like?

```
head(rolls)
```

```
[1]  7 12  8  5  5  6
```

```
length(rolls)
```

```
[1] 100
```

```
mean(rolls)
```

```
[1] 6.99
```

```
summary(rolls)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.00	5.00	7.00	6.99	8.00	12.00

This looks like it roughly agrees with our sketched out ideal histogram in Figure 5.2. However, now that we've loaded the `qplot` function from the *ggplot2* package, we can make a histogram of the data themselves.

```
qplot(rolls, binwidth=1)
```

Warning: ``qplot()`` was deprecated in *ggplot2* 3.4.0.

5. Packages and more dice

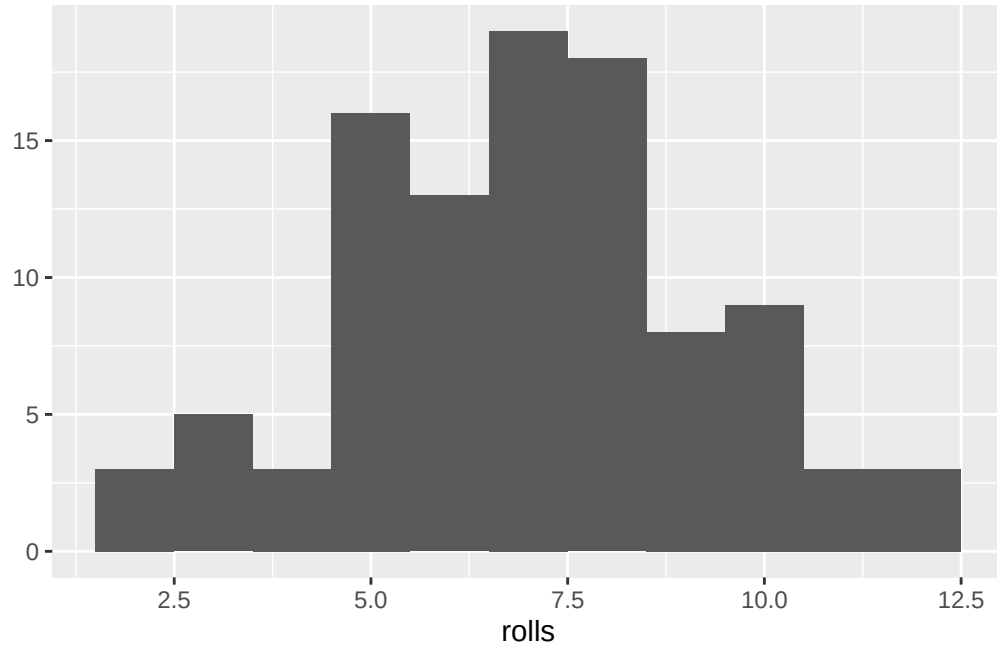


Figure 5.3.: Histogram of the sums from 100 rolls of our fair dice

How does your histogram look (and yours will be different from mine since we are sampling random values)? Is it what you expect?

What happens to our histogram as we increase the number of replicates?

```
rolls = replicate(n = 100000, roll2())  
qplot(rolls, binwidth=1)
```

5. Packages and more dice

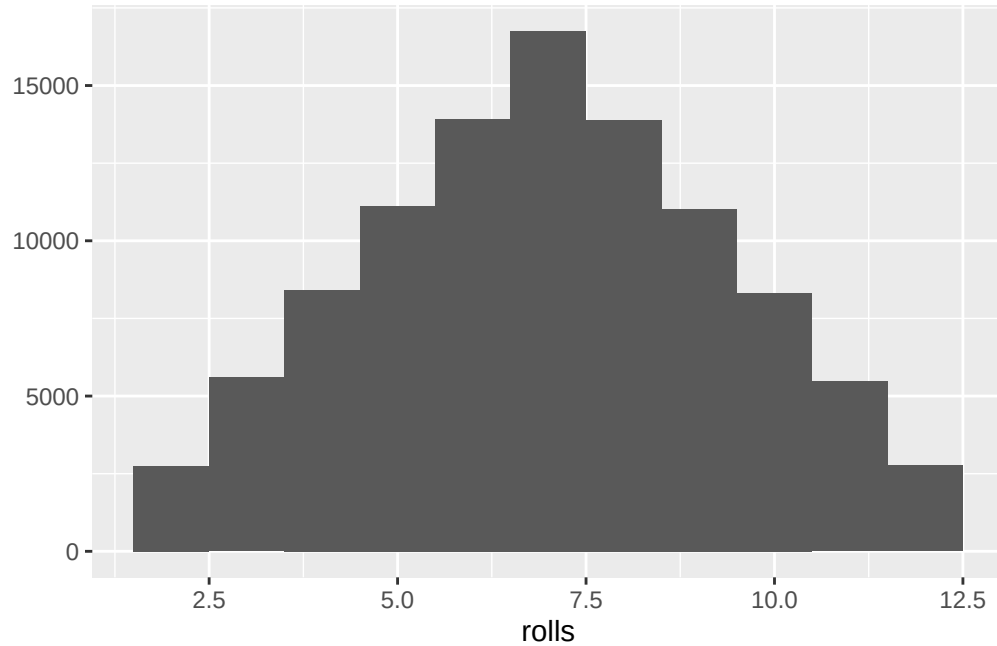


Figure 5.4.: Histogram with 100000 rolls much more closely approximates the pyramidal shape we anticipated

5.3. Bonus exercise

How would you change the `roll2` function to weight the dice?

i Hint

Read the help page for `sample` (i.e., `help("sample")`).

6. Reading and writing data files

Almost no analysis starts inside R. Your data lives somewhere else first — an expression matrix exported from an instrument, a sample sheet a collaborator typed into Excel, a public dataset posted on a government website. Before you can do anything useful, you have to get that tabular data *into* R, and once you’ve cleaned or summarized it, you’ll often need to get it back *out* to share with someone who does not use R.

This chapter is about that round trip. You’ll write a small table of genes and their expression values to disk, read it back, pull a real dataset straight from a URL, and do the same with Excel files. None of it is hard once you know which function to reach for — and knowing that is most of the battle.

6.1. What you’ll learn

- Organize an analysis in an RStudio Project so your file paths stay portable.
- Write a data frame to a CSV file with `write_csv()` and read it back with `read_csv()`.
- Read a CSV directly from a URL, without downloading it by hand.
- Read and write Excel files with `read_excel()` and `write_xlsx()`.
- Recognize when other formats (Google Sheets, JSON, databases) need a different package.

6.2. Organizing your work with RStudio Projects

Before you read or write a single file, it helps to decide *where* those files live. An RStudio Project is a folder that holds everything for one analysis — scripts, data, and outputs — together in one place. When you create a project, RStudio drops a `.Rproj` file into the folder, and that file anchors your workspace.

Working in a project buys you a few things:

- **A consistent working directory.** Open the project and your working directory is automatically the project folder. No more guessing where R thinks it is.

6. Reading and writing data files

- **Everything in one place.** Scripts, data, and outputs travel together.
- **Reproducibility.** A collaborator can open your project and run your code without rewriting any paths.
- **Version control.** Projects work cleanly with Git and GitHub.

6.2.1. Creating a project

You can make a new project from **File > New Project...**, or from the project dropdown in the top-right corner of RStudio. You'll be offered three choices:

- **New Directory** — start a fresh project folder.
- **Existing Directory** — turn a folder you already have into a project.
- **Version Control** — clone a repository from GitHub.

6.2.2. A sensible project layout

A good project has a predictable shape — one that *you* define and then stick to. A common starting point:

```
my_analysis_project/  
  my_analysis_project.Rproj  
  data/  
    raw/  
    processed/  
  scripts/  
  notebooks/  
  outputs/  
    figures/  
    tables/  
  README.md  
  .gitignore
```

This keeps raw data separate from processed data, scripts in one place, and outputs where you can find them later.

6.2.3. Why this fixes file paths

The biggest practical payoff is path management. When you open a project, RStudio sets the working directory to the project folder, so you can refer to files *relative* to that folder instead of spelling out the full path from the root of your hard drive.

```
# An absolute path is tied to one machine and one user:
expr <- read.csv("/Users/username/Documents/my_analysis/data/dataset.csv")

# A relative path works for anyone who opens the project:
expr <- read.csv("data/dataset.csv")
```

Absolute paths break reproducibility

An absolute path like `/Users/username/Documents/...` only exists on *your* computer. Email that script to a colleague and the very first line fails for them. Relative paths (`data/dataset.csv`), anchored by a project, are the habit that keeps your code portable.

The **here** package for robust paths

In larger projects — especially ones with notebooks tucked into subfolders — the **here** package makes paths bulletproof. `here()` always resolves from the project root (the folder with the `.Rproj` file), no matter which subfolder your script sits in:

```
install.packages("here")
library(here)

expr <- read.csv(here("data", "dataset.csv"))
```

When you share a project folder (or push it to GitHub), a collaborator can clone it, open the `.Rproj` file, and run your scripts with no path edits at all. That seamless hand-off is what makes an analysis credible and repeatable.

Now that you have a place to put files, let's read and write some.

6.3. CSV files

A CSV (“comma-separated values”) file is the lingua franca of tabular data: plain text, one row per line, columns separated by commas. Nearly every tool — Excel, Python,

6. Reading and writing data files

databases, sequencing pipelines — can read and write CSV, which is exactly why it's so useful for moving data around.

Base R reads and writes CSV with `read.csv()` and `write.csv()`. We'll use the `readr` package instead, which provides `read_csv()` and `write_csv()`. They are faster, give cleaner output, and guess column types more sensibly. Install it once:

```
install.packages("readr")
```

Then load it at the start of your session:

```
library(readr)
```

6.3.1. Writing a CSV file

We need something to write, so let's build a small biological data frame: a handful of genes, each with a symbol and a measured expression value.

```
expr <- data.frame(  
  gene_id   = c("YAL001C", "YAL002W", "YAL003W", "YAL005C", "YAL007C"),  
  symbol    = c("TFC3", "VPS8", "EFB1", "SSA1", "ERP2"),  
  expression = c(8.2, 5.1, 11.7, 9.4, 6.8)  
)  
expr
```

	gene_id	symbol	expression
1	YAL001C	TFC3	8.2
2	YAL002W	VPS8	5.1
3	YAL003W	EFB1	11.7
4	YAL005C	SSA1	9.4
5	YAL007C	ERP2	6.8

Each row is one gene; the columns mix text (`gene_id`, `symbol`) and numbers (`expression`), which is the everyday shape of real expression data. Now write it to disk with `write_csv()`:

```
write_csv(expr, "data.csv")
```

6. Reading and writing data files

That call created a file named `data.csv` in your working directory. You can confirm it landed where you expect:

```
# Where am I working right now?  
getwd()
```

```
[1] "/_w/RBiocBook/RBiocBook"
```

```
# Did the file get created?  
dir(pattern = "data.csv")
```

```
[1] "data.csv"
```

If you wanted the file somewhere else, you'd pass a fuller path (for example `"outputs/tables/data.csv"`). Inside an RStudio Project, that path is relative to the project folder.

6.3.2. Reading a CSV file

Reading the file back is the mirror image — `read_csv()`:

```
expr_in <- read_csv("data.csv")
```

```
Rows: 5 Columns: 3
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): gene_id, symbol
```

```
dbl (1): expression
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Notice the message `readr` prints: it tells you how it guessed each column's type (text vs. number). Let's look at what came back:

```
expr_in
```

6. Reading and writing data files

```
# A tibble: 5 x 3
  gene_id symbol expression
  <chr>   <chr>         <dbl>
1 YAL001C TFC3           8.2
2 YAL002W VPS8           5.1
3 YAL003W EFB1          11.7
4 YAL005C SSA1           9.4
5 YAL007C ERP2           6.8
```

The genes and their expression values round-tripped intact. The `gene_id` and `symbol` columns came back as character, `expression` as a number (`dbl`) — `readr` inferred all of that automatically.

6.3.3. Reading a CSV straight from the web

`read_csv()` is not limited to files on your own disk. Hand it a URL and it will fetch and parse the file in one step. This matters because a great deal of useful data lives on public websites, and reading it directly — rather than downloading by hand — keeps your analysis reproducible: anyone who runs your script pulls the same data from the same place.

As an example, the `palmerpenguins` project publishes its penguin body measurements as a public CSV on GitHub:

```
penguins <- read_csv("https://raw.githubusercontent.com/allisonhorst/palmerpenguins/main/in
```

```
Rows: 344 Columns: 8
-- Column specification -----
Delimiter: ","
chr (3): species, island, sex
dbl (5): bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, year

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
penguins
```

```
# A tibble: 344 x 8
  species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <chr>   <chr>         <dbl>         <dbl>         <dbl>         <dbl>
```

6. Reading and writing data files

```
1 Adelie Torgersen      39.1      18.7      181      3750
2 Adelie Torgersen      39.5      17.4      186      3800
3 Adelie Torgersen      40.3      18       195      3250
4 Adelie Torgersen      NA       NA       NA       NA
5 Adelie Torgersen      36.7      19.3      193      3450
6 Adelie Torgersen      39.3      20.6      190      3650
7 Adelie Torgersen      38.9      17.8      181      3625
8 Adelie Torgersen      39.2      19.6      195      4675
9 Adelie Torgersen      34.1      18.1      193      3475
10 Adelie Torgersen      42       20.2      190      4250
# i 334 more rows
# i 2 more variables: sex <chr>, year <dbl>
```

The dataset — bill and flipper measurements for three penguin species — is documented [here](#). The point is not the penguins themselves: it’s that `read_csv()` detected the file’s structure over the network and handed you a data frame, exactly as it did for your local file. Any public CSV, biological or otherwise, comes in the same way.

6.4. Excel files

Excel spreadsheets (`.xlsx`) are everywhere in biology — sample sheets, plate layouts, hand-curated gene lists. Unlike CSV, an Excel file can hold multiple sheets, formulas, and formatting, so it needs a dedicated package. Reading is handled by `readxl`; writing by `writexl`.

6.4.1. Reading an Excel file

Install and load `readxl`:

```
install.packages("readxl")
```

```
library(readxl)
```

We don’t have an Excel file lying around, and — unlike `read_csv()` — `readxl` can’t read straight from a URL. So we first *download* the file, then read it from disk. Here’s a small public example spreadsheet:

6. Reading and writing data files

```
download.file(  
  "https://www.w3resource.com/python-exercises/pandas/excel/SaleData.xlsx",  
  "SaleData.xlsx",  
  mode = "wb"  
)
```

The `mode = "wb"` (“write binary”) matters for non-text files like `.xlsx`; without it, the download can be corrupted on some platforms. Confirm it arrived:

```
dir(pattern = "SaleData.xlsx")
```

```
[1] "SaleData.xlsx"
```

Now read it with `read_excel()`:

```
sales <- read_excel("SaleData.xlsx")  
sales
```

```
# A tibble: 45 x 8
```

	OrderDate	Region	Manager	SalesMan	Item	Units	Unit_price	Sale_amt
	<dtm>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	2018-01-06 00:00:00	East	Martha	Alexander	Tele~	95	1198	113810
2	2018-01-23 00:00:00	Central	Hermann	Shell	Home~	50	500	25000
3	2018-02-09 00:00:00	Central	Hermann	Luis	Tele~	36	1198	43128
4	2018-02-26 00:00:00	Central	Timothy	David	Cell~	27	225	6075
5	2018-03-15 00:00:00	West	Timothy	Stephen	Tele~	56	1198	67088
6	2018-04-01 00:00:00	East	Martha	Alexander	Home~	60	500	30000
7	2018-04-18 00:00:00	Central	Martha	Steven	Tele~	75	1198	89850
8	2018-05-05 00:00:00	Central	Hermann	Luis	Tele~	90	1198	107820
9	2018-05-22 00:00:00	West	Douglas	Michael	Tele~	32	1198	38336
10	2018-06-08 00:00:00	East	Martha	Alexander	Home~	60	500	30000

```
# i 35 more rows
```

`read_excel()` has options for choosing a specific sheet (`sheet =`), a cell range (`range =`), or how column types are guessed — handy when a spreadsheet has more than one tab or some messy header rows.

6.4.2. Writing an Excel file

To go the other direction, use `write_xlsx()` from `writexl`. Install and load it:

```
install.packages("writexl")
```

```
library(writexl)
```

Then write our gene expression data frame to an `.xlsx` file:

```
write_xlsx(expr, "data.xlsx")
```

And confirm it was created:

```
getwd()
```

```
[1] "/_w/RBiocBook/RBiocBook"
```

```
dir(pattern = "data.xlsx")
```

```
[1] "data.xlsx"
```

If you open `data.xlsx` in Excel, you'll see the same five genes and their expression values you started with.

6.5. Additional formats

CSV and Excel cover most of what you'll meet, but R reads and writes many other formats too. When you hit one, reach for the matching package:

- **Google Sheets** — `googlesheets4` reads and writes sheets directly from your Google account.
- **JSON** — `jsonlite` converts between R objects and JSON, common for data pulled from web APIs.
- **Databases** — DBI together with a backend like `RSQLite` lets you query a database and pull results straight into a data frame.

The pattern is always the same: find the package for the format, then look for its `read_*` and `write_*` functions.

6.6. Exercises

Use the `expr` data frame from this chapter (five yeast genes with a `symbol` and an `expression` value) unless a problem says otherwise.

1. **Round-trip a CSV.** Write `expr` to a file called `genes.csv`, then read it into a new object called `genes_in`. Confirm the file exists with `dir()`.

i Solution

```
write_csv(expr, "genes.csv")
genes_in <- read_csv("genes.csv")
```

```
Rows: 5 Columns: 3
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): gene_id, symbol
```

```
dbl (1): expression
```

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
dir(pattern = "genes.csv")
```

```
[1] "genes.csv"
```

`write_csv()` saves it; `read_csv()` brings it back. The file name is just a string, so you can call it anything you like.

2. **Read a column back.** From `genes_in`, print just the `symbol` column.

i Solution

```
genes_in$symbol
```

```
[1] "TFC3" "VPS8" "EFB1" "SSA1" "ERP2"
```

The `$` operator pulls one column out of a data frame as a vector — the same trick you used in the data frames chapter.

3. **Save to Excel.** Write `expr` to `genes.xlsx` and confirm it was created.

i Solution

```
write_xlsx(expr, "genes.xlsx")
dir(pattern = "genes.xlsx")
```

```
[1] "genes.xlsx"
```

`write_xlsx()` from `writexl` writes a `.xlsx` file with no need for Excel to be installed.

4. **Why a relative path?** A colleague sends you a script whose first line is `read.csv("/Users/dana/project/counts.csv")`. It fails on your machine. Why, and what would you change it to?

i Solution

That absolute path points to a folder on Dana's computer that doesn't exist on yours. Inside an RStudio Project, a relative path like `read.csv("data/counts.csv")` works for anyone who opens the project, because the working directory is the project folder.

6.7. Summary

You can now move tabular data in and out of R in both directions:

- **Organize first.** An RStudio Project gives you a stable working directory so relative paths (`data/file.csv`) stay portable; absolute paths do not.
- **CSV** — `write_csv()` saves a data frame, `read_csv()` reads it back, and `read_csv()` will even read straight from a URL for reproducible access to public data.
- **Excel** — `read_excel()` reads `.xlsx` (download first; it can't read a URL), and `write_xlsx()` writes one.
- **Other formats** — Google Sheets, JSON, and databases each have a package (`googlesheets4`, `jsonlite`, `DBI/RSQLite`) that follows the same `read_*/write_*` pattern.

With data flowing freely in and out, you're ready to spend your time on the part that matters: the analysis.

Part II.

R Data Structures

Welcome to the section on R data structures! As you begin your journey in learning R, it is essential to understand the fundamental building blocks of this powerful programming language. R offers a variety of data structures to store and manipulate data, each with its unique properties and capabilities. In this section, we will cover the core data structures in R, including:

- Vectors
- Matrices
- Lists
- Data.frames

By the end of this section, you will have a solid understanding of these data structures, and you will be able to choose and utilize the appropriate data structure for your specific data manipulation and analysis tasks.

In each chapter, we will delve into the properties and usage of each data structure, starting with their definitions and moving on to their practical applications. We will provide examples, exercises, and active learning approaches to help you better understand and apply these concepts in your work.

Chapter overview

- **Vectors** In this chapter, we will introduce you to the simplest data structure in R, the vector. We will cover how to create, access, and manipulate vectors, as well as discuss their unique properties and limitations.
- **Matrices** Next, we will explore matrices, which are two-dimensional data structures that extend vectors. You will learn how to create, access, and manipulate matrices, and understand their usefulness in mathematical operations and data organization.
- **Lists** The third chapter will focus on lists, a versatile data structure that can store elements of different types and sizes. We will discuss how to create, access, and modify lists, and demonstrate their flexibility in handling complex data structures.
- **Data.frames** Next, we will examine data.frames, a widely-used data structure for organizing and manipulating tabular data. You will learn how to create, access, and manipulate data.frames, and understand their advantages over other data structures for data analysis tasks.

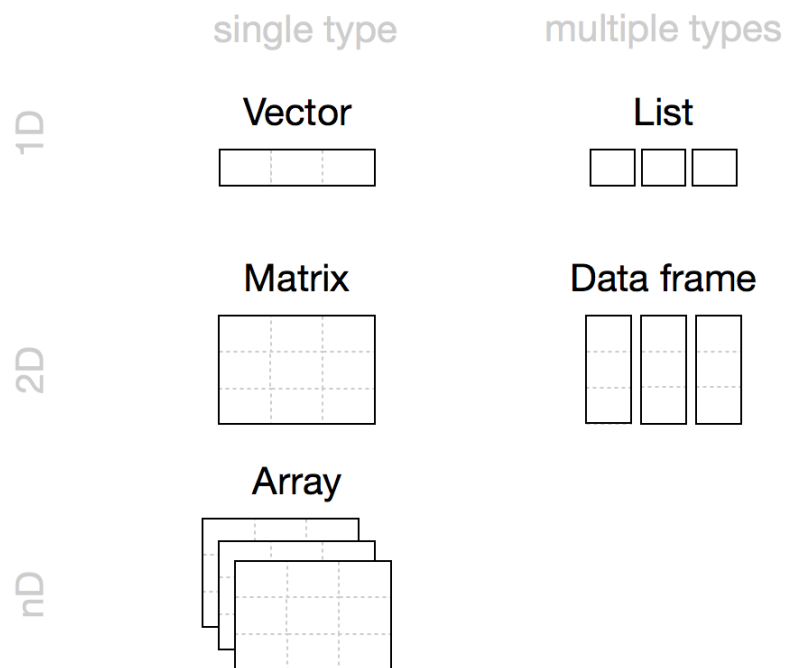


Figure 6.1.: A pictorial representation of R's most common data structures are vectors, matrices, arrays, lists, and dataframes. Figure from [Hands-on Programming with R](#).

Chapter overview

- **Factors** Finally, we will look at factors, R's way of representing categorical data (such as treatment groups or sample conditions). You will learn how factors store their categories as **levels**, why that matters for statistical modeling and plotting, and how to create and reorder them.
- **Arrays** While we will not focus directly on the **array** data type, which are multidimensional data structures that extend matrices, they are very similar to matrices, but with a third dimension.

As you progress through these chapters, practice the examples and exercises provided, engage in discussion, and collaborate with your peers to deepen your understanding of R data structures. This solid foundation will serve as the basis for more advanced data manipulation, analysis, and visualization techniques in R.

7. Vectors

Suppose you’ve just measured the expression of one gene across a dozen samples, or jotted down the ages of every patient in a study, or listed the gene names you want to pull out of a results table. Each of those is the same shape: a single, ordered run of values of one kind. R has a container built exactly for that shape, and it is the workhorse you’ll reach for more than any other. It’s called a **vector**.

Almost everything in R is built on vectors. A column of a data frame is a vector. The output of most calculations is a vector. Even a single number, as you’ll see in a moment, is just a vector of length one. So getting comfortable here pays off everywhere downstream. Let’s build that comfort.

7.1. What you’ll learn

- Create vectors with `c()`, the colon operator `:`, and `seq()`.
- Do arithmetic on whole vectors at once, and recognize the **recycling** rule.
- Build logical vectors with comparisons like `>` and `==`.
- Pull elements out of a vector by position, by name, and by a logical test.
- Work with character vectors using `paste()`, `nchar()`, `substr()`, and `grep()`.
- Spot and handle missing values (`NA`).

7.2. A vector is a column of measurements

A vector is a one-dimensional, ordered collection of elements that are all the same type. That last part matters: every value in a vector is numeric, *or* every value is text, *or* every value is logical — never a mix. Figure 7.1 shows three vectors side by side: a numeric one, a logical one, and a character one. Notice that each element has a position (an *index*) and can optionally carry a *name*.

Here’s the first surprise. In R, even a lone value is a vector — one of length 1:

7. Vectors

Index	1	2	3	4	5	6	7
Vector	3	7	10	NA	932	127	-3
Vector	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	NA
Vector	"Cat"	"Dog"	"A"	"C"	"T"	NA	"G"
Names (Optional)	"H"	"I"	"L"	"Z"	"This"	"That"	"Other"

Figure 7.1.: “Pictorial representation of three vector examples. The first vector is a numeric vector. The second is a ‘logical’ vector. The third is a character vector. Vectors also have indices and, optionally, names.”

```
gene_count <- 1
gene_count
```

```
[1] 1
```

```
length(gene_count)
```

```
[1] 1
```

We *assigned* the value 1 to a variable named `gene_count`. Typing `gene_count` by itself is an *expression*: it returns whatever is stored there. The `length()` function reports how many elements an object holds — here, just one. R gives you many ways to interrogate an object, and `length()` is one of the friendliest.

A vector can hold numbers, text (character data), or logical values (`TRUE` and `FALSE`), or other “atomic” types from Table 7.1. What it *cannot* do is hold a mix of types. When you need a grab-bag of different types in one container, R offers a different structure, the `list`, which you’ll meet in a later chapter.

7. Vectors

Table 7.1.: Atomic (simplest) data types in R.

Data type	Stores
numeric	floating point numbers
integer	integers
complex	complex numbers
factor	categorical data
character	strings
logical	TRUE or FALSE
NA	missing
NULL	empty
function	function type

! One type per vector

If you try to mix types, R doesn't error — it quietly *coerces* everything to a single type that can hold all the values. Watch what happens when a logical meets a string: the TRUE becomes the text "TRUE", quotes and all. Keep this in the back of your mind; a column that “looks numeric” but secretly contains one stray word will arrive as character, and your math will fail in confusing ways.

7.3. Creating vectors

The most direct way to build a vector is `c()`, short for *combine*. Character values go in matching quotes — single or double, your choice — and R always displays them back with double quotes.

```
# a few vectors of different types
c("BRCA1", "TP53")
```

```
[1] "BRCA1" "TP53"
```

```
c(1, 3, 4, 5, 1, 2)
```

```
[1] 1 3 4 5 1 2
```

7. Vectors

```
c(1.12341e7, 78234.126)
```

```
[1] 11234100.00    78234.13
```

```
c(TRUE, FALSE, TRUE, TRUE)
```

```
[1] TRUE FALSE TRUE TRUE
```

```
# mixing a logical and a string forces BOTH to character:  
c(TRUE, "hello")
```

```
[1] "TRUE" "hello"
```

That last line is the coercion we just warned about: `TRUE` came back as `"TRUE"`.

When you want a regular run of integers, the colon operator `:` is the quickest tool. It counts from the first number to the second, in either direction.

```
# the integers 1 through 10  
sample_ids <- 1:10  
sample_ids
```

```
[1] 1 2 3 4 5 6 7 8 9 10
```

```
# ...and counting back down  
countdown <- 10:1  
countdown
```

```
[1] 10 9 8 7 6 5 4 3 2 1
```

For sequences with a custom step size, `seq()` is more flexible. Here we step by 0.3 instead of 1:

```
# from 1 to 4 in steps of 0.3  
grid <- seq(1, 4, by = 0.3)  
grid
```

7. Vectors

```
[1] 1.0 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.0
```

And you can build a new vector by gluing existing ones together with `c()`:

```
# concatenate two vectors into one
combined <- c(grid, sample_ids)
combined
```

```
[1] 1.0 1.3 1.6 1.9 2.2 2.5 2.8 3.1 3.4 3.7 4.0 1.0 2.0 3.0 4.0
[16] 5.0 6.0 7.0 8.0 9.0 10.0
```

7.4. Vector operations

Here is where vectors earn their keep. Arithmetic on a vector happens **element-by-element**. Add 2 to a vector and R adds 2 to every element, handing back a new vector of the same length. You never have to write a loop.

```
expression <- 1:10
expression + 2
```

```
[1] 3 4 5 6 7 8 9 10 11 12
```

Think of how often that's what you want: shift every measurement, scale every value, take the log of a whole column. One short line does it.

When two vectors are involved, the rules depend on their lengths. If they're the **same length**, R simply pairs up matching elements and operates on each pair:

```
expression + expression
```

```
[1] 2 4 6 8 10 12 14 16 18 20
```

If the lengths differ but the longer is a **multiple** of the shorter, R reuses the shorter vector from the start as many times as needed. Here the two-element vector `c(1, 2)` is repeated five times to line up with the ten-element vector:

7. Vectors

```
expression <- 1:10  
pair <- c(1, 2)  
expression * pair
```

```
[1] 1 4 3 8 5 12 7 16 9 20
```

If the longer length is **not** a clean multiple of the shorter, R still recycles the short vector — but warns you, because this is usually a sign of a mistake:

```
expression <- 1:10  
triple <- c(2, 3, 4)  
expression * triple
```

Warning in expression * triple: longer object length is not a multiple of shorter object length

```
[1] 2 6 12 8 15 24 14 24 36 20
```

The usual arithmetic is all here — multiplication (*), addition (+), subtraction (-), division (/), exponentiation (^) — and because each acts on whole vectors, we call these operations *vectorized*.

The recycling rule can bite

When you combine vectors of different lengths, R silently stretches the shorter one to match. If that wasn't your intent, you'll get plausible-looking but wrong numbers and no error. The “not a multiple” case at least prints a warning; the “clean multiple” case prints nothing at all. When two vectors should be the same length, check that they are — `length(x)` is your friend.

7.5. Logical vectors

A logical vector holds only **TRUE** and **FALSE** (all uppercase, no quotes). These turn up constantly, because asking a yes/no question of a whole vector returns one.

7. Vectors

```
keep <- c(TRUE, FALSE, TRUE)

# a logical vector can also be built from numbers:
# 0 becomes FALSE, anything else becomes TRUE
counts <- c(1, 0, 217)
as_flags <- as.logical(counts)
as_flags
```

```
[1] TRUE FALSE TRUE
```

```
# do keep and as_flags match element-by-element?
all.equal(keep, as_flags)
```

```
[1] TRUE
```

```
# and logical converts back to numeric: TRUE -> 1, FALSE -> 0
as.numeric(keep)
```

```
[1] 1 0 1
```

7.5.1. Logical operators

The comparison operators `<`, `>`, `==`, `>=`, `<=`, and `!=` each take a vector and return a logical vector — one TRUE/FALSE per element. This is the engine behind filtering data.

```
expression <- 1:10
# which elements are greater than 5?
expression > 5
```

```
[1] FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE TRUE
```

```
expression <= 5
```

```
[1] TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE
```

7. Vectors

```
expression != 5
```

```
[1] TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE
```

```
expression == 5
```

```
[1] FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
```

You can save that logical result in a variable to reuse later:

```
is_five <- (expression == 5)
is_five
```

```
[1] FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
```

 = assigns, == compares

A single = is assignment; a double == is a test for equality. Writing `x = 5` *changes* `x`, while `x == 5` *asks whether* `x` equals 5. Mixing them up is one of the most common beginner errors. In this book we always assign with `<-`, which keeps the two jobs visually distinct.

7.6. Indexing vectors

An *index* lets you reach into a vector and pull out one element or a handful. R uses square brackets, [and], for this.

 R counts from 1

Unlike many programming languages that start counting at 0, R starts at **1**. The first element of `x` is `x[1]`, the second is `x[2]`, and so on. If you've programmed before, this is a frequent source of off-by-one surprises.

```
x <- seq(0, 1, by = 0.1)
# the 4th element
x[4]
```

7. Vectors

```
[1] 0.3
```

You can index with a *vector* of positions to grab several elements at once:

```
x[c(3, 5, 6)]
```

```
[1] 0.2 0.4 0.5
```

```
positions <- 3:6  
x[positions]
```

```
[1] 0.2 0.3 0.4 0.5
```

Now combine indexing with logical vectors, and you have one of the most powerful patterns in all of R. Here we generate ten random values, flag the ones above 0.25, and keep only those:

```
# rnorm() draws random numbers; see help('rnorm')  
measurements <- rnorm(10)  
  
# a logical vector: TRUE where measurements exceed 0.25  
above <- (measurements > 0.25)  
# sum() of a logical vector counts the TRUEs  
sum(above)
```

```
[1] 5
```

```
# keep only the values where 'above' is TRUE  
measurements[above]
```

```
[1] 1.5952808 0.3295078 0.4874291 0.7383247 0.5757814
```

```
# or the complement, using the logical NOT operator "!"  
measurements[!above]
```

```
[1] -0.6264538 0.1836433 -0.8356286 -0.8204684 -0.3053884
```

7. Vectors

```
# the same idea in one compact line
measurements[measurements > 0.25]
```

```
[1] 1.5952808 0.3295078 0.4874291 0.7383247 0.5757814
```

That last line — *subset a vector by a condition on itself* — is something you’ll write hundreds of times. Read it as “the measurements where measurements exceed 0.25.”

7.7. Named vectors

Elements can carry names as well as positions, which often makes code far more readable. Imagine a small set of patient ages keyed by patient ID. You attach names with `names()`:

```
patient_ages <- c(54, 39, 71)
names(patient_ages) <- c("p01", "p02", "p03")
patient_ages
```

```
p01 p02 p03
 54  39  71
```

Once a vector is named, you can access elements by name instead of remembering their position:

```
patient_ages["p02"]
```

```
p02
 39
```

```
# and update by name, too
patient_ages["p01"] <- 55
patient_ages
```

```
p01 p02 p03
 55  39  71
```

Names are just labels riding alongside the values; the vector is still a plain numeric vector underneath.

7.8. Character vectors, a.k.a. strings

Text data — gene names, sample labels, file paths — lives in character vectors, and R has a toolkit for manipulating them. To glue strings together, use `paste()`:

```
paste("BRCA", "1")
```

```
[1] "BRCA 1"
```

```
paste("BRCA", "1", sep = "_")
```

```
[1] "BRCA_1"
```

```
paste0("BRCA", "1")          # paste0 uses no separator
```

```
[1] "BRCA1"
```

```
paste(c("sample", "control"), 1:10)
```

```
[1] "sample 1"  "control 2" "sample 3"  "control 4" "sample 5"
[6] "control 6" "sample 7"  "control 8" "sample 9"  "control 10"
```

```
paste(c("sample", "control"), 1:10, sep = "_")
```

```
[1] "sample_1"  "control_2" "sample_3"  "control_4" "sample_5"
[6] "control_6" "sample_7"  "control_8" "sample_9"  "control_10"
```

Notice `paste()` is vectorized too: in the last two lines it recycles the two-element vector across the ten numbers.

To count characters in each string, use `nchar()`:

```
nchar("BRCA1")
```

```
[1] 5
```

7. Vectors

```
nchar(c("BRCA1", "TP53", "EGFR"))
```

```
[1] 5 4 4
```

To pull a piece out of a string by position, use `substr()`:

```
substr("chr7:55019021", start = 1, stop = 4)
```

```
[1] "chr7"
```

To find-and-replace within a string, use `sub()`:

```
sub("chr", "Chromosome ", "chr7:55019021")
```

```
[1] "Chromosome 7:55019021"
```

And to find *which* strings in a vector contain a pattern, use `grep()` — the name comes from “globally search a regular expression and print.” By default it returns the matching *positions*; add `value = TRUE` to get the matching strings themselves:

```
genes <- c("BRCA1", "BRCA2", "TP53", "EGFR", "BARD1")
# positions of genes containing "BR"
grep("BR", genes)
```

```
[1] 1 2
```

```
# the matching gene names themselves
grep("BR", genes, value = TRUE)
```

```
[1] "BRCA1" "BRCA2"
```

A close relative, `grep1()`, answers the yes/no version of that question — you’ll meet it in the exercises.

7.9. Missing values, a.k.a. NA

Real data has holes: an assay that failed, a measurement never taken. R marks such gaps with the special value `NA` (*Not Available*). Crucially, `NA` is still an element — it occupies a position and counts toward `length()`.

```
x <- 1:5
x
```

```
[1] 1 2 3 4 5
```

```
length(x)
```

```
[1] 5
```

The function `is.na()` tests each element and returns a logical vector. Let's poke a hole in `x` by setting its second element to `NA`:

```
is.na(x)
```

```
[1] FALSE FALSE FALSE FALSE FALSE
```

```
x[2] <- NA
x
```

```
[1] 1 NA 3 4 5
```

The length hasn't changed — `x` still has five slots — but one of them is now flagged as missing:

```
length(x)
```

```
[1] 5
```

7. Vectors

```
is.na(x)
```

```
[1] FALSE TRUE FALSE FALSE FALSE
```

To drop the missing values, lean on the indexing pattern from earlier. `is.na(x)` gives a logical vector that is `TRUE` at the gaps; the `!` operator flips it to `TRUE` everywhere there's a real value; indexing with that keeps only the real values:

```
x[!is.na(x)]
```

```
[1] 1 3 4 5
```

7.10. Exercises

Each solution names the key idea and a one-line *why*, because the reasoning transfers further than the answer.

1. Create a numeric vector called `temperatures` containing the values 72, 75, 78, 81, 76, and 73.

i Solution

```
temperatures <- c(72, 75, 78, 81, 76, 73)
temperatures
```

```
[1] 72 75 78 81 76 73
```

`c()` combines individual values into one numeric vector — the most basic way to build a vector by hand.

2. Create a character vector called `days` containing “Monday”, “Tuesday”, “Wednesday”, “Thursday”, “Friday”, “Saturday”, and “Sunday”.

i Solution

```
days <- c("Monday", "Tuesday", "Wednesday", "Thursday",
          "Friday", "Saturday", "Sunday")
days
```

7. Vectors

```
[1] "Monday"    "Tuesday"    "Wednesday"  "Thursday"   "Friday"     "Saturday"
[7] "Sunday"
```

The same `c()`, now with quoted strings, so R makes a character vector instead of a numeric one.

3. Calculate the average of `temperatures` and store it in `average_temperature`.

i Solution

```
average_temperature <- mean(temperatures)
average_temperature
```

```
[1] 75.83333
```

`mean()` reduces a whole numeric vector to a single summary value.

4. Build a named vector `weekly_temperatures` whose names are the (first six) days and whose values are the six temperatures.

i Solution

```
weekly_temperatures <- temperatures
names(weekly_temperatures) <- days[1:6]
weekly_temperatures
```

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
72	75	78	81	76	73

`names()` attaches labels to existing elements; we use `days[1:6]` so the six names line up with the six temperatures.

5. Create a numeric vector `ages` with the values 25, 30, 35, 40, 45, 50, 55, and 60.

i Solution

```
ages <- c(25, 30, 35, 40, 45, 50, 55, 60)
ages
```

```
[1] 25 30 35 40 45 50 55 60
```

Again `c()` — the same tool builds every vector regardless of the numbers inside.

6. Create a logical vector `is_adult` that is `TRUE` where `ages` is at least 18.

7. Vectors

i Solution

```
is_adult <- ages >= 18
is_adult
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

A comparison applied to a vector returns a logical vector, one TRUE/FALSE per element.

7. Calculate both the sum and the product of `ages`.

i Solution

```
sum_ages <- sum(ages)
product_ages <- prod(ages)
sum_ages
```

```
[1] 340
```

```
product_ages
```

```
[1] 7.79625e+12
```

`sum()` and `prod()` are reductions: each collapses the whole vector to one number.

8. Extract the ages greater than or equal to 40 into a vector called `older_ages`.

i Solution

```
older_ages <- ages[ages >= 40]
older_ages
```

```
[1] 40 45 50 55 60
```

This is logical indexing: `ages >= 40` builds a TRUE/FALSE mask, and `ages[...]` keeps only the elements where it is TRUE.

9. The `grepl()` function (read `?grepl`) returns a logical vector — TRUE where a pattern is found, FALSE where it isn't. Using the vector below, return a logical vector that is TRUE for each entry containing the letter "a".

```
words <- c("abcdef", "abcd", "bcde", "cdef", "defg")
```

i Solution

```
grepl("a", words)
```

```
[1] TRUE TRUE FALSE FALSE FALSE
```

Where `grep()` returns *positions*, `grepl()` returns a logical vector you can feed straight into indexing — for example, `words[grepl("a", words)]` keeps only the matching entries.

10. From `measurements <- rnorm(20)`, count how many values are negative without writing a loop.

i Solution

```
set.seed(1)
measurements <- rnorm(20)
sum(measurements < 0)
```

```
[1] 8
```

`measurements < 0` is a logical vector, and `sum()` counts its TRUEs because TRUE equals 1 — the same count-by-summing trick from the indexing section.

7.11. Summary

You can now create and wield the container R uses for almost everything:

- **Create** vectors with `c()`, the colon operator `:`, and `seq()` — and recall that even a single value is a length-1 vector, all of one type.
- **Operate** on whole vectors at once; arithmetic is element-by-element, and the **recycling rule** stretches a shorter vector to match a longer one (with a warning when the lengths don't divide evenly).
- **Compare** with `>`, `==`, `!=`, and friends to produce logical vectors.
- **Index** with positions, with names, or with a logical test — and remember R counts from 1.
- **Manipulate strings** with `paste()`, `nchar()`, `substr()`, `sub()`, and `grep()/grepl()`.
- **Handle gaps** with NA and `is.na()`, dropping them via `x[!is.na(x)]`.

The logical-indexing pattern — ask a yes/no question of a vector, then use the answer to subset it — is the single most useful idea here. It reappears every time you filter data, so the practice you put in now pays off in every chapter to come.

8. Matrices

Picture the result of a genomics experiment: you measured the activity of twenty thousand genes across a dozen samples. The natural way to hold that is a grid — one row per gene, one column per sample, a number in every cell. That grid is a **matrix**, and it is the central object you’ll meet again and again in genomics. The gene-expression matrix you build here is the same shape that later powers a `SummarizedExperiment`, so the moves you learn now pay off for the rest of the book.

A matrix is just a rectangle of values that are **all the same type** — all numbers, all characters, or all logicals (see Figure 8.1). You can read it as a stack of columns (each column a vector of the same length and type) or as a stack of rows. If you’ve already met vectors, a matrix is the natural next step: take several vectors of equal length and the same type, and line them up side by side.

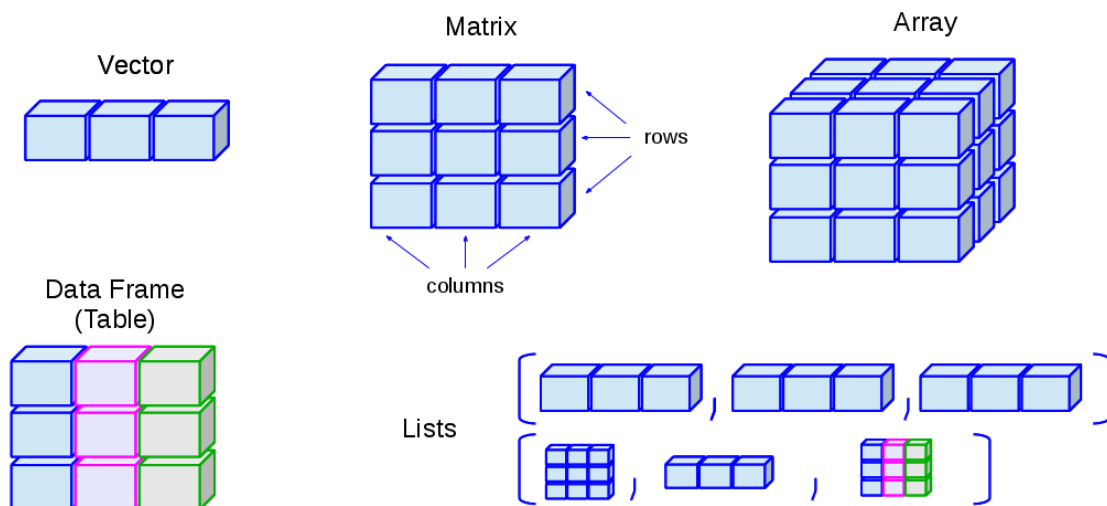


Figure 8.1.: A matrix is a collection of column vectors, all the same length and the same type.

i Matrix versus data frame

A *data frame* is also a rectangle, and it's easy to confuse the two. The difference is one rule: a matrix needs every value to be the **same type**, while a data frame lets each **column** have its own type — numbers in one, text in another. Use a matrix when your whole grid is numeric (like expression values); reach for a data frame when columns mix types (a sample table with names, ages, and treatment labels). The two even name their rows and columns differently under the hood, which is a classic source of confusion — but that's a problem for the data-frame chapter, not this one.

8.1. What you'll learn

- Build a matrix with `matrix()`, `rbind()`, and `cbind()`.
- Inspect a matrix's shape and labels with `dim()`, `nrow()`, `ncol()`, `rownames()`, and `colnames()`.
- Pull out elements, rows, and columns with `[row, col]` indexing.
- Change values in part of a matrix by combining subsetting with assignment.
- Summarize across rows and columns with `rowMeans()`, `colMeans()`, and `apply()` — the per-gene and per-sample summaries you'll use constantly in genomics.

8.2. Creating a matrix

There are many ways to build a matrix in R. The most direct is the `matrix()` function. Below we pour the numbers 1:16 into a four-row grid.

```
mat1 <- matrix(1:16, nrow = 4)
mat1
```

```
      [,1] [,2] [,3] [,4]
[1,]    1    5    9   13
[2,]    2    6   10   14
[3,]    3    7   11   15
[4,]    4    8   12   16
```

By default R fills the matrix **column by column** — walk down the first column, then the second, and so on. If you'd rather fill it row by row, set `byrow = TRUE`.

8. Matrices

```
mat1 <- matrix(1:16, nrow = 4, byrow = TRUE)
mat1
```

```
      [,1] [,2] [,3] [,4]
[1,]     1     2     3     4
[2,]     5     6     7     8
[3,]     9    10    11    12
[4,]    13    14    15    16
```

Look closely at where the numbers land. Same sixteen values, same shape, but a completely different arrangement. Whenever you create a matrix from a flat vector, ask yourself which direction it filled.

We can also assemble a matrix from parts by *binding* vectors together. Let's make two numeric vectors of length 10.

```
gene_a <- 1:10
gene_b <- rnorm(10)
```

Both are numeric and both have length 10, so they can sit together in a matrix. `rbind()` stacks them as rows (think “r for row”):

```
mat <- rbind(gene_a, gene_b)
mat
```

```
      [,1] [,2] [,3] [,4] [,5] [,6] [,7]
gene_a 1.0000000 2.0000000 3.0000000 4.0000000 5.0000000 6.0000000 7.0000000
gene_b -0.6264538 0.1836433 -0.8356286 1.595281 0.3295078 -0.8204684 0.4874291
      [,8] [,9] [,10]
gene_a 8.0000000 9.0000000 10.0000000
gene_b 0.7383247 0.5757814 -0.3053884
```

The companion function `cbind()` glues vectors together as columns:

```
mat <- cbind(gene_a, gene_b)
mat
```

8. Matrices

```
      gene_a    gene_b
[1,]      1 -0.6264538
[2,]      2  0.1836433
[3,]      3 -0.8356286
[4,]      4  1.5952808
[5,]      5  0.3295078
[6,]      6 -0.8204684
[7,]      7  0.4874291
[8,]      8  0.7383247
[9,]      9  0.5757814
[10,]     10 -0.3053884
```

Notice that R kept the vector names — `gene_a` and `gene_b` — as labels. Row and column names are worth their weight in gold once they carry real meaning, like gene identifiers or sample names. You can read them back at any time:

```
rownames(mat)
```

```
NULL
```

```
colnames(mat)
```

```
[1] "gene_a" "gene_b"
```

And you can set them by assigning a vector of valid names:

```
colnames(mat) <- c("control", "treated")
colnames(mat)
```

```
[1] "control" "treated"
```

```
mat
```

```
      control    treated
[1,]      1 -0.6264538
[2,]      2  0.1836433
[3,]      3 -0.8356286
[4,]      4  1.5952808
```

8. Matrices

```
[5,]      5  0.3295078
[6,]      6 -0.8204684
[7,]      7  0.4874291
[8,]      8  0.7383247
[9,]      9  0.5757814
[10,]     10 -0.3053884
```

Every matrix also knows its own shape. `dim()` reports rows then columns; `nrow()` and `ncol()` pull out one each.

```
dim(mat)
```

```
[1] 10  2
```

```
nrow(mat)
```

```
[1] 10
```

```
ncol(mat)
```

```
[1] 2
```

8.3. Accessing elements of a matrix

Indexing a matrix works like indexing a vector, except now you give **two** coordinates: the row first, then the column. The pattern is `mat[r, c]`, where `r` and `c` say which rows and columns you want.

! R counts from one

In R, the first element, row, or column is number **1** — not 0. If you’ve written Python, C, or Java, this will trip you up, because those languages count from 0. Reaching for the wrong starting index is one of the most common beginner mistakes, and the worst kind: it often doesn’t error, it just quietly hands you the wrong value.

Let’s pull various pieces out of `mat`:

8. Matrices

```
# the value in row 1, column 2
mat[1, 2]
```

```
      treated
-0.6264538
```

```
# the entire first ROW (leave the column slot empty)
mat[1, ]
```

```
      control      treated
1.0000000 -0.6264538
```

```
# the entire first COLUMN (leave the row slot empty)
mat[, 1]
```

```
[1] 1 2 3 4 5 6 7 8 9 10
```

Leaving a slot blank means “give me all of them.” So `mat[1, ]` is the whole first row and `mat[, 1]` is the whole first column. There’s one more form worth knowing — a single index with **no comma**:

```
# every value greater than 4; note: no comma
mat[mat > 4]
```

```
[1] 5 6 7 8 9 10
```

⚠ A matrix is a vector wearing a costume

When you index with a single subscript and no comma, R forgets the rectangle and treats the matrix as one long vector, returning a flat result. That’s handy for “give me all the big values,” but it’s also a trap: code that *looks* like it should respect rows and columns may silently flatten your matrix instead. Whenever you mean a row or a column, include the comma.

You can also *exclude* rows or columns by prefixing the index with a minus sign:

8. Matrices

```
mat[, -1]          # drop the first column
```

```
[1] -0.6264538  0.1836433 -0.8356286  1.5952808  0.3295078 -0.8204684
[7]  0.4874291  0.7383247  0.5757814 -0.3053884
```

```
mat[-c(1:5), ]     # drop the first five rows
```

	control	treated
[1,]	6	-0.8204684
[2,]	7	0.4874291
[3,]	8	0.7383247
[4,]	9	0.5757814
[5,]	10	-0.3053884

8.4. Changing values in a matrix

Let's make a fresh matrix of random values to experiment on — pretend it's a small expression matrix with measurements drawn from a normal distribution.

```
expr <- matrix(rnorm(20), nrow = 10)
summary(expr)
```

V1	V2
Min. : -2.21470	Min. : -1.989352
1st Qu.: -0.03775	1st Qu.: -0.397561
Median : 0.49187	Median : 0.009218
Mean : 0.24884	Mean : -0.133673
3rd Qu.: 0.91318	3rd Qu.: 0.569355
Max. : 1.51178	Max. : 0.918977

Arithmetic on a matrix behaves much like arithmetic on a vector. Multiplying by a single number scales every cell at once:

```
# multiply every value by 20
expr2 <- expr * 20
summary(expr2)
```

8. Matrices

V1	V2
Min. : -44.294	Min. : -39.7870
1st Qu.: -0.755	1st Qu.: -7.9512
Median : 9.837	Median : 0.1844
Mean : 4.977	Mean : -2.6735
3rd Qu.: 18.264	3rd Qu.: 11.3871
Max. : 30.236	Max. : 18.3795

By combining subsetting with assignment, you can rewrite just *part* of a matrix. Here we bump up only the first column:

```
# add 100 to the first column of expr2
expr2[, 1] <- expr2[, 1] + 100
summary(expr2)
```

V1	V2
Min. : 55.71	Min. : -39.7870
1st Qu.: 99.25	1st Qu.: -7.9512
Median :109.84	Median : 0.1844
Mean :104.98	Mean : -2.6735
3rd Qu.:118.26	3rd Qu.: 11.3871
Max. :130.24	Max. : 18.3795

A common reshaping move is to **transpose** — flip rows into columns and vice versa with `t()`. You'll need this surprisingly often: a lab instrument might write samples in rows and genes in columns, while Bioconductor expects genes in rows and samples in columns. One call sets it right.

```
t(expr2)
```

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	130.23562	107.79686	87.57519	55.70600	122.49862	99.101328	99.67619
[2,]	18.37955	15.64273	1.49130	-39.78703	12.39651	-1.122575	-3.11591
	[,8]	[,9]	[,10]				
[1,]	118.87672	116.424424	111.878026				
[2,]	-29.41505	-9.563001	8.358831				

8.5. Calculations on matrix rows and columns

For this section we need a matrix to play with. We'll use `rnorm()` again, but ask for values centered at 5 with a standard deviation of 2 — the two extra arguments are the mean and the spread.

```
expr3 <- matrix(rnorm(100, 5, 2), ncol = 10)
```

Because these values come from a normal distribution centered at 5, any single column (or row) should have a mean near 5 and a standard deviation near 2. Let's check the first column, then the first row:

```
mean(expr3[, 1])
```

```
[1] 5.24146
```

```
sd(expr3[, 1])
```

```
[1] 1.617129
```

```
# now a row
mean(expr3[1, ])
```

```
[1] 5.319228
```

```
sd(expr3[1, ])
```

```
[1] 2.047589
```

Computing one summary at a time gets tedious fast. R gives you convenience functions that sweep across **all** the rows or **all** the columns at once. In a gene-expression matrix, `rowMeans()` is the average expression *per gene* and `colMeans()` is the average *per sample* — two of the most common summaries in genomics.

8. Matrices

```
colMeans(expr3)
```

```
[1] 5.241460 5.268273 5.286914 5.902420 4.504528 5.254721 5.224683 5.694960  
[9] 4.760354 4.987658
```

```
rowMeans(expr3)
```

```
[1] 5.319228 5.055767 6.074298 4.394364 4.717785 5.863346 4.492896 5.169545  
[9] 5.225423 5.813319
```

```
rowSums(expr3)
```

```
[1] 53.19228 50.55767 60.74298 43.94364 47.17785 58.63346 44.92896 51.69545  
[9] 52.25423 58.13319
```

```
colSums(expr3)
```

```
[1] 52.41460 52.68273 52.86914 59.02420 45.04528 52.54721 52.24683 56.94960  
[9] 47.60354 49.87658
```

We can save the column means and look at how they're spread:

```
cmeans <- colMeans(expr3)  
summary(cmeans)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
4.505	5.047	5.248	5.213	5.282	5.902

Notice the column means cluster tightly around 5 — exactly the center we asked for when we built the matrix. The averaging smooths out the noise in the individual values.

What about the standard deviation of each column? There's no built-in `colSd()`, but you don't need one. The `apply()` function takes any function that works on a vector — like `sd()` — and *applies* it across either dimension: 1 means rows, 2 means columns.

```
csds <- apply(expr3, 2, sd)
summary(csds)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.192   1.261   1.610   1.672   1.990   2.577
```

These per-column standard deviations sit close to 2, the spread we built in. Keep `apply()` in your back pocket: any time R lacks a ready-made `rowX()` or `colX()`, `apply()` lets you roll your own.

💡 Remember the second argument

The number in `apply(x, MARGIN, FUN)` is the dimension you keep: 1 for rows (one result per row), 2 for columns (one result per column). A quick way to remember: it matches the order in `[row, column]` — 1 is rows, 2 is columns.

8.6. Exercises

For these exercises we'll use a dataset that ships with R: monthly sunspot counts from 1749 to 1983. It arrives as a `ts` (time series) object, so the code below reshapes it into a matrix with one **row per year** and one **column per month**. Just run it as written and move on to the questions.

```
data(sunspots)
sunspot_mat <- matrix(as.vector(sunspots), ncol = 12, byrow = TRUE)
colnames(sunspot_mat) <- as.character(1:12)
rownames(sunspot_mat) <- as.character(1749:1983)
```

1. **Get to know the matrix.** What does `sunspot_mat` look like? Find its number of rows, its number of columns, and some basic summary statistics.

i Solution

```
nrow(sunspot_mat)
```

```
[1] 235
```

8. Matrices

```
ncol(sunspot_mat)
```

```
[1] 12
```

```
dim(sunspot_mat)
```

```
[1] 235 12
```

```
summary(sunspot_mat)
```

1		2		3		4	
Min.	: 0.00	Min.	: 0.00	Min.	: 0.00	Min.	: 0.00
1st Qu.:	14.60	1st Qu.:	15.65	1st Qu.:	14.80	1st Qu.:	16.55
Median :	40.60	Median :	44.00	Median :	45.60	Median :	41.00
Mean :	49.11	Mean :	51.17	Mean :	50.04	Mean :	51.00
3rd Qu.:	70.00	3rd Qu.:	74.50	3rd Qu.:	72.25	3rd Qu.:	75.30
Max.	:217.40	Max.	:182.30	Max.	:190.70	Max.	:196.00

5		6		7		8	
Min.	: 0.00	Min.	: 0.00	Min.	: 0.00	Min.	: 0.00
1st Qu.:	19.65	1st Qu.:	15.55	1st Qu.:	16.30	1st Qu.:	16.55
Median :	41.30	Median :	40.50	Median :	41.90	Median :	40.70
Mean :	52.48	Mean :	51.55	Mean :	51.47	Mean :	52.07
3rd Qu.:	76.80	3rd Qu.:	80.95	3rd Qu.:	76.85	3rd Qu.:	73.70
Max.	:238.90	Max.	:200.70	Max.	:191.40	Max.	:200.20

9		10		11		12	
Min.	: 0.00	Min.	: 0.00	Min.	: 0.00	Min.	: 0.00
1st Qu.:	14.60	1st Qu.:	16.60	1st Qu.:	14.75	1st Qu.:	17.25
Median :	42.70	Median :	43.80	Median :	40.50	Median :	41.20
Mean :	52.28	Mean :	51.86	Mean :	50.64	Mean :	51.51
3rd Qu.:	77.00	3rd Qu.:	71.50	3rd Qu.:	68.45	3rd Qu.:	74.70
Max.	:235.80	Max.	:253.80	Max.	:210.90	Max.	:239.40

```
head(sunspot_mat)
```

	1	2	3	4	5	6	7	8	9	10	11	12
1749	58.0	62.6	70.0	55.7	85.0	83.5	94.8	66.3	75.9	75.5	158.6	85.2
1750	73.3	75.9	89.2	88.3	90.0	100.0	85.4	103.0	91.2	65.7	63.3	75.4
1751	70.0	43.5	45.3	56.4	60.7	50.7	66.3	59.8	23.5	23.2	28.5	44.0
1752	35.0	50.0	71.0	59.3	59.7	39.6	78.4	29.3	27.1	46.6	37.6	40.0
1753	44.0	32.0	45.7	38.0	36.0	31.7	22.2	39.0	28.0	25.0	20.0	6.7
1754	0.0	3.0	1.7	13.7	20.7	26.7	18.8	12.3	8.2	24.1	13.2	4.2

dim() confirms 235 years by 12 months, and head() shows the first few years

— always eyeball the shape before you start computing.

2. **Practice subsetting** by pulling out: the first 10 years (rows), the month of July (the 7th column), and the single value for July 1979 using the row name to select it.

i Solution

```
sunspot_mat[1:10, ]
```

	1	2	3	4	5	6	7	8	9	10	11	12
1749	58.0	62.6	70.0	55.7	85.0	83.5	94.8	66.3	75.9	75.5	158.6	85.2
1750	73.3	75.9	89.2	88.3	90.0	100.0	85.4	103.0	91.2	65.7	63.3	75.4
1751	70.0	43.5	45.3	56.4	60.7	50.7	66.3	59.8	23.5	23.2	28.5	44.0
1752	35.0	50.0	71.0	59.3	59.7	39.6	78.4	29.3	27.1	46.6	37.6	40.0
1753	44.0	32.0	45.7	38.0	36.0	31.7	22.2	39.0	28.0	25.0	20.0	6.7
1754	0.0	3.0	1.7	13.7	20.7	26.7	18.8	12.3	8.2	24.1	13.2	4.2
1755	10.2	11.2	6.8	6.5	0.0	0.0	8.6	3.2	17.8	23.7	6.8	20.0
1756	12.5	7.1	5.4	9.4	12.5	12.9	3.6	6.4	11.8	14.3	17.0	9.4
1757	14.1	21.2	26.2	30.0	38.1	12.8	25.0	51.3	39.7	32.5	64.7	33.5
1758	37.6	52.0	49.0	72.3	46.4	45.0	44.0	38.7	62.5	37.7	43.0	43.0

```
sunspot_mat[, 7]
```

1749	1750	1751	1752	1753	1754	1755	1756	1757	1758	1759	1760	1761
94.8	85.4	66.3	78.4	22.2	18.8	8.6	3.6	25.0	44.0	51.0	66.0	94.1
1762	1763	1764	1765	1766	1767	1768	1769	1770	1771	1772	1773	1774
33.8	54.2	30.0	27.0	3.3	21.9	52.6	118.6	109.8	67.7	77.3	27.7	17.5
1775	1776	1777	1778	1779	1780	1781	1782	1783	1784	1785	1786	1787
1.0	1.0	95.0	153.0	143.0	86.0	73.5	37.0	32.2	6.0	36.3	83.0	128.0
1788	1789	1790	1791	1792	1793	1794	1795	1796	1797	1798	1799	1800
141.5	117.0	69.3	71.0	45.8	50.0	41.0	12.9	26.9	4.3	0.0	2.1	21.0
1801	1802	1803	1804	1805	1806	1807	1808	1809	1810	1811	1812	1813
35.0	48.0	48.3	29.8	37.6	30.0	12.7	6.7	0.3	0.0	6.6	0.5	18.3
1814	1815	1816	1817	1818	1819	1820	1821	1822	1823	1824	1825	1826
18.5	35.5	38.8	50.0	28.0	31.4	20.6	2.5	7.9	0.5	0.0	30.9	52.5
1827	1828	1829	1830	1831	1832	1833	1834	1835	1836	1837	1838	1839
42.9	54.3	90.8	43.9	45.2	13.9	7.0	8.7	59.8	116.7	162.8	108.2	84.7
1840	1841	1842	1843	1844	1845	1846	1847	1848	1849	1850	1851	1852
60.7	30.8	12.6	9.5	21.2	30.6	46.5	52.2	139.2	78.0	39.1	36.1	42.0
1853	1854	1855	1856	1857	1858	1859	1860	1861	1862	1863	1864	1865
45.9	18.7	0.4	4.6	22.2	56.7	95.2	116.7	78.0	73.4	32.7	54.7	26.8

8. Matrices

1866	1867	1868	1869	1870	1871	1872	1873	1874	1875	1876	1877	1878
9.3	5.0	28.6	59.2	132.4	103.0	105.5	66.9	67.8	12.5	15.2	5.9	0.1
1879	1880	1881	1882	1883	1884	1885	1886	1887	1888	1889	1890	1891
7.5	21.9	76.9	45.4	80.6	53.1	66.5	30.3	23.3	3.1	9.7	11.6	58.8
1892	1893	1894	1895	1896	1897	1898	1899	1900	1901	1902	1903	1904
76.8	88.8	106.0	47.8	45.0	27.6	9.0	13.5	8.3	0.7	0.9	27.9	50.6
1905	1906	1907	1908	1909	1910	1911	1912	1913	1914	1915	1916	1917
73.0	103.6	49.7	39.5	35.8	14.1	3.5	3.0	1.7	5.4	71.6	53.5	119.8
1918	1919	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929	1930
107.6	64.7	27.5	41.9	10.9	3.5	28.1	38.5	52.3	54.9	98.0	70.2	21.9
1931	1932	1933	1934	1935	1936	1937	1938	1939	1940	1941	1942	1943
17.4	9.6	2.8	9.3	33.9	52.3	145.1	165.3	97.6	67.5	66.9	17.7	13.2
1944	1945	1946	1947	1948	1949	1950	1951	1952	1953	1954	1955	1956
5.0	42.6	116.2	157.9	142.2	125.8	91.0	61.5	39.3	8.6	4.8	26.7	129.1
1957	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969
187.2	191.4	149.6	121.7	70.2	21.8	19.6	3.1	11.9	56.7	91.5	96.1	96.8
1970	1971	1972	1973	1974	1975	1976	1977	1978	1979	1980	1981	1982
112.5	81.0	76.5	23.1	55.8	28.2	1.9	21.4	70.4	159.4	136.3	143.8	106.1
1983												
82.2												

```
sunspot_mat["1979", 7]
```

```
[1] 159.4
```

Because we set row names, you can index by the label "1979" instead of counting rows — far less error-prone than figuring out which row number that is.

3. **Whole-matrix summaries.** A function that expects a vector — like `max()` — will happily treat the whole matrix as one long vector. What is the highest monthly sunspot count in the data?

i Solution

```
max(sunspot_mat)
```

```
[1] 253.8
```

No comma, no row or column — `max()` flattens the matrix and scans every value at once.

4. **And the minimum?**

i Solution

```
min(sunspot_mat)
```

```
[1] 0
```

Same idea as `max()`, in the other direction.

5. **Center of the data.** What are the overall mean and median?

i Solution

```
mean(sunspot_mat)
```

```
[1] 51.26596
```

```
median(sunspot_mat)
```

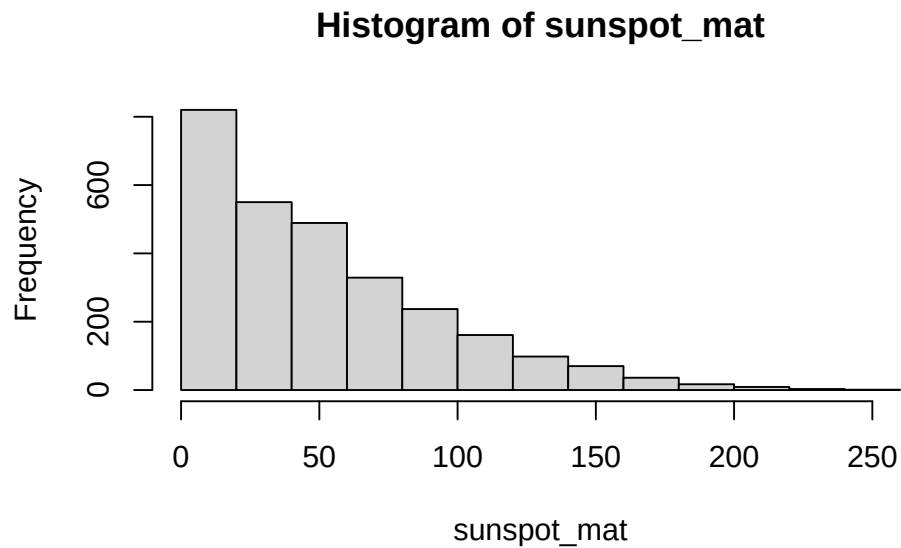
```
[1] 42
```

The mean sits well above the median, a hint that the distribution is skewed — a few very active months pull the average up.

6. **See the shape of the data.** Use `hist()` to look at the distribution of all the monthly counts.

i Solution

```
hist(sunspot_mat)
```



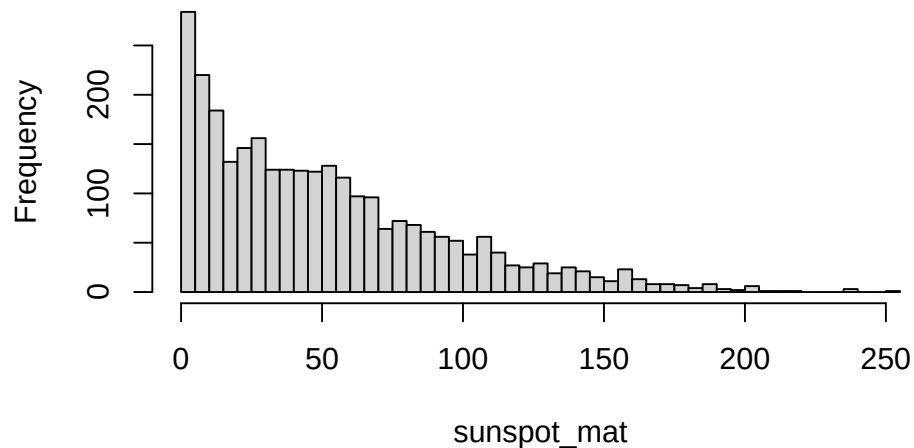
The long right tail confirms the skew we suspected from the mean and median: most months are quiet, a handful are very busy.

7. **Sharpen the histogram.** Read about the `breaks` argument to `hist()` and use it to increase the number of bars for more resolution.

i Solution

```
hist(sunspot_mat, breaks = 40)
```

Histogram of sunspot_mat



More breaks means narrower bars, which reveals finer structure in the distribution.

8. **Average by month.** We'd like the mean number of sunspots for each month of the year. Which summary function gives a result *per column*?

i Solution

```
colMeans(sunspot_mat)
```

```
      1      2      3      4      5      6      7      8
49.11191 51.17489 50.04085 51.00298 52.47830 51.54936 51.46553 52.06511
      9     10     11     12
52.28170 51.86468 50.64170 51.51447
```

```
# apply() does the same thing:
```

```
apply(sunspot_mat, 2, mean)
```

```
      1      2      3      4      5      6      7      8
49.11191 51.17489 50.04085 51.00298 52.47830 51.54936 51.46553 52.06511
      9     10     11     12
52.28170 51.86468 50.64170 51.51447
```

Months are the columns, so `colMeans()` (or `apply(..., 2, mean)`) gives one

8. Matrices

average per month.

9. **Summarize the monthly means.** Save the per-month averages to a variable and summarize it to see the spread across months.

i Solution

```
month_means <- colMeans(sunspot_mat)
summary(month_means)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
49.11	50.91	51.49	51.27	51.91	52.48

The monthly averages are quite close to each other — sunspot activity doesn't depend much on the time of year.

10. **Average by year.** Now play the same game across rows to get one mean per year.

i Solution

```
year_means <- rowMeans(sunspot_mat)
summary(year_means)
```

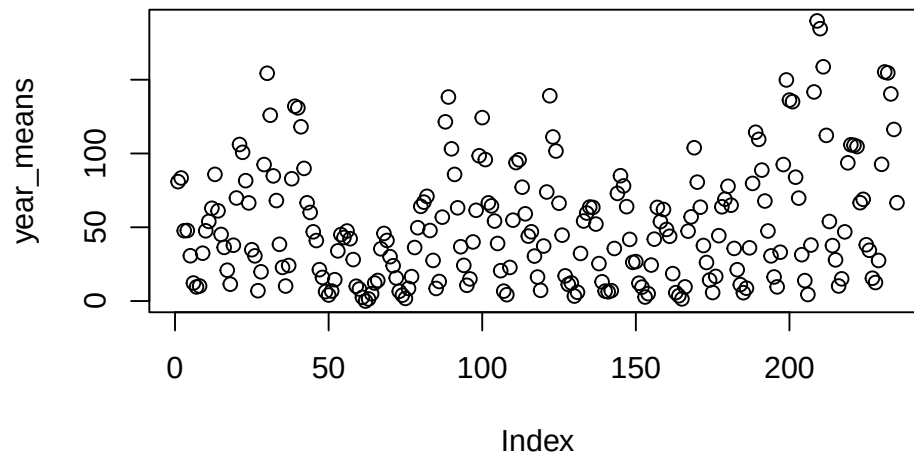
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	16.15	43.05	51.27	70.41	189.85

Years are the rows, so `rowMeans()` collapses each year to a single average.

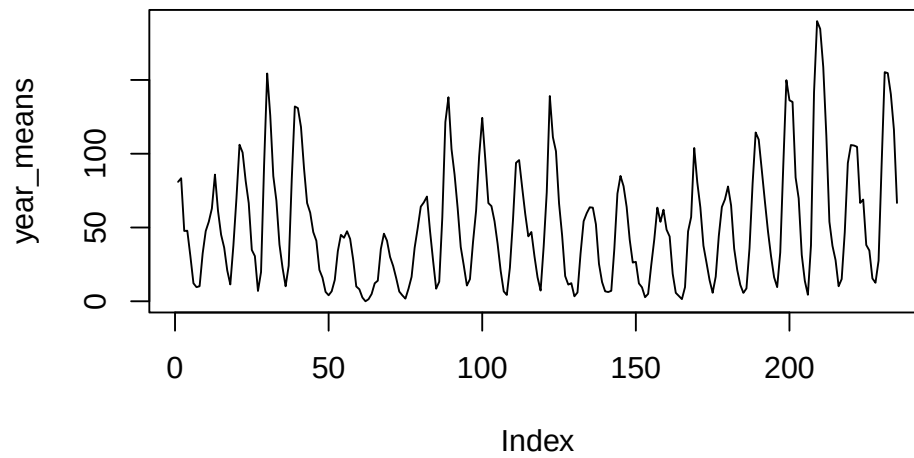
11. **Look for a pattern.** Plot the yearly means. Do you see anything?

i Solution

```
plot(year_means)
```



```
# a line makes the cycle clearer  
plot(year_means, type = "l")
```



The famous ~11-year sunspot cycle jumps out as a regular series of peaks and

troughs — a pattern that’s far easier to see across years (rows) than across months (columns).

8.7. Summary

You can now treat a matrix as the workhorse rectangle it is:

- **Build** one with `matrix()`, or assemble it from vectors with `rbind()` and `cbind()`.
- **Inspect** its shape and labels with `dim()`, `nrow()`, `ncol()`, `rownames()`, and `colnames()`.
- **Index** with `mat[row, col]` — remembering R counts from 1, that a blank slot means “all,” and that a single index with no comma flattens the matrix to a vector (Section 8.3).
- **Edit** part of a matrix by combining subsetting with assignment, and flip its orientation with `t()`.
- **Summarize** across rows and columns with `rowMeans()`, `colMeans()`, and the more general `apply()` — the per-gene and per-sample summaries at the heart of genomics.

That last idea — genes in rows, samples in columns, summaries running in either direction — is exactly the mental model you’ll carry into the **SummarizedExperiment** chapter, where a matrix like this becomes the beating heart of a real genomics container.

9. Lists: a catch-all container

So far in our journey through R's data structures, we've dealt with vectors and matrices. These are fantastic tools, but they have one strict rule: all their elements must be of the *same data type*. You can have a vector of numbers or a matrix of characters, but you can't mix and match.

But what about real-world biological data? A single experiment can generate a dizzying variety of information. Imagine you're studying a particular gene. You might have:

- The gene's name (text).
- Its expression level across several samples (a set of numbers).
- A record of whether it's a known cancer-related gene (a simple TRUE/FALSE).
- The raw fluorescence values from your qPCR machine (a matrix of numbers).
- Some personal notes about the experiment (a paragraph of text).

How could you possibly store all of this related, yet different, information together? You could create many separate variables, but that would be clunky and hard to manage. This is exactly the problem that **lists** are designed to solve.

A list in R is like a flexible, multi-compartment container. It's a single object that can hold a collection of *other* R objects, and those objects can be of any type, length, or dimension. You can put vectors, matrices, logical values, and even other lists inside a single list. This makes them one of the most fundamental and powerful data structures for bioinformatics analysis. In fact, almost every rich result R hands you back — a fitted model, the output of a statistical test, a whole data frame — is a list under the hood. Learn lists and you've learned how to take those results apart.

The key features of lists are:

- **Flexibility:** They can contain a mix of any data type.
- **Organization:** You can and should *name* the elements of a list, making your data self-describing.
- **Hierarchy:** Because lists can contain other lists, you can create complex, nested data structures to represent sophisticated relationships in your data.

9.1. What you'll learn

- Create a list with `list()` and give its elements descriptive names.
- Inspect a list's contents with `str()`, `length()`, `names()`, and `class()`.
- Pull a single element *out* of a list with `[[ ]]` or `$`.
- Take a smaller *sub-list* out with single `[ ]`, and explain why that is different.
- Add, update, and remove list elements.
- Use a list to build a self-contained record for a gene.

9.2. Creating a list

You create a list with the `list()` function. The best practice is to name the elements as you create them. This makes your code infinitely more readable and your data easier to work with.

Let's create a list to store the information for our hypothetical gene study.

```
# An experiment tracking list for the gene TP53
experiment_data <- list(
  experiment_id = "EXP042",
  gene_name = "TP53",
  read_counts = c(120, 155, 98, 210),
  is_control = FALSE,
  sample_matrix = matrix(1:4, nrow = 2, dimnames = list(c("Treated", "Untreated"), c("Repli
)
print(experiment_data)
```

```
$experiment_id
[1] "EXP042"
```

```
$gene_name
[1] "TP53"
```

```
$read_counts
[1] 120 155 98 210
```

```
$is_control
[1] FALSE
```

9. Lists: a catch-all container

```
$sample_matrix
      Replicate1 Replicate2
Treated          1          3
Untreated         2          4
```

Notice how the printout walks through the list one named compartment at a time: `$experiment_id` holds a single string, `$read_counts` holds a four-number vector, and `$sample_matrix` holds an entire 2×2 matrix — all living happily inside one object.

i What does `print()` do?

The `print()` function displays the contents of an R object in the console. For a list, it shows each element and its contents. It's the default action when you just type the variable's name and hit Enter, so `print(experiment_data)` and simply typing `experiment_data` do the same thing.

9.3. Inspecting your list: what's inside?

When someone hands you a tube in the lab, the first thing you do is look at the label. When R gives you a complex object like a list, you need to do the same. R provides several “introspection” functions to help you understand the contents and structure of your lists.

9.3.1. `str()`: the structure function

This is arguably the most useful function for inspecting any R object, especially lists.

```
str(experiment_data)
```

```
List of 5
 $ experiment_id: chr "EXP042"
 $ gene_name    : chr "TP53"
 $ read_counts  : num [1:4] 120 155 98 210
 $ is_control   : logi FALSE
 $ sample_matrix: int [1:2, 1:2] 1 2 3 4
 .. attr(*, "dimnames")=List of 2
 .. ..$ : chr [1:2] "Treated" "Untreated"
 .. ..$ : chr [1:2] "Replicate1" "Replicate2"
```

9. Lists: a catch-all container

The output of `str()` tells us everything we need to know: it's a "List of 5", and for each of the 5 elements, it shows the name (e.g., `experiment_id`), the data type (e.g., `chr` for character, `num` for numeric), and a preview of the content.

💡 `str()` is your best friend for any new object

`str()` provides a compact, human-readable summary of an object's internal **structure**. Whenever a function hands you back something you don't recognize — a list, a model, a complicated result — run `str()` on it first. It's the fastest way to see what's inside without printing pages of output.

9.3.2. `length()`, `names()`, and `class()`

These functions give you more specific information about the list itself.

```
# How many top-level elements does the list contain?  
length(experiment_data)
```

```
[1] 5
```

```
# What are the elements called?  
names(experiment_data)
```

```
[1] "experiment_id" "gene_name"      "read_counts"    "is_control"  
[5] "sample_matrix"
```

```
# What kind of object is this?  
class(experiment_data)
```

```
[1] "list"
```

Read together, these tell a story: the list has 5 elements (`length()`), they are named `experiment_id`, `gene_name`, and so on (`names()`), and the object as a whole is a `list` (`class()`). The `names()` output is especially handy — it's a menu of everything you're allowed to ask for.

9.4. Accessing list elements: getting things out

Okay, you've packed your experimental data into a list. Now, how do you get specific items out? This is a critical concept, and R has a few ways to do it, each with a distinct purpose.

9.4.1. The mighty `[[ ]]` and `$` for single items

To pull out a *single element* from a list in its original form, you use either double square brackets `[[ ]]` or the dollar sign `$` (for named lists). Think of this as carefully reaching into a specific compartment of your container and taking out the item itself.

Let's use our `experiment_data` list.

```
# Get the gene name using [[ ]]
gene <- experiment_data[["gene_name"]]
print(gene)
```

```
[1] "TP53"
```

```
class(gene) # It's a character vector, just as it was when we put it in.
```

```
[1] "character"
```

```
# Get the read counts using the $ shortcut. This is often easier to read.
reads <- experiment_data$read_counts
print(reads)
```

```
[1] 120 155 98 210
```

```
class(reads) # It's a numeric vector.
```

```
[1] "numeric"
```

```
# The [[ ]] has a neat trick: you can use a variable to specify the name.
element_to_get <- "read_counts"
experiment_data[[element_to_get]]
```


9. Lists: a catch-all container

```
[1] 120 155 98 210
```

The key takeaway is that `[[ ]]` and `$` **extract the element**. The result is the object that was stored inside the list — a plain character vector, a plain numeric vector — ready to be used in calculations. Notice that `$` needs the name typed out literally, while `[[ ]]` can take a name stored in a variable, which is invaluable when you write loops over a list's elements.

9.4.2. The subsetting `[ ]` for new lists

The single square bracket `[ ]` behaves differently. It always returns a *new, smaller list* that is a subset of the original list. It's like taking a whole compartment, label and all, out of your larger container — the item stays wrapped inside its list.

```
# Get the gene name using [ ]
gene_sublist <- experiment_data["gene_name"]

print(gene_sublist)
```

```
$gene_name
[1] "TP53"
```

```
# Note the class!
class(gene_sublist)
```

```
[1] "list"
```

See the difference? `experiment_data[["gene_name"]]` gave us a `"character"` vector, but `experiment_data["gene_name"]` gives us back a `"list"` that still has `gene_name` inside it.

! `[[ ]]` extracts; `[ ]` subsets. This trips up everyone.

This single-versus-double-bracket distinction is the most common source of confusion for R beginners — so it's worth burning in:

- `[[ ]]` and `$` **reach in and pull the item out**. You get the vector, matrix, or value itself.

9. Lists: a catch-all container

- `[ ]` returns a smaller *list*. The item is still wrapped inside a list.

This matters the moment you try to compute. To take the `mean()` of the read counts you must *extract* them:

```
mean(experiment_data[["read_counts"]]) # works: operating on a numeric vector
```

```
[1] 145.75
```

But `mean(experiment_data["read_counts"])` would fail, because you can't take the mean of a *list*. When in doubt, ask yourself: do I want the thing itself (`[ ]`), or a smaller list (`[ ]`)?

9.5. Modifying lists

Your data is rarely static. You can easily add, remove, or update elements in a list after you've created it.

9.5.1. Adding and updating elements

You can add a new element or change an existing one by using the `$` or `[ ]` assignment syntax. If the name already exists, R updates it; if it's new, R adds it.

```
# Add the date of the experiment
experiment_data$date <- "2024-06-05"

# Add some notes using the [ ] syntax
experiment_data[["notes"]] <- "Initial pilot experiment. High variance in read counts."

# Let's update the control status
experiment_data$is_control <- TRUE

# Let's look at the structure now
str(experiment_data)
```

```
List of 7
```

```
$ experiment_id: chr "EXP042"
```

9. Lists: a catch-all container

```
$ gene_name      : chr "TP53"
$ read_counts    : num [1:4] 120 155 98 210
$ is_control     : logi TRUE
$ sample_matrix: int [1:2, 1:2] 1 2 3 4
..- attr(*, "dimnames")=List of 2
.. ..$ : chr [1:2] "Treated" "Untreated"
.. ..$ : chr [1:2] "Replicate1" "Replicate2"
$ date           : chr "2024-06-05"
$ notes          : chr "Initial pilot experiment. High variance in read counts."
```

The list has grown from 5 elements to 7 — `date` and `notes` are new — and `is_control` has flipped from `FALSE` to `TRUE`. Same syntax, two different jobs: a name that already exists gets overwritten, a name that doesn't gets created.

9.5.2. Removing elements

To remove an element from a list, you simply assign `NULL` to it. `NULL` is R's special object representing nothingness.

```
# We've decided the matrix isn't needed for this summary object.
experiment_data$sample_matrix <- NULL

# See the final structure of our list
str(experiment_data)
```

List of 6

```
$ experiment_id: chr "EXP042"
$ gene_name     : chr "TP53"
$ read_counts   : num [1:4] 120 155 98 210
$ is_control    : logi TRUE
$ date          : chr "2024-06-05"
$ notes         : chr "Initial pilot experiment. High variance in read counts."
```

The `sample_matrix` element is gone, and the list is back down to 6 elements. Assigning `NULL` is R's idiom for “delete this compartment entirely.”

9.6. A biological example: a self-contained gene record

Let's put this all together. Lists are perfect for creating self-contained records that you can easily pass to functions or combine into larger lists.

```
brca1_gene <- list(  
  gene_symbol = "BRCA1",  
  full_name = "BRCA1 DNA repair associated",  
  chromosome = "17",  
  expression_log2 = log2(c(45, 50, 30, 88, 120)),  
  related_diseases = c("Breast Cancer", "Ovarian Cancer")  
)  
  
# Now we can easily work with this structured information  
cat("Analyzing gene:", brca1_gene$gene_symbol, "\n")
```

Analyzing gene: BRCA1

```
cat("Located on chromosome:", brca1_gene$chromosome, "\n")
```

Located on chromosome: 17

```
# Calculate the average log2 expression  
avg_expression <- mean(brca1_gene$expression_log2)  
cat("Average log2 expression:", avg_expression, "\n")
```

Average log2 expression: 5.881784

Each `cat()` line prints a label followed by a value pulled straight out of the list with `$`, and the average log2 expression comes out around 5.9.

i A couple of helper functions used above

- **cat()** concatenates and prints its arguments to the console. Unlike **print()**, it lets you seamlessly join text and variables on one line, and the `"\n"` character adds a newline (a line break).
- **log2()** calculates the base-2 logarithm. It's very common in gene-expression analysis to transform skewed count data with **log2()** to make it more symmetric

and easier to model.

This simple `brca1_gene` list is now a complete, portable record. You could imagine creating a list of these gene records, building a powerful, hierarchical database for your entire project.

9.7. Exercises

For these exercises, keep the `brca1_gene` list from above handy. Each one practices a move you'll use constantly when pulling apart analysis results.

1. **Extract by name.** Pull the `related_diseases` element *out* of `brca1_gene` so that you end up with a plain character vector (not a list). Confirm its class.

i Solution

```
diseases <- brca1_gene[["related_diseases"]]
diseases
```

```
[1] "Breast Cancer" "Ovarian Cancer"
```

```
class(diseases)
```

```
[1] "character"
```

Using `[[ ]]` (or `brca1_gene$related_diseases`) extracts the element itself, so `class()` reports `"character"`. Had you used single `[ ]`, you'd have gotten a one-element *list* back instead.

2. **Add an element.** A collaborator tells you BRCA1 sits on the *minus* strand. Add a new element called `strand` with the value `"-"` to `brca1_gene`, then confirm it's there.

i Solution

```
brca1_gene$strand <- "-"
names(brca1_gene)
```

```
[1] "gene_symbol"      "full_name"        "chromosome"       "expression_log2"
```

```
[5] "related_diseases" "strand"
```

Assigning to a name that doesn't yet exist *adds* a new compartment. Checking `names()` is a quick way to confirm the addition landed.

9. Lists: a catch-all container

3. **Nest a vector inside a list.** Add an element called `samples` holding the character vector `c("Tumor", "Normal", "Tumor")`. Then count how many samples are tumors.

i Solution

```
brca1_gene$samples <- c("Tumor", "Normal", "Tumor")
sum(brca1_gene$samples == "Tumor")
```

```
[1] 2
```

A list element can be a whole vector. Once you *extract* it with `$`, it behaves like any ordinary vector — here we compare it to "Tumor" and `sum()` the TRUEs to get 2.

4. **Spot the bracket bug.** A labmate runs `mean(brca1_gene["expression_log2"])` and gets an error. Explain why, then fix it.

i Solution

```
mean(brca1_gene[["expression_log2"]])
```

```
[1] 5.881784
```

Single `[ ]` returns a *list* containing the vector, and `mean()` can't average a list. Switching to `[[ ]]` (or `$`) extracts the numeric vector itself, so `mean()` works.

9.8. Summary

You can now reach for a list whenever you need to keep mixed, named things together in one object — which is exactly what a real analysis result looks like. Specifically, you can:

- **Build** a named list with `list(name = value, ...)`.
- **Inspect** it with `str()` (the structure), `length()`, `names()`, and `class()`.
- **Extract** a single element with `[[ ]]` or `$` (you get the item itself), and take a **sub-list** with single `[ ]` (you get a smaller list).
- **Modify** a list by assigning to `$name` / `[[ "name" ]]` to add or update, and by assigning `NULL` to remove.
- **Assemble** a self-contained gene record and pull values back out for calculation.

The one idea to carry forward: `[[ ]]` reaches in and hands you the thing; `[ ]` hands you a smaller list. Get that straight, and lists — and all the model and test results built on

9. *Lists: a catch-all container*

them — stop being mysterious.

10. Data Frames

Almost every dataset you will meet in genomics arrives as a table: genes down the side, samples across the top, measurements in the middle, and a few columns of annotation bolted on. In R, the structure built for exactly this shape is the `data.frame`. Get comfortable with it and most of the rest of the book — loading expression data, filtering genes, summarizing by condition — becomes variations on a theme you already know.

A `data.frame` looks a bit like an R matrix in that it has two dimensions, rows and columns. The difference that matters is this: a matrix **must** hold a single data type, while a `data.frame` may hold a *different* data type in each column — numbers in one, gene names in another, an experimental condition in a third. That is exactly what a real dataset looks like, which is why the `data.frame`, not the matrix, is the workhorse for tabular data in R.

10.1. What you'll learn

By the end of this chapter you will be able to:

- Explain how a `data.frame` differs from a matrix, and why that difference matters.
- Load tabular data from a file or URL with `read.csv()`.
- Inspect a data frame with `head()`, `dim()`, `summary()`, and friends.
- Pull out columns and rows by position, name, and logical condition.
- Summarize measurements by category with `aggregate()`.
- Build a `data.frame` from scratch and save it back to disk.

We'll do all of this on a real yeast gene-expression dataset.

10.2. Dataset

The data used here are borrowed directly from the [fantastic Bioconnector tutorials](#) and are a cleaned up version of the data from [Brauer et al. Coordination of Growth Rate, Cell Cycle, Stress Response, and Metabolic Activity in Yeast \(2008\) Mol Biol Cell 19:352-367](#). These data are from a gene expression microarray, and in this paper the authors examine

the relationship between growth rate and gene expression in yeast cultures limited by one of six different nutrients (glucose, leucine, ammonium, sulfate, phosphate, uracil). If you give yeast a rich media loaded with nutrients except restrict the supply of a single nutrient, you can control the growth rate to any rate you choose. By starving yeast of specific nutrients you can find genes that:

1. Raise or lower their expression in response to growth rate. Growth-rate dependent expression patterns can tell us a lot about cell cycle control, and how the cell responds to stress. The authors found that expression of >25% of all yeast genes is linearly correlated with growth rate, independent of the limiting nutrient. They also found that the subset of negatively growth-correlated genes is enriched for peroxisomal functions, and positively correlated genes mainly encode ribosomal functions.
2. Respond differently when different nutrients are being limited. If you see particular genes that respond very differently when a nutrient is sharply restricted, these genes might be involved in the transport or metabolism of that specific nutrient.

The dataset can be downloaded directly from:

- [brauer2007_tidy.csv](#)

We are going to read this dataset into R and then use it as a playground for learning about data.frames.

10.3. Reading in data

R has many capabilities for reading in data. Many of the functions have names that help us to understand what data format is to be expected. In this case, the filename that we want to read ends in `.csv`, meaning comma-separated-values. The `read.csv()` function reads in `.csv` files. As usual, it is worth reading `help('read.csv')` to get a better sense of the possible bells-and-whistles.

The `read.csv()` function can read directly from a URL, so we do not need to download the file directly. This dataset is relatively large (about 16MB), so this may take a bit depending on your network connection speed.

```
options(width=60)
```

```
url <- paste0(
  'https://raw.githubusercontent.com',
  '/bioconnector/workshops/master/data/brauer2007_tidy.csv'
```

```
)  
ydat <- read.csv(url)
```

Our variable, `ydat`, now “contains” the downloaded and read data. We can check to see what data type `read.csv` gave us:

```
class(ydat)
```

```
[1] "data.frame"
```

R reports `"data.frame"`, so `read.csv()` did what its name promises: it turned a table of text into the structure we want to work with.

10.4. Inspecting data.frames

Your `ydat` variable is a `data.frame`. The dataset is fairly large, so you will not be able to look at it all at once on the screen. Fortunately, R gives you many tools to inspect a `data.frame` a piece at a time.

- Overviews of content
 - `head()` to show first few rows
 - `tail()` to show last few rows
- Size
 - `dim()` for dimensions (rows, columns)
 - `nrow()`
 - `ncol()`
 - `object.size()` for power users interested in the memory used to store an object
- Data and attribute summaries
 - `colnames()` to get the names of the columns
 - `rownames()` to get the “names” of the rows—may not be present
 - `summary()` to get per-column summaries of the data in the `data.frame`.

```
head(ydat)
```

10. Data Frames

```

symbol systematic_name nutrient rate expression
1   SFB2          YNL049C  Glucose 0.05      -0.24
2   <NA>          YNL095C  Glucose 0.05       0.28
3   QRI7          YDL104C  Glucose 0.05      -0.02
4   CFT2          YLR115W  Glucose 0.05      -0.33
5   SS02          YMR183C  Glucose 0.05       0.05
6   PSP2          YML017W  Glucose 0.05      -0.69

```

bp

```

1   ER to Golgi transport
2   biological process unknown
3   proteolysis and peptidolysis
4   mRNA polyadenylation*
5   vesicle fusion*
6   biological process unknown

```

mf

```

1   molecular function unknown
2   molecular function unknown
3   metalloendopeptidase activity
4   RNA binding
5   t-SNARE activity
6   molecular function unknown

```

```
tail(ydat)
```

```

symbol systematic_name nutrient rate expression
198425 DOA1          YKL213C  Uracil 0.3       0.14
198426 KRE1          YNL322C  Uracil 0.3       0.28
198427 MTL1          YGR023W  Uracil 0.3       0.27
198428 KRE9          YJL174W  Uracil 0.3       0.43
198429 UTH1          YKR042W  Uracil 0.3       0.19
198430 <NA>          YOL111C  Uracil 0.3       0.04

```

bp

```

198425 ubiquitin-dependent protein catabolism*
198426 cell wall organization and biogenesis
198427 cell wall organization and biogenesis
198428 cell wall organization and biogenesis*
198429 mitochondrion organization and biogenesis*
198430 biological process unknown

```

mf

```

198425 molecular function unknown

```

10. Data Frames

```
198426 structural constituent of cell wall
198427      molecular function unknown
198428      molecular function unknown
198429      molecular function unknown
198430      molecular function unknown
```

```
dim(ydat)
```

```
[1] 198430      7
```

```
nrow(ydat)
```

```
[1] 198430
```

```
ncol(ydat)
```

```
[1] 7
```

```
colnames(ydat)
```

```
[1] "symbol"      "systematic_name" "nutrient"
[4] "rate"        "expression"      "bp"
[7] "mf"
```

```
summary(ydat)
```

symbol		systematic_name		nutrient	
Length	:198430	Length	:198430	Length	:198430
N.unique	: 4210	N.unique	: 5536	N.unique	: 6
N.blank	: 0	N.blank	: 0	N.blank	: 0
Min.nchar	: 2	Min.nchar	: 5	Min.nchar	: 6
Max.nchar	: 9	Max.nchar	: 9	Max.nchar	: 9
NAs	: 47250				
rate		expression		bp	
Min.	:0.0500	Min.	:-6.500000	Length	:198430
1st Qu.	:0.1000	1st Qu.	:-0.290000	N.unique	: 880

10. Data Frames

```
Median :0.2000   Median : 0.000000   N.blank  :      0
Mean   :0.1752   Mean   : 0.003367   Min.nchar:      7
3rd Qu.:0.2500   3rd Qu.: 0.290000   Max.nchar:     82
Max.   :0.3000   Max.   : 6.640000   NAs      :   7663

mf
Length  :198430
N.unique : 1085
N.blank  :      0
Min.nchar:    11
Max.nchar:   125
NAs      :   7663
```

The `dim()` output tells you the shape at a glance — tens of thousands of rows and a handful of columns — and `summary()` gives a per-column thumbnail: numeric ranges for the measurement columns and counts for the categorical ones.

Look before you leap with `View()`

In RStudio there is one more inspection tool, `View()` (note the capital “V”), that opens the first 1000 rows in a spreadsheet-style window you can scroll and sort. It is a great way to *get a feel* for a dataset before you start writing code against it. We don’t run it here because it only works in interactive RStudio, not when the book is rendered.

```
View(ydat)
```

10.5. Accessing variables (columns) and subsetting

In R, data.frames can be subset similarly to other two-dimensional data structures. The `[` in R is used to denote subsetting of any kind. When working with two-dimensional data, we need two values inside the `[ ]` to specify the details. The specification is `[rows, columns]`. For example, to get the first three rows of `ydat`, use:

```
ydat[1:3, ]
```

```
  symbol systematic_name nutrient rate expression
1  SFB2          YNL049C  Glucose 0.05      -0.24
2  <NA>          YNL095C  Glucose 0.05       0.28
```

10. Data Frames

```
3  QRI7          YDL104C  Glucose 0.05      -0.02
                                bp
1          ER to Golgi transport
2  biological process unknown
3 proteolysis and peptidolysis
                                mf
1  molecular function unknown
2  molecular function unknown
3 metalloendopeptidase activity
```

Note how the second number, the columns, is blank. R takes that to mean “all the columns”. Similarly, we can combine rows and columns specification arbitrarily.

```
ydat[1:3, 1:3]
```

```
  symbol systematic_name nutrient
1  SFB2          YNL049C  Glucose
2  <NA>          YNL095C  Glucose
3  QRI7          YDL104C  Glucose
```

Because selecting a single variable, or column, is such a common operation, there are two shortcuts for doing so *with data.frames*. The first, the `$` operator works like so:

```
# Look at the column names, just to refresh memory
colnames(ydat)
```

```
[1] "symbol"          "systematic_name" "nutrient"
[4] "rate"            "expression"      "bp"
[7] "mf"
```

```
# Note that I am using "head" here to limit the output
head(ydat$symbol)
```

```
[1] "SFB2" NA      "QRI7" "CFT2" "SS02" "PSP2"
```

```
# What is the actual length of "symbol"?
length(ydat$symbol)
```

10. Data Frames

```
[1] 198430
```

The second is related to the fact that, in R, data.frames are also lists. We subset a list by using `[[ ]]` notation. To get the second column of `ydat`, we can use:

```
head(ydat[[2]])
```

```
[1] "YNL049C" "YNL095C" "YDL104C" "YLR115W" "YMR183C"  
[6] "YML017W"
```

Alternatively, we can use the column name:

```
head(ydat[["systematic_name"]])
```

```
[1] "YNL049C" "YNL095C" "YDL104C" "YLR115W" "YMR183C"  
[6] "YML017W"
```

\$ versus [[]]

Both pull out a single column as a plain vector, so why two notations? `$` is the quick, everyday form when you know the column name as you type it (`ydat$symbol`). `[[ ]]` is the form to reach for when the column is chosen by a *value* — a number (`ydat[[2]]`) or a name stored in a variable (`col <- "symbol"; ydat[[col]]`), which `$` cannot do. Reach for `$` by default and `[[ ]]` when you're programming with the column choice.

10.6. Aggregating and exploring

There are a couple of columns that hold numeric values. You can confirm which by asking each column its `class()`:

```
class(ydat$symbol)
```

```
[1] "character"
```

```
class(ydat$rate)
```

```
[1] "numeric"
```

```
class(ydat$expression)
```

```
[1] "numeric"
```

So `symbol` is a character column, while `rate` and `expression` are numeric. We'll put those numeric columns to work in the exercises at the end of the chapter.

10.6.1. More advanced indexing and subsetting

We can use, for example, logical values (TRUE/FALSE) to subset data.frames. Here we keep only the rows where the `symbol` column equals LEU1:

```
head(ydat[ydat$symbol == 'LEU1', ])
```

	symbol	systematic_name	nutrient	rate	expression	bp
NA	<NA>	<NA>	<NA>	NA	NA	<NA>
NA.1	<NA>	<NA>	<NA>	NA	NA	<NA>
NA.2	<NA>	<NA>	<NA>	NA	NA	<NA>
NA.3	<NA>	<NA>	<NA>	NA	NA	<NA>
NA.4	<NA>	<NA>	<NA>	NA	NA	<NA>
NA.5	<NA>	<NA>	<NA>	NA	NA	<NA>
	mf					
NA	<NA>					
NA.1	<NA>					
NA.2	<NA>					
NA.3	<NA>					
NA.4	<NA>					
NA.5	<NA>					

```
tail(ydat[ydat$symbol == 'LEU1', ])
```


10. Data Frames

```

      symbol systematic_name nutrient rate expression
NA.47244   <NA>             <NA>    <NA>    NA        NA
NA.47245   <NA>             <NA>    <NA>    NA        NA
NA.47246   <NA>             <NA>    <NA>    NA        NA
NA.47247   <NA>             <NA>    <NA>    NA        NA
NA.47248   <NA>             <NA>    <NA>    NA        NA
NA.47249   <NA>             <NA>    <NA>    NA        NA
      bp    mf
NA.47244 <NA> <NA>
NA.47245 <NA> <NA>
NA.47246 <NA> <NA>
NA.47247 <NA> <NA>
NA.47248 <NA> <NA>
NA.47249 <NA> <NA>

```

Notice the rows full of NA that came along for the ride. That is not a bug in your code — it is one of the most common surprises in R, and worth understanding once so it never trips you up again.

Logical subsetting and missing values

When you filter with a condition like `ydat$symbol == 'LEU1'`, any row whose `symbol` is NA returns NA for the test — not FALSE — so R keeps it, and you get phantom rows full of NA. The fix is to exclude the missing values explicitly with `!is.na()`:

```
head(ydat[ydat$symbol == 'LEU1' & !is.na(ydat$symbol), ])
```

```

      symbol systematic_name nutrient rate expression
1526    LEU1          YGL009C  Glucose 0.05      -1.12
7043    LEU1          YGL009C  Glucose 0.10      -0.77
12555   LEU1          YGL009C  Glucose 0.15      -0.67
18071   LEU1          YGL009C  Glucose 0.20      -0.59
23603   LEU1          YGL009C  Glucose 0.25      -0.20
29136   LEU1          YGL009C  Glucose 0.30       0.03
      bp
1526 leucine biosynthesis
7043 leucine biosynthesis
12555 leucine biosynthesis
18071 leucine biosynthesis
23603 leucine biosynthesis

```

10. Data Frames

```
29136 leucine biosynthesis
                                     mf
1526 3-isopropylmalate dehydratase activity
7043 3-isopropylmalate dehydratase activity
12555 3-isopropylmalate dehydratase activity
18071 3-isopropylmalate dehydratase activity
23603 3-isopropylmalate dehydratase activity
29136 3-isopropylmalate dehydratase activity
```

Now only the genuine LEU1 rows remain.

Sometimes, looking at the data themselves is not that important. Using `dim()` is one possibility to look at the number of rows and columns after subsetting.

```
dim(ydat[ydat$expression > 3, ])
```

```
[1] 714  7
```

Find the high expressed genes when leucine-starved. For this task we can also use `subset` which allows us to treat column names as R variables (no `$` needed).

```
subset(ydat, nutrient == 'Leucine' & rate == 0.05 & expression > 3)
```

	symbol	systematic_name	nutrient	rate	expression
133768	QDR2	YIL121W	Leucine	0.05	4.61
133772	LEU1	YGL009C	Leucine	0.05	3.84
133858	BAP3	YDR046C	Leucine	0.05	4.29
135186	<NA>	YPL033C	Leucine	0.05	3.43
135187	<NA>	YLR267W	Leucine	0.05	3.23
135288	HXT3	YDR345C	Leucine	0.05	5.16
135963	TP02	YGR138C	Leucine	0.05	3.75
135965	YR02	YBR054W	Leucine	0.05	4.40
136102	GPG1	YGL121C	Leucine	0.05	3.08
136109	HSP42	YDR171W	Leucine	0.05	3.07
136119	HXT5	YHR096C	Leucine	0.05	4.90
136151	<NA>	YJL144W	Leucine	0.05	3.06
136152	MOH1	YBL049W	Leucine	0.05	3.43
136153	<NA>	YBL048W	Leucine	0.05	3.95
136189	HSP26	YBR072W	Leucine	0.05	4.86

10. Data Frames

136231	NCA3	YJL116C	Leucine	0.05	4.03
136233	<NA>	YBR116C	Leucine	0.05	3.28
136486	<NA>	YGR043C	Leucine	0.05	3.07
137443	ADH2	YMR303C	Leucine	0.05	4.15
137448	ICL1	YER065C	Leucine	0.05	3.54
137451	SFC1	YJR095W	Leucine	0.05	3.72
137569	MLS1	YNL117W	Leucine	0.05	3.76

bp

133768		multidrug transport
133772		leucine biosynthesis
133858		amino acid transport
135186		meiosis*
135187		biological process unknown
135288		hexose transport
135963		polyamine transport
135965		biological process unknown
136102		signal transduction
136109		response to stress*
136119		hexose transport
136151		response to dessication
136152		biological process unknown
136153		<NA>
136189		response to stress*
136231	mitochondrion	organization and biogenesis
136233		<NA>
136486		biological process unknown
137443		fermentation*
137448		glyoxylate cycle
137451		fumarate transport*
137569		glyoxylate cycle

mf

133768		multidrug efflux pump activity
133772	3-isopropylmalate	dehydratase activity
133858		amino acid transporter activity
135186		molecular function unknown
135187		molecular function unknown
135288		glucose transporter activity*
135963		spermine transporter activity
135965		molecular function unknown
136102		signal transducer activity
136109		unfolded protein binding

```

136119      glucose transporter activity*
136151      molecular function unknown
136152      molecular function unknown
136153      <NA>
136189      unfolded protein binding
136231      molecular function unknown
136233      <NA>
136486      transaldolase activity
137443      alcohol dehydrogenase activity
137448      isocitrate lyase activity
137451 succinate:fumarate antiporter activity
137569      malate synthase activity

```

10.7. Aggregating data

Aggregating data, or summarizing by category, is a common way to look for trends or differences in measurements between categories. Use `aggregate` to find the mean expression by gene symbol.

```
head(aggregate(ydat$expression, by = list(ydat$symbol), mean))
```

```

  Group.1      x
1   AAC1 0.52888889
2   AAC3 -0.21628571
3  AAD10 0.43833333
4  AAD14 -0.07166667
5  AAD16 0.24194444
6   AAD4 -0.79166667

```

```

# or
head(aggregate(expression ~ symbol, mean, data = ydat))

```

```

  symbol expression
1   AAC1 0.52888889
2   AAC3 -0.21628571
3  AAD10 0.43833333
4  AAD14 -0.07166667
5  AAD16 0.24194444
6   AAD4 -0.79166667

```

Both calls do the same thing: collapse the many measurements for each gene into a single mean expression value, one row per `symbol`. The formula form (`expression ~ symbol`) reads as “expression *by* symbol” and is usually the easier to type and to read.

10.8. Creating a data.frame from scratch

Sometimes it is useful to combine related data into one object. For example, let’s simulate some data.

```
set.seed(42)
smoker <- factor(rep(c("smoker", "non-smoker"), each = 50))
smoker_numeric <- as.numeric(smoker)
x <- rnorm(100)
risk <- x + 2 * smoker_numeric
```

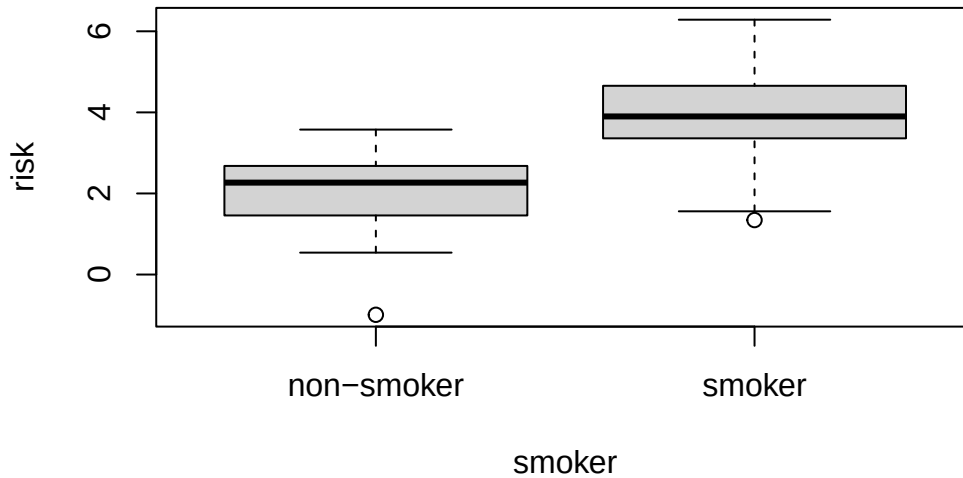
We set a seed first with `set.seed(42)` so the random draws are reproducible — you and a reader will see the same numbers. We now have two related variables, `risk` and `smoker`. We can make a `data.frame` out of them:

```
smoker_risk <- data.frame(smoker = smoker, risk = risk)
head(smoker_risk)
```

```
  smoker    risk
1 smoker 5.370958
2 smoker 3.435302
3 smoker 4.363128
4 smoker 4.632863
5 smoker 4.404268
6 smoker 3.893875
```

R also has plotting shortcuts that work with `data.frames`. Because `smoker` is a factor, `plot()` recognizes the `risk ~ smoker` formula and draws a boxplot of risk in each group:

```
plot(risk ~ smoker, data = smoker_risk)
```



As we built the data, smokers carry the higher simulated risk — and the boxplot shows exactly that.

10.9. Saving a data.frame

Once we have a `data.frame` of interest, we may want to save it. The most portable way to save a `data.frame` is to use one of the `write` functions. In this case, let's save the data as a `.csv` file.

```
write.csv(smoker_risk, "smoker_risk.csv")
```

10.10. Exercises

These all use the yeast dataset, `ydat`, from earlier in the chapter.

1. Which columns of `ydat` are numeric? Check `rate` and `expression`.
2. Make a histogram of the `expression` values, then of the `rate` values.
3. `rate` has only a handful of distinct values. Use `table()` to count how many rows fall at each rate. Which `rate` corresponds to the most nutrient-starved condition?

10. Data Frames

i Solutions

```
class(ydat$rate); class(ydat$expression) # both numeric
```

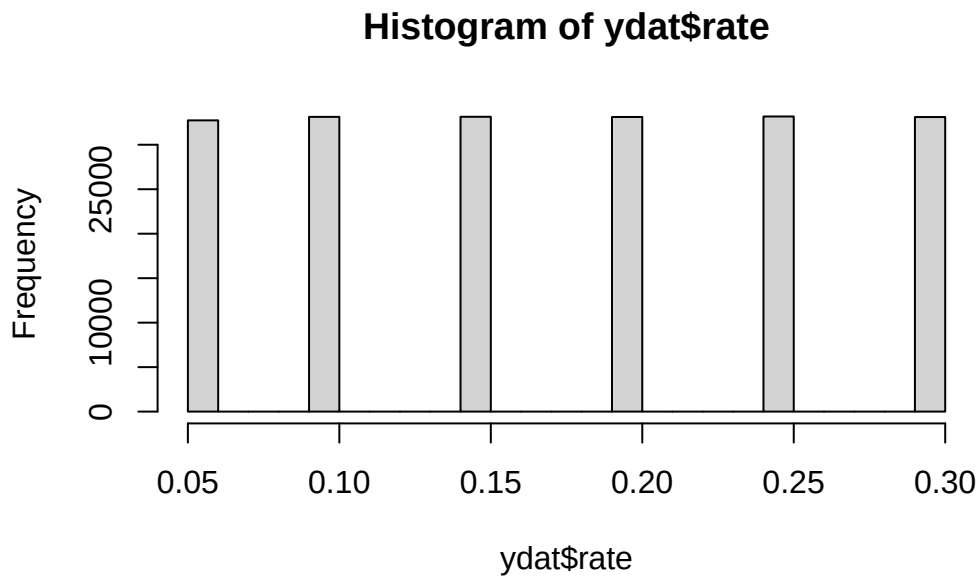
```
[1] "numeric"
```

```
[1] "numeric"
```

```
hist(ydat$expression)
```



```
hist(ydat$rate)
```



```
table(ydat$rate) # the smallest rate = most starved
```

```
0.05  0.1  0.15  0.2  0.25  0.3
32741 33132 33145 33121 33177 33114
```

Growth rate is the *limiting nutrient* dialed down, so the **lowest rate** (0.05) is the most starved condition.

10.11. Summary

You can now load a real dataset into a `data.frame`, inspect its shape and contents, pull out the rows and columns you care about (by position, name, or a logical test), summarize values by category with `aggregate()`, and write results back to disk. These are the moves you'll repeat on every dataset for the rest of the book.

10.12. Reflection

- Can you state one thing a `data.frame` can do that a matrix cannot?
- Given a column `ydat$rate`, can you write the subset that keeps only rows where `rate < 0.1`?
- When would you reach for `aggregate()` instead of subsetting?

11. Factors

Open almost any biological dataset and you will meet columns that are not numbers but *labels*: a sample is **treated** or **control**, a mouse is **wild-type**, **heterozygous**, or **homozygous**, a tumor is stage **I**, **II**, **III**, or **IV**. These categorical variables are the backbone of experiments — they say which group each sample belongs to. R has a special tool built exactly for them: the **factor**.

You could store these labels as ordinary text, and R would let you. But the moment you want to count how many samples are in each group, run an analysis of variance, or fit a model that compares treatment to control, R needs to know that “treated” and “control” are *categories* — a fixed set of possible values — not free-form strings. A factor is how you tell it that. This chapter shows you what factors are, how they behave, and the one quirk that trips up nearly every beginner.

11.1. What you’ll learn

- Create a factor from a character vector with `factor()`.
- Explain how a factor stores its data as integer codes plus a lookup table of **levels**.
- Use `table()` to count how many observations fall in each category.
- Control the set of levels, and understand how missing values arise.
- Avoid the classic `as.numeric()` trap that returns codes instead of labels.

11.2. A vector with a built-in dictionary

Imagine you are recording the country of origin for nine samples in a study. The raw data is just text:

```
# country of origin for nine samples
citizen <- c("uk", "us", "no", "au", "uk", "us", "us", "no", "au")
citizen
```

```
[1] "uk" "us" "no" "au" "uk" "us" "us" "no" "au"
```

11. Factors

That is a plain character vector — nine strings, with the quotation marks R always uses to show text. Now turn it into a factor:

```
citizen_f <- factor(citizen)
citizen_f
```

```
[1] uk us no au uk us us no au
Levels: au no uk us
```

Two things changed. The values printed *without* quotation marks, and a new line appeared: Levels: au no uk us. That Levels: line is the heart of the whole idea.

! The mental model: codes plus a lookup table

A factor *looks* like text but is stored as **integers** behind a **lookup table**. Think of a coat check at a theater: you hand over your coat and get a numbered ticket. The tickets (1, 2, 3, ...) are small and easy to handle; the coats themselves stay on a rack, each hanging at the number on its ticket.

A factor works the same way. The **levels** are the rack — the unique categories, stored once, in alphabetical order by default (**au**, **no**, **uk**, **us**). Each element of your data is just a *ticket number* pointing at one of those levels. So "uk" is not stored nine times; it is stored as the integer 3, and R looks up level 3 to show you **uk**.

We can pull back the curtain and see the tickets directly. Every R object can carry hidden **attributes**, and a factor's class and levels live there:

```
attributes(citizen_f)
```

```
$levels
[1] "au" "no" "uk" "us"
```

```
$class
[1] "factor"
```

```
class(citizen_f)
```

```
[1] "factor"
```

11. Factors

The `attributes()` call shows the `levels` (the rack) and confirms the object's `class` is `"factor"`. If we strip that class away with `unclass()`, the disguise drops and the raw ticket numbers appear:

```
unclass(citizen_f)
```

```
[1] 3 4 2 1 3 4 4 2 1
attr(,"levels")
[1] "au" "no" "uk" "us"
```

Read that output against the levels `au no uk us`. The first sample was `"uk"`, which is the **3rd** level, so its code is 3. The third sample was `"no"`, the **2nd** level, so its code is 2. Each number is simply a pointer into the lookup table.

11.3. Counting categories with `table()`

The single most useful thing you will do with a factor is count how many samples fall in each category. That is exactly what `table()` does:

```
table(citizen_f)
```

```
citizen_f
au no uk us
 2  2  2  3
```

At a glance you can see the shape of your sample set: two samples each from Australia and Norway, two from the UK, and three from the US. For a real experiment this is how you check that your treatment and control groups are balanced before you analyze anything.

11.4. Converting a factor back

You will often need to turn a factor back into something else — and here is where people get burned. There are two ways to convert, and they give very different answers.

To recover the original labels, use `as.character()`:

11. Factors

```
as.character(citizen_f)
```

```
[1] "uk" "us" "no" "au" "uk" "us" "us" "no" "au"
```

That gives back the text you started with — "uk", "us", and so on. But watch what happens if you reach for `as.numeric()` instead:

```
as.numeric(citizen_f)
```

```
[1] 3 4 2 1 3 4 4 2 1
```

 The classic gotcha: `as.numeric()` returns the codes, not the labels

`as.numeric()` on a factor returns the **ticket numbers**, not the values. You get 3 4 2 1 3 4 4 2 1 — the integer codes — *not* the countries, and certainly not any number the labels might have looked like.

This bites hardest when a factor's labels are themselves numbers. Suppose a dose column reads "10", "20", "50" but was accidentally stored as a factor. `as.numeric()` would return 1, 2, 3 (the codes), not 10, 20, 50 (the doses) — a silent, dangerous error. The safe recipe when labels are numeric is to go through character first: `as.numeric(as.character(dose_f))`.

11.5. Controlling the levels

By default, the levels are every unique value in your data, sorted alphabetically. But you are allowed to specify the levels yourself — and this is genuinely useful. Maybe you only care about the US and UK samples for one analysis:

```
citizen_f2 <- factor(citizen, levels = c("us", "uk"))
citizen_f2
```

```
[1] uk   us   <NA> <NA> uk   us   us   <NA> <NA>
Levels: us uk
```

11. Factors

Look closely: the values that were not in your chosen levels — the **no** and **au** samples — turned into `<NA>`, R’s marker for a missing value. By naming only **us** and **uk** as legal levels, you told R that every other value has no category to belong to, so it became missing.

```
table(citizen_f2)
```

```
citizen_f2
us uk
3  2
```

By default, `table()` quietly drops those missing values, so the count reflects only the four samples that matched. If you *want* to see the missing ones, add them as an explicit level with `addNA()`:

```
table(addNA(citizen_f2))
```

```
us  uk <NA>
3   2     4
```

Now the `<NA>` category appears with its own count of 5 — the five samples that fell outside your chosen levels.

i Why specifying levels matters

Setting the levels yourself does more than filter. It also fixes their **order**, which controls how categories appear in tables and plots and which group a model treats as the baseline. For a treatment study you might write `factor(group, levels = c("control", "treated"))` so that “control” is the reference everything else is compared against. The default settings are not always what your analysis needs — knowing what they are, and overriding them on purpose, is part of doing the analysis correctly.

11.6. Exercises

1. **Genotypes.** A small genotyping study records these calls for eight mice: `"wt", "het", "hom", "wt", "wt", "het", "hom", "wt"`. Make a factor from this vector and use `table()` to count how many mice carry each genotype.

i Solution

```
genotype <- c("wt", "het", "hom", "wt", "wt", "het", "hom", "wt")
genotype_f <- factor(genotype)
table(genotype_f)
```

```
genotype_f
het hom wt
  2  2  4
```

`table()` counts each category for you: four `wt`, two `het`, and two `hom`. The factor turned a list of labels into a tidy summary of the sample set.

2. **Read the codes.** Using the `genotype_f` factor from Exercise 1, predict what `unclass(genotype_f)` will print *before* you run it. Then run it and check.

i Solution

```
unclass(genotype_f)
```

```
[1] 3 1 2 3 3 1 2 3
attr("levels")
[1] "het" "hom" "wt"
```

The levels are alphabetical — `het`, `hom`, `wt` — so `het` is code 1, `hom` is 2, and `wt` is 3. The first mouse (`wt`) prints as 3. Each integer is a ticket pointing into the levels table, not a measurement.

3. **The numeric trap.** A colleague stored a dose column as a factor with labels "5", "10", "20" and ran `as.numeric()` on it to get the doses back. Why are their numbers wrong, and what should they do instead?

i Solution

```
dose_f <- factor(c("5", "10", "20", "5", "20"))
as.numeric(dose_f) # WRONG: returns the integer codes
```

```
[1] 3 1 2 3 2
```

```
as.numeric(as.character(dose_f)) # RIGHT: recovers the real doses
```

```
[1] 5 10 20 5 20
```

`as.numeric()` returns the level codes, and because levels are sorted, "10" sorts

before "5" — so the codes do not even match the order you expect. Converting to character first turns each label back into its text, which `as.numeric()` can then read as the true number.

4. **Set the reference group.** You have a `group` vector of "treated" and "control" labels. Build a factor whose levels are ordered so that `control` comes first (the baseline a model would compare against).

i Solution

```
group <- c("treated", "control", "treated", "control", "treated")
group_f <- factor(group, levels = c("control", "treated"))
group_f
```

```
[1] treated control treated control treated
Levels: control treated
```

By naming `control` first in `levels`, you make it level 1 — the reference category. Most modeling functions in R compare every other group against the first level, so setting it deliberately keeps your results interpretable.

11.7. Summary

You can now recognize and build the categorical variables at the center of nearly every biological experiment:

- **Create** a factor from a character vector with `factor()`, and recognize it by the `Levels:` line and the missing quotation marks.
- **Picture** it as codes plus a lookup table — integer tickets pointing at a rack of levels — which explains every behavior in this chapter.
- **Count** categories with `table()` to check group sizes and spot imbalance.
- **Control** the levels to filter categories, set a reference group, and decide how missing values are handled.
- **Convert** safely: `as.character()` gives the labels back, while `as.numeric()` gives the codes — so reach for `as.numeric(as.character(x))` whenever a factor's labels are really numbers.

Keep the coat-check picture in mind and factors stop being mysterious: a factor is just a vector of tickets with a table of what each ticket stands for.

Part III.

Exploratory data analysis

What you'll learn in this part

Imagine you're on an adventure, about to embark on a journey into the unknown. You've just been handed a treasure map, with the promise of valuable insights waiting to be discovered. This map is your data set, and the journey is exploratory data analysis (EDA).

As you begin your exploration, you start by getting a feel for the terrain. You take a broad, bird's-eye view of the data, examining its structure and dimensions. Are you dealing with a vast landscape or a small, confined area? Are there any missing pieces in the map that you'll need to account for? Understanding the overall context of your data set is crucial before venturing further.

With a sense of the landscape, you now zoom in to identify key landmarks in the data. You might look for unusual patterns, trends, or relationships between variables. As you spot these landmarks, you start asking questions: What's causing that spike in values? Are these two factors related, or is it just a coincidence? By asking these questions, you're actively engaging with the data and forming hypotheses that could guide future analysis or experiments.

As you continue your journey, you realize that the map alone isn't enough to fully understand the terrain. You need more tools to bring the data to life. You start visualizing the data using charts, plots, and graphs. These visualizations act as your binoculars, allowing you to see patterns and relationships more clearly. Through them, you can uncover the hidden treasures buried within the data.

EDA isn't a linear path from start to finish. As you explore, you'll find yourself circling back to previous points, refining your questions, and digging deeper. The process is iterative, with each new discovery informing the next. And as you go, you'll gain a deeper understanding of the data's underlying structure and potential.

Finally, after your thorough exploration, you'll have a solid foundation to build upon. You'll be better equipped to make informed decisions, test hypotheses, and draw meaningful conclusions. The insights you've gained through EDA will serve as a compass, guiding you towards the true value hidden within your data. And with that, you've successfully completed your journey through exploratory data analysis.

What you'll learn in this part

- Get a feel for the **shape** of a dataset — its size, variables, and missing pieces — before drawing any conclusions.
- **Transform and summarize** tabular data with the `dplyr` verbs (filter, select, mutate, group, summarize).
- Work through a complete EDA on a real survey dataset, from raw table to summary.

Where to begin

- **Visualize** distributions and relationships with `ggplot2`, and apply a few design principles that make any plot more readable.
- Treat EDA as an **iterative loop**: ask a question, make a summary or plot, refine the question, repeat.

Where to begin

We start gently, with the `dplyr` verbs on a small, friendly dataset in [Introduction to dplyr](#), then put them to work in a full [case study on survey data](#) before turning to visualization with `ggplot2` and a [self-guided tour of data visualization](#).

12. Introduction to dplyr: mammal sleep dataset

Almost every analysis you'll do in biology starts with a table: a sample sheet with one row per sample, a count matrix with one row per gene, a results table with one row per differentially expressed transcript. Over and over you'll want to do the same handful of things to those tables — keep only the rows that pass a filter, pull out a few columns, sort by a statistic, add a derived column, or collapse many rows into a per-group summary.

That recurring pattern has a name: **split-apply-combine**. You *split* a table into groups (say, by treatment condition), *apply* a calculation to each group (say, the mean expression), and *combine* the answers back into one tidy table. The **dplyr** package gives you a small, consistent vocabulary of verbs for exactly these moves, and they read almost like English.

We'll learn those verbs on a gentle, friendly dataset — the sleep habits of mammals — before you turn them loose on genomics tables. The dataset is small enough to see the whole thing, so you can watch each verb do its job. Every move you make here (filtering rows, selecting columns, summarizing by group) is the same move you'll later make on a count matrix or a sample sheet.

12.1. What you'll learn

- Filter rows of a table with `filter()` and select columns with `select()`.
- Re-order rows with `arrange()` and add derived columns with `mutate()`.
- Collapse a table to summary statistics with `summarise()`.
- Use `group_by()` to compute those summaries *per group* — the split-apply-combine pattern.
- Chain verbs together with the pipe operator, `|>`, to read an analysis left to right.

💡 Why a sleep dataset for a genomics book?

`msleep` is a warm-up, chosen because it is small and the columns are easy to reason about. Keep the *biological* payoff in view as you go: filtering mammals that sleep more

12. Introduction to dplyr: mammal sleep dataset

than 16 hours is the same operation as filtering genes with an adjusted p-value below 0.05; selecting the `name` and `sleep_total` columns is the same as pulling `sample_id` and `condition` out of a sample sheet; and summarizing sleep by taxonomic order is the same as summarizing mean expression by treatment group. Learn the verbs here; reuse them on your own data.

12.2. What is dplyr?

The *dplyr* package is a specialized package for working with `data.frames` (and the closely related `tibble`) to transform and summarize tabular data with rows and columns. For another explanation, see the package vignette, [Introduction to dplyr](#).

12.3. Why is dplyr useful?

dplyr provides a set of functions — commonly called the *dplyr* “verbs” — that perform the common data manipulations: filtering rows, selecting columns, re-ordering rows, adding new columns, and summarizing data. It also makes the split-apply-combine pattern straightforward through grouping.

Compared with base R functions, the *dplyr* verbs are often easier to work with. They are more consistent in their syntax and are designed around data frames rather than individual vectors, which is exactly the shape biological data usually arrives in.

12.4. Data: mammals sleep

The `msleep` (mammals sleep) dataset is an updated and expanded version of the mammals sleep dataset, with sleep times and weights taken from Savage and West¹. It contains the sleep times and weights for a set of mammals, with 83 rows and 11 variables. It ships as a dataset inside the *ggplot2* package, so there is nothing to download — you just install *ggplot2* once if you don’t already have it.

```
install.packages("ggplot2")
```

¹A quantitative, theoretical framework for understanding mammalian sleep. Van M. Savage, Geoffrey B. West. *Proceedings of the National Academy of Sciences* Jan 2007, 104 (3) 1051-1056; DOI: [10.1073/pnas.0610080104](https://doi.org/10.1073/pnas.0610080104)

12. Introduction to dplyr: mammal sleep dataset

Then load the library, which makes `msleep` available.

```
library(ggplot2)
data(msleep)
```

As with many datasets in R, a help page describes the dataset itself.

```
?msleep
```

The columns are described on that help page, and are listed here for convenience.

column name	Description
name	common name
genus	taxonomic rank
vore	carnivore, omnivore or herbivore?
order	taxonomic rank
conservation	the conservation status of the mammal
sleep_total	total amount of sleep, in hours
sleep_rem	rem sleep, in hours
sleep_cycle	length of sleep cycle, in hours
awake	amount of time spent awake, in hours
brainwt	brain weight in kilograms
bodywt	body weight in kilograms

12.5. The dplyr verbs

dplyr has many functions, but the six verbs below do the bulk of everyday work. The rest of this chapter takes them one at a time.

dplyr verbs	Description
<code>select()</code>	select columns
<code>filter()</code>	filter rows
<code>arrange()</code>	re-order or arrange rows
<code>mutate()</code>	create new columns
<code>summarise()</code>	summarise values
<code>group_by()</code>	allows for group operations in the “split-apply-combine” concept

12. Introduction to dplyr: mammal sleep dataset

Before going further, install dplyr (once) and load it.

```
install.packages("dplyr")
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

12.6. Selecting columns with select()

`select()` keeps only the columns you name. Here we keep the `name` and `sleep_total` columns — think of pulling just the sample identifier and one measurement out of a wider table.

```
sleepData <- select(msleep, name, sleep_total)
head(sleepData)
```

```
# A tibble: 6 x 2
  name                sleep_total
  <chr>              <dbl>
1 Cheetah             12.1
2 Owl monkey          17
3 Mountain beaver     14.4
4 Greater short-tailed shrew 14.9
5 Cow                  4
6 Three-toed sloth    14.4
```

12. Introduction to dplyr: mammal sleep dataset

To keep all columns *except* one, put a minus sign in front of it (negative indexing). For example, every column except `name`:

```
head(select(msleep, -name))
```

```
# A tibble: 6 x 10
  genus      vore order conservation sleep_total sleep_rem sleep_cycle awake
  <chr>      <chr> <chr>      <chr>          <dbl>      <dbl>      <dbl> <dbl>
1 Acinonyx   carni Carnivo~ lc          12.1        NA         NA     11.9
2 Aotus      omni  Primates <NA>         17         1.8        NA      7
3 Aplodontia herbi Rodentia nt          14.4        2.4        NA     9.6
4 Blarina    omni  Soricom~ lc          14.9        2.3        0.133   9.1
5 Bos        herbi Artioda~ domesticated 4         0.7        0.667   20
6 Bradypus   herbi Pilosa  <NA>         14.4        2.2        0.767   9.6
# i 2 more variables: brainwt <dbl>, bodywt <dbl>
```

To select a range of columns by name, use the `:` operator. Notice that dplyr lets you write the column names without quotes, almost as if they were variables:

```
head(select(msleep, name:order))
```

```
# A tibble: 6 x 4
  name      genus      vore order
  <chr>      <chr>      <chr> <chr>
1 Cheetah   Acinonyx   carni Carnivora
2 Owl monkey Aotus      omni  Primates
3 Mountain beaver Aplodontia herbi Rodentia
4 Greater short-tailed shrew Blarina    omni  Soricomorpha
5 Cow       Bos        herbi Artiodactyla
6 Three-toed sloth Bradypus   herbi Pilosa
```

To select every column whose name *starts with* “sl”, use the helper `starts_with()`:

```
head(select(msleep, starts_with("sl")))
```

```
# A tibble: 6 x 3
  sleep_total sleep_rem sleep_cycle
  <dbl>      <dbl>      <dbl>
```


12. Introduction to dplyr: mammal sleep dataset

1	12.1	NA	NA
2	17	1.8	NA
3	14.4	2.4	NA
4	14.9	2.3	0.133
5	4	0.7	0.667
6	14.4	2.2	0.767

Other helpers let you select columns by other criteria:

1. `ends_with()` — columns that end with a character string
2. `contains()` — columns that contain a character string
3. `matches()` — columns that match a regular expression
4. `one_of()` — columns whose names are in a given group of names

12.7. Selecting rows with `filter()`

`filter()` keeps only the rows that *match* a condition. For example, keep the mammals that sleep a total of 16 or more hours:

```
filter(msleep, sleep_total >= 16)
```

```
# A tibble: 8 x 11
  name      genus vore order conservation sleep_total sleep_rem sleep_cycle awake
  <chr>    <chr> <chr> <chr> <chr>          <dbl>      <dbl>      <dbl> <dbl>
1 Owl mo~ Aotus omni Prim~ <NA>          17        1.8        NA      7
2 Long-n~ Dasy~ carni Cing~ lc          17.4       3.1      0.383   6.6
3 North ~ Dide~ omni Dide~ lc          18        4.9      0.333   6
4 Big br~ Epte~ inse~ Chir~ lc          19.7       3.9      0.117   4.3
5 Thick-- Lutr~ carni Dide~ lc          19.4       6.6        NA      4.6
6 Little~ Myot~ inse~ Chir~ <NA>          19.9       2        0.2     4.1
7 Giant ~ Prio~ inse~ Cing~ en          18.1       6.1        NA      5.9
8 Arctic~ Sper~ herbi Rode~ lc          16.6       NA        NA      7.4
# i 2 more variables: brainwt <dbl>, bodywt <dbl>
```

You can give `filter()` more than one condition, separated by commas; a row must satisfy *all* of them. Here we want mammals that sleep at least 16 hours *and* weigh more than one kilogram:

12. Introduction to dplyr: mammal sleep dataset

```
filter(msleep, sleep_total >= 16, bodywt >= 1)
```

```
# A tibble: 3 x 11
```

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Long-n~	Dasy~	carni	Cing~	lc	17.4	3.1	0.383	6.6
2	North ~	Dide~	omni	Dide~	lc	18	4.9	0.333	6
3	Giant ~	Prio~	inse~	Cing~	en	18.1	6.1	NA	5.9

```
# i 2 more variables: brainwt <dbl>, bodywt <dbl>
```

You can also match against a set of values with the `%in%` operator, which returns TRUE for each element of the first vector that appears *in* the second. Here we keep mammals in either the Perissodactyla or Primates taxonomic order:

```
filter(msleep, order %in% c("Perissodactyla", "Primates"))
```

```
# A tibble: 15 x 11
```

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Owl m~	Aotus	omni	Prim~	<NA>	17	1.8	NA	7
2	Grivet	Cerc~	omni	Prim~	lc	10	0.7	NA	14
3	Horse	Equus	herbi	Peri~	domesticated	2.9	0.6	1	21.1
4	Donkey	Equus	herbi	Peri~	domesticated	3.1	0.4	NA	20.9
5	Patas~	Eryt~	omni	Prim~	lc	10.9	1.1	NA	13.1
6	Galago	Gala~	omni	Prim~	<NA>	9.8	1.1	0.55	14.2
7	Human	Homo	omni	Prim~	<NA>	8	1.9	1.5	16
8	Mongo~	Lemur	herbi	Prim~	vu	9.5	0.9	NA	14.5
9	Macaq~	Maca~	omni	Prim~	<NA>	10.1	1.2	0.75	13.9
10	Slow ~	Nyct~	carni	Prim~	<NA>	11	NA	NA	13
11	Chimp~	Pan	omni	Prim~	<NA>	9.7	1.4	1.42	14.3
12	Baboon	Papio	omni	Prim~	<NA>	9.4	1	0.667	14.6
13	Potto	Pero~	omni	Prim~	lc	11	NA	NA	13
14	Squir~	Saim~	omni	Prim~	<NA>	9.6	1.4	NA	14.4
15	Brazi~	Tapi~	herbi	Peri~	vu	4.4	1	0.9	19.6

```
# i 2 more variables: brainwt <dbl>, bodywt <dbl>
```

Any of the comparison operators (`>`, `<`, `>=`, `<=`, `!=`, `%in%`) can form the logical test. This is exactly how you would later keep only the genes passing a significance cutoff.

12.8. Chaining verbs with the pipe, |>

Real analyses string several of these operations together. The pipe operator, |>, lets you “pipe” the output of one function straight into the first argument of the next. There is nothing magical about it — but it lets you read a chain of operations from left to right, in the order they happen.

💡 Read the pipe as “and then”

`x |> f() |> g()` means: take `x`, **and then** apply `f`, **and then** apply `g`. Without the pipe you’d either save each step in a temporary variable and pass it along, or *nest* the calls inside out — `g(f(x))` — which is hard to read once the chain grows. The pipe keeps the steps in the order you think about them.

Here is an example we have already written the nested way:

```
head(select(msleep, name, sleep_total))
```

```
# A tibble: 6 x 2
  name                sleep_total
  <chr>                <dbl>
1 Cheetah              12.1
2 Owl monkey           17
3 Mountain beaver     14.4
4 Greater short-tailed shrew 14.9
5 Cow                   4
6 Three-toed sloth    14.4
```

With the pipe, you read the same thing top to bottom: take `msleep`, **and then** select two columns (`name` and `sleep_total`), **and then** show the head of the result.

```
msleep |>
  select(name, sleep_total) |>
  head()
```

```
# A tibble: 6 x 2
  name                sleep_total
  <chr>                <dbl>
1 Cheetah              12.1
```

12. Introduction to dplyr: mammal sleep dataset

2 Owl monkey	17
3 Mountain beaver	14.4
4 Greater short-tailed shrew	14.9
5 Cow	4
6 Three-toed sloth	14.4

You'll see how much this helps once we start combining many verbs. From here on, we'll use the pipe throughout.

12.9. Re-ordering rows with `arrange()`

`arrange()` sorts the rows by one or more columns. To sort by the taxonomic order, name the column:

```
msleep |> arrange(order) |> head()
```

```
# A tibble: 6 x 11
  name      genus vore  order conservation sleep_total sleep_rem sleep_cycle awake
  <chr>    <chr> <chr> <chr> <chr>          <dbl>    <dbl>    <dbl> <dbl>
1 Tenrec  Tenr~ omni  Afro~ <NA>          15.6     2.3     NA     8.4
2 Cow     Bos   herbi Arti~ domesticated    4       0.7     0.667  20
3 Roe de~ Capr~ herbi Arti~ lc              3       NA     NA     21
4 Goat    Capri herbi Arti~ lc              5.3     0.6     NA     18.7
5 Giraffe Gira~ herbi Arti~ cd              1.9     0.4     NA     22.1
6 Sheep   Ovis  herbi Arti~ domesticated    3.8     0.6     NA     20.2
# i 2 more variables: brainwt <dbl>, bodywt <dbl>
```

Now let's combine verbs: select three columns, sort the rows first by taxonomic order and then by `sleep_total`, and finally show the head of the result.

```
msleep |>
  select(name, order, sleep_total) |>
  arrange(order, sleep_total) |>
  head()
```

```
# A tibble: 6 x 3
  name      order      sleep_total
  <chr>    <chr>          <dbl>
1 Tenrec  Afro~          15.6
2 Cow     Arti~           4
3 Roe de~ Arti~           3
4 Goat    Arti~           5.3
5 Giraffe Arti~           1.9
6 Sheep   Arti~           3.8
```

12. Introduction to dplyr: mammal sleep dataset

	<chr>	<chr>	<dbl>
1	Tenrec	Afrosoricida	15.6
2	Giraffe	Artiodactyla	1.9
3	Roe deer	Artiodactyla	3
4	Sheep	Artiodactyla	3.8
5	Cow	Artiodactyla	4
6	Goat	Artiodactyla	5.3

The same chain, but instead of showing the head we filter to the mammals that sleep 16 or more hours:

```
msleep |>
  select(name, order, sleep_total) |>
  arrange(order, sleep_total) |>
  filter(sleep_total >= 16)
```

```
# A tibble: 8 x 3
  name                order      sleep_total
  <chr>              <chr>      <dbl>
1 Big brown bat      Chiroptera 19.7
2 Little brown bat   Chiroptera 19.9
3 Long-nosed armadillo Cingulata 17.4
4 Giant armadillo     Cingulata 18.1
5 North American Opossum Didelphimorphia 18
6 Thick-tailed opossum Didelphimorphia 19.4
7 Owl monkey         Primates   17
8 Arctic ground squirrel Rodentia    16.6
```

To sort a column in *descending* order, wrap it in `desc()`. Here we sort by order ascending, then by `sleep_total` descending:

```
msleep |>
  select(name, order, sleep_total) |>
  arrange(order, desc(sleep_total)) |>
  filter(sleep_total >= 16)
```

```
# A tibble: 8 x 3
  name                order      sleep_total
  <chr>              <chr>      <dbl>
```

12. Introduction to dplyr: mammal sleep dataset

1	Little brown bat	Chiroptera	19.9
2	Big brown bat	Chiroptera	19.7
3	Giant armadillo	Cingulata	18.1
4	Long-nosed armadillo	Cingulata	17.4
5	Thick-tailed opossum	Didelphimorphia	19.4
6	North American Opossum	Didelphimorphia	18
7	Owl monkey	Primates	17
8	Arctic ground squirrel	Rodentia	16.6

12.10. Creating new columns with mutate()

`mutate()` adds new columns computed from existing ones. Here we add a column `rem_proportion`, the ratio of REM sleep to total sleep:

```
msleep |>
  mutate(rem_proportion = sleep_rem / sleep_total) |>
  head()
```

```
# A tibble: 6 x 12
```

	name	genus	vore	order	conservation	sleep_total	sleep_rem	sleep_cycle	awake
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Cheetah	Acin~	carni	Carn~	lc	12.1	NA	NA	11.9
2	Owl mo~	Aotus	omni	Prim~	<NA>	17	1.8	NA	7
3	Mounta~	Aplo~	herbi	Rode~	nt	14.4	2.4	NA	9.6
4	Greate~	Blar~	omni	Sori~	lc	14.9	2.3	0.133	9.1
5	Cow	Bos	herbi	Arti~	domesticated	4	0.7	0.667	20
6	Three~~	Brad~	herbi	Pilo~	<NA>	14.4	2.2	0.767	9.6

```
# i 3 more variables: brainwt <dbl>, bodywt <dbl>, rem_proportion <dbl>
```

You can add several columns at once, separated by commas. Here we add `rem_proportion` and also `bodywt_grams`, the body weight expressed in grams:

```
msleep |>
  mutate(rem_proportion = sleep_rem / sleep_total,
         bodywt_grams = bodywt * 1000) |>
  head()
```

12. Introduction to dplyr: mammal sleep dataset

```
# A tibble: 6 x 13
  name      genus vore  order conservation sleep_total sleep_rem sleep_cycle awake
  <chr>    <chr> <chr> <chr> <chr>          <dbl>    <dbl>    <dbl> <dbl>
1 Cheetah Acin~ carni Carn~ lc          12.1      NA      NA     11.9
2 Owl mo~ Aotus omni  Prim~ <NA>       17        1.8    NA      7
3 Mounta~ Aplo~ herbi Rode~ nt         14.4      2.4    NA     9.6
4 Greate~ Blar~ omni  Sori~ lc         14.9      2.3    0.133   9.1
5 Cow      Bos   herbi Arti~ domesticated  4         0.7    0.667   20
6 Three~~ Brad~ herbi Pilo~ <NA>       14.4      2.2    0.767   9.6
# i 4 more variables: brainwt <dbl>, bodywt <dbl>, rem_proportion <dbl>,
#   bodywt_grams <dbl>
```

Computing a derived column like this is the same move as turning raw counts into counts-per-million, or a fold change into a log fold change.

12.11. Summarizing with summarise()

`summarise()` collapses a column down to a single summary statistic. To get the average amount of sleep, apply `mean()` to `sleep_total` and name the result `avg_sleep`:

```
msleep |>
  summarise(avg_sleep = mean(sleep_total))
```

```
# A tibble: 1 x 1
  avg_sleep
  <dbl>
1    10.4
```

Mammals in this dataset sleep about ten and a half hours a day, on average. You can compute several summaries at once. Besides `mean()`, useful summary functions include `sd()`, `min()`, `max()`, `median()`, `sum()`, `n()` (the number of rows), `first()`, `last()`, and `n_distinct()` (the number of distinct values):

```
msleep |>
  summarise(avg_sleep = mean(sleep_total),
            min_sleep = min(sleep_total),
            max_sleep = max(sleep_total),
            total = n())
```

12. Introduction to dplyr: mammal sleep dataset

```
# A tibble: 1 x 4
  avg_sleep min_sleep max_sleep total
  <dbl>      <dbl>      <dbl> <int>
1    10.4      1.9      19.9    83
```

That single row tells you the average, the extremes, and the sample size all at once.

12.12. Grouping with `group_by()`

`group_by()` is what makes `summarise()` powerful. On its own, `summarise()` collapses the whole table to one row. Combined with `group_by()`, it gives you one summary row *per group* — this is split-apply-combine in action.

i Split-apply-combine

Picture the table sorted into piles, one pile per group (here, one per taxonomic order). dplyr **splits** the data into those piles, **applies** your summary to each pile independently, and **combines** the per-pile answers into one tidy result table. It's the same idea as base R's `aggregate()`, but it fits naturally into a dplyr pipe.

Let's split `msleep` by taxonomic order, then ask for the same summary statistics as before. We expect one row of statistics per order:

```
msleep |>
  group_by(order) |>
  summarise(avg_sleep = mean(sleep_total),
            min_sleep = min(sleep_total),
            max_sleep = max(sleep_total),
            total = n())
```

```
# A tibble: 19 x 5
  order      avg_sleep min_sleep max_sleep total
  <chr>      <dbl>      <dbl>      <dbl> <int>
1 Afrosoricida    15.6      15.6      15.6     1
2 Artiodactyla     4.52       1.9       9.1     6
3 Carnivora      10.1       3.5      15.8    12
4 Cetacea         4.5       2.7       5.6     3
5 Chiroptera     19.8      19.7      19.9     2
6 Cingulata      17.8      17.4      18.1     2
```


12. Introduction to dplyr: mammal sleep dataset

7	Didelphimorphia	18.7	18	19.4	2
8	Diprotodontia	12.4	11.1	13.7	2
9	Erinaceomorpha	10.2	10.1	10.3	2
10	Hyracoidea	5.67	5.3	6.3	3
11	Lagomorpha	8.4	8.4	8.4	1
12	Monotremata	8.6	8.6	8.6	1
13	Perissodactyla	3.47	2.9	4.4	3
14	Pilosa	14.4	14.4	14.4	1
15	Primates	10.5	8	17	12
16	Proboscidea	3.6	3.3	3.9	2
17	Rodentia	12.5	7	16.6	22
18	Scandentia	8.9	8.9	8.9	1
19	Soricomorpha	11.1	8.4	14.9	5

Each row is one taxonomic order, with its average, minimum, and maximum sleep and the number of species. Swap “order” for “treatment condition” and “sleep” for “expression” and you have a standard genomics summary.

12.13. Exercises

Use the `msleep` dataset (with `ggplot2` and `dplyr` loaded) for each of these. Try to write the pipe before opening the solution.

1. **Pick a verb.** You want only the mammals that sleep *fewer than* 5 hours, showing just their `name` and `sleep_total`. Which two verbs do you need, and in what order? Write the pipe.

Solution

```
msleep |>
  filter(sleep_total < 5) |>
  select(name, sleep_total)

# A tibble: 11 x 2
  name          sleep_total
  <chr>          <dbl>
1 Cow              4
2 Roe deer         3
3 Asian elephant  3.9
4 Horse           2.9
```

12. Introduction to dplyr: mammal sleep dataset

5	Donkey	3.1
6	Giraffe	1.9
7	Pilot whale	2.7
8	African elephant	3.3
9	Sheep	3.8
10	Caspian seal	3.5
11	Brazilian tapir	4.4

You need `filter()` to choose the rows and `select()` to choose the columns. Either order works here, but filtering first means `select()` operates on fewer rows — the habit that scales to large tables.

2. **Sleepiest carnivores.** Keep only the carnivores (`vore == "carni"`), then sort them from most to least total sleep. Show `name`, `vore`, and `sleep_total`.

Solution

```
msleep |>
  filter(vore == "carni") |>
  select(name, vore, sleep_total) |>
  arrange(desc(sleep_total))
```

A tibble: 19 x 3

	name	vore	sleep_total
	<chr>	<chr>	<dbl>
1	Thick-tailed opossum	carni	19.4
2	Long-nosed armadillo	carni	17.4
3	Tiger	carni	15.8
4	Northern grasshopper mouse	carni	14.5
5	Lion	carni	13.5
6	Domestic cat	carni	12.5
7	Arctic fox	carni	12.5
8	Cheetah	carni	12.1
9	Slow loris	carni	11
10	Jaguar	carni	10.4
11	Dog	carni	10.1
12	Red fox	carni	9.8
13	Northern fur seal	carni	8.7
14	Genet	carni	6.3
15	Gray seal	carni	6.2
16	Common porpoise	carni	5.6

12. Introduction to dplyr: mammal sleep dataset

```
17 Bottle-nosed dolphin      carni      5.2
18 Caspian seal              carni      3.5
19 Pilot whale               carni      2.7
```

`filter()` chooses the rows, `select()` trims the columns, and `arrange(desc(...))` sorts largest-first. Reading the pipe top to bottom tells the whole story.

3. **A derived column.** Is there a relationship between `rem_proportion` (REM sleep divided by total sleep) and body weight? Add a `rem_proportion` column, then sort the table so the mammals with the *highest* REM proportion come first, and look at their `bodywt`.

Solution

```
msleep |>
  mutate(rem_proportion = sleep_rem / sleep_total) |>
  select(name, bodywt, rem_proportion) |>
  arrange(desc(rem_proportion)) |>
  head()
```

A tibble: 6 x 3

	name	bodywt	rem_proportion
	<chr>	<dbl>	<dbl>
1	European hedgehog	0.77	0.347
2	Thick-tailed opossum	0.37	0.340
3	Giant armadillo	60	0.337
4	Tree shrew	0.104	0.292
5	Dog	14	0.287
6	North American Opossum	1.7	0.272

`mutate()` builds the new column, then `arrange(desc(...))` ranks by it. The heaviest mammals are not at the top, hinting that large-bodied animals tend to spend a *smaller* fraction of their sleep in REM — a pattern worth a proper plot to confirm.

4. **Summarize by group.** For each feeding type (`vore`), compute the average total sleep and the number of species. Which group sleeps the most on average?

i Solution

```
msleep |>
  group_by(vore) |>
  summarise(avg_sleep = mean(sleep_total),
            n = n())
```

```
# A tibble: 5 x 3
  vore      avg_sleep     n
<chr>      <dbl> <int>
1 carni      10.4     19
2 herbi       9.51     32
3 insecti    14.9       5
4 omni       10.9     20
5 <NA>      10.2       7
```

This is the split-apply-combine pattern: split by `vore`, apply `mean()` and `n()`, combine into one row per group. The NA row collects mammals whose feeding type is unknown — a reminder to check for missing values in real data.

12.14. Summary

You now have the core dplyr toolkit, and every verb maps onto something you'll do with biological tables:

- `select()` keeps the columns you want (e.g. pulling `sample_id` and `condition` from a sample sheet).
- `filter()` keeps the rows that pass a condition (e.g. genes below a p-value cutoff).
- `arrange()` sorts rows, with `desc()` for largest-first.
- `mutate()` adds derived columns (e.g. log fold changes).
- `summarise()` collapses a column to a summary statistic.
- `group_by()` turns `summarise()` into per-group summaries — the split-apply-combine pattern at the heart of so many analyses.
- The pipe, `|>`, chains these verbs so you can read an analysis left to right, as a sequence of “and then” steps.

Master these six verbs on `msleep` and you can carry them straight over to count matrices, sample sheets, and results tables.

13. Case Study: Behavioral Risk Factor Surveillance System

The Behavioral Risk Factor Surveillance System (BRFSS) is a large health survey run every year by the U.S. Centers for Disease Control and Prevention (CDC). Each year it phones hundreds of thousands of adults and asks about their health behaviors, chronic conditions, and use of preventive care. Public-health researchers lean on it to track risk factors, shape policy, and judge whether prevention programs are working.

In this chapter you take on the role of one of those researchers. You'll work with a small slice of the BRFSS data and walk it through a full **exploratory data analysis (EDA)**: load it, look at it, summarize it, plot it, and ask whether the variables relate to one another. EDA is the first thing you do with any new dataset — before any fancy modeling — because it's how you learn the shape of your data and catch problems early. By the end you'll go one step further and fit a simple statistical model to a question the data can actually answer.

Everything here uses **base R** — no extra packages — so you can follow along on a fresh R install.

13.1. What you'll learn

- Download a dataset and load a CSV into R with `download.file()` and `read.csv()`.
- Inspect a data frame's size and structure with `head()`, `dim()`, and `summary()`.
- Visualize single variables with `hist()` and `boxplot()`, and pairs of variables with `plot()`.
- Measure the strength of a linear relationship with `cor()`, and handle the missing-value trap that makes it return `NA`.
- Compare a measurement between two groups with a `t.test()`, and fit and read a simple linear regression with `lm()`.

13.2. Loading the dataset

First you need the data. You can download it from [this link](https://raw.githubusercontent.com/seandavi/ITR/master/BRFSS-subset.csv) by hand, or — better — let R fetch it for you straight from the command line. The `download.file()` function takes a URL and a local filename to save it under:

```
download.file('https://raw.githubusercontent.com/seandavi/ITR/master/BRFSS-subset.csv',
              destfile = 'BRFSS-subset.csv')
```

That writes a file called `BRFSS-subset.csv` into your working directory. Now you point R at that file and read it in with `read.csv()`, storing the result in a data frame called `brfss`:

```
path <- "BRFSS-subset.csv"
stopifnot(file.exists(path))
brfss <- read.csv(path)
```

The `stopifnot(file.exists(path))` line is a small safety check: if the file isn't where you said it is, R stops immediately with a clear message instead of failing in some confusing way two steps later.

💡 Working interactively? Let R pop up a file picker

If you downloaded the file by hand and aren't sure of its exact path, you can let R open a file-chooser dialog instead of typing the path:

```
path <- file.choose() # opens a dialog; navigate to BRFSS-subset.csv
brfss <- read.csv(path)
```

That's handy when you're poking around interactively, but for a script or a reproducible document you want a fixed path like the one above so it runs the same way every time.

13.3. Inspecting the data

Whenever you load a dataset, your very first move should be to *look at it*. Start with `head()`, which prints the first six rows:

13. Case Study: Behavioral Risk Factor Surveillance System

```
head(brfss)
```

```
   Age  Weight    Sex Height Year
1  31 48.98798 Female 157.48 1990
2  57 81.64663 Female 157.48 1990
3  43 80.28585   Male 177.80 1990
4  72 70.30682   Male 170.18 1990
5  31 49.89516 Female 154.94 1990
6  58 54.43108 Female 154.94 1990
```

That gives you a feel for what each column holds and what type of value it is. Here you can see an *Age*, a *Weight*, a *Height*, a *Sex*, and a *Year*.

Next, check how big the dataset is with `dim()`, which reports the number of rows and columns:

```
dim(brfss)
```

```
[1] 20000      5
```

Knowing the size matters: it tells you whether you have a handful of records or many thousands, which shapes what analyses make sense.

13.4. Summary statistics

Now that you know the structure, get a quick numeric overview of every column at once with `summary()`:

```
summary(brfss)
```

Age		Weight		Sex		Height	
Min.	:18.00	Min.	: 34.93	Length	:20000	Min.	:105.0
1st Qu.:	:36.00	1st Qu.:	: 61.69	N.unique	: 2	1st Qu.:	:162.6
Median	:51.00	Median	: 72.57	N.blank	: 0	Median	:168.0
Mean	:50.99	Mean	: 75.42	Min.nchar:	: 4	Mean	:169.2
3rd Qu.:	:65.00	3rd Qu.:	: 86.18	Max.nchar:	: 6	3rd Qu.:	:177.8
Max.	:99.00	Max.	:278.96			Max.	:218.0

13. Case Study: Behavioral Risk Factor Surveillance System

```
NAs      :139      NAs      :649      NAs      :184
Year
Min.     :1990
1st Qu.  :1990
Median   :2000
Mean     :2000
3rd Qu.  :2010
Max.     :2010
```

For each **numeric** column (**Age**, **Weight**, **Height**) this prints the minimum, first quartile, median, mean, third quartile, and maximum. For text/categorical columns you get a simpler count. Two things to notice right away: the **Weight** and **Height** columns list a number of NA's (missing values), and the spread between the median and the maximum hints at the *shape* of each distribution. Both of those observations will matter shortly.

13.5. Visualizing single variables

Numbers in a summary table only take you so far. A picture of a single variable — its **distribution** — tells you far more about its shape, center, and spread. Start with a histogram of **Age** using `hist()`:

```
hist(brfss$Age, main = "Age Distribution",
      xlab = "Age", col = "lightblue")
```

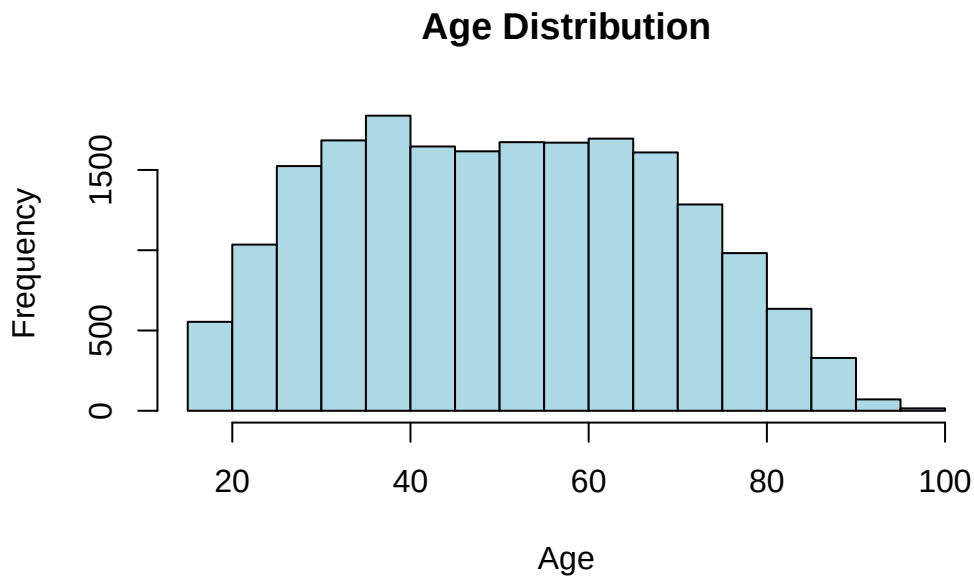



Figure 13.1.: Distribution of respondent ages in the BRFSS subset.

Each bar counts how many respondents fall in that age range. You can read the center, the spread, and whether the distribution is roughly symmetric or lopsided straight off the picture.

💡 A function has many options — read its help page

`hist()` has many options: the number of bins, the bar color, the title, the axis labels, and more. You can pull up the full list any time by typing `?hist` in the R console. This is a habit worth building for *every* function you use. The documentation for any function `f` is one keystroke away with `?f` or `help("f")`, and it's the fastest way to learn what a function can do.

Next, compare a variable *across groups*. A boxplot is perfect for this. Here you compare the `Weight` distribution between the two sexes. The formula `Weight ~ Sex` reads as “weight broken down by sex”:

```
boxplot(brfss$Weight ~ brfss$Sex, main = "Weight Distribution by Sex",
        xlab = "Sex", ylab = "Weight", col = c("pink", "lightblue"))
```

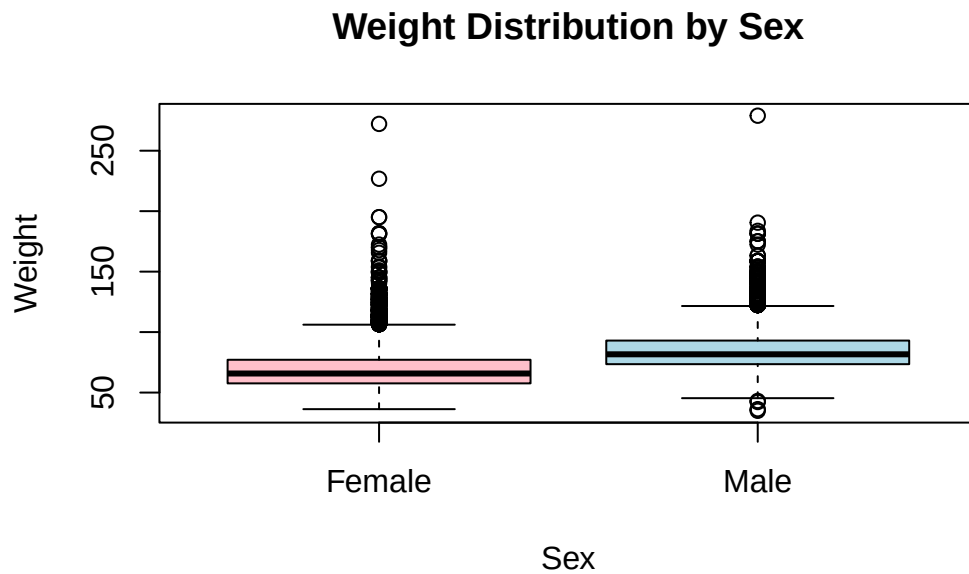


Figure 13.2.: Weight distribution compared between the sexes.

Each box spans the middle 50% of weights, the thick line is the median, and the whiskers reach out toward the extremes. At a glance you can see whether one group tends to weigh more than the other and whether the spreads differ.

13.6. Analyzing relationships between variables

So far you've looked at variables one at a time. EDA gets really interesting when you ask whether *two* variables move together. Start with a scatterplot of age against weight using `plot()`:

```
plot(brfss$Age, brfss$Weight, main = "Scatterplot of Age and Weight",  
     xlab = "Age", ylab = "Weight", col = "darkblue")
```

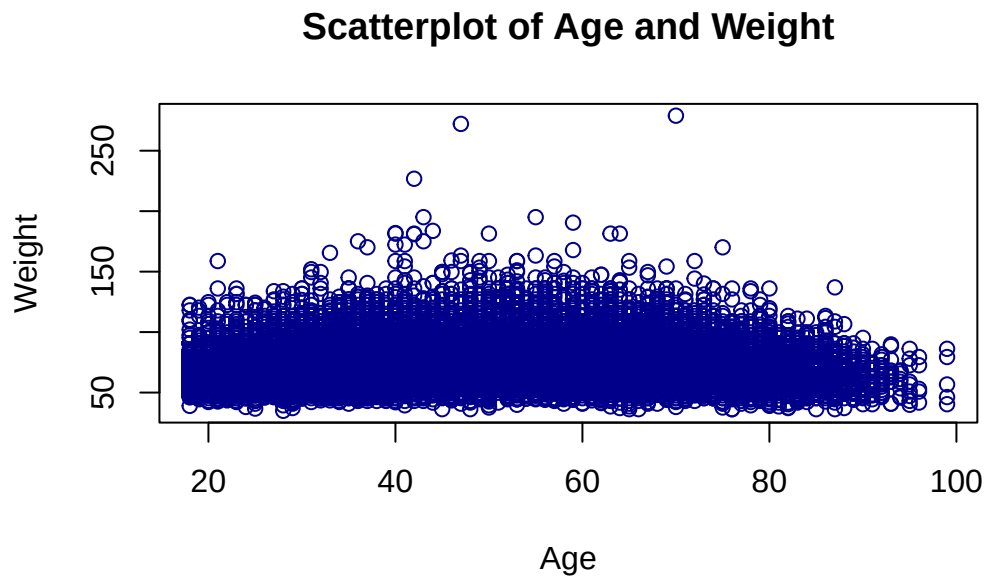


Figure 13.3.: Scatterplot of age versus weight.

Each point is one respondent. A cloud with no obvious slope means little relationship; a tilted, tightening band means the two variables tend to rise or fall together.

To put a single number on that, compute the **correlation coefficient** with `cor()`. It runs from -1 (perfect negative relationship) through 0 (none) to $+1$ (perfect positive relationship):

```
cor(brfss$Age, brfss$Weight)
```

```
[1] NA
```

That's not a number — it's NA. This is one of the most common surprises in R, so it's worth pausing on.

⚠ Missing values make `cor()` return NA

Remember those NA's the `summary()` flagged in the `Weight` column? By default `cor()` refuses to guess what a missing value should be, so if *any* pair has a missing value the

13. Case Study: Behavioral Risk Factor Surveillance System

whole result comes back NA. A quick look at `help("cor")` points to the fix: the `use` argument. Setting `use = "complete.obs"` tells `cor()` to drop the rows with missing values and compute the correlation on the rest.

```
cor(brfss$Age, brfss$Weight, use = "complete.obs")
```

```
[1] 0.02699989
```

Now you get a real number. It's small and positive, telling you age and weight are only weakly related in this dataset.

13.7. Comparing two groups: a t-test

EDA naturally raises questions that a picture alone can't settle. Here's one: the BRFSS subset spans two survey years, 1990 and 2010. **Did the average weight of female respondents change between those years?**

R read the `Year` column as a plain integer, but for grouping it's really a **category** with two levels. Convert it to a factor so R treats it that way:

```
brfss$Year <- factor(brfss$Year)
```

Now pull out just the female respondents into their own data frame so you can compare them across years:

```
brfssFemale <- brfss[brfss$Sex == "Female", ]  
summary(brfssFemale)
```

Age		Weight		Sex		Height		Year	
Min.	:18.00	Min.	: 36.29	Length	:12039	Min.	:105.0	1990:	5718
1st Qu.:	37.00	1st Qu.:	57.61	N.unique	: 1	1st Qu.:	157.5	2010:	6321
Median	:52.00	Median	: 65.77	N.blank	: 0	Median	:163.0		
Mean	:51.92	Mean	: 69.05	Min.nchar:	6	Mean	:163.3		
3rd Qu.:	67.00	3rd Qu.:	77.11	Max.nchar:	6	3rd Qu.:	168.0		
Max.	:99.00	Max.	:272.16			Max.	:200.7		
NAs	:103	NAs	:560			NAs	:140		

13. Case Study: Behavioral Risk Factor Surveillance System

A boxplot of weight by year shows the comparison at a glance. The formula `Weight ~ Year` again reads “weight broken down by year”:

```
plot(Weight ~ Year, brfssFemale)
```

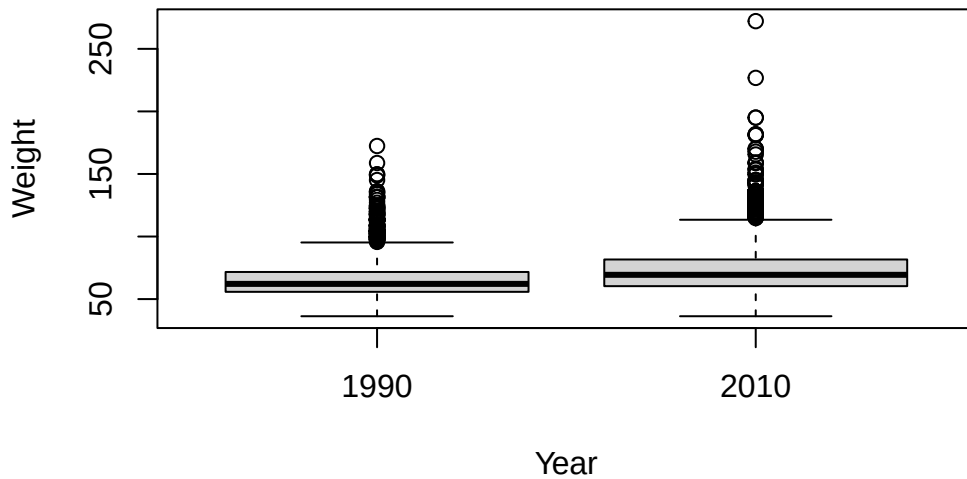


Figure 13.4.: Female respondent weight in 1990 versus 2010.

The 2010 box looks shifted up from the 1990 box — average weight appears higher in the later year. But “appears higher” isn’t an answer. To decide whether that gap is larger than you’d expect from chance alone, use a **two-sample t-test**. (We cover the t-test in depth in Section 17.6; here we just put it to work.)

```
t.test(Weight ~ Year, brfssFemale)
```

Welch Two Sample t-test

data: Weight by Year

t = -27.133, df = 11079, p-value < 2.2e-16

alternative hypothesis: true difference in means between group 1990 and group 2010 is not e

13. Case Study: Behavioral Risk Factor Surveillance System

```
95 percent confidence interval:
-8.723607 -7.548102
sample estimates:
mean in group 1990 mean in group 2010
64.81838          72.95424
```

Read the output from the bottom up. The two group means show 2010 females weigh more on average than 1990 females. The **p-value** is tiny — far below the usual 0.05 threshold — so the difference is *statistically significant*: a gap this large is very unlikely to be a fluke of sampling. The **confidence interval** for the difference doesn't include zero, telling the same story. You can now say, with evidence, that average female weight was higher in 2010 than in 1990.

13.8. Fitting a model: weight and height

A t-test compares groups. A different kind of question asks how one *continuous* variable depends on another: **for 2010 males, how does weight change with height?** That's a job for **linear regression**.

First, subset to 2010 males with `subset()`, then look at the pieces:

```
brfss2010Male <- subset(brfss, Year == 2010 & Sex == "Male")
summary(brfss2010Male)
```

Age		Weight		Sex		Height		Year	
Min.	:18.00	Min.	: 36.29	Length	:3679	Min.	:135	1990:	0
1st Qu.:	45.00	1st Qu.:	77.11	N.unique	: 1	1st Qu.:	173	2010:	3679
Median	:57.00	Median	: 86.18	N.blank	: 0	Median	:178		
Mean	:56.25	Mean	: 88.85	Min.nchar:	4	Mean	:178		
3rd Qu.:	68.00	3rd Qu.:	99.79	Max.nchar:	4	3rd Qu.:	183		
Max.	:99.00	Max.	:278.96			Max.	:218		
NAs	:30	NAs	:49			NAs	:31		

Look at each variable on its own first, then look at the relationship:

```
hist(brfss2010Male$Weight)
```

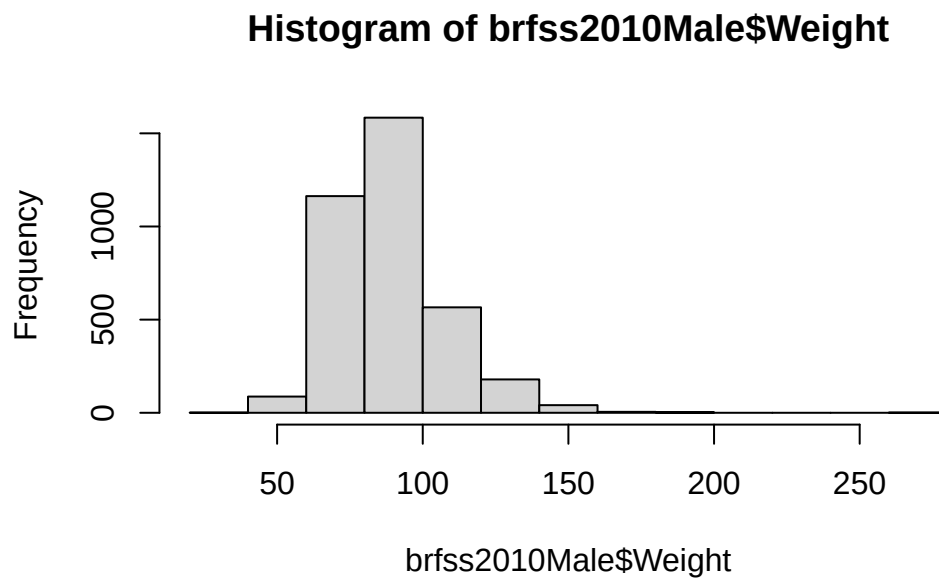


Figure 13.5.: Weight distribution for 2010 male respondents.

```
hist(brfss2010Male$Height)
```

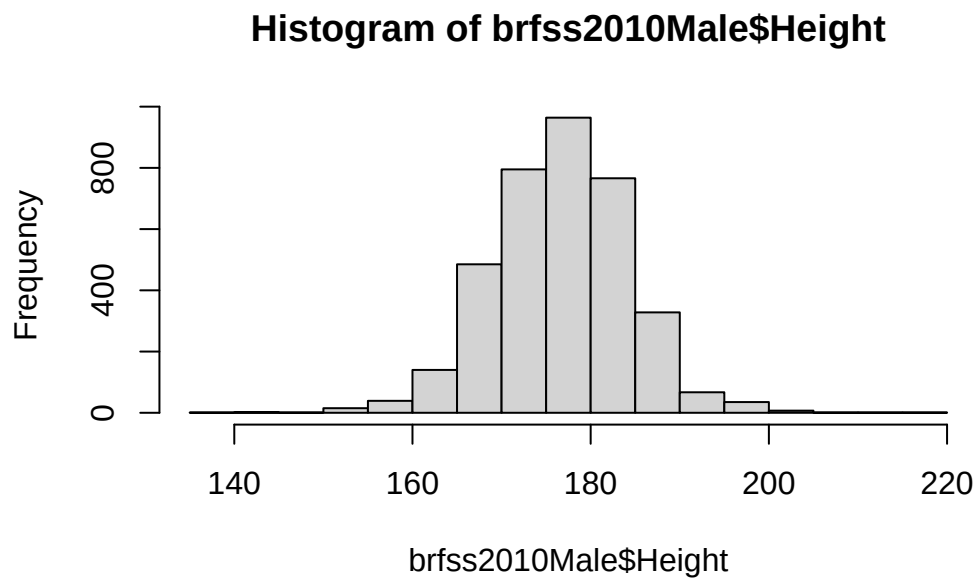


Figure 13.6.: Height distribution for 2010 male respondents.

```
plot(Weight ~ Height, brfss2010Male)
```


13. Case Study: Behavioral Risk Factor Surveillance System

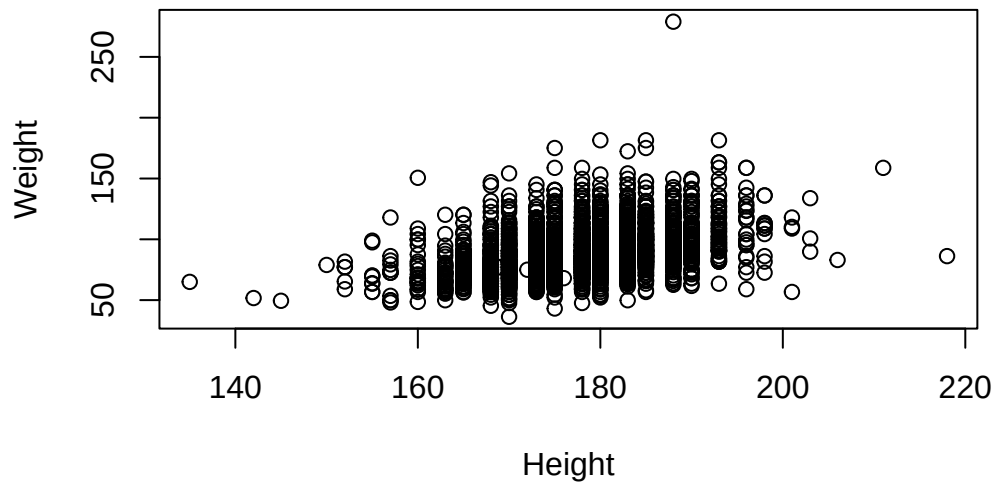


Figure 13.7.: Weight versus height for 2010 male respondents.

The scatterplot shows the expected upward trend: taller men tend to weigh more. Now quantify that trend by fitting a straight line through the cloud with `lm()` (“linear model”). The formula `Weight ~ Height` reads “model weight as a function of height”:

```
fit <- lm(Weight ~ Height, brfss2010Male)
fit
```

Call:

```
lm(formula = Weight ~ Height, data = brfss2010Male)
```

Coefficients:

(Intercept)	Height
-86.8747	0.9873

The printout gives two **coefficients**. The **(Intercept)** is where the line crosses zero height (not physically meaningful here, just where the math anchors the line). The **Height**

13. Case Study: Behavioral Risk Factor Surveillance System

coefficient is the **slope**: for each additional centimeter of height, predicted weight goes up by that many kilograms. That single number is the relationship, summarized.

You can lay out the same fit as an **ANOVA table**, which partitions the variability in weight and tests whether height explains a significant share of it:

```
anova(fit)
```

Analysis of Variance Table

Response: Weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Height	1	197664	197664	693.8	< 2.2e-16 ***
Residuals	3617	1030484	285		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The small p-value on the **Height** row confirms what the scatterplot suggested: height explains a statistically significant amount of the variation in weight.

To *see* the fitted model, redraw the scatterplot and superimpose the regression line with `abline()`, which knows how to read the line straight out of a fitted model object:

```
plot(Weight ~ Height, brfss2010Male)
abline(fit, col = "blue", lwd = 2)
# Substitute your own weight and height to see where you land...
points(73 * 2.54, 178 / 2.2, col = "red", cex = 4, pch = 20)
```

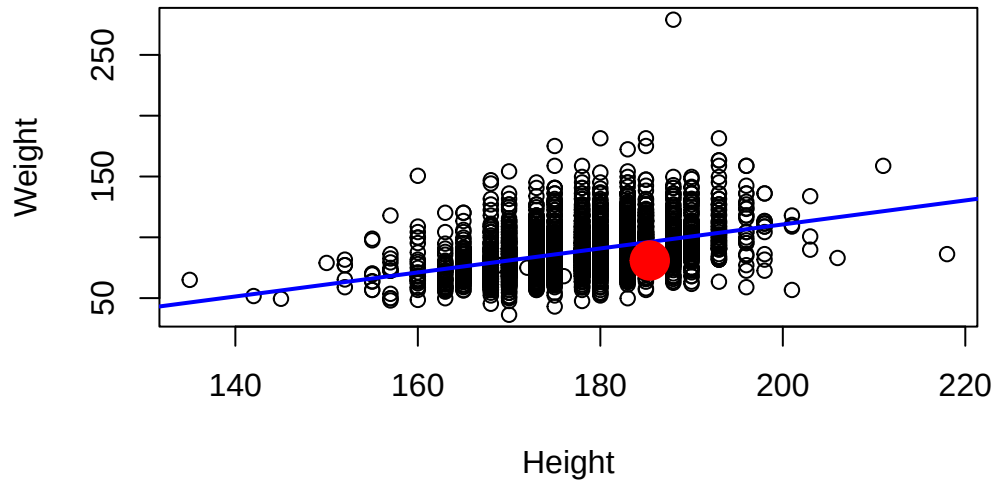


Figure 13.8.: The fitted regression line through the weight-versus-height data, with one point marked in red.

The blue line is the model's best guess at average weight for any given height. The red dot is one person's height and weight (converted to centimeters and kilograms); swap in your own numbers to see where you'd fall relative to the trend.

i Reading the model's diagnostics

A fitted model is a *noun*; the things you can do to it are *verbs* (R calls them **methods**). When you call `plot()` on a linear model, R quietly routes it to a special version, `plot.lm()`, that draws diagnostic plots instead of a scatterplot. The first one, **Residuals vs Fitted**, is the most useful: it plots the model's errors against its predictions. You want a shapeless cloud centered on zero — any strong pattern (a curve, a funnel) is a sign the straight-line model is missing something.

```
plot(fit, which = 1)
```

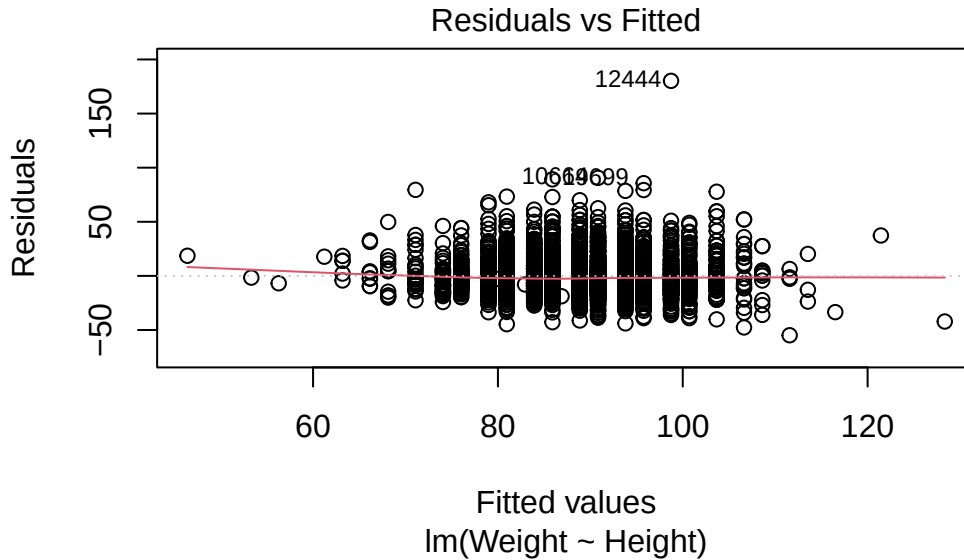


Figure 13.9.: Residuals-versus-fitted diagnostic plot for the weight-versus-height model.

You can see the full set of diagnostics with `plot(fit)` (it draws four), and read about them with `?plot.lm`.

13.9. Exercises

1. What is the mean weight in this dataset? How about the median? What is the difference between the two, and what does that tell you about the shape of the weight distribution?

```
mean(brfss$Weight, na.rm = TRUE)
```

```
[1] 75.42455
```

```
median(brfss$Weight, na.rm = TRUE)
```

```
[1] 72.57478
```

13. Case Study: Behavioral Risk Factor Surveillance System

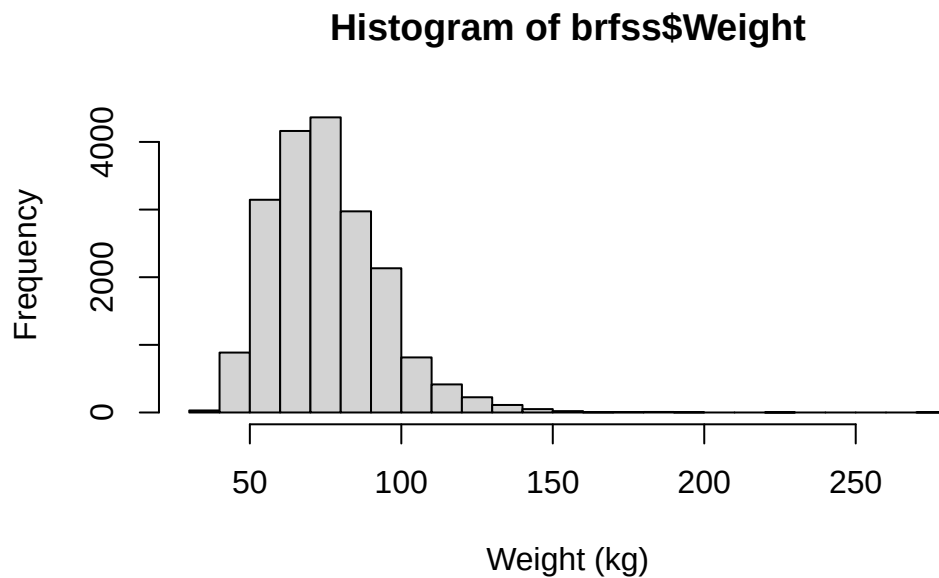
```
mean(brfss$Weight, na.rm = TRUE) - median(brfss$Weight, na.rm = TRUE)
```

```
[1] 2.849774
```

The mean sits above the median, which is the signature of a distribution with a long right tail — a few very heavy individuals pull the mean up. (Note the `na.rm = TRUE`: just like `cor()`, these functions need to be told to ignore the missing values.)

2. Using what you just found about the mean and median, draw a histogram of the weight distribution with `hist()`. How would you describe its shape?

```
hist(brfss$Weight, xlab = "Weight (kg)", breaks = 30)
```

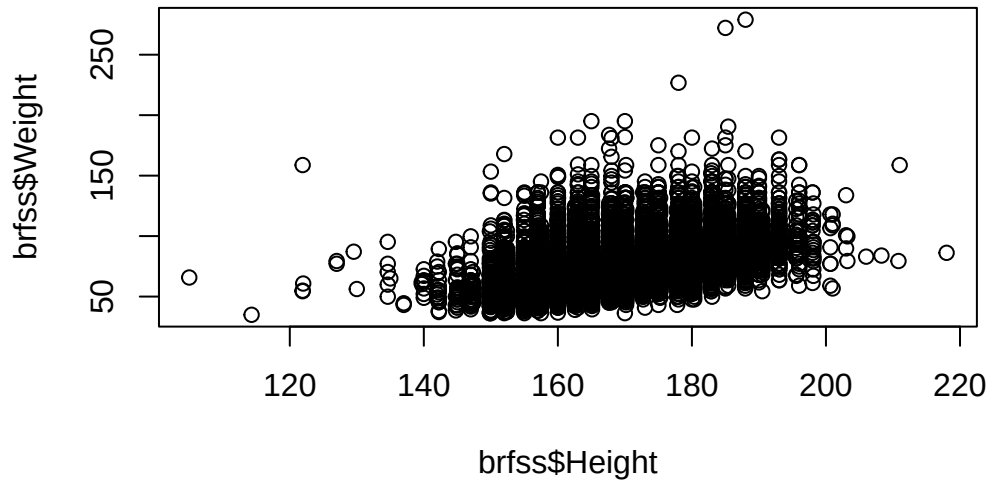


The histogram is **right-skewed**: most respondents cluster at lower weights with a tail stretching toward the high end — exactly what the mean-above-median result predicted.

3. Use `plot()` to examine the relationship between height and weight.

```
plot(brfss$Height, brfss$Weight)
```

13. Case Study: Behavioral Risk Factor Surveillance System



Taller people tend to be heavier, so the points drift upward to the right.

4. What is the correlation between height and weight? What does it tell you about their relationship?

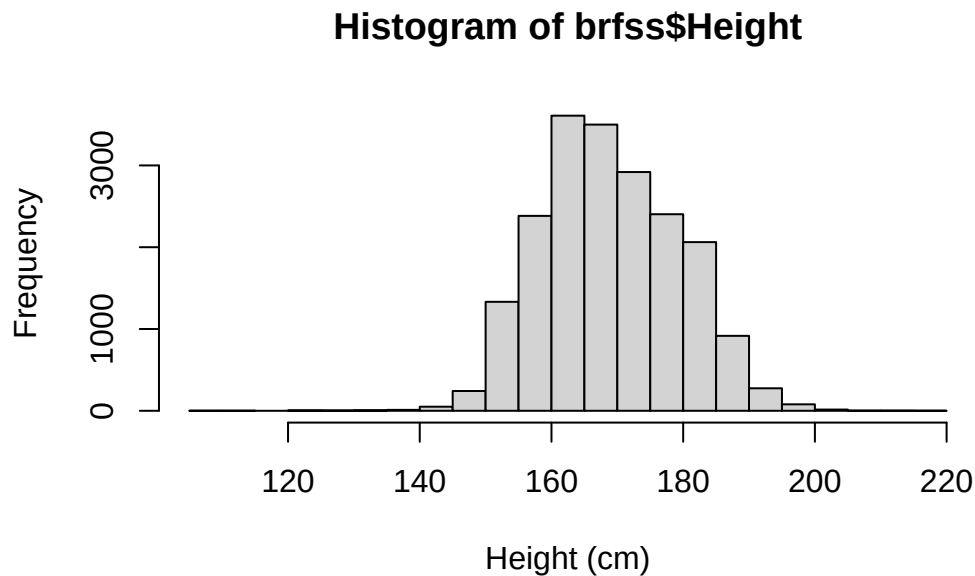
```
cor(brfss$Height, brfss$Weight, use = "complete.obs")
```

```
[1] 0.5140928
```

A moderate-to-strong positive correlation, confirming numerically what the scatter-plot showed. (Don't forget `use = "complete.obs"` to skip the missing values.)

5. Draw a histogram of the height distribution. How would you describe its shape compared to weight?

```
hist(brfss$Height, xlab = "Height (cm)", breaks = 30)
```



Height is far more symmetric than weight — closer to the bell-shaped normal curve, with no strong tail to either side.

13.10. Summary

You've now run a complete exploratory data analysis on a real public-health dataset, end to end:

- You **loaded** the data with `download.file()` and `read.csv()`, and guarded the load with `stopifnot(file.exists(path))`.
- You **inspected** it with `head()`, `dim()`, and `summary()`, which flagged the missing values that mattered later.
- You **visualized** single variables with `hist()` and `boxplot()`, and pairs of variables with `plot()`.
- You **measured** a relationship with `cor()` — and learned that missing values force you to add `use = "complete.obs"`.
- You went **beyond description** to answer real questions: a `t.test()` showed female weight rose between 1990 and 2010, and an `lm()` quantified how male weight rises with height.

13. Case Study: Behavioral Risk Factor Surveillance System

EDA is the beginning of the data-analysis process, not the end — but it’s the part that keeps every later step honest. Knowing the shape of your data, its missing values, and its obvious relationships is what lets you choose the right model and trust the answer it gives. For more on the t-test you used here, see [Section 17.6](#).

14. Exploring data with ggplot2

A plot answers questions a table of numbers cannot. Does a cost rise with age? Do two groups behave differently? Are there outliers worth a second look? `ggplot2` is the package most R users reach for to ask those questions, and it does so with a small, consistent vocabulary you can combine to build almost any graphic. In this chapter you'll start from a blank canvas and, one layer at a time, build up a single plot that shows four variables at once.

This chapter is based on the [Intro to ggplot2 chapter](#) from the book [Modern Data Visualization with R](#) by Robert Kabacoff, which is licensed under the [Creative Commons Attribution-NonCommercial 4.0 International License](#). The original chapter has been modified to fit the context of this book.

The [insurance dataset](#) is described in the book *Machine Learning with R* by Brett Lantz. A cleaned version of the dataset is also available on Kaggle. The dataset describes medical information and costs billed by health insurance companies in 2013, as compiled by the United States Census Bureau. Variables include age, sex, body mass index, number of children covered by health insurance, smoker status, US region, and individual medical costs billed by health insurance for 1338 individuals.

14.1. What you'll learn

- Initialize a plot with `ggplot()` and tell it which data frame to use.
- Add geometric layers (points, lines) with `geom_*()` functions.
- Map variables to aesthetics such as the x-axis, y-axis, and colour, and use colour to group observations.
- Adjust how variables are displayed with `scale_*()` functions.
- Split a plot into small multiples with `facet_wrap()`.
- Add informative labels and titles with `labs()`, and restyle the plot with a `theme_*`.

i The grammar of graphics

ggplot2 is built on an idea called the *grammar of graphics*: a plot is assembled from independent pieces you stack with `+`. The pieces are always the same — **data** (the data frame), **aesthetics** (which variable maps to x, to y, to colour, ...), **geoms** (the shapes that get drawn: points, lines, bars), and optional **scales**, **facets**, **labels**, and **themes**. You don't memorize a separate function for every chart type; you learn this handful of building blocks and recombine them. That is why the same code patterns produce a scatterplot here and, with a different data frame, a plot of gene-expression values.

To get started, you need to install and load the **ggplot2** package. If you haven't installed it yet, you can do so with:

```
install.packages("ggplot2")
```

Once installed, load the package:

```
library(ggplot2)
```

Next, read the insurance dataset into R. We'll use a convenient online version and the `read.csv()` function to load it:

```
# load the data
url <- "https://tinyurl.com/mtktm8e5"
insurance <- read.csv(url)
```

In RStudio you can use the `View()` function to inspect the dataset in a spreadsheet-style window. We don't run it here because it only works in interactive RStudio, not when the book is rendered:

```
# view the dataset
View(insurance)
```

Next, we'll add a variable indicating whether a patient is obese. We'll define obesity as a body mass index greater than or equal to 30:

14. Exploring data with ggplot2

```
# create an obesity variable
insurance$obese <- ifelse(insurance$bmi >= 30,
                          "obese", "not obese")
```

When you build a `ggplot2` graph, only the first two pieces below — `ggplot()` and at least one `geom_*()` — are required. The rest are optional and can appear in any order.

14.2. `ggplot()`

The `ggplot()` function initializes the plot. It takes a data frame as its first argument and can also include aesthetic mappings (`aes()`) that define how variables in the data map to visual properties such as the x and y axes, colour, and size.

```
# initialize the plot
ggplot(data = insurance, aes(x = age, y = expenses))
```

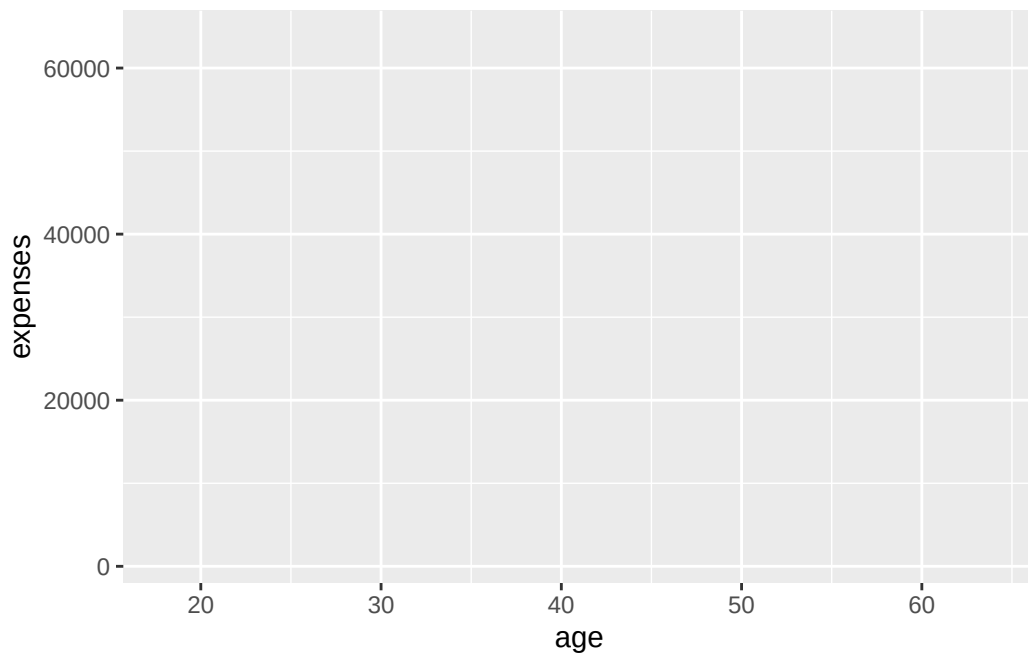


Figure 14.1.: Initialize the plot with `ggplot()`.

Why is Figure 14.1 showing an empty panel? Because we haven't added any layers yet. The `ggplot()` function only sets up the canvas; it doesn't draw anything until we add a geom. We told it to map `age` to the x-axis and `expenses` to the y-axis, but we haven't yet said *what shape* to put on the graph.

14.3. `geom_*()`

The `geom_*`() functions add layers to the plot. Each one corresponds to a specific geometric object: `geom_point()` adds points, `geom_line()` adds lines, `geom_bar()` adds bars, and so on. Let's add points.

```
# add points to the plot
ggplot(data = insurance,
       mapping = aes(x = age, y = expenses)) +
  geom_point()
```

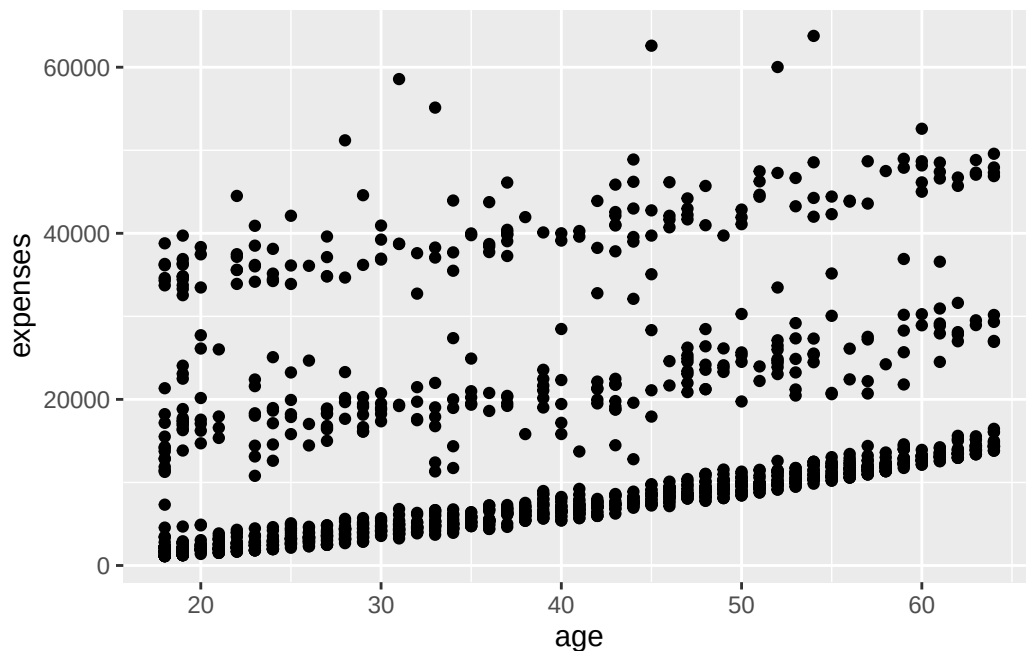


Figure 14.2.: Add points to the plot with `geom_point()`.

In Figure 14.2 we added points with `geom_point()`. The `+` operator stacks layers onto the plot, and the `mapping` argument in `aes()` specifies how variables map to visual properties.

14. Exploring data with ggplot2

You can already see that insurance expenses tend to rise with age, though there is a lot of variability.

A geom can take parameters (options) of its own. For `geom_point()`, the common ones are `color`, `size`, and `alpha`. These control point colour, point size, and transparency. Transparency ranges from 0 (completely transparent) to 1 (completely opaque).

```
# make points blue, larger, and semi-transparent
ggplot(data = insurance,
       mapping = aes(x = age, y = expenses)) +
  geom_point(color = "cornflowerblue",
            alpha = .7,
            size = 2)
```

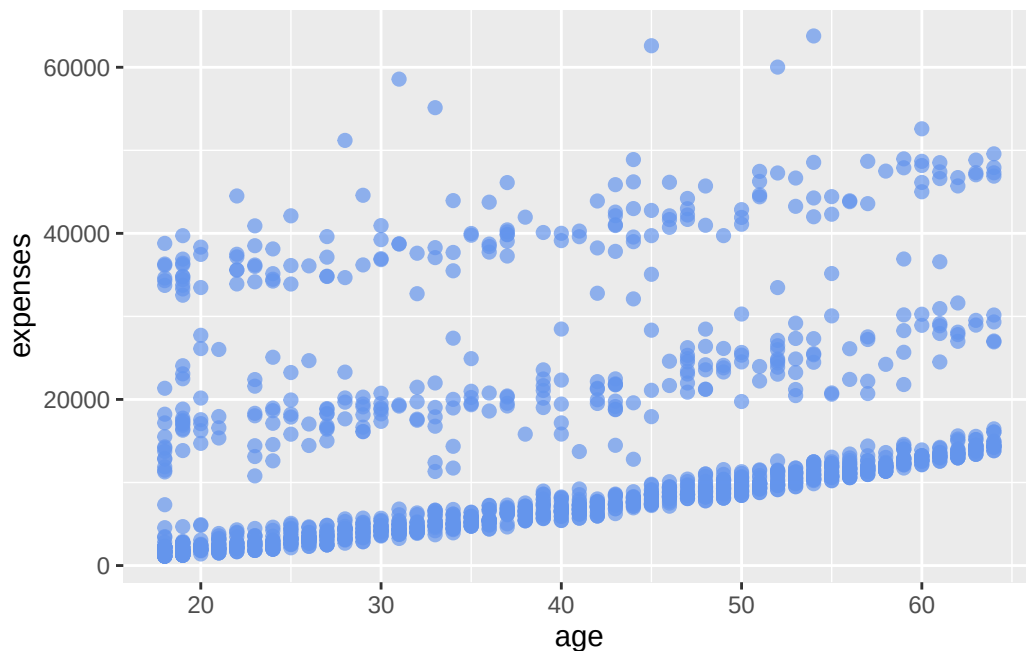


Figure 14.3.: Modify point colour, size, and transparency.

💡 Use `alpha` to see through crowded points

When many points land on top of one another — common with hundreds or thousands of observations — a solid plot hides how the data pile up. Setting `alpha` below 1 makes each point partly transparent, so dense regions render darker and sparse ones lighter.

14. Exploring data with ggplot2

It's the cheapest fix for an overplotted scatterplot, and it works the same way whether your points are patients or single cells.

Next, let's layer on a line of best fit — in other words, a regression fit. We do that with `geom_smooth()`. Its options control the type of line (linear, quadratic, nonparametric), the line's thickness and colour, and whether a confidence band is shown. Here we ask for a linear regression line with `method = "lm"` (where `lm` stands for *linear model*).

```
# add a line of best fit
ggplot(data = insurance,
       mapping = aes(x = age, y = expenses)) +
  geom_point(color = "cornflowerblue",
            alpha = .7,
            size = 2) +
  geom_smooth(method = "lm")
```

``geom_smooth()`` using formula = 'y ~ x'

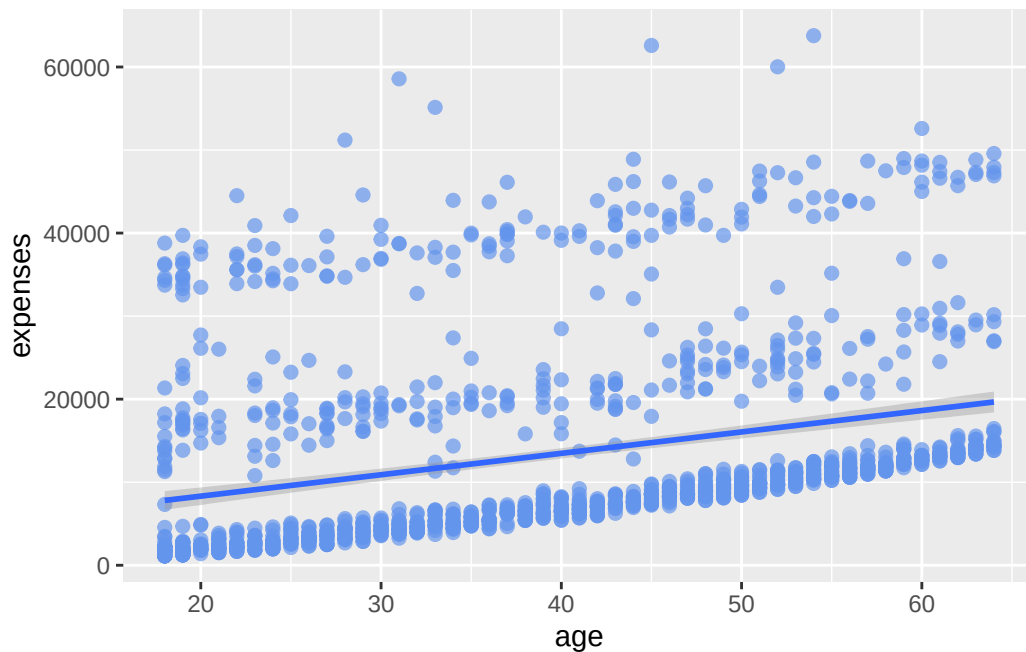


Figure 14.4.: Add a line of best fit with `geom_smooth()`.

In Figure 14.4 we added a fitted line with `geom_smooth(method = "lm")`. The `method` argument chooses the type of fit; here a straight line summarizes the upward trend, and the shaded band around it is the confidence interval.

14.4. Grouping

Beyond the x and y axes, you can map groups of observations to colour, shape, size, transparency, and other visual characteristics. This lets you superimpose several groups in a single graph. Let's add smoker status and show it with colour.

```
# group points by smoker status
ggplot(data = insurance,
       mapping = aes(x = age, y = expenses, color = smoker)) +
  geom_point(alpha = .7, size = 2) +
  geom_smooth(method = "lm", se = FALSE)
```

``geom_smooth()`` using formula = 'y ~ x'

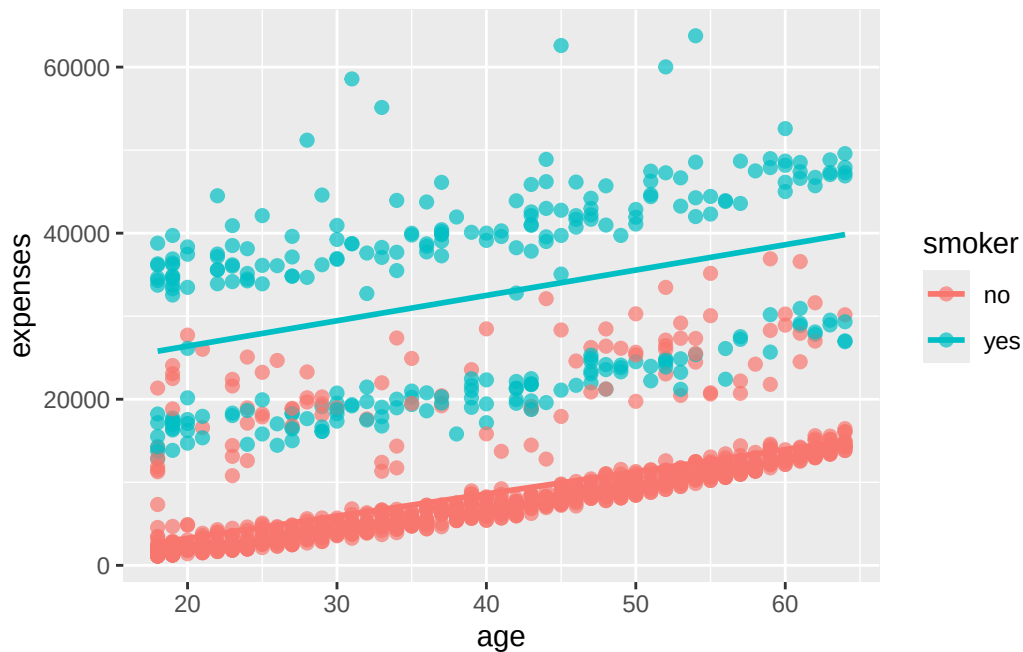


Figure 14.5.: Group points by smoker status.

In Figure 14.5 we moved the `color` aesthetic into `aes()` and mapped it to the `smoker` variable. Because `color` is now part of the mapping, `ggplot2` splits the data by smoker status and draws a separate set of points *and* a separate fitted line for each group. It probably comes as no surprise that smokers appear to incur greater expenses than non-smokers.

14.5. Scales

Scales control how variables are translated into the visual characteristics of the plot. Scale functions (which start with `scale_`) let you adjust that translation. Next we'll change the spacing of the x and y axis tick marks, format the y-axis as dollars, and pick our own colours.

```
# modify the x and y axes and specify the colours to be used
ggplot(data = insurance,
       mapping = aes(x = age,
                     y = expenses,
                     color = smoker)) +
  geom_point(alpha = .5,
            size = 2) +
  geom_smooth(method = "lm",
            se = FALSE,
            linewidth = 1.5) +
  scale_x_continuous(breaks = seq(0, 70, 10)) +
  scale_y_continuous(breaks = seq(0, 60000, 20000),
                    label = scales::dollar) +
  scale_color_manual(values = c("indianred3",
                                "cornflowerblue"))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

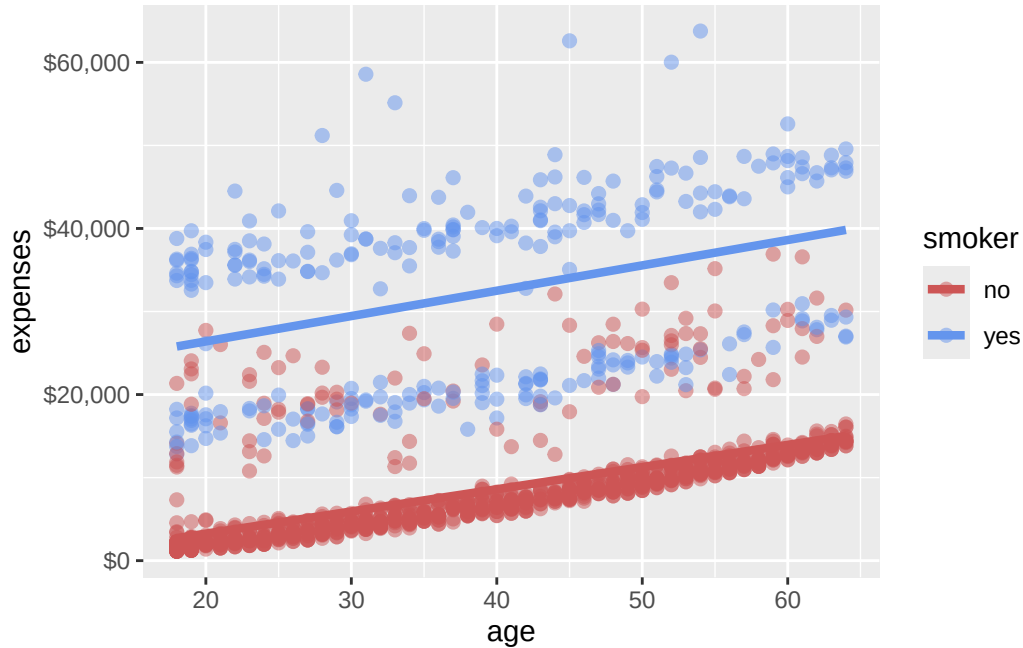



Figure 14.6.: Modify scales for the x and y axes and the colours.

In Figure 14.6 we used `scale_x_continuous()` and `scale_y_continuous()` to set the axis tick marks. The `breaks` argument specifies where the ticks fall, and the `label` argument formats the y-axis values as dollar amounts with `scales::dollar`. `scale_color_manual()` then assigns our chosen colours to the two smoker groups. (Note that line thickness in `geom_smooth()` is set with `linewidth`, the current argument for the width of a line geom.)

14.6. Facets

Faceting splits a plot into small multiples — one panel per level of a categorical variable. This is handy for comparing relationships across groups. The `facet_wrap()` function does this; the `~obese` formula tells it to make one panel for each value of the `obese` variable.

```
# reproduce the plot for obese and non-obese individuals
ggplot(data = insurance,
       mapping = aes(x = age,
                     y = expenses,
```

14. Exploring data with ggplot2

```
color = smoker)) +  
geom_point(alpha = .5) +  
geom_smooth(method = "lm",  
            se = FALSE) +  
scale_x_continuous(breaks = seq(0, 70, 10)) +  
scale_y_continuous(breaks = seq(0, 60000, 20000),  
                  label = scales::dollar) +  
scale_color_manual(values = c("indianred3",  
                              "cornflowerblue")) +  
facet_wrap(~obese)
```

`geom\_smooth()` using formula = 'y ~ x'

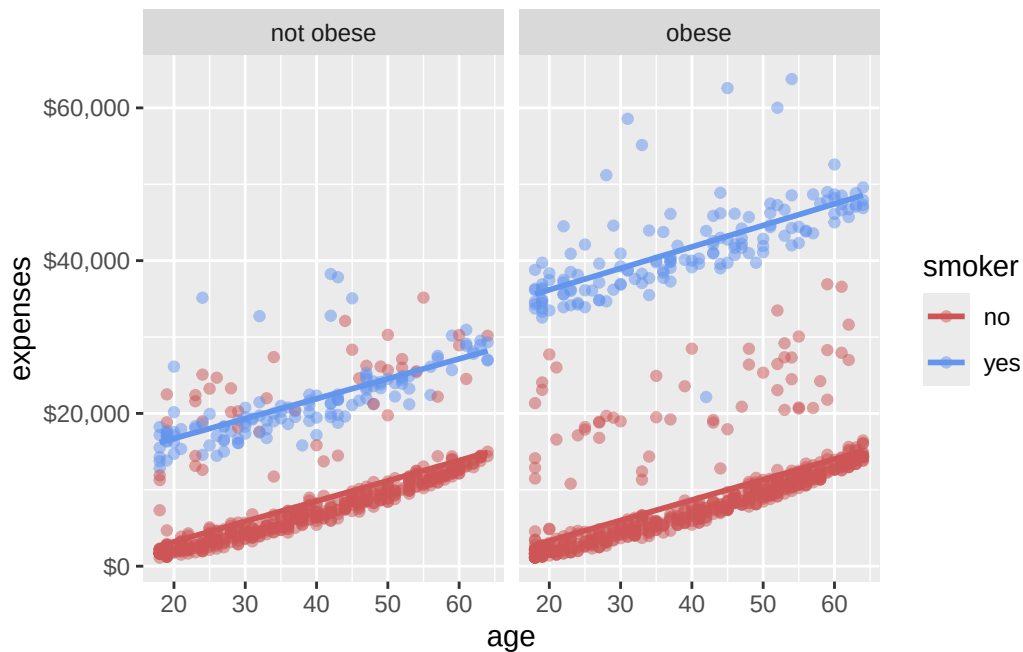


Figure 14.7.: Create facets based on the obese variable.

In Figure 14.7, `facet_wrap(~obese)` produced separate panels for obese and non-obese individuals, letting us compare the age-versus-expenses relationship side by side. We have now packed four dimensions of data — age, smoking status, obesity, and annual expenses — into a single two-dimensional figure.

14.7. Labels and titles

Labels and titles make a plot self-explanatory. The `labs()` function sets the axis labels, the legend title, and the plot title, subtitle, and caption.

```
# add informative labels
ggplot(data = insurance,
       mapping = aes(x = age,
                     y = expenses,
                     color = smoker)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm",
             se = FALSE) +
  scale_x_continuous(breaks = seq(0, 70, 10)) +
  scale_y_continuous(breaks = seq(0, 60000, 20000),
                    label = scales::dollar) +
  scale_color_manual(values = c("indianred3",
                                "cornflowerblue")) +
  facet_wrap(~obese) +
  labs(title = "Relationship between patient demographics and medical costs",
       subtitle = "US Census Bureau 2013",
       caption = "source: http://mosaic-web.org/",
       x = "Age (years)",
       y = "Annual expenses",
       color = "Smoker?")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

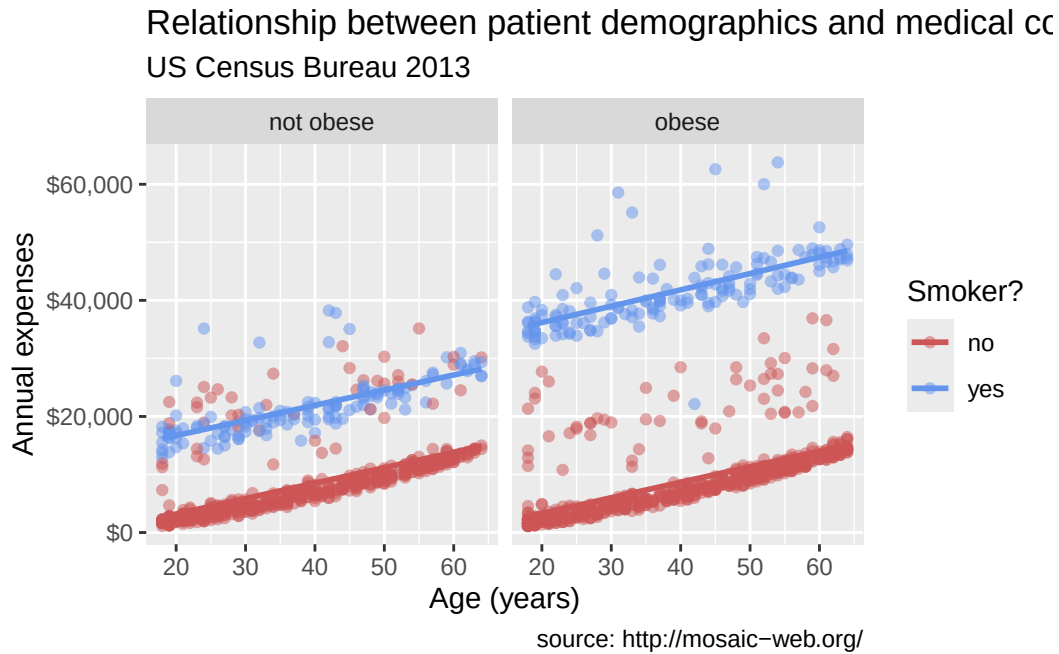


Figure 14.8.: Add labels and a title to the plot.

In Figure 14.8 we used `labs()` to add a title, subtitle, caption, and clearer axis and legend labels. A reader can now understand the figure without hunting for context elsewhere.

14.8. Theming

Finally, you can fine-tune the plot's overall appearance with a theme. Theme functions (which start with `theme_`) control background colours, fonts, grid lines, legend placement, and other non-data features. Let's switch to a cleaner look.

```
# use a minimalist theme
ggplot(data = insurance,
       mapping = aes(x = age,
                     y = expenses,
                     color = smoker)) +
  geom_point(alpha = .5) +
  geom_smooth(method = "lm",
             se = FALSE) +
```

14. Exploring data with ggplot2

```
scale_x_continuous(breaks = seq(0, 70, 10)) +  
scale_y_continuous(breaks = seq(0, 60000, 20000),  
                   label = scales::dollar) +  
scale_color_manual(values = c("indianred3",  
                              "cornflowerblue")) +  
facet_wrap(~obese) +  
labs(title = "Relationship between age and medical expenses",  
      subtitle = "US Census Data 2013",  
      caption = "source: https://github.com/dataspelunking/MLwR",  
      x = "Age (years)",  
      y = "Medical expenses",  
      color = "Smoker?") +  
theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

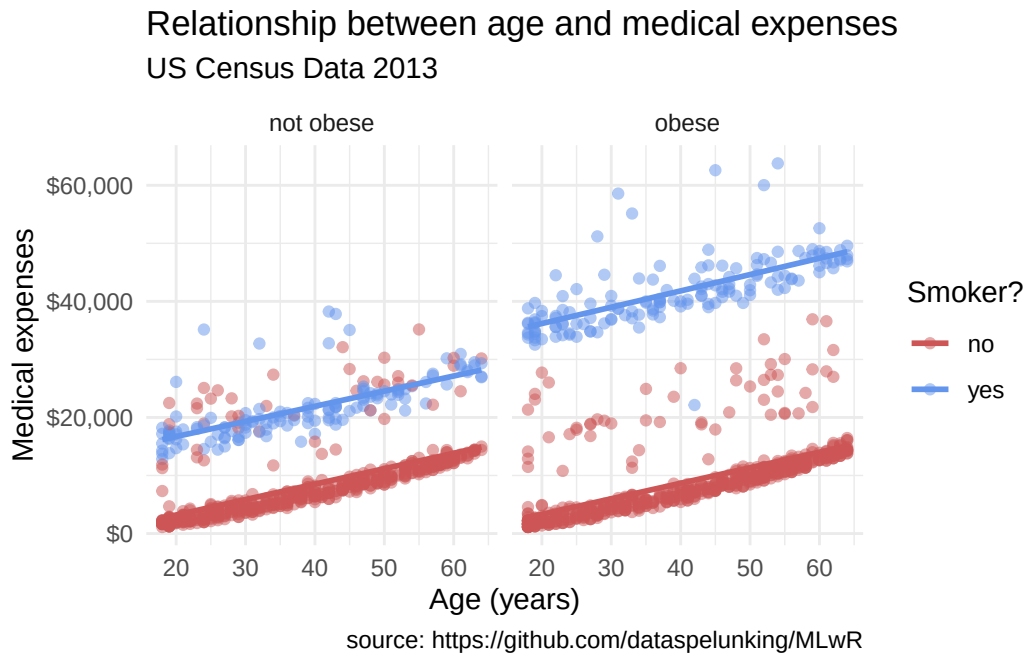


Figure 14.9.: Customize the plot's appearance with a theme.

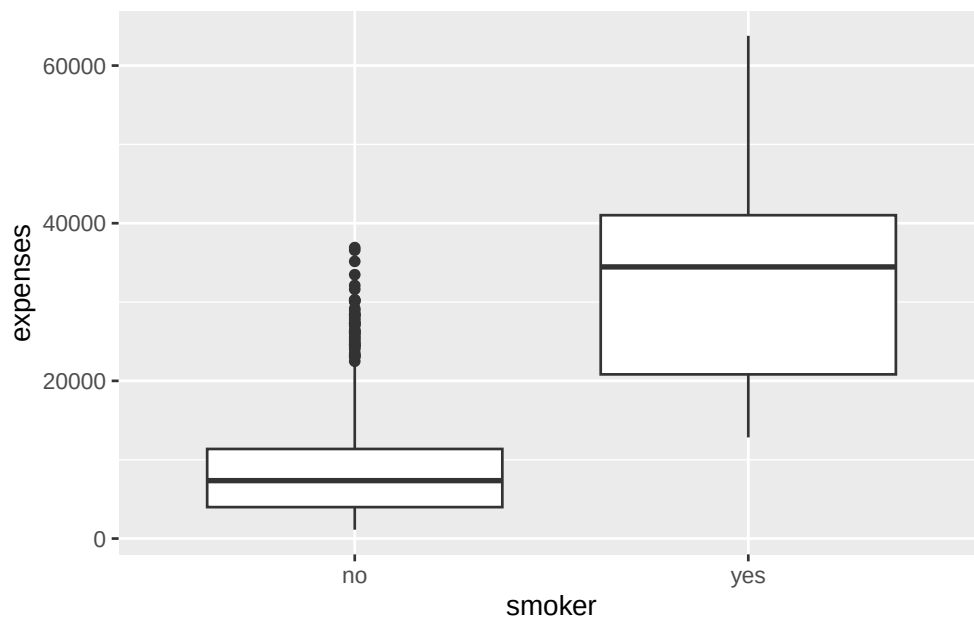
14.9. Exercises

The insurance data is a convenient, non-biological playground, but every technique below is exactly what you'd use to plot a biological dataset — swap **expenses** for a gene's expression level and **region** for a treatment group and the code is unchanged. Try these on the **insurance** data frame you loaded above.

1. **A boxplot by group.** Instead of a scatterplot, draw a boxplot of **expenses** split by **smoker**. Map **smoker** to the x-axis and **expenses** to the y-axis, and use `geom_boxplot()`.

i Solution

```
ggplot(data = insurance,
       mapping = aes(x = smoker, y = expenses)) +
  geom_boxplot()
```



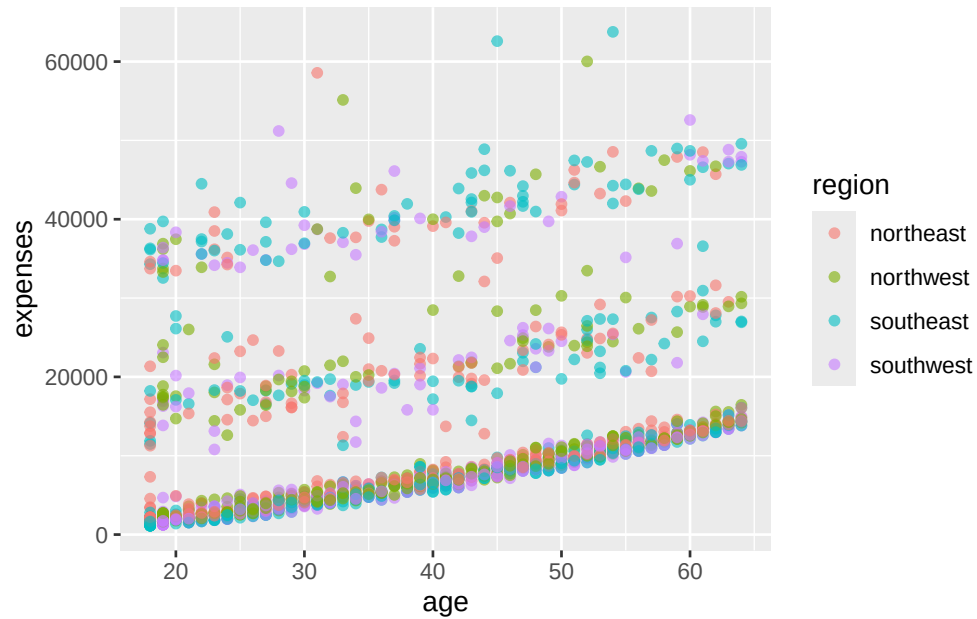
A boxplot is just a different geom on the same `aes()` skeleton. The boxes make it obvious that smokers' expenses are both higher and more spread out than non-smokers'.

2. **Colour by region.** Make a scatterplot of **expenses** against **age** and colour the points by **region**. Add `alpha = .6` so crowded points stay readable.

14. Exploring data with ggplot2

i Solution

```
ggplot(data = insurance,  
       mapping = aes(x = age, y = expenses, color = region)) +  
  geom_point(alpha = .6)
```



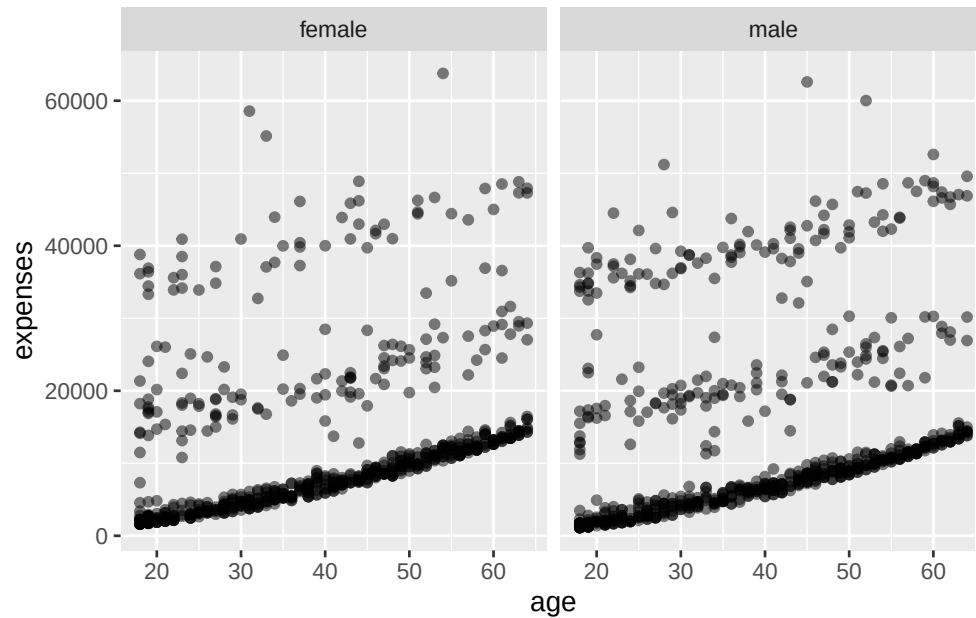
Mapping `region` to `color` inside `aes()` gives each of the four regions its own colour. The clouds overlap heavily, which tells you region alone doesn't explain much of the variation in expenses.

3. **Facet by sex.** Take the age-versus-expenses scatterplot and split it into one panel per sex with `facet_wrap()`.

i Solution

```
ggplot(data = insurance,  
       mapping = aes(x = age, y = expenses)) +  
  geom_point(alpha = .5) +  
  facet_wrap(~sex)
```

14. Exploring data with ggplot2



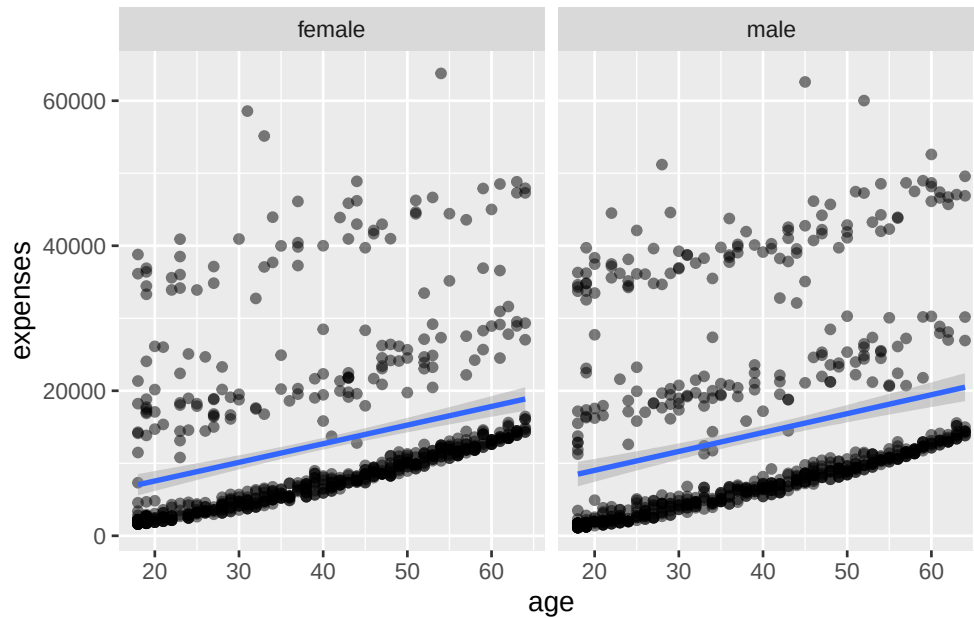
`facet_wrap(~sex)` draws the same plot once for each sex. The two panels look very similar, suggesting sex has little effect on the age-expenses relationship on its own.

4. **Add a fitted line.** Starting from the plot in Exercise 3, add a linear fit with `geom_smooth(method = "lm")` so each panel gets its own trend line.

i Solution

```
ggplot(data = insurance,  
       mapping = aes(x = age, y = expenses)) +  
  geom_point(alpha = .5) +  
  geom_smooth(method = "lm") +  
  facet_wrap(~sex)
```

``geom_smooth()`` using formula = 'y ~ x'



Because faceting splits the data first, `geom_smooth()` fits a separate line within each panel. Both slopes point upward and look nearly parallel, reinforcing that the age trend holds for both groups.

14.10. Summary

You can now build a `ggplot2` graphic from the ground up. You initialized a plot with `ggplot()`, drew data with `geom_point()` and `geom_smooth()`, mapped variables to aesthetics and used colour to separate groups, reshaped the display with `scale_*`() functions, split the plot into panels with `facet_wrap()`, labelled it with `labs()`, and restyled it with `theme_minimal()`. Each was an independent layer added with `+` — the grammar of graphics in action.

Reading the final figure (Figure 14.9), it appears that:

- There is a positive linear relationship between age and expenses, and the slope stays roughly constant across smoking and obesity status.
- Smokers and obese patients have higher medical expenses.
- Smoking and obesity interact. Non-smokers look similar across obesity groups, but among smokers, obese patients have much higher expenses.
- A few very high outliers sit in the obese-smoker group.

14. *Exploring data with ggplot2*

These findings are tentative. They rest on a limited sample and involve no statistical testing to assess whether the differences could be due to chance.

For the design principles behind these plots — the data-ink ratio, colorblind-safe palettes — and specialized plot types for genomics such as UpSet plots and heatmaps, see [A self-guided tour of data visualization in R](#).

15. A self-guided tour of data visualization in R

Imagine you have just finished an experiment. You have a spreadsheet with a thousand rows of penguin measurements, or a matrix of gene-expression values across dozens of samples, and you need to *see* what is going on. Which species are bigger? Are two genes switched on together? Staring at a wall of numbers will not tell you — but the right picture will. Data visualization is the bridge between raw measurements and human understanding, and it is one of the most useful skills a biologist can own.

This chapter is a guided tour. We start with two ideas that make *any* plot better, then build up `ggplot2` — R’s standard plotting system — one layer at a time so you understand *why* the code looks the way it does. From there we tour a few specialized plot types you’ll meet in genomics: UpSet plots for set overlaps, heatmaps for expression matrices, and a pointer to genome-track visualization. The companion chapter [Exploring data with ggplot2](#) walks a single dataset end to end; here we focus on the *grammar* and on the wider toolbox.

15.1. What you’ll learn

- Apply two design principles — the data-ink ratio and clear labeling — to make any plot more readable.
- Read a `ggplot2` call as a stack of **layers**: data → aesthetics (`aes()`) → geoms → scales → facets → themes.
- Build a plot incrementally, adding one layer at a time and interpreting each result.
- Choose a colorblind-safe palette with `RColorBrewer`.
- Reach for the right specialized plot — an UpSet plot for many sets, a heatmap for a data matrix — and produce a runnable example of each.

15.2. Two principles that make any plot better

Before we touch a single `ggplot()` call, it helps to know what we are aiming for. Two ideas carry most of the weight.

15.2.1. Maximize the data-ink ratio

The statistician Edward Tufte coined the term **data-ink ratio**: the proportion of a graphic’s ink that is actually showing your data, rather than decoration. The goal is to keep that ratio high. Put simply, **every mark on the plot should earn its place**.

Common “chart junk” to delete:

- Redundant or heavy grid lines
- Busy backgrounds or unnecessary fill colors
- 3D effects on what is really 2D data
- Drop shadows and other decoration

Let’s see the difference. We’ll use R’s built-in `mpg` dataset (fuel-economy records for various car models) — the relationship is generic, but the lesson about ink transfers directly to a scatterplot of, say, gene length versus expression. First, a cluttered version:

```
library(ggplot2)

ggplot(mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_gray() +
  labs(title = "Fuel efficiency vs. engine displacement",
       subtitle = "This is a very busy plot",
       x = "Engine displacement (litres)",
       y = "Highway miles per gallon",
       color = "Vehicle class")
```

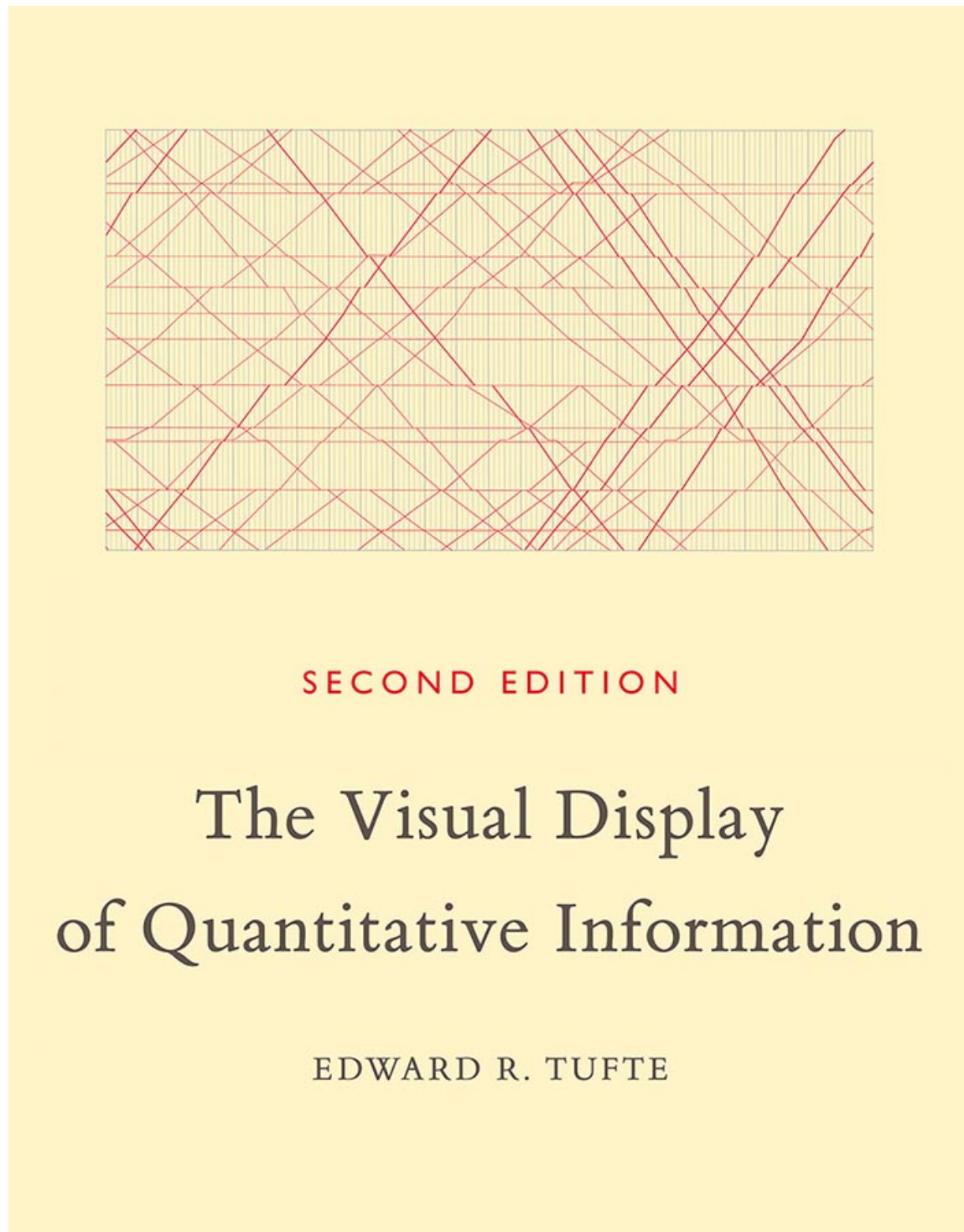


Figure 15.1.: Edward Tufte’s landmark book, *The Visual Display of Quantitative Information*, popularized the data-ink ratio.

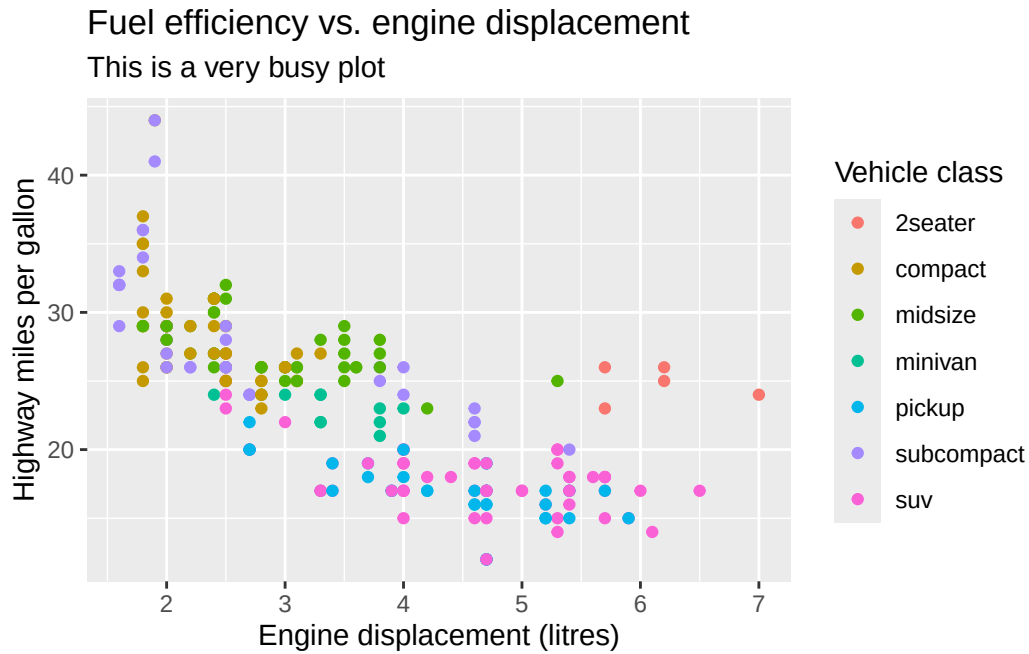


Figure 15.2.: A cluttered plot with a low data-ink ratio. The gray background and heavy gridlines are decoration, not information.

Now the same data with the decoration stripped away:

```
ggplot(mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  theme_minimal() +
  labs(title = "Fuel efficiency vs. engine displacement",
       x = "Engine displacement (litres)",
       y = "Highway miles per gallon",
       color = "Class")
```

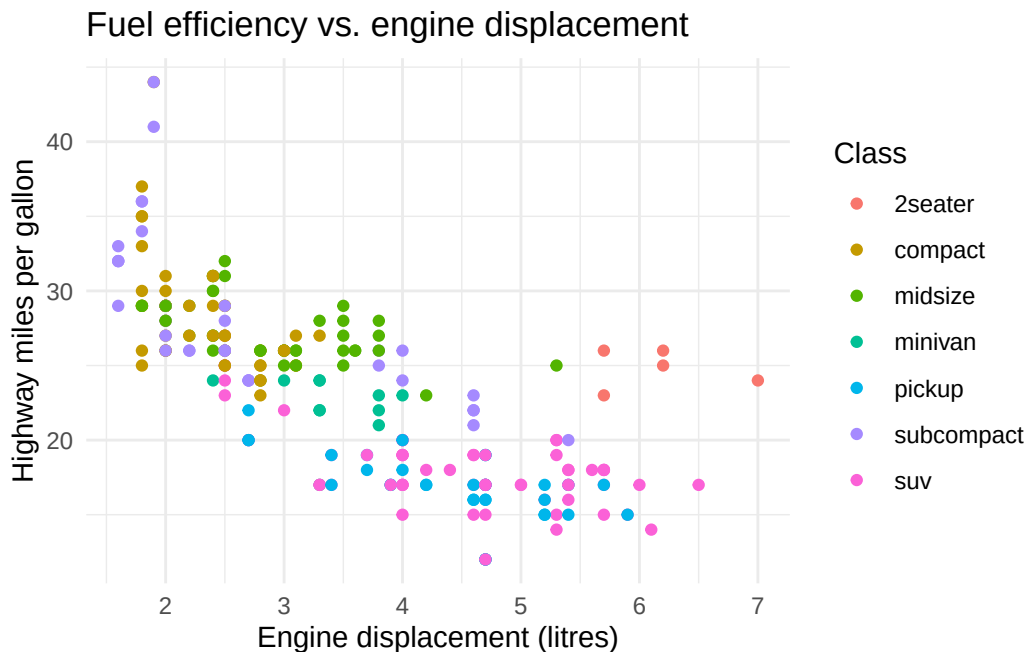


Figure 15.3.: The same data with a high data-ink ratio. The eye goes straight to the points.

The only change between Figure 15.2 and Figure 15.3 is the theme, yet the second reads more easily: the trend (bigger engines get worse mileage) jumps out because nothing competes with it. This is a guideline, not a law — a faint gridline can genuinely help a reader estimate values — but “could I delete this?” is always worth asking.

15.2.2. Label everything

A plot without labels is just a picture. To be analysis, it has to carry its context with it. Every finished plot should have:

- A clear, descriptive title.
- Axis labels *with units* (e.g., “Body mass (g)”).
- A legend whenever color, shape, or size encodes a variable.

💡 Labels are for future-you

The most common audience for your plot is yourself, three months later, trying to remember what it showed. Spelling out units and a title costs seconds now and saves

an embarrassing email later.

15.2.3. Use color deliberately

Color is powerful and easy to overuse. Two rules of thumb: don't use more colors than you have meaningful groups, and pick palettes that survive printing and colorblindness. The `RColorBrewer` package ships curated palettes in three families:

- **Sequential** palettes run light → dark and suit ordered data (a gradient from low to high expression).
- **Qualitative** palettes use distinct hues for *categories* with no order (species, treatment group).
- **Diverging** palettes emphasize a meaningful midpoint with two contrasting directions (log-fold-changes around zero).

```
library(RColorBrewer)

display.brewer.all()
```

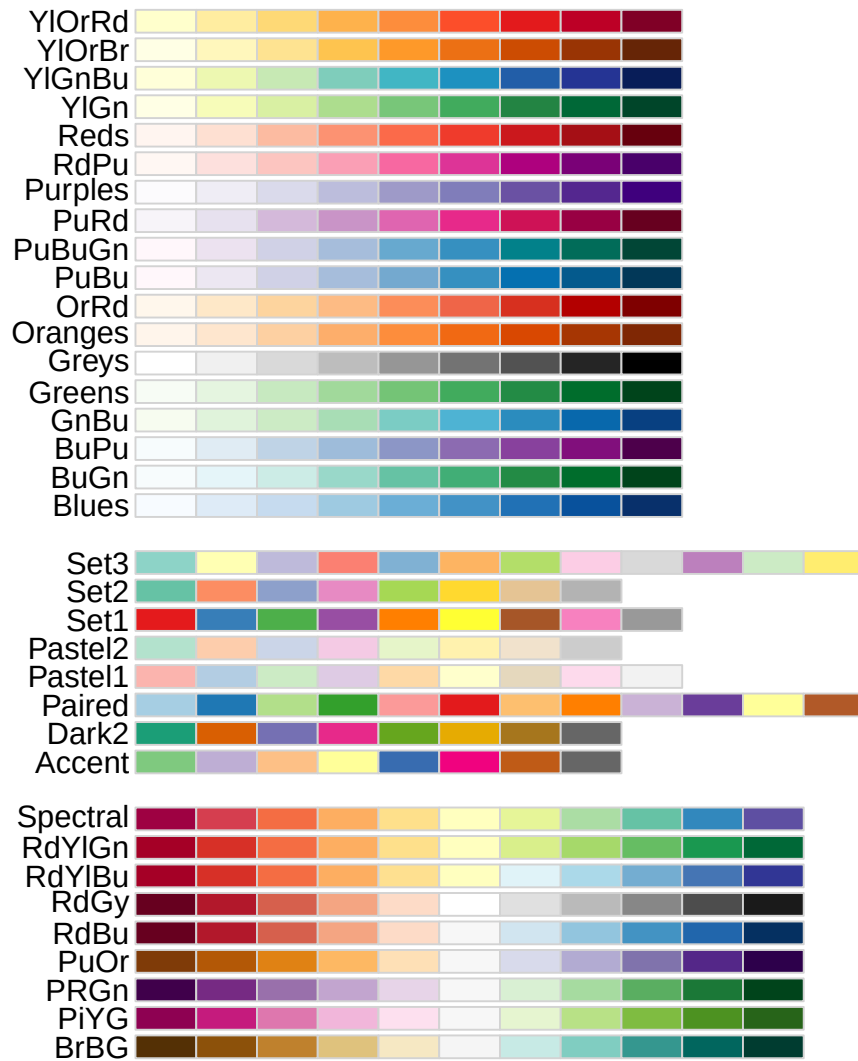



Figure 15.4.: RColorBrewer palettes, grouped into sequential (top), qualitative (middle), and diverging (bottom) families.

Figure 15.4 is a menu: pick a palette name (like "Set2" or "RdBu") and hand it to a `scale_color_brewer()` or `scale_fill_brewer()` layer.

 Roughly 1 in 12 men is red-green colorblind

A red/green scheme that looks fine to you can be unreadable to a chunk of your audience. `RColorBrewer` can show only the palettes that stay distinguishable under common color-vision deficiencies — restrict yourself to these when color carries meaning.

```
display.brewer.all(colorblindFriendly = TRUE)
```

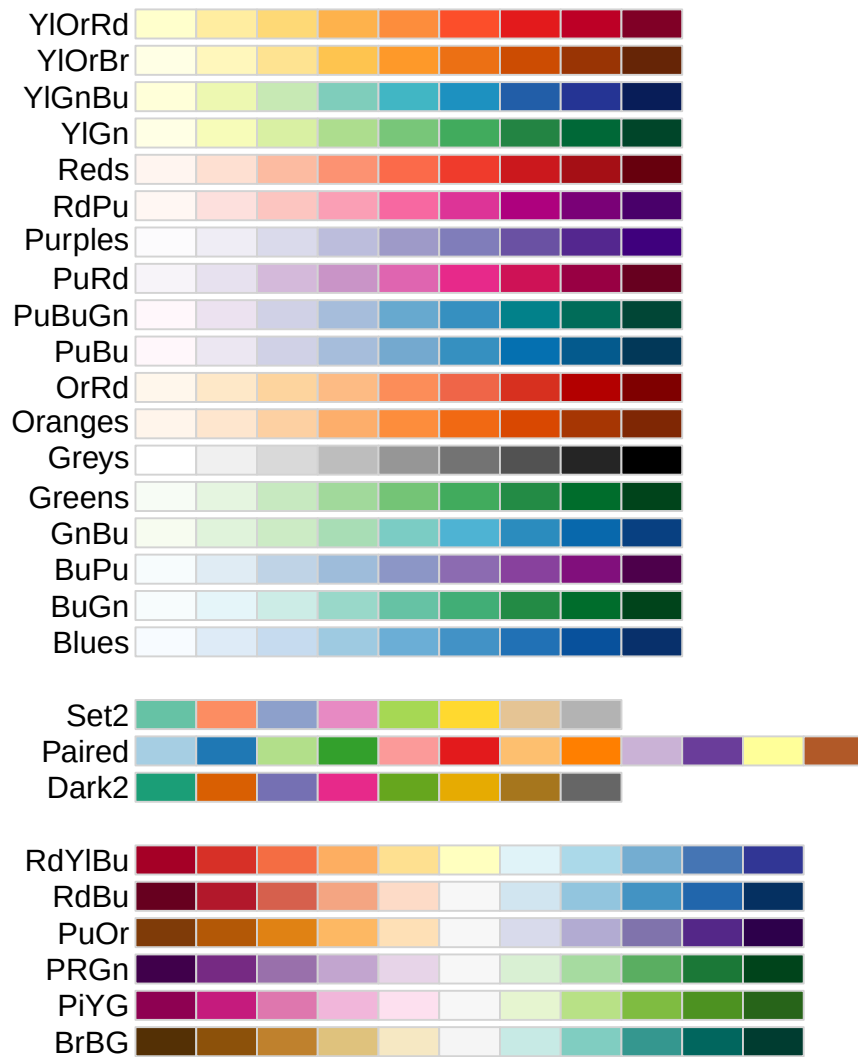


Figure 15.5.: The subset of RColorBrewer palettes that remain distinguishable for viewers with common color-vision deficiencies.

The palettes in Figure 15.5 are the safe default set. When in doubt, choose from here.

15.3. The grammar of graphics, layer by layer

`ggplot2` is built on an idea called the **grammar of graphics**: every plot is assembled from a small set of independent parts, the way a sentence is built from nouns, verbs, and adjectives. Once you know the parts, you can describe an enormous variety of plots with the same handful of “words.” The parts are:

1. **Data** — the data frame you’re plotting.
2. **Aesthetics** (`aes()`) — which *variable* maps to which *visual channel* (x position, y position, color, size, shape).
3. **Geoms** — the geometric shapes that actually get drawn (points, lines, bars).
4. **Scales** — how data values translate into pixels and colors (axis ranges, color palettes).
5. **Facets** — splitting one plot into small multiples by a grouping variable.
6. **Themes** — the non-data styling (fonts, background, gridlines).

Think of it like ordering a coffee: the **data** is the beans, the **aesthetics** say what goes in which cup, the **geom** is whether you get it as espresso or latte, and the **theme** is the mug it’s served in. We’ll build one plot up through these layers, adding exactly one new idea per step.

We’ll use the `palmerpenguins` dataset — body measurements for three penguin species (*Adelie*, *Chinstrap*, *Gentoo*) recorded near Palmer Station, Antarctica. It’s real organismal biology and small enough to reason about.

```
library(palmerpenguins)
library(dplyr)

# drop the few rows with missing measurements so plots don't warn
penguins <- na.omit(penguins)
glimpse(penguins)
```

```
Rows: 333
Columns: 8
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
$ flipper_length_mm <int> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
```

15. A self-guided tour of data visualization in R

```
$ body_mass_g      <int> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~  
$ sex              <fct> male, female, female, female, male, female, male, fe~  
$ year            <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

`glimpse()` shows each penguin has a species, an island, bill and flipper measurements, a body mass, and a sex. We'll ask a biological question: **how do the species differ in body size?**

15.3.1. Step 1: data and aesthetics

We start by telling `ggplot()` *what data* to use and *which variables map to which axes*. Here, flipper length on x and body mass on y:

```
ggplot(data = penguins,  
       mapping = aes(x = flipper_length_mm, y = body_mass_g))
```

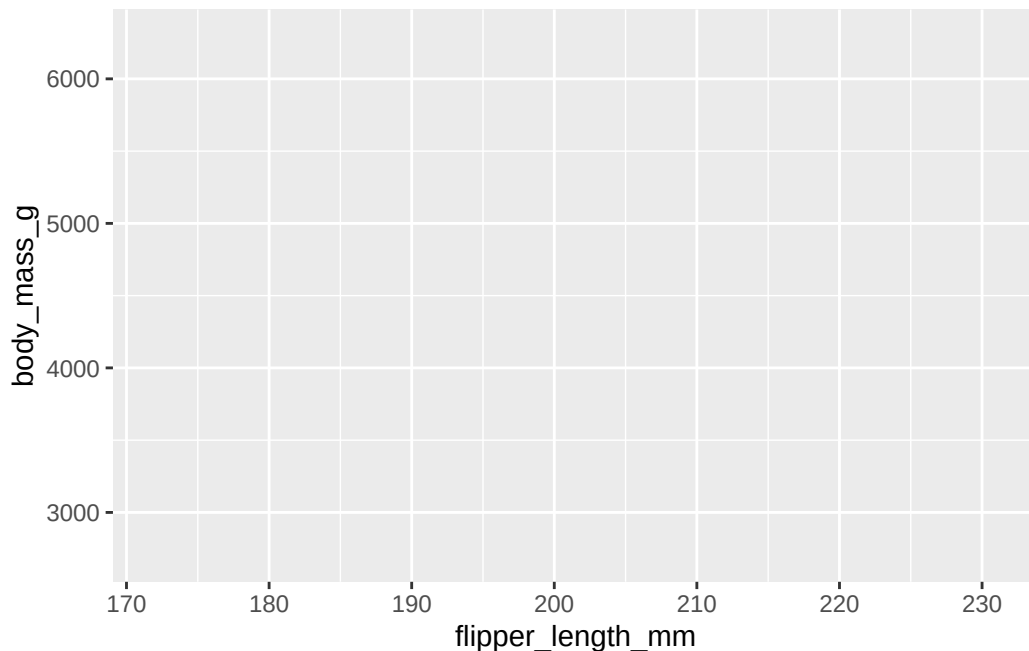


Figure 15.6.: Data plus aesthetics, but no geom: `ggplot()` sets up the axes but draws nothing yet.

Figure 15.6 looks empty — and that’s the point. We’ve declared the axes (`aes()` mapped `flipper_length_mm` to `x` and `body_mass_g` to `y`), but we haven’t said *what shape* to draw. A plot with no geom has nothing to show.

15.3.2. Step 2: add a geom

A **geom** is the layer that draws something. We add layers with `+`. For two continuous variables, points make a scatterplot:

```
ggplot(data = penguins,  
       mapping = aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()
```

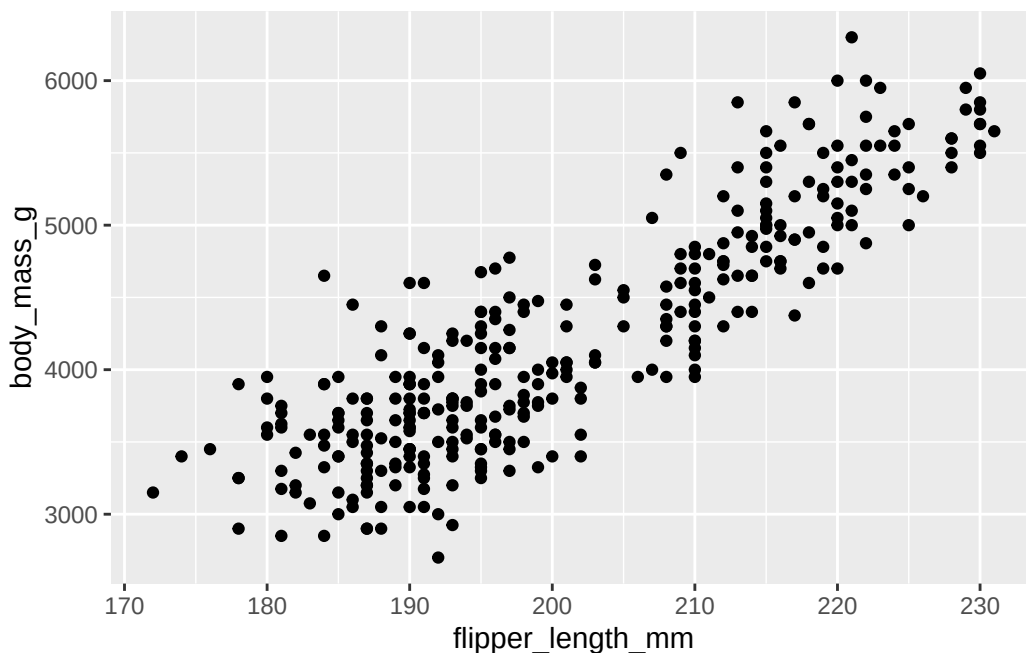


Figure 15.7.: Adding `geom_point()` draws one point per penguin.

Now Figure 15.7 tells a story: heavier penguins tend to have longer flippers, a clear positive relationship. But the points seem to fall into a couple of clusters. Could that be the species?

15.3.3. Step 3: map a third variable to color

Aesthetics aren't limited to x and y. We can map a *categorical* variable onto **color** to encode a third dimension on the same flat plot:

```
ggplot(data = penguins,
       mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point()
```

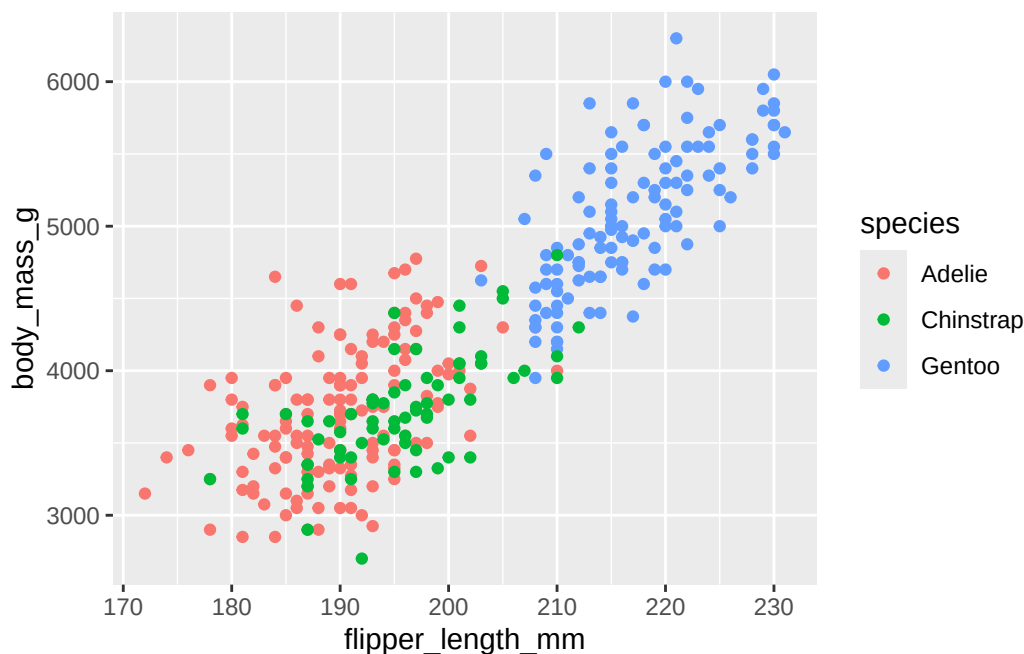


Figure 15.8.: Mapping species to the color aesthetic reveals three clusters.

The clusters in Figure 15.8 resolve cleanly: Gentoo penguins are the large ones (long flippers, heavy), while Adelie and Chinstrap overlap at the smaller end. We answered our biological question by adding *one word* to `aes()`.

i Inside `aes()` vs. outside it

Put a setting **inside** `aes()` when it should vary *with the data* (`color = species`). Put it **outside** `aes()`, in the geom, when it's a fixed constant (`geom_point(color = "blue")` makes every point blue). Mixing these up is the single most common

ggplot2 confusion.

15.3.4. Step 4: layer on a second geom

Layers stack. We can keep the points and add a `geom_smooth()` trend line on top. Because `color = species` is set in the shared `aes()`, we get one line per species for free:

```
ggplot(data = penguins,
       mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE)
```

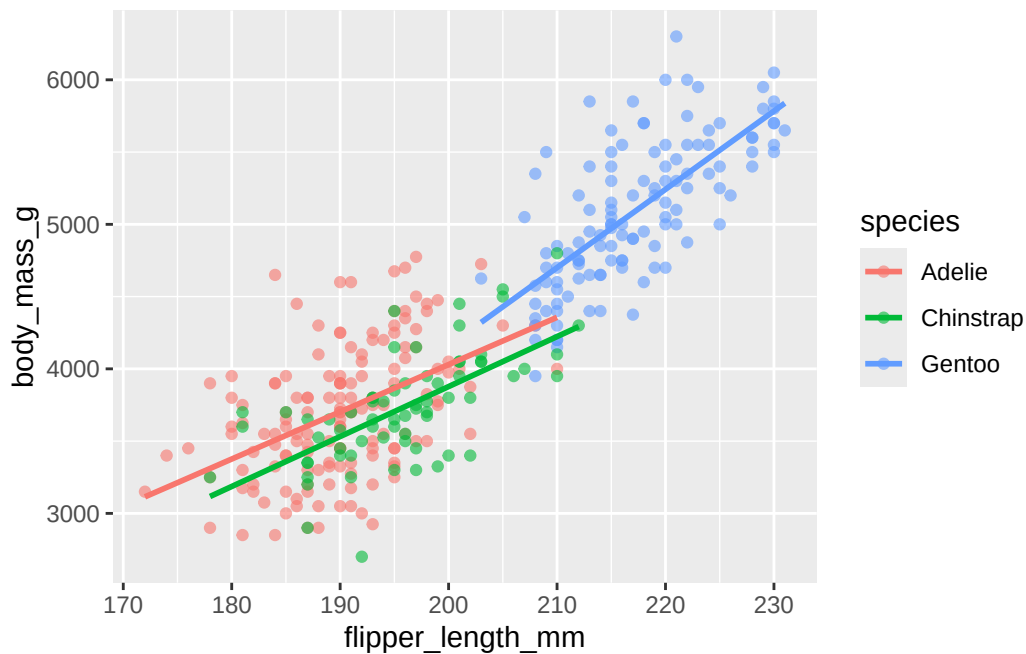


Figure 15.9.: A second geom (a linear trend line) layered over the points, one line per species.

Figure 15.9 adds three regression lines (`method = "lm"`). All three slope upward, confirming the flipper-mass relationship holds *within* each species, not just across the pooled cloud. We also set `alpha = 0.6` (outside `aes()`, so it's constant) to make overlapping points semi-transparent.

15.3.5. Step 5: control scales

Scales govern the translation from data to pixels and colors. Let's swap in a colorblind-safe Brewer palette and label the axis ticks with units:

```
ggplot(data = penguins,
       mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_color_brewer(palette = "Set2") +
  scale_y_continuous(breaks = seq(3000, 6000, 1000))
```

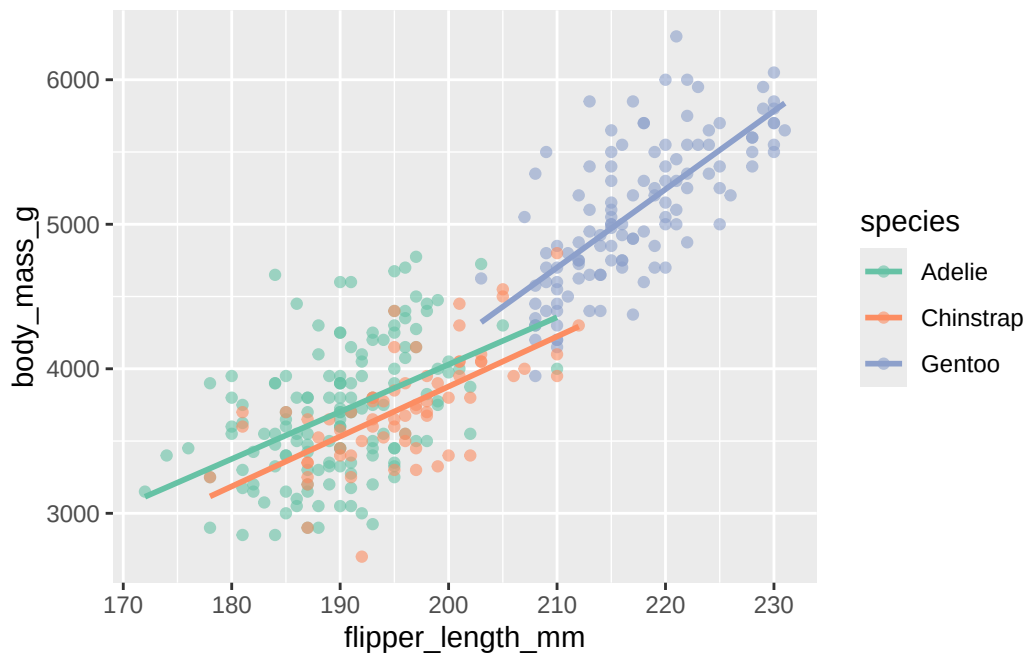


Figure 15.10.: A Brewer qualitative palette via `scale_color_brewer()` and tidy axis breaks.

In Figure 15.10 the colors come from the qualitative "Set2" palette (safe under color blindness, per Section 15.2), and `scale_y_continuous()` sets even gridline breaks. Scales are where you take control of *how* the mapping looks without changing *what* is mapped.

15.3.6. Step 6: facet into small multiples

Faceting splits one plot into a grid of panels, one per group — often clearer than cramming everything together. `facet_wrap(~ species)` gives each species its own panel:

```
ggplot(data = penguins,
       mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_color_brewer(palette = "Set2") +
  facet_wrap(~ species)
```

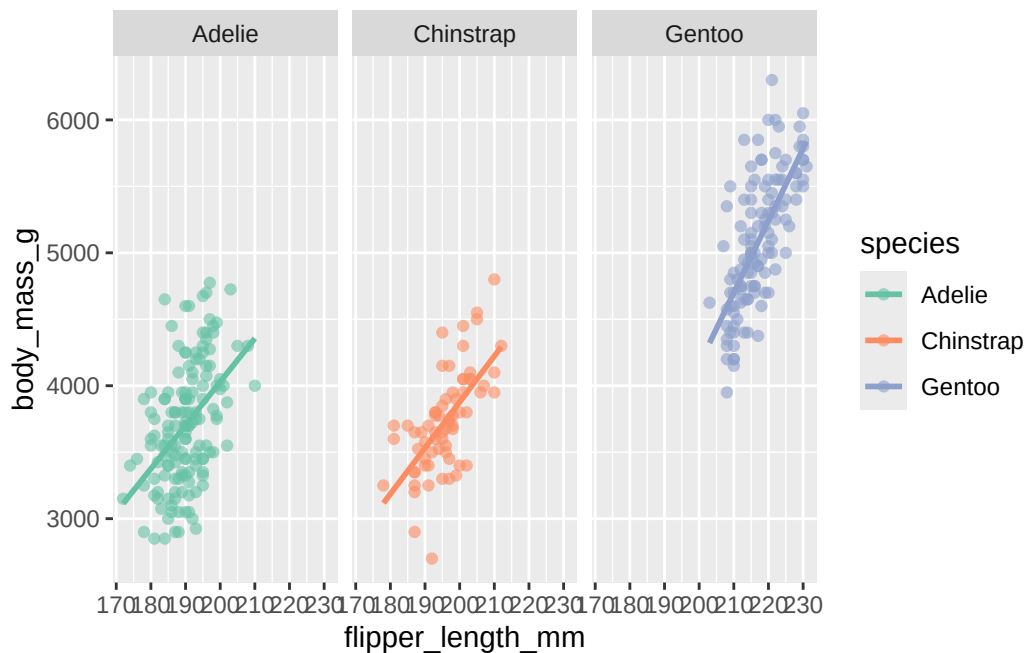


Figure 15.11.: Faceting by species puts each species in its own panel for side-by-side comparison.

Figure 15.11 makes the per-species spread easy to compare: Gentoo not only sit higher and to the right, they also span a wider range of body mass than the other two.

15.3.7. Step 7: labels and theme

Finally, the **theme** controls non-data styling, and `labs()` supplies the human context from Section 15.2. Putting it together:

```
ggplot(data = penguins,
       mapping = aes(x = flipper_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_color_brewer(palette = "Set2") +
  facet_wrap(~ species) +
  labs(title = "Body mass increases with flipper length in all three species",
       x = "Flipper length (mm)",
       y = "Body mass (g)",
       color = "Species") +
  theme_minimal()
```

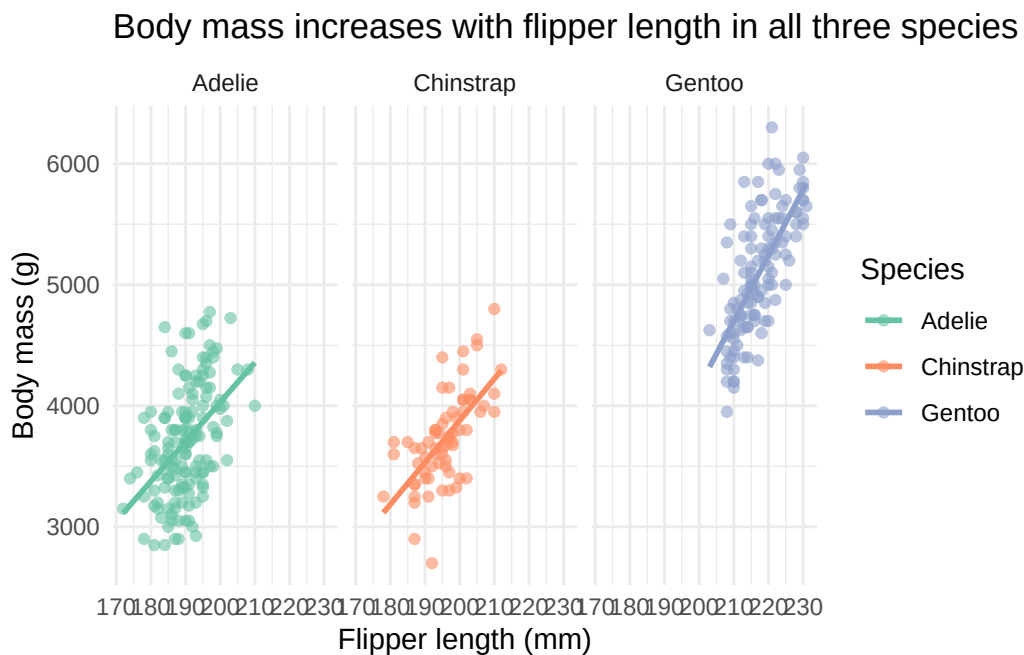


Figure 15.12.: The finished plot: data, aesthetics, two geoms, a scale, facets, labels, and a clean theme.

Figure 15.12 is publication-grade, and you built it one layer at a time. Read the call top

to bottom and you can *name* every part: data, the `aes()` mapping, two geoms, a color scale, facets, labels, and a minimal theme. That is the whole grammar of graphics — every `ggplot2` plot you ever write is some combination of these same pieces.

💡 Combining whole plots: `patchwork`

Faceting splits *one* plot by a variable. To place *different* plots side by side, the `patchwork` package lets you add them with `+` or stack them with `/`, e.g. `plot_a + plot_b`. It's the easiest way to assemble multi-panel figures.

15.4. Specialized plots for biology

`ggplot2` covers most everyday needs, but a few questions call for purpose-built plots. Here are two you'll meet often in genomics.

15.4.1. Sets and intersections: UpSet plots

Suppose you have several gene lists — genes up in treatment A, up in treatment B, bound by a transcription factor — and you want to know which genes are shared. The classic tool is a Venn diagram, but Venn diagrams become unreadable past three sets. An **UpSet plot** scales to many sets: a bar chart shows the size of each intersection, and a dot matrix beneath it shows *which* sets each bar combines.

`UpSetR` takes a data frame of 0s and 1s — one row per element, one column per set, a 1 marking membership. We'll use the `movies` dataset bundled with the package (already in that binary format) as a stand-in for “genes × gene-lists”:

```
library(UpSetR)

movies <- read.csv(system.file("extdata", "movies.csv", package = "UpSetR"),
                  header = TRUE, sep = ";")

upset(movies,
      nsets = 5,
      order.by = "freq",
      mainbar.y.label = "Intersection size",
      sets.x.label = "Total movies in genre")
```

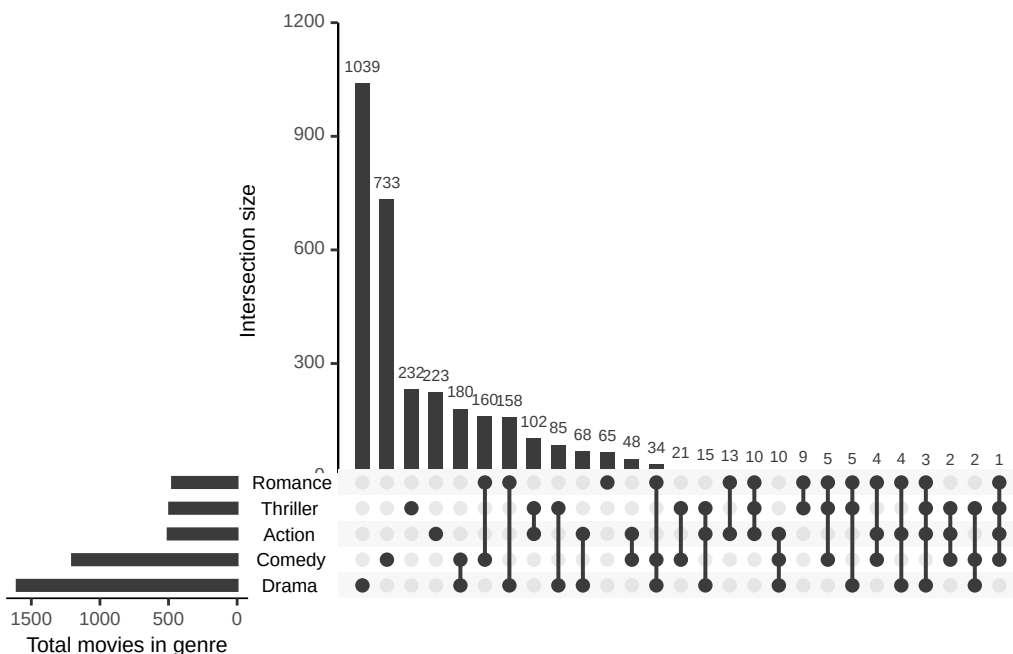


Figure 15.13.: An UpSet plot of movie genres. Each bar is the size of one intersection; the dots below say which genres it combines. Read it as a scalable Venn diagram.

Figure 15.13 reads left to right by decreasing intersection size. The tallest bars with a single filled dot are the “only this set” groups; bars over two or more filled dots are the overlaps. You can see at a glance, for example, how many movies are *both* Drama and Comedy versus Drama alone — a comparison that a five-circle Venn diagram simply cannot show clearly. Swap the movie genres for your gene lists and the reading is identical.

15.4.2. Heatmaps for data matrices

A **heatmap** turns a matrix of numbers into a grid of colored tiles, so the eye can pick out blocks and patterns that a table hides. In genomics, the rows are typically genes, the columns are samples, and the color is expression level. Heatmaps also usually *cluster* similar rows and columns together, so co-behaving genes and similar samples sit next to each other.

i What is a matrix?

A matrix is a two-dimensional grid of values addressed by row and column, and — unlike a data frame — every entry must be the *same* type (here, all numbers). Heatmap functions expect a numeric matrix, not a data frame.

Let’s make a runnable example with **pheatmap** (“pretty heatmap”). We’ll use the **mtcars** data as a numeric matrix — treat the cars as “samples” and the measurements as “features,” exactly as you would genes and samples. The catch: the columns are on wildly different scales (horsepower in the hundreds, number of cylinders in single digits), so we **scale each column** to comparable units first. Scaling is routine for expression data too.

```
library(pheatmap)

# build a numeric matrix and scale each column to mean 0, sd 1
car_matrix <- scale(as.matrix(mtcars))

pheatmap(car_matrix,
  main = "Scaled car measurements",
  color = colorRampPalette(rev(brewer.pal(7, "RdBu")))(100))
```

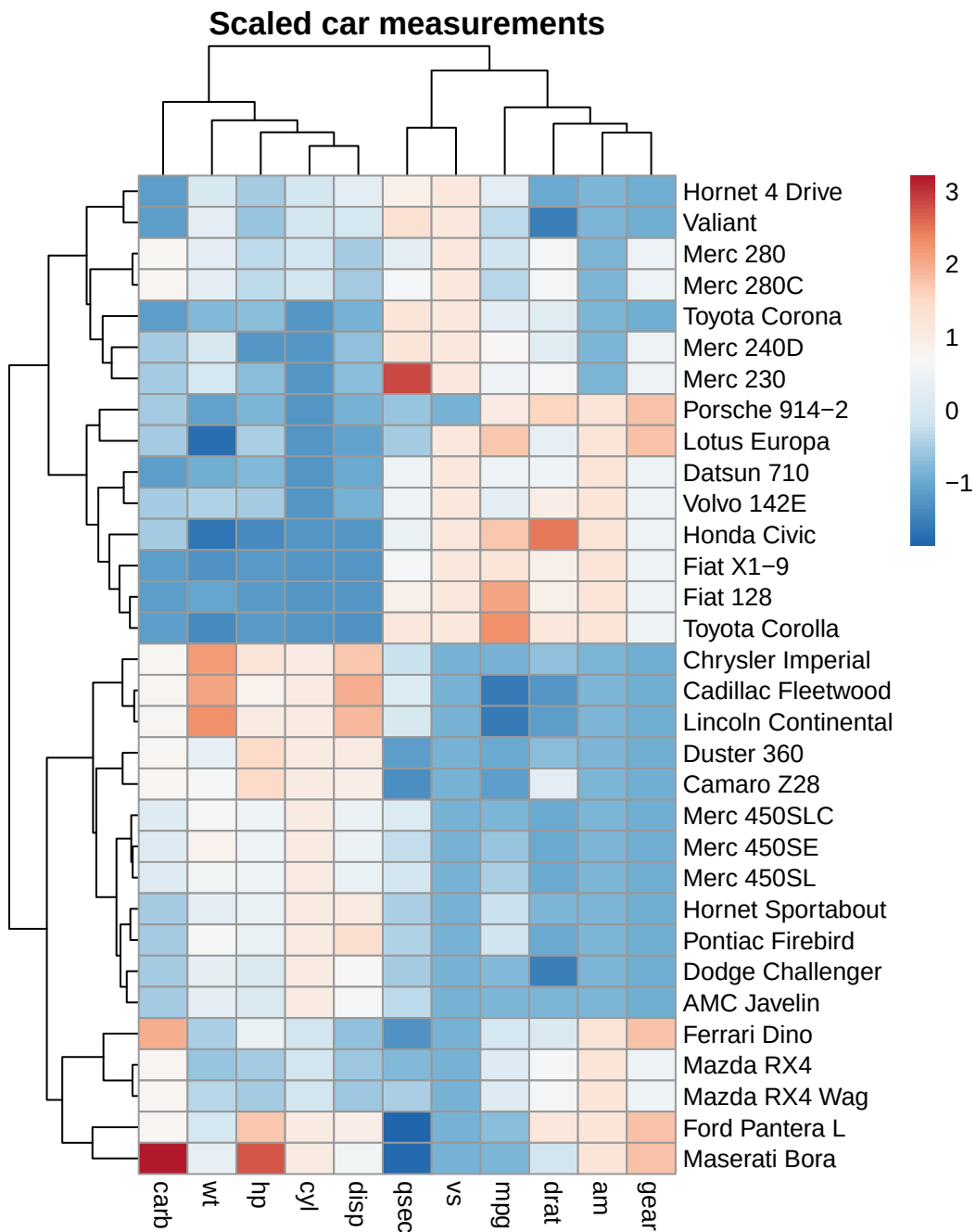


Figure 15.14.: A clustered heatmap of scaled car measurements. Rows (cars) and columns (features) are reordered so similar profiles sit together; the color shows how far above (red) or below (blue) average each value is.

Figure 15.14 clusters the rows and columns automatically. The tree diagrams (**dendrograms**) on the top and left show how cars and features were grouped. You can read off blocks: the fuel-efficiency feature (`mpg`) sits opposite the power/weight features, and the cars split into a heavy-and-powerful block versus a light-and-efficient one. Read genes for cars and samples for features, and this is exactly how an expression heatmap is interpreted.

For richer needs — multiple annotation bars, split heatmaps, precise control over legends — Bioconductor’s `ComplexHeatmap` is the standard. It uses the same mental model (a matrix plus optional row/column annotations) with far more knobs; its [reference book](#) is the place to go when `pheatmap` runs out of room.

15.4.3. Visualizing the genome (further exploration)

The plots above are general-purpose. Some genomic questions need a view organized along **genomic coordinates** — gene models, read coverage, and peaks lined up against a chromosome. This is a heavier topic that leans on Bioconductor, so we flag it as further exploration rather than cover it in depth here:

- The [Gviz package](#) builds publication-quality, multi-track genome browser views — stacking gene annotations, alignments, and quantitative data over a coordinate axis.
- The [GenomicDistributions package](#) summarizes *where* a set of genomic regions falls — for example, how ChIP-seq or ATAC-seq peaks distribute relative to transcription start sites.

When you reach a project that needs these, start from their vignettes; the grammar-of-graphics intuition from Section 15.3 still helps you read the layered tracks they produce.

15.5. Exercises

Use the cleaned `penguins` data frame from Section 15.3 for the first three exercises (`library(palmerpenguins)`
`library(ggplot2); penguins <- na.omit(penguins)`).

1. **A different geom.** Make a scatterplot of bill length (`bill_length_mm`, x) versus bill depth (`bill_depth_mm`, y), colored by species. What do you notice?

15. A self-guided tour of data visualization in R

i Solution

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +  
  geom_point() +  
  theme_minimal()
```

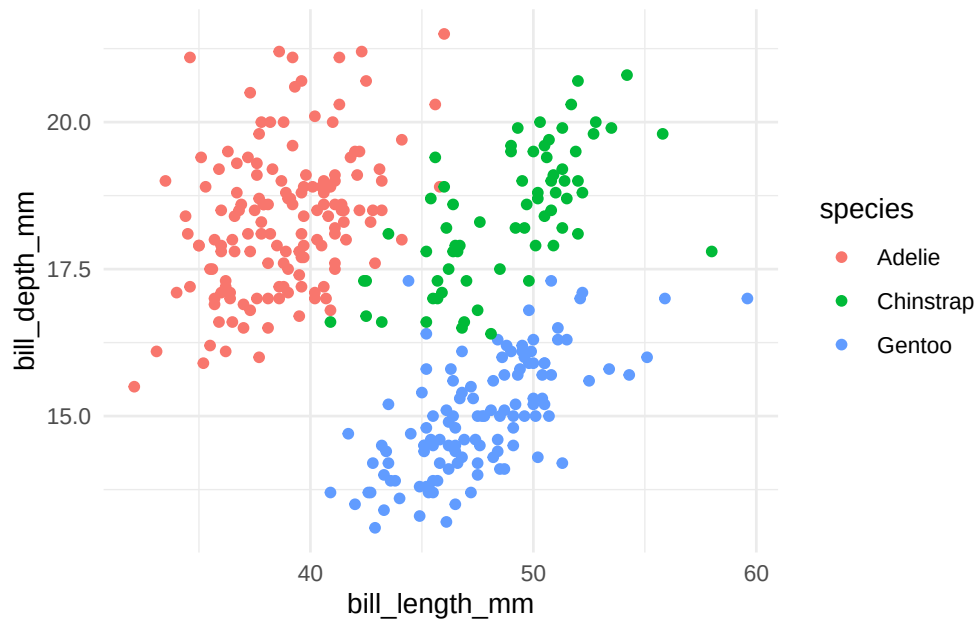


Figure 15.15.: Bill length vs. depth, colored by species.

Within each species, deeper bills go with longer bills, but the three species sit in clearly separated regions — bill shape is a good species fingerprint.

2. **A distribution, not a relationship.** Use `geom_boxplot()` to compare body mass across the three species (species on x, `body_mass_g` on y). Which species is heaviest?

i Solution

```
ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +  
  geom_boxplot() +  
  theme_minimal()
```

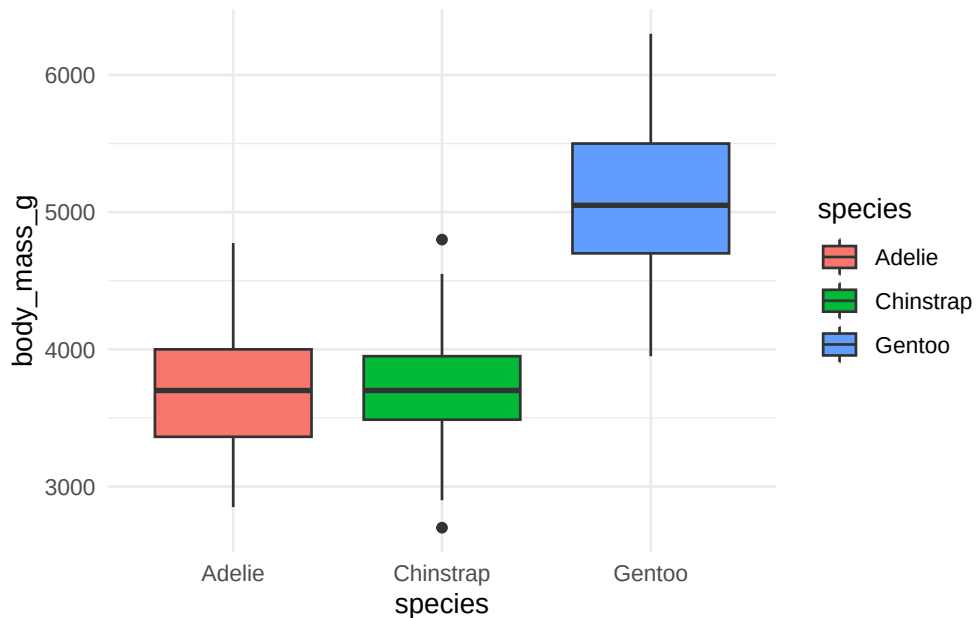


Figure 15.16.: Body-mass distribution by species.

Gentoo penguins are clearly the heaviest; Adelie and Chinstrap have similar, lower body masses. A boxplot summarizes a *distribution* (median, spread, outliers) rather than a point-to-point relationship.

3. **Add the labels.** Take your boxplot from Exercise 2 and give it a title and properly-labeled axes with units, following Section 15.2.

i Solution

```
ggplot(penguins, aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot() +
  labs(title = "Gentoo penguins are the heaviest of the three species",
       x = "Species",
       y = "Body mass (g)",
       fill = "Species") +
  theme_minimal()
```

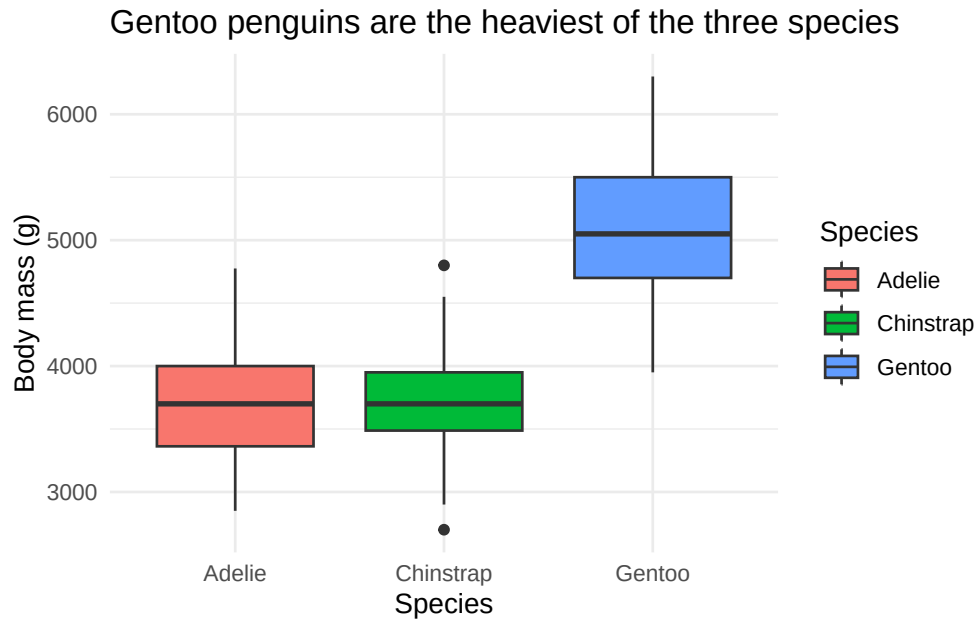


Figure 15.17.: The boxplot, now properly labeled.

The title states the *finding*, not just the variables — a good title does the reader's work for them.

4. **Pick the right palette family.** You're plotting log-fold-changes that range from strongly negative to strongly positive, with zero as the meaningful midpoint. Which `RColorBrewer` palette *family* should you use, and why?

i Solution

A **diverging** palette (such as "RdBu"). Diverging palettes give two contrasting colors that meet at a neutral midpoint, so zero reads as "no change" and the two directions of change are immediately distinguishable. A sequential palette would hide the sign of the change.

15.6. Summary

You now have both the principles and the toolbox for visualizing data in R:

15. A self-guided tour of data visualization in R

- You can make any plot more readable by **maximizing the data-ink ratio** (deleting decoration) and **labeling everything** with a clear title and units.
- You can read and write a `ggplot2` call as a stack of **layers** — data, aesthetics, geoms, scales, facets, themes — and you built one up one layer at a time on real penguin data.
- You can map extra variables onto **color** and other aesthetics, and you know to put data-driven settings *inside* `aes()` and constants *outside* it.
- You can choose a **colorblind-safe palette** from `RColorBrewer`.
- You can reach for a specialized plot when you need one: an **UpSet plot** for intersections of many sets, and a clustered **heatmap** for an expression-style matrix.

The grammar of graphics is the throughline: once a plot is just data plus a stack of layers, the way to build *any* new plot is to ask which layer you need to add next. For more depth on a single dataset, see [Exploring data with ggplot2](#); for inspiration, browse the [R Graph Gallery](#) and [From Data to Viz](#).

Part IV.
statistics

16. Working with distribution functions

Heights, blood pressures, measurement errors, log-transformed gene-expression values — measure enough of almost anything in biology and the values tend to pile up into the same bell-shaped curve: the **normal distribution**. Once your data follow that curve, you'll want to ask questions of it. *What fraction of samples fall below some cutoff? Which value marks the top 5%? Give me twenty random draws to simulate an experiment.* R answers each of those with a different function.

The catch is remembering which function does which job. This chapter is a **visual guide**: for every function we pair the R call with a picture of exactly what it measures on the bell curve, because the difference between a *probability*, a *density*, and a *quantile* is far easier to see than to memorize. It's genuinely hard to reason about these functions without a shape to point at — so a shape is what we'll build for each one.

16.1. What you'll learn

- Use `pnorm()` to turn a value into a probability (the area under the curve to its left).
- Use `dnorm()` to get the height (density) of the curve at a value.
- Use `qnorm()` to go the other direction — from a probability back to a value.
- Use `rnorm()` to draw random samples from a normal distribution.
- Combine these functions to answer real questions about normally distributed data, such as IQ scores.

💡 One naming rule for *every* distribution

R builds these names from a one-letter prefix plus the distribution: **d** for density, **p** for probability, **q** for quantile, **r** for random. Learn it once and it transfers everywhere — `dbinom/pbinom/qbinom/rbinom` for the binomial, `dt/pt/qt/rt` for the t-distribution we meet in the next chapter, `dpois/ppois/...` for the Poisson. Here we use the `...norm` family for the normal.

The four functions, at a glance:

16. Working with distribution functions

Table 16.1.: Functions for the normal distribution

Function	Input	Output
<code>pnorm</code>	a value <code>q</code>	$P(X < q)$: area under the curve to the left of <code>q</code>
<code>dnorm</code>	a value <code>x</code>	the height of the density curve at <code>x</code>
<code>qnorm</code>	a probability from 0 to 1	the value <code>x</code> where $P(X < x)$ equals that probability
<code>rnorm</code>	a count <code>n</code>	<code>n</code> random samples from the distribution

In one sentence each: `pnorm` gives the area to the left of a point, `dnorm` gives the height of the curve at a point, `qnorm` is the inverse of `pnorm` (probability back to value), and `rnorm` generates random samples. The rest of the chapter unpacks them one at a time.

16.2. `pnorm`

`pnorm` answers the question “what fraction of the distribution lies to the **left** of this value?” If you don’t supply a mean and standard deviation, R assumes the *standard normal* (mean 0, standard deviation 1). The help page gives this template:

```
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
```

You hand it `q`, a value on the x-axis, and it returns the area under the curve to the left of that point. The mean sits dead centre and splits the curve in half, so:

```
pnorm(0)      # half the distribution is below the mean
```

```
[1] 0.5
```

```
pnorm(1)      # ~84% is below one standard deviation above the mean
```

```
[1] 0.8413447
```

16. Working with distribution functions

```
pnorm(-1) # ~16% is below one standard deviation below it
```

```
[1] 0.1586553
```

The figures below show each of those as a shaded area.

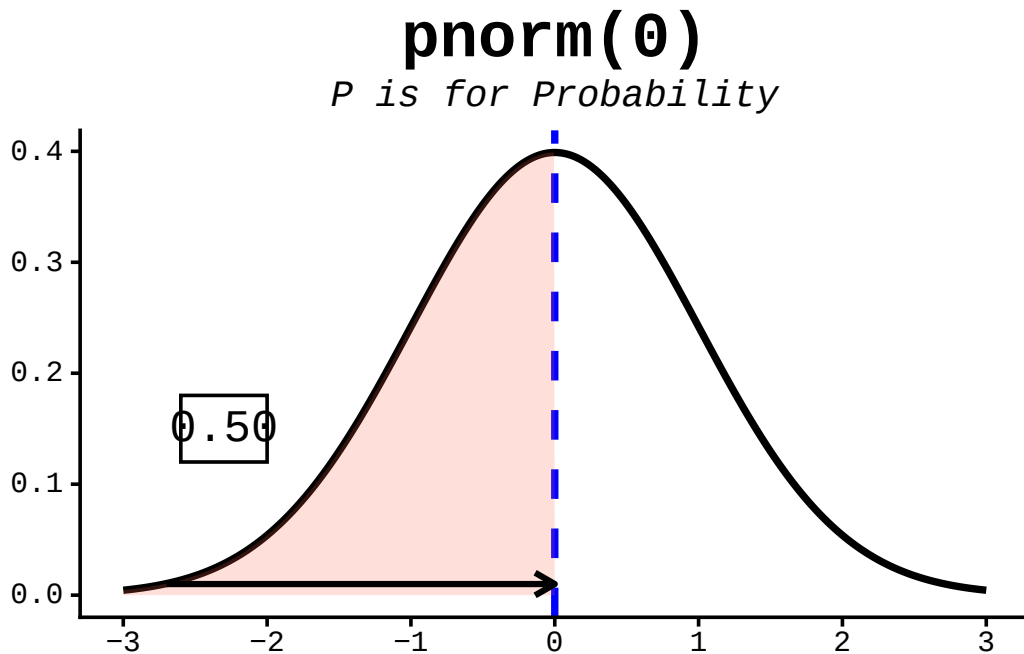


Figure 16.1.: The `pnorm` function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.

The option `lower.tail = TRUE` tells R to use the area to the *left* of the given point — the default. To get the area to the **right** instead, either switch to `lower.tail = FALSE` or subtract from 1:

```
pnorm(1, lower.tail = FALSE) # area to the RIGHT of 1
```

```
[1] 0.1586553
```

```
1 - pnorm(1) # exactly the same thing
```

```
[1] 0.1586553
```

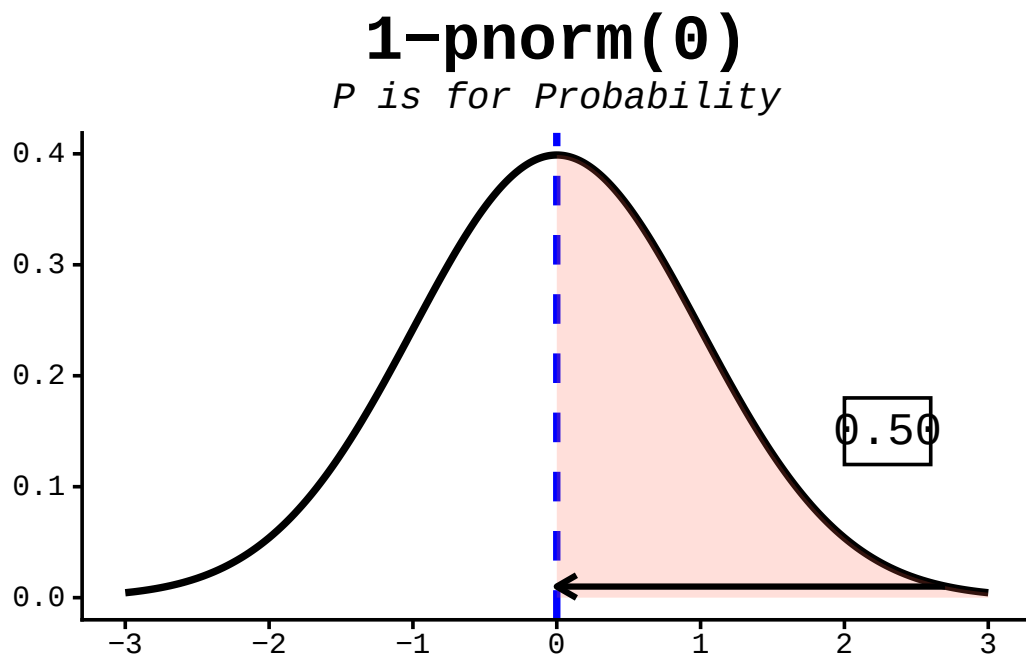



Figure 16.2.: The pnorm function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.

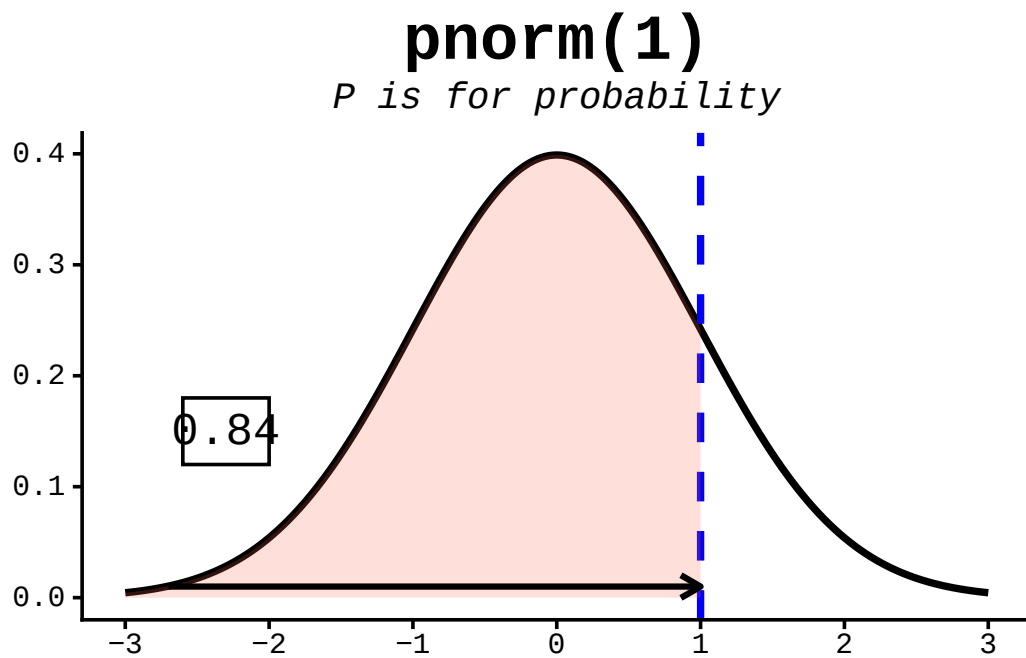


Figure 16.3.: The pnorm function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.

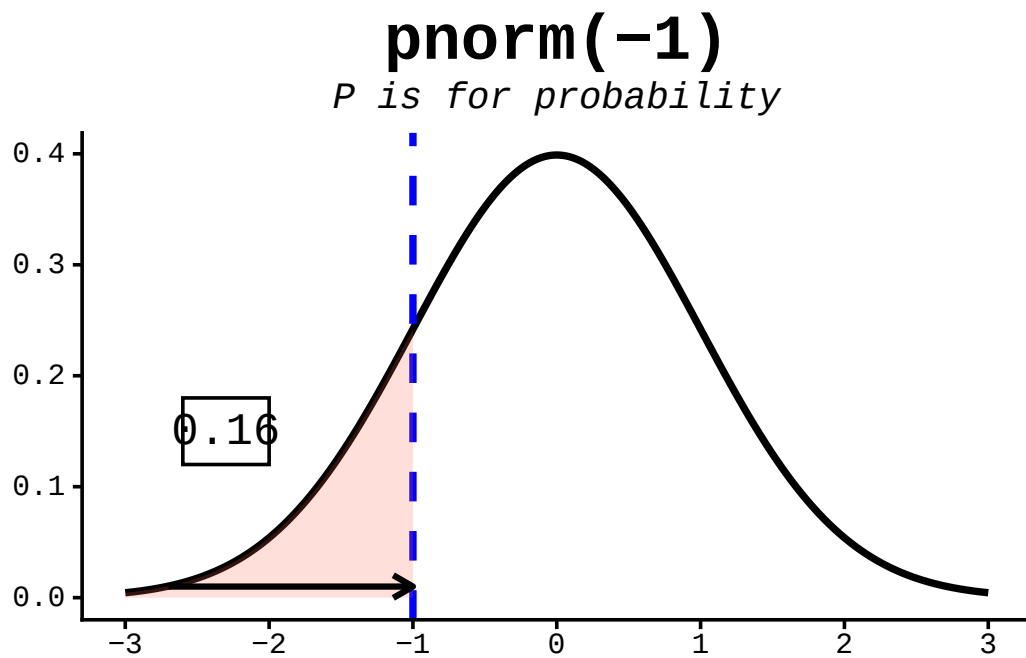


Figure 16.4.: The pnorm function takes a quantile (value on the x-axis) and returns the area under the curve to the left of that value.

16.3. dnorm

`dnorm` gives the **height** of the bell curve at a value — the probability *density*. The curve is tallest at the mean and falls off symmetrically on either side. Unlike `pnorm`, the result is not a probability you can read as a percentage; it's the height of the curve, used mainly for *drawing* the distribution or in likelihood calculations.

```
dnorm(0)      # the peak of the standard normal
```

```
[1] 0.3989423
```

```
dnorm(1)      # lower, one SD out...
```

```
[1] 0.2419707
```

```
dnorm(-1)     # ...and the same height on the other side, by symmetry
```

```
[1] 0.2419707
```

The figures below mark each of these heights on the curve.

16.4. qnorm

`qnorm` runs `pnorm` in reverse: hand it a probability and it returns the value with that much area to its left. Since `pnorm(1)` is about 0.84, `qnorm(0.84)` is about 1 — the two functions undo each other. This is how you find **percentiles**.

```
qnorm(0.5)     # the median: half the area lies to its left
```

```
[1] 0
```

```
qnorm(0.25)    # the first quartile
```

```
[1] -0.6744898
```

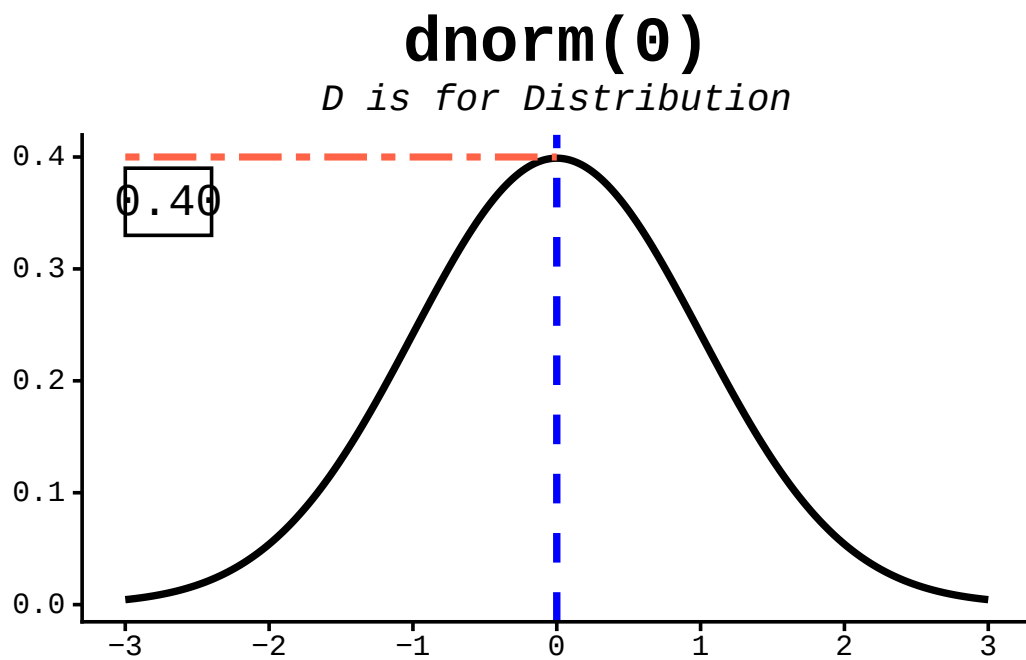


Figure 16.5.: The `dnorm` function returns the height of the normal distribution at a given point.

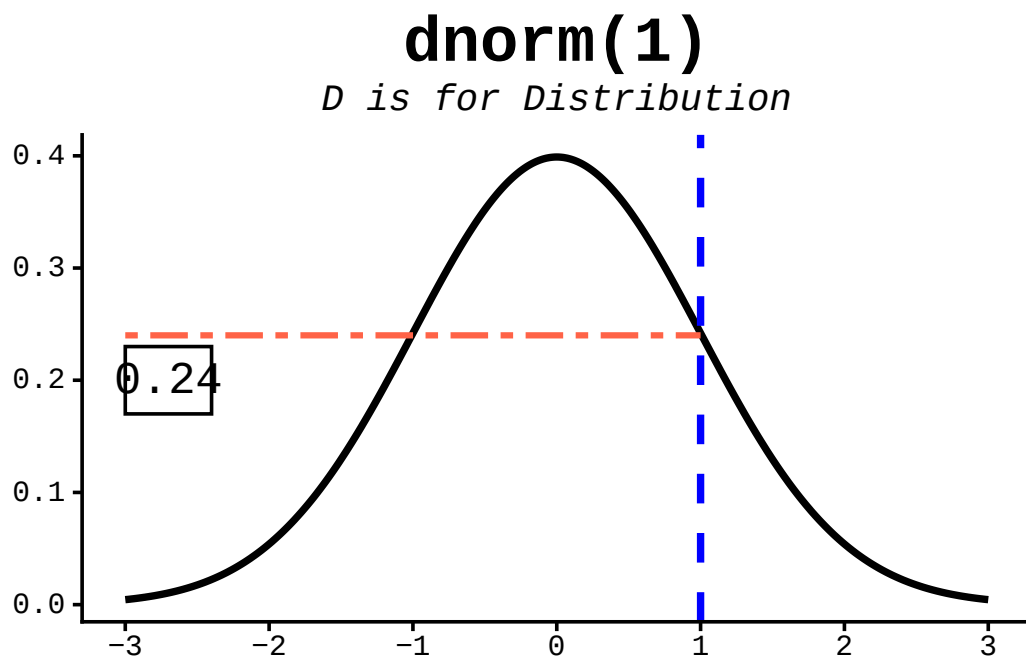


Figure 16.6.: The `dnorm` function returns the height of the normal distribution at a given point.

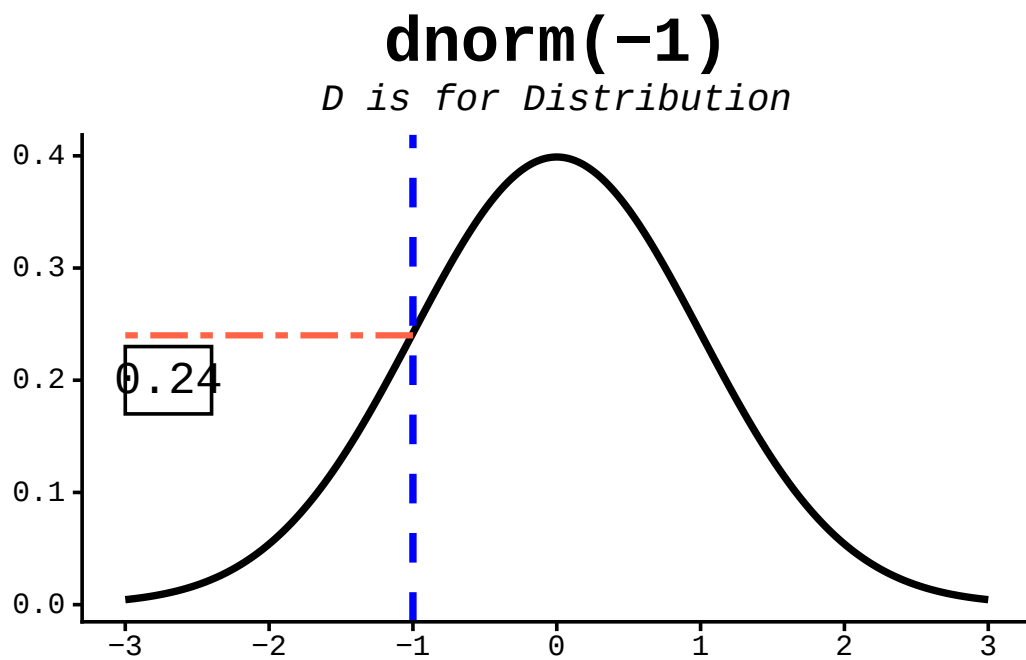


Figure 16.7.: The `dnorm` function returns the height of the normal distribution at a given point.

16. Working with distribution functions

```
qnorm(0.975) # the value cutting off the top 2.5% - the famous 1.96
```

```
[1] 1.959964
```

The figures below show several quantiles as shaded areas.

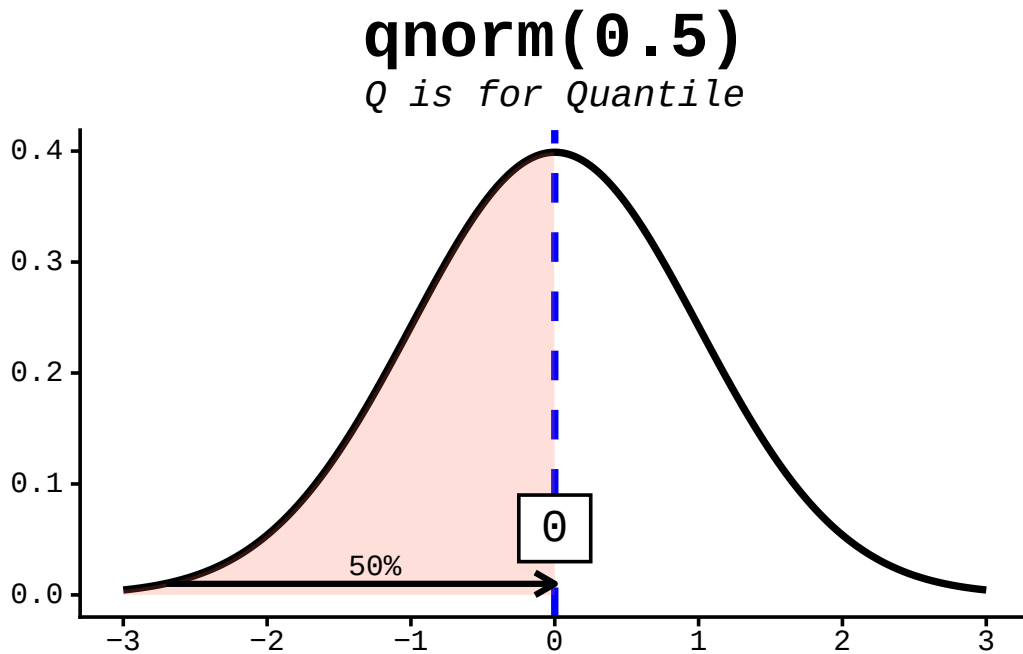


Figure 16.8.: The `qnorm` function is the inverse of the `pnorm` function in that it takes a probability and gives the quantile.

16.5. `rnorm`

`rnorm` draws random samples from the distribution — the workhorse for *simulating* data (we leaned on it heavily in the t-distribution chapter). Give it a count `n` and it hands back that many random values:

```
rnorm(5)
```

```
[1] -0.46271341  0.13850783 -0.07110809 -1.39765804  0.09130481
```

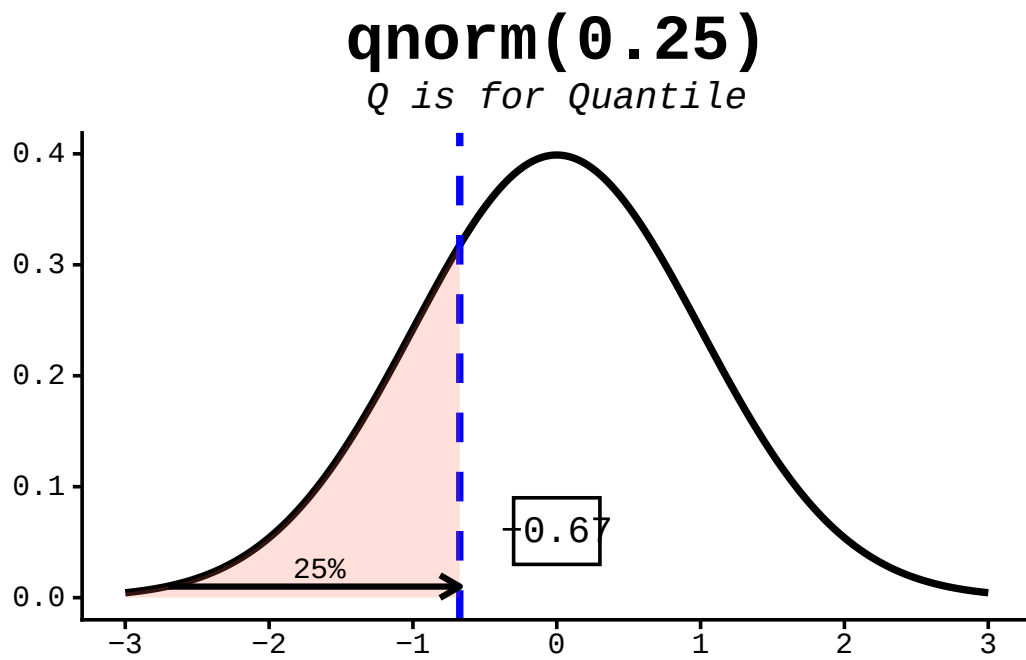



Figure 16.9.: The qnorm function is the inverse of the pnorm function in that it takes a probability and gives the quantile.

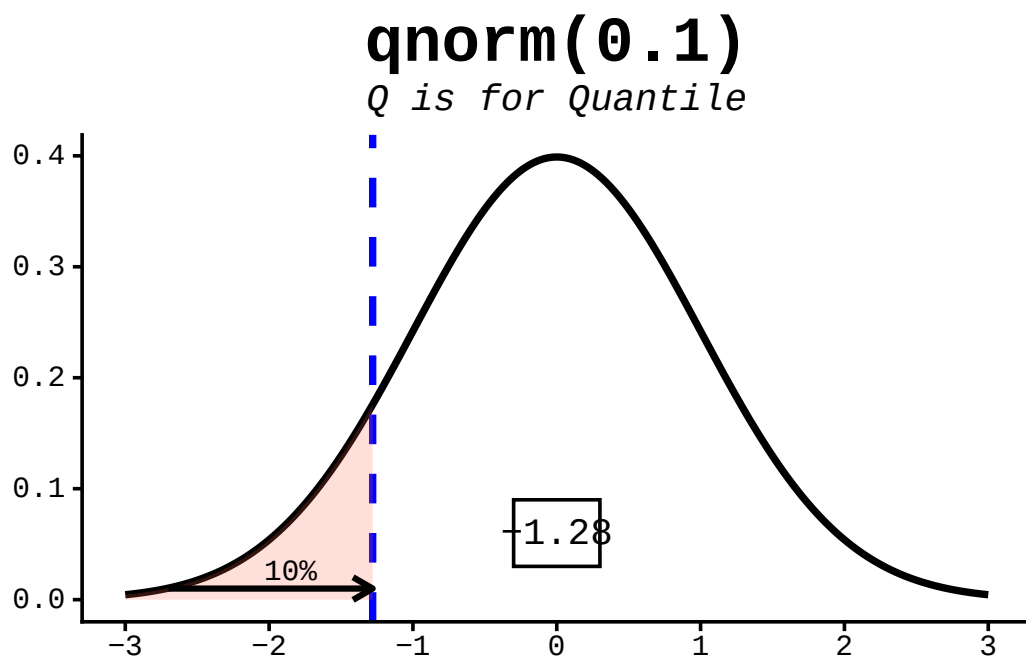


Figure 16.10.: The qnorm function is the inverse of the pnorm function in that it takes a probability and gives the quantile.

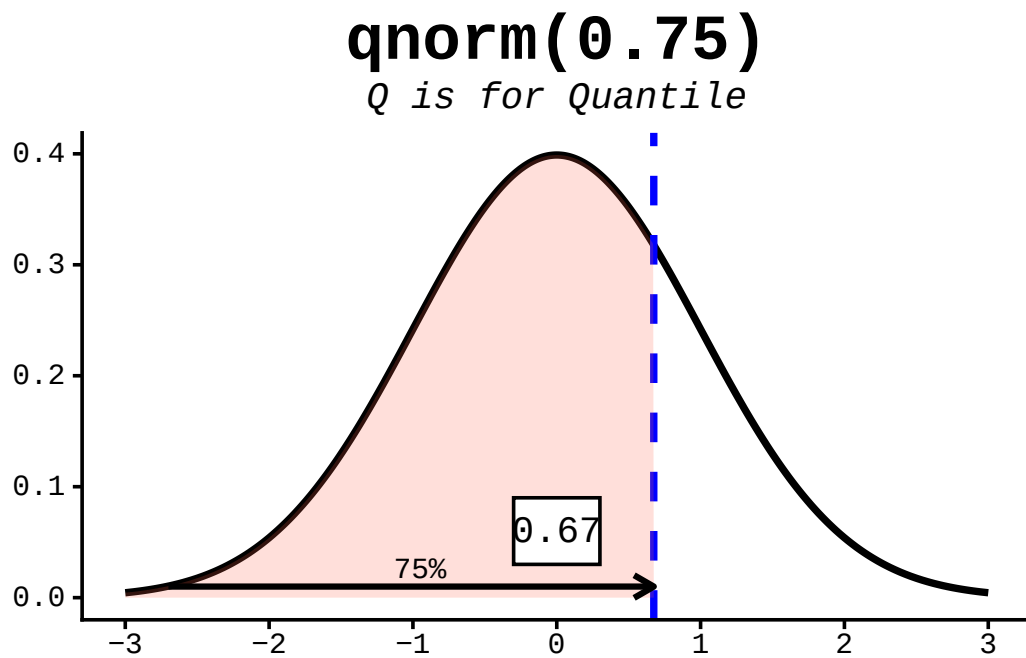


Figure 16.11.: The qnorm function is the inverse of the pnorm function in that it takes a probability and gives the quantile.

16. Working with distribution functions

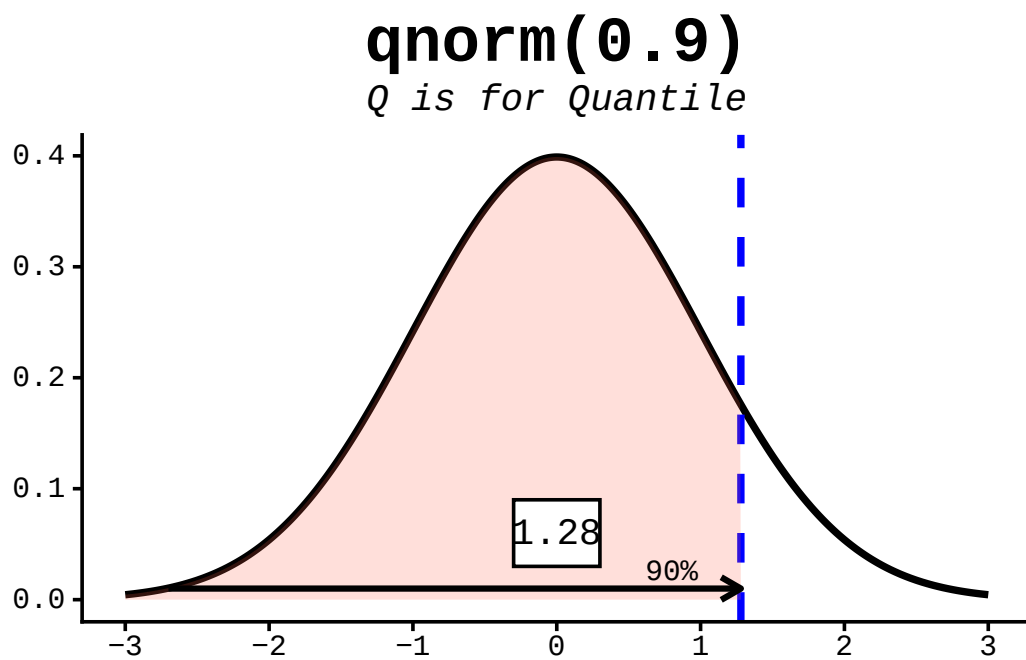


Figure 16.12.: The qnorm function is the inverse of the pnorm function in that it takes a probability and gives the quantile.

16. Working with distribution functions

Run that again and you'll get five *different* numbers. To make a simulation reproducible — so you and a reader see the same “random” draws — set a seed first:

```
set.seed(42)
rnorm(5)
```

```
[1]  1.3709584 -0.5646982  0.3631284  0.6328626  0.4042683
```

And just like the other functions, `mean` and `sd` shift and scale the draws:

```
rnorm(5, mean = 100, sd = 15) # e.g. five simulated IQ scores
```

```
[1]  98.40813 122.67283  98.58011 130.27636  99.05929
```

```
print(r1)
```

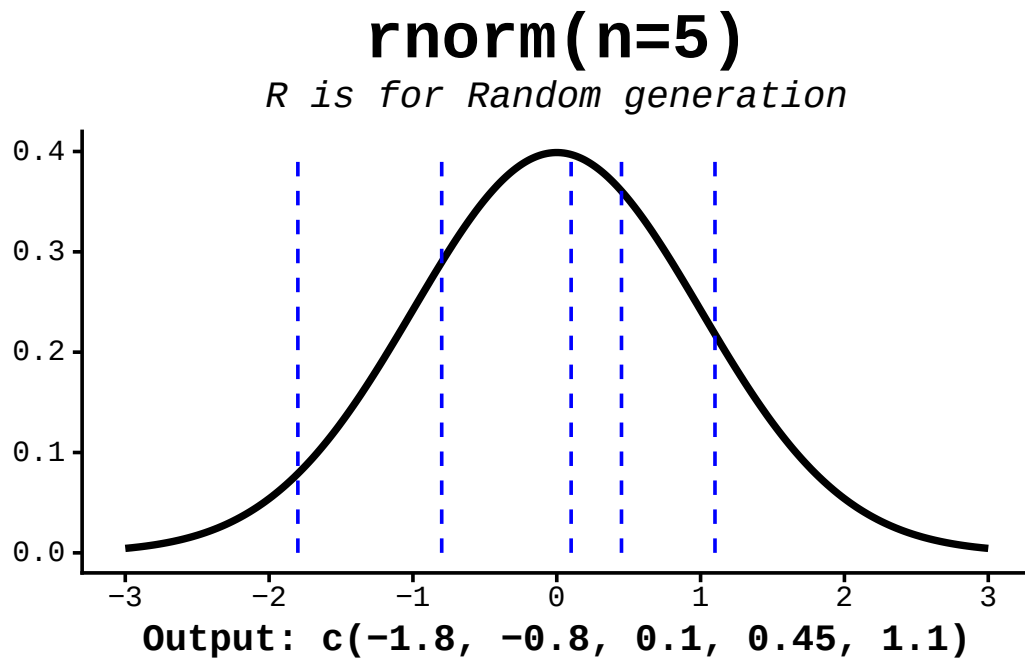


Figure 16.13.: The `rnorm` function takes a number of samples and returns a vector of random numbers from the normal distribution (with `mean=0`, `sd=1` as defaults)

16.6. Worked example: IQ scores

The normal distribution (also called the Gaussian distribution) is defined by two numbers: the **mean** (μ), which sets the centre, and the **standard deviation** (σ), which sets the spread. IQ scores are deliberately built to be normal with a **mean of 100** and a **standard deviation of 15**, which makes them a perfect playground for the four functions.

A handy rule of thumb — the “68–95–99.7 rule” — says about 68% of values fall within one standard deviation of the mean (here, 85 to 115) and about 95% within two (70 to 130). Let’s check the 68% with `pnorm`. The area *between* 85 and 115 is the area below 115 minus the area below 85:

```
pnorm(115, mean = 100, sd = 15) - pnorm(85, mean = 100, sd = 15)
```

```
[1] 0.6826895
```

About 0.68 — exactly what the rule promises. Notice the pattern: a “between” question becomes one `pnorm` minus another.

16.7. Exercises

Use the IQ distribution (mean 100, sd 15) for each. Each solution names *which* of the four functions fits the question — that choice is the real skill.

1. **Top 10%.** Which IQ score marks the 90th percentile — the cutoff above which only the top 10% fall?

i Solution

```
qnorm(0.9, mean = 100, sd = 15)
```

```
[1] 119.2233
```

We want a *value* from a *probability*, so this is a `qnorm` job. About 119.

2. **Above 130.** What fraction of people have an IQ above 130 (two standard deviations up)?

16. Working with distribution functions

i Solution

```
1 - pnorm(130, mean = 100, sd = 15)
```

```
[1] 0.02275013
```

```
# or, equivalently:
```

```
pnorm(130, mean = 100, sd = 15, lower.tail = FALSE)
```

```
[1] 0.02275013
```

About 2.3% — consistent with the 95% rule leaving roughly 2.5% in each tail.

3. **Below 70.** What fraction have an IQ below 70?

i Solution

```
pnorm(70, mean = 100, sd = 15)
```

```
[1] 0.02275013
```

Also about 2.3%. The curve is symmetric, so the lower tail past 70 mirrors the upper tail past 130.

4. **Simulate a classroom.** Draw 30 random IQ scores and count how many land above 115. (Set a seed so your answer is reproducible.)

i Solution

```
set.seed(1)
scores <- rnorm(30, mean = 100, sd = 15)
sum(scores > 115)
```

```
[1] 3
```

Theory predicts about 16% above 115 — roughly 5 of 30 — but any single small sample wobbles around that. Re-run with a different seed to feel the sampling variability for yourself.

16.8. Summary

You now have the four-function toolkit for the normal distribution — and, thanks to R's naming rule, the blueprint for every other distribution too:

16. Working with distribution functions

- `pnorm(x)` — area to the **left** of `x` (a probability). Use `lower.tail = FALSE` or `1 - pnorm(x)` for the right tail, and one `pnorm` minus another for a range.
- `dnorm(x)` — the **height** of the curve at `x` (a density).
- `qnorm(p)` — the **value** at probability `p` (the inverse of `pnorm`); your tool for percentiles.
- `rnorm(n)` — `n` random **draws**; pair it with `set.seed()` for reproducible simulations.

Keep the pictures in mind: `pnorm` is an area, `dnorm` is a height, `qnorm` reads the area axis backwards to a value, and `rnorm` scatters points along the curve.

17. The t-statistic and t-distribution

Suppose you measure a gene's expression in five treated samples and five controls, and the treated mean comes out higher. Is that a real effect, or just the kind of gap you'd expect by chance from five noisy measurements? With only a handful of samples you can't lean on the normal distribution as-is, because you had to *estimate* the variability from the same tiny sample you're testing. The **t-distribution** is the honest accounting for that extra uncertainty — and in this chapter we'll *build* it by simulation so you can see exactly where it comes from, then use R's `t.test()` to put it to work.

The distribution that the t-statistic follows was described in a famous 1908 paper (Student 1908) by “Student” — a pseudonym for [William Sealy Gosset](#), who worked at the Guinness brewery and published anonymously because his employer treated statistical methods as a trade secret.

17.1. What you'll learn

- Explain why estimating the standard deviation from a small sample forces us off the normal distribution and onto the t-distribution.
- Read a Z-score and a t-score and say what each one measures.
- Use simulation to show that t-based p-values are calibrated (uniform under the null hypothesis) when the normal-based ones are not.
- Run one-sample, two-sample, and formula-based tests with `t.test()`.
- Estimate statistical power with `power.t.test()` and by simulation.

17.2. Background

A **t-test** is a [statistical hypothesis test](#) you reach for when you want to compare means and you only have a sample to work from. It comes in three everyday flavors:

- **One-sample** — compare a sample mean to a single hypothesized or target value.
- **Two-sample** — compare the means of two groups.

17. The *t*-statistic and *t*-distribution

- **Paired** — compare two measurements on the same units (e.g., before-and-after on the same patients).

A *t*-test combines three ingredients — the **t-statistic**, the **t-distribution**, and the **degrees of freedom** — to decide whether a difference is bigger than chance would comfortably produce. (To compare three or more means at once you’d reach instead for an analysis of variance, or ANOVA.)

We’ll get to `t.test()`, but the point of this chapter is to first see *why* the *t*-distribution exists. That story starts with something you may already have met: the Z-score.

17.3. The Z-score and probability

Before we talk about *t*-scores, let’s review the Z-score, its relationship to the normal distribution, and how it turns into a probability.

The Z-score is defined as:

$$Z = \frac{x - \mu}{\sigma} \quad (17.1)$$

where μ is the population mean that x is drawn from and σ is the population standard deviation — **taken as known**, not estimated from the data. In words: a Z-score says how many standard deviations a value sits from the mean.

The probability of observing a Z-score of z or smaller can be calculated with `pnorm(z, mu, sigma)`.

For example, suppose our “population” is known and truly has a mean of 0 and a standard deviation of 1 (the *standard normal*). For any observation drawn from that population, we can assign a probability of seeing a value that extreme by random chance — *under the assumption that the null hypothesis is **TRUE***.

```
zscore <- seq(-5, 5, 1)
```

For each value of `zscore`, let’s calculate the probability and collect the results in a `data.frame`. We’ll name it `pdat` (for “probability data”) so it doesn’t collide with the `df` we use later for degrees of freedom.

17. The *t*-statistic and *t*-distribution

```
pdat <- data.frame(  
  zscore = zscore,  
  pval   = pnorm(zscore, 0, 1)  
)  
pdat
```

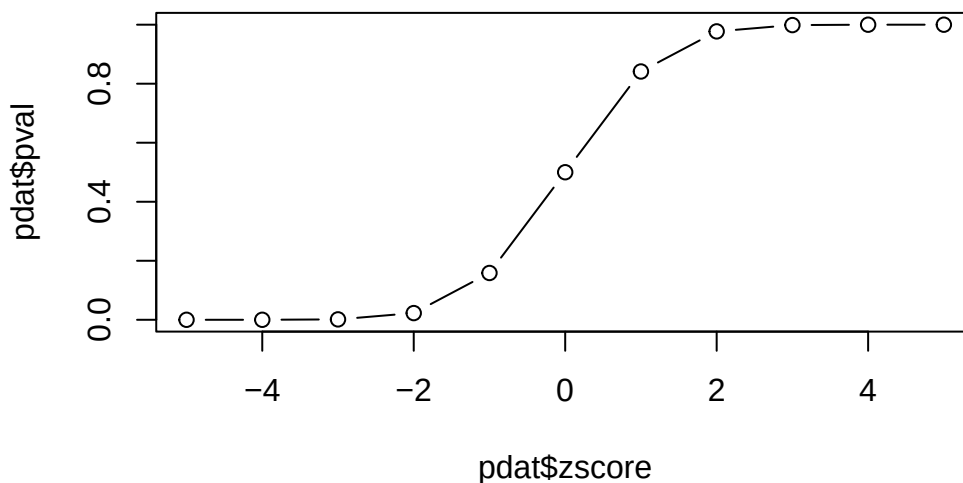
	zscore	pval
1	-5	2.866516e-07
2	-4	3.167124e-05
3	-3	1.349898e-03
4	-2	2.275013e-02
5	-1	1.586553e-01
6	0	5.000000e-01
7	1	8.413447e-01
8	2	9.772499e-01
9	3	9.986501e-01
10	4	9.999683e-01
11	5	9.999997e-01

Why is the p-value for something 5 standard deviations *above* the mean (`zscore = 5`) nearly 1 here? Because `pnorm` reports the *lower* tail by default — the probability of a value *at or below* *z*. A value 5 standard deviations above the mean has almost the entire distribution below it, so that probability is nearly 1.

Let's plot probability against Z-score:

```
plot(pdat$zscore, pdat$pval, type = 'b')
```

17. The *t*-statistic and *t*-distribution



This curve is the *cumulative distribution function* (CDF). Read it like a lookup table: pick a Z-score on the x-axis and read off the probability of seeing a value that small. Because we built our example to follow the standard normal, this is also the CDF of the standard normal distribution.

💡 What a p-value is (and isn't)

A p-value is the probability of a result *at least this extreme* **if the null hypothesis were true**. It is *not* the probability that the null hypothesis is true, and a small p-value is not a measure of effect size. For an accessible, authoritative take on the common misreadings, see the American Statistical Association's statement on p-values (Wasserstein and Lazar 2016).

17.3.1. Small diversion: a two-sided `pnorm`

`pnorm` returns a *one-sided* probability using the lower tail by default. Often we don't care which direction a value deviates — only that it's far from the mean — so we want a *two-sided* p-value. We can build one in three steps:

1. Take the absolute value of `x`, so direction no longer matters.
2. Call `pnorm` with `lower.tail = FALSE`, so larger `|x|` gives smaller probabilities.

17. The *t*-statistic and *t*-distribution

3. Multiply by 2 to count *both* tails.

```
twosidedpnorm <- function(x, mu = 0, sd = 1) {  
  2 * pnorm(abs(x), mu, sd, lower.tail = FALSE)  
}
```

Now we can ask how surprising it is to land 2 or 3 standard deviations from the mean in either direction:

```
twosidedpnorm(2)
```

```
[1] 0.04550026
```

```
twosidedpnorm(3)
```

```
[1] 0.002699796
```

17.4. The **t**-distribution

Everything above leaned on one big assumption: that we *know* the standard deviation σ . Recall the Z-score:

$$Z = \frac{x - \mu}{\sigma}$$

and the *population* standard deviation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (17.2)$$

When we genuinely know σ , the Z-score is the right tool. But usually we only have a *sample*, not the whole population. Then we swap the Z-score for the **t-score**:

$$t = \frac{x - \bar{x}}{s} \quad (17.3)$$

This looks almost identical to the Z-score, with two changes: we use the *sample* mean $\bar{x}$ in place of μ , and we *estimate* the standard deviation as s from the data:

17. The *t*-statistic and *t*-distribution

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (17.4)$$

In words: the sample standard deviation s is the typical distance of a data point from the sample mean. The only change from the population formula is the divisor — $N-1$ instead of N — which corrects for the fact that we used the data twice: once to estimate the mean, and again to measure the spread around it.

! The whole point of the *t*-distribution

Because we *estimate* the standard deviation from a small sample, our test statistic is more uncertain than a Z-score. The *t*-distribution captures that by having **fatter tails** for small samples — so it takes a more extreme result to reach significance. As the sample grows, the estimate sharpens and the *t*-distribution slides back toward the normal.

We can *see* those fatter tails. Let's plot the *t*-distribution densities for 3, 6, 10, and 20 degrees of freedom (smaller degrees of freedom means a smaller sample) alongside the normal:

```
library(dplyr)
library(ggplot2)
t_values <- seq(-6, 6, 0.01)
tdens <- data.frame(
  value = t_values,
  t_3   = dt(t_values, 3),
  t_6   = dt(t_values, 6),
  t_10  = dt(t_values, 10),
  t_20  = dt(t_values, 20),
  Normal = dnorm(t_values)
) |>
  tidyr::pivot_longer(-value, names_to = "Distribution", values_to = "density")
ggplot(tdens, aes(x = value, y = density, color = Distribution)) +
  geom_line()
```

17. The t -statistic and t -distribution

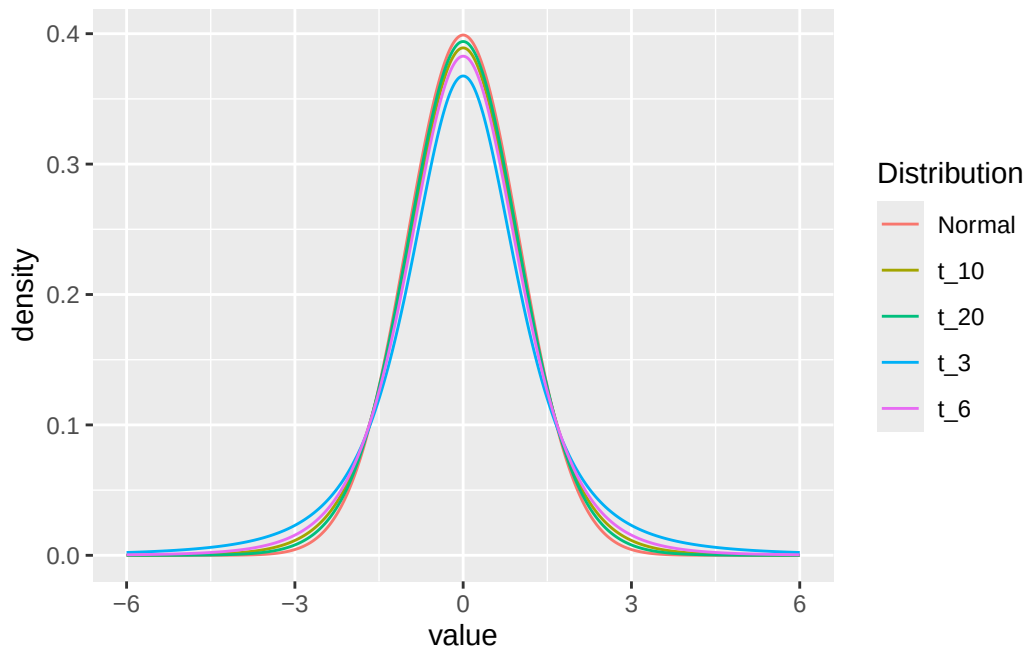


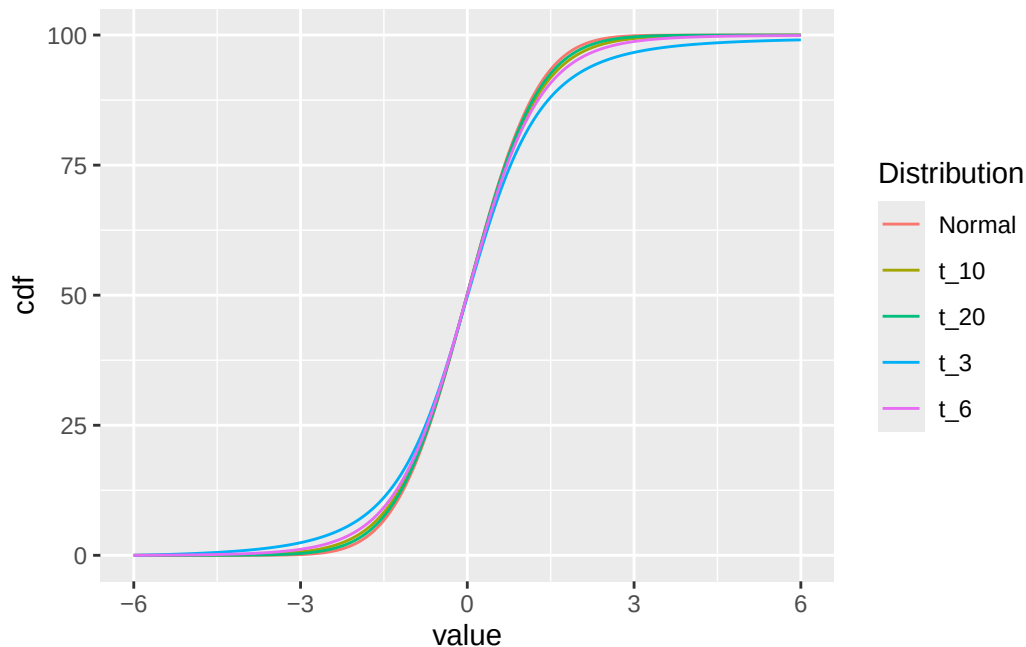
Figure 17.1.: t -distributions for various degrees of freedom. The tails are fatter for smaller degrees of freedom — the visible cost of estimating the standard deviation from the data.

The `dt` and `dnorm` functions give the *density* (the height of the curve) at each point. Notice how the `t_3` curve sits lowest in the middle and highest in the tails — that extra tail weight is the uncertainty from a tiny sample, drawn to scale.

We can also look at the cumulative version of each curve:

```
tcdf <- tdens |>
  group_by(Distribution) |>
  arrange(value) |>
  mutate(cdf = cumsum(density))
ggplot(tcdf, aes(x = value, y = cdf, color = Distribution)) +
  geom_line()
```

17. The t-statistic and t-distribution



17.4.1. p-values based on Z vs t

When we have a *sample* and want to test how far its mean sits from a population mean, we scale by the standard error (the standard deviation of the sample mean), $\sigma/\sqrt{n}$:

$$z = \frac{x - \mu}{\sigma/\sqrt{n}}$$

Let's compare the p-value we'd get from the normal distribution against the one from the t-distribution, for a single small sample.

```
set.seed(5432)
samp <- rnorm(5, mean = 0.5)
z <- sqrt(length(samp)) * mean(samp) # simplifying assumption: sigma = 1, mu = 0
```

The p-value if we *pretend* we know the standard deviation:

```
pnorm(z, lower.tail = FALSE)
```

```
[1] 0.02428316
```


17. The *t*-statistic and *t*-distribution

In reality we don't know it, so we estimate it from the data with `sd()` and use the *t*-distribution (with `length(samp) - 1` degrees of freedom):

```
ts <- sqrt(length(samp)) * mean(samp) / sd(samp)
pnorm(ts, lower.tail = FALSE) # wrong: treats the estimate as known
```

```
[1] 0.0167297
```

```
pt(ts, df = length(samp) - 1, lower.tail = FALSE) # right: accounts for the estimate
```

```
[1] 0.0503001
```

The two numbers differ, and that difference is the whole story: using the normal distribution on an *estimated* standard deviation quietly understates our uncertainty. The next section makes that mistake visible at scale.

17.4.2. Experiment: does the choice of distribution actually matter?

Here's a claim worth testing for yourself rather than taking on faith: *if you estimate the standard deviation but compute p-values from the normal distribution, your p-values will be wrong*. Let's find out how wrong.

There's a tidy way to check. If a test is working correctly, then **under the null hypothesis its p-values should be uniformly distributed between 0 and 1** — every value equally likely. So we'll simulate many samples where the null is *true*, compute p-values three ways, and look at the histograms. A flat histogram means the test is honest; a lopsided one means it isn't.

The plan:

1. Draw many samples of size `n` from the standard normal distribution.
2. Score each with the **Z**-statistic (pretending we know σ).
3. Score each with the **t**-statistic (estimating σ from the data), then compute p-values *two* ways — from the normal, and from the *t*-distribution.

First, a function that draws a sample and returns its Z-score:

```
zf <- function(n) {
  samp <- rnorm(n)
  sqrt(length(samp)) * mean(samp) / 1 # simplifying assumption: sigma = 1, mu = 0
}
```

17. The *t*-statistic and *t*-distribution

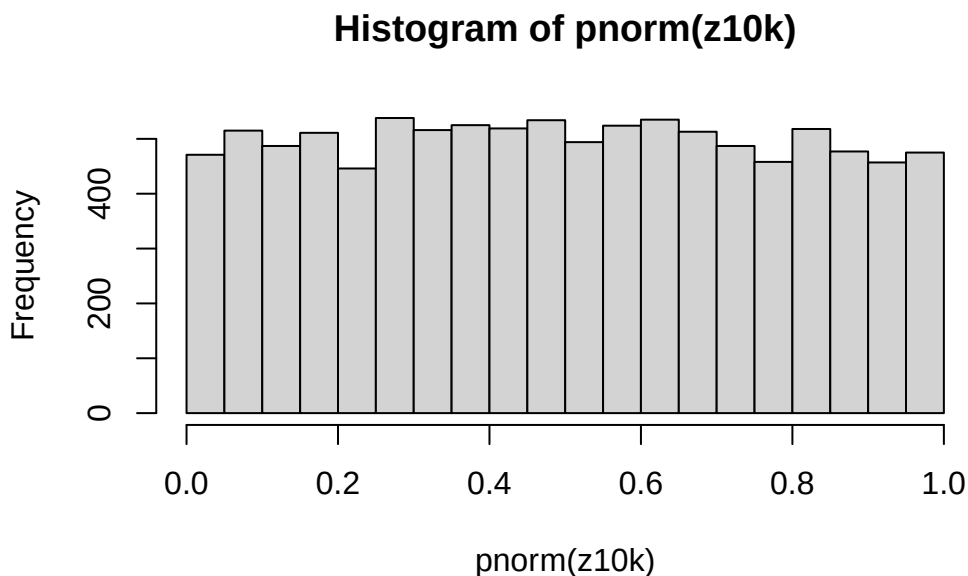
Give it a try:

```
zf(5)
```

```
[1] 0.7406094
```

Run 10,000 replicates and look at the p-value histogram. Here we *do* know the population standard deviation (we're sampling from the standard normal), so the normal distribution is the correct choice — and the histogram should be flat:

```
z10k <- replicate(10000, zf(5))  
hist(pnorm(z10k))
```



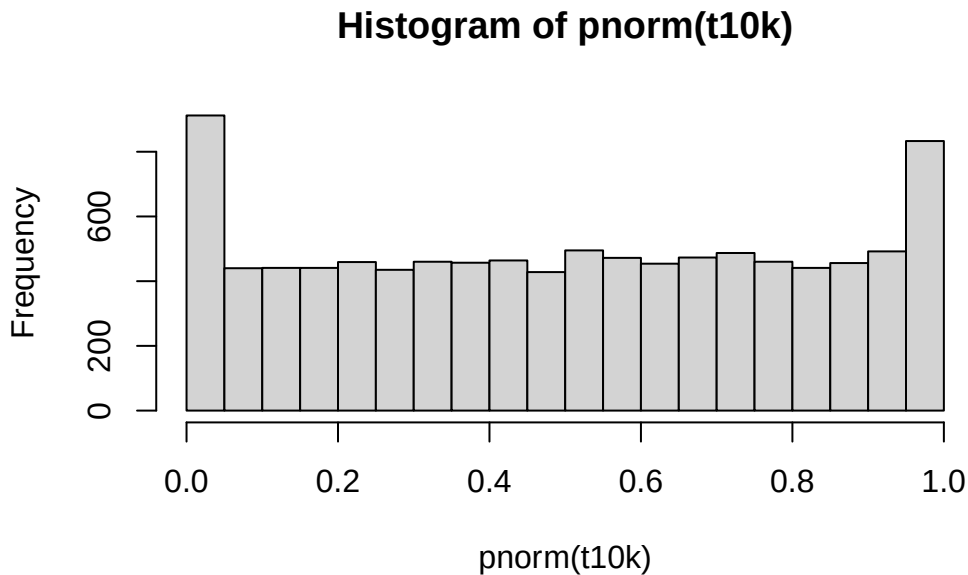
Flat, as promised. Now the *t*-score function — same idea, but we *estimate* the standard deviation with `sd()` instead of assuming it:

```
tf <- function(n) {  
  samp <- rnorm(n)  
  # use the sample standard deviation, since we "don't know" the population one  
  sqrt(length(samp)) * mean(samp) / sd(samp)  
}
```

17. The *t*-statistic and *t*-distribution

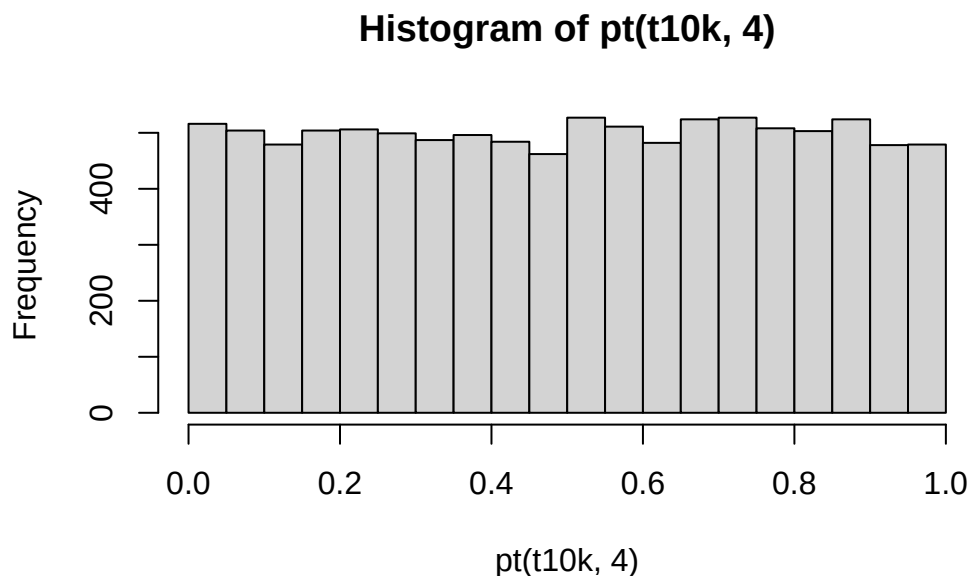
If we feed those *t*-scores to the **normal** distribution — the common mistake — the histogram is no longer flat:

```
t10k <- replicate(10000, tf(5))  
hist(pnorm(t10k))
```



The pile-ups near 0 and 1 mean this test rejects the null too often: it's overconfident, exactly because it treats an *estimated* standard deviation as if it were known. Now feed the same *t*-scores to the **t-distribution** (with $n - 1 = 4$ degrees of freedom):

```
hist(pt(t10k, 4))
```



Flat again. The *t*-distribution restores honest, uniform p-values — *that* is the payoff for all the math above.

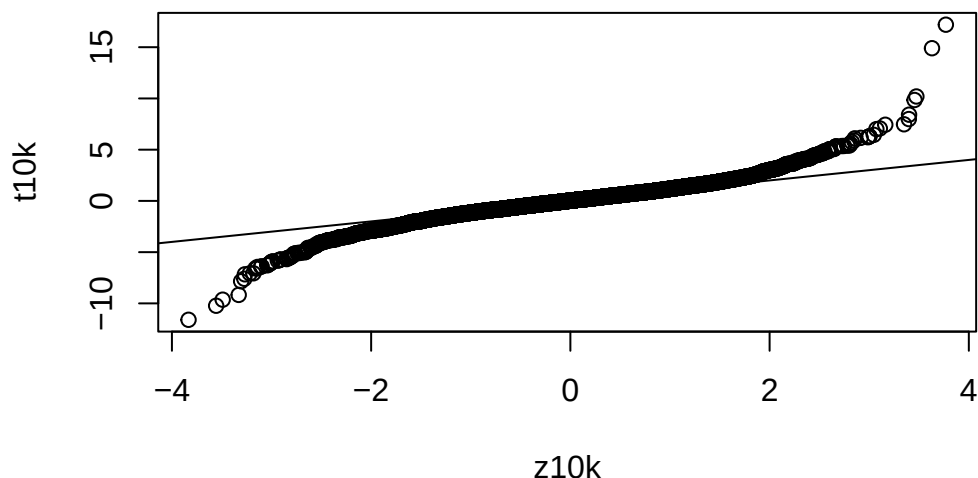
We can also compare the two sets of scores directly with a Q-Q plot.

A **Q-Q plot** plots the quantiles of two distributions against each other. If the two distributions are identical, the points fall on a straight line; where they differ, the points peel away from it.

Here we compare our simulated z-scores against our simulated t-scores. Because the t-scores carry the extra uncertainty from estimating the standard deviation, we expect them to disagree with the z-scores in the tails.

```
qqplot(z10k, t10k)
abline(0, 1)
```

17. The *t*-statistic and *t*-distribution



The points bow away from the line at both ends — the *t*-distribution’s fatter tails, made visible. What would happen if you raised the sample size from 5 to 50? Try it: the bow should shrink as the *t*-distribution approaches the normal.

17.5. Checkpoint: **t** vs normal

Before we start using `t.test()`, let’s pin down the one idea this chapter has been circling.

The *t*-distribution is a *family* of distributions indexed by the **degrees of freedom**, which track the sample size. It has heavier tails than the normal for small samples and approaches the normal as the sample grows. Those heavier tails make it more conservative — it takes a more extreme result to reach significance — which is the correct response to the uncertainty introduced by *estimating* the standard deviation from the data. That correction matters most exactly when you have the least data.

17.6. `t.test`

Now the practical part. R’s `t.test()` packages everything above into one function — you hand it data, it returns a *t*-statistic, degrees of freedom, a *p*-value, and a confidence

interval.

17.6.1. One-sample

A one-sample test compares a sample mean to a single value (zero, by default). We'll simulate a sample whose true mean is 1, so we *know* the answer the test should be reaching for.

```
x <- rnorm(20, 1)
# small sample: just the first 5 values
t.test(x[1:5])
```

One Sample t-test

```
data: x[1:5]
t = 0.97599, df = 4, p-value = 0.3843
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -1.029600  2.145843
sample estimates:
mean of x
0.5581214
```

We rigged this so the null hypothesis (mean = 0) is **false** — the true mean is 1. But with only 5 values the evidence is thin, so the p-value is modest. Add more data and the effect comes through clearly:

```
t.test(x[1:20])
```

One Sample t-test

```
data: x[1:20]
t = 3.8245, df = 19, p-value = 0.001144
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.3541055 1.2101894
sample estimates:
```

17. The *t*-statistic and *t*-distribution

```
mean of x  
0.7821474
```

17.6.2. Two-sample

A two-sample test compares the means of two groups:

```
x <- rnorm(10, 0.5)  
y <- rnorm(10, -0.5)  
t.test(x, y)
```

Welch Two Sample t-test

```
data: x and y  
t = 3.4296, df = 17.926, p-value = 0.003003  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 0.5811367 2.4204048  
sample estimates:  
 mean of x mean of y  
 0.7039205 -0.7968502
```

17.6.3. From a data frame

Often your data and group labels live in two columns of a data frame rather than in two separate vectors:

```
tdata <- data.frame(  
  value = c(x, y),  
  group = as.factor(rep(c('g1', 'g2'), each = 10))  
)  
tdata
```

```
      value group  
1  1.12896674   g1  
2 -1.26838101   g1  
3  1.04577597   g1
```

17. The *t*-statistic and *t*-distribution

```
4  1.69075585  g1
5  0.18672204  g1
6  1.99715092  g1
7  1.15424947  g1
8  0.37671442  g1
9  -0.09565723 g1
10 0.82290783  g1
11 -1.48530261 g2
12 -1.29200440 g2
13 -0.18778362 g2
14 0.59205742  g2
15 -2.10065248 g2
16 -0.29961560 g2
17 -0.38985115 g2
18 -2.47126235 g2
19 -0.63654380 g2
20 0.30245611  g2
```

R lets you run the same test with **formula notation**:

```
t.test(value ~ group, data = tdat)
```

Welch Two Sample t-test

```
data: value by group
t = 3.4296, df = 17.926, p-value = 0.003003
alternative hypothesis: true difference in means between group g1 and group g2 is not equal
95 percent confidence interval:
 0.5811367 2.4204048
sample estimates:
mean in group g1 mean in group g2
    0.7039205      -0.7968502
```

Read `value ~ group` as “*value is a function of group*” — in practice, do a *t*-test of the values in `g1` against those in `g2`.

17.6.4. Equivalence to a linear model

A two-sample *t*-test (with equal variances assumed) is really a linear model in disguise. Run the test:

```
t.test(value ~ group, data = tdat, var.equal = TRUE)
```

Two Sample t-test

```
data: value by group
t = 3.4296, df = 18, p-value = 0.002989
alternative hypothesis: true difference in means between group g1 and group g2 is not equal
95 percent confidence interval:
 0.5814078 2.4201337
sample estimates:
mean in group g1 mean in group g2
    0.7039205      -0.7968502
```

and compare it to fitting a line:

```
res <- lm(value ~ group, data = tdat)
summary(res)
```

Call:

```
lm(formula = value ~ group, data = tdat)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-1.9723 -0.5600  0.2511  0.5252  1.3889
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.7039      0.3094   2.275  0.03538 *
groupg2       -1.5008      0.4376  -3.430  0.00299 **
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

17. The *t*-statistic and *t*-distribution

```
Residual standard error: 0.9785 on 18 degrees of freedom
Multiple R-squared:  0.3952,    Adjusted R-squared:  0.3616
F-statistic: 11.76 on 1 and 18 DF,  p-value: 0.002989
```

The `groupg2` coefficient is the difference in means, and its p-value matches the t-test's. Same question, same answer, two front doors.

17.7. Power calculations

The **power** of a test is the probability that it *correctly* rejects the null hypothesis when the alternative is true — in other words, the probability of *not* missing a real effect (not making a Type II error). Power rises with the significance level (**alpha**), the sample size, and the effect size. It's what you compute *before* an experiment to ask “how many samples do I need?”

`power.t.test()` ties these quantities together. From `help("power.t.test")`, the main arguments are:

- **n** — sample size
- **delta** — effect size (difference in means)
- **sd** — standard deviation
- **sig.level** — significance level
- **power** — power

You supply any four and it solves for the fifth. For example, the power of a one-sample test with `n = 5`, `sd = 1`, and an effect size of 1:

```
power.t.test(n = 5, delta = 1, sd = 1, sig.level = 0.05)
```

Two-sample t test power calculation

```
      n = 5
  delta = 1
      sd = 1
sig.level = 0.05
  power = 0.2859276
alternative = two.sided
```

NOTE: `n` is number in *each* group

17. The *t*-statistic and *t*-distribution

That prints a full summary. To grab just the power value, use `$`:

```
power.t.test(n = 5, delta = 1, sd = 1,
             sig.level = 0.05, type = 'one.sample')$power
```

```
[1] 0.4013203
```

Tip

When a function returns a result that doesn't look “computable” — like the summary from `power.t.test` — you can pull out a single piece with `$`. How do you know what's available to extract? Use `names()` for the labels, or `str()` for the full structure:

```
names(power.t.test(n = 5, delta = 1, sd = 1,
                  sig.level = 0.05, type = 'one.sample'))
```

```
[1] "n"          "delta"      "sd"         "sig.level"  "power"
[6] "alternative" "note"       "method"
```

```
# or
str(power.t.test(n = 5, delta = 1, sd = 1,
                sig.level = 0.05, type = 'one.sample'))
```

List of 8

```
$ n      : num 5
$ delta  : num 1
$ sd     : num 1
$ sig.level : num 0.05
$ power  : num 0.401
$ alternative: chr "two.sided"
$ note    : NULL
$ method  : chr "One-sample t test power calculation"
- attr(*, "class")= chr "power.htest"
```

Just as often you'll flip the question around: *how big a sample* do I need to hit a target power? Solve for `n` by supplying `power` instead:

17. The *t*-statistic and *t*-distribution

```
power.t.test(delta = 1, sd = 1, sig.level = 0.05, power = 0.8, type = "one.sample")
```

One-sample t test power calculation

```
      n = 9.937864
delta = 1
sd = 1
sig.level = 0.05
power = 0.8
alternative = two.sided
```

`power.t.test` is convenient and fast. But as we saw with the *t*-distribution itself, sometimes the distribution of a test statistic is **not** easily calculated in closed form. In those cases we can estimate power by *simulation* — run the experiment thousands of times and count how often it rejects the null:

```
sim_t_test_pval <- function(n = 5, delta = 1, sd = 1, sig.level = 0.05) {
  x <- rnorm(n, delta, sd)
  t.test(x)$p.value <= sig.level
}
pow <- mean(replicate(1000, sim_t_test_pval()))
pow
```

```
[1] 0.405
```

Step by step: `sim_t_test_pval` simulates one sample of size `n` from a normal distribution with mean `delta` and standard deviation `sd`, runs a one-sample *t*-test, and returns `TRUE` when the *p*-value clears the significance threshold. `replicate` repeats that 1000 times, and `mean` turns the resulting `TRUE/FALSE` vector into the *proportion* of significant results — our estimate of power.

The two approaches should land in the same place:

```
power.t.test(n = 5, delta = 1, sd = 1, sig.level = 0.05, type = 'one.sample')$power
```

```
[1] 0.4013203
```

17. The *t*-statistic and *t*-distribution

```
mean(replicate(1000, sim_t_test_pval(n = 5, delta = 1, sd = 1, sig.level = 0.05)))
```

```
[1] 0.414
```

Close — and the simulation route keeps working in situations where no neat formula exists.

17.8. Exercises

Note

Several exercises reuse objects (`samp`, `t10k`, `tdat`) built earlier in the chapter. Run the chapter’s chunks first, or recreate the objects in your session.

1. **One-sided vs two-sided.** Use `twosidedpnorm()` to find the two-sided p-value for a Z-score of 1.96. Does the answer match the “95%” rule of thumb you may have heard?

Solution

```
twosidedpnorm(1.96)
```

```
[1] 0.04999579
```

It’s almost exactly 0.05 — which is *why* 1.96 is the classic cutoff for a two-sided test at the 5% level.

2. **Degrees of freedom matter.** Compute the two-sided p-value for $t = 2.5$ from the *t*-distribution with 4 degrees of freedom and again with 30. Which is larger, and does that fit the “fatter tails” story?

Solution

```
2 * pt(2.5, df = 4, lower.tail = FALSE)
```

```
[1] 0.06676654
```

```
2 * pt(2.5, df = 30, lower.tail = FALSE)
```

```
[1] 0.01811565
```

The 4-df p-value is larger. Fewer degrees of freedom means fatter tails, so the same *t*-statistic is *less* surprising — it takes more extreme evidence to reach significance with a small sample.

3. **A paired test.** Imagine the same 8 patients measured before and after a treatment. Simulate the pairs below and run a paired *t*-test with `t.test(after, before, paired = TRUE)`. How does the result compare to treating the two groups as independent (`paired = FALSE`)?

```
set.seed(11)
before <- rnorm(8, 10, 2)
after <- before + rnorm(8, 1, 0.5) # a real, consistent bump per patient
```

i Solution

```
t.test(after, before, paired = TRUE)
```

Paired t-test

```
data: after and before
t = 6.7665, df = 7, p-value = 0.0002611
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.4408169 0.9144199
sample estimates:
mean difference
 0.6776184
```

```
t.test(after, before, paired = FALSE)
```

Welch Two Sample t-test

```
data: after and before
t = 0.62439, df = 13.955, p-value = 0.5424
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.650735  3.005972
sample estimates:
mean of x mean of y
```

17. The t-statistic and t-distribution

```
10.364923  9.687305
```

The paired test has a much smaller p-value. By matching each patient to themselves it removes the patient-to-patient variability, leaving only the change due to treatment — so it detects the effect that the unpaired test can barely see.

4. **Power and sample size.** Using `power.t.test`, find the sample size needed to detect an effect of `delta = 0.5` (with `sd = 1`) at 80% power for a one-sample test. Then halve the effect size to 0.25 and recompute. Roughly how does the required `n` change?

i Solution

```
power.t.test(delta = 0.5, sd = 1, sig.level = 0.05, power = 0.8, type = "one.sample")
```

```
[1] 33.3672
```

```
power.t.test(delta = 0.25, sd = 1, sig.level = 0.05, power = 0.8, type = "one.sample")
```

```
[1] 127.5161
```

Halving the effect size roughly *quadruples* the required sample size — small effects are expensive to detect. This is the single most useful intuition to carry out of a power calculation.

17.9. Summary

You can now:

- Explain why estimating the standard deviation from a small sample pushes us from the normal distribution onto the t-distribution, and what “fatter tails” buys you.
- Distinguish a Z-score from a t-score and turn either into a p-value with `pnorm` and `pt`.
- Use simulation to *check* that a test is honest — uniform p-values under the null — rather than taking it on trust.
- Run one-sample, two-sample, paired, and formula-based tests with `t.test()`, and recognize the two-sample t-test as a linear model.
- Estimate power and required sample size with `power.t.test()` and by simulation.

The thread tying it all together: when you estimate uncertainty from your data instead of knowing it, you have to pay for that estimate — and the t-distribution is how statistics

keeps the books honest.

17.10. Resources

- The [pwr package](#) for a fuller toolkit of power calculations.
- The American Statistical Association’s statement on p-values (Wasserstein and Lazar 2016) for what a p-value does and doesn’t tell you.

18. K-means clustering

When yeast cells run low on glucose, they flip their entire metabolic program — switching from fermentation to respiration in an event called the **diauxic shift**. DeRisi et al. (1997) watched this happen by measuring the expression of roughly 6,400 genes at seven time points across the shift. That leaves you with a wall of numbers: 6,400 genes, each with its own little trajectory over time. Some genes ramp up as the cells switch fuels, some shut down, and most barely move at all. How do you find the *groups* of genes that rise and fall together — the ones that might be co-regulated, taking part in the same biological response?

That is exactly the kind of question **k-means clustering** answers. It is an unsupervised method: you don't tell it which genes belong together, you just ask it to sort the genes into a handful of groups so that genes with similar behavior land in the same group. In this chapter you'll use k-means on the DeRisi yeast data — the same dataset you met in the SummarizedExperiment chapter — to pull out clusters of co-regulated genes and then read their temporal profiles back as biology.

18.1. What you'll learn

- Explain in plain language what k-means clustering does and when it's useful.
- Pull a public dataset from NCBI GEO with `getGEO()` and convert it to a `SummarizedExperiment`.
- Filter a gene-expression matrix down to its most variable (most informative) genes using the standard deviation.
- Run `kmeans()` on real data and make the result **reproducible** with `set.seed()`.
- Plot each cluster's expression profile and interpret what the clusters say about the diauxic shift.

18.2. What k-means does

K-means clustering is a method for finding groups in a dataset. It is an *unsupervised* learning technique: it doesn't rely on previously labeled data. Instead, it finds structure directly from the data, based on how similar the data points are to one another (Figure 18.1).

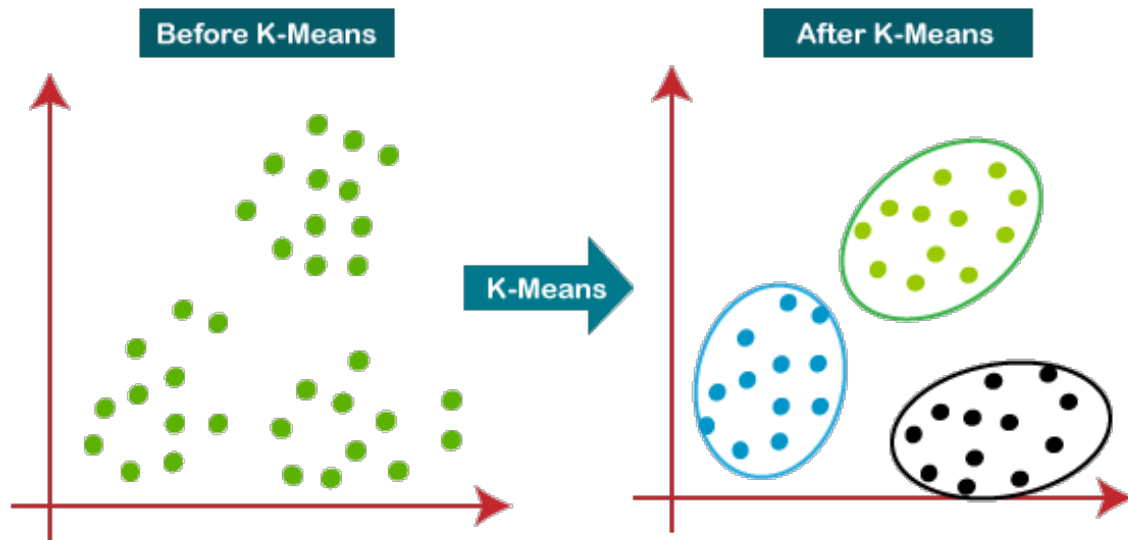


Figure 18.1.: K-means clustering takes a dataset and divides it into k clusters.

The idea is to split the data into k groups, called **clusters**, so that each point belongs to the cluster whose average it is closest to. The algorithm tries to make points within a cluster as similar as possible while keeping different clusters as distinct as possible. For our yeast data, a “point” is a single gene, and “similar” means *follows a similar expression pattern over the time course*.

💡 An analogy for the algorithm

Imagine k party hosts scattered around a room full of guests. Each guest walks to the nearest host (**assignment**). Each host then moves to the middle of their own group (**update**). Guests re-check who is now closest and move again, and the hosts re-center. Repeat until nobody needs to move. The final huddles are your clusters, and each host’s position is a **centroid** — the average of its group.

18.3. The k-means algorithm

Spelled out, k-means follows these steps:

1. Choose the number of clusters, k .
2. Place k centroids at random (often by picking k of the data points).

18. K-means clustering

3. **Assign** each data point to its nearest centroid.
4. **Update** each centroid to the mean of the points assigned to it.
5. Repeat steps 3 and 4 until the centroids stop moving (or a maximum number of iterations is reached).

When the centroids stabilize, the algorithm has *converged*, and the final clusters represent the patterns in the data.

 k-means is random — seed it

Step 2 starts from a **random** placement of centroids, so two runs can give slightly different clusters (and the cluster *numbers* can be shuffled). Whenever you want a result you can reproduce — and a frozen figure that doesn't change on every render — call `set.seed()` before `kmeans()`.

18.4. Pros and cons of k-means clustering

Compared to other clustering algorithms, k-means has several advantages:

- **Simplicity** — the algorithm is straightforward and easy to implement, even on large datasets.
- **Scalability** — it adapts well to large datasets with optimization or parallel processing.
- **Speed** — it is generally fast, especially when **k** is small.
- **Interpretability** — the results are easy to read: every point gets a single cluster label.

It also has real limitations:

- **Choice of k** — you have to pick the number of clusters yourself, and a poor choice gives poor results. Domain knowledge and experimentation help.
- **Sensitivity to initial conditions** — different random starts can give different answers, which is why you seed and sometimes re-run.
- **Assumes round, even clusters** — k-means expects roughly spherical, similarly sized clusters, and struggles when the real groups have other shapes.
- **Sensitivity to outliers** — a few extreme points can drag a centroid around, so cleaning or filtering the data first helps.

Despite these limits, k-means remains a popular, widely used tool for exploring data — and it is especially handy in biology, where spotting groups of co-behaving genes can hint at shared function or regulation.

18.5. A worked example: the diauxic shift

18.5.1. The data and experimental background

The data come from DeRisi et al. (1997). From their abstract:

DNA microarrays containing virtually every gene of *Saccharomyces cerevisiae* were used to carry out a comprehensive investigation of the temporal program of gene expression accompanying the metabolic shift from fermentation to respiration. The expression profiles observed for genes with known metabolic functions pointed to features of the metabolic reprogramming that occur during the diauxic shift, and the expression patterns of many previously uncharacterized genes provided clues to their possible functions.

These data are available from NCBI GEO as [GSE28](#).

Here is the biology in brief. When baker's yeast grows on glucose with plenty of oxygen, it still *prefers* to ferment the glucose rather than respire it — even though respiration is more efficient. Once the glucose runs low, the cells switch metabolic gears to respiration. That switch is the [diauxic shift](#). This experiment profiles about 6,400 genes across a time course spanning the shift.

These data have no replicates; they are a single time course. Watching how *groups* of genes behave across that course can reveal biology — both about the system and about individual genes. So we'll use clustering to group genes with similar expression patterns over time, then look at those groups.

Your goal: use `kmeans()` to divide the most variable (most informative) genes into a few groups, then visualize each group's temporal profile.

18.5.2. Getting the data

These data were deposited at NCBI GEO back in 2002. `GEOquery` can pull them out in one call.

```
library(GEOquery)
gse <- getGEO("GSE28")[[1]]
class(gse)
```

```
[1] "ExpressionSet"
attr(,"package")
[1] "Biobase"
```

18. K-means clustering

GEOquery predates the SummarizedExperiment class, so it hands you an older ExpressionSet. Bioconductor can convert that into the SummarizedExperiment you already know how to work with.

```
library(SummarizedExperiment)
gse <- as(gse, "SummarizedExperiment")
gse
```

```
class: SummarizedExperiment
dim: 6400 7
metadata(3): experimentData annotation protocolData
assays(1): exprs
rownames(6400): 1 2 ... 6399 6400
rowData names(20): ID ORF ... FAILED IS_CONTAMINATED
colnames(7): GSM887 GSM888 ... GSM892 GSM893
colData names(33): title geo_accession ... supplementary_file
                  data_row_count
```

A quick look at the sample descriptions in colData() suggests the columns aren't in time order:

```
colData(gse)$title
```

```
[1] "diauxic shift timecourse: 15.5 hr" "diauxic shift timecourse: 0 hr"
[3] "diauxic shift timecourse: 18.5 hr" "diauxic shift timecourse: 9.5 hr"
[5] "diauxic shift timecourse: 11.5 hr" "diauxic shift timecourse: 13.5 hr"
[7] "diauxic shift timecourse: 20.5 hr"
```

So reorder them by hand so the columns run in time-course order:

```
gse <- gse[, c(2, 4, 5, 6, 1, 3, 7)]
```

18.5.3. Preprocessing: keep only the informative genes

Most of those 6,400 genes barely change across the time course. Feeding all of them to the clustering algorithm is slow and, worse, lets thousands of flat, noisy genes drown out the few hundred that actually respond to the diauxic shift.

18. K-means clustering

So before clustering, you **filter** down to the genes that carry signal. A simple, effective filter is the **standard deviation** of each gene across the samples: a gene whose expression swings widely over the time course has a high standard deviation, while a gene that stays flat has a low one.

! Why filter on variability first

Genes that change a lot across conditions are the ones most likely to be doing something biologically interesting. Dropping the flat, low-variance genes does two good things at once: it shrinks the dataset so clustering is faster, and it removes noise so real patterns stand out. Filtering on a gene-level statistic like the standard deviation — one that ignores the grouping you'll later test — can also genuinely *increase statistical power* and reduce false positives (Bourgon et al. 2010).

The recipe has three steps:

1. Compute each gene's standard deviation across the samples.
2. Pick a cutoff (this is somewhat arbitrary — try a few and see how sensitive your results are).
3. Keep only the genes above the cutoff.

Start with step 1 and look at the distribution of standard deviations:

```
sds <- apply(assays(gse)[[1]], 1, sd)
hist(sds)
```

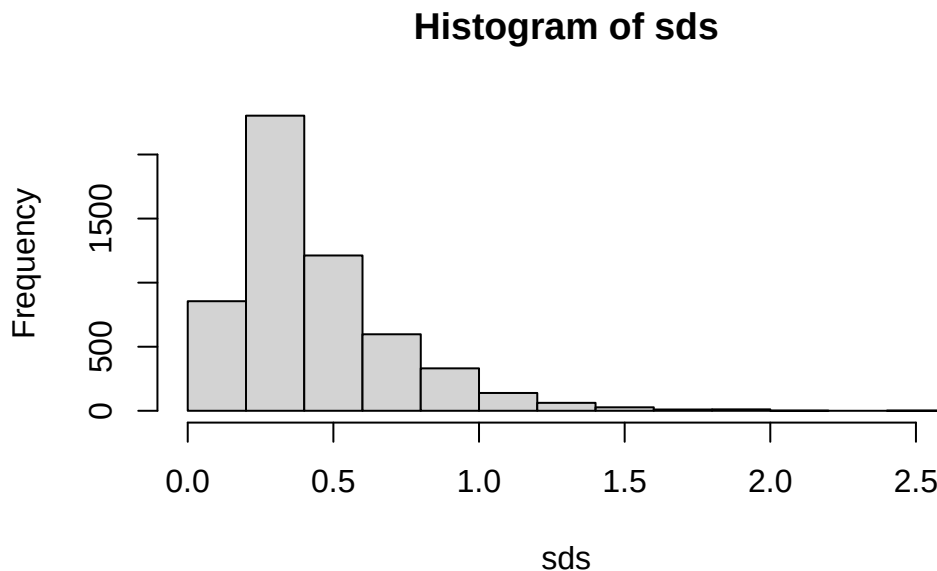


Figure 18.2.: Histogram of per-gene standard deviations across the time course. Most genes barely vary (the tall bars on the left); a long right tail marks the genes that change the most.

Figure 18.2 tells the story directly. The bulk of the genes pile up near zero — they hardly change across the time course and carry little information. A long tail stretches off to the right: these are the genes whose expression swings the most. A cutoff of about `sd > 0.8` sits past the main pile and keeps that informative right tail. The exact value is a judgment call, which is why it's worth re-running with a different cutoff later (one of the exercises does just that).

Now apply the cutoff (step 3) and build a smaller `SummarizedExperiment` holding just the highly variable genes:

```
idx <- sds > 0.8 & !is.na(sds)
gse_sub <- gse[idx, ]
nrow(gse_sub)
```

```
[1] 582
```

That trims thousands of genes down to a few hundred — a far more workable set.

18.5.4. Clustering

`gse_sub` now holds just the informative genes. The `kmeans()` function takes a matrix (rows = genes, columns = samples) and the number of clusters. Remember to seed first so the result is reproducible.

```
k <- 4
set.seed(42)
km <- kmeans(assays(gse_sub)[[1]], centers = k)
```

The result, `km`, carries a vector `km$cluster` that gives the cluster label (1 through `k`) for each gene. With four clusters, you can plot every cluster's genes side by side and see how each group moves over time.

```
expression_values <- assays(gse_sub)[[1]]
par(mfrow = c(2, 2), mar = c(3, 4, 1, 2)) # 2x2 grid of plots
for (i in 1:k) {
  matplot(t(expression_values[km$cluster == i, ]), type = "l",
          ylim = c(-3, 3), ylab = paste("cluster", i))
}
```


18. K-means clustering

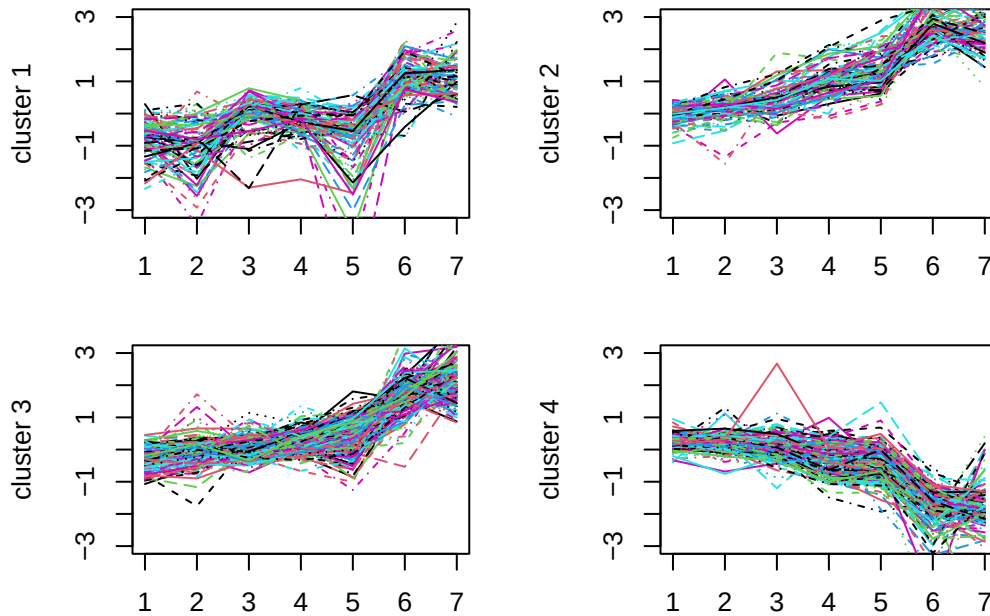


Figure 18.3.: Expression profiles for the four clusters found by k-means. Each line is one gene; the x-axis runs left to right across the time course, and the y-axis is the (standardized) expression level. Genes within a cluster share a trend, which suggests they may be co-regulated.

18.5.5. Reading the clusters as biology

This is the step that the numbers were building toward. Each panel in Figure 18.3 is a group of genes that move *together* across the time course — and “together” is the whole point, because co-regulated genes are often part of the same pathway or controlled by the same switch.

Look at the *shapes* rather than the exact lines:

- Some clusters **rise** toward the late time points. Those are genes that get switched **on** as the glucose runs out and the cells shift to respiration — candidate members of the respiratory and stress-response programs.
- Other clusters **fall** across the course. Those genes are being switched **off** during the shift — for example, the fermentation and growth machinery the cell no longer needs once glucose is scarce.

18. K-means clustering

- The lines within each panel hug one another tightly, which is the visual signature of co-regulation: many genes responding in step to the same metabolic cue.

That is the payoff of clustering. You started with 6,400 noisy trajectories and ended with a handful of clean, interpretable patterns — rising groups, falling groups — that map directly onto the biology of the diauxic shift. If you wanted to go further, you could pull the gene names out of each cluster (`rownames(gse_sub)[km$cluster == i]`) and ask whether the “rising” cluster is enriched for respiration genes.

i Why standardized expression?

The y-axis runs from about -3 to 3 because these microarray values are log-ratios centered near zero: positive means “higher than the reference,” negative means “lower.” Clustering on values like these groups genes by the *shape* of their response over time rather than by their absolute brightness, which is usually what you want.

18.6. Exercises

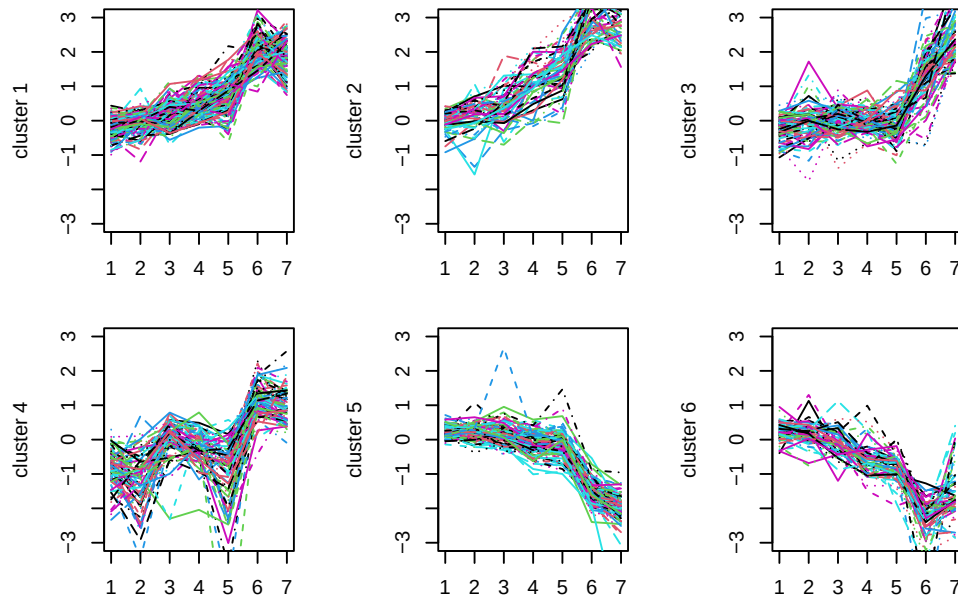
For each exercise, reuse the `gse` and `gse_sub` objects you built above.

1. **Try a different k.** Re-run the clustering with `k <- 6` instead of 4 and re-make the cluster plots. Do the extra clusters reveal new patterns, or do they mostly split the existing ones?

i Solution

```
k6 <- 6
set.seed(42)
km6 <- kmeans(assays(gse_sub)[[1]], centers = k6)
par(mfrow = c(2, 3), mar = c(3, 4, 1, 2))
for (i in 1:k6) {
  matplot(t(expression_values[km6$cluster == i, ]), type = "l",
          ylim = c(-3, 3), ylab = paste("cluster", i))
}
```

18. K-means clustering



With more clusters, the broad “rising” and “falling” groups tend to break into finer sub-patterns (e.g. early vs. late risers). There is no single correct k ; the right number depends on how fine a distinction is biologically useful.

2. **Try a different cutoff.** Lower the standard-deviation cutoff to 0.5 so you keep *more* genes, rebuild the subset, and cluster again with $k = 4$. How many genes do you keep now, and do the cluster shapes get cleaner or noisier?

Solution

```
idx2 <- sds > 0.5 & !is.na(sds)
gse_sub2 <- gse[idx2, ]
nrow(gse_sub2)
```

```
[1] 1657
```

```
set.seed(42)
km2 <- kmeans(assays(gse_sub2)[[1]], centers = 4)
```

A lower cutoff keeps many more genes, including less-variable ones. The extra genes are noisier, so the cluster profiles usually look a little messier. This is the sensitivity check the chapter warned about: the cutoff is a knob, and it’s worth turning it to see whether your conclusions hold.

3. **How big is each cluster?** Using your original `km` result, count how many genes fall in each of the four clusters.

i Solution

```
table(km$cluster)
```

```
 1   2   3   4
98 102 203 179
```

`table()` tallies the cluster labels. Uneven cluster sizes are normal and fine — a cluster of co-regulated genes is just as real whether it has 30 members or

300.

4. **Reproducibility.** Run `kmeans()` on `gse_sub` twice *without* calling `set.seed()` first, and compare `table()` of the cluster labels between the two runs. What changes, and why does that matter for a printed result?

i Solution

```
km_a <- kmeans(assays(gse_sub)[[1]], centers = 4)
km_b <- kmeans(assays(gse_sub)[[1]], centers = 4)
table(km_a$cluster)
```

```
 1   2   3   4
91  88 165 238
```

```
table(km_b$cluster)
```

```
 1   2   3   4
165  91  88 238
```

The two runs start from different random centroids, so the cluster *labels* (and sometimes the exact membership) differ. Seeding with `set.seed()` pins the random start so anyone re-running your code sees the same clusters — which is exactly why we seeded the worked example.

18.7. Summary

You set out to find groups of genes that behave alike across the yeast diauxic shift, and you did it with k-means. Looking back at the objectives:

- You can describe k-means as an **unsupervised** method that sorts points into **k** groups around moving **centroids**, and you know its main trade-offs (you pick **k**; it likes round, even clusters; it's sensitive to outliers and to the random start).
- You pulled a real dataset from GEO with `getGEO()` and converted it to a `SummarizedExperiment`.
- You **filtered** the genes down to the high-variance, informative ones using their standard deviation, and you saw on Figure 18.2 why a cutoff past the main pile makes sense.
- You ran `kmeans()` reproducibly with `set.seed()` and plotted each cluster.
- You read Figure 18.3 as biology: clusters that **rise** are genes switched on for respiration as glucose runs out, and clusters that **fall** are the fermentation/growth machinery being switched off — co-regulated genes moving in step through the diauxic shift.

That wraps up the statistics part of the book. Clustering is your first taste of unsupervised learning; the [machine-learning chapters](#) pick up that thread and build on it.

Part V.

Machine Learning

19. Introduction to machine learning

Suppose you have gene-expression profiles from a hundred tumour biopsies and a hundred matched normal tissues, and a new, unlabelled sample arrives. Can you tell whether it is tumour or normal from its expression pattern alone? Or imagine you have DNA-methylation measurements from thousands of people of known age, and you want to estimate the biological age of someone new. These are not problems you can solve by sitting down and writing an explicit rule — there is no single gene, no simple `if` statement, that separates tumour from normal across every patient. The signal is spread across thousands of features in ways that are too subtle to codify by hand.

Machine learning is the set of tools for exactly this situation: instead of programming the rule yourself, you let an algorithm *learn* the rule from labelled examples, then apply it to new cases it has never seen. That last part — performing well on *new* data, not just the data you trained on — is the whole point. It is called **generalization**, and most of this chapter is really about how to get it and how to avoid fooling yourself into thinking you have it.

Machine learning in biology is a broad and fast-moving field. Greener et al. (2022) give an approachable overview of the methods used in biology, and Libbrecht and Noble (2015) review their early use in genetics and genomics. This chapter is the conceptual on-ramp: what machine learning is, the three families of problems it solves, the workflow you follow, and the single most important pitfall (overfitting) to watch for. The chapters that follow put these ideas to work — a [tour of supervised algorithms](#) on a shared dataset, the [mlr3verse framework](#) for running them cleanly, and worked biological examples — including the [tumour-classification](#) and [age-from-methylation](#) problems we just sketched.

19.1. What you'll learn

- Recognize the three paradigms of machine learning — **supervised**, **unsupervised**, and **reinforcement** learning — and match a biological question to the right one.
- Distinguish **classification** (predicting a category) from **regression** (predicting a number) within supervised learning.
- Walk through the stages of a machine learning **workflow**, from problem definition to evaluation.

- Explain why you must **split your data** into training, validation, and test sets, and what each set is for.
- Recognize **overfitting** and **underfitting**, and describe how **cross-validation** and the **bias–variance tradeoff** help you steer between them.

19.2. The three kinds of machine learning

Machine learning problems come in a few distinct flavours, and the first skill to develop is recognizing which flavour you are facing. The flavour determines what data you need and what kind of answer you can expect.

Supervised learning is the most common in practice, and it covers most of the biological examples above. Here you have both the inputs (the **features** — gene expression values, methylation levels, clinical measurements) *and* the correct answers (the **labels** or **targets** — tumour vs normal, the patient’s age) for a set of training examples. The algorithm learns to map inputs to outputs by studying these example pairs. Supervised problems split further into two types:

- **Classification** predicts a category. *Is this sample tumour or normal? Which of five cancer subtypes is this?* The answer is one of a fixed set of labels.
- **Regression** predicts a number. *What is this patient’s age? What is the expression level of this gene?* The answer is a continuous value.

Unsupervised learning applies when you have the features but *no* labels — you do not have a known answer to predict. Instead, the algorithm looks for structure hidden in the data itself. **Clustering** groups similar samples together; you might cluster single cells by their expression profiles to discover cell types nobody labelled in advance. The [k-means clustering chapter](#) works through a complete clustering example on real yeast gene-expression data. **Dimensionality reduction** (like PCA) finds the few underlying factors that explain most of the variation across thousands of genes. These methods are often the first thing you reach for when you want to *explore* a new dataset before you commit to a specific prediction.

Reinforcement learning is a different shape of problem altogether. Rather than learning from a fixed pile of examples, an agent takes actions in an environment and receives rewards or penalties, improving its decisions through trial and error. It powers game-playing systems and robotics, and is appearing in areas like experimental design and molecule generation, but it needs specialized techniques and a lot of compute. We mention it for completeness; the rest of this part of the book focuses on supervised and unsupervised learning, which cover the great majority of everyday biological data analysis.

19. Introduction to machine learning

Learning type	What you give it	What it gives back	Biological example
Supervised	Features and labels	Predicted labels/values	Tumour vs normal from expression; age from methylation
Unsupervised	Features only	Discovered structure	Clustering cells into types; PCA of samples
Reinforcement	An environment to act in	A decision policy	Sequential experiment design

When thinking about machine learning, it can help to have a simple framework in mind. Figure 19.1 shows a tidy view of the field from the [scikit-learn](#) project, organized around the same questions: are you predicting a category, predicting a quantity, or looking for structure?

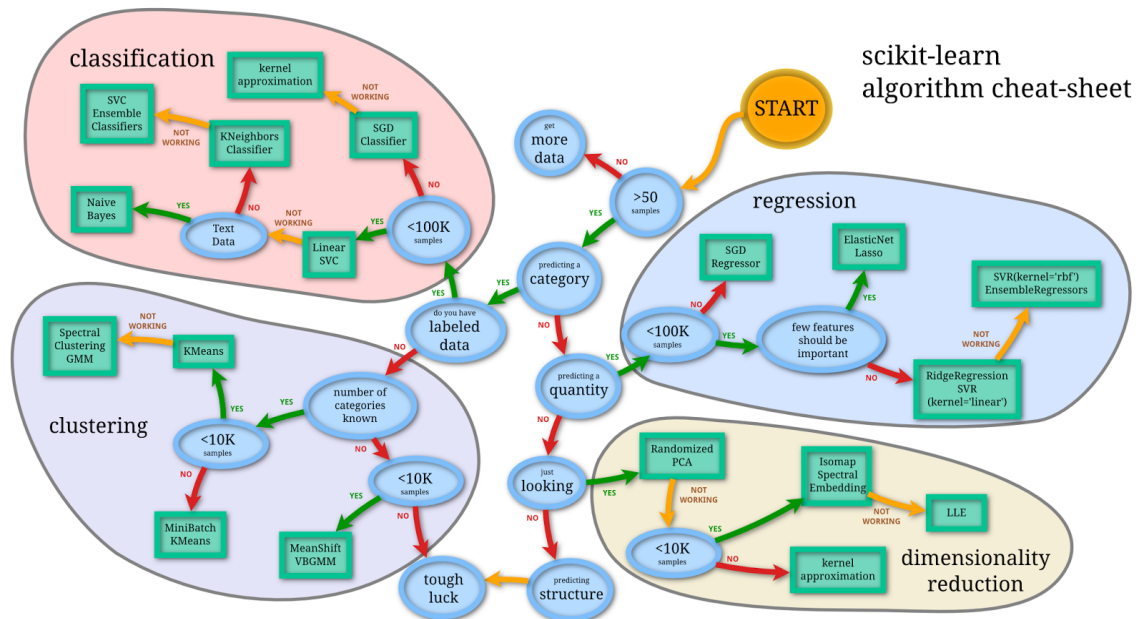


Figure 19.1.: A simple view of machine learning according to scikit-learn.

i Terminology and concepts

- **Data** The foundation of machine learning. It can be structured (tabular) or unstructured (text, images, audio), and is usually divided into training, validation, and testing sets for model development and evaluation.

- **Features** The variables or attributes used to describe each data point (for example, the expression level of each gene). Feature engineering and selection are crucial steps for improving model performance and interpretability.
- **Models and algorithms** A model is a mathematical representation of the relationship between features and the target. An algorithm is the method used to fit a model to data, such as linear regression, decision trees, and neural networks.
- **Hyperparameters and tuning** Hyperparameters are adjustable settings that control the learning process (for example, how deep a decision tree may grow). Tuning is the search for the values that give the best performance.
- **Evaluation metrics** Numbers that quantify how well a model does — accuracy, precision, recall, and F1-score for classification; mean squared error and R-squared for regression.

19.3. The machine learning workflow

Successful machine learning projects follow a structured workflow. The process begins long before any algorithm is trained and extends well beyond the first model you fit.

i The workflow at a glance

1. **Problem definition** — state clearly what you want to predict and decide whether machine learning is even the right tool.
2. **Data collection and preparation** — gather relevant data and clean it into a form a model can use.
3. **Data splitting** — divide the data into training, validation, and test sets so you can evaluate honestly.
4. **Model selection** — choose an algorithm and configure its hyperparameters.
5. **Training** — fit the model to the training data so it can learn the patterns.
6. **Evaluation** — measure performance on held-out data using appropriate metrics.
7. **Deployment** — put the model to work on new, real data.

The journey starts with **problem definition**. You need to articulate what you are trying to achieve and whether machine learning is the appropriate solution. Not every problem needs it; sometimes a simple statistical test or a known biological rule gives a better answer with far less complexity.

Data collection and preparation usually consume most of the time in a real project. Raw data rarely arrives ready to model. You will spend effort handling missing values, encoding categorical variables, scaling numerical features, and dealing with outliers and batch effects. The quality of this preparation often decides whether the whole project succeeds.

Data splitting deserves special attention because it directly controls how trustworthy your results are. The training set teaches the algorithm, but if you then judge performance on that same data, you get a flattering picture that will not hold up on new samples. It is like letting students see the exam questions while they study, then grading them on those exact questions.

! Why splitting your data is non-negotiable

Splitting your data into separate sets is the foundation of honest evaluation, and it is the single best defence against fooling yourself.

- **Training data** is used to fit the model's parameters.
- **Validation data** helps you choose between models and tune hyperparameters.
- **Test data** gives a final, unbiased estimate of performance on new data.

The golden rule: never use the test data for any decision during model development. Reserve it for the final evaluation only.

A common starting point is an 80% training / 20% testing split. For more involved projects, a three-way split might use 60% for training, 20% for validation (model selection), and 20% for final testing. Cross-validation, which we return to below, offers a more data-efficient alternative while still keeping a test set untouched.

Model selection means choosing both the type of algorithm and its specific settings. Different algorithms make different assumptions and shine in different situations. Linear models work well when relationships are roughly linear and you value interpretability. Tree-based methods handle nonlinear relationships and interactions naturally but can overfit with limited data. Neural networks can capture very complex patterns but need large datasets and careful tuning. Hyperparameters describe the shape of the model — the depth of a decision tree, the k in k -nearest neighbours, the learning rate in gradient descent — and hyperparameter tuning is the search for good values, often using grid or random search together with cross-validation.

Training is where the algorithm actually learns. The model adjusts its internal parameters to reduce its errors on the training set. But the goal is not a perfect score on the training data; a model that fits the training data too closely often fails on new examples, a problem called **overfitting**.

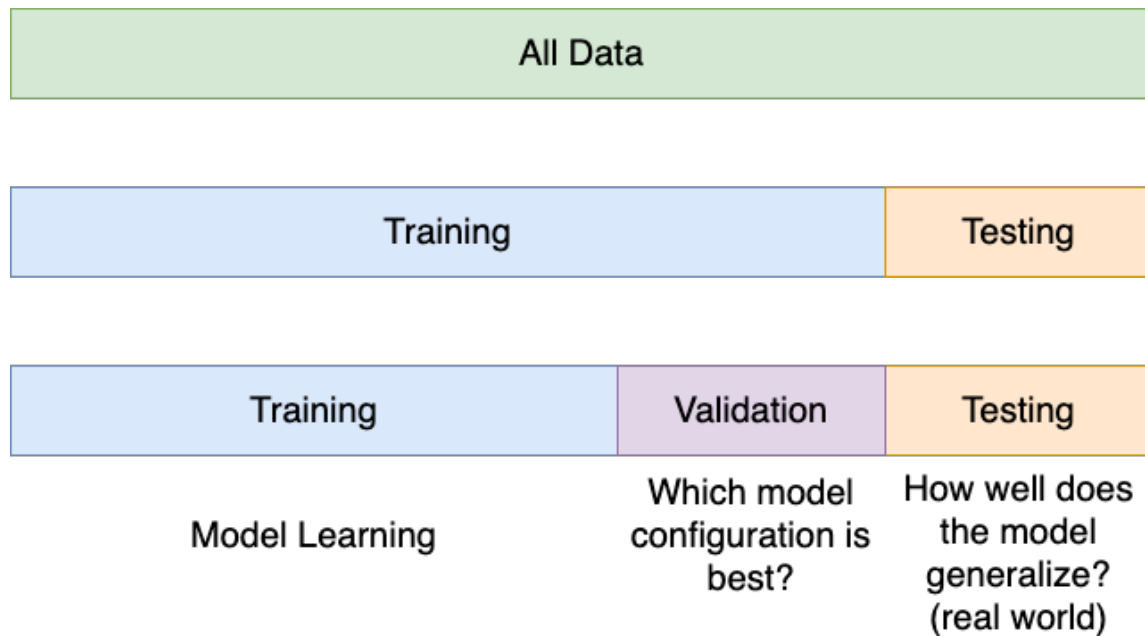


Figure 19.2.: Data splitting and the train/validate/test paradigm. For models without hyperparameters, only the training and testing sets are needed. For models that must learn their *structure* (such as the k in k -nearest neighbours) as well as their parameters (like the slopes in linear regression), a separate validation set is essential.

Evaluation tells you whether the model is ready. This means looking past a single accuracy number — precision, recall, and F1-score for classification, or mean squared error and R-squared for regression — and checking that performance holds up across the different subgroups in your data.

19.4. Overfitting and underfitting

Recall that the goal is a model that generalizes well to new, unseen data. Given enough flexibility, almost any model can fit the data it was trained on perfectly. The real challenge is balancing a good fit on the training data against good performance on *new examples* — and it is that second part, generalization, that we actually care about.

When a model **overfits**, it learns the training data too well, memorizing the specific examples instead of the general pattern. The result is excellent performance on training data and poor performance on anything new.

A familiar analogy: imagine teaching someone to recognize cats by showing them 100 photos. An overfitted learner memorizes every pixel of those particular photos rather than learning the general features — whiskers, pointed ears, fur. Shown a new cat, they are lost, because they fixated on details unique to the training images. The genomics version is the same story: a model that learns to separate your 200 tumour and normal samples by latching onto a few quirks of *those* 200 patients (or a batch effect in *that* sequencing run) will collapse the moment you hand it samples from a new hospital.

Underfitting is the opposite failure. An underfitted model is too simple to capture the real pattern, so it does poorly on both training and new data. In the cat analogy, an underfitted model might only look at overall image brightness and miss every feature that actually distinguishes a cat.

```
set.seed(123)
sinsim <- function(n, sd = 0.1) {
  x <- seq(0, 1, length.out = n)
  y <- sin(2 * pi * x) + rnorm(n, 0, sd)
  return(data.frame(x = x, y = y))
}
dat <- sinsim(100, 0.25)
library(ggplot2)
library(patchwork)
p_base <- ggplot(dat, aes(x = x, y = y)) +
  geom_point(alpha = 0.7) +
  theme_bw()
```

19. Introduction to machine learning

```
p_lm <- p_base +  
  geom_smooth(method = "lm", se = FALSE, alpha = 0.6, formula = y ~ x)  
p_lmsin <- p_base +  
  geom_smooth(method = "lm", formula = y ~ sin(2 * pi * x), se = FALSE, alpha = 0.6)  
p_loess_wide <- p_base +  
  geom_smooth(method = "loess", span = 0.5, se = FALSE, alpha = 0.6, formula = y ~ x)  
p_loess_narrow <- p_base +  
  geom_smooth(method = "loess", span = 0.05, se = FALSE, alpha = 0.6, formula = y ~ x)  
p_lm + p_lmsin + p_loess_wide + p_loess_narrow + plot_layout(ncol = 2) +  
  plot_annotation(tag_levels = "A") &  
  theme(plot.tag = element_text(size = 8))
```

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: k-d tree limited by memory. ncmx= 200

Warning in sqrt(sum.squares/one.delta): NaNs produced

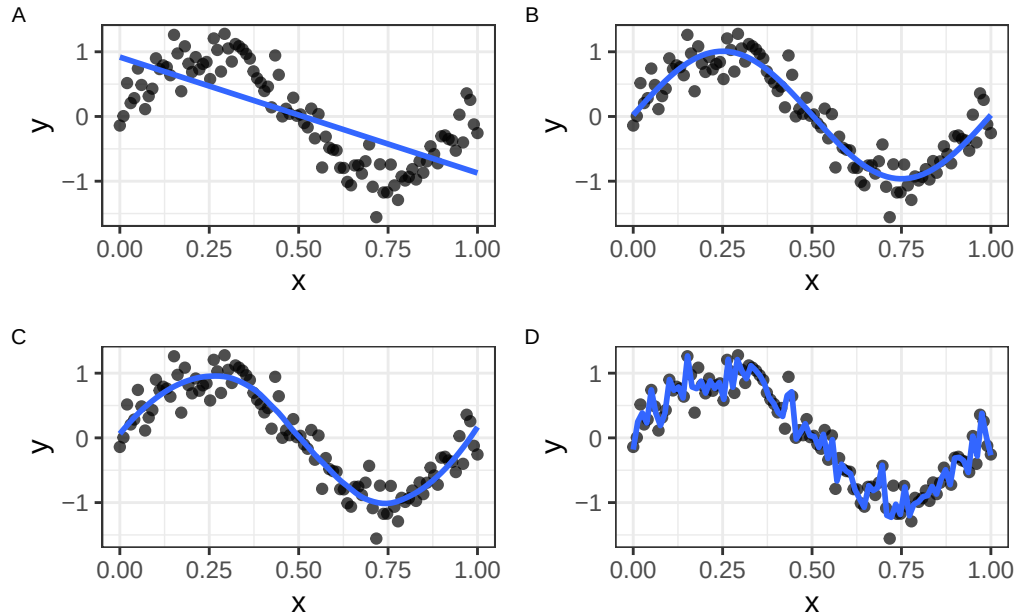


Figure 19.3.: Data simulated according to the function $f(x) = \sin(2\pi x) + N(0, 0.25)$ fitted with four different models. A) A simple linear model demonstrates *underfitting*. B) A linear model with a sin term ($y = \sin(2\pi x)$) and C) a loess model with a wide span (0.5) demonstrate *good fits*. D) A loess model with a narrow span (0.05) is a good example of *overfitting*.

In Figure 19.3 we simulate data from the function $f(x) = \sin(2\pi x) + N(0, 0.25)$ and fit four models of increasing flexibility. A model that is too simple (A) **underfits**: the straight line cannot follow the wave at all. A model that is too flexible (D) **overfits**: the narrow-span loess curve chases every wiggle of the random noise instead of the smooth trend beneath it. Models (B) and (C) sit in the sweet spot — flexible enough to capture the real pattern, restrained enough to ignore the noise.

The relationship between model complexity and performance tends to follow a U shape. Very simple models do badly because they underfit. As you add complexity, performance improves as the model captures more of the real signal. But past an optimal point, extra complexity starts fitting noise, and performance on new data gets worse again.

💡 Detecting overfitting

Overfitting shows up as a widening gap between training and validation performance. If your model scores 95% accuracy on training data but only 70% on validation data, it is almost certainly overfitting.

Watch both numbers throughout development. The best models perform *similarly* on training and validation data — a sign they have learned the pattern, not the particular examples.

Several strategies push back against overfitting. **Regularization** adds a penalty for complexity, nudging the model toward simpler solutions. **Cross-validation** (next section) gives more reliable performance estimates by validating on several data splits. **Early stopping** halts training once validation performance starts to slip. **Feature selection** trims the input down to the variables that actually matter — often crucial in genomics, where you may start with tens of thousands of genes.

The **bias–variance tradeoff** is the theory behind all of this. *Bias* is systematic error from a model that is too simple to capture the truth. *Variance* is sensitivity to the particular training sample you happened to draw. High-bias models underfit; high-variance models overfit. The model that generalizes best is the one that balances these two competing sources of error.

19.5. Cross-validation and model selection

Cross-validation answers a practical question: how do you reliably estimate a model’s performance when you only have so much data? A single train–test split can mislead you, especially with small datasets, because the estimate depends heavily on which samples happen to land in which set.

K-fold cross-validation is the standard fix. You split the training data into k equal parts (folds). You then train the model k times; each time, one fold is held out for validation and the other $k - 1$ are used for training. Averaging the k resulting scores gives a much steadier estimate of model quality than any single split.

Five-fold and ten-fold cross-validation are the usual choices, trading off computation against reliability. *Leave-one-out* cross-validation is the extreme case where k equals the number of training examples — it uses the data maximally but can be slow and, for some models, overly optimistic.

A couple of biological wrinkles are worth flagging. **Stratified** cross-validation keeps the same proportion of each class in every fold, which matters whenever your classes are im-

balanced — say, many more normal samples than tumours. And when samples are *not* independent — repeated measures from the same patient, or cells from the same donor — you must keep related samples together in the same fold, or the model can “peek” and your estimate becomes too rosy.

Cross-validation does triple duty: it helps you pick the best algorithm from a set of candidates, tune hyperparameters within an algorithm, and estimate realistic performance on new data. But remember the golden rule from earlier — every one of those cross-validation results still comes from the *training* data. The final verdict always comes from a completely separate test set.

19.6. Exercises

These are concept-checking exercises; no coding required. Try each one before opening the solution.

1. **Name the paradigm and the task.** For each scenario, say whether it is supervised, unsupervised, or reinforcement learning. If it is supervised, say whether it is classification or regression.
 - a. You have RNA-seq profiles from 300 breast-tumour samples, each labelled with one of four molecular subtypes, and you want to assign subtypes to new tumours.
 - b. You have DNA-methylation profiles from 1,000 people of known chronological age, and you want to predict the age of a new person.
 - c. You have single-cell expression profiles from a tissue with no prior labels, and you want to discover how many distinct cell populations are present.

i Solution

- a. **Supervised, classification** — the answer is one of a fixed set of categories (the four subtypes).
- b. **Supervised, regression** — age is a continuous number.
- c. **Unsupervised** (clustering) — there are no labels to predict; you are discovering structure in the data.

2. **Spot the leakage.** A colleague reports 99% accuracy classifying tumour vs normal. You discover they tuned the model and reported accuracy on the *same* set of samples. What is wrong, and what should they have done instead?

i Solution

They evaluated on data the model had already seen during development, so the 99% is optimistic and will not hold on new samples — exactly the “exam questions seen while studying” trap. They should have held out a **test set** that played no role in training or tuning, and reported accuracy on that. Cross-validation on the training portion could guide the tuning, but the final number must come from the untouched test set.

3. **Underfit or overfit?** During development you see 98% accuracy on the training data but only 62% on the validation data. Is this underfitting or overfitting? Name one thing you could try.

i Solution

This large train-versus-validation gap is the signature of **overfitting**: the model has learned the training examples rather than the general pattern. You could add **regularization**, **select fewer features**, choose a simpler model, or gather more training data. (Underfitting would instead show *poor* scores on *both* sets.)

4. **Why fold at all?** You have only 80 labelled samples. Why might five-fold cross-validation give you a more trustworthy performance estimate than a single 80/20 train-test split?

i Solution

With only 80 samples, a single test set of 16 is small, and its score depends heavily on which samples happen to land in it. Five-fold cross-validation lets *every* sample serve as validation data exactly once and averages five estimates, giving a steadier, less luck-dependent picture of how the model performs.

19.7. Summary

You can now place a biological question into the right machine-learning family — **supervised** when you have labelled examples to predict from (and within that, **classification** for categories, **regression** for numbers), **unsupervised** when you are exploring unlabelled data for structure, and **reinforcement** for learning by interaction. You have walked the **workflow** from problem definition through data preparation, splitting, model selection, training, and evaluation, and you understand why the **test set** must stay untouched until the very end. You can recognize **overfitting** and **underfitting**, read them off the

19. Introduction to machine learning

gap between training and validation scores, and explain how **cross-validation** and the **bias–variance tradeoff** help you find the model that generalizes.

With these concepts in hand, you are ready for the [tour of supervised algorithms](#), where each of these ideas becomes runnable R code on real biological data.

20. Supervised Learning

Imagine you have measured a few hundred penguins on a remote island — the length and depth of each bird’s bill, the length of its flipper. Now a new bird shows up. Can you predict its **body mass** from those measurements alone? Or guess which **species** it belongs to? That is exactly the kind of question supervised learning answers: you have examples where you know the answer, and you want a rule that predicts the answer for a new case you haven’t seen.

The same shape of problem shows up everywhere in biology. Predict a tumour’s response to a drug from its gene-expression profile. Predict a patient’s blood pressure from a panel of clinical measurements. Classify a cell as healthy or diseased from its morphology. In every case you have **features** (the inputs you measure) and a **target** (the answer you want to predict), and a pile of past examples to learn from.

In the [previous chapter](#) we met the three paradigms of machine learning and the all-important habit of splitting data to avoid fooling ourselves. This chapter is a guided tour of the workhorse **supervised** algorithms — linear regression, k-nearest neighbours, ridge/LASSO/elastic-net, decision trees, and random forests. For each one you’ll get a plain-language intuition and a small, runnable example on a single shared dataset, so you can see the algorithms as variations on one theme rather than a zoo of unrelated tricks. (The later [mlr3verse chapter](#) shows how to run all of these through one tidy interface; here we stay close to base R so the moving parts stay visible.)

20.1. What you’ll learn

- Describe any supervised algorithm in terms of its **hypothesis space**, **objective function**, and **learning algorithm**.
- Fit and interpret a **linear regression** with `lm()`.
- Make predictions with **k-nearest neighbours** using `class::knn()`, and explain why feature scaling matters.
- Fit **ridge**, **LASSO**, and **elastic-net** models with `glmnet()`, and read off which features each one keeps.
- Grow and read a **decision tree** with `rpart`, and a **random forest** with `ranger`.

20.2. A framework for understanding supervised algorithms

Every supervised algorithm in this chapter — and almost every one you’ll meet elsewhere — can be taken apart into the same three pieces. Once you can name those three pieces for an algorithm, you understand most of what makes it tick, and you can compare any two algorithms cleanly.

! The three pillars of a supervised algorithm

1. **Hypothesis space (the model family).** The set of all candidate models the algorithm is allowed to consider. For linear regression it’s “all straight lines”; for a decision tree it’s “all branching yes/no rule sets.” This choice bakes in an assumption — a *bias* — about what the answer can look like.
2. **Objective function (the goal).** A number that says how *good* a candidate model is on the training data — usually how badly it errs, so smaller is better. For regression that’s often the *mean squared error*. Learning means finding the model that makes this number as small as possible.
3. **Learning algorithm (the optimiser).** The actual procedure that searches the hypothesis space for the model with the best objective value. Sometimes it’s a one-shot formula (ordinary linear regression); sometimes it’s a long iterative grind (neural networks).

Here’s the payoff of this framing. Ridge and LASSO regression, which we’ll meet shortly, use the *same* hypothesis space and *same* style of optimiser as plain linear regression. The *only* thing they change is the objective function — they add a penalty for being too complicated. That one small change is what lets them avoid overfitting. Keep the three pillars in mind and each new algorithm becomes a small variation, not a whole new world.

20.3. Our shared dataset: Palmer penguins

To keep the tour concrete, we’ll use one small, friendly biological dataset throughout: measurements of penguins from three species on islands in the Palmer Archipelago, Antarctica. It lives in the `palmerpenguins` package.

```
library(palmerpenguins)
library(dplyr)

penguins_clean <- penguins |>
```

20. Supervised Learning

```
na.omit()          # drop the handful of rows with missing measurements

glimpse(penguins_clean)
```

```
Rows: 333
Columns: 8
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, 36.7, 39.3, 38.9, 39.2, 41.1, 38.6~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, 19.3, 20.6, 17.8, 19.6, 17.6, 21.2~
$ flipper_length_mm <int> 181, 186, 195, 193, 190, 181, 195, 182, 191, 198, 18~
$ body_mass_g   <int> 3750, 3800, 3250, 3450, 3650, 3625, 4675, 3200, 3800~
$ sex          <fct> male, female, female, female, male, female, male, fe~
$ year         <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

Each row is one bird. We'll use two prediction problems throughout:

- **Regression:** predict `body_mass_g` (a continuous number) from the body measurements.
- **Classification:** predict `species` (one of three categories) from the same measurements.

As we insisted in the previous chapter, we never judge a model on the same data we trained it on. So let's split once, here, into a **training** set and a **test** set, and reuse that split for every algorithm.

```
set.seed(42)
n <- nrow(penguins_clean)
train_idx <- sample(seq_len(n), size = 0.7 * n) # 70% for training
train <- penguins_clean[train_idx, ]
test <- penguins_clean[-train_idx, ]
c(train = nrow(train), test = nrow(test))
```

```
train  test
233    100
```

We now have 233 birds to learn from and 100 held back to test on honestly.

20.4. Linear regression

Plain-language intuition. Linear regression draws the best straight-line relationship between your features and a continuous target. “Best” means the line that makes the total squared prediction error as small as possible. Each feature gets a *coefficient* — a slope — telling you how much the target changes when that feature goes up by one unit. It is the simplest, most interpretable model in the toolbox, and a sensible first thing to try.

In the language of the three pillars: the hypothesis space is “all straight-line combinations of the features,” the objective function is the sum of squared errors (Equation 20.1), and the learning algorithm is a tidy formula that R solves instantly.

$$L = \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (20.1)$$

Here $\hat{y}_i$ is the predicted value, y_i is the true value, and n is the number of observations. Let’s predict penguin body mass from flipper length and bill dimensions:

```
fit_lm <- lm(body_mass_g ~ flipper_length_mm + bill_length_mm + bill_depth_mm,
             data = train)
coef(fit_lm)
```

(Intercept)	flipper_length_mm	bill_length_mm	bill_depth_mm
-6359.49030	50.52663	1.88153	18.08102

Read the coefficients as “holding the others fixed, one extra millimetre of flipper length is associated with about 51 grams more body mass.” That is the chief virtue of linear regression: the model *is* its explanation.

Now the honest test — how far off are the predictions on the held-out birds? We summarise that with the **root mean squared error** (RMSE), roughly the typical prediction error in grams:

```
pred_lm <- predict(fit_lm, newdata = test)
rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
rmse(test$body_mass_g, pred_lm)
```

```
[1] 387.8589
```

So the model’s predictions are typically off by around that many grams — for birds weighing 3,000–6,000 g, not bad for three measurements and a straight line.

20.5. K-nearest neighbours

Plain-language intuition. k-nearest neighbours (k-NN) is the “you are who your neighbours are” algorithm. To predict the answer for a new penguin, it finds the k most similar penguins in the training data — the ones closest to it in measurement space — and lets them vote. For classification, the new bird is assigned the most common species among its k neighbours. For regression, it gets the average of their values.

k-NN is unusual: it has no real “training” step and no equation to fit. It just stores the data and measures distances at prediction time. Its one knob is k : a small k (say 1) follows every wiggle in the data and can overfit; a large k smooths things out and can underfit. k is a **hyperparameter** you tune, exactly like we discussed in the previous chapter.

Scale your features first

k-NN decides “closeness” using distance, so a feature measured in large numbers will dominate one measured in small numbers — purely because of its units, not its importance. Flipper length (200 mm) would swamp bill depth (17 mm) if we left them as-is. The fix is to **standardise** every feature to a common scale before computing distances. Forgetting this is the single most common k-NN mistake.

Let’s classify penguin **species** from the three measurements. We standardise using the training set’s mean and standard deviation, then apply the *same* shift and scale to the test set:

```
library(class)

features <- c("flipper_length_mm", "bill_length_mm", "bill_depth_mm")

train_means <- colMeans(train[features])
train_sds   <- apply(train[features], 2, sd)

scale_with <- function(df) scale(df[features], center = train_means, scale = train_sds)
train_x <- scale_with(train)
test_x  <- scale_with(test)

set.seed(42)
knn_pred <- knn(train = train_x, test = test_x, cl = train$species, k = 5)

mean(knn_pred == test$species)  # accuracy: fraction classified correctly
```



```
[1] 0.98
```

With $k = 5$ neighbours, k-NN gets the species right for the large majority of held-out birds. The three penguin species really do cluster into distinct regions of measurement space, which is exactly the situation where “ask your neighbours” works beautifully.

20.6. Penalized regression

Some material in this section is adapted from the [STHDA penalized-regression tutorial](#).

Plain linear regression has a weakness: when you have many features — especially more features than observations, the common situation in genomics — it can chase noise and overfit. **Penalized regression** fixes this by tweaking the *objective function* (pillar two). It keeps the squared-error term but adds a **penalty** that grows as the coefficients get large, gently pressuring the model to stay simple.

A single dial, **lambda** (λ), controls how hard that pressure pushes. When $\lambda = 0$ there’s no penalty and you’re back to ordinary linear regression. As λ grows, coefficients are squeezed toward zero. The three methods below — ridge, LASSO, and elastic net — differ only in the *shape* of the penalty.

We’ll fit all three with the `glmnet` package. `glmnet` does not take a formula; it wants a numeric **matrix** of features $\mathbf{x}$ and a response vector $\mathbf{y}$. The `model.matrix()` step below builds that matrix (and drops the intercept column with `[, -1]`):

```
library(glmnet)

x_train <- model.matrix(body_mass_g ~ flipper_length_mm + bill_length_mm +
                        bill_depth_mm, data = train)[, -1]
y_train <- train$body_mass_g

x_test <- model.matrix(body_mass_g ~ flipper_length_mm + bill_length_mm +
                      bill_depth_mm, data = test)[, -1]
y_test <- test$body_mass_g
```

20.6.1. Ridge regression

Plain-language intuition. Ridge adds an **L2 penalty**: the sum of the *squared* coefficients (Equation 20.2). This shrinks every coefficient toward zero but never quite *to* zero

20. Supervised Learning

— so ridge keeps all your features, just turns down their volume. It's the right choice when you believe many features each contribute a little.

$$L = \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^k \beta_j^2 \quad (20.2)$$

In `glmnet`, ridge is `alpha = 0`. We use `cv.glmnet()` to let cross-validation pick a good λ automatically:

```
set.seed(42)
fit_ridge <- cv.glmnet(x_train, y_train, alpha = 0)
coef(fit_ridge, s = "lambda.min")
```

```
4 x 1 sparse Matrix of class "dgCMatrix"
               lambda.min
(Intercept)   -4592.10682
flipper_length_mm  41.68954
bill_length_mm   13.28295
bill_depth_mm   -10.47499
```

Notice that all three coefficients are present and non-zero — ridge shrank them but kept them. Its test error:

```
pred_ridge <- predict(fit_ridge, newx = x_test, s = "lambda.min")
rmse(y_test, pred_ridge)
```

```
[1] 396.0116
```

20.6.2. LASSO regression

Plain-language intuition. LASSO (Least Absolute Shrinkage and Selection Operator) adds an **L1 penalty**: the sum of the *absolute* coefficients (Equation 20.3). The difference from ridge sounds small but matters a lot: the L1 penalty can push a coefficient *exactly* to zero, effectively deleting that feature from the model. So LASSO does **automatic feature selection** — it hands you a simpler model that uses only the features that earn their keep.

20. Supervised Learning

$$L = \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^k |\beta_j| \quad (20.3)$$

LASSO is `alpha = 1` in `glmnet`:

```
set.seed(42)
fit_lasso <- cv.glmnet(x_train, y_train, alpha = 1)
coef(fit_lasso, s = "lambda.min")
```

```
4 x 1 sparse Matrix of class "dgCMatrix"
      lambda.min
(Intercept) -5499.365152
flipper_length_mm 47.897811
bill_length_mm 1.414023
bill_depth_mm .
```

Any coefficient shown as `.` has been zeroed out and dropped. With only three strong features here, LASSO may keep all of them — but on a genomics dataset with thousands of features, this is how you go from “20,000 genes” to “the handful that actually predict the outcome.”

💡 Ridge or LASSO? Let the data decide

Neither always wins. LASSO shines when a few features carry the signal and the rest are noise (so zeroing them helps). Ridge shines when *many* features each contribute a bit. When you’re unsure — which is most of the time — fit both with cross-validation and compare their test error, exactly as we’re doing here.

20.6.3. Elastic net

Plain-language intuition. Why choose? Elastic net **blends** the L1 and L2 penalties, so it can zero out useless features (the LASSO trick) *while* gracefully handling groups of correlated features (the ridge strength). When two features are highly correlated, pure LASSO tends to arbitrarily keep one and drop the other; elastic net is happier keeping both, which is common and useful in biology where genes in the same pathway move together.

20. Supervised Learning

The mixing dial is `alpha`, between 0 and 1: `alpha = 0` is pure ridge, `alpha = 1` is pure LASSO, and anything in between is elastic net. Let's use a 50/50 blend, `alpha = 0.5`:

```
set.seed(42)
fit_enet <- cv.glmnet(x_train, y_train, alpha = 0.5)

coef(fit_enet, s = "lambda.min")
```

```
4 x 1 sparse Matrix of class "dgCMatrix"
               lambda.min
(Intercept)    -6166.200623
flipper_length_mm  49.790369
bill_length_mm    2.389242
bill_depth_mm     14.153060
```

```
pred_enet <- predict(fit_enet, newx = x_test, s = "lambda.min")
rmse(y_test, pred_enet)
```

```
[1] 388.1342
```

Elastic net gives you a coefficient profile somewhere between ridge's “keep everything, shrunk” and LASSO's “keep a few.” On this small three-feature problem all four regression models land in roughly the same place; the penalized methods earn their keep when features are many and correlated.

20.7. Classification and regression trees (CART)

Plain-language intuition. A decision tree predicts by asking a series of simple yes/no questions about the features, like a flow chart or the game of Twenty Questions. “Is the flipper longer than 206 mm? If yes, is the bill deeper than 17 mm?” Each question splits the data into purer and purer groups, until a final **leaf** gives the prediction — the majority species (for classification) or the average value (for regression).

In three-pillar terms: the hypothesis space is “all such question trees,” the objective at each split is to make the resulting groups as *pure* as possible (measured by the Gini index or entropy), and the learning algorithm greedily picks the best split at each step. Trees are prized because you can *read* them — they match how people naturally reason.

We'll grow a classification tree for species with `rpart` (which ships with R):

20. Supervised Learning

```
library(rpart)

set.seed(42)
fit_tree <- rpart(species ~ flipper_length_mm + bill_length_mm + bill_depth_mm,
                  data = train, method = "class")

tree_pred <- predict(fit_tree, newdata = test, type = "class")
mean(tree_pred == test$species)  # test accuracy
```

```
[1] 0.94
```

The tree learned a short list of measurement thresholds that separate the species, and it classifies the held-out birds with high accuracy. We can even print the learned rules:

```
fit_tree
```

```
n= 233
```

```
node), split, n, loss, yval, (yprob)
```

```
  * denotes terminal node
```

```
1) root 233 131 Adelie (0.43776824 0.19742489 0.36480687)
  2) flipper_length_mm< 206.5 142 42 Adelie (0.70422535 0.29577465 0.00000000)
    4) bill_length_mm< 44.2 101 3 Adelie (0.97029703 0.02970297 0.00000000) *
    5) bill_length_mm>=44.2 41 2 Chinstrap (0.04878049 0.95121951 0.00000000) *
  3) flipper_length_mm>=206.5 91 6 Gentoo (0.02197802 0.04395604 0.93406593)
    6) bill_depth_mm>=17.2 7 3 Chinstrap (0.28571429 0.57142857 0.14285714) *
    7) bill_depth_mm< 17.2 84 0 Gentoo (0.00000000 0.00000000 1.00000000) *
```

Figure 20.1 shows the same idea drawn out for a clinical example: each internal node tests one feature, and following the branches leads to a leaf with a predicted class.

A single tree is twitchy

Decision trees have a real flaw: they're **unstable**. Change a few training rows and you can get a noticeably different tree. A single deep tree also overfits easily. The fix — averaging many trees together — is exactly the idea behind random forests, next.

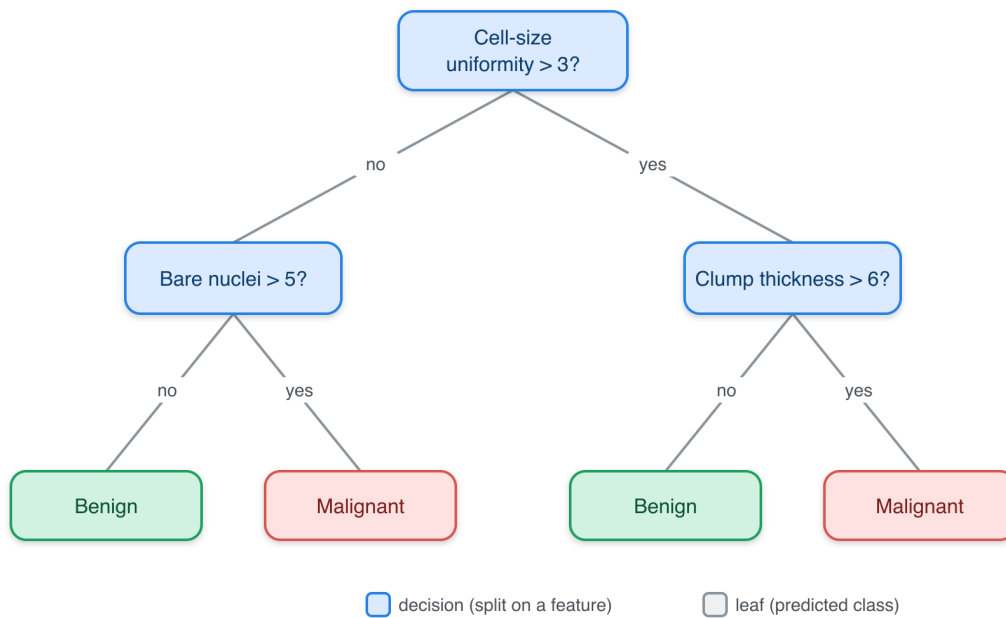


Figure 20.1.: A classification tree for a breast-cancer screening example. Each internal (blue) node tests one feature; following the branch that matches a sample leads, eventually, to a leaf (green/red) that gives the predicted class.

20.8. Random forests

Plain-language intuition. If one decision tree is twitchy, build *hundreds* of them and let them vote. A **random forest** grows many trees, each on a random bootstrap sample of the data and each allowed to consider only a random subset of features at every split. That deliberate randomness makes the individual trees disagree in their mistakes — and when you average many imperfect-but-different predictions, the errors cancel out. The result is far more stable and accurate than any single tree, while needing very little tuning. Figure 20.2 sketches the idea: many trees, one combined vote.

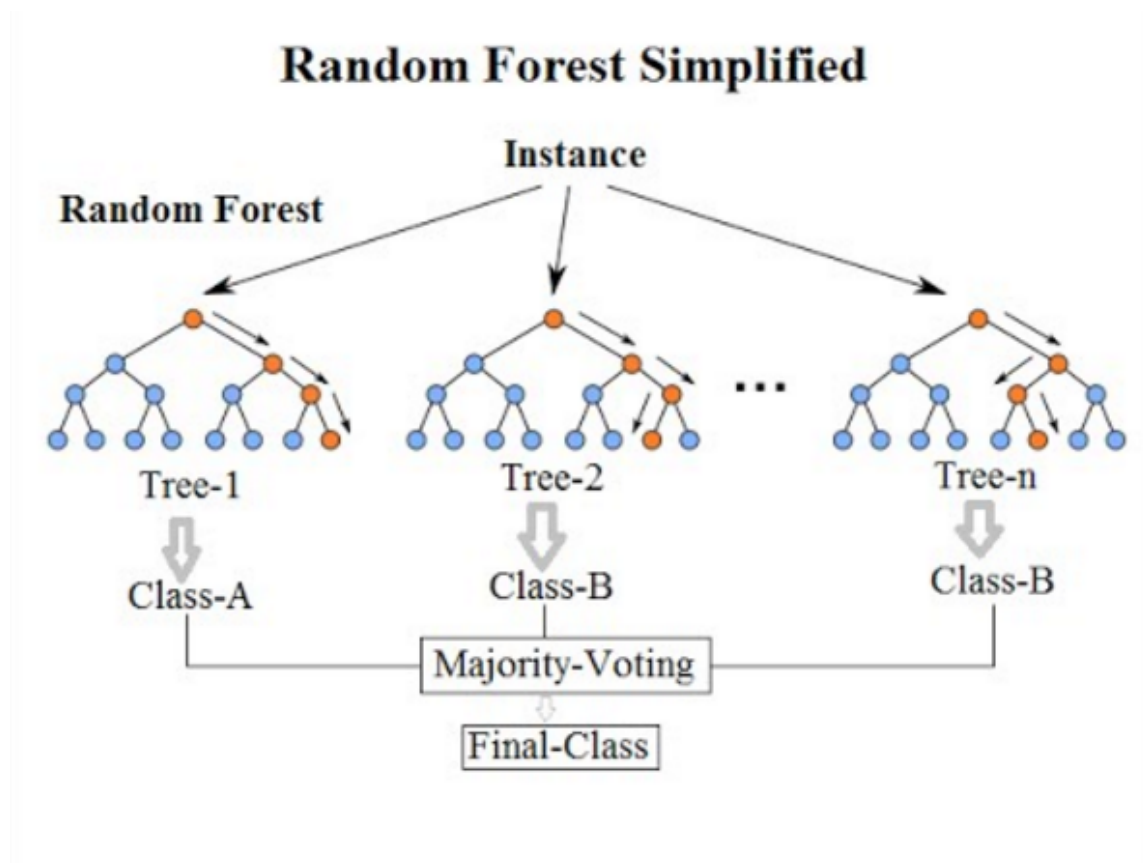


Figure 20.2.: Graphical representation of a random forest: many decision trees, each grown on a slightly different view of the data, combine their votes into one prediction.

We'll use the fast **ranger** package. Random forests are randomised, so we set a seed for a reproducible result:

```
library(ranger)

set.seed(42)
fit_rf <- ranger(species ~ flipper_length_mm + bill_length_mm + bill_depth_mm,
                 data = train)

rf_pred <- predict(fit_rf, data = test)$predictions
mean(rf_pred == test$species) # test accuracy
```

```
[1] 0.97
```

The forest typically matches or beats the single tree on the held-out birds, with no manual tuning — which is exactly why random forests are such a popular, reliable default for tabular biological data.

20.9. Summary

You now have a working map of the supervised-learning landscape, and you’ve run every algorithm on it with your own hands:

- Every algorithm is a **hypothesis space** + an **objective function** + a **learning algorithm**. Name those three and you understand it.
- **Linear regression** (`lm()`) fits an interpretable straight-line model; coefficients tell you how each feature moves the target.
- **k-nearest neighbours** (`class::knn()`) predicts from the most similar training examples — remember to **scale** your features first.
- **Penalized regression** (`glmnet()`) adds a complexity penalty to fight overfitting: **ridge** (`alpha = 0`) shrinks all coefficients, **LASSO** (`alpha = 1`) zeroes some out for feature selection, and **elastic net** (`0 < alpha < 1`) blends the two.
- **Decision trees** (`rpart`) are readable flow charts but unstable on their own; **random forests** (`ranger`) average many trees into a stable, accurate, low-fuss predictor.

In every case we trained on one split and judged on a held-out split — the habit that keeps you honest. The next chapters show how to run all of these through the unified `mlr3verse` interface, so swapping one algorithm for another becomes a one-line change.

20.10. Exercises

1. **A simpler linear model.** Refit the linear regression for `body_mass_g` using *only* `flipper_length_mm`. Compute its test RMSE and compare it to the three-feature model. Did dropping two features hurt much?

i Solution

```
fit_simple <- lm(body_mass_g ~ flipper_length_mm, data = train)
pred_simple <- predict(fit_simple, newdata = test)
rmse(test$body_mass_g, pred_simple)
```

```
[1] 389.7472
```

Flipper length alone is a strong predictor of body mass, so the simpler model is usually only a little worse. That's a useful reminder: more features are not automatically better.

2. **Tuning `k` in `k`-NN.** Try `k = 1`, `k = 5`, and `k = 25` for the species classifier. Which gives the best test accuracy? What goes wrong at the extremes?

i Solution

```
set.seed(42)
for (k in c(1, 5, 25)) {
  pred_k <- knn(train = train_x, test = test_x, cl = train$species, k = k)
  cat("k =", k, " accuracy =", round(mean(pred_k == test$species), 3), "\n")
}
```

```
k = 1  accuracy = 0.98
```

```
k = 5  accuracy = 0.98
```

```
k = 25 accuracy = 0.96
```

Very small `k` (like 1) follows noise and can overfit; very large `k` blurs the class boundaries and underfits. A middle value usually wins — which is why `k` is a hyperparameter worth tuning.

3. **Watch `LASSO` select.** Push the LASSO penalty up by using `lambda.1se` (a more aggressive, simpler model) instead of `lambda.min`, and look at the coefficients. Are any features dropped?

i Solution

```
coef(fit_lasso, s = "lambda.1se")
```

```
4 x 1 sparse Matrix of class "dgCMatrix"
```

```
              lambda.1se
(Intercept)   -4070.4745
flipper_length_mm  41.1108
bill_length_mm      .
bill_depth_mm      .
```

`lambda.1se` is the largest penalty whose error is still within one standard error of the best — a deliberately simpler model. Any coefficient printed as `.` has been set to exactly zero and removed. With strong features here it may keep them all, but the mechanism is the same one that prunes thousands of genes down to a few.

4. **Feature importance from the forest.** Refit the random forest with `importance = "impurity"` and print `fit_rf$variable.importance`. Which measurement matters most for telling the species apart?

i Solution

```
set.seed(42)
fit_rf_imp <- ranger(species ~ flipper_length_mm + bill_length_mm + bill_depth_mm,
                     data = train, importance = "impurity")
sort(fit_rf_imp$variable.importance, decreasing = TRUE)
```

```
      bill_length_mm flipper_length_mm      bill_depth_mm
      58.71576         47.88220         40.17413
```

The feature with the largest importance score contributed most to splitting the species apart across the forest's trees. Importance scores are one of the friendliest things random forests give you — a ranked list of which measurements carry the signal.

5. **Regression, three ways.** You have test RMSE for linear regression, ridge, and elastic net. Compute it for LASSO too, then state which model you'd ship and why.

i Solution

```
pred_lasso <- predict(fit_lasso, newx = x_test, s = "lambda.min")
c(linear = rmse(y_test, pred_lm),
  ridge  = rmse(y_test, pred_ridge),
  lasso  = rmse(y_test, pred_lasso),
  enet   = rmse(y_test, pred_enet))
```

```
linear    ridge    lasso    enet
387.8589  396.0116  390.8471  388.1342
```

On this small, low-dimensional problem the four are nearly tied, so you'd pick the simplest and most interpretable — plain linear regression. The penalized methods pull ahead only when features are numerous and correlated, as in real genomic data.

21. The mlr3verse framework

In the [previous chapter](#) you ran a whole tour of supervised algorithms — linear regression, k-nearest neighbours, penalized regression, decision trees, random forests — on the Palmer penguins. Each one came from a different R package with its own habits: `lm()` wants a formula, `glmnet()` wants a matrix, `knn()` wants the training and test sets handed in separately. Swapping one algorithm for another meant rewriting the surrounding plumbing every time.

That friction is exactly what the **mlr3verse** removes. It wraps every algorithm in one consistent interface, so the question “what happens if I try a random forest instead of a tree?” becomes a one-line change instead of a rewrite. This chapter is a guided tour of the four object types that interface is built from. We’ll meet them on the same penguins you already know — predicting a bird’s **species** from its morphological measurements — so you can focus on the framework rather than on a new dataset.

21.1. What you’ll learn

- Explain the four core mlr3 object types — **Task**, **Learner**, **Measure**, **Resampling** — and what each one is responsible for.
- Build a classification **Task** from a data frame and read off its target and features with `task$target_names` and `task$feature_names`.
- Discover available algorithms in the **learner registry** with `mlr_learners$keys()` and create one with `lrn()`.
- Inspect a learner’s tunable **hyperparameters** with `learner$param_set`.
- Choose a **Measure** with `msr()` and a **Resampling** strategy with `rsmp()`.

21.2. Why a framework at all?

The R ecosystem offers dozens of machine-learning packages, each with its own interface and conventions. That diversity is powerful, but it also means you spend time relearning a new API for every algorithm and rewriting the same data-preparation code over and over.

21. The *mlr3*verse framework

The *mlr3*verse fixes this by giving you one consistent, object-oriented framework that all the algorithms plug into.

The design follows a simple idea: **separate the moving parts**. Instead of one giant function that does everything from preprocessing to evaluation, *mlr3* splits the workflow into four kinds of object, each with one job.

- **Tasks** hold the data and define the problem (what are we predicting?).
- **Learners** are the algorithms (how do we learn the rule?).
- **Measures** score performance (how good is the result?).
- **Resamplings** control how data is split for honest validation (did it generalize?).

! The big idea: interchangeable parts

Think of *mlr3*'s four object types as the standardized, swappable components of a machine. A Task is the workpiece, a Learner is a tool head, a Measure is the gauge you check the result with, and a Resampling is the jig that holds everything for a fair test. Because every component shares the same fittings, you can pop out the decision-tree tool head and drop in a random forest without rebuilding the machine. That single design choice — common fittings everywhere — is what makes “try another algorithm” a one-line edit instead of a rewrite. Learn the four sockets once and every algorithm in the ecosystem clicks into them the same way.

The framework is built on R6 classes, which is why you'll see the `object$method()` syntax (a dollar sign and parentheses) throughout, rather than the `function(object)` style you may be used to. It also takes reproducibility seriously: every random step can be pinned down with `set.seed()`, which we'll do whenever a step involves chance.

21.3. The *mlr3*verse ecosystem

*mlr3*verse is a meta-package: loading it pulls in a family of focused packages that build on the core. You rarely need to load them individually.

Package	Purpose
<i>mlr3</i>	Core framework: tasks, learners, measures, resampling
<i>mlr3learners</i>	Interfaces to popular algorithms (random forest, glmnet, xgboost, ...)
<i>mlr3pipelines</i>	Preprocessing and model-composition pipelines

21. The *mlr3verse* framework

Package	Purpose
<code>mlr3tuning</code>	Hyperparameter optimization
<code>mlr3measures</code>	Additional evaluation metrics
<code>mlr3viz</code>	Plotting utilities for models and results

For this tour we only need the core ideas, all reachable through `mlr3verse`. Let's load it along with the penguins.

```
library(mlr3verse)
library(palmerpenguins)
library(data.table)

data(penguins)
penguins <- na.omit(penguins) # drop the few rows with missing measurements
```

We dropped the rows with missing values up front so the examples stay simple; in a real analysis you might instead impute them.

21.4. Tasks: the problem and the data

A **Task** bundles your data together with metadata about the problem: which column is the **target** (the answer you want to predict) and which columns are the **features** (the inputs you predict from). A *classification* task has a categorical target; a *regression* task has a continuous one. The Task also figures out feature types for you and offers helpers for filtering rows and selecting columns.

Our biological question is the same one from the previous chapter: can we predict a penguin's **species** from its body measurements? That makes **species** the target.

```
task <- as_task_classif(penguins, target = "species")

task$nrow # number of penguins (rows)
```

```
[1] 333
```

```
task$ncol # number of columns, including the target
```

```
[1] 8
```

```
task$feature_names # the predictor columns
```

```
[1] "bill_depth_mm"      "bill_length_mm"    "body_mass_g"
[4] "flipper_length_mm" "island"             "sex"
[7] "year"
```

```
task$target_names # the column we're predicting
```

```
[1] "species"
```

Reading that output top to bottom: the task holds 333 penguins, the `feature_names` are the morphological and contextual measurements we predict *from* (bill length and depth, flipper length, body mass, island, sex, year), and `target_names` confirms that `species` is what we’re predicting. Notice we never told mlr3 which columns were features — it took every column except the target automatically. That bookkeeping is the Task’s job, so you don’t have to track it yourself.

21.5. Learners: the algorithms

A **Learner** is an algorithm wrapped in the standard mlr3 interface. Every learner declares what it can do — which task types it supports, whether it tolerates missing values, and which hyperparameters it exposes — and provides the same `train()` and `predict()` methods regardless of the maths underneath.

You discover learners through the **registry**, `mlr_learners`. Querying it for classification learners answers the practical question “what algorithms can I even try on this task?”

```
classif_learners <- mlr_learners$keys("classif")
head(classif_learners, 10)
```

```
[1] "classif.cv_glmnet"      "classif.debug"      "classif.featureless"
[4] "classif.glmnet"         "classif.kknn"        "classif.lda"
[7] "classif.log_reg"        "classif.multinom"    "classif.naive_bayes"
[10] "classif.nnet"
```

21. The *mlr3*verse framework

Each entry is the id of an algorithm you can use on a classification task — for example `classif.rpart` (a decision tree) or `classif.ranger` (a random forest). This is the menu the framework offers for our species-prediction problem.

i Learner naming convention

`mlr3` names learners as `[task_type].[algorithm]`. So `classif.rpart` is a decision tree for classification, while `regr.lm` is linear regression for a regression task. Once you know this scheme you can guess a learner's name before you look it up.

Let's create the decision-tree learner and inspect its hyperparameters — the dials you can turn to change how it learns.

```
rpart_learner <- lrn("classif.rpart")
rpart_learner$param_set$ids() # the hyperparameters you can tune
```

```
[1] "cp"           "keep_model"    "maxcompete"    "maxdepth"
[5] "maxsurrogate" "minbucket"     "minsplit"      "surrogatestyle"
[9] "usesurrogate" "xval"
```

These are `rpart`'s tuning dials. A few you'll meet again: `cp` is the complexity penalty that controls how aggressively the tree is pruned (larger means a simpler tree), `maxdepth` caps how deep the tree may grow, and `minsplit` sets the smallest node the algorithm is willing to split. You don't have to set any of them — the learner ships with sensible defaults — but this is where you'd reach to fight overfitting.

21.6. Measures: scoring the result

A **Measure** is the yardstick you use to judge predictions. The choice matters, because different measures reward different things, and the measure you optimize is the behaviour you'll get. For classification you'll commonly reach for accuracy or its mirror image, classification error.

```
acc_measure <- msr("classif.acc") # accuracy: fraction predicted correctly
ce_measure  <- msr("classif.ce")  # classification error: fraction wrong (1 - accuracy)

acc_measure
```



```
-- <MeasureClassifSimple> (classif.acc): Classification Accuracy -----
* Packages: mlr3 and mlr3measures
* Range: [0, 1]
* Minimize: FALSE
* Average: macro
* Parameters: list()
* Properties: weights and obs_loss
* Predict type: response
* Predict sets: test
* Aggregator: mean()
```

Printing a measure tells you its key facts at a glance: its id (`classif.acc`), the range of values it can take (`[0, 1]` for accuracy), and `Minimize: FALSE` — meaning for *this* measure bigger is better, so the framework knows to maximize it when tuning. Classification error would show `Minimize: TRUE`, because there you want the number as small as possible.

21.7. Resamplings: honest validation

The cardinal rule from the previous chapters: never judge a model on the same data it trained on. A **Resampling** object encodes *how* you split the data to keep yourself honest. The common strategies are a single **holdout** split, **k-fold cross-validation** (split into k parts, train on $k-1$ and test on the held-out part, rotate), and the **bootstrap** (resample with replacement many times).

```
set.seed(42)
holdout <- rsmp("holdout", ratio = 0.8)      # 80% train, 20% test
cv5     <- rsmp("cv", folds = 5)            # 5-fold cross-validation
bootstrap <- rsmp("bootstrap", repeats = 30) # 30 bootstrap resamples

cv5$param_set
```

```
<ParamSet(1)>
      id    class lower upper nlevels    default  value
<char> <char> <int> <num>   <num>      <list> <list>
1: folds ParamInt      2   Inf      Inf <NoDefault[0]>      5
```

We seeded the chunk because holdout and bootstrap make random splits, and we want the same split every time the book is rebuilt. The printed `param_set` for `cv5` lists the strategy’s own settings: `folds` is the number of cross-validation folds (we set it to 5), and `repeats` is how many times the whole k-fold procedure is repeated (1 by default). Together these control the trade-off between a reliable performance estimate and how much computation it costs.

21.8. How the four fit together

Each object owns one part of the workflow, and they snap together in a predictable way: a **Learner** trains on a **Task** to produce a fitted model, that model makes predictions, a **Measure** scores those predictions, and a **Resampling** strategy wraps the whole thing so the score reflects performance on data the model has never seen. Because all learners share the same interface, swapping `classif.rpart` for `classif.ranger` changes one string and nothing else.

! Why this design pays off

- **Consistency** — every algorithm uses the same `train()/predict()` interface, so there’s one set of habits to learn, not one per package.
- **Flexibility** — mix and match tasks, learners, measures, and resamplings to run experiments by editing a single component.
- **Reproducibility** — randomness is controllable, so results are repeatable.
- **Best practices baked in** — the design nudges you toward train/test splitting and cross-validation rather than leaving them as an afterthought.

You’ve now met all four mlr3 building blocks on a problem you already understand. In the [next chapter](#) these pieces come together on real biological data — classifying cancer types from gene expression, predicting age from DNA methylation — where you’ll see a Learner actually train on a Task and have its predictions scored by a Measure under a Resampling. The concepts are the same; only the data gets bigger and more interesting.

21.9. Summary

You can now:

- Name the four core mlr3 objects and the single job each one does — **Task** (data + problem), **Learner** (algorithm), **Measure** (score), **Resampling** (validation strategy).

- Build a classification Task with `as_task_classif()` and inspect it with `task$feature_names` and `task$target_names`.
- List available algorithms with `mlr_learners$keys()`, create one with `lrn()`, and see its tunable dials with `learner$param_set$ids()`.
- Pick a Measure with `msr()` and a Resampling strategy with `rsmp()`, and read what their printouts are telling you.

The payoff is the interchangeable-parts design: learn the four sockets once, and every algorithm in the mlr3 ecosystem plugs into them the same way.

21.10. Exercises

These exercises reuse the penguins. Recall that in the previous chapter we also framed a **regression** problem — predicting a penguin's `body_mass_g` from its other measurements. Let's set that up in mlr3.

1. **Build a regression task.** Create a regression task that predicts `body_mass_g`, using `as_task_regr()`. Confirm the target and list the features.

i Solution

```
task_mass <- as_task_regr(penguins, target = "body_mass_g")
task_mass$target_names
```

```
[1] "body_mass_g"
```

```
task_mass$feature_names
```

```
[1] "bill_depth_mm"      "bill_length_mm"     "flipper_length_mm"
[4] "island"              "sex"                 "species"
[7] "year"
```

`as_task_regr()` is the regression counterpart of `as_task_classif()`. The target is now the continuous `body_mass_g`, and `species` has joined the feature list — any column that isn't the target becomes a feature.

2. **Find regression learners.** Query the registry for learners that work on a regression task, and show the first ten.

i Solution

```
regr_learners <- mlr_learners$keys("regr")
head(regr_learners, 10)
```

```
[1] "regr.cv_glmnet"      "regr.debug"          "regr.featureless"    "regr.glmnet"
[5] "regr.kknn"           "regr.km"              "regr.lm"              "regr.nnet"
[9] "regr.ranger"         "regr.rpart"
```

These are the `regr.*` algorithms — for example `regr.lm` (linear regression) and `regr.rpart` (a regression tree). The naming convention tells you each one's task type at a glance.

3. **Set up 10-fold cross-validation.** Create a resampling object for 10-fold cross-validation and confirm the fold count from its parameter set.

i Solution

```
cv10 <- rsmp("cv", folds = 10)
cv10$param_set$values
```

```
$folds
[1] 10
```

`rsmp("cv", folds = 10)` builds the strategy; `param_set$values` reports the settings you actually chose, confirming `folds` is 10. More folds give a more stable estimate of performance at the cost of more computation.

4. **Pick a regression measure.** Create the root-mean-squared-error measure with `msr("regr.rmse")` and print it. Does the framework want it minimized or maximized?

i Solution

```
rmse_measure <- msr("regr.rmse")
rmse_measure
```

```
-- <MeasureRegrSimple> (regr.rmse): Root Mean Squared Error -----
* Packages: mlr3 and mlr3measures
* Range: [0, Inf]
* Minimize: TRUE
* Average: macro
```

21. The *mlr3verse* framework

```
* Parameters: list()
* Properties: weights and obs_loss
* Predict type: response
* Predict sets: test
* Aggregator: mean()
```

The printout shows `Minimize: TRUE` — RMSE is an error, so smaller is better, and the framework will minimize it when comparing models. Contrast this with accuracy, which it maximizes.

22. Worked example: classifying cancer types from gene expression

The earlier machine learning chapters introduced the *ideas*: the difference between supervised and unsupervised learning ([Introduction](#)), the workhorse algorithms ([Supervised Learning](#)), and the `mlr3` framework that wraps all of them in one tidy interface ([The mlr3verse](#)). This is the first of three chapters where those ideas meet real data.

Here we tackle a **classification** problem: given a tumour’s gene-expression profile, can we predict which **cancer type** it is? This is the first of three worked examples — the next two predict age from DNA methylation and gene expression from histone marks — and because it comes first, it also lays down the habits the others reuse: how to set up an `mlr3` analysis, how to split the data, how to select features, and how to read the results without fooling yourself.

Three ideas from the [introduction](#) do the heavy lifting in every one of these examples, so it is worth stating them plainly here, in the concrete terms of gene-expression classification:

- **Feature selection.** A tumour profile measures tens of thousands of genes, but most of them barely change from one sample to the next. A gene that stays flat across every tumour cannot help tell cancer types apart — it is noise. So we *select* the genes that vary the most and discard the rest, shrinking the problem to the handful of features that actually carry signal.
- **The train/test split.** We hold back a slice of tumours that the model never sees while it learns. Scoring it on those unseen tumours is the only honest estimate of how it will classify a *new* patient’s biopsy — the thing we actually care about.
- **Overfitting.** A model given too much freedom — a decision tree grown too deep, say — can memorize the exact training tumours instead of learning a rule that generalizes. We catch this by watching the gap between the training score and the test score.

The next two chapters revisit all three from different biological angles; this is where we meet them as first principles.

i The through-line for all three worked examples

Feature selection, an honest train/test split, and vigilance against overfitting are the spine of every worked example. Here they take their plainest form: keep the most variable genes, hold out some tumours, and distrust a model that aces the training set but stumbles on the test set. Hold on to this picture — the methylation and histone-mark chapters return to each idea and add a new twist.

22.1. What you'll learn

- Turn a real genomics dataset into an `mlr3` **task** with `as_task_classif()`.
- **Split** data into training and test sets, and explain why that split protects you from fooling yourself.
- **Select features** by variance, and say why selection must happen on the training data only.
- **Train and predict** with several learners — kNN, a classification tree, and a random forest — using the same handful of `mlr3` calls.
- **Interpret** a confusion matrix and an accuracy, and recognize overfitting when the test score falls below the training score.

22.2. A quick recap of the `mlr3` workflow

You met the `mlr3` building blocks in [The `mlr3`verse](#); here is the one-paragraph reminder so you can keep your eyes on the biology. Every analysis in these worked-example chapters follows the same four-step rhythm (Figure [22.1](#)):

1. **Task** — bundle your data with the name of the column you want to predict (the *target*). Classification tasks predict a category; regression tasks predict a number.
2. **Learner** — pick an algorithm by name with `lrn()`, e.g. `lrn("classif.kknn")`.
3. **Train and predict** — call `learner$train(task, ...)` then `learner$predict(task, ...)`.
4. **Measure** — score the predictions with `msrs()`, e.g. accuracy for classification or R^2 for regression.

The power of `mlr3` is that this rhythm is *identical* no matter which algorithm you use. Swapping a k-nearest-neighbours model for a random forest is a one-line change to the `lrn()` call; everything else stays the same. For the full tour of tasks, learners, measures, and resamplings, see [The `mlr3`verse](#); the official [mlr3 book](#) is the deep reference.

22. Worked example: classifying cancer types from gene expression

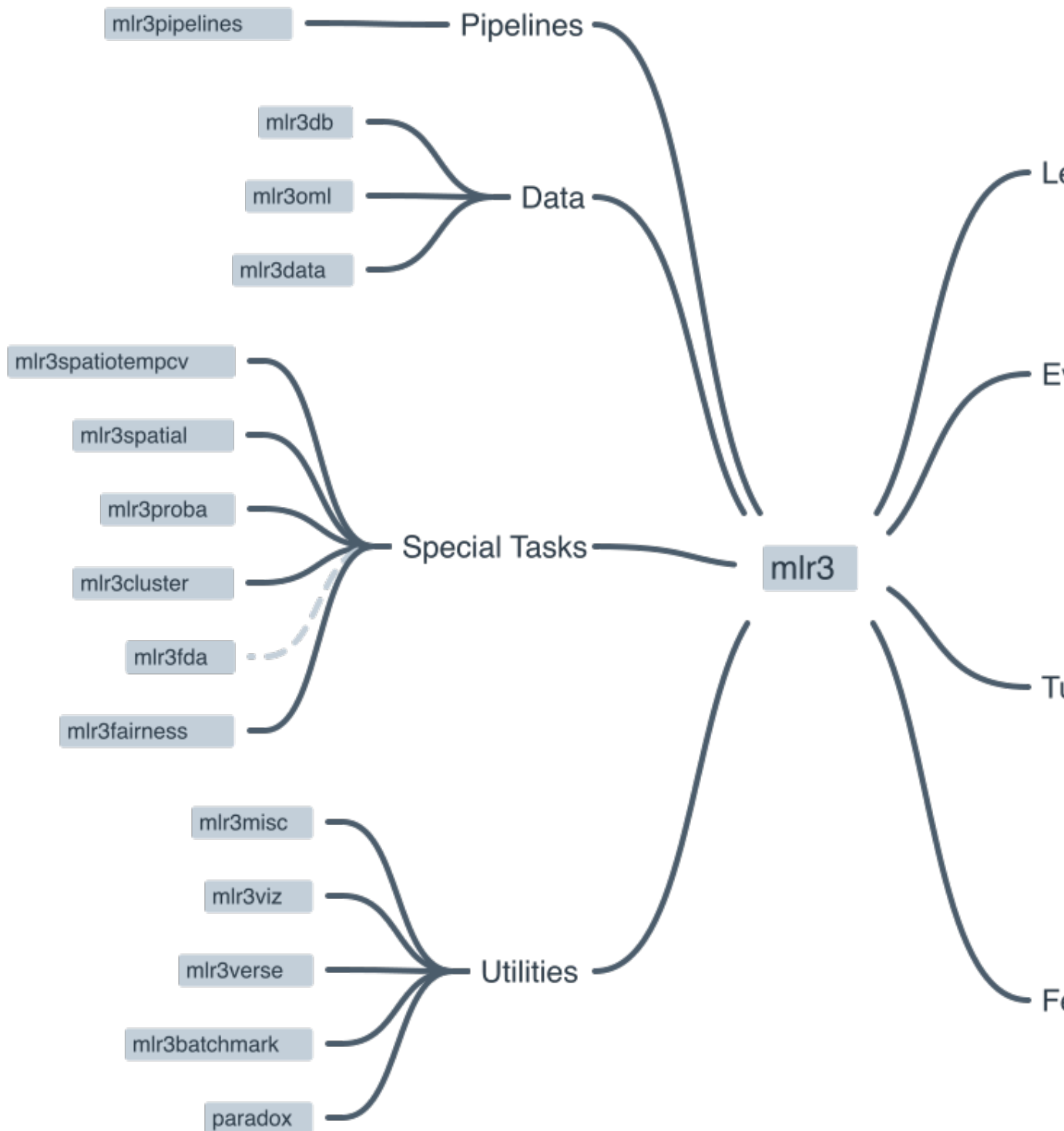


Figure 22.1.: The mlr3 ecosystem. A consistent set of objects — tasks, learners, measures, and resampling strategies — that snap together the same way for every algorithm.

22. Worked example: classifying cancer types from gene expression

! The single most important habit: split your data

A model that is *tested* on the same data it was *trained* on will look far better than it really is — it can simply memorize the answers. We always hold out a **test set** that the model never sees during training, and we report performance from that held-out set. This is the only honest estimate of how the model will behave on new patients or new genes.

22.3. Setup

```
library(mlr3verse)
library(GEOquery)
library(mlr3learners) # for knn
library(ranger)       # for random forests
set.seed(789)
```

22.4. Understanding the problem

In this example we classify **cancer types** from gene-expression data. The data come from Brouwer-Visser et al. (2018), available through the Gene Expression Omnibus (GEO), a public repository of functional-genomics data. The study profiled gene expression in tumours of several types, and our question is: **given a tumour's expression profile, can we predict which cancer type it is?**

Before building anything, it pays to answer four questions about the task:

- **What are the features?** The expression level of each gene.
- **What is the target?** The cancer type (a category).
- **What kind of task is this?** Classification — we are predicting a label, not a number.
- **What is the goal?** A model that assigns the correct cancer type to a tumour it has never seen.

22.5. Data preparation

We use the `GEQquery` package to fetch dataset [GSE103512](#). This downloads over the network, so give it a moment.

```
gse <- getGEO("GSE103512")[[1]]
```

`GEQquery` was written in 2007 and returns an older Bioconductor structure, the `ExpressionSet`. As a first housekeeping step we convert it to the modern `SummarizedExperiment` (covered in the [SummarizedExperiment chapter](#)).

```
library(SummarizedExperiment)
se <- as(gse, "SummarizedExperiment")
```

Let's look at two variables of interest: the cancer type and whether each sample is tumour or normal tissue.

```
with(colData(se), table(`cancer.type.ch1`, `normal.ch1`))
```

	normal.ch1	
cancer.type.ch1	no	yes
BC	65	10
CRC	57	12
NSCLC	60	9
PCA	60	7

This table tells us how many samples we have of each cancer type, split by tumour versus normal status. Most samples are tumours (**normal: no**), with a smaller set of matched normal tissues. We'll keep only the tumour samples for classification.

Always explore before you model

Missing values, outliers, and unexpected categories can quietly break a machine learning model. Plots, summaries, PCA, and clustering all help you understand the data first. Time spent here saves time later — and saves you from confidently reporting a result that was an artifact all along.

22.6. Feature selection and data cleaning

This dataset measures *thousands* of genes, but most of them barely change between samples and carry little signal. We will keep the 200 genes with the **highest variance** across samples, on the reasoning that a gene that never changes can't help us tell cancer types apart. (Other selection strategies exist, such as ranking genes by their correlation with the outcome.)

! Select features on the training data only

In a fully rigorous workflow, feature selection must use the **training data only** — never the test data. If you peek at the test data while choosing features, information leaks across the split and your test score is no longer honest. (For simplicity this example selects on the full matrix; in a real analysis, fold the selection step inside the training split.)

The `apply()` function applies a function to each row or column of a matrix. Here we apply `sd()` (standard deviation) to each row of the expression matrix to get a vector of per-gene variability, then keep the 200 most variable genes among the tumour samples.

```
sds <- apply(assay(se, 'exprs'), 1, sd)
## keep the 200 most-variable genes, tumour samples only
se_small <- se[order(sds, decreasing = TRUE)[1:200],
               colData(se)$characteristics_ch1.1 == 'normal: no']
# remove genes with no gene symbol
se_small <- se_small[rowData(se_small)$Gene.Symbol != '', ]
```

To make the data easier to work with, we use one of the `rowData` columns (the gene symbol) as the row names. `make.names()` ensures those names are valid, unique R variable names.

```
## convert to a matrix for later use
expr_mat <- assay(se_small, 'exprs')
rownames(expr_mat) <- make.names(rowData(se_small)$Gene.Symbol)
```

`mlr3` expects samples in rows and features in columns, so we **transpose** the matrix and attach the cancer type as a column.

```
feat_dat <- t(expr_mat)
tumor <- data.frame(tumor_type = colData(se_small)$cancer.type.ch1, feat_dat)
```

22. Worked example: classifying cancer types from gene expression

This is another good moment to check the data: confirm the dimensions, the column names, and the data types are what you expect before moving on.

22.7. Creating the task

The first `mlr3` step is to create a **task** — a dataset paired with the name of the target column. Because the target (cancer type) is a category, this is a *classification* task, so we use `as_task_classif()`. The target must be a factor.

```
tumor$tumor_type <- as.factor(tumor$tumor_type)
task <- as_task_classif(tumor, target = 'tumor_type')
```

22.8. Splitting the data

We randomly assign two-thirds of the samples to a **training set** and one-third to a **test set**. The model learns from the training set; we judge it on the test set it never saw. We seed the split so the result is reproducible.

```
set.seed(7)
train_set <- sample(task$row_ids, 0.67 * task$nrow)
test_set <- setdiff(task$row_ids, train_set)
```

In the next sections we train three different learners on this task — k-nearest-neighbours, a classification tree, and a random forest — and compare how each one does at predicting cancer type.

22.9. k-nearest-neighbours

The first learner is **k-nearest-neighbours** (kNN). Its idea is simple: a new sample is classified by looking at the k most similar samples and taking a vote. The number of neighbours, k , is a tunable setting; we'll use the default of 7. (`mlr3` can tune such settings automatically, but that's beyond this chapter.)

22.9.1. Create the learner

In `mlr3` an algorithm is a **learner**, created with `lrn()` by name.

```
learner <- lrn("classif.kknn")
```

Calling `lrn()` with no arguments lists every available learner.

```
lrn()
```

<DictionaryLearner> with 55 stored values

Keys: `classif.cv_glmnet`, `classif.debug`, `classif.featureless`,
`classif.glmnet`, `classif.kknn`, `classif.lda`, `classif.log_reg`,
`classif.multinom`, `classif.naive_bayes`, `classif.nnet`, `classif.qda`,
`classif.ranger`, `classif.rpart`, `classif.svm`, `classif.xgboost`,
`clust.agnes`, `clust.ap`, `clust.bico`, `clust.birch`, `clust.clara`,
`clust.cmeans`, `clust.cobweb`, `clust.dbscan`, `clust.dbscan_fpc`,
`clust.diana`, `clust.em`, `clust.fanny`, `clust.featureless`, `clust.ff`,
`clust.hclust`, `clust.hdbscan`, `clust.kkmeans`, `clust.kmeans`,
`clust.kproto`, `clust.MBatchKMeans`, `clust.mclust`, `clust.meanshift`,
`clust.optics`, `clust.pam`, `clust.protoclust`, `clust.SimpleKMeans`,
`clust.specc`, `clust.xmeans`, `regr.cv_glmnet`, `regr.debug`,
`regr.featureless`, `regr.glmnet`, `regr.kknn`, `regr.km`, `regr.lm`,
`regr.nnet`, `regr.ranger`, `regr.rpart`, `regr.svm`, `regr.xgboost`

22.9.2. Train

`learner$train()` takes the task and the row IDs of the training data.

```
learner$train(task, row_ids = train_set)
```

We *could* inspect the trained model, but for kNN the printout is large, so this chunk is left for you to run on your own.

```
# output is large, so do this on your own  
learner$model
```

22.9.3. Predict

Let's first predict the classes of the **training** data. We already know those answers, but predicting on the training set gives us a baseline we can compare against the test set to detect overfitting.

```
pred_train <- learner$predict(task, row_ids = train_set)
```

And now the **test** data — the honest evaluation:

```
pred_test <- learner$predict(task, row_ids = test_set)
```

22.9.4. Assess

A **confusion matrix** cross-tabulates the true class (rows) against the predicted class (columns). Correct predictions land on the diagonal; off-diagonal entries are mistakes.

```
pred_train$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	42	0	0	0
CRC	0	40	0	0
NSCLC	1	0	44	0
PCA	0	0	0	35

On the training data, almost every sample falls on the diagonal — the model classifies the data it learned from very well, as expected.

Accuracy summarizes this in one number: the fraction of samples classified correctly.

```
measures <- msrs(c('classif.acc'))
pred_train$score(measures)
```

```
classif.acc
0.9938272
```

Now the test set, which the model never saw during training:

22. Worked example: classifying cancer types from gene expression

```
pred_test$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	22	0	0	0
CRC	0	17	1	0
NSCLC	0	0	15	0
PCA	0	0	0	25

```
pred_test$score(measures)
```

```
classif.acc  
0.9875
```

The test accuracy is the number to trust. If it is noticeably lower than the training accuracy, the model has *overfit* — it memorized the training samples rather than learning a rule that generalizes. Compare the two numbers above: how big is the gap?

22.10. Classification tree

A **classification tree** asks a series of yes/no questions about the features — for example, “is gene *X* expressed above some threshold?” — and follows the answers down branches to a predicted class. At each split it picks the feature that best separates the classes, and it keeps splitting until the groups are reasonably pure. Trees are appealing because you can *read* them.

22.10.1. Train

We set `keep_model = TRUE` so we can inspect the fitted tree afterwards.

```
learner <- lrn("classif.rpart", keep_model = TRUE)  
learner$train(task, row_ids = train_set)
```

Here is the trained model:

22. Worked example: classifying cancer types from gene expression

```
learner$model
```

```
n= 162
```

```
node), split, n, loss, yval, (yprob)
```

```
* denotes terminal node
```

```
1) root 162 118 NSCLC (0.26543210 0.24691358 0.27160494 0.21604938)
  2) CDHR5>=5.101625 40 0 CRC (0.00000000 1.00000000 0.00000000 0.00000000) *
  3) CDHR5< 5.101625 122 78 NSCLC (0.35245902 0.00000000 0.36065574 0.28688525)
    6) ACPP< 6.088431 87 43 NSCLC (0.49425287 0.00000000 0.50574713 0.00000000)
      12) GATA3>=4.697803 41 1 BC (0.97560976 0.00000000 0.02439024 0.00000000) *
      13) GATA3< 4.697803 46 3 NSCLC (0.06521739 0.00000000 0.93478261 0.00000000) *
        7) ACPP>=6.088431 35 0 PCA (0.00000000 0.00000000 0.00000000 1.00000000) *
```

Trees are easy to visualize when they are small. This one has only a couple of splits, which makes the decision logic easy to follow.

```
library(mlr3viz)
```

```
library(ggparty)
```

```
Loading required package: ggplot2
```

```
Loading required package: partykit
```

```
Loading required package: grid
```

```
Loading required package: libcoin
```

```
Loading required package: mvtnorm
```

```
Attaching package: 'partykit'
```

```
The following object is masked from 'package:SummarizedExperiment':
```

```
width
```


22. Worked example: classifying cancer types from gene expression

The following object is masked from 'package:GenomicRanges':

```
width
```

The following object is masked from 'package:IRanges':

```
width
```

The following object is masked from 'package:S4Vectors':

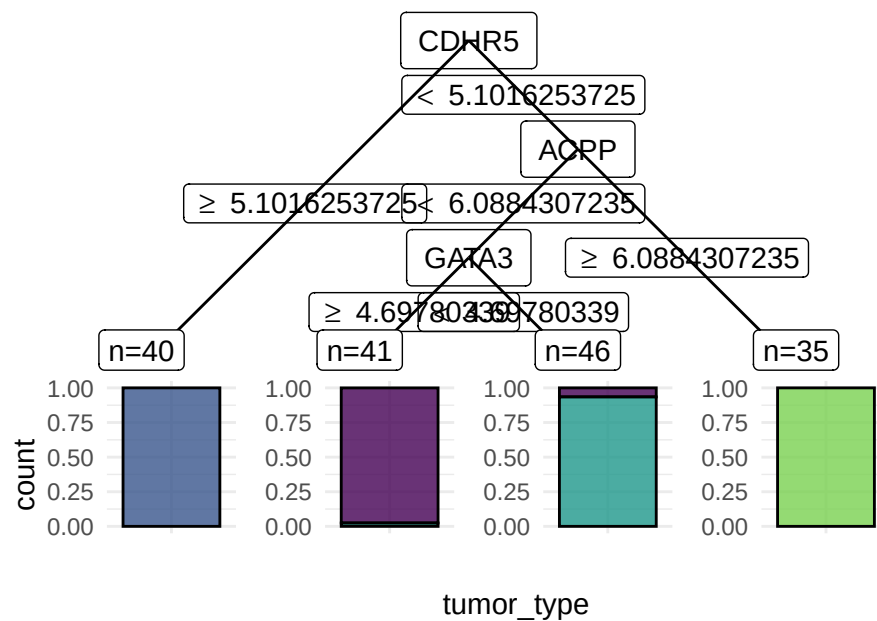
```
width
```

The following object is masked from 'package:BiocGenerics':

```
width
```

```
autoplot(learner, type = 'ggparty')
```

Warning in ggparty::geom_edge_label(): Ignoring unknown parameters:
`label.size`



22. Worked example: classifying cancer types from gene expression

22.10.2. Predict

As before, we predict on both the training and test sets.

```
pred_train <- learner$predict(task, row_ids = train_set)
pred_test  <- learner$predict(task, row_ids = test_set)
```

22.10.3. Assess

! Important

Performance should **always** be reported from an independent test set, never from the training set.

```
pred_train$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	40	0	1	0
CRC	0	40	0	0
NSCLC	3	0	43	0
PCA	0	0	0	35

```
measures <- msrs(c('classif.acc'))
pred_train$score(measures)
```

```
classif.acc
0.9753086
```

```
pred_test$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	20	0	1	0
CRC	0	17	3	0
NSCLC	2	0	12	0
PCA	0	0	0	25

22. Worked example: classifying cancer types from gene expression

```
pred_test$score(measures)
```

```
classif.acc  
0.925
```

A single tree is a simple model, so it often classifies the training data less perfectly than kNN did, but it may *generalize* about as well to the test set. Look at the test accuracy and the test confusion matrix: which cancer types does the tree confuse with each other?

Overfitting

When the test score is much worse than the training score, the model is likely **overfit** — it has learned the training data's quirks rather than a general rule. How you fix overfitting depends on the model (simpler trees, more regularization, more data), but a large train/test gap is always a signal to look more carefully at model choice and settings.

22.11. Random forest

A **random forest** grows many decision trees, each on a random subset of the data and features, and averages their votes. Because the trees disagree in different ways, their combined prediction is usually more accurate and more stable than any single tree. We set `importance = "impurity"` so the model also reports how useful each gene was.

22.11.1. Train

```
set.seed(789)  
learner <- lrn("classif.ranger", importance = "impurity")  
learner$train(task, row_ids = train_set)
```

We can look at the fitted forest:

```
learner$model
```

22. Worked example: classifying cancer types from gene expression

Ranger result

Call:

```
ranger::ranger(dependent.variable.name = task$target_names, data = task$data(),
```

proba

```
Type: Classification
Number of trees: 500
Sample size: 162
Number of independent variables: 192
Mtry: 13
Target node size: 1
Variable importance mode: impurity
Splitrule: gini
OOB prediction error: 0.00 %
```

The printout reports an **OOB (out-of-bag) error**. Because each tree is trained on a bootstrap sample, the samples it *didn't* see act as a free held-out set; the OOB error estimates test performance without needing a separate split. A low OOB error is a good sign.

Random forests also give us a **variable importance** score — roughly, how often and how usefully each gene was used across the trees.

```
head(learner$importance(), 15)
```

CDHR5	FABP1	KRT20	GATA3	LGALS4	TRPS1	SFTPB.1	EFHD1
4.668165	4.042106	3.722492	3.470107	3.271688	3.047672	2.925743	2.895864
EPS8L3	TRPS1.1	GATA3.1	SLC45A3.1	TSPAN8	KLK3.1	AZGP1	
2.894204	2.442827	2.432394	2.242252	2.163998	2.134062	2.119718	

The most important genes here are the ones the forest leaned on most heavily to separate the cancer types. It's worth looking a few of them up to see whether they make biological sense — and comparing them to the gene the single tree split on.

22.11.2. Predict and assess

```
pred_train <- learner$predict(task, row_ids = train_set)
pred_test <- learner$predict(task, row_ids = test_set)
```

22. Worked example: classifying cancer types from gene expression

```
pred_train$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	43	0	0	0
CRC	0	40	0	0
NSCLC	0	0	44	0
PCA	0	0	0	35

```
measures <- msrs(c('classif.acc'))  
pred_train$score(measures)
```

```
classif.acc  
1
```

```
pred_test$confusion
```

	truth			
response	BC	CRC	NSCLC	PCA
BC	22	0	0	0
CRC	0	17	0	0
NSCLC	0	0	16	0
PCA	0	0	0	25

```
pred_test$score(measures)
```

```
classif.acc  
1
```

Compare all three test accuracies — kNN, the single tree, and the random forest. On most genomics classification problems the random forest is the strongest of the three, and it gives you the importance ranking as a bonus.

22.12. Summary

You have now run a complete classification analysis on real gene-expression data, all through the `mlr3` workflow:

- You turned a GEO dataset into an `mlr3` classification **task**, **split** it into training and test sets, and **selected** the 200 most-variable genes as features.
- You trained three learners — kNN, a single classification tree, and a random forest — and read each one's **confusion matrix** and **accuracy**.
- On most genomics classification problems the random forest is the strongest of the three, and it hands you a gene-importance ranking for free.

The disciplines you practised here recur in every analysis: **split** the data, **select features** thoughtfully, **train then predict**, and judge every model by its **test-set** score. That last habit — never trusting the training score — is the single most important thing to carry forward to the next two chapters.

22.13. Exercises

Each exercise reuses objects you already built, so the solutions are quick to run.

1. **Swap a learner.** The random forest above used impurity-based importance. Train a **support vector machine** instead and compare its test accuracy to the random forest. (You may need `library(e1071)`; if it isn't installed, fall back to comparing kNN with a different `k`.)

Solution

```
svm_learner <- lrn("classif.svm")
svm_learner$train(task, row_ids = train_set)
svm_pred <- svm_learner$predict(task, row_ids = test_set)
svm_pred$score(msrs(c('classif.acc')))
```

The point is that swapping algorithms is a one-line change to `lrn()` — everything else in the workflow is identical. Compare the test accuracy to the forest's; on this small, high-variance dataset the ranking can shift, which is why you always judge on the held-out test set.

2. **Change the feature-selection cutoff.** We kept the 200 most-variable genes. Conceptually, what would you expect to happen to the random forest's test accuracy if you kept only the top 50, or the top 1000?

22. Worked example: classifying cancer types from gene expression

i Solution

Too few features (top 50) may discard genes that help separate the rarer cancer types, lowering accuracy. Too many (top 1000) reintroduces low-variance, low-signal genes that add noise and computation without much benefit — and they raise the risk of overfitting. The “right” cutoff is a balance, best chosen by comparing test-set performance across a few values rather than guessing.

23. Worked example: predicting age from DNA methylation

How old is a tissue sample? Not the donor’s age on a birthday cake, but the biological age written into the DNA itself. It turns out you can read it off the **DNA-methylation** pattern: at certain sites the methylation level drifts steadily with age, and a model trained on those sites becomes an **epigenetic clock**. This example builds one, following the work of [Naue et al. \(2017\)](#).

This is the second of three worked examples. Unlike the [cancer-classification example](#), where the target was a category, here the target — a person’s age — is a *number*, so this is a **regression** problem. We use the same `mlr3` framework throughout; if you need the refresher on tasks, learners, measures, and the train/test split, see [The mlr3verse](#) and the [cancer example](#) for the general workflow. Here we keep our eyes on the biology and on how regression differs from classification.

The three core ideas from the cancer chapter all return, but each takes on a new shade once the goal is an epigenetic clock rather than a cancer label:

- **The split is about *validity*, not just accuracy.** A clock that predicts age beautifully on the very people it was trained on is worthless as a biomarker — of course it fits them, it learned them. The whole point of an epigenetic clock is to estimate the age of *someone new*, so the held-out test set is not a nicety here; it is the only thing that tells you whether you have built a real biomarker or an elaborate lookup table.
- **Feature selection is forced on you.** A methylation array measures hundreds of thousands of CpG sites, but a study has only dozens or hundreds of people. Far more features than samples — the $p \gg n$ regime — is the normal state of affairs in genomics, and it means selecting or limiting features is not an optional tidy-up but a requirement (see the warning below).
- **The yardstick changes.** Classification used accuracy and a confusion matrix. Regression has no “correct/incorrect” — instead we measure how far off the predictions are, with **RMSE** (typical error, in years) and **R^2** (the fraction of the variation in age the model explains).

23. Worked example: predicting age from DNA methylation

⚠ More features than samples: the $p \gg n$ problem

When the number of features p dwarfs the number of samples n , a flexible model can find some combination of CpG sites that fits the training ages *perfectly* by chance alone — pure memorization dressed up as signal. This is overfitting in its most dangerous form, because the training score looks fantastic. The defences are the same ones you already know: limit the features (keep only the CpG sites most associated with age), prefer simpler models, and *always* judge on the held-out test set. The next chapter shows a second defence — letting the model shrink its own coefficients.

23.1. What you'll learn

- Build a **regression** task from a real methylation dataset with `as_task_regr()`.
- Fit and compare three regression learners — **linear regression**, a **regression tree**, and a **random forest** — through the same `mlr3` calls.
- Read **regression diagnostic plots** and a **truth-versus-prediction** scatter.
- Interpret **R^2** as the fraction of variation explained (and **RMSE** as the typical error in years), and spot overfitting when the test R^2 falls below the training R^2 .
- Explain why having far more CpG features than people (the $p \gg n$ problem) makes feature selection and an honest test set non-negotiable for a biomarker.

23.2. Setup

```
library(mlr3verse)
library(ranger)      # for random forests
library(data.table)
library(ggplot2)
set.seed(789)
```

23.3. The data

The biology behind it: within each individual, the level of [DNA methylation](#) at certain sites changes steadily with age. The [CpG sites](#) whose methylation correlates most strongly with age make excellent **biomarkers** — and excellent features for a regression model. The data come from [this Galaxy tutorial](#).

23. Worked example: predicting age from DNA methylation

```
meth_age <- rbind(
  fread('https://zenodo.org/record/2545213/files/test_rows_labels.csv'),
  fread('https://zenodo.org/record/2545213/files/train_rows.csv')
)
```

A quick look at the data:

```
head(meth_age)
```

	RPA2_3	ZYG11A_4	F5_2	HOXC4_1	NKIRAS2_2	MEIS1_1	SAMD10_2	GRM2_9	TRIM59_5
	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>	<num>
1:	65.96	18.08	41.57	55.46	30.69	63.42	40.86	68.88	44.32
2:	66.83	20.27	40.55	49.67	29.53	30.47	37.73	53.30	50.09
3:	50.30	11.74	40.17	33.85	23.39	58.83	38.84	35.08	35.90
4:	65.54	15.56	33.56	36.79	20.23	56.39	41.75	50.37	41.46
5:	59.01	14.38	41.95	30.30	24.99	54.40	37.38	30.35	31.28
6:	81.30	14.68	35.91	50.20	26.57	32.37	32.30	55.19	42.21

	LDB2_3	ELOVL2_6	DDO_1	KLF14_2	Age
	<num>	<num>	<num>	<num>	<int>
1:	56.17	62.29	40.99	2.30	40
2:	58.40	61.10	49.73	1.07	44
3:	58.81	50.38	63.03	0.95	28
4:	58.05	50.58	62.13	1.99	37
5:	65.80	48.74	41.88	0.90	24
6:	70.15	61.36	33.62	1.87	43

Each row is a person, each `cg...` column is the methylation level at one CpG site, and `Age` is what we want to predict. Because the target is a *number*, this is a **regression** task — so we use `as_task_regr()`.

```
task <- as_task_regr(meth_age, target = 'Age')
```

We split into training and test sets exactly as in the [cancer example](#), seeding for reproducibility.

```
set.seed(7)
train_set <- sample(task$row_ids, 0.67 * task$nrow)
test_set <- setdiff(task$row_ids, train_set)
```

23.4. Linear regression

We start with the simplest regression model: a straight-line fit relating methylation to age.

```
learner <- lrn("regr.lm")
```

23.4.1. Train

```
learner$train(task, row_ids = train_set)
```

After fitting a linear model, the standard **diagnostic plots** (Figure 23.1) help check whether the model's assumptions hold.

```
par(mfrow = c(2, 2))  
plot(learner$model)
```

23. Worked example: predicting age from DNA methylation

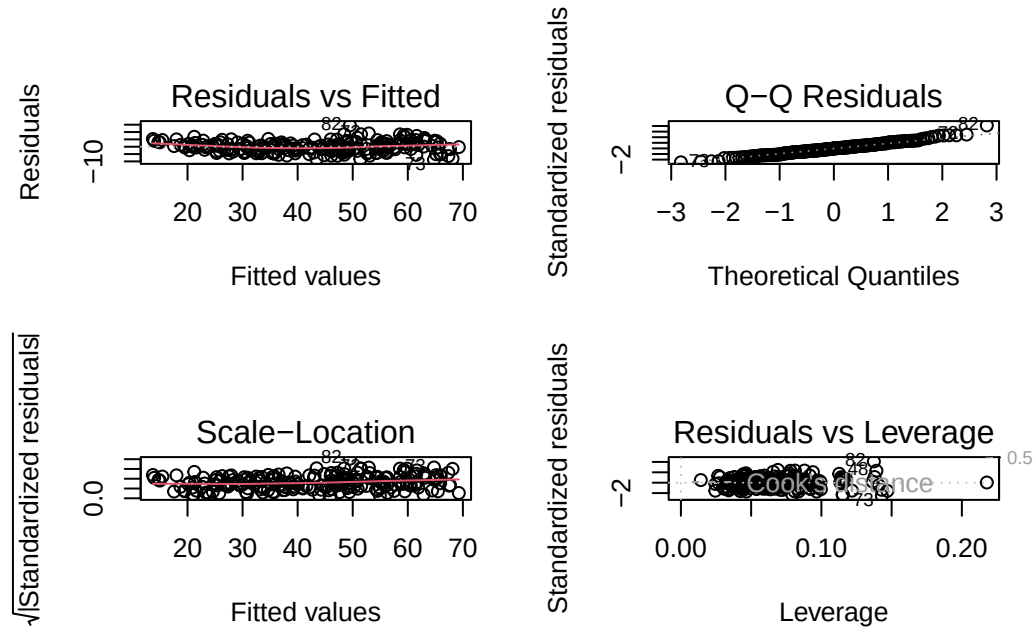


Figure 23.1.: Regression diagnostic plots. Top left: residuals vs. fitted values (look for no pattern). Top right: normal Q-Q plot (points should follow the line). Bottom left: scale-location. Bottom right: residuals vs. leverage.

23.4.2. Predict

```
pred_train <- learner$predict(task, row_ids = train_set)
pred_test  <- learner$predict(task, row_ids = test_set)
```

23.4.3. Assess

Printing a regression prediction shows the predicted (**response**) and true (**truth**) values side by side.

```
pred_train
```

```
-- <PredictionRegr> for 209 observations: -----
```

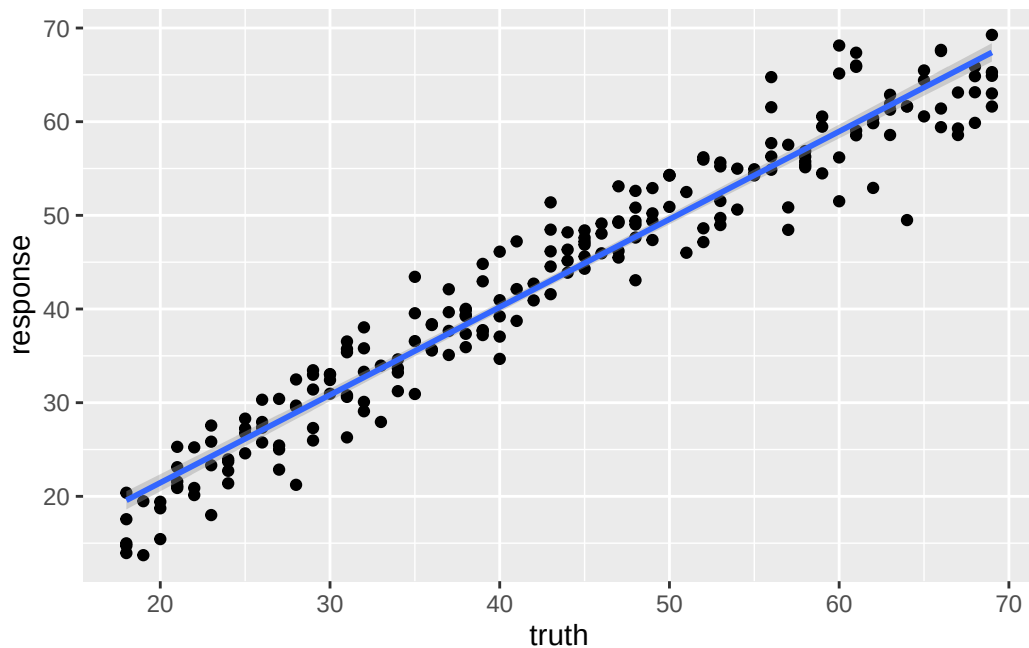
23. Worked example: predicting age from DNA methylation

row_ids	truth	response
298	29	31.40565
103	58	56.26019
194	53	48.96480
---	---	---
312	48	52.61195
246	66	67.66312
238	38	39.38414

A scatter plot of truth versus prediction makes the fit visible: a perfect model would put every point on the diagonal.

```
ggplot(pred_train, aes(x = truth, y = response)) +  
  geom_point() +  
  geom_smooth(method = 'lm')
```

`geom\_smooth()` using formula = 'y ~ x'



We summarize the fit with R^2 (the fraction of variation in age the model explains; 1.0 is perfect, 0 is no better than guessing the mean).

23. Worked example: predicting age from DNA methylation

```
measures <- msrs(c('regr.rsq'))  
pred_train$score(measures)
```

```
regr.rsq  
0.9376672
```

```
pred_test
```

```
-- <PredictionRegr> for 103 observations: -----  
row_ids truth response  
    4    37 37.64301  
    5    24 28.34777  
    7    34 33.22419  
---    ---    ---  
  306    42 41.65864  
  307    63 58.68486  
  309    68 70.41987
```

```
pred_test$score(measures)
```

```
regr.rsq  
0.9363526
```

The training R^2 is high, but the **test R^2** is the honest measure of how well the clock would predict the age of a new person. A drop from training to test is, again, the signature of overfitting.

23.5. Regression tree

We can use a tree for regression too. Instead of voting on a class, each leaf of the tree reports a predicted *number*.

```
learner <- lrn("regr.rpart", keep_model = TRUE)  
learner$train(task, row_ids = train_set)  
learner$model
```

23. Worked example: predicting age from DNA methylation

n= 209

```
node), split, n, deviance, yval
  * denotes terminal node
```

```
1) root 209 45441.4500 43.27273
  2) ELOVL2_6< 56.675 98 5512.1220 30.24490
    4) ELOVL2_6< 47.24 47 866.4255 24.23404
      8) GRM2_9< 31.3 34 289.0588 22.29412 *
      9) GRM2_9>=31.3 13 114.7692 29.30769 *
    5) ELOVL2_6>=47.24 51 1382.6270 35.78431
      10) F5_2>=39.295 35 473.1429 33.28571 *
      11) F5_2< 39.295 16 213.0000 41.25000 *
  3) ELOVL2_6>=56.675 111 8611.3690 54.77477
    6) ELOVL2_6< 65.365 63 3101.2700 49.41270
      12) KLF14_2< 3.415 37 1059.0270 46.16216 *
      13) KLF14_2>=3.415 26 1094.9620 54.03846 *
    7) ELOVL2_6>=65.365 48 1321.3120 61.81250 *
```

Notice something odd: a single regression tree can only output a handful of **discrete** age values — one per leaf. It cannot predict, say, 43.7 if no leaf sits there.

23.5.1. Predict

```
pred_train <- learner$predict(task, row_ids = train_set)
pred_test  <- learner$predict(task, row_ids = test_set)
```

23.5.2. Assess

```
pred_train
```

```
-- <PredictionRegr> for 209 observations: -----
row_ids truth response
  298    29 33.28571
  103    58 61.81250
```

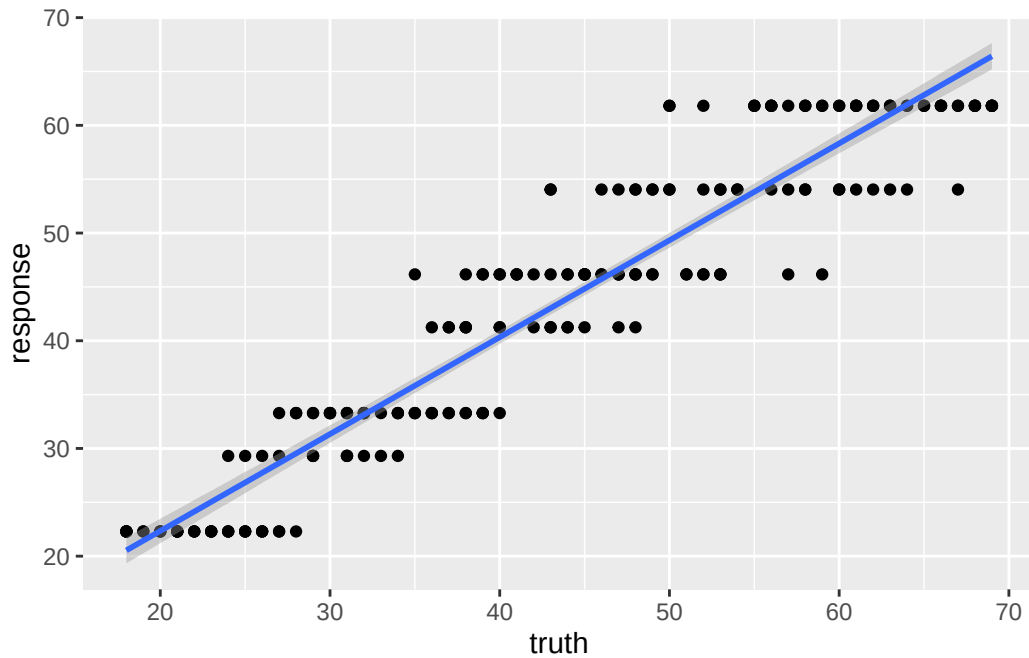
23. Worked example: predicting age from DNA methylation

194	53	46.16216
---	---	---
312	48	54.03846
246	66	61.81250
238	38	41.25000

The discreteness is obvious in the scatter plot: predictions stack up into horizontal bands, one per leaf.

```
ggplot(pred_train, aes(x = truth, y = response)) +  
  geom_point() +  
  geom_smooth(method = 'lm')
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
measures <- msrs(c('regr.rsq'))  
pred_train$score(measures)
```

```
regr.rsq  
0.8995351
```


23. Worked example: predicting age from DNA methylation

```
pred_test
```

```
-- <PredictionRegr> for 103 observations: -----
row_ids truth response
    4    37 41.25000
    5    24 33.28571
    7    34 33.28571
---    ---    ---
 306    42 46.16216
 307    63 61.81250
 309    68 61.81250
```

```
pred_test$score(measures)
```

```
regr.rsq
0.8545402
```

The R^2 differs noticeably from the linear model — the single tree’s blocky predictions are a poor fit for a continuous quantity like age.

23.6. Random forest

A random forest is also tree-based, but averaging over many trees smooths away the blocky, discrete behaviour of a single tree, giving genuinely continuous predictions.

```
set.seed(789)
learner <- lrn("regr.ranger", mtry = 2, min.node.size = 20)
learner$train(task, row_ids = train_set)
learner$model
```

```
$model
Ranger result
```

Call:

```
ranger::ranger(dependent.variable.name = task$target_names, data = data, min.node.size = 20)
```

23. Worked example: predicting age from DNA methylation

Type:	Regression
Number of trees:	500
Sample size:	209
Number of independent variables:	13
Mtry:	2
Target node size:	20
Variable importance mode:	none
Splitrule:	variance
OOB prediction error (MSE):	17.72152
R squared (OOB):	0.918883

23.6.1. Predict

```
pred_train <- learner$predict(task, row_ids = train_set)
pred_test  <- learner$predict(task, row_ids = test_set)
```

23.6.2. Assess

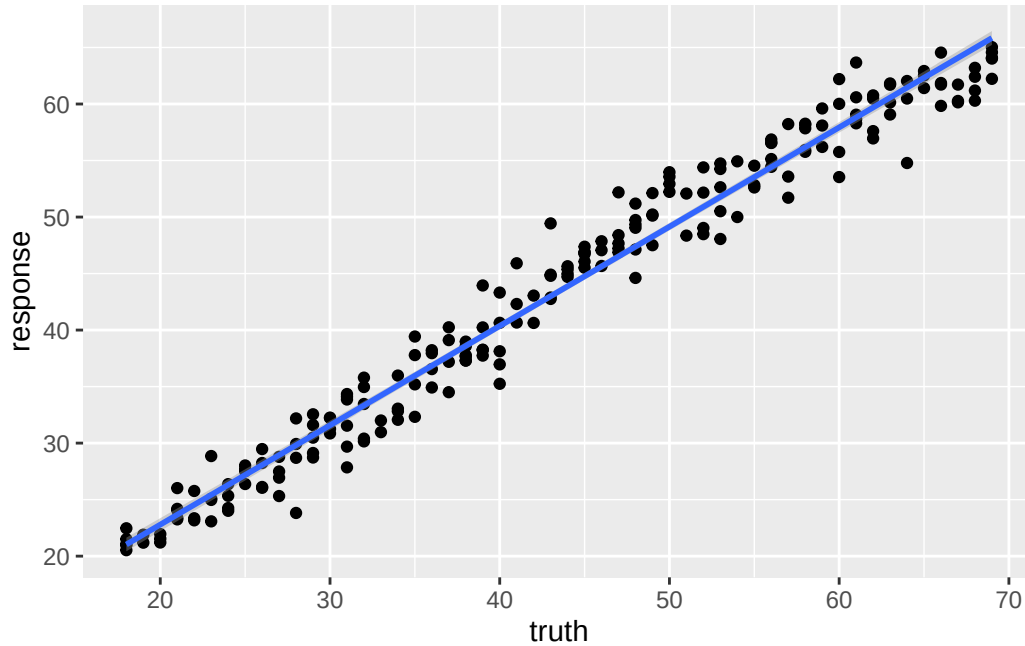
```
pred_train
```

```
-- <PredictionRegr> for 209 observations: -----
row_ids truth response
  298    29 30.48368
  103    58 58.25574
  194    53 48.05929
  ---    ---      ---
  312    48 51.18667
  246    66 64.54711
  238    38 37.65249
```

```
ggplot(pred_train, aes(x = truth, y = response)) +
  geom_point() +
  geom_smooth(method = 'lm')
```

```
`geom_smooth()` using formula = 'y ~ x'
```

23. Worked example: predicting age from DNA methylation



The points now scatter smoothly around the diagonal — no more horizontal bands.

```
measures <- msrs(c('regr.rsq'))  
pred_train$score(measures)
```

```
regr.rsq  
0.9623924
```

```
pred_test
```

```
-- <PredictionRegr> for 103 observations: -----  
row_ids truth response  
    4    37 37.10820  
    5    24 29.29540  
    7    34 33.02805  
---    ---    ---  
 306    42 40.41656  
 307    63 58.32185  
 309    68 63.00852
```

23. Worked example: predicting age from DNA methylation

```
pred_test$score(measures)
```

```
regr.rsq  
0.9226935
```

Compare the test R^2 across the three regression models. The forest typically predicts age more accurately than either the linear model or the single tree, and without the single tree's discrete artifacts.

23.7. Summary

You have now built an epigenetic clock — a complete **regression** analysis on real methylation data, through the same **mlr3** workflow you used for cancer classification:

- You turned the methylation table into an **mlr3** regression **task** with `as_task_regr()` and split it into training and test sets.
- You fit a **linear regression**, a **regression tree**, and a **random forest**, and judged each with R^2 and a truth-versus-prediction scatter.
- The single tree's predictions came in discrete bands — a poor fit for a continuous target — while the random forest's smooth predictions typically beat both the straight line and the single tree.

The change from the cancer example was only the *kind* of target: a number instead of a category, so `as_task_regr()` instead of `as_task_classif()` and R^2 instead of accuracy. The discipline of trusting only the test-set score carries over unchanged.

23.8. Exercises

Each exercise reuses objects you already built, so the solutions are quick to run.

1. **Tune the forest.** The random forest used `mtry = 2` and `min.node.size = 20`. Refit it with the defaults (drop both arguments) and compare the test R^2 . Did the hand-chosen settings help?

23. Worked example: predicting age from DNA methylation

i Solution

```
rf_default <- lrn("regr.ranger")
rf_default$train(task, row_ids = train_set)
rf_default$predict(task, row_ids = test_set)$score(msrs(c('regr.rsq')))
```

Changing `mtry` and `min.node.size` is a one-line edit to the `lrn()` call; everything else is identical. On this dataset the defaults are usually close, which is part of why random forests are a low-fuss default — but the honest comparison is always the held-out test R^2 .

2. **Add a second measure.** Alongside R^2 , score the test predictions with root-mean-squared error, `msr("regr.rmse")`, which reports the typical prediction error in *years*. Which number is easier to explain to a biologist?

i Solution

```
pred_test$score(msrs(c('regr.rsq', 'regr.rmse')))
```

R^2 is unitless and says what *fraction* of the variation in age the model explains; RMSE is in the target's own units (years) and says how far off a typical prediction is. For an audience that thinks in years, RMSE is often the more intuitive headline number.

24. Worked example: predicting gene expression from histone marks

A gene’s DNA sequence doesn’t change from one cell type to the next, yet the same gene can be switched loudly on in one tissue and silent in another. Much of that control is written in **histone modifications** — chemical marks on the proteins that DNA wraps around, concentrated near each gene’s promoter. This example asks a question at the heart of gene regulation: **can the pattern of histone modifications near a gene predict that gene’s expression level?** (Figure 24.1)

This is the third of three worked examples, and like the [methylation age example](#) it is a **regression** problem — the target is a continuous expression level. We use the same `mlr3` framework throughout; for the four building blocks see [The mlr3verse](#), and for the general train/predict/assess workflow see the [cancer example](#). Here the new idea is **penalized regression** with cross-validation, and a peek under the hood at how R^2 is actually computed.

This chapter views the same three ideas through the lens of **regularization**:

- **Feature selection, done automatically.** In the cancer chapter you hand-picked features — you ranked genes by variance and kept the top 200. Penalized regression flips that around: instead of you choosing which features to drop, the model’s **L1 (lasso) penalty** shrinks the coefficients of useless features all the way to zero, selecting features *for* you as a side effect of fitting. Same goal — fewer, more useful features — but the work moves from your hands into the algorithm.
- **Cross-validation, inside the training set.** A penalty has a strength, λ , and we have to choose it. The trick is to choose it by **cross-validation** *within the training data* — carving the training set into folds, never touching the test set. This is a fresh facet of the same golden rule from the cancer and methylation chapters: the test set stays sealed until the very end.
- **Overfitting, as a tunable knob.** Regularization reframes overfitting as something you can dial. The penalty λ is a knob that trades a tighter fit on the training data for better generalization to new genes: turn it up and the model gets simpler and steadier; turn it down and it bends closer to the training points, at the risk of memorizing noise.

i From a manual filter to an automatic penalty

The cancer chapter selected features by hand (keep the most variable genes); this chapter lets the model do it. A lasso penalty is, in effect, automatic feature selection baked into the fit — and cross-validation, run *only* on the training data, is how we pick how aggressive that penalty should be. The held-out test set is never consulted while we tune; it is saved for the single honest score at the end.

24.1. What you'll learn

- Center and scale features, and visualize them with boxplots and a heatmap.
- Build a regression task and split it with `mlr3`'s `partition()` helper.
- Fit **penalized regression** (ridge) with `regr.cv_glmnet`, and explain how **cross-validation** *within the training set* chooses the penalty strength .
- Read a penalized model's **coefficients**, contrast ridge with lasso, and see how the lasso penalty performs **automatic feature selection** — the job you did by hand in the cancer chapter.
- Compute **R² by hand** and confirm it matches the `regr.rsq` measure.

24.2. Setup

```
library(mlr3verse)
library(mlr3learners) # for cv_glmnet
library(mlr3viz)      # for plotting predictions
library(ComplexHeatmap)
set.seed(789)
```

$$y \sim h_1 + h_2 + h_3 + \dots \quad (24.1)$$

The data summarize the histone marks within each gene's **promoter** (here, the region 2 kb upstream of the transcription start site), paired with that gene's measured expression. The target is the expression level; the features are the histone-mark summaries. As Equation 24.1 sketches, we are modelling expression y as a combination of histone marks $h_1, h_2, \dots$. We'll try two approaches: **penalized regression** and a **random forest**.

Relationship between chromatin marks and gene expression

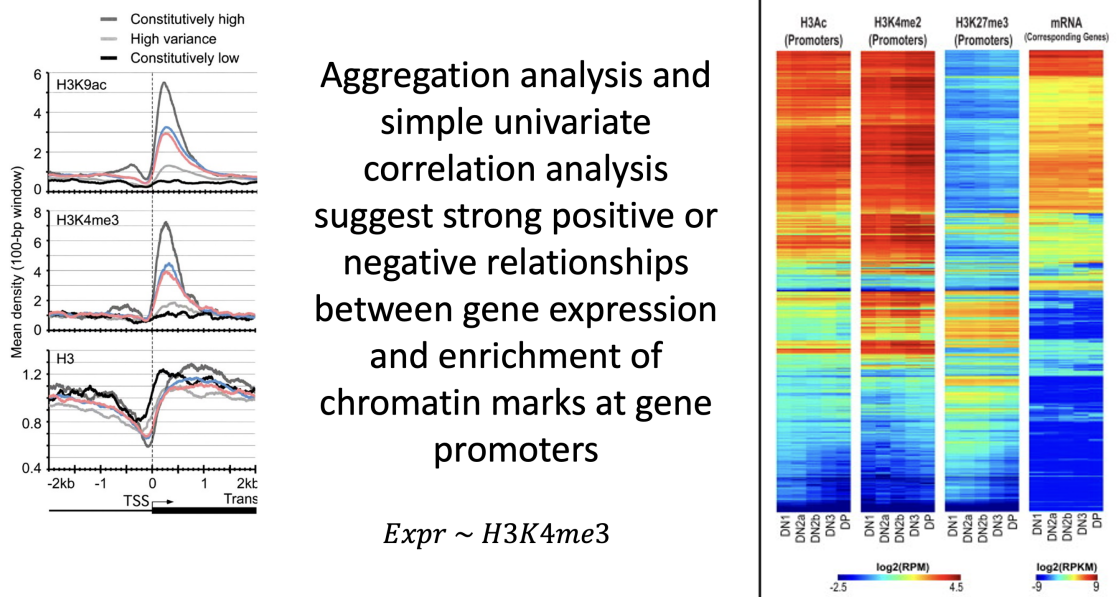


Figure 24.1.: What is the combined effect of histone marks on gene expression?

24.3. The data

The data were developed by Anshul Kundaje. We won't dwell on the collection details; we'll focus on the modelling.

```
feature_set <- read.table("http://seandavi.github.io/ITR/expression-prediction/features.txt")
```

What are the predictor columns?

```
colnames(feature_set)
```

```
[1] "Control"  "Dnase"    "H2az"     "H3k27ac"  "H3k27me3" "H3k36me3"
[7] "H3k4me1"  "H3k4me2"  "H3k4me3"  "H3k79me2" "H3k9ac"   "H3k9me1"
[13] "H3k9me3"  "H4k20me1"
```

These histone-mark summaries are our predictors. Some learners assume the predictors are on a similar scale, so we **center and scale** each column (subtract its mean, divide by its standard deviation) by converting to a matrix and using `scale()`.

```
par(mfrow = c(1, 2))
scaled_features <- scale(as.matrix(feature_set))
boxplot(feature_set, main = 'Original data')
boxplot(scaled_features, main = 'Centered and scaled data')
```

24. Worked example: predicting gene expression from histone marks

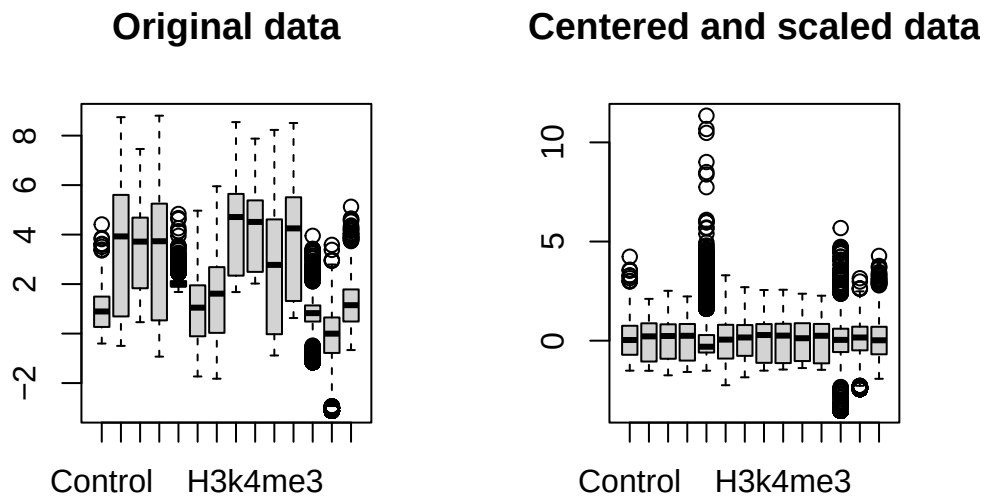


Figure 24.2.: Boxplots of the original and the centered-and-scaled features. After scaling, every feature sits on a comparable scale.

After scaling, every feature is centered near zero with comparable spread, as the right-hand boxplot shows. We can also spot structure across genes with a heatmap (Figure 24.3).

```
set.seed(789)
sampled_features <- scaled_features[sample(nrow(scaled_features), 500), ]
Heatmap(sampled_features, name = 'histone marks', show_row_names = FALSE)
```

24. Worked example: predicting gene expression from histone marks

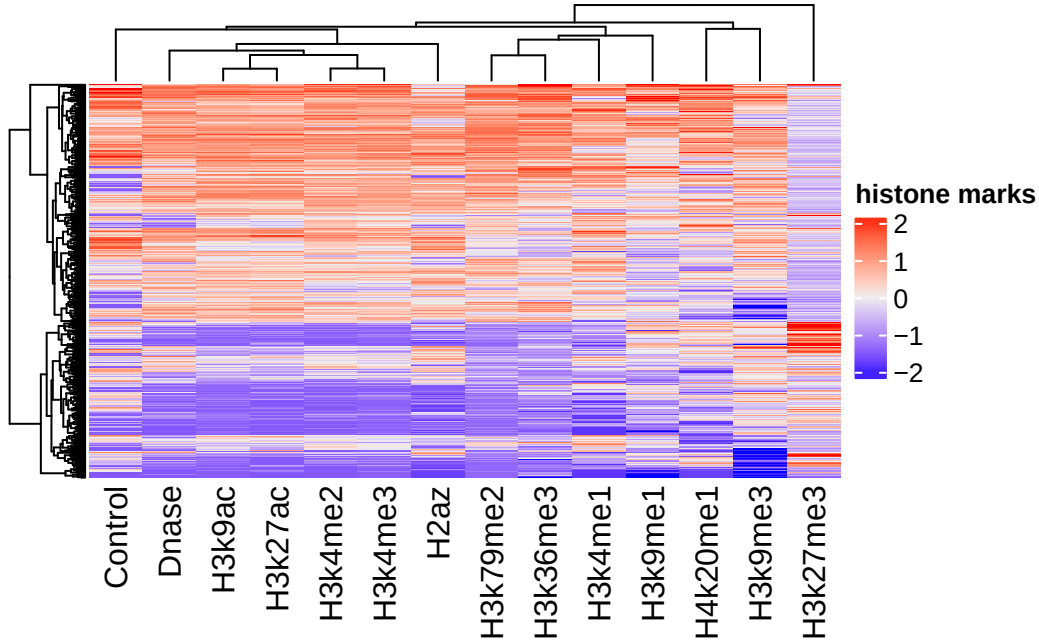


Figure 24.3.: Heatmap of 500 randomly sampled genes (rows) across the histone marks (columns). Blocks of similar color reveal groups of genes with similar mark patterns.

Now we read in the matching gene-expression values — our prediction **target** — and combine them with the scaled features into one data frame.

```
target <- scan(url("http://seandavi.github.io/ITR/expression-prediction/target.txt"), skip
# combine target and features into one data frame
exp_pred_data <- data.frame(gene_expression = target, scaled_features)
```

The first few rows:

```
head(exp_pred_data, 3)
```

	gene_expression	Control	Dnase	H2az
ENSG000000000419.7.49575069	6.082343	0.7452926	0.7575546	1.0728432
ENSG000000000457.8.169863093	2.989145	1.9509786	1.0216546	0.3702787
ENSG000000000938.7.27961645	-5.058894	-0.3505542	-1.4482958	-1.0390775
		H3k27ac	H3k27me3	H3k36me3
				H3k4me1

24. Worked example: predicting gene expression from histone marks

ENSG000000000419.7.49575069	1.0950440	-0.5125312	1.1334793	0.4127984
ENSG000000000457.8.169863093	0.7142157	-0.4079244	0.8739005	1.1649282
ENSG000000000938.7.27961645	-1.0173283	1.4117293	-0.5157582	-0.5017450
	H3k4me2	H3k4me3	H3k79me2	H3k9ac
ENSG000000000419.7.49575069	1.2136176	1.1202901	1.5155803	1.2468256
ENSG000000000457.8.169863093	0.6456572	0.6508561	0.7976487	0.5792891
ENSG000000000938.7.27961645	-0.1878255	-0.6560973	-1.3803974	-1.0067972
	H3k9me1	H3k9me3	H4k20me1	
ENSG000000000419.7.49575069	0.1426980	1.185622	1.9599992	
ENSG000000000457.8.169863093	0.3630902	1.014923	-0.2695111	
ENSG000000000938.7.27961645	0.6564520	-1.370871	-1.8773178	

24.4. Create the task and split

The target is a number, so this is a regression task. This time we use `mlr3`'s own `partition()` helper to make the train/test split, which by default holds out one-third for testing.

```
set.seed(7)
exp_pred_task <- as_task_regr(exp_pred_data, target = 'gene_expression')
split <- partition(exp_pred_task)
```

24.5. Linear regression

As a baseline, fit an ordinary linear regression.

```
learner <- lrn("regr.lm")
```

24.5.1. Train and predict

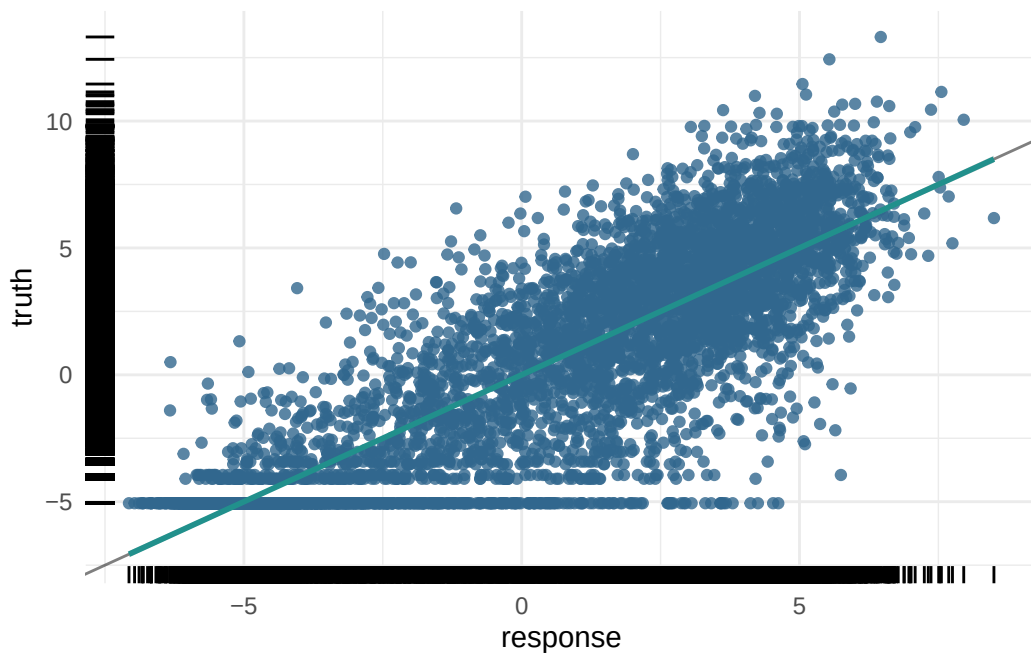
```
learner$train(exp_pred_task, split$train)
pred_train <- learner$predict(exp_pred_task, split$train)
pred_test <- learner$predict(exp_pred_task, split$test)
```

24.5.2. Assess

```
pred_train
```

```
-- <PredictionRegr> for 5789 observations: -----
row_ids      truth  response
      2  2.989145  2.996674
      3 -5.058894 -5.364407
      5  5.558023  4.105793
    ---      ---      ---
 8639 -5.058894 -5.238131
 8640 -3.114148 -4.756313
 8641  5.731075  4.198905
```

```
plot(pred_train)
```



The training R^2 :

24. Worked example: predicting gene expression from histone marks

```
measures <- msrs(c('regr.rsq'))  
pred_train$score(measures)
```

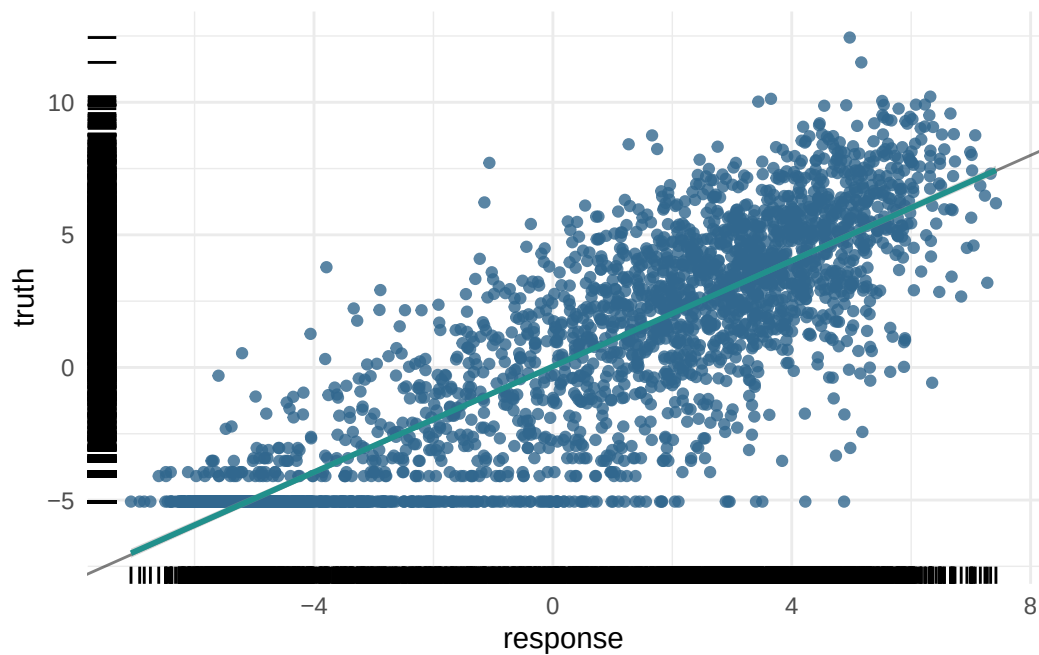
```
regr.rsq  
0.7507266
```

And the test R^2 — the number that matters:

```
pred_test$score(measures)
```

```
regr.rsq  
0.7505519
```

```
plot(pred_test)
```



A moderate R^2 here is a real biological result: histone marks in the promoter carry genuine, but partial, information about how strongly a gene is expressed.

24.6. Penalized regression

When a model has many predictors, it can become *too* flexible and start fitting noise — the problem of **overfitting**. **Penalized regression** combats this by adding a penalty for complexity, nudging the model toward simpler, more general solutions.

i Three flavours of penalty

- **Ridge (L2):** shrinks all coefficients toward zero, like a rubber band pulling them in. It keeps every feature but tames large coefficients.
- **Lasso (L1):** can shrink some coefficients *all the way* to zero, like scissors that cut the least useful features out of the model entirely. This doubles as automatic **feature selection**.
- **Elastic net:** a blend of the two — rubber band *and* scissors — combining shrinkage with feature elimination.

In `mlr3`, the `alpha` setting picks the flavour: `alpha = 0` is ridge, `alpha = 1` is lasso, and values in between are elastic net.

We'll use the `regr.cv_glmnet` learner, which automatically uses **cross-validation** to choose the penalty strength.

i What is cross-validation?

Cross-validation estimates how a model will perform on unseen data by splitting the training data into k equal **folds**. The model trains on $k - 1$ folds and is validated on the held-out fold, repeating until each fold has served as the validation set once. Averaging the k scores gives a stable performance estimate, and `cv_glmnet` uses it to pick the best penalty (`lambda`). The `nfolds` setting controls k ; 5 or 10 are common choices.

With `alpha = 0` we fit a **ridge** model with 10-fold cross-validation:

```
set.seed(789)
learner <- lrn("regr.cv_glmnet", nfolds = 10, alpha = 0)
```

24.6.1. Train

24. Worked example: predicting gene expression from histone marks

```
learner$train(exp_pred_task, split$train)
```

For a penalized model we can inspect the **coefficients** directly:

```
coef(learner$model)
```

```
15 x 1 sparse Matrix of class "dgCMatrix"
      lambda.1se
(Intercept)  0.08832251
Control      -0.05020310
Dnase         0.85954551
H2az          0.31828260
H3k27ac       0.26566165
H3k27me3      -0.27170626
H3k36me3       0.65845540
H3k4me1       -0.02832563
H3k4me2        0.17902035
H3k4me3        0.37064260
H3k79me2       0.86371398
H3k9ac         0.48125738
H3k9me1       -0.05440077
H3k9me3       -0.16204169
H4k20me1       0.13611422
```

Each coefficient tells us how strongly its histone mark pushes predicted expression up or down. With lasso ($\alpha = 1$) many of these would be exactly zero — the model would have *selected* a subset of marks for us, which is useful not just for prediction but for understanding which marks matter.

24.6.2. Predict and assess

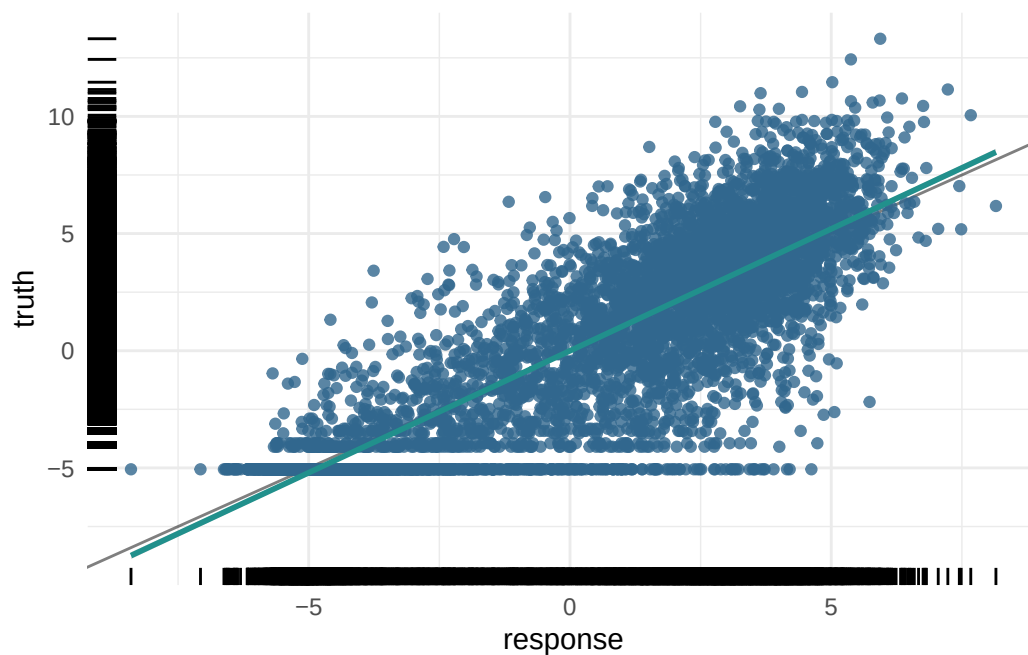
```
pred_train <- learner$predict(exp_pred_task, split$train)
pred_test  <- learner$predict(exp_pred_task, split$test)
```

```
pred_train
```


24. Worked example: predicting gene expression from histone marks

```
-- <PredictionRegr> for 5789 observations: -----  
row_ids      truth  response  
    2  2.989145  2.933043  
    3 -5.058894 -4.471610  
    5  5.558023  4.221675  
    ---      ---      ---  
 8639 -5.058894 -5.581904  
 8640 -3.114148 -4.105914  
 8641  5.731075  4.427548
```

```
plot(pred_train)
```



The training R^2 :

```
measures <- msrs(c('regr.rsq'))  
pred_train$score(measures)
```

```
regr.rsq  
0.740439
```

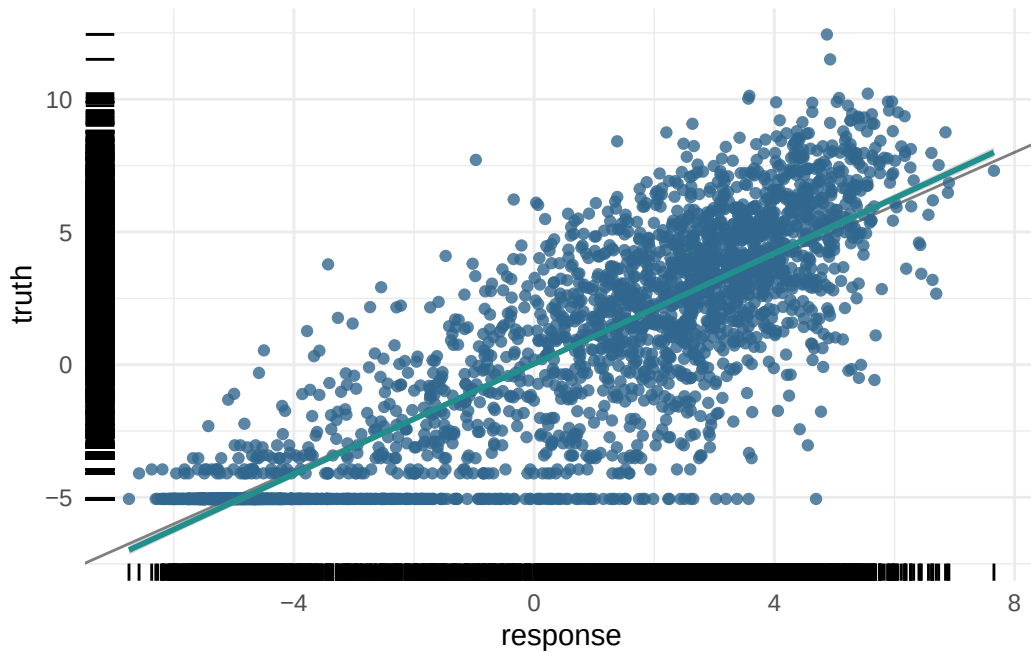
24. Worked example: predicting gene expression from histone marks

And the test R^2 :

```
pred_test$score(measures)
```

```
regr.rsq  
0.7405317
```

```
plot(pred_test)
```



We can also compute the test R^2 by hand, to demystify the measure: it is one minus the ratio of the residual sum of squares to the total sum of squares.

```
# calculate the R-squared value by hand  
truth <- pred_test$truth  
predicted <- pred_test$response  
rss <- sum((truth - predicted)^2)           # residual sum of squares  
tss <- sum((truth - mean(truth))^2)        # total sum of squares  
r_squared <- 1 - (rss / tss)  
r_squared
```

24. Worked example: predicting gene expression from histone marks

```
[1] 0.7405317
```

This matches the `regr.rsq` measure above — reassuring confirmation that you understand exactly what R^2 is counting.

24.7. Summary

You have now used machine learning to probe a question in gene regulation — whether promoter histone marks predict expression — through the same `mlr3` workflow as the previous two chapters:

- You **scaled** the histone-mark features, visualized them with boxplots and a **heatmap**, and built a regression task split with `partition()`.
- A baseline **linear regression** gave a moderate R^2 — a real, if partial, biological signal that marks carry information about expression.
- **Penalized regression** (ridge via `regr.cv_glmnet`) used **cross-validation** to choose its penalty and fight overfitting; with lasso it would also perform feature selection, turning the model into a tool for understanding which marks matter.
- Computing R^2 **by hand** confirmed exactly what the `regr.rsq` measure counts.

Across all three worked examples the same disciplines recurred: **split** the data, **select features** thoughtfully, **train then predict**, and judge every model by its **test-set** score. That last habit — never trusting the training score — is the single most important thing to carry forward.

24.8. Exercises

Each exercise reuses objects you already built, so the solutions are quick to run.

1. **Change the number of CV folds.** The ridge model used `nfolds = 10`. Refit it with `nfolds = 5` and see whether the test R^2 changes much.

i Solution

```
ridge5 <- lrn("regr.cv_glmnet", nfolds = 5, alpha = 0)
ridge5$train(exp_pred_task, split$train)
ridge5$predict(exp_pred_task, split$test)$score(msrs(c('regr.rsq')))
```

Fewer folds means each training fold is a little smaller and the chosen penalty is

a little noisier, but the test R^2 usually shifts only slightly. The penalty selection is fairly robust to the exact number of folds.

2. **Try lasso instead of ridge.** We used `alpha = 0` (ridge). Refit with `alpha = 1` (lasso) and inspect the coefficients. How many histone marks does lasso drop to exactly zero?

i Solution

```
lasso <- lrn("regr.cv_glmnet", nfolds = 10, alpha = 1)
lasso$train(exp_pred_task, split$train)
coef(lasso$model)           # zeros are the dropped marks
lasso$predict(exp_pred_task, split$test)$score(msrs(c('regr.rsq')))
```

Lasso sets the coefficients of the least useful marks to exactly zero, so the non-zero entries are the marks it “selected.” This is feature selection built into the model, and it makes the result easier to interpret biologically.

3. **Add a random forest.** The chapter opening promised a random forest too. Fit `regr.ranger` on the same task and split, and compare its test R^2 to ridge.

i Solution

```
rf <- lrn("regr.ranger")
rf$train(exp_pred_task, split$train)
rf$predict(exp_pred_task, split$test)$score(msrs(c('regr.rsq')))
```

Swapping the learner is a one-line change to `lrn()`. The forest can capture non-linear interactions between marks that a linear model cannot, so it sometimes edges out ridge — but, as always, the honest comparison is the held-out test R^2 .

Part VI.

Bioconductor

25. Accessing public omics data with GEOquery

Someone has already run the experiment you need. Thousands of expression studies — microarray and RNA-seq, across nearly every tissue and disease — sit in a public archive called the **Gene Expression Omnibus (GEO)**, free to download and reanalyze. Suppose you read a paper comparing tumor and normal tissue across several cancers and you want to *see the data for yourself*: pull it down, check how the samples group, and ask whether tumor and normal really separate. That is exactly the kind of reuse GEO exists for.

In this chapter you'll do that end to end. You'll fetch a real published dataset with the **GEOquery** package, load it into a **SummarizedExperiment** (the container from the previous chapter), keep the most variable genes, and run a **principal component analysis (PCA)** to visualize how the samples relate. The data come from a study of immune gene expression in matched normal and tumor tissue (Brouwer-Visser et al. 2018), deposited as accession [GSE103512](#).

25.1. What you'll learn

By the end of this chapter you will be able to:

- Download a published dataset from GEO with `getGEO()`.
- Convert the result into a **SummarizedExperiment** and inspect its parts.
- Summarize sample annotations with `table()`.
- Filter an expression matrix down to its most variable genes.
- Run PCA with `prcomp()` and plot the samples to see how they group.

This chapter builds directly on the **SummarizedExperiment** chapter. If the words `assay()`, `colData()`, or “rows are genes, columns are samples” don't feel familiar yet, read that chapter first — here we'll lean on it rather than re-explain the object's anatomy.

 Installing GEOquery

GEOquery is a Bioconductor package. Install it once with BiocManager; you won't need to run this again:

```
BiocManager::install("GEOquery")
```

25.2. Fetching a dataset from GEO

GEOquery has essentially one function you'll reach for: `getGEO()`. You hand it a GEO **accession number** — the unique ID for a dataset, here "GSE103512" — and it downloads the data over the network.

```
library(GEOquery)
gse <- getGEO("GSE103512")[[1]]
```

 Why the `[[1]]` and the conversion later?

Two small historical quirks are worth a moment, because they trip up everyone the first time.

First, `getGEO()` returns a **list** of datasets, not a single one. In the early days a single accession could bundle several datasets, so the function always returns a list. That's uncommon now, but the convention stayed because so much older data is still useful — hence the `[[1]]` to grab the first (and here, only) element.

Second, GEOquery was written in 2007 and hands you data in the older Bioconductor container, the `ExpressionSet`. We'll convert it to the modern `SummarizedExperiment` in the next step so we can use the accessors you already know.

Now convert the `ExpressionSet` into a `SummarizedExperiment` with `as()`:

```
library(SummarizedExperiment)
se <- as(gse, "SummarizedExperiment")
se
```

```
class: SummarizedExperiment
dim: 54715 280
metadata(3): experimentData annotation protocolData
```

```
assays(1): exprs
rownames(54715): 1007_PM_s_at 1053_PM_at ... AFFX-TrpnX-5_at
  AFFX-TrpnX-M_at
rowData names(16): ID GB_ACC ... Gene.Ontology.Cellular.Component
  Gene.Ontology.Molecular.Function
colnames(280): GSM2772660 GSM2772661 ... GSM2772938 GSM2772939
colData names(72): title geo_accession ... tumorsize.ch1 weight.ch1
```

Printing `se` confirms what you have: a `SummarizedExperiment` whose rows are genes (features) and whose columns are samples. The `assay()` holds the expression matrix, and `colData()` holds the per-sample annotations — the same three-part structure from the previous chapter.

```
dim(se)
```

```
[1] 54715 280
```

So this experiment has 54715 genes measured across 280 samples. The expression values live in the assay named `"exprs"` (a leftover name from the `ExpressionSet` days), and you reach them with `assay(se, "exprs")`.

25.3. Summarizing the sample annotations

Before any analysis, look at what the samples *are*. The interesting variables here are the cancer type and whether each sample is tumor or normal tissue. They live in `colData(se)` under the names `cancer.type.ch1` and `normal.ch1`.

The `table()` function counts unique values, and `with()` lets you refer to those column names directly instead of typing `colData(se)$...` for each one:

```
with(colData(se), table(`cancer.type.ch1`, `normal.ch1`))
```

```

              normal.ch1
cancer.type.ch1 no yes
BC              65  10
CRC             57  12
NSCLC          60   9
PCA            60   7
```


Read this as a grid: each row is a cancer type, and the two columns split it into tumor (no, i.e. *not* normal) and normal samples. Most cancer types here have both tumor and matched normal tissue, with tumor samples outnumbering normals — exactly the design you’d want for a tumor-vs-normal comparison. A quick table like this is the cheapest possible sanity check that the dataset contains what the paper claimed.

25.4. Keeping the most variable genes

We want to see how the samples group. Most of the genes barely change from sample to sample, and those flat genes only add noise. A standard move before dimensionality reduction is to keep just the **most variable** genes — the ones actually carrying the signal that distinguishes samples.

i Why filter to high-variance genes before PCA?

PCA looks for the directions in which your samples differ most. A gene whose expression is nearly constant everywhere contributes almost nothing to those differences — it’s dead weight that dilutes the signal and slows the computation. Ranking genes by their **standard deviation** and keeping the top slice focuses PCA on the genes that vary, which usually makes the biological structure (tumor vs. normal, one cancer type vs. another) far easier to see. There’s no single correct cutoff; 500 genes is a reasonable starting point, and you’re encouraged to try others.

Compute each gene’s standard deviation with `apply()` (which applies a function across the rows or columns of a matrix — here, `sd` to each row, `1` meaning “by row”), then use `order()` to rank genes from most to least variable and keep the top 500:

```
sds <- apply(assay(se, "exprs"), 1, sd)
dat <- assay(se, "exprs")[order(sds, decreasing = TRUE)[1:500], ]
```

`order(sds, decreasing = TRUE)` returns the row positions sorted by standard deviation, largest first; `[1:500]` takes the top 500; and we use those to subset the expression matrix. `dat` is now a 500-gene by 280-sample matrix.

25.5. Running and interpreting the PCA

PCA finds new axes (principal components) that capture the largest patterns of variation in the data. `prcomp()` expects **samples as rows**, so we transpose the matrix with `t()`

before handing it over:

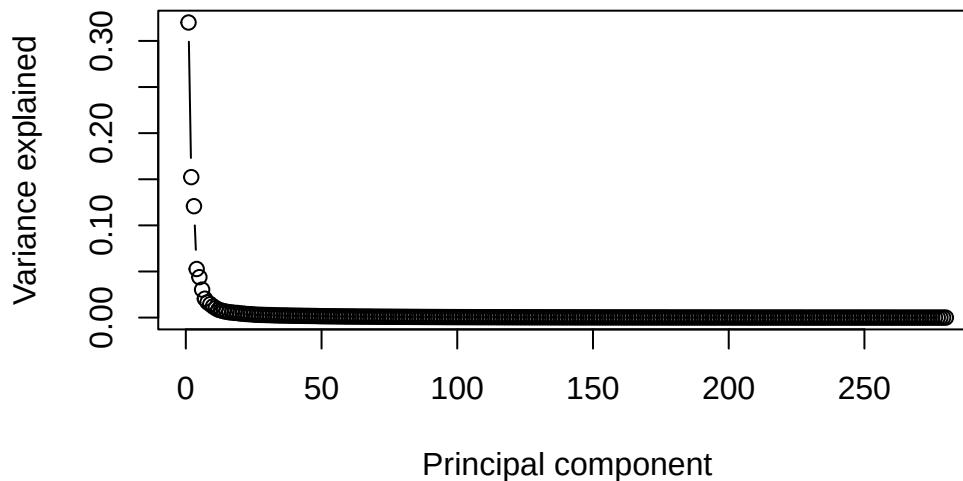
```
pca_results <- prcomp(t(dat))
pca_df <- as.data.frame(pca_results$x)
pca_df$Type <- factor(colData(se)[, "cancer.type.ch1"])
pca_df$Normal <- factor(colData(se)[, "normal.ch1"])
```

`pca_results$x` holds each sample's coordinates along every principal component. We pull those into a data frame and attach the cancer type and tumor/normal label so we can color and shape the plot.

How much of the variation does each component capture? Each component's variance is its squared standard deviation; divide by the total to get a fraction:

```
pca_var <- pca_results$sdev^2 / sum(pca_results$sdev^2)
plot(pca_var, xlab = "Principal component", ylab = "Variance explained",
     main = "Variance explained by each principal component", type = "b")
```

Variance explained by each principal component



The curve drops off steeply: the first principal component (PC1) explains far more of the variation than any other, and after the first few components each one adds little. That steep drop is good news — it means a two-dimensional plot of PC1 and PC2 captures most of what's going on, so it's worth plotting.

```
library(ggplot2)
ggplot(pca_df, aes(x = PC1, y = PC2, shape = Normal, color = Type)) +
  geom_point(alpha = 0.6) +
  theme(text = element_text(size = 18))
```

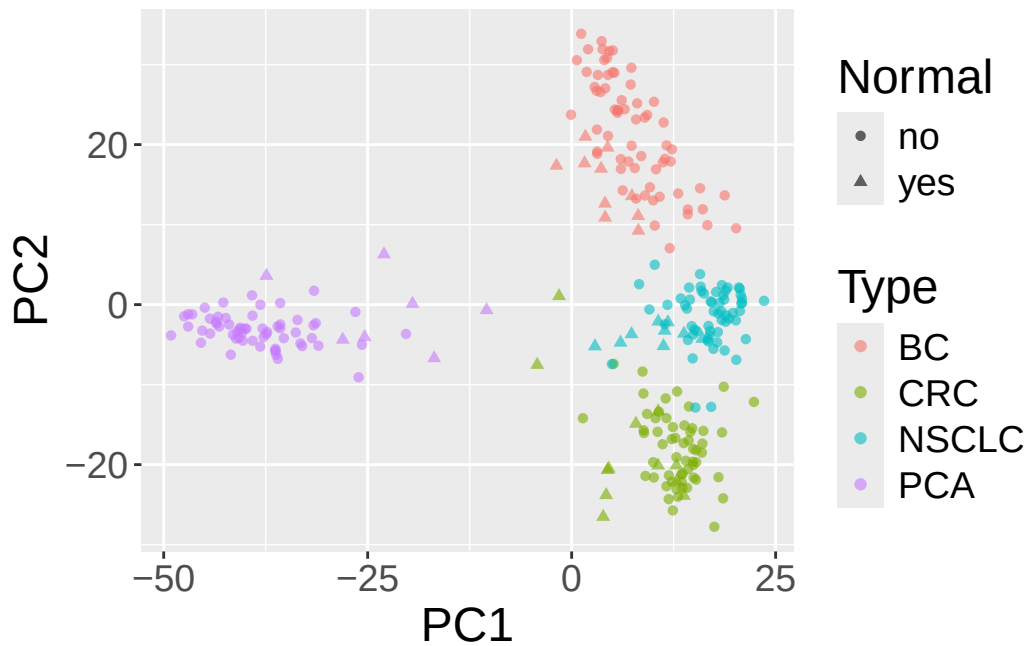


Figure 25.1.: Samples projected onto the first two principal components, colored by cancer type and shaped by tumor/normal status.

In Figure 25.1 the x-axis is PC1 and the y-axis is PC2, each point is one sample, color marks the cancer type, and the shape marks tumor versus normal. The dominant pattern is **cancer type**: samples of the same color cluster together, so PC1 and PC2 are mostly separating tissues of origin from one another. Within each cancer type, the tumor and normal shapes tend to occupy slightly different regions, reflecting the tumor-vs-normal expression differences the original study set out to characterize. In other words, a handful of unsupervised components, computed from just 500 genes, already recover biology you'd expect: different cancers look different, and within a cancer, tumor looks different from normal.

25.6. Summary

You just reused a published dataset from start to finish. You:

- Downloaded GSE103512 from GEO with `getGEO()` and grabbed the dataset with `[[1]]`.
- Converted the legacy `ExpressionSet` into a `SummarizedExperiment` and inspected its dimensions and sample annotations.
- Summarized the cancer-type-by-tumor/normal design with `table()`.
- Filtered to the 500 most variable genes, ran PCA with `prcomp()`, and read the variance-explained plot.
- Plotted PC1 vs. PC2 and saw the samples separate by cancer type, with tumor and normal differing within each type.

The same recipe — fetch, load into a `SummarizedExperiment`, filter, reduce, plot — works for thousands of other GEO datasets.

25.7. Exercises

i 1. Inspect the object

What is the class of `se`, and what are the dimensions of its `assay` and `colData`? Which variables are stored in `colData`?

```
class(se)
dim(assay(se, "exprs")) # genes x samples
dim(colData(se))        # samples x annotation variables
colnames(colData(se))   # the available sample variables
```

`se` is a `SummarizedExperiment`. The assay has one row per gene and one column per sample; `colData` has one row per sample and one column per annotation variable, including `cancer.type.ch1` and `normal.ch1`.

i 2. Change the number of top genes

Re-run the analysis keeping the top 100 genes, and again with the top 2000. Does the PC1-vs-PC2 plot change much?

```
dat100 <- assay(se, "exprs")[order(sds, decreasing = TRUE)[1:100], ]
pca100 <- prcomp(t(dat100))
plot(pca100$x[, 1], pca100$x[, 2], xlab = "PC1", ylab = "PC2")
```

The broad structure — samples separating by cancer type — is usually robust to the exact cutoff. Too few genes can lose finer distinctions; too many let low-variance noise creep back in. The stability of the big picture across cutoffs is itself reassuring.

i 3. Plot more components with `ggpairs()`

Use the `GGally` package's `ggpairs()` to view the first few principal components together, colored by cancer type.

```
library(GGally)
ggpairs(pca_df, columns = 1:4, mapping = aes(color = Type))
```

`ggpairs()` draws every pairwise combination of the chosen columns (here PC1–PC4) in a grid, so you can spot structure that PC1-vs-PC2 alone might hide — for example, a cancer type that only separates along PC3 or PC4.

26. Storing experiments with SummarizedExperiment

An RNA-seq or microarray experiment hands you several things at once: a matrix of numbers (counts or intensities), a description of every feature down the side (which gene is in each row), and a description of every sample across the top (which condition, which patient). Keep these in three separate variables and they *will* drift out of sync — you drop a bad sample from the count matrix, forget to drop it from the sample table, and now your “treated vs. control” labels point at the wrong columns. Mismatches exactly like this have produced retracted papers.

SummarizedExperiment is Bioconductor’s answer: one object that holds the assay matrix, the feature data, and the sample data together, and keeps them aligned for you whenever you subset.

It is, in many ways, a modern replacement for the historical **ExpressionSet**. The main difference is that **SummarizedExperiment** is more flexible about its feature (row) information: features can be described either by genomic ranges (**GRanges**) or by an arbitrary table (**DataFrame**). That flexibility makes it a natural fit for sequencing experiments such as RNA-seq and ChIP-seq.

26.1. What you’ll learn

By the end of this chapter you will be able to:

- Describe the three coordinated parts of a **SummarizedExperiment**: `assays()`, `rowData()/rowRanges()`, and `colData()`.
- Subset an experiment by sample or feature and trust that the metadata follows.
- Reach into assays, sample data, and experiment-wide metadata with the accessor functions.
- Build a **SummarizedExperiment** from raw matrices and data frames.

This chapter assumes you’ve met `GRanges` and `DataFrame`. If “genomic ranges” isn’t familiar yet, the *GenomicRanges* chapters come earlier in the Bioconductor part of the book — skim those first.

💡 Installing the packages

The examples below use the `SummarizedExperiment` class and the `airway` example dataset. Install both once with `BiocManager`:

```
BiocManager::install(c("SummarizedExperiment", "airway"))
```

26.2. Anatomy of a *SummarizedExperiment*

The *SummarizedExperiment* package contains two classes: `SummarizedExperiment` and `RangedSummarizedExperiment`.

`SummarizedExperiment` is a matrix-like container where rows represent features of interest (e.g. genes, transcripts, exons, etc.) and columns represent samples. The objects contain one or more assays, each represented by a matrix-like object of numeric or other mode. The rows of a `SummarizedExperiment` object represent features of interest. Information about these features is stored in a `DataFrame` object, accessible using the function `rowData()`. Each row of the `DataFrame` provides information on the feature in the corresponding row of the `SummarizedExperiment` object. Columns of the `DataFrame` represent different attributes of the features of interest, e.g., gene or transcript IDs, etc.

`RangedSummarizedExperiment` is the “child” of the `SummarizedExperiment` class, which means that all the methods on `SummarizedExperiment` also work on a `RangedSummarizedExperiment`.

The fundamental difference between the two classes is that the rows of a `RangedSummarizedExperiment` object represent genomic ranges of interest instead of a `DataFrame` of features. The `RangedSummarizedExperiment` ranges are described by a `GRanges` or a `GRangesList` object, accessible using the `rowRanges()` function.

Figure 26.1 displays the class geometry and highlights the vertical (column) and horizontal (row) relationships.

i A mental model: three linked sheets

Picture a spreadsheet workbook with three sheets that are *locked together*:

- the **assay** sheet — the matrix of numbers, features (genes) $\times$ samples;

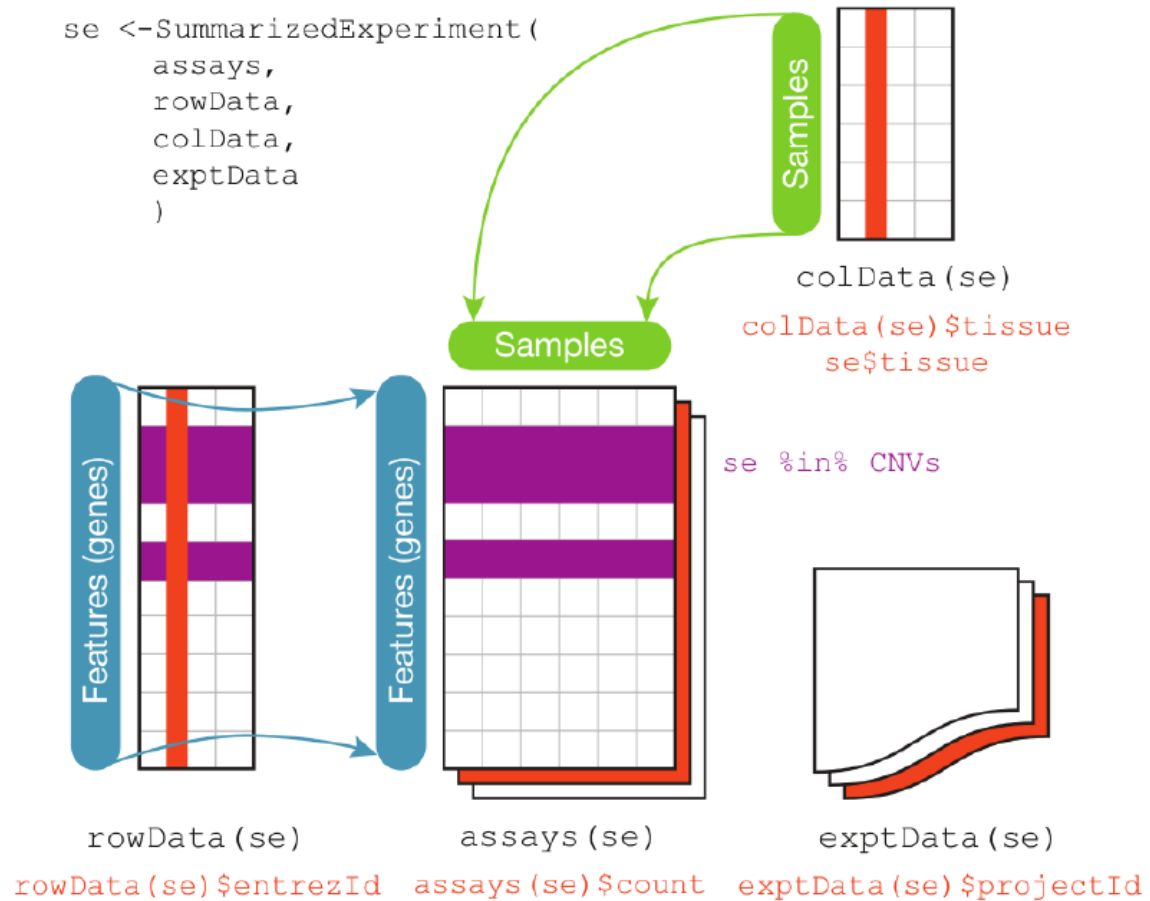


Figure 26.1.: Summarized Experiment. There are three main components, the `colData()`, the `rowData()` and the `assays()`. The accessors for the various parts of a complete `SummarizedExperiment` object match the names.

26. Storing experiments with *SummarizedExperiment*

- the **row** sheet (`rowData/rowRanges`) — one row of annotation per feature, lined up with the assay's rows;
- the **column** sheet (`colData`) — one row of annotation per sample, lined up with the assay's columns.

The magic is the lock: when you delete a sample, all three sheets update together, so the numbers and their labels can never disagree. Every accessor below is just a way of looking at one of these sheets.

26.2.1. Assays

The `airway` package contains an example dataset from an RNA-seq experiment of read counts per gene for airway smooth muscle cells. These data are stored in a `RangedSummarizedExperiment` object that holds read counts for tens of thousands of gene transcripts across 8 samples (we'll confirm the exact dimensions in a moment).

Loading required package: `airway`

```
library(SummarizedExperiment)
data(airway, package="airway")
se <- airway
se
```

```
class: RangedSummarizedExperiment
dim: 63677 8
metadata(1): ''
assays(1): counts
rownames(63677): ENSG000000000003 ENSG000000000005 ... ENSG00000273492
               ENSG00000273493
rowData names(10): gene_id gene_name ... seq_coord_system symbol
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(9): SampleName cell ... Sample BioSample
```

Read that printout like a label on the box: `dim` tells you it's 63,677 features across 8 samples, and the `assays`, `rowRanges`, and `colData` lines preview the three linked sheets from the model above. The sections that follow open each one in turn.

To retrieve the experiment data from a `SummarizedExperiment` object one can use the `assays()` accessor. An object can have multiple assay datasets each of which can be

26. Storing experiments with SummarizedExperiment

accessed using the `$` operator. The `airway` dataset contains only one assay (`counts`). Here each row represents a gene transcript and each column one of the samples.

```
assays(se)$counts
```

	SRR103951	SRR103952	SRR103953	SRR103954	SRR103955	SRR103956	SRR103957
ENSG000000000000379	448	873	408	1138	1047	770	572
ENSG00000000000050	0	0	0	0	0	0	0
ENSG0000000000004467	515	621	365	587	799	417	508
ENSG0000000000004360	211	263	164	245	331	233	229
ENSG0000000000004660	55	40	35	78	63	76	60
ENSG0000000000009380	0	2	0	1	0	0	0
ENSG0000000000009251	3679	6177	4252	6721	11027	5176	7995
ENSG00000000000010333	1062	1733	881	1424	1439	1359	1109
ENSG00000000000010519	380	595	493	820	714	696	704
ENSG00000000000011694	236	464	175	658	584	360	269

26.2.2. 'Row' (regions-of-interest) data

The `rowRanges()` accessor is used to view the range information for a `RangedSummarizedExperiment`. (Note if this were the parent `SummarizedExperiment` class we'd use `rowData()`). The data are stored in a `GRangesList` object, where each list element corresponds to one gene transcript and the ranges in each `GRanges` correspond to the exons in the transcript.

```
rowRanges(se)
```

```
GRangesList object of length 63677:
```

\$ENSG000000000003

GRanges object with 17 ranges and 2 metadata columns:

seqnames		ranges	strand	exon_id	exon_name
<Rle>	<IRanges>	<Rle>	<integer>	<character>	
[1]	X 99883667-99884983	-	667145	ENSE00001459322	
[2]	X 99885756-99885863	-	667146	ENSE00000868868	
[3]	X 99887482-99887565	-	667147	ENSE00000401072	
[4]	X 99887538-99887565	-	667148	ENSE00001849132	
[5]	X 99888402-99888536	-	667149	ENSE00003554016	
...	...	...	...	...	...
[13]	X 99890555-99890743	-	667156	ENSE00003512331	

26. Storing experiments with SummarizedExperiment

```
[14]      X 99891188-99891686      - |      667158 ENSE00001886883
[15]      X 99891605-99891803      - |      667159 ENSE00001855382
[16]      X 99891790-99892101      - |      667160 ENSE00001863395
[17]      X 99894942-99894988      - |      667161 ENSE00001828996
-----
seqinfo: 722 sequences (1 circular) from an unspecified genome

...
<63676 more elements>
```

26.2.3. 'Column' (sample) data

Sample meta-data describing the samples can be accessed using `colData()`, and is a `DataFrame` that can store any number of descriptive columns for each sample row.

```
colData(se)
```

DataFrame with 8 rows and 9 columns

	SampleName	cell	dex	albut	Run	avgLength
	<factor>	<factor>	<factor>	<factor>	<factor>	<integer>
SRR1039508	GSM1275862	N61311	untrt	untrt	SRR1039508	126
SRR1039509	GSM1275863	N61311	trt	untrt	SRR1039509	126
SRR1039512	GSM1275866	N052611	untrt	untrt	SRR1039512	126
SRR1039513	GSM1275867	N052611	trt	untrt	SRR1039513	87
SRR1039516	GSM1275870	N080611	untrt	untrt	SRR1039516	120
SRR1039517	GSM1275871	N080611	trt	untrt	SRR1039517	126
SRR1039520	GSM1275874	N061011	untrt	untrt	SRR1039520	101
SRR1039521	GSM1275875	N061011	trt	untrt	SRR1039521	98
	Experiment	Sample	BioSample			
	<factor>	<factor>	<factor>			
SRR1039508	SRX384345	SRS508568	SAMN02422669			
SRR1039509	SRX384346	SRS508567	SAMN02422675			
SRR1039512	SRX384349	SRS508571	SAMN02422678			
SRR1039513	SRX384350	SRS508572	SAMN02422670			
SRR1039516	SRX384353	SRS508575	SAMN02422682			
SRR1039517	SRX384354	SRS508576	SAMN02422673			
SRR1039520	SRX384357	SRS508579	SAMN02422683			
SRR1039521	SRX384358	SRS508580	SAMN02422677			

26. Storing experiments with *SummarizedExperiment*

Each row here is one sample, and each column is something we know about that sample — including `dex`, which records whether the cells were treated with dexamethasone. That `dex` column is what makes the next operation possible.

This sample metadata can be accessed using the `$` accessor, which makes it easy to subset the entire object by a given phenotype.

```
# subset for only those samples treated with dexamethasone
se[, se$dex == "trt"]

class: RangedSummarizedExperiment
dim: 63677 4
metadata(1): ''
assays(1): counts
rownames(63677): ENSG000000000003 ENSG000000000005 ... ENSG00000273492
               ENSG00000273493
rowData names(10): gene_id gene_name ... seq_coord_system symbol
colnames(4): SRR1039509 SRR1039513 SRR1039517 SRR1039521
colData names(9): SampleName cell ... Sample BioSample
```

26.2.4. Experiment-wide metadata

Meta-data describing the experimental methods and publication references can be accessed using `metadata()`.

```
metadata(se)

[[1]]
Experiment data
  Experimenter name: Himes BE
  Laboratory: NA
  Contact information:
  Title: RNA-Seq transcriptome profiling identifies CRISPLD2 as a glucocorticoid responsive
  URL: http://www.ncbi.nlm.nih.gov/pubmed/24926665
  PMIDs: 24926665

  Abstract: A 226 word abstract is available. Use 'abstract' method.
```

Note that `metadata()` is just a simple list, so it is appropriate for *any* experiment wide metadata the user wishes to save, such as storing model formulas.

26. Storing experiments with SummarizedExperiment

```
metadata(se)$formula <- counts ~ dex + albut
```

```
metadata(se)
```

```
[[1]]
```

```
Experiment data
```

```
  Experimenter name: Himes BE
```

```
  Laboratory: NA
```

```
  Contact information:
```

```
  Title: RNA-Seq transcriptome profiling identifies CRISPLD2 as a glucocorticoid responsive
```

```
  URL: http://www.ncbi.nlm.nih.gov/pubmed/24926665
```

```
  PMIDs: 24926665
```

```
  Abstract: A 226 word abstract is available. Use 'abstract' method.
```

```
$formula
```

```
counts ~ dex + albut
```

26.3. Common operations on SummarizedExperiment

26.3.1. Subsetting

- [Performs two dimensional subsetting, just like subsetting a matrix or data frame.

```
# subset the first five transcripts and first three samples
se[1:5, 1:3]
```

```
class: RangedSummarizedExperiment
```

```
dim: 5 3
```

```
metadata(2): '' formula
```

```
assays(1): counts
```

```
rownames(5): ENSG00000000003 ENSG00000000005 ENSG000000000419
```

```
  ENSG000000000457 ENSG000000000460
```

```
rowData names(10): gene_id gene_name ... seq_coord_system symbol
```

```
colnames(3): SRR1039508 SRR1039509 SRR1039512
```

```
colData names(9): SampleName cell ... Sample BioSample
```

- \$ operates on colData() columns, for easy sample extraction.

```
se[, se$cell == "N61311"]
```

```
class: RangedSummarizedExperiment
dim: 63677 2
metadata(2): '' formula
assays(1): counts
rownames(63677): ENSG000000000003 ENSG000000000005 ... ENSG00000273492
               ENSG00000273493
rowData names(10): gene_id gene_name ... seq_coord_system symbol
colnames(2): SRR1039508 SRR1039509
colData names(9): SampleName cell ... Sample BioSample
```

26.3.2. Getters and setters

- `rowRanges()` / `(rowData(), colData(), metadata())`

```
counts <- matrix(1:15, 5, 3, dimnames=list(LETTERS[1:5], LETTERS[1:3]))

dates <- SummarizedExperiment(assays=list(counts=counts),
                             rowData=DataFrame(month=month.name[1:5], day=1:5))

# Subset all January assays
dates[rowData(dates)$month == "January", ]
```

```
class: SummarizedExperiment
dim: 1 3
metadata(0):
assays(1): counts
rownames(1): A
rowData names(2): month day
colnames(3): A B C
colData names(0):
```

- `assay()` versus `assays()` There are two accessor functions for extracting the assay data from a `SummarizedExperiment` object. `assays()` operates on the entire list of assay data as a whole, while `assay()` operates on only one assay at a time. `assay(x, i)` is simply a convenience function which is equivalent to `assays(x)[[i]]`.

26. Storing experiments with SummarizedExperiment

```
assays(se)
```

```
List of length 1
names(1): counts
```

```
assays(se)[[1]][1:5, 1:5]
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	679	448	873	408	1138
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	60	55	40	35	78

```
# assay defaults to the first assay if no i is given
assay(se)[1:5, 1:5]
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	679	448	873	408	1138
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	60	55	40	35	78

```
assay(se, 1)[1:5, 1:5]
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	679	448	873	408	1138
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	60	55	40	35	78

26.3.3. Range-based operations

- `subsetByOverlaps()` `SummarizedExperiment` objects support all of the `findOverlaps()` methods and associated functions. This includes `subsetByOverlaps()`, which makes it easy to subset a `SummarizedExperiment` object by an interval.

In the next code block, we define a region of interest (a single genomic range) and then subset our `SummarizedExperiment` to the genes that overlap it.

```
# Subset for only the genes that fall in the interval 100,000 to 1,100,000
# of chromosome 1
roi <- GRanges(seqnames="1", ranges=IRanges(start=100000, end=1100000))
sub_se <- subsetByOverlaps(se, roi)
sub_se
```

```
class: RangedSummarizedExperiment
dim: 74 8
metadata(2): '' formula
assays(1): counts
rownames(74): ENSG00000131591 ENSG00000177757 ... ENSG00000272512
             ENSG00000273443
rowData names(10): gene_id gene_name ... seq_coord_system symbol
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521
colData names(9): SampleName cell ... Sample BioSample
```

```
dim(sub_se)
```

```
[1] 74 8
```

Notice that `subsetByOverlaps()` worked on the *ranges* (`rowRanges`) but returned a full `SummarizedExperiment` — the assay matrix and `colData` came along automatically, kept in sync with the smaller set of genes.

26.4. Constructing a SummarizedExperiment

To construct a `SummarizedExperiment` object, you need to provide the following components: - **assays**: A list of matrices or matrix-like objects containing the data. - **rowData**: A `DataFrame` containing information about the features (rows). - **colData**: A `DataFrame`

26. Storing experiments with SummarizedExperiment

containing information about the samples (columns). - +/- **metadata**: A list containing additional metadata about the experiment.

For a nearly real example, we will use the DeRisi dataset. We'll start with the original data, which is a **data.frame** with the first couple of columns containing the gene information and the rest of the columns containing the data.

```
# Load the DeRisi dataset
deRisi <- read.csv("https://raw.githubusercontent.com/seandavi/RBiocBook/refs/heads/main/data/deRisi.csv")
head(deRisi)
```

	ORF	Name	R1	R2	R3	R4	R5	R6	R7	R1.Bkg	R2.Bkg	R3.Bkg
1	YHR007C	ERG11	7896	7484	14679	14617	9853	7599	6490	1155	1984	1323
2	YBR218C	PYC2	12144	11177	10241	4820	4950	7047	17035	1074	1694	1243
3	YAL051W	FUN43	4478	6435	6230	6848	5111	7180	4497	1140	1950	1649
4	YAL053W		6343	8243	6743	3304	3556	4694	3849	1020	1897	1196
5	YAL054C	ACS1	1542	3044	2076	1695	1753	4806	10802	1082	1940	1504
6	YAL055W		1769	3243	2094	1367	1853	3580	1956	975	1821	1185
	R4.Bkg	R5.Bkg	R6.Bkg	R7.Bkg	G1	G2	G3	G4	G5	G6	G7	G1.Bkg
1	1171	914	2445	981	8432	7173	11736	16798	12315	16111	13931	2404
2	876	1211	2444	742	11509	10226	13372	6500	6255	9024	6904	2148
3	1183	898	2637	927	5865	5895	5345	6302	5400	7933	5026	2422
4	881	1045	2518	697	6762	7454	6323	3595	4689	5660	4145	2107
5	1108	902	2610	980	3138	3785	2419	2114	2763	3561	1897	2405
6	851	1047	2536	698	2844	4069	2583	1651	2530	3484	1550	1674
	G2.Bkg	G3.Bkg	G4.Bkg	G5.Bkg	G6.Bkg	G7.Bkg						
1	2561	1598	1506	1696	2667	1244						
2	2527	1641	1196	1553	2569	848						
3	2496	1902	1501	1644	2808	1154						
4	2663	1607	1162	1577	2544	857						
5	2528	1847	1445	1713	2767	1142						
6	2648	1591	1114	1528	2668	870						

To convert this to a **SummarizedExperiment**, we need to extract the assay data, row data, and column data. The assay data will be the numeric values in the data frame, the row data will be the gene information, and the column data will be the sample information.

26.4.1. rowData, or feature information

Let's start with the **rowData**, which will be a **DataFrame** containing the gene information. We can use the first two columns of the data frame for this purpose.

```
rdata <- deRisi[, 1:2]
head(rdata)
```

```
      ORF  Name
1 YHR007C ERG11
2 YBR218C  PYC2
3 YAL051W FUN43
4 YAL053W
5 YAL054C  ACS1
6 YAL055W
```

26.4.2. colData, or sample information

Next, we will create the `colData`, which will be a `DataFrame` containing the sample information. Since the sample information really isn't in the dataset, we will create a simple `DataFrame` with sample names.

```
cdata <- DataFrame(sample=paste("Sample", 0:6), timepoint = 0:6,
  hours = c(0, 9.5, 11.5, 13.5, 15.5, 18.5, 20.5))
head(cdata)
```

DataFrame with 6 rows and 3 columns

```
      sample timepoint      hours
<character> <integer> <numeric>
1   Sample 0         0        0.0
2   Sample 1         1         9.5
3   Sample 2         2        11.5
4   Sample 3         3        13.5
5   Sample 4         4        15.5
6   Sample 5         5        18.5
```

26.4.3. assays, or the data

Remember that the DeRisi dataset has four *different* assays,

assay	description
R	Red fluorescence

26. Storing experiments with *SummarizedExperiment*

assay	description
G	Green fluorescence
Rb	Red background fluorescence
Gb	Green background fluorescence

We will create a list of matrices, one for each assay. The matrices will be the numeric values in the data frame, excluding the first two columns.

```
R <- as.matrix(deRisi[, 3:9])
G <- as.matrix(deRisi[, 10:16])
Rb <- as.matrix(deRisi[, 17:23])
Gb <- as.matrix(deRisi[, 24:30])
```

When we create a *SummarizedExperiment* object, the “constructor” will check to see that the colnames of the matrices in the list are the same as the *rownames* of the *colData* *DataFrame*, and that the rownames of the matrices in the list are the same as the *rownames* of the *rowData* *DataFrame*.

So, we need to fix that all up. Let’s start with the rownames of the *rowData* *DataFrame*:

```
rownames(rdata) <- rdata$ORF
```

Now, let’s set the rownames of the *coldata* *DataFrame* to the sample names:

```
rownames(cdata) <- cdata$sample
```

Now, we can fix the rownames and colnames of the matrices for our R, G, Rb, and Gb assays:

```
rownames(R) <- rdata$ORF
rownames(G) <- rdata$ORF
rownames(Rb) <- rdata$ORF
rownames(Gb) <- rdata$ORF
colnames(R) <- cdata$sample
colnames(G) <- cdata$sample
colnames(Rb) <- cdata$sample
colnames(Gb) <- cdata$sample
```

Take a look at the matrices to make sure they look right.

26.4.4. Putting it all together

```
se <- SummarizedExperiment(assays=list(R=R, G=G, Rb=Rb, Gb=Gb), rowData=rdata, colData=cdat
```

26.4.5. Getting logRatios

Now that we have a `SummarizedExperiment` object, we can easily compute the log ratios of the Red and Green foreground fluorescence. This is a common operation in microarray data analysis.

```
logRatios <- log2(
  (assay(se, "R") - assay(se, "Rb")) / (assay(se, "G") - assay(se, "Gb"))
)
```

Warning: NaNs produced

```
assays(se)$logRatios <- logRatios
```

Now, we've added a new assay to the `SummarizedExperiment` object called `logRatios`. This assay contains the log ratios of the Red and Green foreground fluorescence.

```
assays(se)
```

```
List of length 5
names(5): R G Rb Gb logRatios
```

And if we want to access the log ratios, we can do so using the `assay()` method:

```
head(assay(se, "logRatios"))
```

	Sample 0	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
YHR007C	-1.2204686	NaN	NaN	2.70275677	1.6545902	5.260867
YBR218C	NaN	NaN	2.9757832	2.39231742	1.9319816	3.983313
YAL051W	0.1135715	NaN	NaN	NaN	-1.3681061	2.138654
YAL053W	-1.3753298	NaN	NaN	0.05044902	1.0906497	5.215440
YAL054C	0.2706476	0.333657387	0.0000000	0.31420165	0.3165815	NaN

26. Storing experiments with SummarizedExperiment

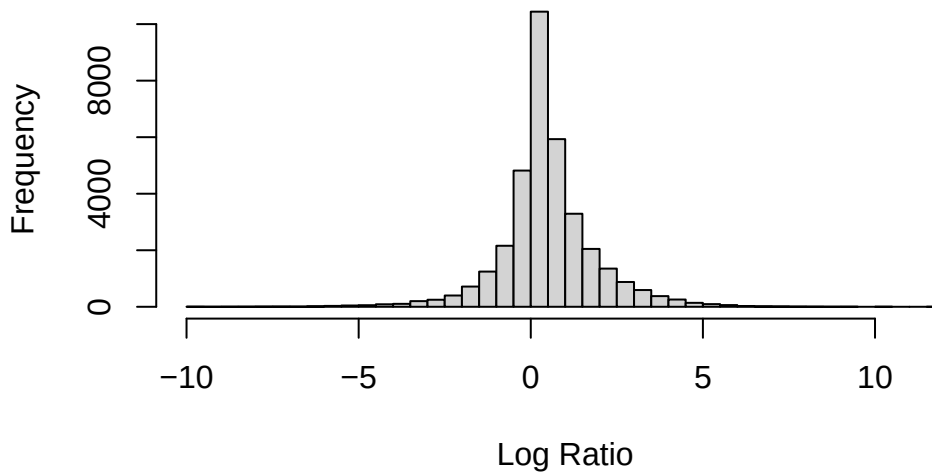
```
YAL055W 0.6209723 -0.001745548 0.2683547 0.11082813 0.4931189      NaN
      Sample 6
YHR007C 4.8223618
YBR218C      NaN
YAL051W 1.2205754
YAL053W 0.8875253
YAL054C      NaN
YAL055W      NaN
```

```
## OR
head(assays(se)$logRatios)
```

```
      Sample 0      Sample 1 Sample 2 Sample 3 Sample 4 Sample 5
YHR007C -1.2204686      NaN      NaN 2.70275677 1.6545902 5.260867
YBR218C      NaN      NaN 2.9757832 2.39231742 1.9319816 3.983313
YAL051W 0.1135715      NaN      NaN      NaN -1.3681061 2.138654
YAL053W -1.3753298      NaN      NaN 0.05044902 1.0906497 5.215440
YAL054C 0.2706476 0.333657387 0.0000000 0.31420165 0.3165815      NaN
YAL055W 0.6209723 -0.001745548 0.2683547 0.11082813 0.4931189      NaN
      Sample 6
YHR007C 4.8223618
YBR218C      NaN
YAL051W 1.2205754
YAL053W 0.8875253
YAL054C      NaN
YAL055W      NaN
```

```
hist(assays(se)$logRatios, breaks=50, main="Log Ratios of Red and Green Foreground Fluoresc
```

Log Ratios of Red and Green Foreground Fluorescence



26.5. Summary

A `SummarizedExperiment` bundles an assay matrix together with aligned feature data (`rowData/rowRanges`) and sample data (`colData`) into a single object that stays internally consistent when you subset. In this chapter you toured the accessors (`assays()`, `rowData()`, `colData()`, `metadata()`), subset an experiment by phenotype and by genomic range, and built a `SummarizedExperiment` from scratch out of the DeRisi microarray data. From here on, most Bioconductor analysis packages will hand you — or expect — an object shaped exactly like this.

26.6. Exercises

The `airway` dataset is a good sandbox. Load a fresh copy so you're not relying on any object created earlier in the chapter:

```
data(airway, package = "airway")
air <- airway
```

26. Storing experiments with *SummarizedExperiment*

1. How many genes (rows) and samples (columns) does `air` have? Which accessor tells you each sample's treatment status?
2. Subset `air` to just the untreated samples (`dex == "untrt"`) and confirm that both the assay matrix and the `colData` shrank to match.
3. Pull the `counts` assay for the first 5 genes and first 3 samples into a plain matrix.

i Solutions

```
# 1. shape and the treatment column
dim(air)           # genes x samples
```

```
[1] 63677      8
```

```
colData(air)$dex    # treatment per sample
```

```
[1] untrt trt   untrt trt   untrt trt   untrt trt
Levels: trt untrt
```

```
# 2. subsetting carries the metadata along
untrt <- air[, air$dex == "untrt"]
dim(untrt)           # fewer columns; same number of rows
```

```
[1] 63677      4
```

```
ncol(colData(untrt)) # colData has the same columns, fewer rows
```

```
[1] 9
```

```
# 3. reach into one assay and slice it
assay(air, "counts")[1:5, 1:3]
```

	SRR1039508	SRR1039509	SRR1039512
ENSG000000000003	679	448	873
ENSG000000000005	0	0	0
ENSG000000000419	467	515	621
ENSG000000000457	260	211	263
ENSG000000000460	60	55	40

The key idea in exercise 2: you subset the *object*, and `colData` follows in lockstep — you never have to remember to subset the sample table separately.

27. Analyzing microbiome data

The human microbiome is a crowded ecosystem of bacteria, viruses, fungi, and other microorganisms living in and on the human body. These microbes shape human health and disease, influencing everything from digestion and metabolism to immune function and mood. Working out *which* microbes are present, in *what* proportions, and how those proportions shift between healthy and sick people is one of the most active areas in biology today.

In this chapter you'll take a curated gut-microbiome dataset from a colorectal cancer study, load it into R, and ask the kinds of questions a microbiome researcher asks: How diverse is each sample? Do cancer patients and controls separate when you compare whole communities? Which microbes dominate? You'll do all of this without leaving a single, self-describing data object.

We'll assume the raw sequencing data has already been processed into a tidy abundance table, so you can focus on the analysis rather than the bioinformatics that came before it.

27.1. What you'll learn

By the end of this chapter you will be able to:

- Load a curated microbiome dataset as a `TreeSummarizedExperiment` object.
- Reach into its parts with `assays()`, `colData()`, `rowData()`, and `rowTree()`.
- Subset samples and agglomerate features to a coarser taxonomic rank.
- Compute alpha diversity (within a sample) and beta diversity (between samples).
- Visualize community composition with abundance plots, an ordination, and a heatmap.

This chapter builds directly on the [SummarizedExperiment chapter](#). A `TreeSummarizedExperiment` is a `SummarizedExperiment` with a couple of extra parts, so if the three-linked-sheets idea (`assays`, `rowData`, `colData`) isn't fresh, skim that chapter first.

27.2. Getting started

We'll lean on the `mia` package (short for *microbiome analysis*) throughout this chapter. It provides a suite of tools for loading, transforming, and visualizing microbiome data, all built around the `SummarizedExperiment` and `TreeSummarizedExperiment` classes you'll meet below.

`mia` is a released Bioconductor package, so you install it the usual way with `BiocManager`. If you don't already have `BiocManager` itself, install that first (see the Bioconductor installation chapter), then run:

```
# Install the mia package (a normal Bioconductor install)
BiocManager::install("mia")
```

Note

`mia` is a standard, released Bioconductor package. The `BiocManager::install("mia")` call above pulls the stable version that matches your Bioconductor release — the same way you install every other Bioconductor package in this book.

Once `mia` is installed, load it into your R session:

```
# Load the mia package
library(mia)
```

Now that the `mia` package is loaded, we can load the microbiome dataset that we will be working with.

27.3. Accessing microbiome datasets

There are many publicly available microbiome datasets you can use for analysis. Bioconductor provides several packages that bundle curated microbiome datasets, making it easy to load real data into R without first wrangling raw sequencing files.

[curatedMetagenomicData](#) is a large collection of curated human microbiome datasets, provided as `TreeSE` objects (Pasolli et al. 2017). The resource provides curated human microbiome data including gene families, marker abundance, marker presence, pathway abundance, pathway coverage, and relative abundance for samples from different body sites. See the package homepage for more details on data availability and access.

The `microbiomeDataSets` package provides a collection of curated microbiome datasets for teaching and research purposes. The package contains several example datasets that can be used to explore different aspects of microbiome analysis, such as alpha and beta diversity, differential abundance analysis, and visualization.

The `mia` package and several other packages provide example datasets for learning and testing the functions in the package. These datasets are typically small and easy to work with, making them ideal for learning how to analyze microbiome data in R.

27.4. Data containers

Bioconductor provides several specialized data structures for storing and working with microbiome data in R. These data structures are designed to handle the complex and high-dimensional nature of microbiome data and provide a convenient and efficient way to store, manipulate, and analyze microbiome data in R.

`SummarizedExperiment` (SE) (Morgan et al. 2020) is a generic, highly optimized container for complex experimental data. It has become a common choice for many kinds of biomedical profiling data, such as RNA-seq, ChIP-seq, microarrays, flow cytometry, proteomics, and single-cell sequencing.

`TreeSummarizedExperiment` (Huang 2020) extends that container to carry hierarchical information — phylogenetic trees and sample hierarchies — alongside the usual assays and metadata.

i A mental model: three linked sheets, plus a tree

In the `SummarizedExperiment` chapter you met the *three linked sheets* picture: an **assay** sheet of numbers (features $\times$ samples), a **row** sheet (`rowData`) describing each feature, and a **column** sheet (`colData`) describing each sample, all locked together so they can never drift out of sync.

A `TreeSummarizedExperiment` is exactly those three sheets with one more piece pinned to the side: a **phylogenetic tree** (`rowTree`) that records how the features — the microbes — are related to one another by evolution. When you drop a microbe from the object, it drops from the tree too. So everything you already know about reaching into a `SummarizedExperiment` carries over here; the tree is the only genuinely new idea.

27.5. Loading a microbiome dataset

We will be using the `curatedMetagenomicData` package to load the `FengQ_2015` dataset. This dataset contains microbiome data from a study by [Feng et al. \(2015\)](#). The `curatedMetagenomicData` package provides a convenient interface for accessing microbiome datasets that have been pre-processed and curated for analysis.

i Note

Take a moment to read at least the title and abstract of the paper. We won't reproduce its analysis, but connecting the code in this workflow to the underlying biology and research question will make everything that follows more meaningful.

```
# Again, before using a package, we must install it
# just once.
BiocManager::install('curatedMetagenomicData')
```

Once installed, we can use the package.

Ignore the details of the next few lines of code for now, but suffice it to say that what the code will do for us is to load a dataset for downstream analysis.

```
# Load the FengQ_2015 dataset
library(curatedMetagenomicData)
tse <- curatedMetagenomicData("FengQ_2015.relative_abundance", dryrun = FALSE, rownames = "
```

```
$`2021-03-31.FengQ_2015.relative_abundance`
dropping rows without rowTree matches:
k__Bacteria|p__Actinobacteria|c__Coriobacteriia|o__Coriobacteriales|f__Atopobiaceae|g__01
k__Bacteria|p__Actinobacteria|c__Coriobacteriia|o__Coriobacteriales|f__Coriobacteriaceae|
k__Bacteria|p__Actinobacteria|c__Coriobacteriia|o__Coriobacteriales|f__Coriobacteriaceae|
k__Bacteria|p__Firmicutes|c__Bacilli|o__Bacillales|f__Bacillales_unclassified|g__Gemella|
k__Bacteria|p__Firmicutes|c__Bacilli|o__Lactobacillales|f__Carnobacteriaceae|g__Granulica
k__Bacteria|p__Firmicutes|c__Clostridia|o__Clostridiales|f__Ruminococcaceae|g__Ruminococc
k__Bacteria|p__Firmicutes|c__Erysipelotrichia|o__Erysipelotrichales|f__Erysipelotrichacea
k__Bacteria|p__Proteobacteria|c__Betaproteobacteria|o__Burkholderiales|f__Sutterellaceae|
k__Bacteria|p__Synergistetes|c__Synergistia|o__Synergistales|f__Synergistaceae|g__Cloacib
```

Duplicated labels were made unique.

27. Analyzing microbiome data

This code goes out to the internet and downloads the dataset for you. The `tse` object is a `TreeSummarizedExperiment` object that contains the microbiome data from the Feng et al. (2015) study. The `TreeSummarizedExperiment` class is a specialized data structure for storing microbiome data in R, and it is used by the `mia` package for microbiome analysis.

Tip

If you are working with your own microbiome data, you will often need to load it into R using functions like `read.csv()` or `read.table()` to read tabular data files. Make sure your data is properly formatted and cleaned before loading it into R for analysis.

Once the data is loaded, you can start exploring its structure and contents. The `tse` object is a `TreeSummarizedExperiment`. You can think of it as a richer relative of a `data.frame`: it has many specialized slots for storing the parts of a microbiome experiment, laid out in Figure 27.1. We'll use a handful of accessor functions to look inside it. See the [TreeSummarizedExperiment](#) package for the full details.

In microbiome data science, containers like this one link a taxonomic abundance table to rich side information about both the features (the microbes) and the samples. That abundance data can come from 16S rRNA amplicon sequencing, metagenomic sequencing, phylogenetic microarrays, or other methods. A single experiment often holds several versions of the data at once — raw counts plus transformed or agglomerated versions derived from them — and the container keeps them all together.

To get an overview of the data, we can simply type the name of the object in the R console:

```
# Print the object
tse
```

```
class: TreeSummarizedExperiment
dim: 606 154
metadata():
assays(1): relative_abundance
rownames(606): species:Faecalibacterium prausnitzii
               species:Streptococcus salivarius ... species:Lactobacillus
               kullabergensis species:Lactobacillus kimbladii
rowData names(7): superkingdom phylum ... genus species
colnames(154): SID31004 SID31009 ... SID532832 SID532915
colData names(28): study_name subject_id ... triglycerides hba1c
reducedDimNames(0):
mainExpName: NULL
```

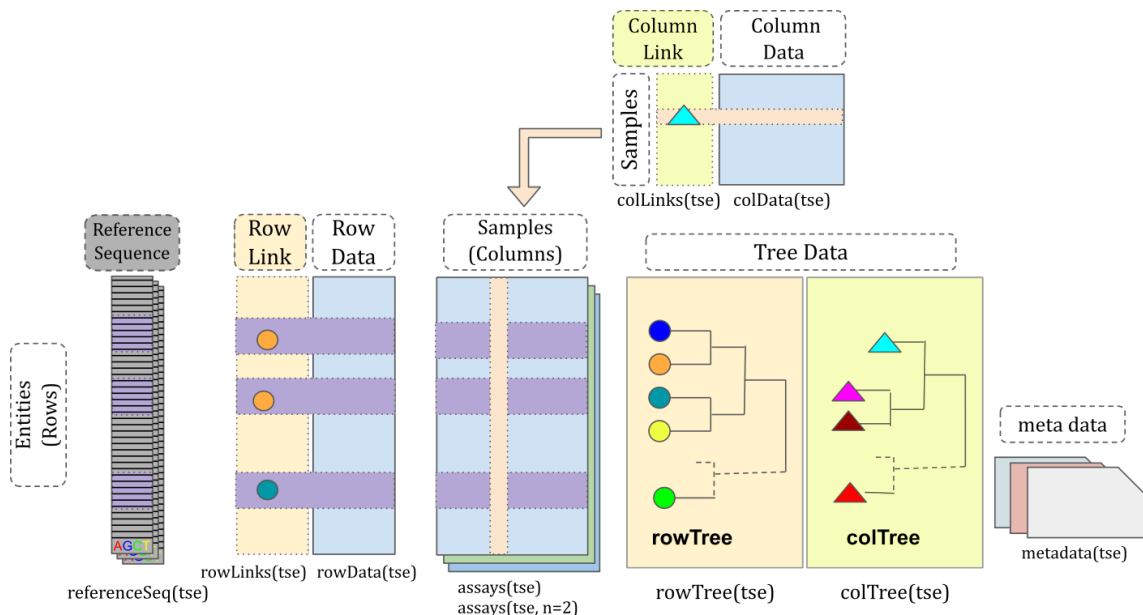


Figure 27.1.: Anatomy of a `TreeSummarizedExperiment` object. Compared to the `SingleCellExperiment` objects, `TreeSummarizedExperiment` has five additional slots. **rowTree**: the hierarchical structure on the rows of the assays. **rowLinks**: the link information between rows of the assays and the `rowTree`. **colTree**: the hierarchical structure on the columns of the assays. **colLinks**: the link information between columns of the assays and the `colTree`. **referenceSeq** (optional): the reference sequence data per feature (row).

27. Analyzing microbiome data

```
altExpNames(0):  
rowLinks: a LinkDataFrame (606 rows)  
rowTree: 1 phylo tree(s) (10430 leaves)  
collinks: NULL  
colTree: NULL
```

Read that printout like the label on a box. The `dim` line tells you how many features (rows) and samples (columns) you have; the `assays`, `colData`, `rowData`, and `rowTree` lines preview the parts described above. Notice that the *numbers* themselves are not printed — the dataset is far too large for that — so R shows you the shape and the index of contents instead.

Next, we'll look at some of the key components of the `TreeSummarizedExperiment` object and how to access and manipulate them.

27.5.1. Experimental measurements: assays

When you work with high-throughput biological data, the result of the workflow usually resembles an Excel spreadsheet: samples down the columns, features (genes, microbes, CpG sites, ...) down the rows. The numbers in that grid are called *assay* data, because they come from running an assay — a measurement of the presence and abundance of something. For a microbiome study, an assay tells you how much of each microbe is in each sample.

The original measurements are count-based, but those counts are often transformed — into logarithms, Centered Log-Ratio (CLR) values, or relative abundances — for different analyses. A single object can hold all of these versions at once, so the `assays` slot is a *list* of matrices rather than a single one.

```
assays(tse)
```

```
List of length 1  
names(1): relative_abundance
```

An individual assay is reached with the `assay()` function (singular), which returns a single matrix. Recall that to subset a matrix you use `[<rows>, <columns>]`, so here we ask for just the first five rows and columns to keep the output readable:

27. Analyzing microbiome data

```
# just the first 5 rows and columns to make the results easier to see
assay(tse, "relative_abundance")[1:5, 1:5]
```

	SID31004	SID31009	SID31021	SID31030
species:Faecalibacterium prausnitzii	13.58862	5.18307	6.09213	2.79834
species:Streptococcus salivarius	11.00755	0.13897	0.17498	0.15692
species:Anaerostipes hadrus	8.26264	7.55176	8.26219	1.07529
species:Bacteroides stercoris	5.21961	0.00000	0.01189	0.59787
species:Collinsella aerofaciens	4.58481	3.68522	2.67519	2.19600
	SID31071			
species:Faecalibacterium prausnitzii	12.96581			
species:Streptococcus salivarius	0.26242			
species:Anaerostipes hadrus	3.07124			
species:Bacteroides stercoris	0.06529			
species:Collinsella aerofaciens	7.94793			

The values are relative abundances — each column sums to 1 across all microbes in that sample. Choose a different number of rows or columns if you like.

27.5.2. Sample information: colData

The `colData` holds everything you know about the *samples*. For a study like this one, that can include the sample ID, the type of sample (fecal, soil, mock, ...), the subject's age and sex, the disease status, and any experimental details you care to track (prep kit lot number, sequencing run, and so on).

```
colData(tse)
```

DataFrame with 154 rows and 28 columns

	study_name	subject_id	body_site	antibiotics_current_use
	<character>	<character>	<character>	<character>
SID31004	FengQ_2015	SID31004	stool	no
SID31009	FengQ_2015	SID31009	stool	no
SID31021	FengQ_2015	SID31021	stool	no
SID31030	FengQ_2015	SID31030	stool	no
SID31071	FengQ_2015	SID31071	stool	no
...	...	...	...	...
SID532796	FengQ_2015	SID532796	stool	no

27. Analyzing microbiome data

SID532802	FengQ_2015	SID532802	stool	no
SID532826	FengQ_2015	SID532826	stool	no
SID532832	FengQ_2015	SID532832	stool	no
SID532915	FengQ_2015	SID532915	stool	no
	study_condition	disease	age	age_category
	<character>	<character>	<integer>	<character>
SID31004	CRC	CRC;fatty_liver;hype..	64	adult
SID31009	control	fatty_liver;hyperten..	68	senior
SID31021	control	healthy	60	adult
SID31030	adenoma	adenoma;fatty_liver;..	70	senior
SID31071	control	fatty_liver	68	senior
...	...	...	...	...
SID532796	control	fatty_liver;hyperten..	73	senior
SID532802	control	healthy	68	senior
SID532826	control	T2D;fatty_liver;hype..	78	senior
SID532832	adenoma	adenoma;fatty_liver;..	68	senior
SID532915	control	healthy	43	adult
	gender	country	non_westernized	sequencing_platform
	<character>	<character>	<character>	<character>
SID31004	male	AUT	no	IlluminaHiSeq
SID31009	male	AUT	no	IlluminaHiSeq
SID31021	female	AUT	no	IlluminaHiSeq
SID31030	male	AUT	no	IlluminaHiSeq
SID31071	male	AUT	no	IlluminaHiSeq
...	...	...	...	...
SID532796	female	AUT	no	IlluminaHiSeq
SID532802	male	AUT	no	IlluminaHiSeq
SID532826	male	AUT	no	IlluminaHiSeq
SID532832	female	AUT	no	IlluminaHiSeq
SID532915	female	AUT	no	IlluminaHiSeq
	DNA_extraction_kit	PMID	number_reads	number_bases
	<character>	<character>	<integer>	<numeric>
SID31004	MoBio	25758642	40898340	3649611221
SID31009	MoBio	25758642	66107961	6196998053
SID31021	MoBio	25758642	60789126	5708593447
SID31030	MoBio	25758642	50300253	4741158330
SID31071	MoBio	25758642	51945426	4913627034
...	...	...	...	...
SID532796	MoBio	25758642	50845712	4754848864
SID532802	MoBio	25758642	41480415	3868696049
SID532826	MoBio	25758642	35346002	3348368467

27. Analyzing microbiome data

SID532832	MoBio	25758642	42184599	3951224696
SID532915	MoBio	25758642	51594677	4886998833
	minimum_read_length	median_read_length	NCBI_accession	
	<integer>	<integer>	<character>	
SID31004	30	93	ERR688505;ERR688358	
SID31009	30	96	ERR688506;ERR688359	
SID31021	30	96	ERR688507;ERR688360	
SID31030	30	96	ERR688508;ERR688361	
SID31071	30	97	ERR688509;ERR688362	
...	...	...	...	
SID532796	30	95	ERR710428;ERR710419	
SID532802	30	96	ERR710429;ERR710420	
SID532826	30	97	ERR710430;ERR710421	
SID532832	30	96	ERR710431;ERR710422	
SID532915	30	96	ERR710432;ERR710423	
	curator	BMI	disease_subtype	diet
	<character>	<numeric>	<character>	<character>
SID31004	Paolo_Manghi;Marisa_..	29.35	carcinoma	vegetarian
SID31009	Paolo_Manghi;Marisa_..	32.00	NA	omnivore
SID31021	Paolo_Manghi;Marisa_..	22.10	NA	omnivore
SID31030	Paolo_Manghi;Marisa_..	34.11	advancedadenoma	omnivore
SID31071	Paolo_Manghi;Marisa_..	23.45	NA	omnivore
...	...	...	...	...
SID532796	Paolo_Manghi;Marisa_..	26.56	NA	omnivore
SID532802	Paolo_Manghi;Marisa_..	23.53	NA	omnivore
SID532826	Paolo_Manghi;Marisa_..	31.22	NA	omnivore
SID532832	Paolo_Manghi;Marisa_..	27.55	advancedadenoma	omnivore
SID532915	Paolo_Manghi;Marisa_..	22.65	NA	omnivore
	hdl	ldl	tnm	triglycerides
	<numeric>	<numeric>	<character>	<numeric>
SID31004	28	92	tlmOm0	172
SID31009	50	157	NA	101
SID31021	60	122	NA	53
SID31030	74	146	NA	89
SID31071	40	231	NA	258
...	...	...	...	...
SID532796	61	114	NA	100
SID532802	67	158	NA	80
SID532826	32	90	NA	212
SID532832	51	184	NA	141
SID532915	80	132	NA	51

27. Analyzing microbiome data

There is a lot here. Scan the column names: alongside identifiers you'll find biologically interesting variables such as `study_condition` (whether the subject had colorectal cancer or was a control), `age`, and `gender`. These are exactly the variables you'll group and color by when you compare diversity and composition later in the chapter.

💡 Treat `colData` as your sample database

By keeping a reasonably complete `colData`, you can answer questions about your findings without flipping back to an external Excel spreadsheet or a lab notebook. The metadata travels with the data.

27.5.3. Measured feature information: `rowData`

The `rowData` describes the *features* — here, the microbes. In microbiome work this is where taxonomic information lives: the kingdom, phylum, class, order, family, genus, and species of each detected organism. That taxonomy is what lets you identify which microbes are present and connect community composition to environmental or health factors.

```
head(rowData(tse))
```

DataFrame with 6 rows and 7 columns

	superkingdom	phylum	class
	<character>	<character>	<character>
species:Faecalibacterium prausnitzii	Bacteria	Firmicutes	Clostridia
species:Streptococcus salivarius	Bacteria	Firmicutes	Bacilli
species:Anaerostipes hadrus	Bacteria	Firmicutes	Clostridia
species:Bacteroides stercoris	Bacteria	Bacteroidota	Bacteroidia
species:Collinsella aerofaciens	Bacteria	Actinobacteria	Coriobacteriia
species:Bifidobacterium longum	Bacteria	Actinobacteria	Actinomycetia
	order	family	
	<character>	<character>	
species:Faecalibacterium prausnitzii	Eubacteriales	Oscillospiraceae	
species:Streptococcus salivarius	Lactobacillales	Streptococcaceae	
species:Anaerostipes hadrus	Eubacteriales	Lachnospiraceae	
species:Bacteroides stercoris	Bacteroidales	Bacteroidaceae	
species:Collinsella aerofaciens	Coriobacteriales	Coriobacteriaceae	
species:Bifidobacterium longum	Bifidobacteriales	Bifidobacteriaceae	
	genus	species	
	<character>	<character>	

27. Analyzing microbiome data

```
species:Faecalibacterium prausnitzii Faecalibacterium Faecalibacterium pra..  
species:Streptococcus salivarius      Streptococcus Streptococcus saliva..  
species:Anaerostipes hadrus           Anaerostipes   Anaerostipes hadrus  
species:Bacteroides stercoris         Bacteroides   Bacteroides stercoris  
species:Collinsella aerofaciens       Collinsella   Collinsella aerofaci..  
species:Bifidobacterium longum        Bifidobacterium Bifidobacterium longum
```

Each row is one microbe, and the columns walk down the taxonomic hierarchy from broad (phylum) to specific (species).

27.5.4. rowTree

A phylogenetic tree records evolutionary relationships, and microbiome analysis leans on it heavily. A `TreeSummarizedExperiment` tracks features and their positions in the tree through two accessors, `rowTree()` and `rowLinks()`.

The tree itself is reached with `rowTree()`, which returns a `phylo` object — R’s standard representation of a tree.

i What is this tree of which we speak?

The “tree” in a `TreeSummarizedExperiment` is a phylogenetic tree: it describes the inferred evolutionary relationships among species (or other entities) based on similarities and differences in their genetic or physical characteristics. The famous “tree of life” is one such tree, showing how all living organisms on Earth are related.

In microbiome analysis, phylogenetic trees represent how the different microbial taxa in a sample are related by descent. This matters because it lets you ask not just *which* microbes are present, but how closely related they are — two samples can share no exact species yet still host very similar branches of the tree.

The `phylo` class is R’s standard structure for holding such trees, and the `ggtree` package is the popular tool for drawing them.

```
rowTree(tse)
```

Phylogenetic tree with 10430 tips and 10429 internal nodes.

Tip labels:

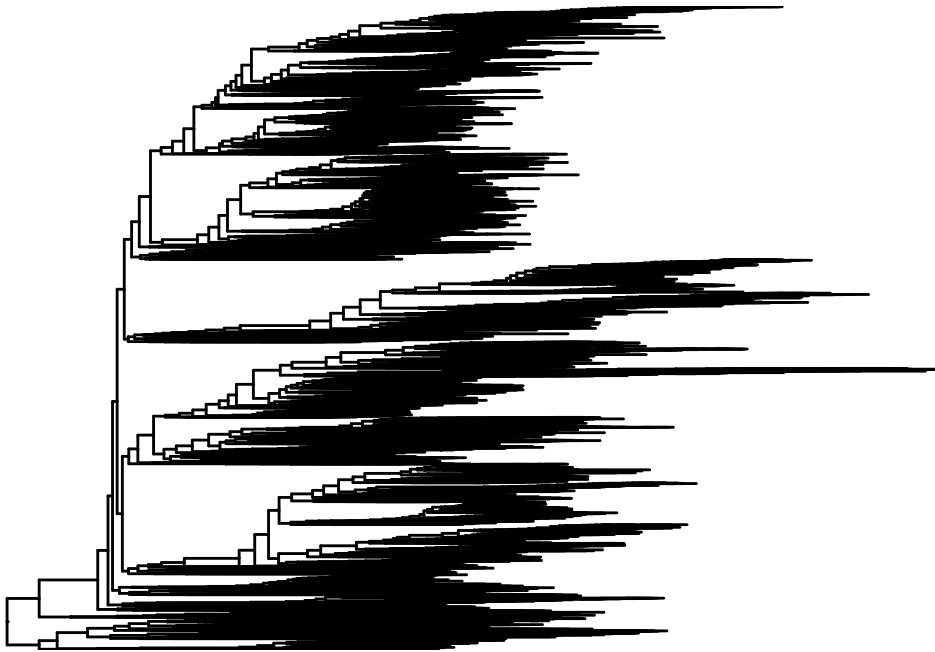
```
k__Archaea|p__Candidatus_Micrarchaeota|c__Candidatus_Micrarchaeota_unclassified|o__Candid
```

27. Analyzing microbiome data

Rooted; includes `branch length(s)`.

The printed `phylo` object summarizes the tree: how many tips (one per microbe) and internal nodes it has. We can draw it with the `ggtree` package, which renders trees in several layouts (rectangular, circular, radial). Here's the default rectangular view of this study's tree:

```
library(ggtree)
ggtree(rowTree(tse))
```



Don't try to read individual tips — at this scale the value is the overall shape, which shows how the detected microbes fan out across the branches of life. See the [ggtree](#) package and the [ggtree book](#) for much more on tree visualization.

The companion accessor `rowLinks()` records *which* tip of the tree each feature corresponds to — the bookkeeping that keeps the assay rows and the tree in step:

```
rowLinks(tse)
```

27. Analyzing microbiome data

LinkDataFrame with 606 rows and 5 columns

	nodeLab	nodeLab_alias
	<character>	<character>
species:Faecalibacterium prausnitzii	k__Bacteria p__Firmi..	alias_3547
species:Streptococcus salivarius	k__Bacteria p__Firmi..	alias_4750
species:Anaerostipes hadrus	k__Bacteria p__Firmi..	alias_3579
species:Bacteroides stercoris	k__Bacteria p__Bacte..	alias_5690
species:Collinsella aerofaciens	k__Bacteria p__Actin..	alias_3019
...	...	...
species:Streptococcus gallolyticus	k__Bacteria p__Firmi..	alias_4745
species:Lactobacillus apis	k__Bacteria p__Firmi..	alias_4851
species:Anaerostipes sp. 494a	k__Bacteria p__Firmi..	alias_3580
species:Lactobacillus kullabergensis	k__Bacteria p__Firmi..	alias_4855
species:Lactobacillus kimbladii	k__Bacteria p__Firmi..	alias_4854

	nodeNum	isLeaf	whichTree
	<integer>	<logical>	<character>
species:Faecalibacterium prausnitzii	3547	TRUE	phylo
species:Streptococcus salivarius	4750	TRUE	phylo
species:Anaerostipes hadrus	3579	TRUE	phylo
species:Bacteroides stercoris	5690	TRUE	phylo
species:Collinsella aerofaciens	3019	TRUE	phylo
...	...	...	...
species:Streptococcus gallolyticus	4745	TRUE	phylo
species:Lactobacillus apis	4851	TRUE	phylo
species:Anaerostipes sp. 494a	3580	TRUE	phylo
species:Lactobacillus kullabergensis	4855	TRUE	phylo
species:Lactobacillus kimbladii	4854	TRUE	phylo

Both `rowTree` and `rowLinks` are optional parts of a `TreeSummarizedExperiment`, but when present they carry the evolutionary structure that distinguishes this class from a plain `SummarizedExperiment`.

27.6. Wrangling and subsetting

A major advantage of the `SummarizedExperiment` family is that a single object encapsulates everything needed to describe the samples (`colData`), the measured features (`rowData`, `rowTree`), and the measurements themselves (`assays`). When you subset or manipulate the object, every one of those parts is subset to match — automatically, in lockstep.

27. Analyzing microbiome data

The `mia` package adds functions for filtering, transforming, and reshaping microbiome data on top of that foundation.

27.6.1. Subsetting samples

You subset the samples in a `TreeSummarizedExperiment` much as you would the columns of a `data.frame`. Say you want to focus on older subjects. First, look at the `age` column in `colData`:

```
head(colData(tse)$age)
```

```
[1] 64 68 60 70 68 66
```

Because reaching into `colData` is such a common operation, you can also grab a column directly with `$`, as if the object were a data frame:

```
head(tse$age)
```

```
[1] 64 68 60 70 68 66
```

```
# and get a summary of all patient ages
summary(tse$age)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
43.00	63.00	68.00	66.85	72.00	86.00

Age is a numeric variable. Let's focus on older patients.

```
# Include all rows, but only columns (samples) who are older than 50
tse_subset_by_age <- tse[, tse$age > 50]
tse_subset_by_age
```

```
class: TreeSummarizedExperiment
dim: 606 146
metadata(0):
assays(1): relative_abundance
rownames(606): species:Faecalibacterium prausnitzii
```

27. Analyzing microbiome data

```
species:Streptococcus salivarius ... species:Lactobacillus
kullabergensis species:Lactobacillus kimbladii
rowData names(7): superkingdom phylum ... genus species
colnames(146): SID31004 SID31009 ... SID532826 SID532832
colData names(28): study_name subject_id ... triglycerides hba1c
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
rowLinks: a LinkDataFrame (606 rows)
rowTree: 1 phylo tree(s) (10430 leaves)
colLinks: NULL
colTree: NULL
```

`tse_subset_by_age` is a copy of the original `tse` holding only the 50-and-older subjects. To double-check, look at the ages in the new object — the minimum should now be above 50:

```
summary(tse_subset_by_age$age)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
52.00	63.25	68.00	68.02	72.00	86.00

27.6.2. Agglomerating data

The microbiome features (organisms) that are measured may be measured at different taxonomic levels. For example, the data may be available at the species, genus, family, or phylum level. Agglomerating data is the process of summarizing the data at a higher taxonomic level by combining the abundances of the lower-level taxa. This can be useful for simplifying the data and reducing the dimensionality of the data.

The `agglomerateByRank` function in the `mia` package can be used to agglomerate the data at a specified taxonomic rank. For example, to agglomerate the data at the phylum level, you can use the following code:

```
new_tse <- agglomerateByRank(tse, rank = "phylum")
# observe the new object, particularly the number of features (rows)
new_tse
```

27. Analyzing microbiome data

```
class: TreeSummarizedExperiment
dim: 13 154
metadata(1): agglomerated_by_rank
assays(1): relative_abundance
rownames(13): Actinobacteria Ascomycota ... Synergistetes
               Verrucomicrobia
rowData names(7): superkingdom phylum ... genus species
colnames(154): SID31004 SID31009 ... SID532832 SID532915
colData names(28): study_name subject_id ... triglycerides hba1c
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
rowLinks: a LinkDataFrame (13 rows)
rowTree: 1 phylo tree(s) (13 leaves)
colLinks: NULL
colTree: NULL
```

Compare the number of features (rows) in `new_tse` to the original `tse`: collapsing hundreds of species into a few dozen phyla shrinks the data dramatically. That makes the data easier to plot and interpret, at the cost of fine-grained detail — a trade-off you’ll make often.

27.7. Community indices

Microbiome ecology offers many indices that summarize a sample, or compare a community of samples. Here we give just a brief introduction to two broad flavors: *alpha* diversity (diversity within a single sample) and *beta* diversity (how different two samples are from each other).

27.7.1. Alpha diversity

Diversity within a sample is captured by several related indices:

- **species richness** — the total number of species present;
- **equitability** — how evenly the species are distributed;
- **diversity** — a combination of the two.

`mia` provides functions for all of these. The `addAlpha()` function calculates one or more indices and stores the results automatically in `colData`, right alongside your sample meta-data.

i Why do we care about alpha diversity?

Alpha diversity is a single number that captures how complex or “diverse” one microbiome sample is. Plenty of things can shift it — antibiotic use, chemotherapy, autoimmune conditions, dietary changes, and more.

When you compare two groups (say, antibiotic users versus non-users), a significant difference in alpha diversity between them can point you toward real biology, or even toward a useful biomarker.

In the code below we calculate the Shannon and observed diversity indices.

```
tse <- addAlpha(tse, index = "shannon", assay.type = "relative_abundance")
tse <- addAlpha(tse, index = "observed", assay.type = "relative_abundance")
colnames(colData(tse))
```

```
[1] "study_name"           "subject_id"
[3] "body_site"            "antibiotics_current_use"
[5] "study_condition"      "disease"
[7] "age"                  "age_category"
[9] "gender"               "country"
[11] "non_westernized"      "sequencing_platform"
[13] "DNA_extraction_kit"   "PMID"
[15] "number_reads"         "number_bases"
[17] "minimum_read_length"  "median_read_length"
[19] "NCBI_accession"       "curator"
[21] "BMI"                  "disease_subtype"
[23] "diet"                 "hdl"
[25] "ldl"                  "tnm"
[27] "triglycerides"        "hba1c"
[29] "shannon"              "observed"
```

The `observed` index simply counts the species present in a sample, while the `shannon` index also accounts for how evenly abundant those species are. Notice from the printed column names that both new indices now live in `colData`, ready to plot or model.

Look at the full `colData()` and you’ll see the new diversity columns sitting beside the original sample metadata:

```
colData(tse)
```

27. Analyzing microbiome data

DataFrame with 154 rows and 30 columns

	study_name	subject_id	body_site	antibiotics_current_use	
	<character>	<character>	<character>	<character>	
SID31004	FengQ_2015	SID31004	stool	no	
SID31009	FengQ_2015	SID31009	stool	no	
SID31021	FengQ_2015	SID31021	stool	no	
SID31030	FengQ_2015	SID31030	stool	no	
SID31071	FengQ_2015	SID31071	stool	no	
...	...	...	...	...	
SID532796	FengQ_2015	SID532796	stool	no	
SID532802	FengQ_2015	SID532802	stool	no	
SID532826	FengQ_2015	SID532826	stool	no	
SID532832	FengQ_2015	SID532832	stool	no	
SID532915	FengQ_2015	SID532915	stool	no	
	study_condition		disease	age	age_category
	<character>		<character>	<integer>	<character>
SID31004	CRC	CRC	CRC;fatty_liver;hype..	64	adult
SID31009	control	fatty_liver	hyperten..	68	senior
SID31021	control		healthy	60	adult
SID31030	adenoma	adenoma	adenoma;fatty_liver;..	70	senior
SID31071	control		fatty_liver	68	senior
...	...	...	...	...	...
SID532796	control	fatty_liver	hyperten..	73	senior
SID532802	control		healthy	68	senior
SID532826	control	T2D	fatty_liver;hype..	78	senior
SID532832	adenoma	adenoma	adenoma;fatty_liver;..	68	senior
SID532915	control		healthy	43	adult
	gender	country	non_westernized	sequencing_platform	
	<character>	<character>	<character>	<character>	
SID31004	male	AUT	no	IlluminaHiSeq	
SID31009	male	AUT	no	IlluminaHiSeq	
SID31021	female	AUT	no	IlluminaHiSeq	
SID31030	male	AUT	no	IlluminaHiSeq	
SID31071	male	AUT	no	IlluminaHiSeq	
...	...	...	...	...	
SID532796	female	AUT	no	IlluminaHiSeq	
SID532802	male	AUT	no	IlluminaHiSeq	
SID532826	male	AUT	no	IlluminaHiSeq	
SID532832	female	AUT	no	IlluminaHiSeq	
SID532915	female	AUT	no	IlluminaHiSeq	
	DNA_extraction_kit		PMID	number_reads	number_bases

27. Analyzing microbiome data

	<character>	<character>	<integer>	<numeric>		
SID31004	MoBio	25758642	40898340	3649611221		
SID31009	MoBio	25758642	66107961	6196998053		
SID31021	MoBio	25758642	60789126	5708593447		
SID31030	MoBio	25758642	50300253	4741158330		
SID31071	MoBio	25758642	51945426	4913627034		
...	...	...	...	...		
SID532796	MoBio	25758642	50845712	4754848864		
SID532802	MoBio	25758642	41480415	3868696049		
SID532826	MoBio	25758642	35346002	3348368467		
SID532832	MoBio	25758642	42184599	3951224696		
SID532915	MoBio	25758642	51594677	4886998833		
	minimum_read_length	median_read_length	NCBI_accession			
	<integer>	<integer>	<character>			
SID31004	30	93	ERR688505;ERR688358			
SID31009	30	96	ERR688506;ERR688359			
SID31021	30	96	ERR688507;ERR688360			
SID31030	30	96	ERR688508;ERR688361			
SID31071	30	97	ERR688509;ERR688362			
...	...	...	...			
SID532796	30	95	ERR710428;ERR710419			
SID532802	30	96	ERR710429;ERR710420			
SID532826	30	97	ERR710430;ERR710421			
SID532832	30	96	ERR710431;ERR710422			
SID532915	30	96	ERR710432;ERR710423			
	curator	BMI	disease_subtype	diet		
	<character>	<numeric>	<character>	<character>		
SID31004	Paolo_Manghi;Marisa_..	29.35	carcinoma	vegetarian		
SID31009	Paolo_Manghi;Marisa_..	32.00	NA	omnivore		
SID31021	Paolo_Manghi;Marisa_..	22.10	NA	omnivore		
SID31030	Paolo_Manghi;Marisa_..	34.11	advancedadenoma	omnivore		
SID31071	Paolo_Manghi;Marisa_..	23.45	NA	omnivore		
...	...	...	...	...		
SID532796	Paolo_Manghi;Marisa_..	26.56	NA	omnivore		
SID532802	Paolo_Manghi;Marisa_..	23.53	NA	omnivore		
SID532826	Paolo_Manghi;Marisa_..	31.22	NA	omnivore		
SID532832	Paolo_Manghi;Marisa_..	27.55	advancedadenoma	omnivore		
SID532915	Paolo_Manghi;Marisa_..	22.65	NA	omnivore		
	hdl	ldl	tnm	triglycerides	hba1c	shannon
	<numeric>	<numeric>	<character>	<numeric>	<numeric>	<numeric>
SID31004	28	92	t1n0m0	172	5.2	3.28935

27. Analyzing microbiome data

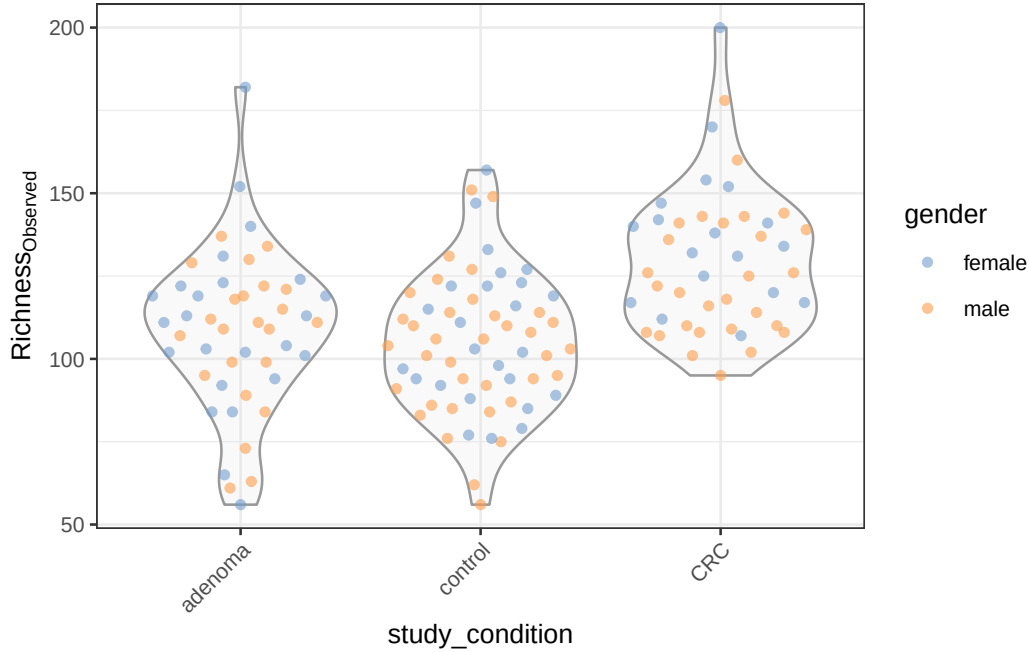
SID31009	50	157	NA	101	NA	3.11014
SID31021	60	122	NA	53	NA	3.38207
SID31030	74	146	NA	89	NA	2.60265
SID31071	40	231	NA	258	NA	3.14500
...	...	...	...	...	...	...
SID532796	61	114	NA	100	5.4	3.66184
SID532802	67	158	NA	80	5.8	3.19522
SID532826	32	90	NA	212	6.9	3.16254
SID532832	51	184	NA	141	5.6	3.21467
SID532915	80	132	NA	51	5.1	3.48822

	observed
	<numeric>
SID31004	101
SID31009	106
SID31021	116
SID31030	107
SID31071	113
...	...
SID532796	133
SID532802	111
SID532826	103
SID532832	94
SID532915	127

The `scater` package provides convenient plotting functions designed to work directly with the `SummarizedExperiment` family, so they can read straight from `colData`.

```
library(scater)
plotColData(tse,
  "observed",          # the y-axis value that we just calculated
  "study_condition",   # The x-axis
  colour_by = "gender" # Color the points by gender
) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(y = expression(Richness[Observed])) # and give a meaningful label
```

27. Analyzing microbiome data



Each point is one sample, grouped along the x-axis by study condition. Look at where the clouds of points sit: a noticeable shift in observed richness between the cancer and control groups would be the first hint that disease status is associated with how many species live in the gut.

27.7.2. Beta diversity

Where alpha diversity looks *within* a single sample, beta diversity compares two samples and asks how *different* their communities are. In microbiome analysis the most common measure is the Bray-Curtis dissimilarity between two samples A and B, defined as:

$$BC_{ij} = \frac{\sum_k |A_k - B_k|}{\sum_k (A_k + B_k)}$$

where A_k and B_k are the abundances of taxon k in samples A and B, respectively. The Bray-Curtis dissimilarity ranges from 0 (identical communities) to 1 (completely different communities).

The `mia` package provides functions for calculating beta diversity indices, such as the Bray-Curtis dissimilarity, Jaccard similarity, and UniFrac distance. These indices can be used

27. Analyzing microbiome data

to compare the microbial communities between different samples and to identify patterns and relationships between samples based on their microbial composition.

```
# Run PCoA on relabundance assay with Bray-Curtis distances
library(vegan)

tse <- runMDS(tse,
  FUN = vegdist,
  method = "bray",
  assay.type = "relative_abundance",
  name = "MDS_bray"
)
```

This code is a bit to unpack. It calculates the Bray-Curtis dissimilarity between every pair of samples in the `tse` object using the `vegdist` function from the `vegan` package. The resulting dissimilarity matrix is then used to perform a Principal Coordinate Analysis (PCoA) using the `runMDS` function from the `scater` package. The PCoA is a dimensionality reduction technique that projects the high-dimensional Bray-Curtis dissimilarity matrix onto a lower-dimensional space while preserving the pairwise distances between samples.

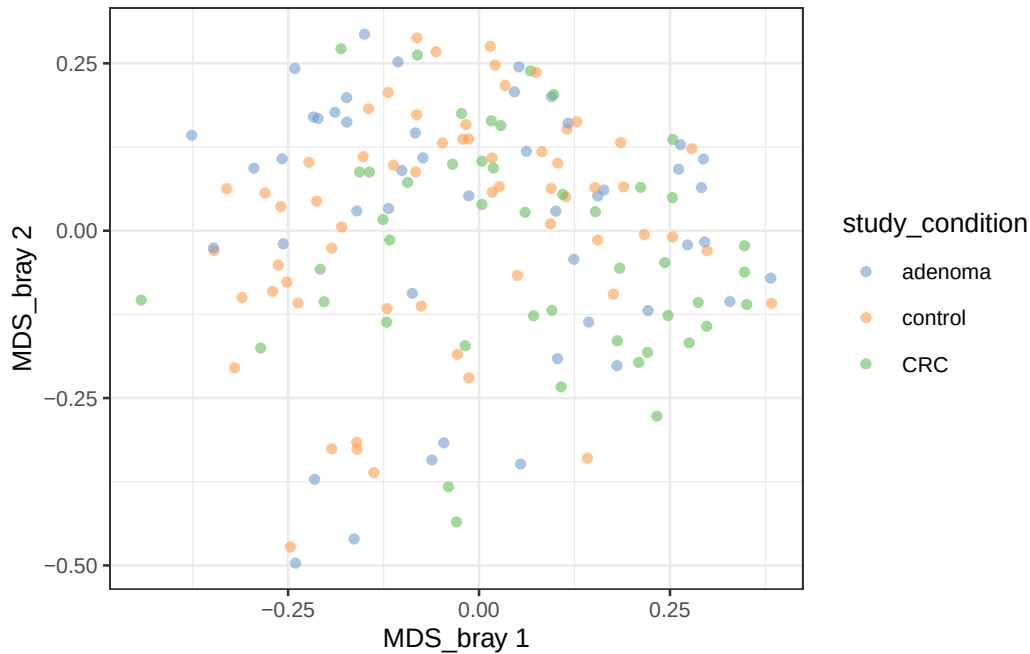
The resulting `TreeSummarizedExperiment` object now contains the Bray-Curtis dissimilarity matrix in the `MDS_bray` slot, which can be used to visualize the relationships between samples based on their microbial composition.

We can visualize the result with `plotReducedDim()` from the `scater` package, coloring each sample by its study condition:

```
# Create ggplot object
p <- plotReducedDim(tse, "MDS_bray", colour_by = "study_condition")

print(p)
```

27. Analyzing microbiome data



Each point is now a whole microbial community, and points that sit close together have similar compositions. The question to ask of this plot is whether the colors *separate*: if cancer and control samples fell into distinct regions, that would suggest disease status leaves a fingerprint on the community as a whole, not just on a handful of species. Try coloring by other `colData` variables to explore what else structures the data.

27.8. Microbial composition

Let's now look at the microbial composition of the samples in the `tse` object. The microbial composition refers to the relative abundances of different microbial taxa in each sample. This information can provide insights into the diversity and structure of the microbial community in each sample and help identify patterns and relationships between samples based on their microbial composition.

The `mia` package provides functions for visualizing the microbial composition of samples, such as bar plots, heatmaps, and stacked bar plots. These plots can help you explore the relative abundances of different microbial taxa in each sample and identify patterns and relationships between samples based on their microbial composition.

27.8.1. Abundance

We can start with the 40 most abundant taxa. This plot shows, for each of those taxa, the spread of its relative abundance across all samples (one jittered point per sample), sorted by abundance.

```
library(miaViz)
```

```
Loading required package: ggraph
```

```
Attaching package: 'miaViz'
```

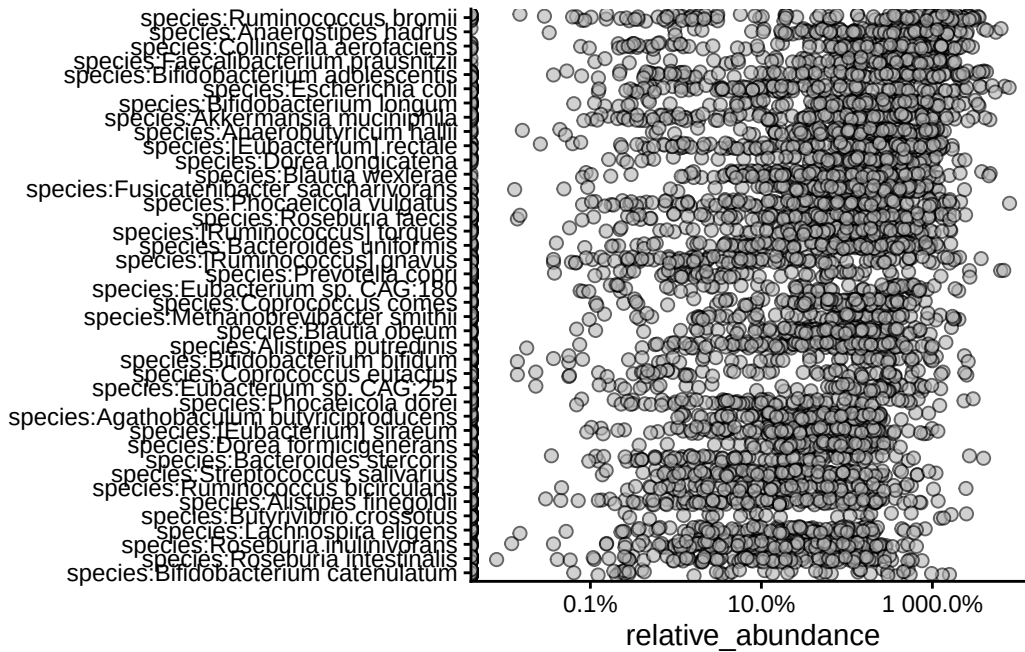
```
The following object is masked from 'package:mia':
```

```
plotNMDS
```

```
plotAbundanceDensity(tse,  
  layout = "jitter",  
  assay.type = "relative_abundance",  
  n = 40  
) +  
  scale_x_log10(label = scales::percent)
```

```
Warning in scale_x_log10(label = scales::percent): log-10 transformation  
introduced infinite values.
```


27. Analyzing microbiome data



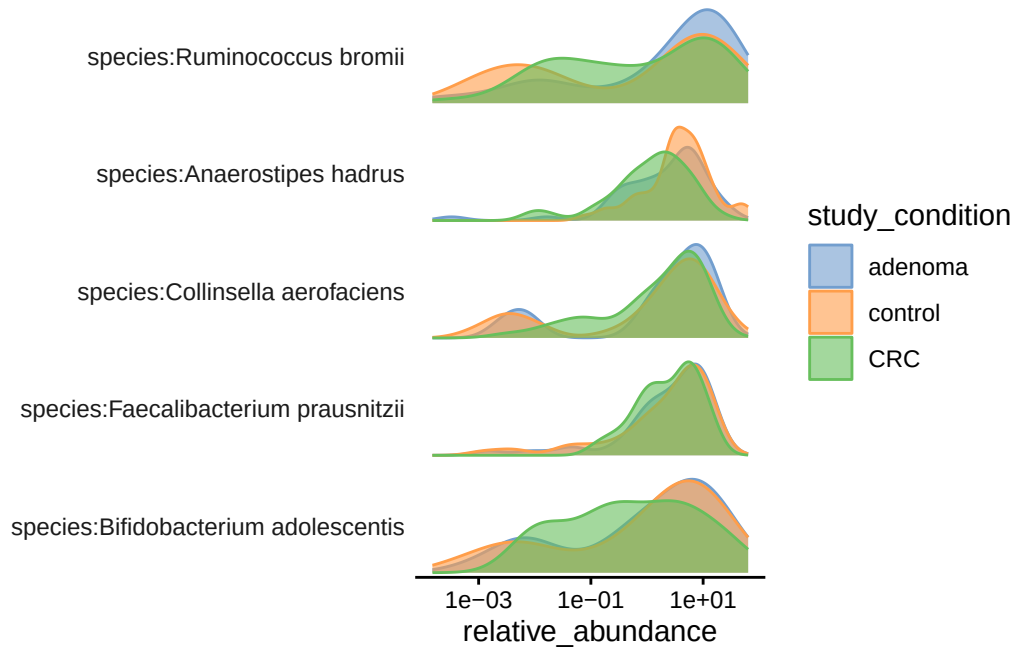
A density layout is handy for zooming in on a few taxa and splitting them by group. Here we take the top five and color by study condition — separated curves would suggest that taxon's abundance differs between cancer and control samples.

```
plotAbundanceDensity(tse,
  layout = "density",
  assay.type = "relative_abundance",
  n = 5, colour_by = "study_condition"
) +
  scale_x_log10()
```

Warning in scale_x_log10(): log-10 transformation introduced infinite values.

Warning: Removed 53 rows containing non-finite outside the scale range (`stat\_density()`).

27. Analyzing microbiome data



27.8.2. Prevalence

Abundance asks how *much* of a taxon is present; *prevalence* asks in how *many* samples it appears at all. A microbe can be abundant in a few people yet rare across the cohort, or modestly abundant but present in nearly everyone — prevalence tells these apart.

Let's agglomerate to the genus level and list the most prevalent genera, counting a genus as "present" wherever its relative abundance exceeds 1%:

```
tse_genus <- agglomerateByRank(tse, rank = "genus")
```

```
head(getPrevalence(tse_genus,
  detection = 1 / 100,
  sort = TRUE, assay.type = "relative_abundance",
  as.relative = TRUE
))
```

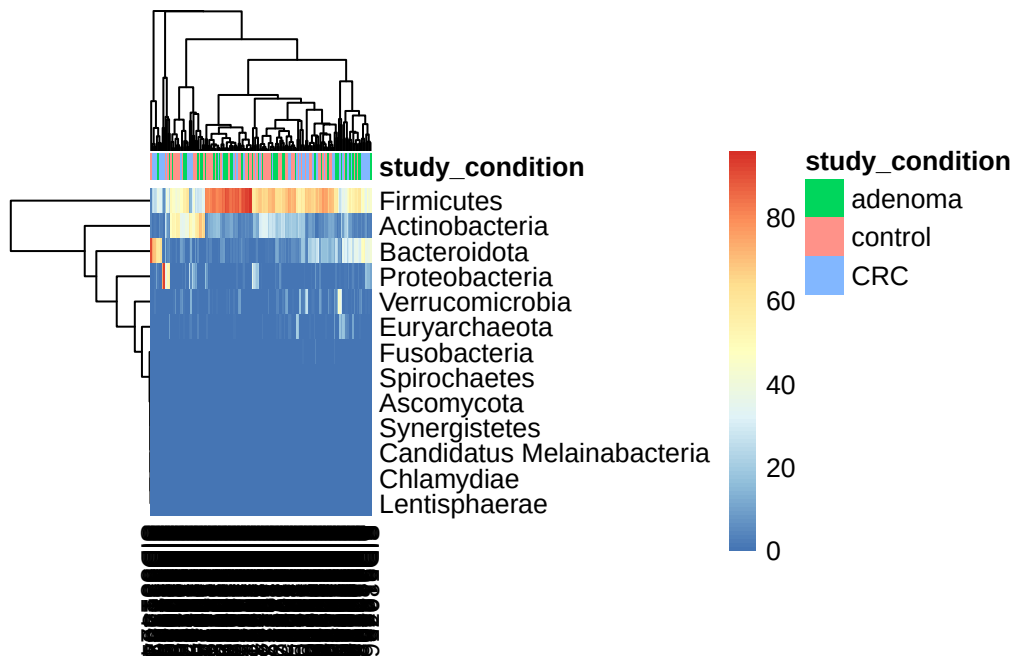
Bifidobacterium	Dorea	Faecalibacterium	Mediterraneibacter
0.7857143	0.7662338	0.7337662	0.7272727
Anaerostipes	Blautia		
0.7272727	0.7077922		

Each value is the fraction of samples in which that genus clears the 1% threshold. A value near 1 marks a near-universal member of the gut community; a small value marks a genus that is only occasionally abundant.

27.9. Heatmaps

A heatmap lets you see abundance across every taxon and every sample at once. To keep it readable, we'll agglomerate to the phylum level first — a handful of phyla, rather than hundreds of species — and then draw the grid, annotating each sample column with its study condition.

```
library(pheatmap)
tse_phylum <- agglomerateByRank(tse, rank = "phylum")
pheatmap(
  assay(tse_phylum, "relative_abundance"),
  annotation_col = as.data.frame(colData(tse_phylum)[, "study_condition", drop = FALSE])
)
```



Read the heatmap as a grid: each row is a phylum, each column a sample, and the color is that phylum's relative abundance. The colored bar across the top marks each sam-

ple’s study condition. The clustering reorders rows and columns so that similar profiles sit together — if columns of the same condition tend to cluster, you’re seeing the same community-level signal the PCoA hinted at, now broken down phylum by phylum.

27.10. Summary

In this chapter you took a real curated dataset and ran a complete first-pass microbiome analysis without ever splitting your data across loose variables. You loaded the `FengQ_2015` study as a `TreeSummarizedExperiment` — a `SummarizedExperiment` with a phylogenetic tree attached — and reached into its parts with `assays()`, `colData()`, `rowData()`, and `rowTree()`. You subset samples by age and agglomerated features up to coarser taxonomic ranks, trusting the object to keep every part aligned. Then you summarized the communities: alpha diversity within each sample, beta diversity (Bray-Curtis) between samples visualized as a PCoA, and composition shown as abundance plots and a heatmap. Each of these is a doorway to a real research question about how the gut community differs between health and disease.

27.11. Exercises

These exercises reuse the `tse` object built earlier in the chapter. If you’ve started a fresh session, re-run the loading chunk first.

1. How many samples in the dataset come from subjects older than 60? (Hint: subset on `tse$age` as you did for the 50+ subjects, then check the dimensions.)
2. Agglomerate `tse` to the `family` rank and report how many features remain. How does that compare to the phylum-level count?
3. Add the `shannon` alpha-diversity index to `tse` and make a `plotColData()` plot of it against `study_condition`. Does the picture match what you saw for observed richness?

i Solutions

```
# 1. subset the samples (columns) and read off the new dimensions
tse_over60 <- tse[, tse$age > 60]
ncol(tse_over60)
```

```
[1] 128
```

27. Analyzing microbiome data

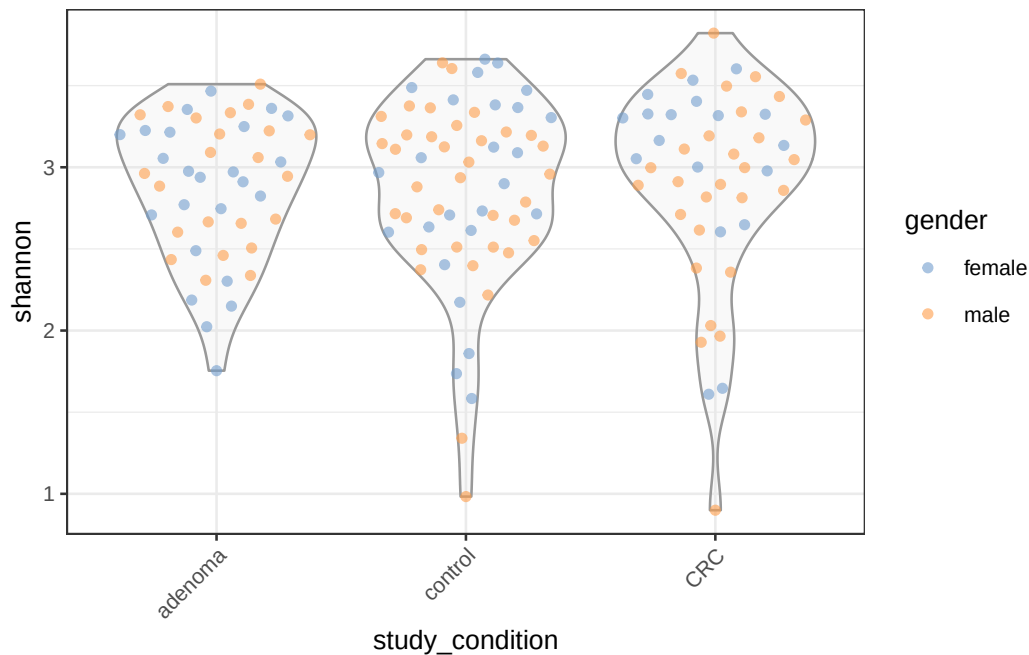
```
# 2. agglomerate to family level and count features (rows)
tse_family <- agglomerateByRank(tse, rank = "family")
nrow(tse_family) # fewer than at species level, more than at phylum level
```

```
[1] 66
```

```
# 3. add Shannon diversity, then plot it
tse <- addAlpha(tse, index = "shannon", assay.type = "relative_abundance")
```

Warning: The following values are already present in `colData` and will be overwritten: 'shannon'. Consider using the 'name' argument to specify alternative names.

```
plotColData(tse, "shannon", "study_condition", colour_by = "gender") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The key idea in exercises 1 and 2: you subset or agglomerate the *whole object*, and every linked part — assays, `colData`, `rowData`, the tree — follows along automatically. You never have to remember to update them separately.

27.12. Resources

The `mia` developers maintain [Orchestrating Microbiome Analysis with Bioconductor \(OMA\)](#), an online book that goes well beyond this introduction — data loading, preprocessing, differential abundance, and a wide range of visualizations, all worked through with detailed examples. It's the natural next step once you're comfortable with the basics here.

For the phylogenetic-tree side of things, the [ggtree](#) package and its companion [ggtree book](#) cover tree visualization in depth.

28. Genomic ranges with real ChIP-seq peaks

CTCF is one of the most important architectural proteins in the human genome. It binds thousands of sites across the chromosomes, where it helps fold DNA into loops and marks the boundaries between active and silent regions. So a natural question for a biologist is: **where does CTCF actually bind?**

A ChIP-seq experiment answers that question by handing you a list of genomic intervals — the *peaks* where the protein was detected. In this chapter you’ll take a real CTCF peak set from the ENCODE project, load it into R, and learn the tools you need to explore it: how genomic coordinates are stored on disk, how to import them, and how to ask questions like “how wide are these peaks?” and “where exactly is the center of each one?”

This chapter focuses on the **real-data workflow** — file formats, importing, and exploring. For the full catalog of range operations (set operations, overlaps, nearest-feature searches, gene models), see the *Genomic ranges and features* chapter, which builds **GRanges** objects by hand and works through those methods systematically. Here we keep our hands on a real dataset.

28.1. What you’ll learn

- Read the BED / narrowPeak file format and explain its coordinate system.
- Import a peak file from a URL into a **GRanges** object with **rtracklayer**.
- Use the accessor functions (**seqnames()**, **start()**, **width()**, **mcols()**) to pull information out of a **GRanges**.
- Interpret the distribution of peak widths and what it tells you about the peak caller.
- Manipulate ranges with **shift()**, **resize()**, and **flank()** to build the windows downstream analyses need.
- Compute and visualize read coverage along a chromosome.

28.2. The BED file format

Genomic intervals are most often stored in **BED** (Browser Extensible Data) format: a tab-delimited text file where each line is one feature. The first three columns are required and define the interval itself:

- **chrom** — chromosome name (e.g. `chr1`, `chrX`).
- **chromStart** — start position.
- **chromEnd** — end position.

Optional columns can follow: a feature **name**, a **score**, the **strand**, and several more. Peak callers extend BED with extra columns (signal values, p-values); that flavor is called **narrowPeak**, and it's what our ENCODE file uses.

 BED is 0-based; R is 1-based

This is the single most common source of off-by-one bugs in genomics, so it is worth burning into memory. BED files use a **0-based, half-open** coordinate system: counting starts at 0, the start is *included*, and the end is *excluded*. R — and the **GRanges** objects you'll build from these files — use a **1-based, fully-closed** system: counting starts at 1 and both ends are included.

Take the BED line `chr1 1000 2000`. In BED that means the 1000 bases spanning the interval `[1000, 2000)`. When `rtracklayer` imports it, the start becomes **1001** (shifted up by one) and the end stays **2000**, giving the same 1000 bases but now written as `1001-2000` in 1-based coordinates. The width is identical ($2000 - 1000 = 1000$); only the *numbering* differs. The good news: `rtracklayer` does this conversion for you. The trap is doing the arithmetic yourself on the raw file and being off by one.

28.3. Loading the libraries

```
library(rtracklayer)
library(GenomicRanges)
library(ggplot2)
library(dplyr)

theme_set(theme_minimal())
```


28.4. Importing the CTCF peaks

We'll use a CTCF ChIP-seq peak file from ENCODE. It lives at a URL, and `rtracklayer::import()` can read straight from that URL — no need to download it by hand first.

Before we use the convenient importer, let's peek at the raw file so you can see that it really is just a table of numbers and text:

```
bed_url <- "https://www.encodeproject.org/files/ENCFF960ZGP/@@download/ENCFF960ZGP.bed.gz"

# Read the raw file as a plain table just to see its shape
ctcf_peaks_raw <- readr::read_table(bed_url, col_names = FALSE)
```

```
-- Column specification -----
cols(
  X1 = col_character(),
  X2 = col_double(),
  X3 = col_double(),
  X4 = col_character(),
  X5 = col_double(),
  X6 = col_character(),
  X7 = col_double(),
  X8 = col_double(),
  X9 = col_double(),
  X10 = col_double()
)
```

```
head(ctcf_peaks_raw)
```

```
# A tibble: 6 x 10
  X1      X2      X3 X4      X5 X6      X7      X8      X9      X10
  <chr>   <dbl>   <dbl> <chr> <dbl> <chr> <dbl> <dbl> <dbl> <dbl>
1 chr2  238305323 238305539 .    1000 .    7.70   -1 0.274   108
2 chr20 49982706 49982922 .    541 .    8.72   -1 0.466   108
3 chr15 39921015 39921231 .    672 .    9.08   -1 0.547   108
4 chr8   6708273 6708489 .    560 .    9.86   -1 0.662   108
5 chr9  136645956 136646172 .    584 .   10.2   -1 0.751   108
6 chr7   47669294 47669510 .    614 .   10.5   -1 0.807   108
```

28. Genomic ranges with real ChIP-seq peaks

Each row is one peak, and you can see the chromosome, start, and end in the first three columns followed by the narrowPeak extras. Now let's import it properly into a `GRanges` object, which is the data structure built for genomic intervals:

```
ctcf_peaks <- rtracklayer::import(bed_url, format = "narrowPeak")
ctcf_peaks
```

`GRanges` object with 43865 ranges and 6 metadata columns:

	seqnames	ranges	strand	name	score
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>
[1]	chr2	238305324-238305539	*	<NA>	1000
[2]	chr20	49982707-49982922	*	<NA>	541
[3]	chr15	39921016-39921231	*	<NA>	672
[4]	chr8	6708274-6708489	*	<NA>	560
[5]	chr9	136645957-136646172	*	<NA>	584
...	...	...	...	...	...
[43861]	chr11	66222736-66222972	*	<NA>	1000
[43862]	chr10	75235956-75236225	*	<NA>	1000
[43863]	chr16	57649100-57649347	*	<NA>	1000
[43864]	chr17	37373596-37373838	*	<NA>	1000
[43865]	chr12	53676108-53676355	*	<NA>	1000

	signalValue	pValue	qValue	peak
	<numeric>	<numeric>	<numeric>	<integer>
[1]	7.70385	-1	0.27378	108
[2]	8.71626	-1	0.46571	108
[3]	9.07638	-1	0.54697	108
[4]	9.86234	-1	0.66189	108
[5]	10.15488	-1	0.75082	108
...	...	...	...	...
[43861]	478.640	-1	4.87574	124
[43862]	480.183	-1	4.87574	141
[43863]	491.081	-1	4.87574	116
[43864]	491.991	-1	4.87574	127
[43865]	494.303	-1	4.87574	126

seqinfo: 26 sequences from an unspecified genome; no seqlengths

i What `format = "narrowPeak"` gives you

Telling `import()` that the file is `narrowPeak` does two helpful things. First, it applies the 0-based-to-1-based conversion described above, so the `start`, `end`, and `width` you see are already in R's coordinate system. Second, it parses the peak-caller columns into named metadata: `signalValue`, `pValue`, `qValue`, and `peak` (the offset of the summit within the peak). Those land in the metadata columns of the `GRanges`, where you can reach them with `$`.

28.5. Looking inside a `GRanges`

A `GRanges` keeps the *ranges* (chromosome, start, end, strand) separate from the *metadata* (everything else). You pull each part out with an accessor function.

```
length(ctcf_peaks)      # how many peaks?
```

```
[1] 43865
```

```
head(seqnames(ctcf_peaks)) # which chromosomes
```

```
factor-Rle of length 6 with 6 runs
  Lengths:      1      1      1      1      1      1
  Values  : chr2  chr20 chr15 chr8  chr9  chr7
Levels(26): chr2 chr20 chr15 ... chr1_KI270714v1_random chrUn_GL000219v1
```

```
head(start(ctcf_peaks))  # start positions (1-based, after import)
```

```
[1] 238305324 49982707 39921016 6708274 136645957 47669295
```

```
head(end(ctcf_peaks))    # end positions
```

```
[1] 238305539 49982922 39921231 6708489 136646172 47669510
```

28. Genomic ranges with real ChIP-seq peaks

```
head(width(ctcf_peaks))    # width = end - start + 1
```

```
[1] 216 216 216 216 216 216
```

So this file contains 43865 CTCF peaks. The widths vary, but let's look at the whole distribution rather than the first six values:

```
hist(width(ctcf_peaks), breaks = 50,  
     main = "CTCF peak widths", xlab = "Width (bp)", col = "lightblue")
```

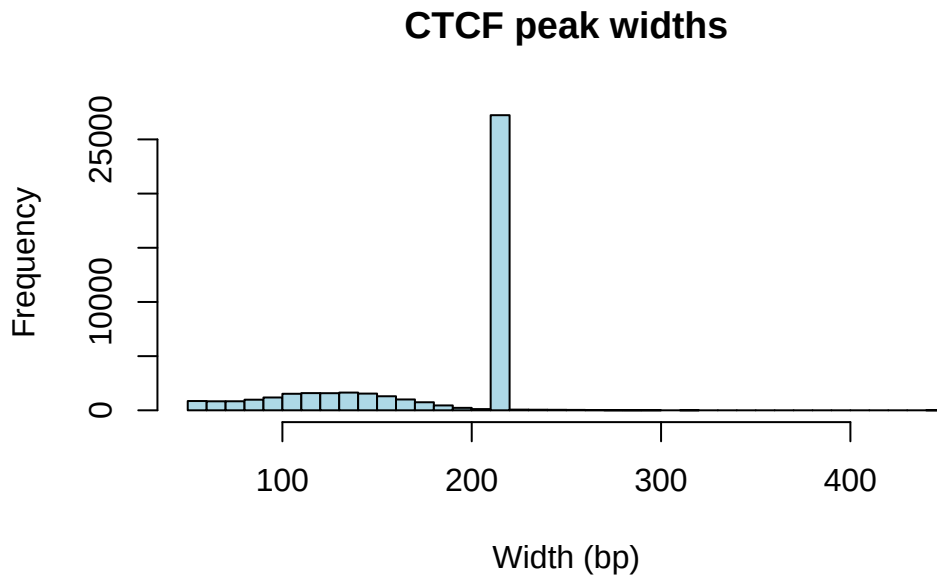


Figure 28.1.: Distribution of CTCF peak widths. The sharp spike reflects the peak caller's fixed-width output.

Figure 28.1 has a striking feature: a tall spike at one particular width rather than a smooth spread. That's not biology — it's the peak caller. Many ChIP-seq pipelines report peaks at a **fixed width** (here around 200 bp) centered on the detected summit, so the same number shows up over and over. When you see a histogram like this, read it as a signature of how the data were processed, not as a property of CTCF itself.

The metadata sits in a separate table you reach with `mcols()`:

28. Genomic ranges with real ChIP-seq peaks

```
head(mcols(ctcf_peaks))
```

DataFrame with 6 rows and 6 columns

	name	score	signalValue	pValue	qValue	peak
	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<integer>
1	NA	1000	7.70385	-1	0.27378	108
2	NA	541	8.71626	-1	0.46571	108
3	NA	672	9.07638	-1	0.54697	108
4	NA	560	9.86234	-1	0.66189	108
5	NA	584	10.15488	-1	0.75082	108
6	NA	614	10.46015	-1	0.80676	108

These are the narrowPeak columns. You can pull one out directly with `$` — for example the `signalValue`, which measures how strong the binding signal was at each peak:

```
hist(mcols(ctcf_peaks)$signalValue, breaks = 50,  
     main = "CTCF peak signal values", xlab = "Signal value", col = "lightblue")
```

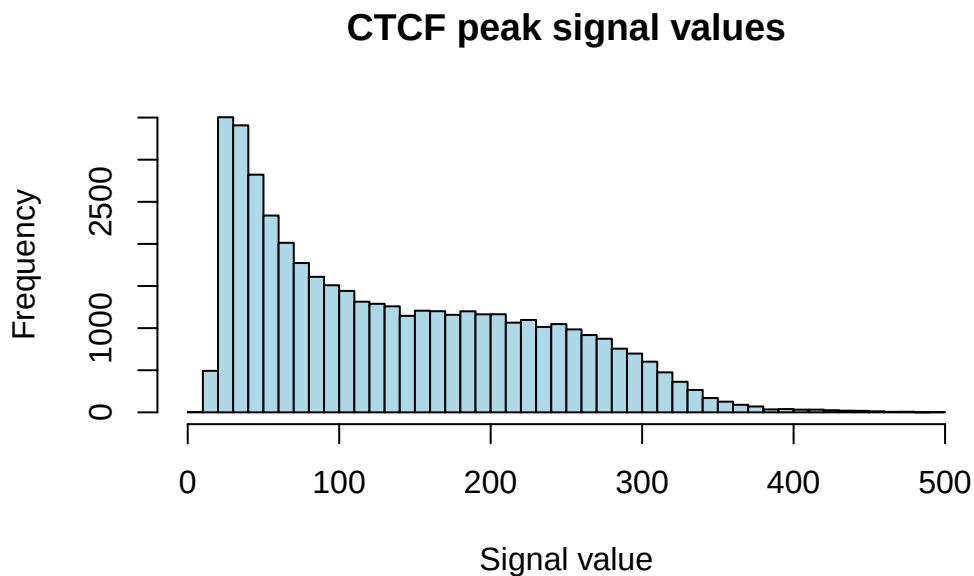


Figure 28.2.: Distribution of CTCF peak signal values. Most peaks are weak; a long tail of strong sites stretches to the right.

Figure 28.2 is **right-skewed**: most peaks carry a modest signal, with a long tail of much stronger binding sites. That shape is typical of ChIP-seq — a handful of high-confidence sites and many weaker ones — and it’s why people often filter on `signalValue` before downstream work.

28.6. Some quick summaries

How many peaks, on how many chromosomes, and how wide? A few small expressions answer this more clearly than a wall of printed text:

```
length(ctcf_peaks)          # total peaks
```

```
[1] 43865
```

```
length(unique(seqnames(ctcf_peaks)))  # chromosomes represented
```

```
[1] 26
```

```
range(width(ctcf_peaks))      # narrowest and widest peak
```

```
[1] 54 442
```

```
median(width(ctcf_peaks))     # typical peak width
```

```
[1] 216
```

The median width sits right at the spike we saw in Figure 28.1, confirming that most peaks share the caller’s fixed width. We can also count peaks per chromosome:

```
peaks_per_chr <- table(seqnames(ctcf_peaks))
head(as.data.frame(peaks_per_chr), 10)
```

28. Genomic ranges with real ChIP-seq peaks

	Var1	Freq
1	chr2	3332
2	chr20	1236
3	chr15	1431
4	chr8	1935
5	chr9	1820
6	chr7	2140
7	chr4	2030
8	chr12	2274
9	chr5	2355
10	chr22	873

Larger chromosomes generally carry more peaks, which makes sense: more DNA means more potential binding sites.

28.7. Where is each peak?

Three quantities you'll reach for constantly are the start, the end, and the **center** of each peak. The center is especially useful — motif analyses and fixed-width windows are usually built around it.

```
peak_starts <- start(ctcf_peaks)
peak_ends   <- end(ctcf_peaks)
peak_centers <- start(ctcf_peaks) + width(ctcf_peaks) / 2
head(peak_centers, 5)
```

```
[1] 238305432 49982815 39921124 6708382 136646065
```

`peak_centers` now holds the midpoint of every peak, ready to anchor a window.

28.8. Reshaping ranges

Three functions cover most of the range-resaping you'll do on a peak set: `shift()` slides a range, `resize()` changes its width, and `flank()` builds a neighbouring region. They all return a new `GRanges` and leave the original untouched.

28.8.1. Shifting

`shift()` moves every range by a fixed number of bases without changing its width:

```
peaks_shifted <- shift(ctcf_peaks, 100) # slide right by 100 bp
ctcf_peaks[1]
```

GRanges object with 1 range and 6 metadata columns:

	seqnames	ranges	strand	name	score	signalValue
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>
[1]	chr2	238305324-238305539	*	<NA>	1000	7.70385
	pValue	qValue	peak			
	<numeric>	<numeric>	<integer>			
[1]	-1	0.27378	108			

seqinfo: 26 sequences from an unspecified genome; no seqlengths

```
peaks_shifted[1]
```

GRanges object with 1 range and 6 metadata columns:

	seqnames	ranges	strand	name	score	signalValue
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>
[1]	chr2	238305424-238305639	*	<NA>	1000	7.70385
	pValue	qValue	peak			
	<numeric>	<numeric>	<integer>			
[1]	-1	0.27378	108			

seqinfo: 26 sequences from an unspecified genome; no seqlengths

The start and end of the first peak both moved up by 100, and the width is unchanged — exactly what “shift” should do.

28.8.2. Resizing

`resize()` sets every range to a chosen width. The `fix` argument says which point to hold still while the width changes. Resizing all peaks to a common width is the standard way to build comparable windows:

28. Genomic ranges with real ChIP-seq peaks

```
peaks_200bp <- resize(ctcf_peaks, width = 200, fix = "center")
```

```
peaks_200bp[1]
```

GRanges object with 1 range and 6 metadata columns:

	seqnames	ranges	strand	name	score	signalValue
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>
[1]	chr2	238305332-238305531	*	<NA>	1000	7.70385
	pValue	qValue	peak			
	<numeric>	<numeric>	<integer>			
[1]	-1	0.27378	108			

seqinfo: 26 sequences from an unspecified genome; no seqlengths

```
all(width(peaks_200bp) == 200) # every peak is now exactly 200 bp
```

```
[1] TRUE
```

Fixing on "center" keeps each peak's midpoint in place and grows or shrinks it symmetrically — so a downstream tool sees uniform 200 bp windows centered on the binding sites.

28.8.3. Flanking

`flank()` returns the region *beside* a range — useful for grabbing the DNA just upstream or downstream of a peak:

```
upstream_1kb <- flank(ctcf_peaks, width = 1000, start = TRUE)
```

```
ctcf_peaks[1]
```

GRanges object with 1 range and 6 metadata columns:

	seqnames	ranges	strand	name	score	signalValue
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>
[1]	chr2	238305324-238305539	*	<NA>	1000	7.70385
	pValue	qValue	peak			
	<numeric>	<numeric>	<integer>			

28. Genomic ranges with real ChIP-seq peaks

```
[1]      -1    0.27378      108
```

```
-----
```

```
seqinfo: 26 sequences from an unspecified genome; no seqlengths
```

```
upstream_1kb[1]
```

GRanges object with 1 range and 6 metadata columns:

	seqnames	ranges	strand	name	score	signalValue
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>	<numeric>
[1]	chr2	238304324-238305323	*	<NA>	1000	7.70385
	pValue	qValue	peak			
	<numeric>	<numeric>	<integer>			
[1]	-1	0.27378	108			

```
-----
```

```
seqinfo: 26 sequences from an unspecified genome; no seqlengths
```

The flanking range is the 1000 bp sitting immediately before the start of the first peak — the kind of window you might scan for a neighbouring promoter.

i Overlaps and grouped operations live next door

Once you have peaks and other features (genes, promoters, other peak sets), the next question is usually *which ones overlap?* The `findOverlaps()` and `nearest()` functions, set operations like `intersect()`, and grouped operations over a `GRangesList` are covered in depth in the *Genomic ranges and features* chapter. We don't repeat them here so we can stay focused on the real-data import workflow.

28.9. Coverage

Coverage counts how many ranges overlap each position along a chromosome. It's the foundation of peak calling: pile up sequencing reads, and the tall stacks are where the protein bound. Let's build a toy example by scattering random reads across a one-megabase stretch and counting the pile-up.

```
set.seed(42)    # reproducible random reads
chrom_length <- 1000000
num_reads    <- 100000
read_length  <- 50
```

28. Genomic ranges with real ChIP-seq peaks

```
random_starts <- sample(seq_len(chrom_length - read_length + 1),
                        num_reads, replace = TRUE)
random_reads <- GRanges(seqnames = "chr1",
                        ranges = IRanges(start = random_starts,
                                         end = random_starts + read_length - 1))
random_reads
```

GRanges object with 100000 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	61413-61462	*
[2]	chr1	54425-54474	*
[3]	chr1	623844-623893	*
[4]	chr1	74362-74411	*
[5]	chr1	46208-46257	*
...	...	...	...
[99996]	chr1	663489-663538	*
[99997]	chr1	886082-886131	*
[99998]	chr1	933222-933271	*
[99999]	chr1	486340-486389	*
[100000]	chr1	177556-177605	*

seqinfo: 1 sequence from an unspecified genome; no seqlengths

The `coverage()` function turns those reads into a per-base count. It returns an `Rle` (run-length encoded) object — a compact way to store long stretches of repeated values:

```
random_coverage <- coverage(random_reads, width = chrom_length)
head(random_coverage)
```

RleList of length 1

\$chr1

integer-Rle of length 1000000 with 173047 runs

Lengths:	7	1	6	3	8	8	1	1	22	1	4	...	3	1	2	8	12	23	5	2	8	6
Values :	0	1	3	4	5	6	7	8	9	8	6	...	3	2	3	4	5	4	3	2	1	0

Let's look at the coverage across the first 10,000 bases:

```
plot(random_coverage[[1]][1:10000], type = "h", col = "blue",
     main = "Coverage of random reads", xlab = "Position", ylab = "Coverage")
```

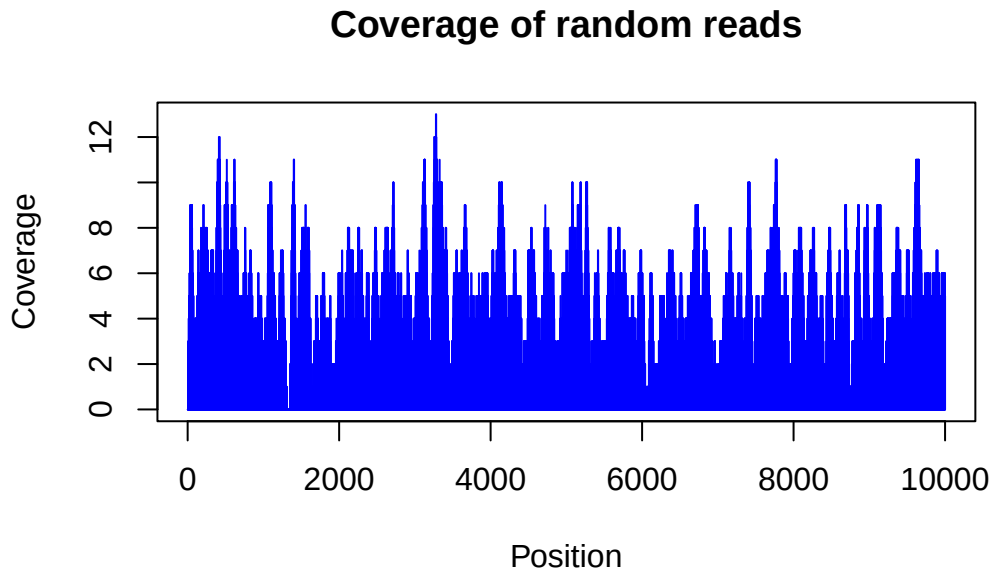


Figure 28.3.: Per-base coverage from randomly placed reads across the first 10,000 bases of a simulated chromosome.

Figure 28.3 looks like noise, and that’s the point: because we placed the reads *at random*, the coverage hovers around a constant average with no real peaks. Real ChIP-seq data would show tall, localized spikes where the protein bound — those spikes are what a peak caller hunts for, and their absence here is exactly what “no signal” should look like.

We can also ask how the coverage is distributed across all positions:

```
hist(as.numeric(random_coverage[[1]]),
     main = "Coverage distribution", xlab = "Coverage", ylab = "Frequency",
     col = "lightblue")
```

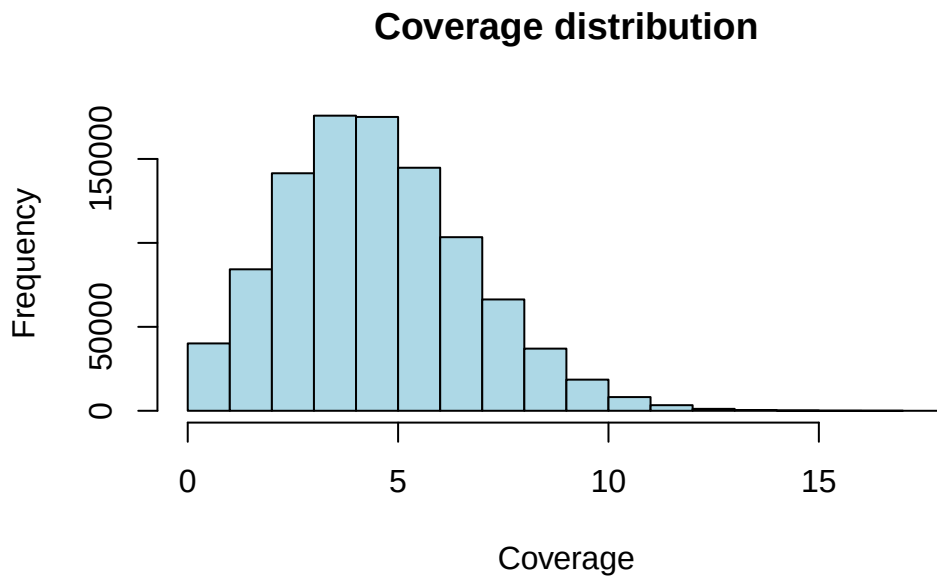


Figure 28.4.: How often each coverage depth occurs across the simulated chromosome.

Figure 28.4 is a tidy bell-ish hump centered on the average depth. With $1e+05$ reads of length 50 spread over $1e+06$ bases, the expected coverage is about 5x — right where the hump sits. Random reads give random, evenly-spread coverage; biological signal is what makes that distribution lumpy.

28.10. Exercises

Use the `ctcf_peaks` object you imported above. Each solution names the accessor or function that does the job.

1. **Wide peaks.** How many CTCF peaks are wider than 250 bp?

i Solution

```
sum(width(ctcf_peaks) > 250)
```

`width()` returns a numeric vector, `> 250` turns it into TRUE/FALSE, and `sum()` counts the TRUES. Most peaks sit at the caller's fixed width, so only a

28. Genomic ranges with real ChIP-seq peaks

minority exceed 250 bp.

2. **Fixed-width windows.** Build a new `GRanges` of 500 bp windows centered on each peak, and confirm every window is exactly 500 bp wide.

i Solution

```
windows_500 <- resize(ctcf_peaks, width = 500, fix = "center")
all(width(windows_500) == 500)
```

`fix = "center"` keeps each peak's midpoint fixed while the width grows to 500, giving uniform windows for a downstream analysis.

3. **Peak centers.** Compute the center position of every peak and report the median center.

i Solution

```
centers <- start(ctcf_peaks) + width(ctcf_peaks) / 2
median(centers)
```

The center is the start plus half the width. (You could also use `start(resize(ctcf_peaks, width = 1, fix = "center"))` to let `GRanges` do the arithmetic.)

4. **Fraction on chr1.** What fraction of all CTCF peaks fall on `chr1`?

i Solution

```
mean(seqnames(ctcf_peaks) == "chr1")
```

Comparing `seqnames()` to `"chr1"` gives a TRUE/FALSE vector; `mean()` of a logical vector is the *proportion* that are TRUE — a handy idiom for “what fraction satisfy this condition?”

28.11. Summary

You can now take a real peak set from file to insight:

- **Read a BED/narrowPeak file** and explain its 0-based, half-open coordinate system — and why R's 1-based coordinates differ.

28. Genomic ranges with real ChIP-seq peaks

- **Import peaks into a GRanges** straight from a URL with `rtracklayer::import(..., format = "narrowPeak")`, which handles the coordinate conversion and parses the metadata columns.
- **Explore the object** with `seqnames()`, `start()`, `end()`, `width()`, and `mcols()`, and read its width and signal distributions as fingerprints of the peak caller and the biology.
- **Reshape ranges** with `shift()`, `resize()`, and `flank()` to build the windows downstream tools expect.
- **Compute coverage** and recognize that random reads give flat coverage, while biological signal makes it lumpy.

For overlaps, nearest-feature searches, set operations, and gene models, continue to the [Genomic ranges and features](#) chapter, which treats those operations in full.

29. Genomic ranges and features

Almost everything you study in genomics lives at a *place* on the genome. A gene sits between two coordinates on a chromosome. An exon is a stretch within that gene. A ChIP-seq peak marks where a protein bound. A SNP is a single position. As soon as you have two such sets of places, the most interesting biological questions become questions about *location*: Which peaks fall inside a promoter? What is the nearest gene to this variant? Where do these two experiments agree?

In this chapter you'll learn to represent those places as **genomic ranges** and to ask how they relate to one another. The workhorse is Bioconductor's [GenomicRanges](#) package (Lawrence, Huber, Pagès, Aboyoun, Carlson, Gentleman, Martin T. Morgan, et al. 2013a), which gives you a compact object to hold ranges plus any data attached to them, and a vocabulary of fast operations — overlap, nearest-neighbor, flank, reduce — that scale to millions of regions. We'll build small example objects by hand so you can see exactly what each operation does, then finish with real gene models pulled from a genome annotation.

i A note on coordinate systems

Different tools count positions differently. The UCSC Genome Browser uses 0-based, half-open coordinates internally, while Ensembl and Bioconductor use 1-based, inclusive coordinates (position 1 is the first base, and both endpoints are included). **GRanges** objects are always 1-based and inclusive, which is one fewer thing to track once you stay inside Bioconductor.

29.1. What you'll learn

- Build a **GRanges** object and read its two-part display (coordinates on the left, meta-data on the right).
- Subset and extract parts of a **GRanges** object with accessors like `seqnames()`, `ranges()`, `strand()`, and `mcols()`.
- Apply intra-range operations (`flank()`, `shift()`, `resize()`) and inter-range operations (`reduce()`, `disjoin()`, `gaps()`, `coverage()`).

- Find relationships between two sets of ranges with `findOverlaps()`, `subsetByOverlaps()`, `nearest()`, and `distanceToNearest()`.
- Group ranges into a `GRangesList` (for example, the exons of a transcript).
- Pull gene, transcript, and exon models out of a `TxDb` annotation package.

29.2. Bioconductor and GenomicRanges

```
library(GenomicRanges)
```

The Bioconductor `GenomicRanges` package is a comprehensive toolkit designed to handle and manipulate genomic intervals and variables systematized on these intervals (Lawrence, Huber, Pagès, Aboyoun, Carlson, Gentleman, Martin T. Morgan, et al. 2013a). Developed by Bioconductor, this package simplifies the complexity of managing genomic data, facilitating the efficient exploration, manipulation, and visualization of such data. `GenomicRanges` aids in dealing with the challenges of genomic data, including its massive size, intricate relationships, and high dimensionality.

The `GenomicRanges` package in Bioconductor covers a wide range of use cases related to the management and analysis of genomic data. Here are some key examples:

- **Genomic Feature Manipulation** The `GenomicRanges` and `GRanges` classes can be used to represent and manipulate various genomic features such as genes, transcripts, exons, or single-nucleotide polymorphisms (SNPs). Users can query, subset, resize, shift, or sort these features based on their genomic coordinates.
- **Genomic Interval Operations** The `GenomicRanges` package provides functions for performing operations on genomic intervals, such as finding overlaps, nearest neighbors, or disjoint intervals. These operations are fundamental to many types of genomic data analyses, such as identifying genes that overlap with ChIP-seq peaks, or finding variants that are in close proximity to each other.
- **Alignments and Coverage** The `GAlignments` and `GAlignmentPairs` classes can be used to represent alignments of sequencing reads to a reference genome, such as those produced by a read aligner. Users can then compute coverage of these alignments over genomic ranges of interest, which is a common task in RNA-seq or ChIP-seq analysis.
- **Annotation and Metadata Handling** The metadata column of a `GRanges` object can be used to store various types of annotation data associated with genomic ranges, such as gene names, gene biotypes, or experimental scores. This makes

29. Genomic ranges and features

it easy to perform analyses that integrate genomic coordinates with other types of biological information.

- **Genome Arithmetic** The GenomicRanges package supports a version of “genome arithmetic”, which includes set operations (union, intersection, set difference) as well as other operations (like coverage, complement, or reduction) that are adapted to the specificities of genomic data.
- **Efficient Data Handling** The CompressedGRangesList class provides a space-efficient way to represent a large list of GRanges objects, which is particularly useful when working with large genomic datasets, such as whole-genome sequencing data.

The GenomicRanges package in Bioconductor uses the S4 class system (see Table 29.1), which is a part of the methods package in R. The S4 system is a more rigorous and formal approach to object-oriented programming in R, providing enhanced capabilities for object design and function dispatch.

Table 29.1.: Classes within the GenomicRanges package. Each class has a slightly different use case.

Class Name	Description	Potential Use
GRanges	Represents a collection of genomic ranges and associated variables.	ChIPSeq peaks, CpG islands, etc.
GRangesList	Represents a list of GenomicRanges objects.	transcript models (exons, introns)
RangesList	Represents a list of Ranges objects.	
IRanges	Represents a collection of integer ranges.	used mainly to <i>build</i> GRanges, etc.
GPos	Represents genomic positions.	SNPs or other single-nucleotide locations
GAlignments	Represents alignments against a reference genome.	Sequence read locations from a BAM file
GAlignment-Pairs	Represents pairs of alignments, typically representing a single fragment of DNA.	Paired-end sequence alignments

In the context of the GenomicRanges package, the S4 class system allows for the creation of complex, structured data objects that can effectively encapsulate genomic intervals and

29. Genomic ranges and features

associated data. This system enables the package to handle the complexity and intricacy of genomic data.

For example, the `GenomicRanges` class in the package is an S4 class that combines several basic data types into a composite object. It includes slots for sequence names (`seqnames`), ranges (start and end positions), strand information, and metadata. Each slot in the S4 class corresponds to a specific component of the genomic data, and methods (see Table 29.2 and Table 29.3) can be defined to interact with these slots in a structured and predictable way.

Table 29.2.: Methods for accessing, manipulating single objects

Method	Description
length	Returns the number of ranges in the <code>GRanges</code> object.
seqnames	Retrieves the sequence names of the ranges.
ranges	Retrieves the start and end positions of the ranges.
strand	Retrieves the strand information of the ranges.
elementMetadata	Retrieves the metadata associated with the ranges.
seqlevels	Returns the levels of the factor that the <code>seqnames</code> slot is derived from.
seqinfo	Retrieves the <code>Seqinfo</code> (sequence information) object associated with the <code>GRanges</code> object.
start, end, width	Retrieve or set the start or end positions, or the width of the ranges.
resize	Resizes the ranges.
subset, [, [[, \$	Subset or extract elements from the <code>GRanges</code> object.
sort	Sorts the <code>GRanges</code> object.
shift	Shifts the ranges by a certain number of base pairs.

The S4 class system also supports inheritance, which allows for the creation of specialized subclasses that share certain characteristics with a parent class but have additional features or behaviors.

The S4 system's formalism and rigor make it well-suited to the complexities of bioinformatics and genomic data analysis. It allows for the creation of robust, reliable code that can handle complex data structures and operations, making it a key part of the `GenomicRanges` package and other Bioconductor packages.

29. Genomic ranges and features

Table 29.3.: Methods for comparing and combining multiple GenomicRanges-class objects

Method	Description
findOverlaps	Finds overlaps between different sets of ranges.
countOverlaps	Counts overlaps between different sets of ranges.
subsetByOverlaps	Subsets the ranges based on overlaps.
distanceToNearest	Computes the distance to the nearest range in another set of ranges.

To get going, we can construct a **GRanges** object by hand as an example.

The **GRanges** class represents a collection of genomic ranges that each have a single start and end location on the genome. It can be used to store the location of genomic features such as contiguous binding sites, transcripts, and exons. These objects can be created by using the **GRanges** constructor function. The following code just creates a **GRanges** object from scratch.

```
gr <- GRanges(
  seqnames = Rle(c("chr1", "chr2", "chr1", "chr3"), c(1, 3, 2, 4)),
  ranges = IRanges(101:110, end = 111:120, names = head(letters, 10)),
  strand = Rle(strand(c("-", "+", "*", "+", "-")), c(1, 2, 2, 3, 2)),
  score = 1:10,
  GC = seq(1, 0, length=10))
gr
```

GRanges object with 10 ranges and 2 metadata columns:

	seqnames	ranges	strand	score	GC
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>
a	chr1	101-111	-	1	1.000000
b	chr2	102-112	+	2	0.888889
c	chr2	103-113	+	3	0.777778
d	chr2	104-114	*	4	0.666667
e	chr1	105-115	*	5	0.555556
f	chr1	106-116	+	6	0.444444
g	chr3	107-117	+	7	0.333333
h	chr3	108-118	+	8	0.222222
i	chr3	109-119	-	9	0.111111
j	chr3	110-120	-	10	0.000000

```
-----
```

```
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

This creates a `GRanges` object with 10 genomic ranges. The output of the `GRanges show()` method separates the information into a left and right hand region that are separated by `|` symbols (see Figure 29.1). The genomic coordinates (seqnames, ranges, and strand) are located on the left-hand side and the metadata columns are located on the right. For this example, the metadata is comprised of score and GC information, but almost anything can be stored in the metadata portion of a `GRanges` object.

i Reading a `GRanges`: the `|` is the dividing line

Think of a `GRanges` as a spreadsheet with a heavy vertical rule drawn down the middle. **Everything to the left of the `|` answers “where?”** — the chromosome (`seqnames`), the start and end (`ranges`), and the `strand`. These columns are mandatory and have their own special accessors. **Everything to the right of the `|` answers “what do we know about it?”** — the metadata: scores, GC content, gene names, p-values, whatever you attach. These are optional and you reach them all at once with `mcols()`. Keeping that line in mind makes every later operation easier: range operations rearrange the *left* side; your annotations ride along on the *right*.

The components of the genomic coordinates within a `GRanges` object can be extracted using the `seqnames`, `ranges`, and `strand` accessor functions.

```
seqnames(gr)
```

```
factor-Rle of length 10 with 4 runs
 Lengths:    1    3    2    4
  Values : chr1 chr2 chr1 chr3
Levels(3): chr1 chr2 chr3
```

```
ranges(gr)
```

```
IRanges object with 10 ranges and 0 metadata columns:
      start      end      width
  <integer> <integer> <integer>
a         101       111         11
b         102       112         11
```

29. Genomic ranges and features

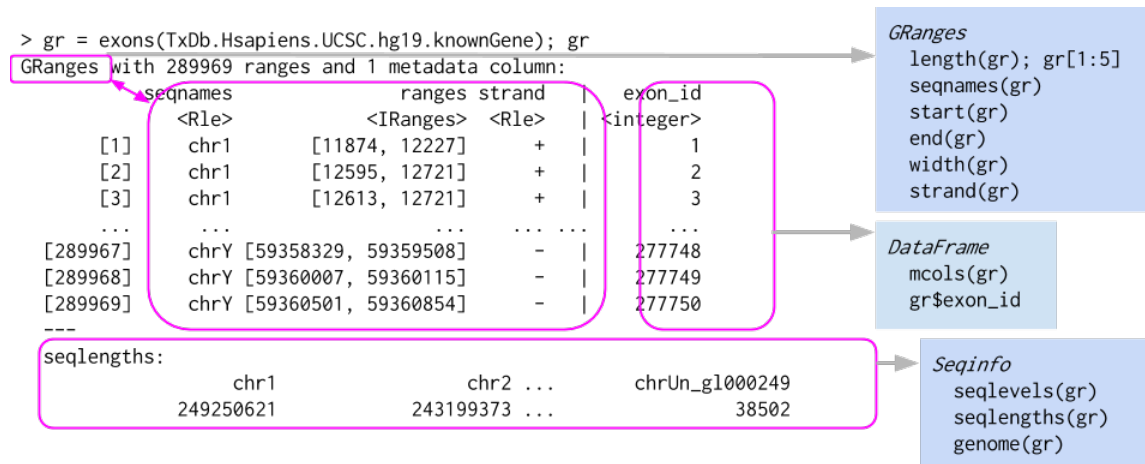


Figure 29.1.: The structure of a **GRanges** object, which behaves a bit like a vector of ranges, although the analogy is not perfect. A **GRanges** object is composed of the “Ranges” part the lefthand box, the “metadata” columns (the righthand box), and a “seqinfo” part that describes the names and lengths of associated sequences. Only the “Ranges” part is required. The figure also shows a few of the “accessors” and approaches to subsetting a **GRanges** object.

```

c      103      113      11
d      104      114      11
e      105      115      11
f      106      116      11
g      107      117      11
h      108      118      11
i      109      119      11
j      110      120      11
  
```

```
strand(gr)
```

```

factor-Rle of length 10 with 5 runs
Lengths: 1 2 2 3 2
Values  : - + * + -
Levels(3): + - *
  
```

Note that the **GRanges** object has information to the “left” side of the | that has special accessors. The information to the right side of the |, when it is present, is the metadata and is accessed using `mcols()`, for “metadata columns”.

29. Genomic ranges and features

```
class(mcols(gr))
```

```
[1] "DFrame"  
attr(,"package")  
[1] "S4Vectors"
```

```
mcols(gr)
```

DataFrame with 10 rows and 2 columns

	score	GC
	<integer>	<numeric>
a	1	1.000000
b	2	0.888889
c	3	0.777778
d	4	0.666667
e	5	0.555556
f	6	0.444444
g	7	0.333333
h	8	0.222222
i	9	0.111111
j	10	0.000000

Since the class of `mcols(gr)` is `DFrame`, we can use our `DataFrame` approaches to work with the data.

```
mcols(gr)$score
```

```
[1] 1 2 3 4 5 6 7 8 9 10
```

We can even assign a new column. Here we add an `AT` column derived from `GC` (since the fractions sum to 1). Notice that the new column appears on the right of the `|`, alongside `score` and `GC`.

```
mcols(gr)$AT <- 1 - mcols(gr)$GC  
gr
```

29. Genomic ranges and features

GRanges object with 10 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333
e	chr1	105-115	*	5	0.555556	0.444444
f	chr1	106-116	+	6	0.444444	0.555556
g	chr3	107-117	+	7	0.333333	0.666667
h	chr3	108-118	+	8	0.222222	0.777778
i	chr3	109-119	-	9	0.111111	0.888889
j	chr3	110-120	-	10	0.000000	1.000000

seqinfo: 3 sequences from an unspecified genome; no seqlengths

Another common way to create a `GRanges` object is to start with a `data.frame`, perhaps created by hand like below or read in using `read.csv` or `read.table`. We can convert from a `data.frame`, when columns are named appropriately, to a `GRanges` object.

```
df_regions <- data.frame(chromosome = rep("chr1", 10),
                        start = seq(1000, 10000, 1000),
                        end = seq(1100, 10100, 1000))
as(df_regions, 'GRanges') # note that names have to match with GRanges slots
```

GRanges object with 10 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	1000-1100	*
[2]	chr1	2000-2100	*
[3]	chr1	3000-3100	*
[4]	chr1	4000-4100	*
[5]	chr1	5000-5100	*
[6]	chr1	6000-6100	*
[7]	chr1	7000-7100	*
[8]	chr1	8000-8100	*
[9]	chr1	9000-9100	*
[10]	chr1	10000-10100	*

seqinfo: 1 sequence from an unspecified genome; no seqlengths

29. Genomic ranges and features

```
## fix column name
colnames(df_regions)[1] <- 'seqnames'
gr2 <- as(df_regions, 'GRanges')
gr2
```

GRanges object with 10 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	1000-1100	*
[2]	chr1	2000-2100	*
[3]	chr1	3000-3100	*
[4]	chr1	4000-4100	*
[5]	chr1	5000-5100	*
[6]	chr1	6000-6100	*
[7]	chr1	7000-7100	*
[8]	chr1	8000-8100	*
[9]	chr1	9000-9100	*
[10]	chr1	10000-10100	*

seqinfo: 1 sequence from an unspecified genome; no seqlengths

The first `as()` call worked even though the chromosome column was named `chromosome` — `as()` recognized `start` and `end` and treated the rest as metadata. After we rename that column to `seqnames`, the conversion places it correctly on the *left* of the `|` as the sequence name.

`GRanges` have one-dimensional-like behavior. For instance, we can check the `length` and even give `GRanges` names.

```
names(gr)
```

```
[1] "a" "b" "c" "d" "e" "f" "g" "h" "i" "j"
```

```
length(gr)
```

```
[1] 10
```

29.2.1. Subsetting GRanges objects

While `GRanges` objects look a bit like a `data.frame`, they can be thought of as vectors with associated ranges. Subsetting, then, works very similarly to vectors. To subset a `GRanges` object to include only second and third regions:

```
gr[2:3]
```

GRanges object with 2 ranges and 3 metadata columns:

	seqnames	ranges	strand		score	GC	AT
	<Rle>	<IRanges>	<Rle>		<integer>	<numeric>	<numeric>
b	chr2	102-112	+		2	0.888889	0.111111
c	chr2	103-113	+		3	0.777778	0.222222

seqinfo: 3 sequences from an unspecified genome; no seqlengths

That said, if the `GRanges` object has metadata columns, we can also treat it like a two-dimensional object kind of like a `data.frame`. Note that the information to the left of the `|` is not like a `data.frame`, so we *cannot* do something like `gr$seqnames`. Here is an example of subsetting with the subset of one metadata column.

```
gr[2:3, "GC"]
```

GRanges object with 2 ranges and 1 metadata column:

	seqnames	ranges	strand		GC
	<Rle>	<IRanges>	<Rle>		<numeric>
b	chr2	102-112	+		0.888889
c	chr2	103-113	+		0.777778

seqinfo: 3 sequences from an unspecified genome; no seqlengths

The usual `head()` and `tail()` also work just fine.

```
head(gr, n=2)
```

GRanges object with 2 ranges and 3 metadata columns:

	seqnames	ranges	strand		score	GC	AT
	<Rle>	<IRanges>	<Rle>		<integer>	<numeric>	<numeric>

29. Genomic ranges and features

```
a      chr1   101-111   - |           1  1.000000  0.000000
b      chr2   102-112   + |           2  0.888889  0.111111
-----
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

```
tail(gr,n=2)
```

GRanges object with 2 ranges and 3 metadata columns:

```
      seqnames      ranges strand |      score      GC      AT
      <Rle> <IRanges> <Rle> | <integer> <numeric> <numeric>
i      chr3    109-119      - |        9  0.111111  0.888889
j      chr3    110-120      - |       10  0.000000  1.000000
-----
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

29.2.2. Interval operations on one GRanges object

29.2.2.1. Intra-range methods

The methods described in this section work *one-region-at-a-time* and are, therefore, called “intra-region” methods. Methods that work across all regions are described below in the [Inter-range methods](#) section.

The `GRanges` class has accessors for the “ranges” part of the data. For example:

```
## Make a smaller GRanges subset
g <- gr[1:3]
start(g) # to get start locations
```

```
[1] 101 102 103
```

```
end(g) # to get end locations
```

```
[1] 111 112 113
```

```
width(g) # to get the "widths" of each range
```

```
[1] 11 11 11
```

29. Genomic ranges and features

```
range(g) # to get the "range" for each sequence (min(start) through max(end))
```

GRanges object with 2 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	101-111	-
[2]	chr2	102-113	+

seqinfo: 3 sequences from an unspecified genome; no seqlengths

The GRanges class also has many methods for manipulating the ranges. The methods can be classified as intra-range methods, inter-range methods, and between-range methods. Intra-range methods operate on each element of a GRanges object independent of the other ranges in the object. For example, the flank method can be used to recover regions flanking the set of ranges represented by the GRanges object. So to get a GRanges object containing the ranges that include the 10 bases *upstream* of the ranges:

```
flank(g, 10)
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	112-121	-	1	1.000000	0.000000
b	chr2	92-101	+	2	0.888889	0.111111
c	chr2	93-102	+	3	0.777778	0.222222

seqinfo: 3 sequences from an unspecified genome; no seqlengths

Note how flank pays attention to “strand”. To get the flanking regions *downstream* of the ranges, we can do:

```
flank(g, 10, start=FALSE)
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	91-100	-	1	1.000000	0.000000
b	chr2	113-122	+	2	0.888889	0.111111

29. Genomic ranges and features

```
c      chr2    114-123      + |          3  0.777778  0.222222
-----
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

⚠ Strand changes what “upstream” means

For `flank()`, `resize()`, `precede()`, and `follow()`, “upstream” and “downstream” are interpreted *relative to the strand*. On the + strand, upstream is toward lower coordinates; on the - strand it is toward higher coordinates. A * (unstranded) range is treated as +. If your ranges have the wrong strand — or you forgot to set it — these operations will silently flank the wrong side. When a result looks backwards, check `strand()` first.

Other examples of intra-range methods include `resize` and `shift`. The `shift` method will move the ranges by a specific number of base pairs, and the `resize` method will extend the ranges by a specified width.

```
shift(g, 5)
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	106-116	-	1	1.000000	0.000000
b	chr2	107-117	+	2	0.888889	0.111111
c	chr2	108-118	+	3	0.777778	0.222222

```
-----
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

```
resize(g, 30)
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	82-111	-	1	1.000000	0.000000
b	chr2	102-131	+	2	0.888889	0.111111
c	chr2	103-132	+	3	0.777778	0.222222

```
-----
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

The [GenomicRanges](#) help page `?intra-range-methods` summarizes these methods.

29.2.2.2. Inter-range methods

Inter-range methods involve comparisons between ranges in a single `GRanges` object. For instance, the `reduce` method will align the ranges and merge overlapping ranges to produce a simplified set.

```
reduce(g)
```

`GRanges` object with 2 ranges and 0 metadata columns:

```

      seqnames      ranges strand
      <Rle> <IRanges> <Rle>
[1]      chr1    101-111      -
[2]      chr2    102-113      +
-----

```

```
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

The `reduce` method could, for example, be used to collapse individual overlapping coding exons into a single set of coding regions.

Sometimes one is interested in the gaps or the qualities of the gaps between the ranges represented by your `GRanges` object. The `gaps` method provides this information:

```
gaps(g)
```

`GRanges` object with 2 ranges and 0 metadata columns:

```

      seqnames      ranges strand
      <Rle> <IRanges> <Rle>
[1]      chr1      1-100      -
[2]      chr2      1-101      +
-----

```

```
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

 `gaps()` needs to know how long each chromosome is

We never told this object how long the chromosomes are, so Bioconductor assumes each chromosome ends at the largest coordinate it has seen. That means the final gap (from the last range to the true end of the chromosome) is missing. In real work you set the chromosome lengths with `seqlengths()` so that `gaps()`, `coverage()`, and friends know the full extent of each sequence. We can ignore that detail for these toy

examples.

The `disjoin` method represents a `GRanges` object as a collection of non-overlapping ranges:

```
disjoin(g)
```

`GRanges` object with 4 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	101-111	-
[2]	chr2	102	+
[3]	chr2	103-112	+
[4]	chr2	113	+

seqinfo: 3 sequences from an unspecified genome; no seqlengths

The `coverage` method quantifies the degree of overlap for all the ranges in a `GRanges` object.

```
coverage(g)
```

RleList of length 3

\$chr1

integer-Rle of length 111 with 2 runs

Lengths: 100 11

Values : 0 1

\$chr2

integer-Rle of length 113 with 4 runs

Lengths: 101 1 10 1

Values : 0 1 2 1

\$chr3

integer-Rle of length 0 with 0 runs

Lengths:

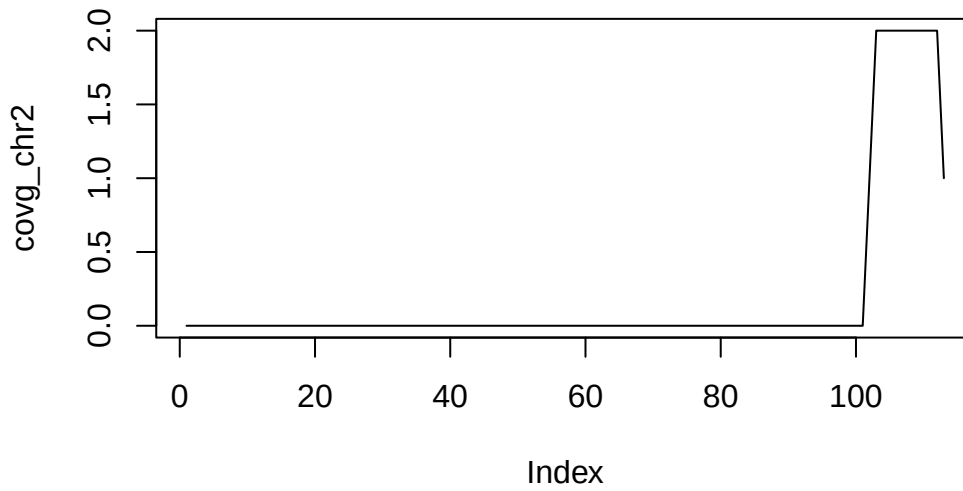
Values :

The coverage is summarized as a `list` of coverages, one for each chromosome. The `Rle` class is used to store the values. Sometimes, one must convert these values to `numeric` using `as.numeric`. In many cases, this will happen automatically, though.

```

covg <- coverage(g)
covg_chr2 <- covg[['chr2']]
plot(covg_chr2, type='l')

```



The plot shows how many ranges cover each base along `chr2`: the line steps up to 2 where two ranges overlap and drops back to 0 in the gaps. This is exactly the kind of per-base signal you compute from aligned sequencing reads in RNA-seq or ChIP-seq.

See the [GenomicRanges](#) help page `? "intra-range-methods"` for more details.

29.2.3. Set operations for GRanges objects

Between-range methods calculate relationships between different `GRanges` objects. Of central importance are `findOverlaps` and related operations; these are discussed below. Additional operations treat `GRanges` as mathematical sets of coordinates; `union(g, g2)` is the union of the coordinates in `g` and `g2`. Here are examples for calculating the union, the intersect and the asymmetric difference (using `setdiff`).

```

g2 <- head(gr, n=2)
GenomicRanges::union(g, g2)

```


29. Genomic ranges and features

GRanges object with 2 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	101-111	-
[2]	chr2	102-113	+

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
GenomicRanges::intersect(g, g2)
```

GRanges object with 2 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	101-111	-
[2]	chr2	102-112	+

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
GenomicRanges::setdiff(g, g2)
```

GRanges object with 1 range and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr2	113	+

seqinfo: 3 sequences from an unspecified genome; no seqlengths

There is extensive additional help available or by looking at the vignettes in at the [GenomicRanges](#) pages.

```
?GRanges
```

There are also many possible `methods` that work with `GRanges` objects. To see a complete list (long), try:

```
methods(class="GRanges")
```

29.3. GRangesList

Some important genomic features, such as spliced transcripts that are comprised of exons, are inherently compound structures. Such a feature makes much more sense when expressed as a compound object such as a **GRangesList**. If we think of each transcript as a set of exons, each transcript would be summarized as a **GRanges** object. However, if we have multiple transcripts, we want to keep them separate, with each transcript holding its own exons. A **GRangesList** is a list of **GRanges** objects. Continuing with the transcripts idea, a **GRangesList** can contain all the transcripts and their exons; each transcript is one element in the list.

```
> grl = exonsBy(TxDb.Hsapiens.UCSC.hg19.knownGene, "tx", use.names=TRUE); grl
GRangesList of length 82960:
$uc001aaa.3
GRanges with 3 ranges and 3 metadata columns:
      seqnames      ranges strand | exon_id exon_name exon_rank
      <Rle>         <IRanges> <Rle> | <integer> <character> <integer>
[1]   chr1 [11874, 12227]      + |         1      <NA>         1
[2]   chr1 [12613, 12721]      + |         3      <NA>         2
[3]   chr1 [13221, 14409]      + |         5      <NA>         3

$uc010nxq.1
GRanges with 3 ranges and 3 metadata columns:
      seqnames      ranges strand | exon_id exon_name exon_rank
      <Rle>         <IRanges> <Rle> | <integer> <character> <integer>
[1]   chr1 [11874, 12227]      + |         1      <NA>         1
[2]   chr1 [12595, 12721]      + |         2      <NA>         2
[3]   chr1 [13403, 14409]      + |         6      <NA>         3

$uc010nxr.1
GRanges with 3 ranges and 3 metadata columns:
      seqnames      ranges strand | exon_id exon_name exon_rank
      <Rle>         <IRanges> <Rle> | <integer> <character> <integer>
[1]   chr1 [11874, 12227]      + |         1      <NA>         1
[2]   chr1 [12646, 12697]      + |         4      <NA>         2
[3]   chr1 [13221, 14409]      + |         5      <NA>         3
...
<82957 more elements>
---
seqinfo: 93 sequences (1 circular) from hg19 genome
```

GRangesList
(list of GRanges)
length(grl)
grl[1:3]
shift(grl, 1)
range(grl)

GRanges
grl[[2]]
grl[["uc010nxq.1"]]

Two kinds of fun!
introns =
 psetdiff(range(grl), grl)

grr = unlist(grl)
transform grr, then...
grl = relist(grr, grl)

↑ 'flesh'
↑ 'skeleton'

Figure 29.2.: The structure of a **GRangesList**, which is a **list** of **GRanges** objects. While the analogy is not perfect, a **GRangesList** behaves a bit like a list. Each element in the **GRangesList** is a **GRanges** object. A common use case for a **GRangesList** is to store a list of transcripts, each of which have exons as the regions in the **GRanges**.

Whenever genomic features consist of multiple ranges that are grouped by a parent feature, they can be represented as a **GRangesList** object. Consider the simple example of the two

29. Genomic ranges and features

transcript `GRangesList` below created using the `GRangesList` constructor.

```
gr1 <- GRanges(  
  seqnames = "chr1",  
  ranges = IRanges(103, 106),  
  strand = "+",  
  score = 5L, GC = 0.45)  
gr2 <- GRanges(  
  seqnames = c("chr1", "chr1"),  
  ranges = IRanges(c(107, 113), width = 3),  
  strand = c("+", "-"),  
  score = 3:4, GC = c(0.3, 0.5))
```

The `gr1` and `gr2` are each `GRanges` objects. We can combine them into a “named” `GRangesList` like so:

```
grl <- GRangesList("txA" = gr1, "txB" = gr2)  
grl
```

`GRangesList` object of length 2:

`$txA`

`GRanges` object with 1 range and 2 metadata columns:

	seqnames	ranges	strand	score	GC
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>
[1]	chr1	103-106	+	5	0.45

seqinfo: 1 sequence from an unspecified genome; no seqlengths

`$txB`

`GRanges` object with 2 ranges and 2 metadata columns:

	seqnames	ranges	strand	score	GC
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>
[1]	chr1	107-109	+	3	0.3
[2]	chr1	113-115	-	4	0.5

seqinfo: 1 sequence from an unspecified genome; no seqlengths

The `show` method for a `GRangesList` object displays it as a named list of `GRanges` objects, where the names of this list are considered to be the names of the grouping feature. In the

example above, the groups of individual exon ranges are represented as separate **GRanges** objects which are further organized into a list structure where each element name is a transcript name. Many other combinations of grouped and labeled **GRanges** objects are possible of course, but this example is a common arrangement.

In some cases, **GRangesLists** behave quite similarly to **GRanges** objects.

29.3.1. Basic *GRangesList* accessors

Just as with **GRanges** object, the components of the genomic coordinates within a **GRangesList** object can be extracted using simple accessor methods. Not surprisingly, the **GRangesList** objects have many of the same accessors as **GRanges** objects. The difference is that many of these methods return a list since the input is now essentially a list of **GRanges** objects. Here are a few examples:

```
seqnames(grl)
```

```
RleList of length 2
$txA
factor-Rle of length 1 with 1 run
  Lengths:    1
  Values  : chr1
Levels(1): chr1

$txB
factor-Rle of length 2 with 1 run
  Lengths:    2
  Values  : chr1
Levels(1): chr1
```

```
ranges(grl)
```

```
IRangesList object of length 2:
$txA
IRanges object with 1 range and 0 metadata columns:
      start      end      width
<integer> <integer> <integer>
[1]      103      106         4
```

29. Genomic ranges and features

```
$txB
IRanges object with 2 ranges and 0 metadata columns:
      start      end      width
  <integer> <integer> <integer>
[1]      107      109         3
[2]      113      115         3
```

```
strand(grl)
```

```
RleList of length 2
$txA
factor-Rle of length 1 with 1 run
  Lengths: 1
  Values  : +
Levels(3): + - *

$txB
factor-Rle of length 2 with 2 runs
  Lengths: 1 1
  Values  : + -
Levels(3): + - *
```

The length and names methods will return the length or names of the list and the seqlengths method will return the set of sequence lengths.

```
length(grl)
```

```
[1] 2
```

```
names(grl)
```

```
[1] "txA" "txB"
```

```
seqlengths(grl)
```

```
chr1
NA
```

29.4. Relationships between region sets

One of the more powerful approaches to genomic data integration is to ask about the relationship between sets of genomic ranges. The key features of this process are to look at overlaps and distances to the nearest feature. These functionalities, combined with the operations like `flank` and `resize`, for instance, allow pretty useful analyses with relatively little code. In general, these operations are *very* fast, even on thousands to millions of regions.

29.4.1. Overlaps

The `findOverlaps` method in the `GenomicRanges` package is a very useful function that allows users to identify overlaps between two sets of genomic ranges.

Here's how it works:

- **Inputs** The function requires two `GRanges` objects, referred to as query and subject.
- **Processing** The function then compares every range in the query object with every range in the subject object, looking for overlaps. An overlap is defined as any instance where the range in the query object intersects with a range in the subject object.
- **Output** The function returns a `Hits` (see `?Hits`) object, which is a compact representation of the matrix of overlaps. Each entry in the `Hits` object corresponds to a pair of overlapping ranges, with the query index and the subject index.

Here is an example of how you might use the `findOverlaps` function:

```
# Create two GRanges objects
gr1 <- gr[1:4]
gr2 <- gr[3:8]
gr1
```

`GRanges` object with 4 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333

29. Genomic ranges and features

```
-----  
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

```
gr2
```

GRanges object with 6 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333
e	chr1	105-115	*	5	0.555556	0.444444
f	chr1	106-116	+	6	0.444444	0.555556
g	chr3	107-117	+	7	0.333333	0.666667
h	chr3	108-118	+	8	0.222222	0.777778

```
-----  
seqinfo: 3 sequences from an unspecified genome; no seqlengths
```

```
# Find overlaps  
overlaps <- findOverlaps(query = gr1, subject = gr2)  
overlaps
```

Hits object with 7 hits and 0 metadata columns:

	queryHits	subjectHits
	<integer>	<integer>
[1]	1	3
[2]	2	1
[3]	2	2
[4]	3	1
[5]	3	2
[6]	4	1
[7]	4	2

```
-----  
queryLength: 4 / subjectLength: 6
```

In the resulting overlaps object, each row corresponds to an overlapping pair of ranges, with the first column giving the index of the range in gr1 and the second column giving the index of the overlapping range in gr2.

29. Genomic ranges and features

If you are interested in only the `queryHits` or the `subjectHits`, there are accessors for those (ie., `queryHits(overlaps)`). To get the actual ranges that overlap, you can use the `subjectHits` or `queryHits` as an index into the original `GRanges` object.

Spend some time looking at these results. Note how the strand comes into play when determining overlaps.

```
gr1[queryHits(overlaps)]
```

GRanges object with 7 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333
d	chr2	104-114	*	4	0.666667	0.333333

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
gr2[subjectHits(overlaps)]
```

GRanges object with 7 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
e	chr1	105-115	*	5	0.555556	0.444444
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333

seqinfo: 3 sequences from an unspecified genome; no seqlengths

As you might expect, the `countOverlaps` method counts the regions in the second `GRanges` that overlap with those that overlap with each element of the first.

29. Genomic ranges and features

```
countOverlaps(gr1, gr2)
```

```
a b c d
1 2 2 2
```

The `subsetByOverlaps` method simply subsets the query `GRanges` object to include *only* those that overlap the subject.

```
subsetByOverlaps(gr1, gr2)
```

GRanges object with 4 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333

seqinfo: 3 sequences from an unspecified genome; no seqlengths

In some cases, you may be interested in only one hit per query range. The `select` parameter handles this: with `select="first"`, `findOverlaps()` returns a plain integer vector with one entry per query range (the index of its first overlapping subject, or `NA` if there is none) instead of a full `Hits` object.

```
findOverlaps(gr1, gr2, select="first")
```

```
[1] 3 1 1 1
```

The `%over%` logical operator allows us to do similar things to `findOverlaps` and `subsetByOverlaps`. It returns a `TRUE/FALSE` for each range, which is handy for subsetting.

```
gr2 %over% gr1
```

```
[1] TRUE TRUE TRUE FALSE FALSE FALSE
```

```
gr1[gr1 %over% gr2]
```

GRanges object with 4 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222
d	chr2	104-114	*	4	0.666667	0.333333

seqinfo: 3 sequences from an unspecified genome; no seqlengths

29.4.2. Nearest feature

There are a number of useful methods that find the nearest feature (region) in a second set for each element in the first set.

We can review our two GRanges toy objects:

```
g
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
gr
```

GRanges object with 10 ranges and 3 metadata columns:

	seqnames	ranges	strand	score	GC	AT
	<Rle>	<IRanges>	<Rle>	<integer>	<numeric>	<numeric>
a	chr1	101-111	-	1	1.000000	0.000000
b	chr2	102-112	+	2	0.888889	0.111111
c	chr2	103-113	+	3	0.777778	0.222222

29. Genomic ranges and features

d	chr2	104-114	*	4	0.666667	0.333333
e	chr1	105-115	*	5	0.555556	0.444444
f	chr1	106-116	+	6	0.444444	0.555556
g	chr3	107-117	+	7	0.333333	0.666667
h	chr3	108-118	+	8	0.222222	0.777778
i	chr3	109-119	-	9	0.111111	0.888889
j	chr3	110-120	-	10	0.000000	1.000000

seqinfo: 3 sequences from an unspecified genome; no seqlengths

- nearest: Performs conventional nearest neighbor finding. Returns an integer vector containing the index of the nearest neighbor range in subject for each range in x. If there is no nearest neighbor NA is returned. For details of the algorithm see the man page in the IRanges package (?nearest).
- precede: For each range in x, precede returns the index of the range in subject that is directly preceded by the range in x. Overlapping ranges are excluded. NA is returned when there are no qualifying ranges in subject.
- follow: The opposite of precede, follow returns the index of the range in subject that is directly followed by the range in x. Overlapping ranges are excluded. NA is returned when there are no qualifying ranges in subject.

Orientation and strand for precede and follow: Orientation is 5' to 3', consistent with the direction of translation. Because positional numbering along a chromosome is from left to right and transcription takes place from 5' to 3', precede and follow can appear to have 'opposite' behavior on the + and - strand. Using positions 5 and 6 as an example, 5 precedes 6 on the + strand but follows 6 on the - strand.

The table below outlines the orientation when ranges on different strands are compared. In general, a feature on * is considered to belong to both strands. The single exception is when both x and subject are * in which case both are treated as +.

	x		subject		orientation
	-----	+	-----	+	-----
a)	+		+		--->
b)	+		-		NA
c)	+		*		--->
d)	-		+		NA
e)	-		-		<---
f)	-		*		<---
g)	*		+		--->

29. Genomic ranges and features

h) * | - | <---
i) * | * | ---> (the only situation where * arbitrarily means +)

```
res <- nearest(g, gr)
res
```

```
[1] 5 4 4
```

Each element of `res` is the index *into* `gr` of the nearest range to the corresponding range in `g`. So `gr[res]` would give you the actual nearest ranges.

While `nearest` and friends give the index of the nearest feature, the distance to the nearest is sometimes also useful to have. The `distanceToNearest` method calculates the nearest feature as well as reporting the distance.

```
res <- distanceToNearest(g, gr)
res
```

Hits object with 3 hits and 1 metadata column:

	queryHits	subjectHits	distance
	<integer>	<integer>	<integer>
[1]	1	5	0
[2]	2	4	0
[3]	3	4	0

queryLength: 3 / subjectLength: 10

This returns a `Hits` object with an extra `distance` column. A distance of 0 means the two ranges overlap or touch; larger values are the gap in base pairs between them.

29.5. Gene models

So far every range has been one we typed by hand. In practice you usually pull ranges from an annotation. A `TxDb` (“transcript database”) package provides a convenient interface to gene models from a variety of sources. The `TxDb.Hsapiens.UCSC.hg38.knownGene` package provides access to the UCSC `knownGene` gene model for the **hg38** build of the human genome. Everything it returns is a `GRanges` (or `GRangesList`), so all the operations from this chapter apply directly.

29. Genomic ranges and features

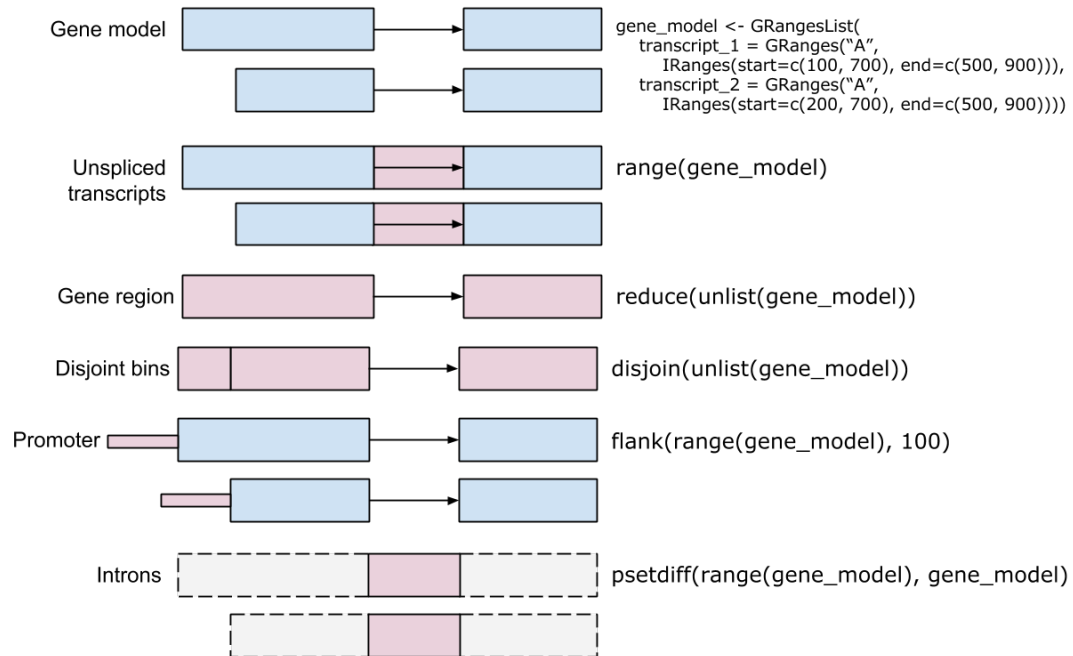


Figure 29.3.: A graphical representation of range operations demonstrated on a gene model.

29. Genomic ranges and features

```
library(TxDb.Hsapiens.UCSC.hg38.knownGene)
txdb <- TxDb.Hsapiens.UCSC.hg38.knownGene
```

The `transcripts` function returns a `GRanges` object with the transcripts for all genes in the database.

```
tx <- transcripts(txdb)
tx
```

`GRanges` object with 412044 ranges and 2 metadata columns:

	seqnames	ranges	strand	tx_id
	<Rle>	<IRanges>	<Rle>	<integer>
[1]	chr1	11121-14413	+	1
[2]	chr1	11125-14405	+	2
[3]	chr1	11410-14413	+	3
[4]	chr1	11411-14413	+	4
[5]	chr1	11426-14409	+	5
...	...	...	...	...
[412040]	chrX_MU273397v1_alt	239036-260095	-	412040
[412041]	chrX_MU273397v1_alt	272358-282686	-	412041
[412042]	chrX_MU273397v1_alt	314193-316302	-	412042
[412043]	chrX_MU273397v1_alt	314813-315236	-	412043
[412044]	chrX_MU273397v1_alt	324527-324923	-	412044

	tx_name
	<character>
[1]	ENST00000832824.1
[2]	ENST00000832825.1
[3]	ENST00000832826.1
[4]	ENST00000832827.1
[5]	ENST00000832828.1
...	...
[412040]	ENST00000710260.1
[412041]	ENST00000710028.1
[412042]	ENST00000710030.1
[412043]	ENST00000710216.1
[412044]	ENST00000710031.1

seqinfo: 711 sequences (1 circular) from hg38 genome

The `exons` function returns a `GRanges` object with the exons.

29. Genomic ranges and features

```
ex <- exons(txdb)
ex
```

GRanges object with 966596 ranges and 1 metadata column:

	seqnames	ranges	strand	exon_id
	<Rle>	<IRanges>	<Rle>	<integer>
[1]	chr1	11121-11211	+	1
[2]	chr1	11125-11211	+	2
[3]	chr1	11410-11671	+	3
[4]	chr1	11411-11671	+	4
[5]	chr1	11426-11671	+	5
...	...	...	...	...
[966592]	chrX_MU273397v1_alt	314193-314248	-	966592
[966593]	chrX_MU273397v1_alt	314813-315236	-	966593
[966594]	chrX_MU273397v1_alt	315258-315407	-	966594
[966595]	chrX_MU273397v1_alt	316254-316302	-	966595
[966596]	chrX_MU273397v1_alt	324527-324923	-	966596

seqinfo: 711 sequences (1 circular) from hg38 genome

The `genes` function returns a GRanges object with one range per gene.

```
gn <- genes(txdb)
gn
```

GRanges object with 35356 ranges and 1 metadata column:

	seqnames	ranges	strand	gene_id
	<Rle>	<IRanges>	<Rle>	<character>
1	chr19	58345178-58362751	-	1
10	chr8	18386273-18401218	+	10
100	chr20	44584896-44652252	-	100
1000	chr18	27932879-28177946	-	1000
100008586	chrX	49551278-49568218	+	100008586
...	...	...	...	...
9990	chr15	34229784-34338060	-	9990
9991	chr9	112215071-112333653	-	9991
9992	chr21	34180729-34371381	+	9992
9993	chr22	19036282-19122454	-	9993
9997	chr22	50523568-50526461	-	9997

```
-----
seqinfo: 711 sequences (1 circular) from hg38 genome
```

Notice the scale: these are real **GRanges** objects with tens of thousands of ranges, yet they print and subset instantly. That is the payoff of the compact representation — the same `findOverlaps()`, `nearest()`, and `subsetByOverlaps()` calls you practiced on the toy **gr** work unchanged here. For example, `subsetByOverlaps(gn, peaks)` would give you the genes hit by a set of ChIP-seq **peaks**, and `nearest(peaks, tx)` would assign each peak to its closest transcript.

💡 Grouping exons by transcript

`exons(txdb)` gives you a flat **GRanges** of every exon. When you instead want the exons *grouped by the transcript they belong to* — a **GRangesList**, one element per transcript — use `exonsBy(txdb, by = "tx")`. That is the natural object for counting reads per transcript and is exactly the **GRangesList** structure we built by hand earlier.

29.6. Summary

You now have a working vocabulary for genomic locations.

- A **GRanges** object holds a set of ranges. Coordinates (`seqnames`, `ranges`, `strand`) sit to the left of the `|`; optional metadata reached with `mcols()` sits to the right. You can build one from scratch or coerce a `data.frame` with `as(df, "GRanges")`.
- **Intra-range** operations (`flank()`, `shift()`, `resize()`) reshape each range on its own and respect strand. **Inter-range** operations (`reduce()`, `disjoin()`, `gaps()`, `coverage()`) summarize a set of ranges together.
- **Between-set** operations answer the integration questions: `findOverlaps()` and `subsetByOverlaps()` for overlap, `nearest()` and `distanceToNearest()` for proximity, plus `union()/intersect()/setdiff()` for set arithmetic.
- A **GRangesList** groups ranges by a parent feature, such as exons within a transcript.
- A **TxDb** package (here [TxDb.Hsapiens.UCSC.hg38.knownGene](#)) hands you real gene, transcript, and exon models as **GRanges**, so every operation above applies to genome-scale annotation with no new syntax.

29.7. Exercises

These reuse the objects built earlier in the chapter: the 10-range `gr`, its 3-range subset `g`, and the `grl` `GRangesList`. Run the chapter's chunks first so those objects exist.

- 1. Read the structure.** Without running any code, which of these belong to the *left* of the `|` in a `GRanges` (the coordinates) and which to the *right* (the metadata): `strand`, `score`, `seqnames`, `GC`, `width`?

i Solution

Left of the `|` (coordinates): `seqnames`, `strand`, and `width` (part of `ranges`). Right of the `|` (metadata): `score` and `GC`. The coordinate columns have dedicated accessors; the metadata columns are reached together with `mcols()`.

- 2. Promoters by hand.** A common task is to grab the region just upstream of a feature's start — a rough promoter. Use `flank()` to get the 50 bases upstream of each range in `g`, then confirm with `width()` that each result is 50 bases wide.

i Solution

```
prom <- flank(g, 50)
prom
```

GRanges object with 3 ranges and 3 metadata columns:

	seqnames	ranges	strand		score	GC	AT
	<Rle>	<IRanges>	<Rle>		<integer>	<numeric>	<numeric>
a	chr1	112-161	-		1	1.000000	0.000000
b	chr2	52-101	+		2	0.888889	0.111111
c	chr2	53-102	+		3	0.777778	0.222222

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
width(prom)
```

```
[1] 50 50 50
```

`flank()` defaults to the upstream side (`start = TRUE`), measured relative to strand. Every flank is exactly 50 bases wide.

3. Collapse overlaps. Use `reduce()` on `gr` to merge overlapping or adjacent ranges into a minimal set, then report how many ranges remain with `length()`.

i Solution

```
reduced <- reduce(gr)
reduced
```

GRanges object with 7 ranges and 0 metadata columns:

	seqnames	ranges	strand
	<Rle>	<IRanges>	<Rle>
[1]	chr1	106-116	+
[2]	chr1	101-111	-
[3]	chr1	105-115	*
[4]	chr2	102-113	+
[5]	chr2	104-114	*
[6]	chr3	107-118	+
[7]	chr3	109-120	-

seqinfo: 3 sequences from an unspecified genome; no seqlengths

```
length(reduced)
```

```
[1] 7
```

`reduce()` works per chromosome and per strand, merging ranges that overlap or abut into single ranges — the standard way to collapse overlapping exons into non-redundant regions.

4. Overlap counting. Build `q <- gr[1:4]` and `s <- gr[3:8]`, then use `countOverlaps(q, s)` to find how many ranges in `s` each range in `q` overlaps. Which range in `q` has the most overlaps?

i Solution

```
q <- gr[1:4]
s <- gr[3:8]
counts <- countOverlaps(q, s)
counts
```

29. Genomic ranges and features

```
a b c d
1 2 2 2
```

```
which.max(counts)
```

```
b
2
```

`countOverlaps()` returns one integer per range in the query. Remember that strand participates in the overlap test, so a + query range will not overlap a - subject range unless you set `ignore.strand = TRUE`.

5. Nearest with distance. Use `distanceToNearest(g, gr)` and pull out the `distance` metadata column with `mcols(...)$distance`. What does a distance of 0 tell you?

i Solution

```
hits <- distanceToNearest(g, gr)
mcols(hits)$distance
```

```
[1] 0 0 0
```

A distance of 0 means the nearest range overlaps or directly touches the query range — here every range in `g` is also present in `gr`, so each finds itself at distance 0.

30. Ranges Exercises

The two ranges chapters — *Genomic ranges and features* and *Genomic ranges with real ChIP-seq peaks* — walked you through the `GenomicRanges` toolkit on small hand-built objects and a single real peak set. This chapter is their hands-on companion. Here you'll point those same tools at *real* ENCODE data and answer the kinds of questions a genomicist actually asks: How many peaks did this experiment call? How wide are they? Do two replicates agree? Are open-chromatin regions enriched near gene promoters? Is one set of histone marks enriched at CpG islands?

The data comes from the [ENCODE project](#) via the `AnnotationHub` package. ENCODE published these datasets as BED-style files; `AnnotationHub` has packaged them into ready-to-use `GRanges` objects, so you can load one with a single line instead of downloading and parsing a file. (If you ever have a peak file of your own — peak calls, enhancer regions, and so on — you can read it into a `GRanges` the same way the ChIP-seq chapter does, with `rtracklayer::import()`.)

Each exercise below is a short prompt followed by a folded solution. Try the prompt yourself first; then click **Show the answer** to reveal the code *and* its output, with a sentence or two on what to notice.

30.1. What you'll learn

- Query `AnnotationHub` to find and load a published dataset as a `GRanges`.
- Inspect a `GRanges` — its class, its metadata columns, and its per-chromosome peak counts.
- Measure and summarize peak widths, and compute how much of the genome a peak set covers.
- Read and switch the `seqlevelsStyle` of an object (the `chr1` vs `1` naming convention) and recognize the silent bug a mismatch causes.
- Compute overlaps between two range sets with `%over%`, and reshape ranges with `resize()`, `flank()`, `promoters()`, and `reduce()`.
- Quantify how enriched one set of ranges is inside another with `intersect/union`.

30.2. Exercise 1: explore a DNase peak set

DNase-seq finds **open chromatin** — stretches of DNA that are accessible to the DNase enzyme, which is where regulatory machinery tends to sit. In this exercise you'll load one DNase hypersensitivity peak set from the K562 cell line and get to know it as a **GRanges** object.

Find and load the peak set. Use `AnnotationHub` to find the record `goldenpath/hg19/encodeDCC/wgEncodeUwDnaseK562PkRep1.narrowPeak.gz` and load it into a variable named `dnase`.

```
library(AnnotationHub)
# GenomicRanges gives us the GRanges accessors and overlap/resize operators;
# GenomeInfoDb provides the seqlevels()/seqlevelsStyle() machinery used below.
library(GenomicRanges)
library(GenomeInfoDb)
ah <- AnnotationHub()
query(ah, "goldenpath/hg19/encodeDCC/wgEncodeUwDnase/wgEncodeUwDnaseK562PkRep1.narrowPeak.gz")
```

```
AnnotationHub with 1 record
# snapshotDate(): 2026-04-23
# names(): AH26801
# $dataprovder: UCSC
# $species: Homo sapiens
# $rdataclass: GRanges
# $rdatadateadded: 2015-03-11
# $title: wgEncodeUwDnaseK562PkRep1.narrowPeak.gz
# $description: narrowPeak file from ENCODE
# $taxonomyid: 9606
# $genome: hg19
# $sourcetype: BED
# $sourceurl: http://hgdownload.cse.ucsc.edu/goldenpath/hg19/encodeDCC/wgEnc...
# $sourcesize: 1734368
# $tags: c("wgEncode", "DnaseSeq", "bam", "K562 cell", "Stam grant")
# retrieve record with 'object[["AH26801"]]'
```

```
# The query above matches exactly one record, so we grab it with [[1]].
dnase <- query(ah, "goldenpath/hg19/encodeDCC/wgEncodeUwDnase/wgEncodeUwDnaseK562PkRep1.narrowPeak.gz")
```

The `query()` call returns a one-row hub listing, and `[[1]]` downloads that one record and hands you the object itself. `AnnotationHub` caches the download, so a second `[[1]]` is instant.

30. Ranges Exercises

What kind of object did you get? Print it and ask for its class.

```
dnase
```

GRanges object with 142986 ranges and 6 metadata columns:

	seqnames	ranges	strand	name	score
	<Rle>	<IRanges>	<Rle>	<character>	<numeric>
[1]	chr1	115601-115750	*	<NA>	0
[2]	chr1	136281-136430	*	<NA>	0
[3]	chr1	138961-139110	*	<NA>	0
[4]	chr1	235041-235190	*	<NA>	0
[5]	chr1	235601-235750	*	<NA>	0
...	...	...	...	...	...
[142982]	chrX	154739261-154739410	*	<NA>	0
[142983]	chrX	154842581-154842730	*	<NA>	0
[142984]	chrX	154871201-154871350	*	<NA>	0
[142985]	chrX	154892021-154892170	*	<NA>	0
[142986]	chrX	154896481-154896630	*	<NA>	0

	signalValue	pValue	qValue	peak
	<numeric>	<numeric>	<numeric>	<numeric>
[1]	121	175.9120	-1	-1
[2]	15	15.9719	-1	-1
[3]	10	5.5961	-1	-1
[4]	7	5.3138	-1	-1
[5]	52	60.7856	-1	-1
...	...	...	...	...
[142982]	9	5.59481	-1	-1
[142983]	30	25.91170	-1	-1
[142984]	13	8.92979	-1	-1
[142985]	14	7.12722	-1	-1
[142986]	23	16.29100	-1	-1

seqinfo: 298 sequences (2 circular) from hg19 genome

```
class(dnase)
```

```
[1] "GRanges"
attr(,"package")
[1] "GenomicRanges"
```

30. Ranges Exercises

It's a **GRanges** — the same class you built by hand in the ranges chapter, only this one was pulled from a public archive. Printing it shows the seqnames, ranges, strand, and a block of metadata columns.

What metadata is attached? The per-range metadata lives in the `mcols`.

```
mcols(dnase)
```

DataFrame with 142986 rows and 6 columns

	name	score	signalValue	pValue	qValue	peak
	<character>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
1	NA	0	121	175.9120	-1	-1
2	NA	0	15	15.9719	-1	-1
3	NA	0	10	5.5961	-1	-1
4	NA	0	7	5.3138	-1	-1
5	NA	0	52	60.7856	-1	-1
...	...	...	...	...	...	...
142982	NA	0	9	5.59481	-1	-1
142983	NA	0	30	25.91170	-1	-1
142984	NA	0	13	8.92979	-1	-1
142985	NA	0	14	7.12722	-1	-1
142986	NA	0	23	16.29100	-1	-1

These columns are the extra fields from the original narrowPeak file — things like the signal value and score for each peak. They ride along with every range so you never lose the link between a location and its data.

How many peaks are on each chromosome? `seqnames()` returns the chromosome of every peak; `table()` tallies them.

```
table(seqnames(dnase))
```

chr1	chr2	chr3
16557	10734	8586
chr4	chr5	chr6
6046	7006	10440
chr7	chr8	chr9
8329	5757	5431
chr10	chr11	chr12

30. Ranges Exercises

6517	7730	7344
chr13	chr14	chr15
2297	3607	4501
chr16	chr17	chr18
4981	6731	2531
chr19	chr20	chr21
6186	3534	1821
chr22	chrX	chrY
2915	3405	0
chrM	chrMT	chr1_gl000191_random
0	0	0
chr1_gl000192_random	chr1_gl383516_fix	chr1_gl383517_fix
0	0	0
chr1_gl383518_alt	chr1_gl383519_alt	chr1_gl383520_alt
0	0	0
chr1_gl949741_fix	chr1_jh636052_fix	chr1_jh636053_fix
0	0	0
chr1_jh636054_fix	chr1_jh806573_fix	chr1_jh806574_fix
0	0	0
chr1_jh806575_fix	chr2_gl383521_alt	chr2_gl383522_alt
0	0	0
chr2_gl582966_alt	chr2_gl877870_fix	chr2_gl877871_fix
0	0	0
chr2_kb663603_fix	chr3_gl383523_fix	chr3_gl383524_fix
0	0	0
chr3_gl383525_fix	chr3_gl383526_alt	chr3_jh159131_fix
0	0	0
chr3_jh159132_fix	chr3_jh636055_alt	chr3_ke332495_fix
0	0	0
chr4_ctg9_hap1	chr4_gl000193_random	chr4_gl000194_random
0	0	0
chr4_gl383527_alt	chr4_gl383528_alt	chr4_gl383529_alt
0	0	0
chr4_gl582967_fix	chr4_gl877872_fix	chr4_ke332496_fix
0	0	0
chr5_gl339449_alt	chr5_gl383530_alt	chr5_gl383531_alt
0	0	0
chr5_gl383532_alt	chr5_gl949742_alt	chr5_jh159133_fix
0	0	0
chr5_ke332497_fix	chr6_apd_hap1	chr6_cox_hap2
0	0	0

30. Ranges Exercises

chr6_dbb_hap3	chr6_gl383533_alt	chr6_jh636056_fix
0	0	0
chr6_jh636057_fix	chr6_jh806576_fix	chr6_kb021644_alt
0	0	0
chr6_kb663604_fix	chr6_ke332498_fix	chr6_mann_hap4
0	0	0
chr6_mcf_hap5	chr6_qbl_hap6	chr6_ssto_hap7
0	0	0
chr7_gl000195_random	chr7_gl383534_alt	chr7_gl582968_fix
0	0	0
chr7_gl582969_fix	chr7_gl582970_fix	chr7_gl582971_fix
0	0	0
chr7_gl582972_fix	chr7_jh159134_fix	chr7_jh636058_fix
0	0	0
chr7_ke332499_fix	chr8_gl000196_random	chr8_gl000197_random
0	0	0
chr8_gl383535_fix	chr8_gl383536_fix	chr8_gl949743_fix
0	0	0
chr8_jh159135_fix	chr8_ke332500_fix	chr9_gl000198_random
0	0	0
chr9_gl000199_random	chr9_gl000200_random	chr9_gl000201_random
0	0	0
chr9_gl339450_fix	chr9_gl383537_fix	chr9_gl383538_fix
0	0	0
chr9_gl383539_alt	chr9_gl383540_alt	chr9_gl383541_alt
0	0	0
chr9_gl383542_alt	chr9_jh636059_fix	chr9_jh806577_fix
0	0	0
chr9_jh806578_fix	chr9_jh806579_fix	chr9_kb663605_fix
0	0	0
chr10_gl383543_fix	chr10_gl383544_fix	chr10_gl383545_alt
0	0	0
chr10_gl383546_alt	chr10_gl877873_fix	chr10_jh591181_fix
0	0	0
chr10_jh591182_fix	chr10_jh591183_fix	chr10_jh636060_fix
0	0	0
chr10_jh806580_fix	chr10_kb663606_fix	chr10_ke332501_fix
0	0	0
chr11_gl000202_random	chr11_gl383547_alt	chr11_gl582973_fix
0	0	0
chr11_gl949744_fix	chr11_jh159136_alt	chr11_jh159137_alt

30. Ranges Exercises

0	0	0
chr11_jh159138_fix	chr11_jh159139_fix	chr11_jh159140_fix
0	0	0
chr11_jh159141_fix	chr11_jh159142_fix	chr11_jh159143_fix
0	0	0
chr11_jh591184_fix	chr11_jh591185_fix	chr11_jh720443_fix
0	0	0
chr11_jh806581_fix	chr12_gl383548_fix	chr12_gl383549_alt
0	0	0
chr12_gl383550_alt	chr12_gl383551_alt	chr12_gl383552_alt
0	0	0
chr12_gl383553_alt	chr12_gl582974_fix	chr12_gl877875_alt
0	0	0
chr12_gl877876_alt	chr12_gl949745_alt	chr12_jh720444_fix
0	0	0
chr12_kb663607_fix	chr13_gl582975_fix	chr14_kb021645_fix
0	0	0
chr15_gl383554_alt	chr15_gl383555_alt	chr15_jh720445_fix
0	0	0
chr16_gl383556_alt	chr16_gl383557_alt	chr16_jh720446_fix
0	0	0
chr17_ctg5_hap1	chr17_gl000203_random	chr17_gl000204_random
0	0	0
chr17_gl000205_random	chr17_gl000206_random	chr17_gl383558_fix
0	0	0
chr17_gl383559_fix	chr17_gl383560_fix	chr17_gl383561_fix
0	0	0
chr17_gl383562_fix	chr17_gl383563_alt	chr17_gl383564_alt
0	0	0
chr17_gl383565_alt	chr17_gl383566_alt	chr17_gl582976_fix
0	0	0
chr17_jh159144_fix	chr17_jh159145_fix	chr17_jh159146_alt
0	0	0
chr17_jh159147_alt	chr17_jh159148_alt	chr17_jh591186_fix
0	0	0
chr17_jh636061_fix	chr17_jh720447_fix	chr17_jh806582_fix
0	0	0
chr17_kb021646_fix	chr17_ke332502_fix	chr18_gl000207_random
0	0	0
chr18_gl383567_alt	chr18_gl383568_alt	chr18_gl383569_alt
0	0	0

30. Ranges Exercises

chr18_gl383570_alt	chr18_gl383571_alt	chr18_gl383572_alt
0	0	0
chr19_gl000208_random	chr19_gl000209_random	chr19_gl383573_alt
0	0	0
chr19_gl383574_alt	chr19_gl383575_alt	chr19_gl383576_alt
0	0	0
chr19_gl582977_fix	chr19_gl949746_alt	chr19_gl949747_alt
0	0	0
chr19_gl949748_alt	chr19_gl949749_alt	chr19_gl949750_alt
0	0	0
chr19_gl949751_alt	chr19_gl949752_alt	chr19_gl949753_alt
0	0	0
chr19_jh159149_fix	chr19_kb021647_fix	chr19_ke332505_fix
0	0	0
chr20_gl383577_alt	chr20_gl582979_fix	chr20_jh720448_fix
0	0	0
chr20_kb663608_fix	chr21_gl000210_random	chr21_gl383578_alt
0	0	0
chr21_gl383579_alt	chr21_gl383580_alt	chr21_gl383581_alt
0	0	0
chr21_ke332506_fix	chr22_gl383582_alt	chr22_gl383583_alt
0	0	0
chr22_jh720449_fix	chr22_jh806583_fix	chr22_jh806584_fix
0	0	0
chr22_jh806585_fix	chr22_jh806586_fix	chr22_kb663609_alt
0	0	0
chrX_gl877877_fix	chrX_jh159150_fix	chrX_jh720451_fix
0	0	0
chrX_jh720452_fix	chrX_jh720453_fix	chrX_jh720454_fix
0	0	0
chrX_jh720455_fix	chrX_jh806587_fix	chrX_jh806588_fix
0	0	0
chrX_jh806589_fix	chrX_jh806590_fix	chrX_jh806591_fix
0	0	0
chrX_jh806592_fix	chrX_jh806593_fix	chrX_jh806594_fix
0	0	0
chrX_jh806595_fix	chrX_jh806596_fix	chrX_jh806597_fix
0	0	0
chrX_jh806598_fix	chrX_jh806599_fix	chrX_jh806600_fix
0	0	0
chrX_jh806601_fix	chrX_jh806602_fix	chrX_jh806603_fix

30. Ranges Exercises

0	0	0
chrX_kb021648_fix	chrUn_gl000211	chrUn_gl000212
0	0	0
chrUn_gl000213	chrUn_gl000214	chrUn_gl000215
0	0	0
chrUn_gl000216	chrUn_gl000217	chrUn_gl000218
0	0	0
chrUn_gl000219	chrUn_gl000220	chrUn_gl000221
0	0	0
chrUn_gl000222	chrUn_gl000223	chrUn_gl000224
0	0	0
chrUn_gl000225	chrUn_gl000226	chrUn_gl000227
0	0	0
chrUn_gl000228	chrUn_gl000229	chrUn_gl000230
0	0	0
chrUn_gl000231	chrUn_gl000232	chrUn_gl000233
0	0	0
chrUn_gl000234	chrUn_gl000235	chrUn_gl000236
0	0	0
chrUn_gl000237	chrUn_gl000238	chrUn_gl000239
0	0	0
chrUn_gl000240	chrUn_gl000241	chrUn_gl000242
0	0	0
chrUn_gl000243	chrUn_gl000244	chrUn_gl000245
0	0	0
chrUn_gl000246	chrUn_gl000247	chrUn_gl000248
0	0	0
chrUn_gl000249		
0		

Each count is the number of DNase peaks called on that chromosome. Bigger chromosomes generally carry more peaks, but accessibility varies by chromosome too.

How wide are the peaks? Summarize the `width()` of every range.

```
summary(width(dnase))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
150.0	150.0	150.0	150.2	150.0	430.0

30. Ranges Exercises

The summary gives you the min, median, mean, and max peak width in base pairs. A tight distribution around a couple hundred bp is typical for a DNase peak caller.

What sequence names were used — chr1 or just 1? Look at the `seqlevels` and the `seqlevelsStyle`, then switch the style to Ensembl and look again.

```
seqlevels(dnase)
```

```
[1] "chr1"          "chr2"          "chr3"
[4] "chr4"          "chr5"          "chr6"
[7] "chr7"          "chr8"          "chr9"
[10] "chr10"         "chr11"         "chr12"
[13] "chr13"         "chr14"         "chr15"
[16] "chr16"         "chr17"         "chr18"
[19] "chr19"         "chr20"         "chr21"
[22] "chr22"         "chrX"          "chrY"
[25] "chrM"          "chrMT"         "chr1_gl000191_random"
[28] "chr1_gl000192_random" "chr1_gl383516_fix" "chr1_gl383517_fix"
[31] "chr1_gl383518_alt"   "chr1_gl383519_alt" "chr1_gl383520_alt"
[34] "chr1_gl949741_fix"   "chr1_jh636052_fix" "chr1_jh636053_fix"
[37] "chr1_jh636054_fix"   "chr1_jh806573_fix" "chr1_jh806574_fix"
[40] "chr1_jh806575_fix"   "chr2_gl383521_alt" "chr2_gl383522_alt"
[43] "chr2_gl582966_alt"   "chr2_gl877870_fix" "chr2_gl877871_fix"
[46] "chr2_kb663603_fix"   "chr3_gl383523_fix" "chr3_gl383524_fix"
[49] "chr3_gl383525_fix"   "chr3_gl383526_alt" "chr3_jh159131_fix"
[52] "chr3_jh159132_fix"   "chr3_jh636055_alt" "chr3_ke332495_fix"
[55] "chr4_ctg9_hap1"      "chr4_gl000193_random" "chr4_gl000194_random"
[58] "chr4_gl383527_alt"   "chr4_gl383528_alt" "chr4_gl383529_alt"
[61] "chr4_gl582967_fix"   "chr4_gl877872_fix" "chr4_ke332496_fix"
[64] "chr5_gl339449_alt"   "chr5_gl383530_alt" "chr5_gl383531_alt"
[67] "chr5_gl383532_alt"   "chr5_gl949742_alt" "chr5_jh159133_fix"
[70] "chr5_ke332497_fix"   "chr6_apd_hap1"     "chr6_cox_hap2"
[73] "chr6_dbb_hap3"       "chr6_gl383533_alt" "chr6_jh636056_fix"
[76] "chr6_jh636057_fix"   "chr6_jh806576_fix" "chr6_kb021644_alt"
[79] "chr6_kb663604_fix"   "chr6_ke332498_fix" "chr6_mann_hap4"
[82] "chr6_mcf_hap5"       "chr6_qbl_hap6"     "chr6_ssto_hap7"
[85] "chr7_gl000195_random" "chr7_gl383534_alt" "chr7_gl582968_fix"
[88] "chr7_gl582969_fix"   "chr7_gl582970_fix" "chr7_gl582971_fix"
[91] "chr7_gl582972_fix"   "chr7_jh159134_fix" "chr7_jh636058_fix"
[94] "chr7_ke332499_fix"   "chr8_gl000196_random" "chr8_gl000197_random"
[97] "chr8_gl383535_fix"   "chr8_gl383536_fix" "chr8_gl949743_fix"
```

30. Ranges Exercises

[100]	"chr8_jh159135_fix"	"chr8_ke332500_fix"	"chr9_gl000198_random"
[103]	"chr9_gl000199_random"	"chr9_gl000200_random"	"chr9_gl000201_random"
[106]	"chr9_gl339450_fix"	"chr9_gl383537_fix"	"chr9_gl383538_fix"
[109]	"chr9_gl383539_alt"	"chr9_gl383540_alt"	"chr9_gl383541_alt"
[112]	"chr9_gl383542_alt"	"chr9_jh636059_fix"	"chr9_jh806577_fix"
[115]	"chr9_jh806578_fix"	"chr9_jh806579_fix"	"chr9_kb663605_fix"
[118]	"chr10_gl383543_fix"	"chr10_gl383544_fix"	"chr10_gl383545_alt"
[121]	"chr10_gl383546_alt"	"chr10_gl877873_fix"	"chr10_jh591181_fix"
[124]	"chr10_jh591182_fix"	"chr10_jh591183_fix"	"chr10_jh636060_fix"
[127]	"chr10_jh806580_fix"	"chr10_kb663606_fix"	"chr10_ke332501_fix"
[130]	"chr11_gl000202_random"	"chr11_gl383547_alt"	"chr11_gl582973_fix"
[133]	"chr11_gl949744_fix"	"chr11_jh159136_alt"	"chr11_jh159137_alt"
[136]	"chr11_jh159138_fix"	"chr11_jh159139_fix"	"chr11_jh159140_fix"
[139]	"chr11_jh159141_fix"	"chr11_jh159142_fix"	"chr11_jh159143_fix"
[142]	"chr11_jh591184_fix"	"chr11_jh591185_fix"	"chr11_jh720443_fix"
[145]	"chr11_jh806581_fix"	"chr12_gl383548_fix"	"chr12_gl383549_alt"
[148]	"chr12_gl383550_alt"	"chr12_gl383551_alt"	"chr12_gl383552_alt"
[151]	"chr12_gl383553_alt"	"chr12_gl582974_fix"	"chr12_gl877875_alt"
[154]	"chr12_gl877876_alt"	"chr12_gl949745_alt"	"chr12_jh720444_fix"
[157]	"chr12_kb663607_fix"	"chr13_gl582975_fix"	"chr14_kb021645_fix"
[160]	"chr15_gl383554_alt"	"chr15_gl383555_alt"	"chr15_jh720445_fix"
[163]	"chr16_gl383556_alt"	"chr16_gl383557_alt"	"chr16_jh720446_fix"
[166]	"chr17_ctg5_hap1"	"chr17_gl000203_random"	"chr17_gl000204_random"
[169]	"chr17_gl000205_random"	"chr17_gl000206_random"	"chr17_gl383558_fix"
[172]	"chr17_gl383559_fix"	"chr17_gl383560_fix"	"chr17_gl383561_fix"
[175]	"chr17_gl383562_fix"	"chr17_gl383563_alt"	"chr17_gl383564_alt"
[178]	"chr17_gl383565_alt"	"chr17_gl383566_alt"	"chr17_gl582976_fix"
[181]	"chr17_jh159144_fix"	"chr17_jh159145_fix"	"chr17_jh159146_alt"
[184]	"chr17_jh159147_alt"	"chr17_jh159148_alt"	"chr17_jh591186_fix"
[187]	"chr17_jh636061_fix"	"chr17_jh720447_fix"	"chr17_jh806582_fix"
[190]	"chr17_kb021646_fix"	"chr17_ke332502_fix"	"chr18_gl000207_random"
[193]	"chr18_gl383567_alt"	"chr18_gl383568_alt"	"chr18_gl383569_alt"
[196]	"chr18_gl383570_alt"	"chr18_gl383571_alt"	"chr18_gl383572_alt"
[199]	"chr19_gl000208_random"	"chr19_gl000209_random"	"chr19_gl383573_alt"
[202]	"chr19_gl383574_alt"	"chr19_gl383575_alt"	"chr19_gl383576_alt"
[205]	"chr19_gl582977_fix"	"chr19_gl949746_alt"	"chr19_gl949747_alt"
[208]	"chr19_gl949748_alt"	"chr19_gl949749_alt"	"chr19_gl949750_alt"
[211]	"chr19_gl949751_alt"	"chr19_gl949752_alt"	"chr19_gl949753_alt"
[214]	"chr19_jh159149_fix"	"chr19_kb021647_fix"	"chr19_ke332505_fix"
[217]	"chr20_gl383577_alt"	"chr20_gl582979_fix"	"chr20_jh720448_fix"
[220]	"chr20_kb663608_fix"	"chr21_gl000210_random"	"chr21_gl383578_alt"

30. Ranges Exercises

```
[223] "chr21_gl1383579_alt" "chr21_gl1383580_alt" "chr21_gl1383581_alt"
[226] "chr21_ke332506_fix" "chr22_gl1383582_alt" "chr22_gl1383583_alt"
[229] "chr22_jh720449_fix" "chr22_jh806583_fix" "chr22_jh806584_fix"
[232] "chr22_jh806585_fix" "chr22_jh806586_fix" "chr22_kb663609_alt"
[235] "chrX_gl1877877_fix" "chrX_jh159150_fix" "chrX_jh720451_fix"
[238] "chrX_jh720452_fix" "chrX_jh720453_fix" "chrX_jh720454_fix"
[241] "chrX_jh720455_fix" "chrX_jh806587_fix" "chrX_jh806588_fix"
[244] "chrX_jh806589_fix" "chrX_jh806590_fix" "chrX_jh806591_fix"
[247] "chrX_jh806592_fix" "chrX_jh806593_fix" "chrX_jh806594_fix"
[250] "chrX_jh806595_fix" "chrX_jh806596_fix" "chrX_jh806597_fix"
[253] "chrX_jh806598_fix" "chrX_jh806599_fix" "chrX_jh806600_fix"
[256] "chrX_jh806601_fix" "chrX_jh806602_fix" "chrX_jh806603_fix"
[259] "chrX_kb021648_fix" "chrUn_gl1000211" "chrUn_gl1000212"
[262] "chrUn_gl1000213" "chrUn_gl1000214" "chrUn_gl1000215"
[265] "chrUn_gl1000216" "chrUn_gl1000217" "chrUn_gl1000218"
[268] "chrUn_gl1000219" "chrUn_gl1000220" "chrUn_gl1000221"
[271] "chrUn_gl1000222" "chrUn_gl1000223" "chrUn_gl1000224"
[274] "chrUn_gl1000225" "chrUn_gl1000226" "chrUn_gl1000227"
[277] "chrUn_gl1000228" "chrUn_gl1000229" "chrUn_gl1000230"
[280] "chrUn_gl1000231" "chrUn_gl1000232" "chrUn_gl1000233"
[283] "chrUn_gl1000234" "chrUn_gl1000235" "chrUn_gl1000236"
[286] "chrUn_gl1000237" "chrUn_gl1000238" "chrUn_gl1000239"
[289] "chrUn_gl1000240" "chrUn_gl1000241" "chrUn_gl1000242"
[292] "chrUn_gl1000243" "chrUn_gl1000244" "chrUn_gl1000245"
[295] "chrUn_gl1000246" "chrUn_gl1000247" "chrUn_gl1000248"
[298] "chrUn_gl1000249"
```

```
seqlevelsStyle(dnase)
```

```
[1] "UCSC"
```

```
seqlevelsStyle(dnase) <- "ensembl"
seqlevelsStyle(dnase)
```

```
[1] "UCSC"
```

```
seqlevels(dnase)
```

30. Ranges Exercises

[1]	"1"	"2"	"3"
[4]	"4"	"5"	"6"
[7]	"7"	"8"	"9"
[10]	"10"	"11"	"12"
[13]	"13"	"14"	"15"
[16]	"16"	"17"	"18"
[19]	"19"	"20"	"21"
[22]	"22"	"X"	"Y"
[25]	"MT"	"chrMT"	"chr1_gl000191_random"
[28]	"chr1_gl000192_random"	"chr1_gl383516_fix"	"chr1_gl383517_fix"
[31]	"chr1_gl383518_alt"	"chr1_gl383519_alt"	"chr1_gl383520_alt"
[34]	"chr1_gl949741_fix"	"chr1_jh636052_fix"	"chr1_jh636053_fix"
[37]	"chr1_jh636054_fix"	"chr1_jh806573_fix"	"chr1_jh806574_fix"
[40]	"chr1_jh806575_fix"	"chr2_gl383521_alt"	"chr2_gl383522_alt"
[43]	"chr2_gl582966_alt"	"chr2_gl877870_fix"	"chr2_gl877871_fix"
[46]	"chr2_kb663603_fix"	"chr3_gl383523_fix"	"chr3_gl383524_fix"
[49]	"chr3_gl383525_fix"	"chr3_gl383526_alt"	"chr3_jh159131_fix"
[52]	"chr3_jh159132_fix"	"chr3_jh636055_alt"	"chr3_ke332495_fix"
[55]	"chr4_ctg9_hap1"	"chr4_gl000193_random"	"chr4_gl000194_random"
[58]	"chr4_gl383527_alt"	"chr4_gl383528_alt"	"chr4_gl383529_alt"
[61]	"chr4_gl582967_fix"	"chr4_gl877872_fix"	"chr4_ke332496_fix"
[64]	"chr5_gl339449_alt"	"chr5_gl383530_alt"	"chr5_gl383531_alt"
[67]	"chr5_gl383532_alt"	"chr5_gl949742_alt"	"chr5_jh159133_fix"
[70]	"chr5_ke332497_fix"	"chr6_apd_hap1"	"chr6_cox_hap2"
[73]	"chr6_dbb_hap3"	"chr6_gl383533_alt"	"chr6_jh636056_fix"
[76]	"chr6_jh636057_fix"	"chr6_jh806576_fix"	"chr6_kb021644_alt"
[79]	"chr6_kb663604_fix"	"chr6_ke332498_fix"	"chr6_mann_hap4"
[82]	"chr6_mcf_hap5"	"chr6_qbl_hap6"	"chr6_ssto_hap7"
[85]	"chr7_gl000195_random"	"chr7_gl383534_alt"	"chr7_gl582968_fix"
[88]	"chr7_gl582969_fix"	"chr7_gl582970_fix"	"chr7_gl582971_fix"
[91]	"chr7_gl582972_fix"	"chr7_jh159134_fix"	"chr7_jh636058_fix"
[94]	"chr7_ke332499_fix"	"chr8_gl000196_random"	"chr8_gl000197_random"
[97]	"chr8_gl383535_fix"	"chr8_gl383536_fix"	"chr8_gl949743_fix"
[100]	"chr8_jh159135_fix"	"chr8_ke332500_fix"	"chr9_gl000198_random"
[103]	"chr9_gl000199_random"	"chr9_gl000200_random"	"chr9_gl000201_random"
[106]	"chr9_gl339450_fix"	"chr9_gl383537_fix"	"chr9_gl383538_fix"
[109]	"chr9_gl383539_alt"	"chr9_gl383540_alt"	"chr9_gl383541_alt"
[112]	"chr9_gl383542_alt"	"chr9_jh636059_fix"	"chr9_jh806577_fix"
[115]	"chr9_jh806578_fix"	"chr9_jh806579_fix"	"chr9_kb663605_fix"
[118]	"chr10_gl383543_fix"	"chr10_gl383544_fix"	"chr10_gl383545_alt"
[121]	"chr10_gl383546_alt"	"chr10_gl877873_fix"	"chr10_jh591181_fix"

30. Ranges Exercises

[124]	"chr10_jh591182_fix"	"chr10_jh591183_fix"	"chr10_jh636060_fix"
[127]	"chr10_jh806580_fix"	"chr10_kb663606_fix"	"chr10_ke332501_fix"
[130]	"chr11_gl000202_random"	"chr11_gl383547_alt"	"chr11_gl582973_fix"
[133]	"chr11_gl949744_fix"	"chr11_jh159136_alt"	"chr11_jh159137_alt"
[136]	"chr11_jh159138_fix"	"chr11_jh159139_fix"	"chr11_jh159140_fix"
[139]	"chr11_jh159141_fix"	"chr11_jh159142_fix"	"chr11_jh159143_fix"
[142]	"chr11_jh591184_fix"	"chr11_jh591185_fix"	"chr11_jh720443_fix"
[145]	"chr11_jh806581_fix"	"chr12_gl383548_fix"	"chr12_gl383549_alt"
[148]	"chr12_gl383550_alt"	"chr12_gl383551_alt"	"chr12_gl383552_alt"
[151]	"chr12_gl383553_alt"	"chr12_gl582974_fix"	"chr12_gl877875_alt"
[154]	"chr12_gl877876_alt"	"chr12_gl949745_alt"	"chr12_jh720444_fix"
[157]	"chr12_kb663607_fix"	"chr13_gl582975_fix"	"chr14_kb021645_fix"
[160]	"chr15_gl383554_alt"	"chr15_gl383555_alt"	"chr15_jh720445_fix"
[163]	"chr16_gl383556_alt"	"chr16_gl383557_alt"	"chr16_jh720446_fix"
[166]	"chr17_ctg5_hap1"	"chr17_gl000203_random"	"chr17_gl000204_random"
[169]	"chr17_gl000205_random"	"chr17_gl000206_random"	"chr17_gl383558_fix"
[172]	"chr17_gl383559_fix"	"chr17_gl383560_fix"	"chr17_gl383561_fix"
[175]	"chr17_gl383562_fix"	"chr17_gl383563_alt"	"chr17_gl383564_alt"
[178]	"chr17_gl383565_alt"	"chr17_gl383566_alt"	"chr17_gl582976_fix"
[181]	"chr17_jh159144_fix"	"chr17_jh159145_fix"	"chr17_jh159146_alt"
[184]	"chr17_jh159147_alt"	"chr17_jh159148_alt"	"chr17_jh591186_fix"
[187]	"chr17_jh636061_fix"	"chr17_jh720447_fix"	"chr17_jh806582_fix"
[190]	"chr17_kb021646_fix"	"chr17_ke332502_fix"	"chr18_gl000207_random"
[193]	"chr18_gl383567_alt"	"chr18_gl383568_alt"	"chr18_gl383569_alt"
[196]	"chr18_gl383570_alt"	"chr18_gl383571_alt"	"chr18_gl383572_alt"
[199]	"chr19_gl000208_random"	"chr19_gl000209_random"	"chr19_gl383573_alt"
[202]	"chr19_gl383574_alt"	"chr19_gl383575_alt"	"chr19_gl383576_alt"
[205]	"chr19_gl582977_fix"	"chr19_gl949746_alt"	"chr19_gl949747_alt"
[208]	"chr19_gl949748_alt"	"chr19_gl949749_alt"	"chr19_gl949750_alt"
[211]	"chr19_gl949751_alt"	"chr19_gl949752_alt"	"chr19_gl949753_alt"
[214]	"chr19_jh159149_fix"	"chr19_kb021647_fix"	"chr19_ke332505_fix"
[217]	"chr20_gl383577_alt"	"chr20_gl582979_fix"	"chr20_jh720448_fix"
[220]	"chr20_kb663608_fix"	"chr21_gl000210_random"	"chr21_gl383578_alt"
[223]	"chr21_gl383579_alt"	"chr21_gl383580_alt"	"chr21_gl383581_alt"
[226]	"chr21_ke332506_fix"	"chr22_gl383582_alt"	"chr22_gl383583_alt"
[229]	"chr22_jh720449_fix"	"chr22_jh806583_fix"	"chr22_jh806584_fix"
[232]	"chr22_jh806585_fix"	"chr22_jh806586_fix"	"chr22_kb663609_alt"
[235]	"chrX_gl877877_fix"	"chrX_jh159150_fix"	"chrX_jh720451_fix"
[238]	"chrX_jh720452_fix"	"chrX_jh720453_fix"	"chrX_jh720454_fix"
[241]	"chrX_jh720455_fix"	"chrX_jh806587_fix"	"chrX_jh806588_fix"
[244]	"chrX_jh806589_fix"	"chrX_jh806590_fix"	"chrX_jh806591_fix"

30. Ranges Exercises

[247] "chrX_jh806592_fix"	"chrX_jh806593_fix"	"chrX_jh806594_fix"
[250] "chrX_jh806595_fix"	"chrX_jh806596_fix"	"chrX_jh806597_fix"
[253] "chrX_jh806598_fix"	"chrX_jh806599_fix"	"chrX_jh806600_fix"
[256] "chrX_jh806601_fix"	"chrX_jh806602_fix"	"chrX_jh806603_fix"
[259] "chrX_kb021648_fix"	"chrUn_gl000211"	"chrUn_gl000212"
[262] "chrUn_gl000213"	"chrUn_gl000214"	"chrUn_gl000215"
[265] "chrUn_gl000216"	"chrUn_gl000217"	"chrUn_gl000218"
[268] "chrUn_gl000219"	"chrUn_gl000220"	"chrUn_gl000221"
[271] "chrUn_gl000222"	"chrUn_gl000223"	"chrUn_gl000224"
[274] "chrUn_gl000225"	"chrUn_gl000226"	"chrUn_gl000227"
[277] "chrUn_gl000228"	"chrUn_gl000229"	"chrUn_gl000230"
[280] "chrUn_gl000231"	"chrUn_gl000232"	"chrUn_gl000233"
[283] "chrUn_gl000234"	"chrUn_gl000235"	"chrUn_gl000236"
[286] "chrUn_gl000237"	"chrUn_gl000238"	"chrUn_gl000239"
[289] "chrUn_gl000240"	"chrUn_gl000241"	"chrUn_gl000242"
[292] "chrUn_gl000243"	"chrUn_gl000244"	"chrUn_gl000245"
[295] "chrUn_gl000246"	"chrUn_gl000247"	"chrUn_gl000248"
[298] "chrUn_gl000249"		

The object started in **UCSC** style (`chr1`, `chr2`, ...). Assigning a new `seqlevelsStyle` rewrites every sequence name in place — after the switch the same chromosomes are now named 1, 2, ... in **Ensembl** style. Nothing about the ranges changed, only the labels.

 The `chr` gotcha that silently returns zero overlaps

`seqlevelsStyle` matters enormously the moment you compare two objects. Overlap operations match on the *exact* sequence name, so a peak on `chr1` will **never** overlap a region on 1 — R reports zero overlaps and raises no error. If an overlap count comes back suspiciously at zero, the first thing to check is whether both objects share the same `seqlevelsStyle`. We switched to Ensembl here to practice; in Exercise 3 we'll switch back before comparing against UCSC-style gene models.

How much of the genome do the peaks cover? Assuming the peaks don't overlap, sum their widths and divide by the total genome length.

```
sum(width(dnase))
```

```
[1] 21472260
```

```
sum(seqlengths(dnase))
```

```
[1] 3234851260
```

```
sum(width(dnase)) / sum(seqlengths(dnase))
```

```
[1] 0.006637789
```

The first number is the total base pairs of open chromatin called; the second is the size of the genome. Their ratio — a small fraction of a percent — tells you that accessible chromatin is a tiny slice of the whole genome, which is exactly why these regions are interesting.

30.3. Exercise 2: compare two replicates

A good experiment is reproducible. ENCODE ran the DNase assay twice on K562, so we have a second replicate. In this exercise you'll load it and ask how well the two replicates agree.

Load the second replicate (`...wgEncodeUwDnaseK562PkRep2.narrowPeak.gz`) as `dnase2`.

```
query(ah, "goldenpath/hg19/encodeDCC/wgEncodeUwDnase/wgEncodeUwDnaseK562PkRep2.narrowPeak.g
```

```
AnnotationHub with 1 record
# snapshotDate(): 2026-04-23
# names(): AH26802
# $dataprovder: UCSC
# $species: Homo sapiens
# $rdataclass: GRanges
# $rdatadateadded: 2015-03-11
# $title: wgEncodeUwDnaseK562PkRep2.narrowPeak.gz
# $description: narrowPeak file from ENCODE
# $taxonomyid: 9606
# $genome: hg19
# $sourcetype: BED
# $sourceurl: http://hgdownload.cse.ucsc.edu/goldenpath/hg19/encodeDCC/wgEnc...
# $sourcesize: 1715821
# $tags: c("wgEncode", "DnaseSeq", "bai", "K562 cell", "Stam grant")
# retrieve record with 'object[["AH26802"]]'
```

30. Ranges Exercises

```
# Again, a single matching record - grab it with [[1]].
dnase2 <- query(ah, "goldenpath/hg19/encodeDCC/wgEncodeUwDnase/wgEncodeUwDnaseK562PkRep2.na
```

How many peaks are in each replicate? Compare the lengths.

```
length(dnase)
```

```
[1] 142986
```

```
length(dnase2)
```

```
[1] 142811
```

The two replicates call a broadly similar number of peaks. A wildly different count between replicates would be a red flag that something went wrong in one of the experiments.

How wide are the dnase2 peaks?

```
summary(width(dnase2))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
150.0	150.0	150.0	150.2	150.0	410.0

The width distribution looks much like the first replicate's — another reassuring sign of reproducibility.

What proportion of the genome does dnase2 cover? Same calculation as before.

```
sum(width(dnase2)) / sum(seqlengths(dnase2))
```

```
[1] 0.006631294
```

A comparable coverage fraction to replicate 1, as you'd hope for two runs of the same assay.

How many dnase peaks overlap a dnase2 peak? The `%over%` operator returns a logical vector — TRUE for each `dnase` peak that hits at least one `dnase2` peak — so summing it counts the overlapping peaks.

30. Ranges Exercises

```
sum(dnase %over% dnase2)
```

```
[1] 0
```

A high overlap count is the quantitative version of “the replicates agree”: the same regions of open chromatin show up both times. This is the kind of number you would report when arguing that a peak set is trustworthy.

Expand every peak by 50 bp on each end. Suppose your peak caller was too strict and you want peaks 100 bp wider. Use `resize()` with `fix="center"` so the expansion is symmetric, and call the result `dnase_wide`.

```
w <- width(dnase)
dnase_wide <- resize(dnase, width = w + 100, fix = "center")
width(dnase_wide)
```

```
[1] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[19] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[37] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[55] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[73] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[91] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[109] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[127] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[145] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[163] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[181] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[199] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[217] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[235] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[253] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[271] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[289] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[307] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[325] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[343] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[361] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[379] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
[397] 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250 250
```

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

30. Ranges Exercises

[illegible]

`resize()` sets each range to a new width; `fix="center"` keeps the midpoint fixed so the peak grows 50 bp in each direction. Comparing `width(dnase_wide)` to the original widths confirms every peak is exactly 100 bp larger.

30.4. Exercise 3: DNase sites relative to gene promoters

Open chromatin should cluster near where genes are regulated — especially at **promoters**, the regions just upstream of a gene's start. In this exercise you'll build promoter regions from a gene model and ask whether the DNase sites land there more than chance would predict.

You'll need the `TxDb.Hsapiens.UCSC.hg19.knownGene` transcript database. If it isn't already installed, install it once (this is a one-time setup step, not part of the analysis):

```
BiocManager::install("TxDb.Hsapiens.UCSC.hg19.knownGene")
library("TxDb.Hsapiens.UCSC.hg19.knownGene")
kg <- TxDb.Hsapiens.UCSC.hg19.knownGene
```

Load the genes from the TxDb. Attach the package, grab the TxDb object, and pull out the genes. What class is the result, and how many genes are there?

```
library("TxDb.Hsapiens.UCSC.hg19.knownGene")
kg <- TxDb.Hsapiens.UCSC.hg19.knownGene
gx <- genes(kg)
class(gx)
```

```
[1] "GRanges"
attr(,"package")
[1] "GenomicRanges"
```

```
length(gx)
```

```
[1] 28622
```

`genes()` returns a **GRanges** — one range per gene, in UCSC (`chr1`) style to match the genome it was built on. The length tells you how many genes the annotation contains.

Get promoter regions with `flank()`. Read `?flank`. The promoter sits *upstream* of a gene, so flanking each gene by 2 kb on its upstream side gives a first approximation.

30. Ranges Exercises

```
flank(gx, 2000)
```

GRanges object with 28622 ranges and 1 metadata column:

	seqnames	ranges	strand	gene_id
	<Rle>	<IRanges>	<Rle>	<character>
1	chr19	58874118-58876117	-	1
10	chr8	18246792-18248791	+	10
100	chr20	43280894-43282893	-	100
1000	chr18	25757911-25759910	-	1000
10000	chr1	244014382-244016381	-	10000
...	...	...	...	...
9991	chr9	115095934-115097933	-	9991
9992	chr21	35551030-35553029	+	9992
9993	chr22	19109968-19111967	-	9993
9994	chr6	90537613-90539612	+	9994
9997	chr22	50964891-50966890	-	9997

seqinfo: 298 sequences (2 circular) from hg19 genome

`flank()` returns a new range immediately upstream of each gene — strand-aware, so “upstream” means the correct direction for genes on either strand. These are 2 kb-wide candidate promoter regions.

Do the same thing with `promoters()`. The TxDb knows where transcripts start, so `promoters()` builds promoter regions directly (see `?promoters` for the default window).

```
proms <- promoters(kg)
```

`promoters()` is the purpose-built tool: it returns a promoter window around each transcript’s start site without you having to reason about strand and flanking yourself.

Do these promoters overlap each other? Genes are dense, and nearby transcripts share promoter neighborhoods, so many of these windows will overlap. `countOverlaps()` of the set against itself counts how many promoters each one touches.

```
summary(countOverlaps(proms))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	2.00	6.00	17.59	16.00	516.00

30. Ranges Exercises

Every promoter overlaps at least itself (count = 1), and the larger counts show that promoters pile up on top of one another — overlapping transcripts and alternative start sites. That redundancy is a problem if we want clean, non-double-counted regions.

Collapse overlapping promoters with `reduce()`. `reduce()` merges every group of overlapping ranges into a single contiguous region, giving us a tidy set of promoter regions to test against.

```
# reduce() takes all overlapping ranges and collapses each cluster
# into one region spanning the whole cluster.
prom_regions <- reduce(proms)

# After reducing, each region overlaps only itself.
summary(countOverlaps(prom_regions))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	2

After `reduce()`, the self-overlap count is 1 everywhere — confirming the regions no longer overlap each other and can be treated as a clean, non-redundant set.

Count DNase sites that overlap a promoter region. Here is where the `seqlevelsStyle` warning above comes due: we switched `dnase` to Ensembl (1) in Exercise 1, but the gene models are UCSC (`chr1`). Watch what happens.

```
# dnase is still in Ensembl style from Exercise 1, while
# prom_regions is UCSC style - so this first count is misleadingly zero.
sum(dnase %over% prom_regions)
```

```
[1] 0
```

```
# Switch dnase back to UCSC so the sequence names match, then recount.
seqlevelsStyle(dnase) <- "UCSC"
sum(dnase %over% prom_regions)
```

```
[1] 0
```

```
sum(dnase2 %over% prom_regions)
```

```
[1] 46357
```

The first count is **zero** — not because the data disagree, but because `chr1` and `1` never match. After switching `dnase` back to UCSC style, the same overlap query returns a large, sensible count. This is the silent bug from the warning, caught and fixed. (`dnase2` was never reassigned, so it was UCSC-style all along.)

Is that overlap surprising? If DNase sites and promoters were placed *independently* on the genome, how many overlaps would you expect by chance? Compare that expectation to what you observed.

```
prop_proms <- sum(width(prom_regions)) / sum(seqlengths(prom_regions))
prop_dnase <- sum(width(dnase)) / sum(seqlengths(prom_regions))
# Under independence, the expected number of DNase peaks landing in a
# promoter is (fraction of genome that is promoter) times the number of peaks.
prop_proms * prop_dnase * length(dnase)
```

```
[1] 91.23822
```

The expected-by-chance number is far smaller than the count you actually observed. That gap is the biology: DNase hypersensitive sites are **not** scattered at random — they concentrate at promoters, exactly where regulatory machinery opens up the chromatin. The overlap count is meaningful precisely because it dwarfs the null expectation.

30.5. Exercise 4: enrichment of histone marks at CpG islands

Histone modifications mark functional regions of the genome: H3K4me3 flags active promoters, H3K4me1 flags enhancers, H3K27me3 flags repressed regions, and so on. In this exercise you'll ask how much each mark **overlaps CpG islands** — and build a small reusable enrichment score along the way.

The data again comes from **AnnotationHub**. ENCODE published these histone ChIP-seq peak calls as BED files; the hub serves them as **GRanges**. We've already found the right hub IDs, so you can load each mark directly.

30. Ranges Exercises

```
library(AnnotationHub)
ah <- AnnotationHub()
```

Here are the `GRanges` for four histone marks plus one extra H3K4me3 replicate:

```
h3k4me1 <- ah[["AH25832"]]
h3k4me3 <- ah[["AH25833"]]
h3k9ac <- ah[["AH25834"]]
h3k27me3 <- ah[["AH25835"]]
h3k4me3_2 <- ah[["AH27284"]]
```

Each variable holds the peak calls from one ChIP-seq experiment pulling down that histone mark. Now grab the hg19 CpG islands the same way:

```
cpg <- query(ah, c("cpg", "UCSC", "hg19"))[[1]]
```

How many H3K4me1 peaks overlap a CpG island? Start with the simple `%over%` count, as in Exercise 2.

```
sum(h3k4me1 %over% cpg)
```

```
[1] 21123
```

That's the number of H3K4me1 peaks touching at least one CpG island — a quick first look at the association.

How many *bases* overlap? A peak count ignores how much of each peak overlaps. `intersect()` returns the overlapping regions themselves, so summing their widths gives the total overlapping base pairs.

```
# intersect() returns only the overlapping stretches as a new range set;
# summing their widths gives the total bases of overlap.
sum(width(intersect(h3k4me1, cpg)))
```

```
[1] 10541975
```

30. Ranges Exercises

This is a finer measure than the peak count: it tells you how many base pairs the two sets actually share, not just how many peaks touch.

Turn that into an enrichment score. Broad marks cover more bases and so overlap CpG islands more just by being big. To correct for that, normalize the shared bases by the *total* bases covered by either set (their *union*). The result is a Jaccard-style ratio between 0 and 1: the fraction of the combined footprint that both sets share.

```
sum(width(union(h3k4me1, cpg)))
```

```
[1] 79640291
```

```
# Normalize the overlap by the combined coverage of both sets.
sum(width(intersect(h3k4me1, cpg))) / sum(width(union(h3k4me1, cpg)))
```

```
[1] 0.1323699
```

A score near 0 means the two sets barely co-occur; a score near 1 means they cover almost exactly the same ground. Because the denominator grows with peak breadth, this score lets you compare a broad mark and a narrow mark on equal footing.

Wrap it in a function so you can score any pair of range sets.

```
range_enrichment_score <- function(r1, r2) {
  i <- sum(width(intersect(r1, r2)))
  u <- sum(width(union(r1, r2)))
  return(i / u)
}
```

The function is just the intersect-over-union calculation with names: *i* for the shared bases, *u* for the combined footprint. Give it a try on H3K4me1 and the CpG islands:

```
range_enrichment_score(h3k4me1, cpg)
```

```
[1] 0.1323699
```

You get back the same normalized score as before. Now you can call `range_enrichment_score()` on any two `GRanges` — try the other histone marks against `cpg` and see which one is most enriched at CpG islands.

30.6. Summary

You’ve now run the core range-analysis workflow end to end on real ENCODE data. You can:

- **Find and load** a published dataset from `AnnotationHub` as a `GRanges`, and inspect its class, metadata, and per-chromosome peak counts.
- **Measure** peak widths and compute the fraction of the genome a peak set covers.
- **Read and switch `seqlevelsStyle`**, and — just as important — recognize the silent zero-overlap bug a `chr1-vs-1` mismatch causes.
- **Compare range sets** with `%over%`, and **reshape** ranges with `resize()`, `flank()`, `promoters()`, and `reduce()`.
- **Quantify enrichment** of one set inside another with an intersect-over-union score.

For the underlying concepts — how `GRanges` objects are built, the full catalog of set operations and overlap methods, and the coordinate-system details — return to the *Genomic ranges and features* chapter. For the real-data import workflow with `rtracklayer`, see *Genomic ranges with real ChIP-seq peaks*. This chapter was the place to put those tools to work.

31. Genomic ranges and ATAC-seq

Where is the DNA in a cell actually *open* for business? Most of the genome is wound tightly around nucleosomes and is functionally silent at any given moment. The stretches a cell is actively using — promoters of active genes, enhancers, transcription-factor binding sites — are comparatively *accessible*. In this chapter you'll take real sequencing data from an experiment that measures exactly that accessibility, read it into Bioconductor, and walk it through the steps that turn a pile of aligned reads into a biological picture: where reads pile up, how long the sequenced fragments are, and how that fragment length reveals the positions of nucleosomes around the start of every gene.

This chapter builds on the genomic-ranges ideas from [the ranges chapter](#) — `GRanges`, coverage, and the run-length encoding that makes genome-wide signals compact. We'll lean on those fundamentals rather than re-teach them, and put them to work on a complete ATAC-seq mini-analysis.

R / *Bioconductor* packages used

- [Rsamtools](#)
- [GenomicRanges](#)
- [GenomicFeatures](#)
- [GenomicAlignments](#)
- [rtracklayer](#)
- [heatmaps](#)

31.1. What you'll learn

After working through this chapter, you'll be able to:

- Read a paired-end ATAC-seq BAM file into Bioconductor with `readGAlignmentPairs()`, filtering reads as you read.
- Summarize reads per chromosome and compute genome-wide **coverage** as a run-length-encoded signal.

- Examine the **fragment-length** distribution and explain what nucleosome-free and mononucleosome fragments tell you about chromatin.
- Split reads by fragment length and build **TSS heatmaps** showing that nucleosome-free reads enrich at transcription start sites.
- Export a coverage track as a BigWig file for viewing in IGV.

31.2. Background

Chromatin accessibility assays measure the extent to which DNA is open and accessible. Such assays now use high throughput sequencing as a quantitative readout. DNase assays, first using microarrays (Crawford, Davis, et al. 2006) and then DNase-Seq (Crawford, Holt, et al. 2006), requires a larger amount of DNA and is labor-intensive and has been largely supplanted by ATAC-Seq (Buenrostro et al. 2013).

The Assay for Transposase Accessible Chromatin with high-throughput sequencing (ATAC-seq) method maps chromatin accessibility genome-wide. This method quantifies DNA accessibility with a hyperactive Tn5 transposase that cuts and inserts sequencing adapters into regions of chromatin that are accessible. High throughput sequencing of fragments produced by the process map to regions of increased accessibility, transcription factor binding sites, and nucleosome positioning. The method is both fast and sensitive and can be used as a replacement for DNase *and* MNase.

An early review of chromatin accessibility assays (Tsompana and Buck 2014) compares the use cases, pros and cons, and expected signals from each of the most common approaches (Figure 31.1).

The first manuscript describing the ATAC-seq protocol showed how ATAC-seq data “line up” with other data types such as ChIP-seq and DNase-seq (Figure 31.2). It also highlighted how fragment length correlates with specific genomic regions and characteristics (Buenrostro et al. 2013, fig. 3).

Buenrostro et al. provide a detailed protocol for performing ATAC-Seq and quality control of results (Buenrostro et al. 2015). Updated and modified protocols that improve on signal-to-noise and reduce input DNA requirements have been described.

31.3. Informatics overview

ATAC-seq protocols typically use paired-end sequencing. The reads are aligned to the relevant *genome* with `bowtie2`, `BWA`, or another short-read aligner. After appropriate ma-

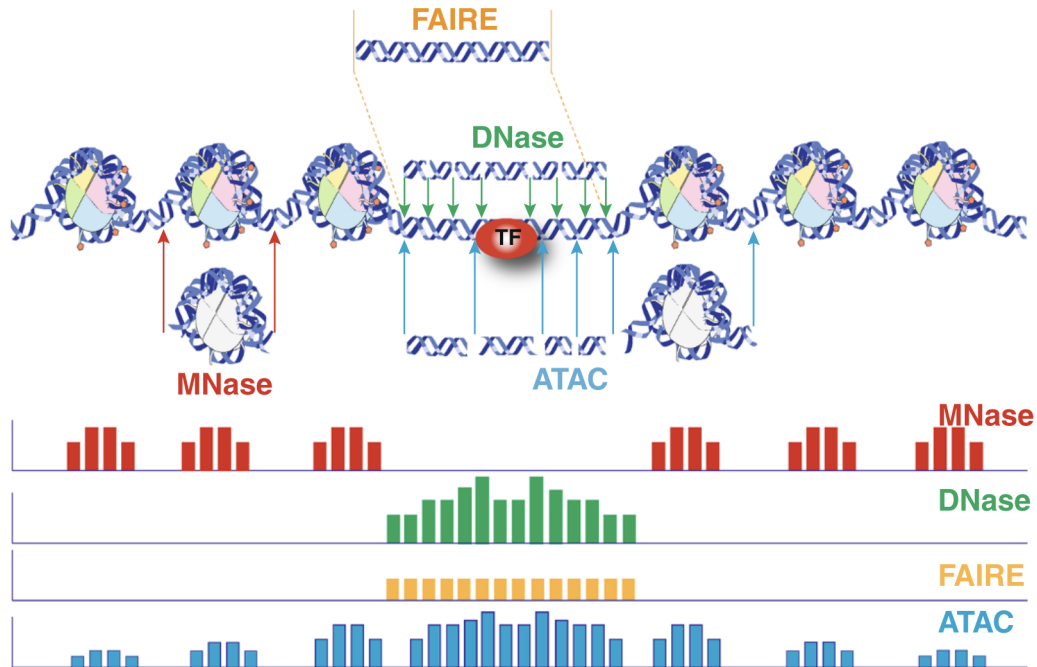


Figure 31.1.: Chromatin accessibility methods, compared. Representative DNA fragments generated by each assay are shown, with end locations within chromatin defined by colored arrows. Bar diagrams represent data signal obtained from each assay across the entire region. The footprint created by a transcription factor (TF) is shown for ATAC-seq and DNase-seq experiments.

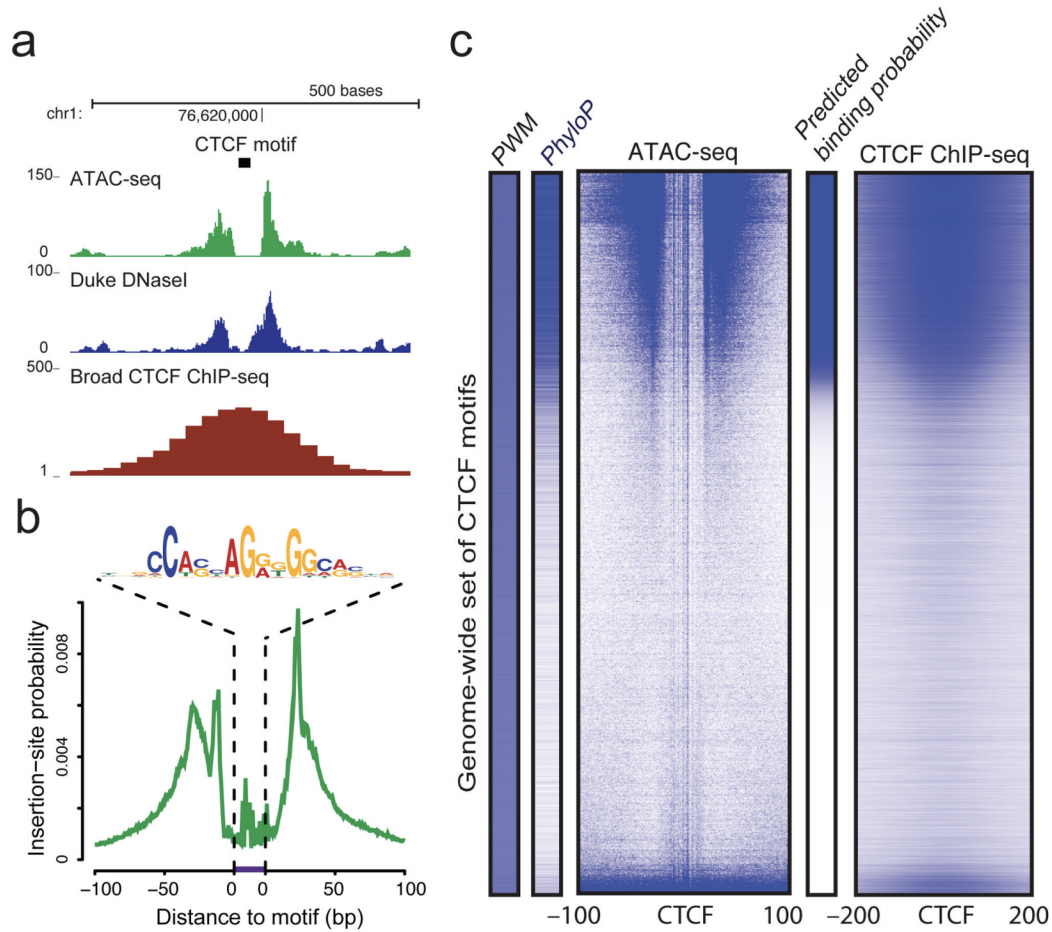


Figure 31.2.: Multimodal chromatin comparisons. From (Buenrostro et al. 2013), Figure 4. (a) CTCF footprints observed in ATAC-seq and DNase-seq data, at a specific locus on chr1. (b) Aggregate ATAC-seq footprint for CTCF (motif shown) generated over binding sites within the genome. (c) CTCF predicted binding probability inferred from ATAC-seq data, position weight matrix (PWM) scores for the CTCF motif, and evolutionary conservation (PhyloP). The right-most column is the CTCF ChIP-seq data (ENCODE) for this GM12878 cell line, demonstrating high concordance with predicted binding probability.

nipulation — often with **samtools** — the result is a BAM file (Figure 31.3). Among other details, the BAM format records, for each read:

```
knitr::include_graphics('imgs/bam_shot.png')
```

NCI-02069526-ML:extdata sdavis2\$ samtools view Sorted_ATAC_21_22.bam head -10							
SRR891269.3150012	99	chr21	9411377	40	50M	=	9411438 111 CTCTGTTCCCTTGCTGACCTC
TTTGTCTATCCTTTTGCTGAGAGGTC			CCCCFFDFHHHHHJJJIIJJJJJJIGJIIFHIJJJJJJIIJJJJJJJGHH				HI:i:1 NH:i:1 NM:i
SRR891269.6181501	99	chr21	9411377	40	50M	=	9411438 111 CTCTGTTCCCTTGCTGACCTC
TTTGTCTATCCTTTTGCTGAGAGGTC			CCCCFFFFHHHHHJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFIIJJHH				HI:i:1 NH:i:1 NM:i
SRR891269.13803406	73	chr21	9411423	0	46M4S	* 0 0	GGTCTGCTTAACCTCCTTT
AGGTAGCTCGATTTTATGCTAAATCT			?81B;D>DFF<+C@E+2AAC<33<CHB<9<+9B#####				HI:i:1 NH:i:1 NM:i
SRR891269.3150012	147	chr21	9411438	20	50M	=	9411377 -111 CTTTTAGTCAGGTAGCTCCA
ATGCTAAGCTTCTTAGTTGCTCACCT			JJJJJJJJJIIIIJJJJIGJJJJIIJJJJIIJJIIHGJGHHHHHFFED?@BB				HI:i:1 NH:i:1 NM:i
SRR891269.6181501	147	chr21	9411438	20	50M	=	9411377 -111 CTTTTAGTCAGGTAGCTCCA
ATGCTAAGCTTCTTAGTTGCTCACCT			JJJIGJHHHHHHFFDDCCC				HI:i:1 NH:i:1 NM:i
SRR891269.52466482	81	chr21	9411533	0	8S42M	=	9411534 59 AAGAGACAGAAATTGCATTG
TACCGGCCCTTTATCAAGCCCTGGCC			JJIJJJJJJJJJJJJGIGGHCJIIHGDJIIJJJJJJJJHHHHHHFDD7FCCC				HI:i:1 NH:i:1 NM:i
SRR891269.52466482	161	chr21	9411534	0	41M9S	=	9411533 -59 AAATTGCATTGTTTCTACCG
TTTATCAAGCCCTGGCCCTGTCTCTT			@BCFFFFFFHHHHHJJJJJJIIJJJJJJIIIGIJJJJJIIJJJJIIHJJJJ				HI:i:1 NH:i:1 NM:i
SRR891269.8220653	163	chr21	9411566	13	50M	=	9411639 123 GCCTGGCCACCATTGATAGT
AATTCCAATTGTTGTCTATGCAGGCC			CCCCFFFHHHHHJJJJJJIIJJJJJJJJJJJJJJJJJJJJJJJJJJJJJE				HI:i:1 NH:i:1 NM:i
SRR891269.8220653	83	chr21	9411639	20	50M	=	9411566 -123 GCTACCATTTTCTTCTTAGC
TGCTCAGCAAATGTATCCAAATGAAA			EJJFHHHHHHFFFFCCC				HI:i:1 NH:i:1 NM:i

Figure 31.3.: A BAM file in text form. The output of `samtools view` is the text representation of a BAM file (called SAM format). Bioconductor and many other tools use BAM files for input. BAM files also often have a companion index `.bai` file that enables random access into the file, so you can read just one genomic region without reading the whole thing.

- sequence name (`chr1`)
- start position (integer)
- a *CIGAR* string that describes the alignment in a compact form
- the sequence to which the pair aligns
- the position to which the pair aligns
- a bit flag field that describes multiple characteristics of the alignment
- the sequence and quality string of the read
- additional tags that tend to be aligner-specific

Duplicate fragments (those with the *same* start and end position of other reads) are marked and likely discarded. Reads that fail to align “properly” are also often excluded from analysis. It is worth noting that most software packages allow simple “marking” of such reads and that there is usually no need to create a special BAM file before proceeding with downstream work.

After alignment and BAM processing, the workflow can switch to *Bioconductor*.

31.4. Working with sequencing data in Bioconductor

The *Bioconductor* project includes several infrastructure packages for dealing with ranges (sequence name, start, end, +/- strand) on sequences (Lawrence, Huber, Pagès, Aboyoun, Carlson, Gentleman, Martin T. Morgan, et al. 2013b) as well as capabilities with working with Fastq files directly (Morgan et al. 2016).

Table 31.1.: Commonly used Bioconductor and their high-level use cases.

Package	Use cases
<i>Rsamtools</i>	low level access to FASTQ, VCF, SAM, BAM, BCF formats
<i>GenomicRanges</i>	Container and methods for handling genomic regions
<i>GenomicFeatures</i>	Work with transcript databases, gff, gtf and BED formats
<i>GenomicAlignments</i>	Reader for BAM format
<i>rtracklayer</i>	import and export multiple UCSC file formats including BigWig and Bed

As noted above, the output of an ATAC-seq experiment is a BAM file. Because paired-end sequencing is the standard for ATAC-seq, `readGAlignmentPairs()` is the right function to read it.

31.5. Data import and quality control

```
library(GenomicAlignments)
```

Reading a paired-end BAM file looks a little involved, but the following code will:

1. Read the included BAM file.
2. Include read pairs only (`isPaired = TRUE`)
3. Include properly paired reads (`isProperPair = TRUE`)
4. Include reads with mapping quality ≥ 1

5. Add a couple of additional fields, `mapq` (mapping quality) and `isize` (insert size) to the default fields.

```
greenleaf <- readGAlignmentPairs(
  "https://github.com/seandavi/RBiocBook/raw/main/atac-seq/extdata/Sorted_ATAC_21_22.bam"
  param = ScanBamParam(
    mapqFilter = 1,
    flag = scanBamFlag(
      isPaired = TRUE,
      isProperPair = TRUE
    ),
    what = c("mapq", "isize")
  )
)
```

The object we just created, `greenleaf`, holds every read pair that passed our filters, with the start and end of each fragment and the two extra columns we asked for. (The [Exercises](#) at the end of the chapter ask you to inspect it more closely.)

Now let's get a first sense of the data. A simple, useful quality-control check is the number of reads mapping to each chromosome.

```
library(ggplot2)
library(dplyr)
chromCounts <- table(seqnames(greenleaf)) %>%
  data.frame() %>%
  dplyr::rename(chromosome = Var1, count = Freq)
```

To keep the example small, this BAM file includes only chromosomes 21 and 22, so [Figure 31.4](#) shows just those two bars. With a full genome you'd see one bar per chromosome, and a chromosome with far fewer (or far more) reads than its neighbors can be an early sign of a problem.

```
ggplot(chromCounts, aes(x = chromosome, y = count)) +
  geom_bar(stat = "identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

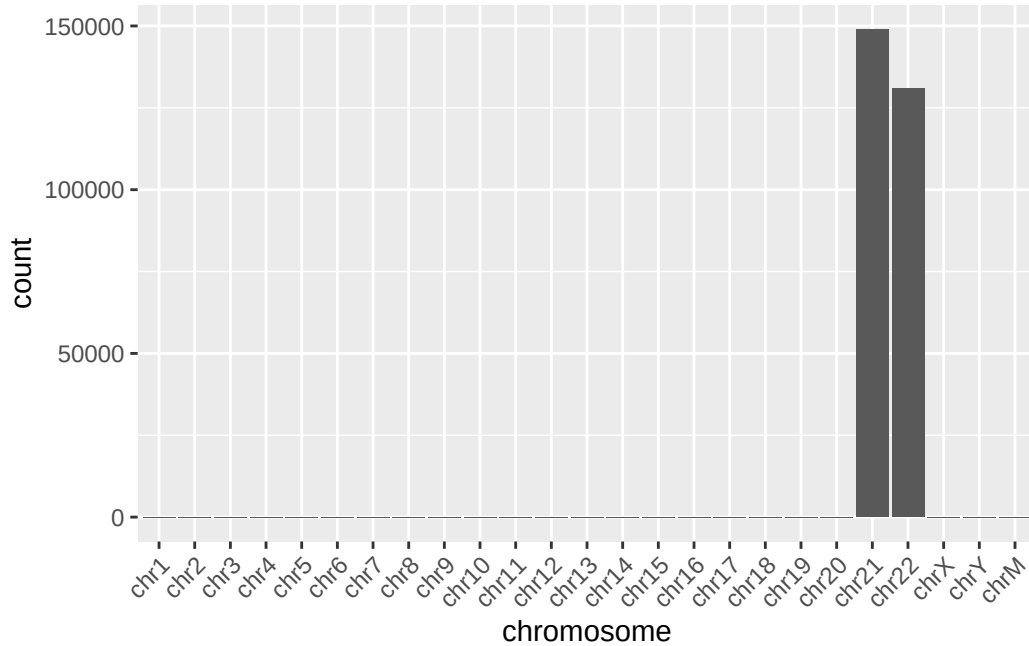


Figure 31.4.: Reads per chromosome. In our example data, only chromosomes 21 and 22 are present.

Raw counts are hard to compare across chromosomes because longer chromosomes naturally collect more reads. Normalizing by chromosome length gives reads per megabase, which should be roughly similar across chromosomes if coverage is even.

```
chromCounts <- chromCounts %>%
  dplyr::mutate(readsPerMb = (count / (seqlengths(greenleaf) / 1e6)))
```

With only two chromosomes the normalized plot (Figure 31.5) is a little underwhelming, but the same idea scales to a whole genome.

```
ggplot(chromCounts, aes(x = chromosome, y = readsPerMb)) +
  geom_bar(stat = "identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  theme_bw()
```

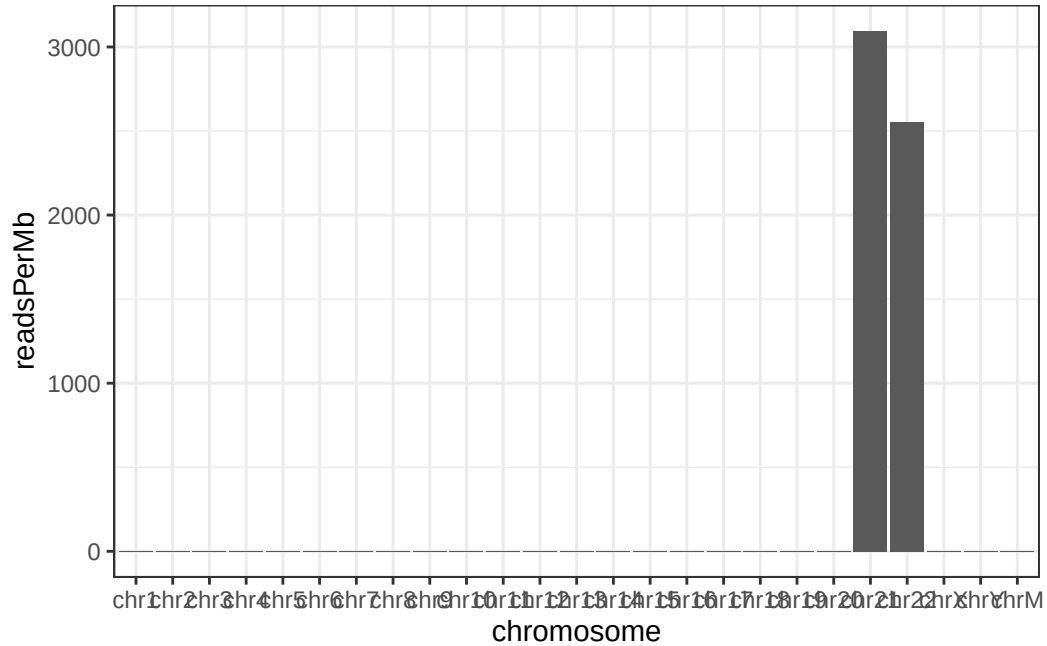


Figure 31.5.: Read counts normalized by chromosome length. Not a critical plot here, but with a full genome it shows each chromosome's relative contribution given its size.

31.6. Coverage

The `coverage()` method for genomic ranges counts, at each base, how many features overlap that position. For our ATAC-seq read pairs, the coverage tells us how reads pile up — and tall piles are the peaks that mark accessible chromatin.

```
cvg <- coverage(greenleaf)
class(cvg)
```

```
[1] "SimpleRleList"
attr(,"package")
[1] "IRanges"
```

Coverage comes back as a `SimpleRleList`. The `names()` function gives us the names of its elements.

```
names(cvg)
```

```
[1] "chr1" "chr2" "chr3" "chr4" "chr5" "chr6" "chr7" "chr8" "chr9"
[10] "chr10" "chr11" "chr12" "chr13" "chr14" "chr15" "chr16" "chr17" "chr18"
[19] "chr19" "chr20" "chr21" "chr22" "chrX" "chrY" "chrM"
```

There is one element per chromosome. Let's look at the chr21 entry:

```
cvg$chr21
```

```
integer-Rle of length 48129895 with 397462 runs
```

```
Lengths: 9411376      50      11      50 ...      36      14      28  10806
Values :          0          2          0          2 ...          1          2          1          0
```

i What is an Rle?

Each chromosome's coverage is stored as an **Rle**, short for **run-length encoding**. Along a chromosome the coverage value stays the same for long stretches — millions of bases in a row might all be 0. Instead of storing each base separately, an **Rle** records (*run length* : *value*) pairs. If the first 9,410,000 bases all have coverage 0, that's stored as a single pair (9,410,000 : 0) rather than nine-million-plus zeros. Across a whole genome this saves an enormous amount of memory, which is why genome-wide signals in Bioconductor are almost always run-length encoded.

The little function `plotCvgHistByChrom()` below plots a histogram of the coverage values seen on a chromosome — how many bases have coverage 0, coverage 1, coverage 2, and so on.

```
library(ggplot2)
plotCvgHistByChrom <- function(cvg, chromosome) {
  cvgcounts <- as.data.frame(table(cvg[[chromosome]]))
  cvgcounts[, 1] <- as.numeric(as.character(cvgcounts[, 1]))
  colnames(cvgcounts) <- c("Coverage", "Count")
  ggplot(cvgcounts, aes(x = Coverage, y = Count)) +
    ggtitle(paste("Chromosome", chromosome)) +
    geom_point(alpha = 0.5) +
    geom_smooth(span = 0.2) +
    scale_y_log10() +
    theme_bw()
}
```

Drawing it for both chromosomes (Figure 31.6):

```
plotCvgHistByChrom(cvg, "chr21")
plotCvgHistByChrom(cvg, "chr22")
```

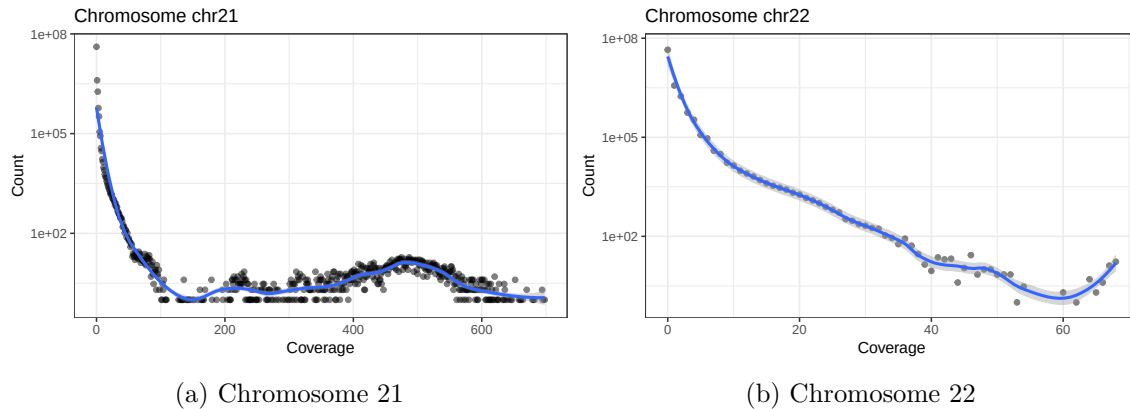


Figure 31.6.: Distribution of per-base coverage on chromosomes 21 and 22 (log-scaled y axis). Most bases have low coverage; the long tail of high-coverage bases corresponds to the peaks where reads pile up.

In Figure 31.6 (note the log-scaled y axis) the vast majority of bases have very low coverage, and the count drops sharply as coverage rises. That long right-hand tail of high-coverage bases is exactly what we care about — those are the accessible regions where reads accumulate into peaks.

31.7. Fragment lengths

The first ATAC-seq manuscript (Buenrostro et al. 2013) highlighted a beautiful relationship between fragment length and nucleosomes (Figure 31.7).

```
knitr::include_graphics("https://cdn.ncbi.nlm.nih.gov/pmc/blobs/0ad9/3959825/fde39a9fb288/n")
```

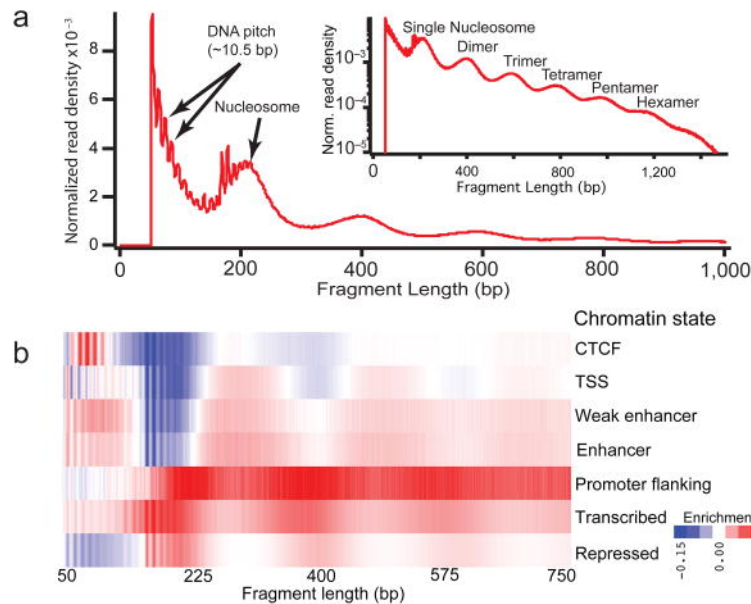


Figure 31.7.: Relationship between fragment length and nucleosome number, from Buenroostro et al. (2013). Short fragments come from nucleosome-free DNA; longer fragments span one or more nucleosomes, producing periodic peaks roughly one nucleosome (~ 147 bp) apart.

! Fragment length is a readout of nucleosome structure

The Tn5 transposase can only cut DNA where it can physically reach it — between nucleosomes, not through them. So the distance between two cut sites (the **fragment length**, also called insert size) tells you how much chromatin sat between them:

- **Nucleosome-free** fragments are short, **shorter than ~ 147 bp** (one nucleosome's worth of DNA). These come from open regions — promoters, enhancers, accessible regulatory DNA.
- **Mononucleosome** fragments are roughly **147–294 bp**: long enough to have wrapped one nucleosome.
- Longer fragments span two or more nucleosomes, giving the periodic ~ 147 bp ladder you see in Figure 31.7.

Because active promoters are nucleosome-free right at the transcription start site (TSS), the short, nucleosome-free reads should pile up *at* the TSS, while mononucleosome reads should flank it. We'll confirm exactly that with a heatmap at the end of the chapter.

We loaded the BAM file with insert sizes (`isize`), so we can read fragment lengths straight off that column. The insert size is the same (in absolute value) for the `first` and `second` read of a pair, so we'll just use `first`.

```
GenomicAlignments::first(greenleaf)
mcols(GenomicAlignments::first(greenleaf))
class(mcols(GenomicAlignments::first(greenleaf)))
head(mcols(GenomicAlignments::first(greenleaf))$isize)
fraglengths <- abs(mcols(GenomicAlignments::first(greenleaf))$isize)
```

Now we can look at the distribution of fragment lengths. The `density()` function gives a smoothed version (Figure 31.8).

```
plot(density(fraglengths, bw = 0.05), xlim = c(0, 1000))
```

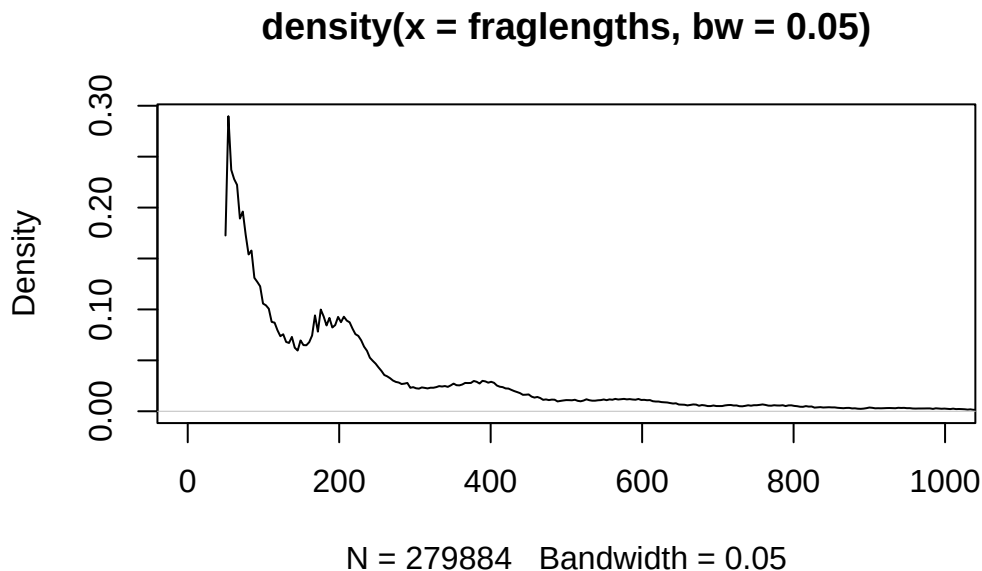


Figure 31.8.: Fragment-length (insert-size) density for our ATAC-seq reads.

A `ggplot2` version of the same distribution (Figure 31.9) makes the periodic structure easier to read:


```
library(dplyr)
library(ggplot2)
fragLenPlot <- table(fraglengths) %>%
  data.frame() |>
  rename(
    InsertSize = fraglengths,
    Count = Freq
  ) |>
  mutate(
    InsertSize = as.numeric(as.vector(InsertSize)),
    Count = as.numeric(as.vector(Count))
  ) |>
  ggplot(aes(x = InsertSize, y = Count)) +
  geom_line()
print(fragLenPlot + theme_bw() + lims(x = c(-1, 500)))
```

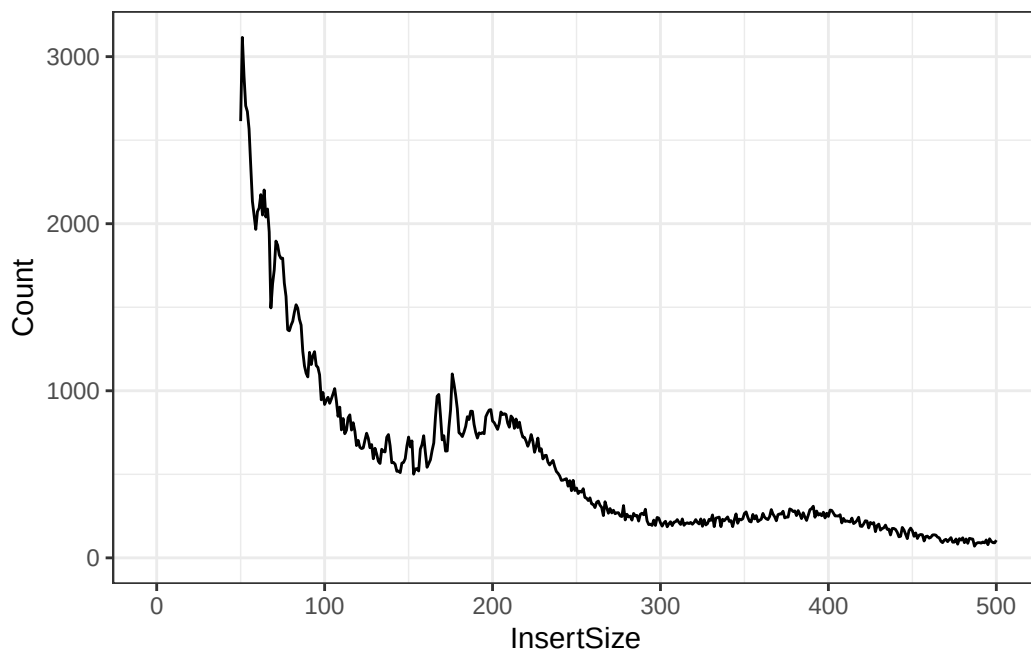


Figure 31.9.: Fragment-length distribution drawn with ggplot2, zoomed to fragments under 500 bp. The large peak of short fragments is the nucleosome-free signal; the bump near ~200 bp is mononucleosome fragments.

Look at the shape: a tall peak at short fragment lengths, then a smaller hump around 200 bp. That first peak is the nucleosome-free DNA, and the hump is the mononucleosome fragments described in the callout above. This bimodal shape is one of the standard quality checks for an ATAC-seq library — a healthy library shows it clearly.

Because we can read each fragment’s length, we can *split* the reads by their biology. First, the nucleosome-free reads, with insert size shorter than one nucleosome:

```
gl_nf <- greenleaf[mcols(GenomicAlignments::first(greenleaf))$isize < 147]
```

And the mononucleosome reads, with insert size between 147 and 294 bp:

```
gl_mn <- greenleaf[mcols(GenomicAlignments::first(greenleaf))$isize > 147 &
  mcols(GenomicAlignments::first(greenleaf))$isize < 294]
```

Now for the payoff. If the biology in the callout holds, the nucleosome-free reads should be enriched right at the TSS, while the mononucleosome reads should not be. We’ll use the [heatmaps](#) package to compare these two sets of reads against the transcription start sites of the human genome.

```
library(TxDb.Hsapiens.UCSC.hg19.knownGene)
proms <- promoters(TxDb.Hsapiens.UCSC.hg19.knownGene, 250, 250)
seqs <- c("chr21", "chr22")
seqlevels(proms, pruning.mode = "coarse") <- seqs # only chromosome 21 and 22
```

Here `proms` is a set of windows reaching 250 bp on each side of every annotated TSS on chromosomes 21 and 22 — one row per promoter for the heatmaps to come. The [heatmaps](#) package vignette describes the package’s full capabilities if you’d like to go further.

We build one heatmap per read set, lining up the coverage over those promoter windows and scaling both to the same range so they’re directly comparable.

```
library(heatmaps)
gl_nf_hm <- CoverageHeatmap(proms, coverage(gl_nf), coords = c(-250, 250))
label(gl_nf_hm) <- "NucFree"
scale(gl_nf_hm) <- c(0, 10)
```

The “meta” plot in Figure 31.10 averages the signal across all promoters, giving one summary curve per read set.

```
gl_mn_hm <- CoverageHeatmap(proms, coverage(gl_mn), coords = c(-250, 250))
label(gl_mn_hm) <- "MonoNuc"
scale(gl_mn_hm) <- c(0, 10)
plotHeatmapMeta(list(gl_nf_hm, gl_mn_hm))
```

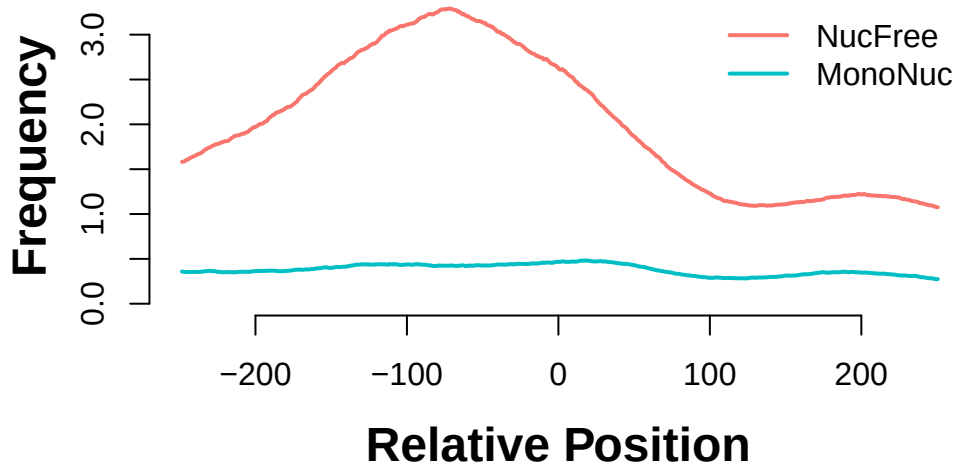


Figure 31.10.: Average signal around the TSS. Nucleosome-free reads peak sharply at position 0 (the TSS); mononucleosome reads stay comparatively flat.

This is the result the callout predicted: the nucleosome-free curve spikes right at the TSS (position 0), while the mononucleosome curve barely moves. The full heatmaps in Figure 31.11 show the same thing promoter-by-promoter — each row is one TSS, and the bright vertical band in the nucleosome-free panel is the accessible signal stacking up at the start of gene after gene.

```
plotHeatmapList(list(gl_mn_hm, gl_nf_hm))
```

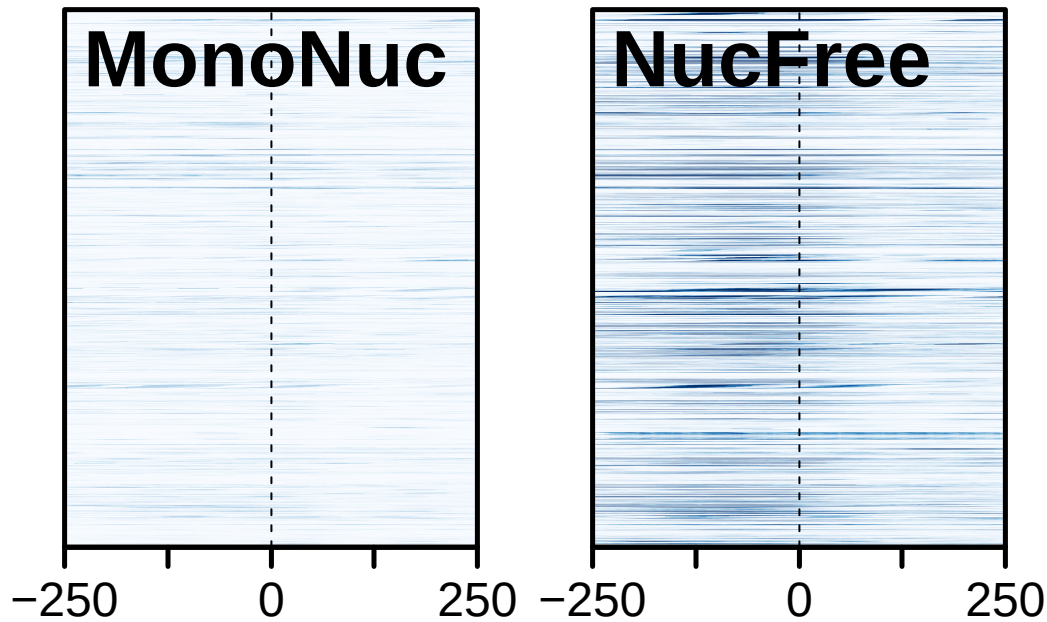


Figure 31.11.: Per-promoter signal at the TSS. Mononucleosome reads on the left, nucleosome-free on the right; both scaled to the same range. Each row is one promoter.

31.8. Viewing data in IGV

A heatmap summarizes thousands of regions at once, but sometimes you want to scroll along the genome and *look* at the pileups. The Integrative Genomics Viewer (IGV) is the standard tool for that. Install it from [the Broad Institute](#).

To load our data into IGV, we export the coverage as a BigWig file — a compact, indexed format IGV reads quickly.

```
library(rtracklayer)
export.bw(coverage(greenleaf), "greenleaf.bw")
```

The [Exercises](#) below ask you to open this track in IGV and explore it.

31.9. Exercises

These exercises build on the `greenleaf` object you read in above. The first three are short coding tasks with solutions; the last two are hands-on exploration in IGV.

1. **What kind of object is `greenleaf`?** Find its class.

i Solution

```
class(greenleaf)

[1] "GAlignmentPairs"
attr("package")
[1] "GenomicAlignments"

readGAlignmentPairs() returns a GAlignmentPairs object — a container that
keeps the two mates of each read pair together.
```

2. **Get the first read of each pair.** Use the `GenomicAlignments::first()` accessor to pull the first read of every pair as a `GAlignments` object, save it as `gl_first_read`, then use `mcols()` to look at its metadata columns.

i Solution

```
gl_first_read <- GenomicAlignments::first(greenleaf)
gl_first_read
```

GAlignments object with 279884 alignments and 2 metadata columns:

	seqnames	strand	cigar	qwidth	start	end	width
	<Rle>	<Rle>	<character>	<integer>	<integer>	<integer>	<integer>
[1]	chr21	+	50M	50	9411377	9411426	50
[2]	chr21	+	50M	50	9411377	9411426	50
[3]	chr21	-	50M	50	9411639	9411688	50
[4]	chr21	-	50M	50	9413594	9413643	50
[5]	chr21	-	50M	50	9413594	9413643	50
...	...	...	...	...	...	...	...
[279880]	chr22	+	50M	50	51238222	51238271	50
[279881]	chr22	+	50M	50	51238594	51238643	50
[279882]	chr22	-	50M	50	51239601	51239650	50
[279883]	chr22	+	50M	50	51240499	51240548	50
[279884]	chr22	+	50M	50	51240499	51240548	50

njunc | mapq isize

```

      <integer> | <integer> <integer>
[1]           0 |          40         111
[2]           0 |          40         111
[3]           0 |          20        -123
[4]           0 |          20        -168
[5]           0 |          20        -168
...           . |          .         .
[279880]        0 |          40         205
[279881]        0 |          13         373
[279882]        0 |          20         -98
[279883]        0 |          13         424
[279884]        0 |          10         424
-----
seqinfo: 25 sequences from an unspecified genome

mcols(gl_first_read)

DataFrame with 279884 rows and 2 columns
      mapq      isize
      <integer> <integer>
1          40         111
2          40         111
3          20        -123
4          20        -168
5          20        -168
...           .         .
279880        40         205
279881        13         373
279882        20         -98
279883        13         424
279884        10         424

The metadata columns hold the extra fields we requested when reading the
BAM — mapq (mapping quality) and isize (insert size, i.e. fragment length).

```

3. **How many read pairs map to each chromosome?** Summarize the counts per chromosome.

i Solution

```
table(seqnames(greenleaf))
```

chr1	chr2	chr3	chr4	chr5	chr6	chr7	chr8	chr9	chr10	chr11
0	0	0	0	0	0	0	0	0	0	0
chr12	chr13	chr14	chr15	chr16	chr17	chr18	chr19	chr20	chr21	chr22
0	0	0	0	0	0	0	0	0	148940	130944
chrX	chrY	chrM								
0	0	0								

This is the same count we plotted in Figure 31.4. Only chromosomes 21 and 22 are present in this trimmed example file.

4. **Explore the full track in IGV.** Open IGV, choose the `hg19` genome, then load the `greenleaf.bw` file you exported. Navigate to chromosomes 21 and 22 and look for the peaks. Do the tallest pileups fall near gene starts?
5. **Compare the nucleosome-free track in IGV.** Export the nucleosome-free reads (`gl_nf`) as their own BigWig file with `export.bw()`, load it alongside the full track, and compare. Where do you expect the strongest signals — and does what you see in IGV match the TSS heatmaps from Figure 31.11?

31.10. Summary

You took a paired-end ATAC-seq BAM file all the way from raw alignments to a biological result. You read and filtered the reads with `readGAlignmentPairs()`, checked reads per chromosome, and computed genome-wide coverage stored compactly as run-length-encoded `Rle` signals. You then used the fragment-length distribution to separate nucleosome-free from mononucleosome reads, and built TSS heatmaps showing that the nucleosome-free reads enrich sharply at transcription start sites — exactly what the biology of open chromatin predicts. Finally, you exported a coverage track for browsing in IGV. The same workflow scales from these two chromosomes to a whole genome.

31.11. Additional work

If you work extensively with ATAC-seq, Thomas Carroll’s workshop is an excellent next step:

MACS2

- https://rockefelleruniversity.github.io/RU_ATAC_Workshop.html

Feel free to work through it. The *ATACseqQC* package vignette also offers a great deal beyond quality control, and several more ATAC-seq packages are available in *Bioconductor*.

MACS2

The MACS2 package is a commonly-used package for calling peaks. Installation and other details are available¹.

```
pip install macs2
```

¹<https://github.com/taoliu/MACS>

32. Single-cell RNA-seq analysis

Bulk RNA-seq hands you one expression profile per sample — the average over every cell in the tube. But a tissue is not one cell type. Mash a piece of brain into a single average and you blur neurons, glia, and blood vessels into one indistinct smear. Single-cell RNA sequencing (scRNA-seq) measures gene expression one cell at a time, so instead of an average you get a catalog: which cell types are present, how many of each, and what makes each one distinct. That shift is what lets you discover a rare cell population, trace a developmental trajectory, or spot the cells that change in disease.

In this chapter you'll take a published mouse-brain dataset from raw counts to a labeled map of cell types, using the Bioconductor single-cell toolkit. Along the way you'll meet the standard workflow — load, normalize, pick informative genes, reduce dimensions, cluster, and find marker genes — that underlies essentially every scRNA-seq analysis.

The data container we'll use, the `SingleCellExperiment`, is built directly on the `SummarizedExperiment` class from the [SummarizedExperiment](#) chapter. If that chapter's three-linked-sheets picture (an assay matrix flanked by aligned row and column metadata) isn't fresh, skim it first — everything here assumes it.

32.1. What you'll learn

By the end of this chapter you will be able to:

- Load a curated scRNA-seq dataset as a `SingleCellExperiment` object.
- Log-normalize raw counts to correct for differences in sequencing depth.
- Select the highly variable genes that carry the biological signal.
- Run PCA and then t-SNE to reduce the data to a 2D map.
- Cluster cells into groups and inspect how those groups line up with known cell types.
- Find and visualize the marker genes that define each cluster.

32.2. Setup and data loading

32.2.1. Loading the libraries

```
# Load essential libraries for single-cell RNA-seq analysis
library(scRNAseq)
library(scran)
library(scater)
```

We start by loading three essential R packages for single-cell analysis:

- **scRNAseq**: Provides access to curated single-cell RNA-seq datasets, making it easy to practice analysis techniques on real data. At the time of writing, it includes more than 50 datasets covering various tissues, organisms, and experimental conditions. See Risso and Cole (2024)
- **scran**: Contains methods for normalization, feature selection, and clustering specifically designed for single-cell data. See Lun et al. (2016).
- **scater**: Offers tools for quality control, visualization, and exploratory data analysis. See McCarthy et al. (2017).

Single-cell data has unique characteristics (sparsity, high dimensionality, technical noise) that require specialized tools. These packages form the backbone of the Bioconductor single-cell analysis ecosystem and provide methods that account for the specific challenges of single-cell data.

32.2.2. Exploring available datasets

We are going to use the **scRNAseq** package to access a curated collection of single-cell datasets. This package provides a convenient way to load and explore publicly available datasets that have been processed and annotated.

```
datasets <- scRNAseq::surveyDatasets()
```

The **surveyDatasets()** function queries the **scRNAseq** package database to show all available datasets. This creates a simple table of publicly available single-cell datasets that have been processed and made easily accessible.

32. Single-cell RNA-seq analysis

Having access to well-curated datasets is useful for learning and benchmarking. These datasets come from published studies and represent different tissues, conditions, and experimental designs, allowing us to practice on real biological data without having to process raw sequencing files.

DataFrame with 10 rows and 15 columns

	name	version	path	object	
	<character>	<character>	<character>	<character>	
1	aztekin-tail-2019	2023-12-14	NA	single_cell_experiment	
2	splicing-demonstrati..	2023-12-20	NA	single_cell_experiment	
3	marques-brain-2016	2023-12-19	NA	single_cell_experiment	
4	grun-bone_marrow-2016	2023-12-14	NA	single_cell_experiment	
5	giladi-hsc-2018	2023-12-21	crispr	single_cell_experiment	
6	giladi-hsc-2018	2023-12-21	rna	single_cell_experiment	
7	macosko-retina-2015	2023-12-19	NA	single_cell_experiment	
8	messmer-esc-2019	2023-12-19	NA	single_cell_experiment	
9	ernst-spermatogenesi..	2023-12-21	cellranger	single_cell_experiment	
10	ernst-spermatogenesi..	2023-12-21	emptydrops	single_cell_experiment	
	title	description	taxonomy_id	genome	rows
	<character>	<character>	<List>	<List>	<integer>
1	Identification of a ..	Identification of a ..	8355	Xenla9.1	31535
2	[reprocessed, subset..	[reprocessed, subset..	10090	GRCm38	54448
3	Oligodendrocyte hete..	Oligodendrocyte hete..	10090	GRCm38	23556
4	De Novo Prediction o..	De Novo Prediction o..	10090	GRCm38	23536
5	Single-cell characte..	Single-cell characte..	10090	MGSCv37	30
6	Single-cell characte..	Single-cell characte..	10090	MGSCv37	27389
7	Highly Parallel Geno..	Highly Parallel Geno..	10090	GRCm38	24658
8	Transcriptional Hete..	Transcriptional Hete..	9606	GRCh38	58302
9	Staged developmental..	Staged developmental..	10090	GRCm38	33226
10	Staged developmental..	Staged developmental..	10090	GRCm38	32105
	columns	assays	column_annotations		
	<integer>	<List>	<List>		
1	13199	counts	sample,DevelopmentalStage,DaysPostAmputation,...		
2	2325	spliced,unspliced	celltype		
3	5069	counts	source_name,age,inferred cell type,...		
4	1915	counts	sample,protocol		
5	24070	counts	amplification.batch		
6	81408	counts	Amp_batch_ID,Seq_batch_ID,tier,...		
7	49300	counts	cluster		
8	1344	counts	Source Name,phenotype,single cell quality,...		
9	53510	counts	Sample,Barcode,Library,...		

32. Single-cell RNA-seq analysis

10	68937	counts	Sample,Barcode,Library,...
	reduced_dimensions	alternative_experiments	
	<List>	<List>	
1	UMAP		
2			
3			
4			
5			
6			
7			
8		ERCC	
9			
10			
		sources	
		<SplitDataFrameList>	
1		ArrayExpress:E-MTAB-7716:NA,PubMed:31097661:NA	
2		GEO:GSE109033:NA,GEO:GSM2928341:NA,SRA:SRR6459157:NA	
3		GEO:GSE75330:NA,PubMed:27284195:NA	
4		GEO:GSE76983:NA,PubMed:27345837:NA	
5		PubMed:29915358:NA,GEO:GSE11349:NA	
6		PubMed:29915358:NA,GEO:GSE92575:NA	
7		GEO:GSE63472:NA,PubMed:26000488:NA,URL:http://mccarrolllab...:2024-02-23	
8		PubMed:30673604:NA,ArrayExpress:E-MTAB-6819:NA	
9		ArrayExpress:E-MTAB-6946:NA,PubMed:30890697:NA	
10		ArrayExpress:E-MTAB-6946:NA,PubMed:30890697:NA	

Each row represents a different study. The ‘columns’ field shows the number of cells, while ‘rows’ shows the number of genes. This helps you choose datasets appropriate for your computational resources and analysis goals. This little lightweight database is not the richest way to explore datasets, but it is a good starting point. In this tutorial, we will use the **ZeiselBrainData** dataset, which contains single-cell RNA-seq data from mouse brain tissue (Zeisel et al. (2015)).

i Abstract: the Zeisel brain dataset

The mammalian cerebral cortex supports cognitive functions such as sensorimotor integration, memory, and social behaviors. Normal brain function relies on a diverse set of differentiated cell types, including neurons, glia, and vasculature. Here, we have used large-scale single-cell RNA sequencing (RNA-seq) to classify cells in the mouse somatosensory cortex and hippocampal CA1 region. We found 47 molecularly

distinct subclasses, comprising all known major cell types in the cortex. We identified numerous marker genes, which allowed alignment with known cell types, morphology, and location. We found a layer I interneuron expressing Pax6 and a distinct postmitotic oligodendrocyte subclass marked by Itpr2. Across the diversity of cortical cell types, transcription factors formed a complex, layered regulatory code, suggesting a mechanism for the maintenance of adult cell type identity.

```
sce <- scRNAseq::ZeiselBrainData()
```

What is the `sce` object? It is a `SingleCellExperiment` object, a data structure designed specifically for single-cell RNA-seq data. It is built directly on top of the `SummarizedExperiment` class you met in the [SummarizedExperiment](#) chapter (see Figure 32.1), so the same mental model applies: an assay matrix of counts, with aligned metadata about the features (genes) down the rows and the cells across the columns. A `SingleCellExperiment` simply adds a few single-cell-specific extras on top, most importantly a slot for dimensionality reductions like PCA and t-SNE.

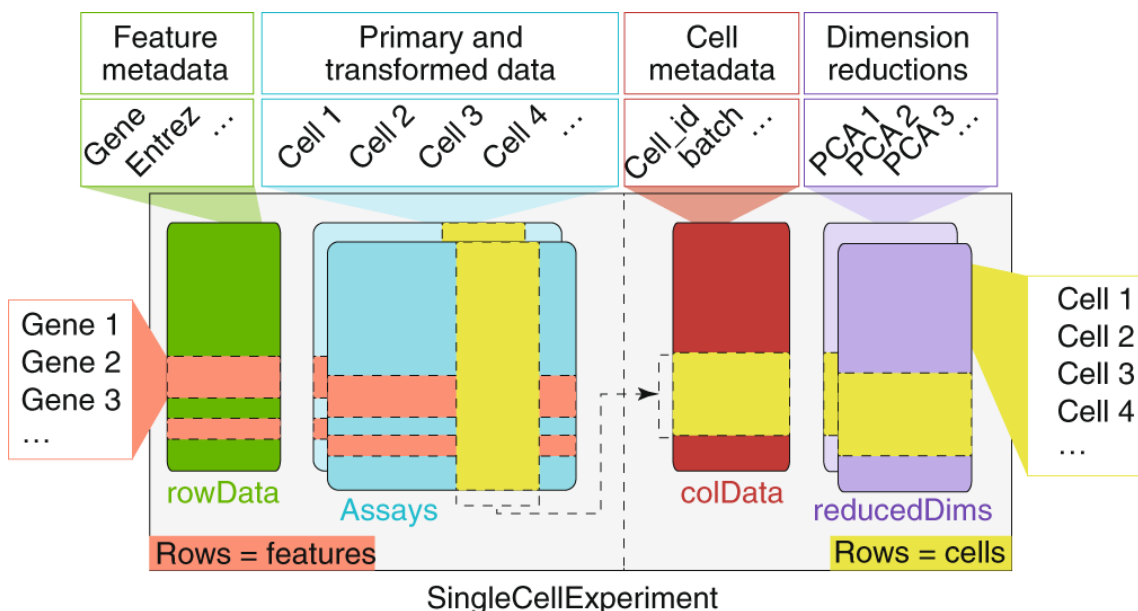


Figure 32.1.: The `SingleCellExperiment` object.

Printing the `sce` object gives us a summary of the dataset:

```
sce
```

```
class: SingleCellExperiment
dim: 20006 3005
metadata(0):
assays(1): counts
rownames(20006): Tspan12 Tshz1 ... mt-Rnr1 mt-Nd41
rowData names(1): featureType
colnames(3005): 1772071015_C02 1772071017_G12 ... 1772066098_A12
               1772058148_F03
colData names(9): tissue group # ... level1class level2class
reducedDimNames(0):
mainExpName: gene
altExpNames(2): repeat ERCC
```

The `SingleCellExperiment` object contains:

- **Assays:** The main data matrix (counts) and any additional matrices (e.g., normalized counts).
- **Row metadata:** Information about genes (e.g., gene names, IDs).
- **Column metadata:** Information about cells (e.g., cell barcodes, tissue type, experimental conditions).

It **can** also contain additional information such as dimensionality reductions (e.g., PCA, t-SNE), clustering results, and quality control metrics.

In fact, we are going to add some of this additional information in the next steps. First, we add a new assay with the log-normalized counts, which is a common preprocessing step in single-cell RNA-seq analysis.

```
sce <- logNormCounts(sce)
```

```
Warning in .library_size_factors(assay(x, assay.type), ...): 'librarySizeFactors' is deprecated
Use 'scrapper::centerSizeFactors' instead.
See help("Deprecated")
```

```
Warning in .local(x, ...): 'normalizeCounts' is deprecated.
Use 'scrapper::normalizeCounts' instead.
See help("Deprecated")
```

32. Single-cell RNA-seq analysis

The `logNormCounts()` function performs log-normalization of the count data. Raw count data from scRNA-seq experiments contains technical variation due to differences in sequencing depth between cells. Some cells might have been sequenced more deeply than others, leading to higher total counts that don't reflect true biological differences. Log-normalization addresses this by:

1. **Size factor normalization:** Scaling each cell's counts by a size factor that accounts for differences in sequencing depth. There are several methods to calculate size factors, but the most common is the median of ratios method, which divides each cell's counts by the median count for each gene across all cells. This helps to adjust for differences in library size.
2. **Log transformation:** Taking $\log(\text{count} + 1)$ to stabilize variance and make the data more normally distributed

For each gene g in cell i , the log-normalized expression becomes:

$$\log_2(\text{count}_{g_i} \times \text{sizefactor}_i + 1)$$

This log transformation is useful because the next steps in the downstream analyses (like clustering and dimensionality reduction) assume that the data is approximately normally distributed (or at least bell-shaped). Log-normalization helps achieve this by reducing the impact of outliers and making the data more suitable for PCA and visualization.

32.3. Dimensionality reduction

Single-cell datasets are extremely high-dimensional - typically containing 15,000-30,000 genes per cell. This creates several challenges: computational complexity, noise dominance, and visualization difficulties. Dimensionality reduction techniques help us identify the most informative patterns in the data while reducing noise and computational burden.

32.3.1. Feature selection: identifying highly variable genes

Before applying dimensionality reduction, we need to identify the most informative genes. Highly variable genes (HVGs) are those that show significant variation across cells, indicating they may be biologically relevant. These genes are often more informative for downstream analyses like clustering and visualization.

Not all genes are equally informative for distinguishing cell types. Genes that show little variation across cells (like housekeeping genes) don't help us identify distinct cell populations. Highly variable genes are more likely to:

32. Single-cell RNA-seq analysis

- Reflect biological differences between cell types
- Capture developmental or activation states
- Represent responses to environmental conditions

```
library(scran)
top_var_genes <- getTopHVGs(sce, n=2000)
```

Warning in getTopHVGs(sce, n = 2000): 'getTopHVGs' is deprecated.
Use 'scrapper::chooseHighlyVariableGenes' instead.
See help("Deprecated")

Warning in fitTrendVar(fm, fv, ...): 'fitTrendVar' is deprecated.
Use 'scrapper::fitVarianceTrend' instead.
See help("Deprecated")

Warning in combineBlocks(collected, method = method, equiweight = equiweight, : 'combineBlo
See help("Deprecated")

The `getTopHVGs()` function identifies the top 2000 highly variable genes based on their variance across cells. Choosing a subset of highly variable genes reduces the dimensionality of the dataset while retaining the most informative features for downstream analyses.

The algorithm models the relationship between gene expression mean and variance, then identifies genes that show more variation than expected based on their expression level. This accounts for the fact that highly expressed genes naturally show more variance.

```
# Visualizing the highly variable genes
library(scran)
dec <- modelGeneVar(sce)
```

Warning in fitTrendVar(fm, fv, ...): 'fitTrendVar' is deprecated.
Use 'scrapper::fitVarianceTrend' instead.
See help("Deprecated")

Warning in combineBlocks(collected, method = method, equiweight = equiweight, : 'combineBlo
See help("Deprecated")

32. Single-cell RNA-seq analysis

```
# Visualizing the fit:
fit.mv <- metadata(dec)
plot(fit.mv$mean, fit.mv$var, xlab="Mean of log-expression",
     ylab="Variance of log-expression", ylim=c(0, 4))
curve(fit.mv$trend(x), col="dodgerblue", add=TRUE, lwd=2)
points(
  fit.mv$mean[top_var_genes], fit.mv$var[top_var_genes],
  col="red", pch=16, cex=0.5
)
```

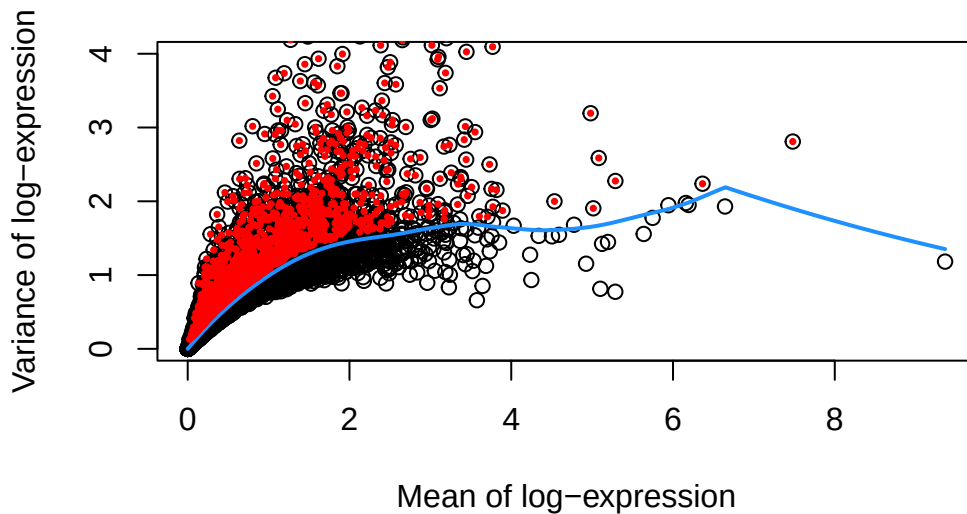


Figure 32.2.: A plot of the mean vs variance of log-expression for all the genes in the dataset, with the highly variable genes highlighted in red. Here, the x-axis represents the mean log-expression of each gene, and the y-axis represents the variance of log-expression. The blue curve represents the trend of variance as a function of mean expression, while the red points represent the highly variable genes.

Figure 32.2 shows the mean vs. variance of log-expression for all genes in the dataset, with highly variable genes highlighted in red. The blue curve represents the trend of variance as a function of mean expression, while the red points represent the highly variable genes.

32. Single-cell RNA-seq analysis

We next proceed to calculating PCA using the subset of highly variable genes. PCA is a linear dimensionality reduction technique that finds the directions of maximum variance in the data. It's computationally efficient and provides a good foundation for further analysis. The first few principal components capture the most important patterns of variation in the dataset.

💡 Why PCA first?

PCA is a widely *linear* dimensionality reduction technique that:

- Reduces dimensionality while preserving variance
- Helps visualize high-dimensional data in 2D or 3D
- Serves as a preprocessing step for other methods like t-SNE or UMAP
- Is computationally efficient and easy to interpret

```
# Perform PCA on the highly variable genes
set.seed(100) # See below.
sce <- fixedPCA(sce, subset.row=top_var_genes)
```

```
Warning in fixedPCA(sce, subset.row = top_var_genes): 'fixedPCA' is deprecated.
Use 'scraper::runPca.se' instead.
See help("Deprecated")
```

```
reducedDimNames(sce)
```

```
[1] "PCA"
```

Note that we assign the output of `fixedPCA()` to the `sce` object. This function performs PCA on the highly variable genes we identified earlier, reducing the dimensionality of the dataset while retaining the most informative features. Because dimensionality reduction is a common step in single-cell RNA-seq analysis, we can store the PCA results in the `reducedDims` slot of the `SingleCellExperiment` object. This allows us to easily access and visualize the PCA coordinates later.

32.3.2. Non-linear dimensionality reduction with t-SNE

After PCA, we can apply non-linear dimensionality reduction techniques like t-SNE (t-distributed Stochastic Neighbor Embedding) to visualize the data in 2D or 3D. t-SNE is particularly useful for revealing local structure and separating distinct cell populations.

💡 Why t-SNE after PCA

This is a two-step approach:

1. **PCA first:** reduces noise and computational burden while preserving global structure.
2. **t-SNE second:** reveals local neighborhood structure and separates distinct cell populations.

t-SNE involves a random initialization, so its output changes from run to run. We call `set.seed()` first so that the figure and the clusters below are reproducible — you'll see exactly the same map we do.

```
library(scater)
set.seed(1234)
sce <- runTSNE(sce, dimred="PCA")
```

The `runTSNE()` function applies t-SNE to the PCA results, creating a 2D representation of the data. The `dimred="PCA"` argument specifies that we want to use the PCA coordinates as input for t-SNE.

Graph-based clustering is also stochastic, so we set a seed before clustering too.

```
set.seed(1234)
nn.clusters <- clusterCells(sce, use.dimred = "TSNE")
```

Warning in `clusterCells(sce, use.dimred = "TSNE")`: 'clusterCells' is deprecated.
Use 'bluster::clusterRows' instead.
See `help("Deprecated")`

```
table(nn.clusters)
```

```
nn.clusters
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20
89 296 153 40 102 192 63 69 143 96 314 133 97 71 91 70 69 112 47 91
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38
101 63 54 36 62 34 42 59 32 39 20 28 15 18 16 19 17 12
```

32. Single-cell RNA-seq analysis

The `clusterCells()` function uses graph-based clustering to identify groups of similar cells. Graph-based clustering was [popularized by the Seurat package](#). It builds a nearest-neighbor graph and then finds communities within this graph. The resulting clusters often correspond to distinct cell types or states. The `use.dimred = "TSNE"` argument specifies that we want to use the t-SNE coordinates for clustering. This is an important choice because t-SNE captures local structure and can reveal distinct cell populations that may not be apparent in the original high-dimensional space. There are other clustering methods available, such as k-means or hierarchical clustering, but graph-based clustering is particularly effective for single-cell data due to its ability to handle complex, non-linear relationships.

The table above is your first concrete result: each entry is the number of cells assigned to a cluster, so you can read off how many groups the algorithm found and roughly how balanced they are. A handful of large clusters plus several small ones is typical — the large ones are usually abundant cell types, the small ones rare populations.

32.3.3. Visualizing cell types and clusters

In this particular dataset, we have pre-defined cell type annotations that we can use to color the t-SNE plot. The `level1class` and `level2class` columns in the metadata contain these annotations. We can use these annotations to color the t-SNE plot and visualize how well the clustering aligns with known cell types.

We can cluster the cells based on their t-SNE coordinates and visualize the results. The `plotReducedDim()` function from the `scater` package allows us to create a scatter plot of the t-SNE coordinates, coloring the points by their cluster assignments or cell type annotations.

```
library(scater)
colLabels(sce) <- nn.clusters
plotReducedDim(sce, "TSNE", colour_by="level1class")
```

32. Single-cell RNA-seq analysis

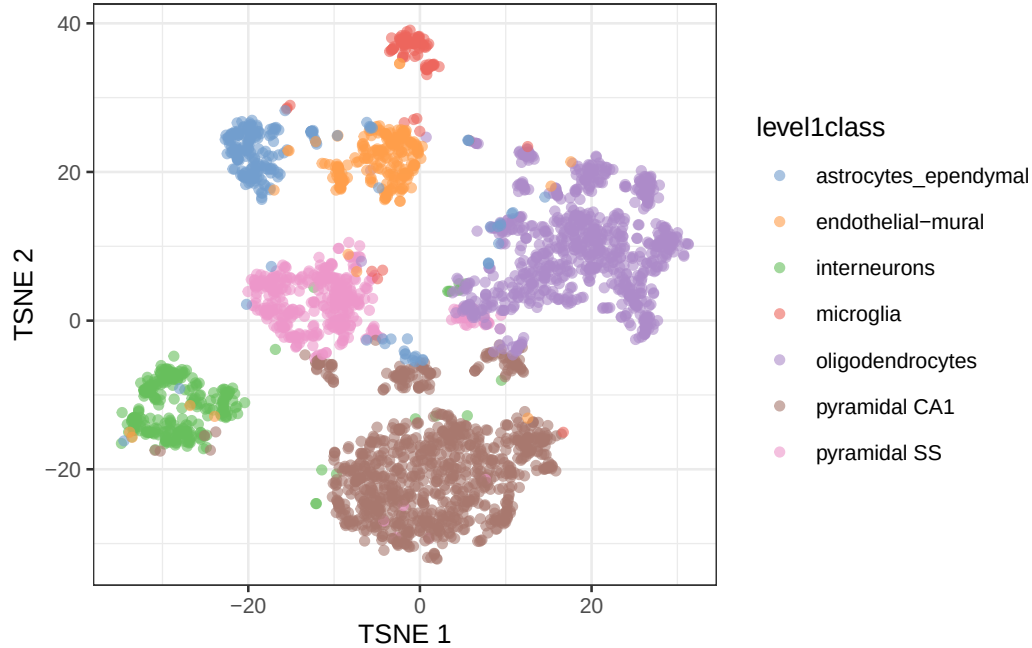


Figure 32.3.

Figure 32.3 is the payoff of the whole workflow so far. Each point is a cell, placed by t-SNE so that similar cells sit close together, and colored by its known `level1class` cell type. Because cells of the same type fall into the same visual island, you can see at a glance that the unsupervised steps — normalization, HVG selection, PCA, t-SNE — recovered real biology: the broad cell types separate cleanly without ever being told what they were.

We can also visualize the second level of cell type annotations.

```
plotReducedDim(sce, "TSNE", colour_by="level2class")
```

32. Single-cell RNA-seq analysis

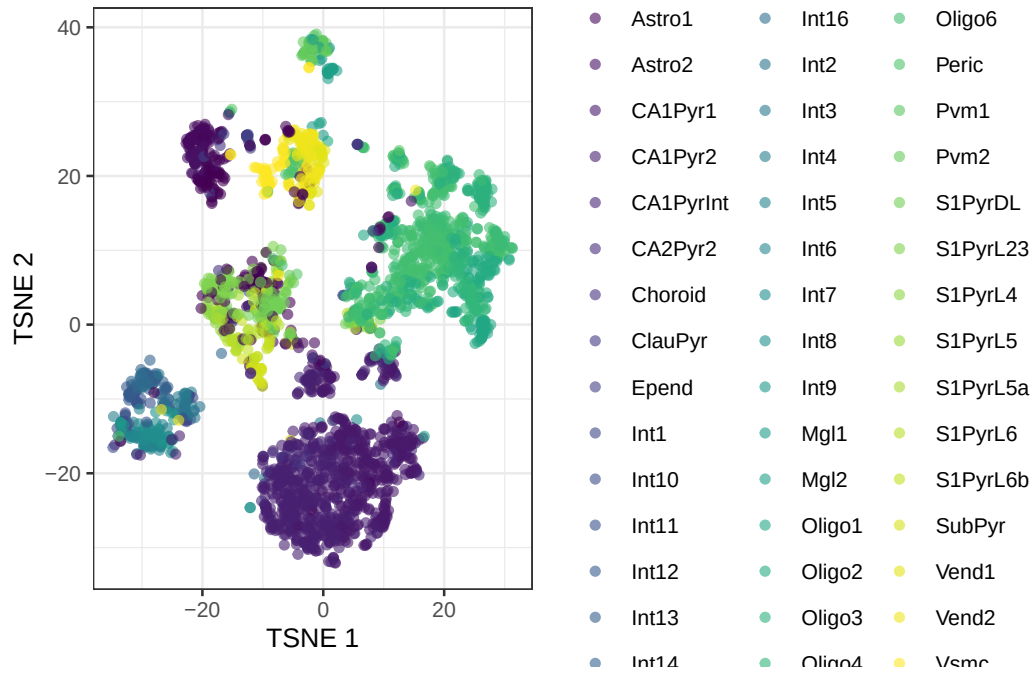


Figure 32.4.

And add additional metadata such as tissue type to the plot. The `shape_by` argument allows us to differentiate points by another categorical variable, such as tissue type.

```
plotReducedDim(sce, "TSNE", colour_by="level1class", shape_by="tissue")
```

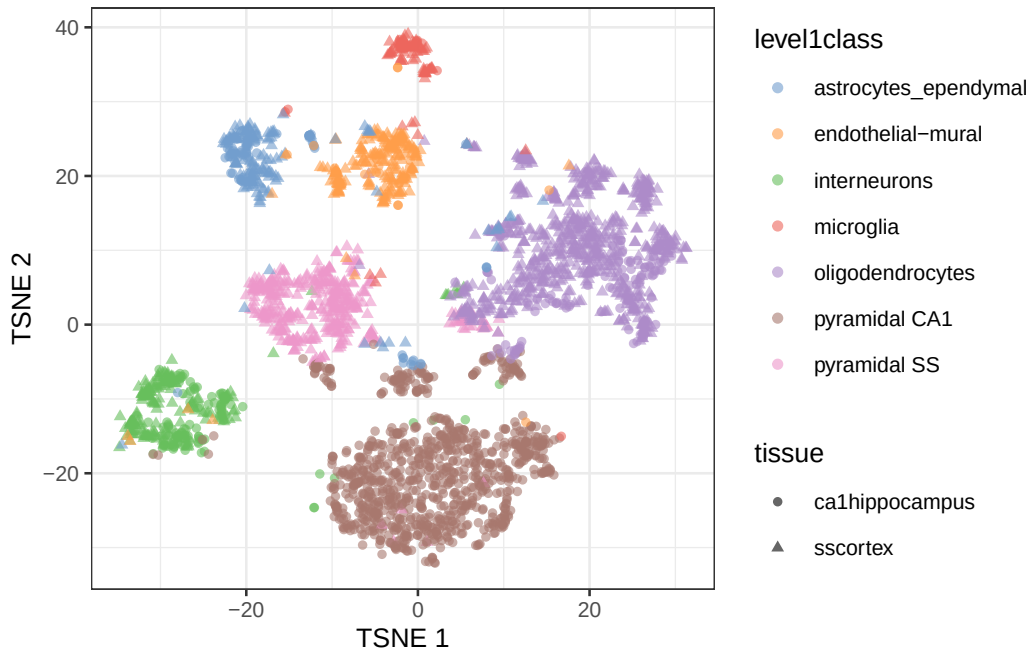


Figure 32.5.

32.4. Finding marker genes

Marker genes are genes that are differentially expressed in a specific cell type or cluster compared to others. Identifying marker genes helps us understand the molecular characteristics of each cell type and can provide insights into their functions.

To find marker genes, we can use the `findMarkers()` function from the `scraper` package. This function performs pairwise differential expression analysis between clusters or cell types. We can specify the clusters we want to compare and the method for calculating differential expression.

```
markers <- findMarkers(sce,
  groups=as.numeric(as.factor(colData(sce)$level1class)), pval.type="all",
  direction="up", subset.row=top_var_genes, test="t")
```

```
Warning in .findMarkers(assay(x, i = assay.type), ...): 'findMarkers' is deprecated.
Use 'scraper::scoreMarkers.se' instead.
See help("Deprecated")
```

32. Single-cell RNA-seq analysis

```
Warning in .local(x, ...): 'pairwiseTTests' is deprecated.  
See help("Deprecated")
```

```
Warning in combineMarkers(fit$statistics, fit$pairs, pval.type = pval.type, : 'combineMarkers' is deprecated.  
Use 'scrapper::summarizeEffects' instead.  
See help("Deprecated")
```

The `findMarkers()` function identifies genes that are significantly upregulated in each cluster compared to all other clusters. The `pval.type="all"` argument specifies that we want to calculate p-values for all comparisons, and the `direction="up"` argument indicates that we are interested in genes that are upregulated in the specified clusters.

The `subset.row=top_var_genes` argument restricts the analysis to the highly variable genes we identified earlier, which are more likely to be informative for distinguishing cell types.

The resulting `markers` object is a list where each element corresponds to a cluster and contains the differential expression results for that cluster. A compact way to see them all at once is a heatmap: take the top genes from every cluster, then plot their expression across all the cells.

The code below pulls the top 10 marker genes for each cluster, collapses them into one unique set, and draws a single heatmap of every cluster side by side. This lets us see which genes are most strongly associated with each cluster and how the clusters differ from one another.

```
# Heatmap of the top marker genes across all clusters  
library(scater)  
top_marker_genes_for_all_clusters <- lapply(markers, function(x) {  
  return(head(rownames(x), 10))  
})  
plotHeatmap(sce, features=unique(unlist(top_marker_genes_for_all_clusters)),  
  cluster_rows=TRUE, cluster_cols=TRUE, show_colnames=FALSE,  
  main="Top marker genes for all clusters",  
  color_columns_by="level1class")
```

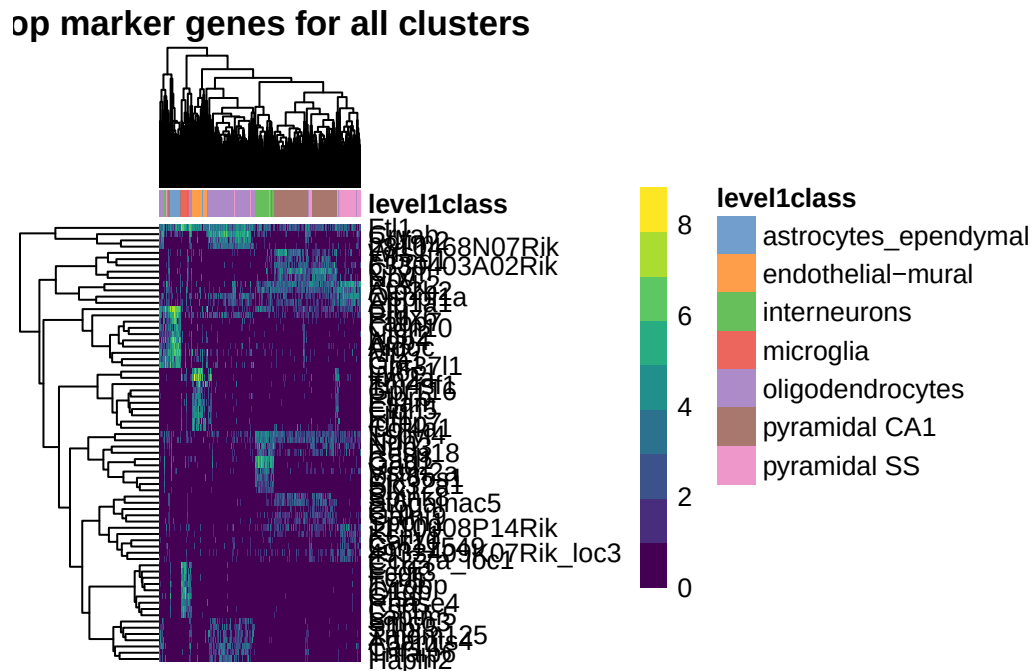



Figure 32.6.

In Figure 32.6 the columns are cells (grouped and colored by cell type) and the rows are marker genes. The blocky structure is what you're after: each set of marker genes lights up in one band of cells and stays dim elsewhere, which is exactly what makes them useful as markers — they cleanly distinguish one population from the rest.

32.5. Interactive visualization with iSEE

The [Interactive SummarizedExperiment Explorer \(iSEE\)](#) is a powerful tool for exploring single-cell data interactively. It allows us to visualize and interact with the data in real-time, making it easier to explore complex datasets and identify patterns.

A key component of iSEE is the `SingleCellExperiment` object, which serves as the data container for single-cell RNA-seq data. iSEE provides a user-friendly interface for exploring the data.

To use iSEE, we first need to install the `iSEE` package and then launch the iSEE app.

32. Single-cell RNA-seq analysis

```
# Install iSEE package if not already installed
iSEE::iSEE(sce)
```

The `iSEE()` function launches the iSEE app, which provides an interactive interface for exploring the `SingleCellExperiment` object.

You may want to play with a few genes to see how they are expressed across the different clusters. You can also explore the metadata associated with the cells, such as tissue type, cell type annotations, and quality control metrics. The set of “marker genes” we identified earlier can also be visualized in iSEE, allowing us to see how these genes are expressed across different clusters and cell types.

For cluster 1, the marker genes are:

```
cat(paste(top_marker_genes_for_all_clusters[[1]], '\n'))
```

```
Aqp4
Cldn10
Clu
Aldoc
Ntsr2
Mt2
Fabp7
Gja1
Gpr37l1
Prdx6
```

Try entering a couple of those gene names into the iSEE feature-expression panels to see how they vary across the clusters.

32.6. Exercises

These build directly on the `sce` object you created above. If you’ve closed your session, re-run the chapter up to the marker-gene step first.

1. **Read the cluster table.** How many clusters did `clusterCells()` find, and which is the largest? Which function’s output answers this?

i Solution

```
table(nn.clusters)
```

```
nn.clusters
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
89 296 153  40 102 192  63  69 143  96 314 133  97  71  91  70  69 112  47  91
21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38
101  63  54  36  62  34  42  59  32  39  20  28  15  18  16  19  17  12
```

```
length(unique(nn.clusters)) # number of clusters
```

```
[1] 38
```

```
which.max(table(nn.clusters)) # the most populous cluster
```

```
11
```

```
11
```

`table()` counts cells per cluster; `length(unique(...))` turns that into the number of groups. The largest cluster is usually an abundant cell type rather than a rare one.

2. **Tune the number of variable genes.** Re-run `getTopHVGs()` asking for only 500 genes instead of 2000. Does the size of the gene set change which features you'd carry into PCA?

i Solution

```
top_500 <- getTopHVGs(sce, n=500)
```

```
Warning in getTopHVGs(sce, n = 500): 'getTopHVGs' is deprecated.
Use 'scraper::chooseHighlyVariableGenes' instead.
See help("Deprecated")
```

```
Warning in fitTrendVar(fm, fv, ...): 'fitTrendVar' is deprecated.
Use 'scraper::fitVarianceTrend' instead.
See help("Deprecated")
```

```
Warning in combineBlocks(collected, method = method, equiweight = equiweight, : 'com
See help("Deprecated")
```

```
length(top_500)
```

```
[1] 500
```

32. Single-cell RNA-seq analysis

```
length(intersect(top_500, top_var_genes)) # overlap with the 2000-gene set
```

```
[1] 500
```

The 500 genes are (almost entirely) a subset of the 2000 — they are simply the *most* variable ones. Picking fewer genes is faster and noisier; picking more keeps weaker signal at the cost of more noise. There is no single right answer, which is why it's a choice you make deliberately.

3. **Switch the coloring.** Redraw the t-SNE plot, but color the cells by their cluster label (`colour_by="label"`) instead of by the known cell type. How well do the data-driven clusters line up with the annotated cell types in Figure 32.3?

i Solution

```
plotReducedDim(sce, "TSNE", colour_by="label")
```

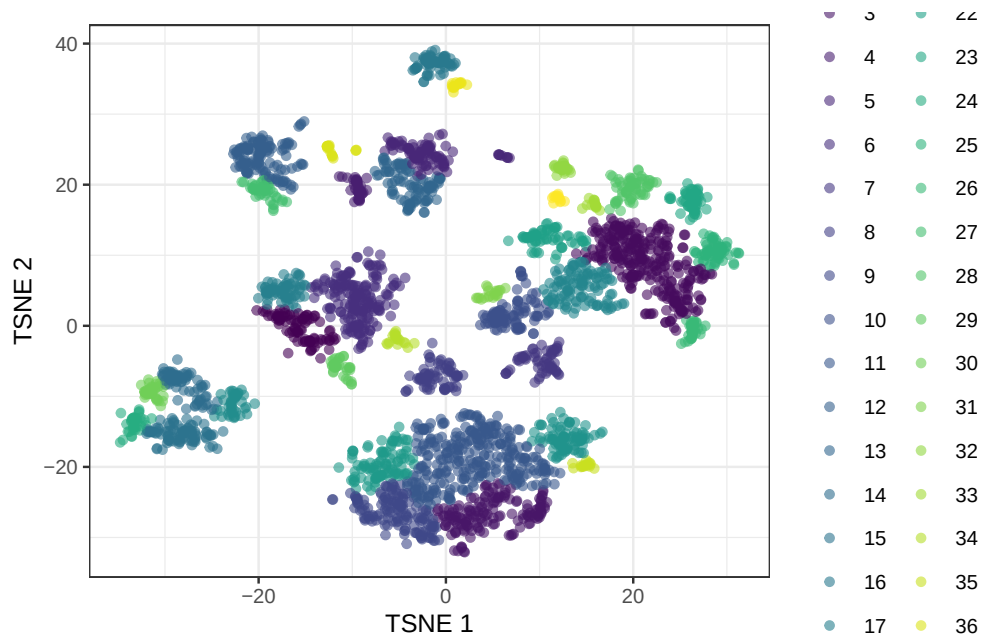


Figure 32.7.

Each cluster occupies its own island, and those islands correspond closely to the cell-type islands you saw earlier — evidence that the unsupervised clustering recovered the same structure as the curated annotations.

32.7. Summary

You took a published mouse-brain dataset all the way from raw counts to a labeled map of cell types, and in doing so walked the standard scRNA-seq workflow:

- **Loaded** the data as a `SingleCellExperiment` — a `SummarizedExperiment` (the [SummarizedExperiment chapter](#)) with single-cell extras like a reduced-dimensions slot.
- **Log-normalized** the counts with `logNormCounts()` to correct for differences in sequencing depth between cells.
- **Selected highly variable genes** with `getTopHVGs()`, focusing the analysis on the genes that carry biological signal.
- **Reduced dimensions** with `fixedPCA()` and then `runTSNE()`, collapsing thousands of genes into a 2D map.
- **Clustered** the cells with `clusterCells()` and read the cluster table to see how many groups emerged.
- **Found and visualized marker genes** with `findMarkers()` and `plotHeatmap()`, confirming that each cluster has its own molecular fingerprint.

The reassuring result was that the unsupervised steps recovered the known cell types: clusters and curated labels landed on the same islands of the t-SNE. From here you could annotate clusters, compare conditions, or hand the object to an interactive explorer like iSEE.

32.8. Bonus: working with 10x Genomics data

The rest of the chapter is an optional appendix. Everything above used a curated dataset that arrives already cleaned and annotated. Real projects rarely start there: most droplet-based experiments come off a 10x Genomics platform as raw output that you must load and quality-control yourself. The code in this section is not executed (the files don't exist in this book), but it shows the shape of that early workflow so you'll recognize it when you meet it.

32.8.1. Understanding 10x Genomics output formats

10x Genomics Cell Ranger produces several output formats:

1. **Matrix Market format** (`.mtx` files): Sparse matrix format with separate files for the matrix, gene names, and cell barcodes

32. Single-cell RNA-seq analysis

2. **HDF5 format** (.h5 files): A single compressed file containing all the data
3. **Filtered vs. unfiltered**: Cell Ranger provides both versions - filtered contains only cells that pass basic quality filters

32.8.2. Loading 10x data from Matrix Market files

```
# Loading from standard 10x output directory
# This assumes you have a directory with:
# - matrix.mtx.gz (or matrix.mtx)
# - features.tsv.gz (or genes.tsv.gz for older versions)
# - barcodes.tsv.gz

library(DropletUtils)
library(SingleCellExperiment)

# Method 1: Using DropletUtils (recommended)
sce_10x <- read10xCounts("path/to/10x/output/", col.names = TRUE)

# Method 2: Using Seurat (if you prefer the Seurat ecosystem)
# library(Seurat)
# seurat_obj <- Read10X("path/to/10x/output/")
# sce_10x <- as.SingleCellExperiment(CreateSeuratObject(seurat_obj))
```

What we're doing: The `read10xCounts()` function from `DropletUtils` reads the three files that comprise 10x output and creates a `SingleCellExperiment` object. The `col.names = TRUE` parameter ensures that cell barcodes are used as column names.

Why `DropletUtils`: This package is specifically designed for handling droplet-based single-cell data (like 10x Genomics). It includes specialized functions for:

- Reading 10x formats efficiently
- Distinguishing empty droplets from cells
- Handling ambient RNA contamination
- Quality control specific to droplet-based methods

32.8.3. Loading from HDF5 files

32. Single-cell RNA-seq analysis

```
# For HDF5 format files
library(rhdf5)

# Method 1: Using DropletUtils
sce_h5 <- read10xCounts("path/to/sample.h5", type = "HDF5")

# Method 2: Direct HDF5 reading (more control)
h5_file <- "path/to/sample.h5"
gene_info <- h5read(h5_file, "matrix/features")
barcodes <- h5read(h5_file, "matrix/barcodes")
matrix_data <- h5read(h5_file, "matrix/data")
# Additional processing needed to reconstruct sparse matrix
```

What we're doing: HDF5 files are more convenient as they contain all information in a single file. The `type = "HDF5"` parameter tells the function to expect this format instead of separate matrix files.

When to use H5 files:

- **Convenience:** Single file is easier to manage and transfer
- **Speed:** Often faster to read than multiple compressed files
- **Storage:** More efficient storage, especially for large datasets

32.8.4. Essential quality control after loading raw data

```
# Add mitochondrial gene information
library(scuttle)
is.mito <- grepl("^MT-", rownames(sce_10x)) # Human mitochondrial genes
# For mouse data, use: is.mito <- grepl("^mt-", rownames(sce_10x))

# Calculate per-cell QC metrics
sce_10x <- addPerCellQC(sce_10x, subsets = list(Mito = is.mito))

# Calculate per-gene QC metrics
sce_10x <- addPerFeatureQC(sce_10x)

# View the QC metrics
colnames(colData(sce_10x))
```

32. Single-cell RNA-seq analysis

What we're doing: We're calculating quality control metrics that are essential for raw 10x data:

- **Mitochondrial genes:** High mitochondrial gene expression often indicates dying cells
- **Per-cell metrics:** Total counts, number of detected genes, mitochondrial percentage
- **Per-gene metrics:** How many cells express each gene

Why QC is crucial for raw data: Unlike curated datasets, raw 10x data contains:

- **Empty droplets:** Droplets that captured ambient RNA but no cells
- **Dying cells:** Cells with compromised membranes
- **Doublets:** Droplets containing multiple cells
- **Low-quality cells:** Cells with very few detected genes

32.8.5. Filtering low-quality cells and genes

```
# Visualize QC metrics
library(scater)
plotColData(sce_10x, x = "sum", y = "detected", colour_by = "subsets_Mito_percent")

# Set filtering thresholds (these are examples - adjust based on your data)
# Filter cells
high_mito <- sce_10x$subsets_Mito_percent > 20 # Remove cells with >20% mitochondrial read
low_lib_size <- sce_10x$sum < 1000 # Remove cells with <1000 total counts
low_n_features <- sce_10x$detected < 500 # Remove cells with <500 detected genes

# Filter genes (remove genes expressed in very few cells)
low_expression <- rowSums(counts(sce_10x) > 0) < 10 # Expressed in <10 cells

# Apply filters
sce_filtered <- sce_10x[!low_expression, !(high_mito | low_lib_size | low_n_features)]

# Check how many cells and genes remain
dim(sce_10x) # Before filtering
dim(sce_filtered) # After filtering
```

What we're doing: We're applying filters to remove low-quality cells and rarely expressed genes. The specific thresholds should be adjusted based on your dataset characteristics.

Understanding the filters:

32. Single-cell RNA-seq analysis

- **Mitochondrial percentage:** Cells with high mitochondrial gene expression are likely dying
- **Library size:** Total number of UMIs per cell - very low values suggest poor capture
- **Number of features:** How many different genes are detected - very low values suggest poor quality
- **Gene expression frequency:** Genes expressed in very few cells are often noise

Critical considerations:

- **Thresholds are dataset-dependent:** What works for one experiment may not work for another
- **Biological vs. technical variation:** Some cell types naturally have different RNA content
- **Visualization first:** Always plot your QC metrics before setting thresholds

32.9. Further reading and resources

For more in-depth information on single-cell RNA-seq analysis, consider the following resources:

- The [Orchestrating Single-Cell RNA-seq Analysis](#) book provides a comprehensive guide to single-cell RNA-seq analysis using Bioconductor.
- The [Seurat](#) package documentation offers extensive tutorials and vignettes for single-cell analysis in the Seurat ecosystem.

References

- Bourgon, Richard, Robert Gentleman, and Wolfgang Huber. 2010. “Independent Filtering Increases Detection Power for High-Throughput Experiments.” *Proceedings of the National Academy of Sciences* 107 (21): 9546–51. <https://doi.org/10.1073/pnas.0914005107>.
- Brouwer-Visser, Jurriaan, Wei-Yi Cheng, Anna Bauer-Mehren, et al. 2018. “Regulatory T-Cell Genes Drive Altered Immune Microenvironment in Adult Solid Cancers and Allow for Immune Contextual Patient Subtyping.” *Cancer Epidemiology, Biomarkers & Prevention* 27 (1): 103–12. <https://doi.org/10.1158/1055-9965.EPI-17-0461>.
- Bućinca, Zana, Maja Barbara Malaya, and Krzysztof Z. Gajos. 2021. “To Trust or to Think.” *Proceedings of the ACM on Human-Computer Interaction* 5 (CSCW1): 1–21. <https://doi.org/10.1145/3449287>.
- Budzyń, Krzysztof, Marcin Romańczyk, Diana Kitala, et al. 2025. “Endoscopist Deskillng Risk After Exposure to Artificial Intelligence in Colonoscopy: A Multicentre, Observational Study.” *The Lancet. Gastroenterology & Hepatology* 10 (10): 896–903. [https://doi.org/10.1016/s2468-1253\(25\)00133-5](https://doi.org/10.1016/s2468-1253(25)00133-5).
- Buenrostro, Jason D, Paul G Giresi, Lisa C Zaba, Howard Y Chang, and William J Greenleaf. 2013. “Transposition of Native Chromatin for Fast and Sensitive Epigenomic Profiling of Open Chromatin, DNA-binding Proteins and Nucleosome Position.” *Nature Methods* 10 (12): 1213–18. <https://doi.org/10.1038/nmeth.2688>.
- Buenrostro, Jason D, Beijing Wu, Howard Y Chang, and William J Greenleaf. 2015. “ATAC-seq: A Method for Assaying Chromatin Accessibility Genome-Wide.” *Current Protocols in Molecular Biology* / Edited by Frederick M. Ausubel ... [Et Al.] 109 (January): 21.29.1–9. <https://doi.org/10.1002/0471142727.mb2129s109>.
- Center, Pew Research. 2016. “Lifelong Learning and Technology.” In *Pew Research Center: Internet, Science & Tech.* <https://www.pewresearch.org/internet/2016/03/22/lifelong-learning-and-technology/>.

References

- Crawford, Gregory E, Sean Davis, Peter C Scacheri, et al. 2006. “DNase-chip: A High-Resolution Method to Identify DNase I Hypersensitive Sites Using Tiled Microarrays.” *Nature Methods* 3 (7): 503–9. <http://www.ncbi.nlm.nih.gov/pubmed/16791207?dopt=AbstractPlus>.
- Crawford, Gregory E, Ingeborg E Holt, James Whittle, et al. 2006. “Genome-Wide Mapping of DNase Hypersensitive Sites Using Massively Parallel Signature Sequencing (MPSS).” *Genome Research* 16 (1): 123–31. <http://www.ncbi.nlm.nih.gov/pubmed/16344561?dopt=AbstractPlus>.
- DeRisi, J. L., V. R. Iyer, and P. O. Brown. 1997. “Exploring the Metabolic and Genetic Control of Gene Expression on a Genomic Scale.” *Science (New York, N. Y.)* 278 (5338): 680–86. <https://doi.org/10.1126/science.278.5338.680>.
- Goddard, Kate, Abdul Roudsari, and Jeremy C Wyatt. 2012. “Automation Bias: A Systematic Review of Frequency, Effect Mediators, and Mitigators.” *Journal of the American Medical Informatics Association* 19 (1): 121–27. <https://doi.org/10.1136/amiajnl-2011-000089>.
- Greener, Joe G., Shaun M. Kandathil, Lewis Moffat, and David T. Jones. 2022. “A Guide to Machine Learning for Biologists.” *Nature Reviews Molecular Cell Biology* 23 (1): 40–55. <https://doi.org/10.1038/s41580-021-00407-0>.
- Knowles, Malcolm S., Elwood F. Holton, and Richard A. Swanson. 2005. *The Adult Learner: The Definitive Classic in Adult Education and Human Resource Development*. 6th ed. Elsevier.
- Lawrence, Michael, Wolfgang Huber, Hervé Pagès, Patrick Aboyoun, Marc Carlson, Robert Gentleman, Martin T. Morgan, et al. 2013a. “Software for Computing and Annotating Genomic Ranges.” *PLoS Computational Biology* 9 (8): e1003118. <https://doi.org/10.1371/journal.pcbi.1003118>.
- Lawrence, Michael, Wolfgang Huber, Hervé Pagès, Patrick Aboyoun, Marc Carlson, Robert Gentleman, Martin T Morgan, et al. 2013b. “Software for Computing and Annotating Genomic Ranges.” *PLoS Computational Biology* 9 (8): e1003118. <https://doi.org/10.1371/journal.pcbi.1003118>.
- Lee, Hao-Ping (Hank), Advait Sarkar, Lev Tankelevitch, et al. 2025. “The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers.” *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, April 25, 1–22. <https://doi.org/10.1145/3702456>.

References

[//doi.org/10.1145/3706598.3713778](https://doi.org/10.1145/3706598.3713778).

- Libbrecht, Maxwell W., and William Stafford Noble. 2015. “Machine Learning Applications in Genetics and Genomics.” *Nature Reviews Genetics* 16 (6): 321–32. <https://doi.org/10.1038/nrg3920>.
- Lun, Aaron T. L., Davis J. McCarthy, and John C. Marioni. 2016. “A Step-by-Step Workflow for Low-Level Analysis of Single-Cell RNA-Seq Data with Bioconductor.” *F1000Res*. 5: 2122. <https://doi.org/10.12688/f1000research.9501.2>.
- McCarthy, Davis J., Kieran R. Campbell, Aaron T. L. Lun, and Quin F. Willis. 2017. “Scater: Pre-Processing, Quality Control, Normalisation and Visualisation of Single-Cell RNA-Seq Data in R.” *Bioinformatics* 33: 1179–86. <https://doi.org/10.1093/bioinformatics/btw777>.
- Meskó, Bertalan, and Eric J. Topol. 2023. “The Imperative for Regulatory Oversight of Large Language Models (or Generative AI) in Healthcare.” *Npj Digital Medicine* 6 (1). <https://doi.org/10.1038/s41746-023-00873-0>.
- Messeri, Lisa, and M. J. Crockett. 2024. “Artificial Intelligence and Illusions of Understanding in Scientific Research.” *Nature* 627 (8002): 49–58. <https://doi.org/10.1038/s41586-024-07146-0>.
- Morgan, Martin, Herve Pages, V Obenchain, and N Hayden. 2016. “Rsamtools: Binary Alignment (BAM), FASTA, Variant Call (BCF), and Tabix File Import.” *R Package Version* 1 (0): 677–89.
- Murdoch, Blake. 2021. “Privacy and Artificial Intelligence: Challenges for Protecting Health Information in a New Era.” *BMC Medical Ethics* 22 (1). <https://doi.org/10.1186/s12910-021-00687-3>.
- Risso, Davide, and Michael Cole. 2024. *scRNAseq: Collection of Public Single-Cell RNA-Seq Datasets*. <https://doi.org/10.18129/B9.bioc.scRNAseq>.
- Student. 1908. “The Probable Error of a Mean.” *Biometrika* 6 (1): 1–25. <https://doi.org/10.2307/2331554>.
- Tsompana, Maria, and Michael J Buck. 2014. “Chromatin Accessibility: A Window into the Genome.” *Epigenetics & Chromatin* 7 (1): 33. <https://doi.org/10.1186/1756-8935-7-33>.

References

- Wasserstein, Ronald L., and Nicole A. Lazar. 2016. “The ASA Statement on p -Values: Context, Process, and Purpose.” *The American Statistician* 70 (2): 129–33. <https://doi.org/10.1080/00031305.2016.1154108>.
- Zeisel, Amit, Ana B. Muñoz-Manchado, Simone Codeluppi, et al. 2015. “Cell Types in the Mouse Cortex and Hippocampus Revealed by Single-Cell RNA-seq.” *Science* 347 (6226): 1138–42. <https://doi.org/10.1126/science.aaa1934>.

A. Git and GitHub

Git is a version control system that allows you to track changes in your code and collaborate with others. GitHub is a web-based platform that hosts Git repositories, making it easy to share and collaborate on projects. Github is NOT the only place to host Git repositories, but it is the most popular and has a large community of users.

You can use git by itself locally for version control. However, if you want to collaborate with others, you will need to use a remote repository, such as GitHub. This allows you to share your code with others, track changes, and collaborate on projects.

Note

It can be confusing to understand the difference between Git and GitHub. In short, Git is the version control system that tracks changes in your code, while GitHub is a platform that hosts your Git repositories and provides additional features for collaboration.

A.1. install Git and GitHub CLI

To use Git and GitHub, you need to have Git installed on your computer. You can download it from git-scm.com. After installation, you can check if Git is installed correctly by running the following command in your terminal:

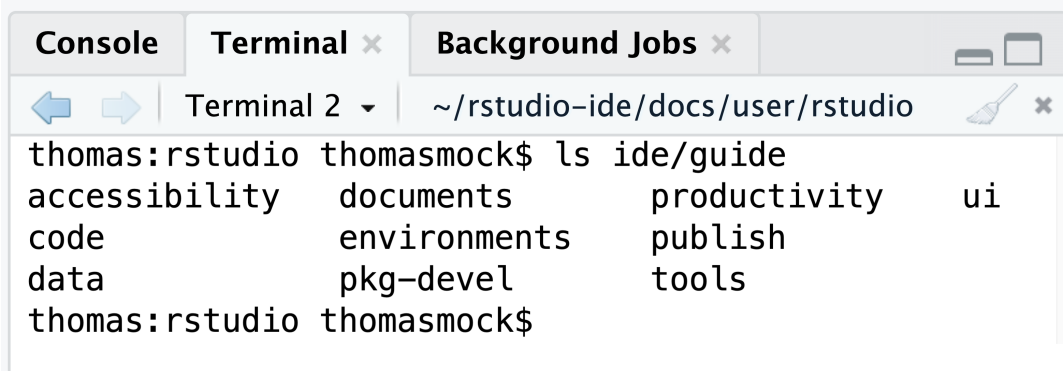
```
git --version
```

We also need the `gh` command line tool to interact with GitHub. You can install it from cli.github.com. To install, go to [the releases page](#) and download the appropriate version for your operating system. For the Mac, it is the file named something like “Macos Universal” and the file will have a `.pkg` extension. You can install it by double-clicking the file after downloading it.

A. Git and GitHub

Using the RStudio Terminal

If you are using RStudio, you can use the built-in terminal to run Git commands. To open the terminal, go to the “Terminal” tab in the bottom pane of RStudio. This allows you to run Git commands directly from RStudio without needing to switch to a separate terminal application.



For more details, see the [RStudio terminal documentation](#).

A.2. Configure Git

After installing Git, you need to configure it with your name and email address. This information will be used to identify you as the author of the commits you make. Run the following commands in your terminal, replacing “Your Name” and “you@example.com” with your actual name and email address:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

A.3. Create a GitHub account

If you don’t already have a GitHub account, you can create one for free at github.com.

A.4. Login to GitHub CLI

After installing the GitHub CLI, you need to log in to your GitHub account. Run the following command in your terminal:

```
gh auth login
```

A.5. Introduction to Version Control with Git

Welcome to the world of version control! Think of Git as a “save” button for your entire project, but with the ability to go back to previous saves, see exactly what you changed, and even work on different versions of your project at the same time. It’s an essential tool for reproducible and collaborative research.

In this tutorial, we’ll learn the absolute basics of Git using the command line directly within RStudio.

A.5.1. Key Git Commands We’ll Learn Today:

- **git init**: Initializes a new Git repository in your project folder. This is the first step to start tracking your files.
 - **git add**: Tells Git which files you want to track changes for. You can think of this as putting your changes into a “staging area.”
 - **git commit**: Takes a snapshot of your staged changes. This is like creating a permanent save point with a descriptive message.
 - **git restore**: Discards changes in your working directory. It’s a way to undo modifications you haven’t committed yet.
 - **git branch**: Allows you to create separate timelines of your project. This is useful for developing new features without affecting your main work.
 - **git merge**: Combines the changes from one branch into another.
-

A.6. The Toy Example: An R Script

First, let's create a simple R script that we can use for our Git exercise. In RStudio, create a new R Script and save it as `data_analysis.R`.

```
# data_analysis.R

# Load necessary libraries
library(ggplot2)
library(dplyr)

# Create some sample data
data <- data.frame(
  x = 1:10,
  y = (1:10) ^ 2
)

# Initial data summary
summary(data)
```

A.7. Let's Get Started with Git!

Open the **Terminal** in RStudio (you can usually find it as a tab next to the Console). We'll be typing all our Git commands here.

A.7.1. Step 1: Initialize Your Git Repository

First, we need to tell Git to start tracking our project folder.

```
git init
```

You'll see a message like `Initialized empty Git repository in....` You might also notice a new `.git` folder in your project directory (it might be hidden). This is where Git stores all its tracking information. Your default branch is automatically named `main`.

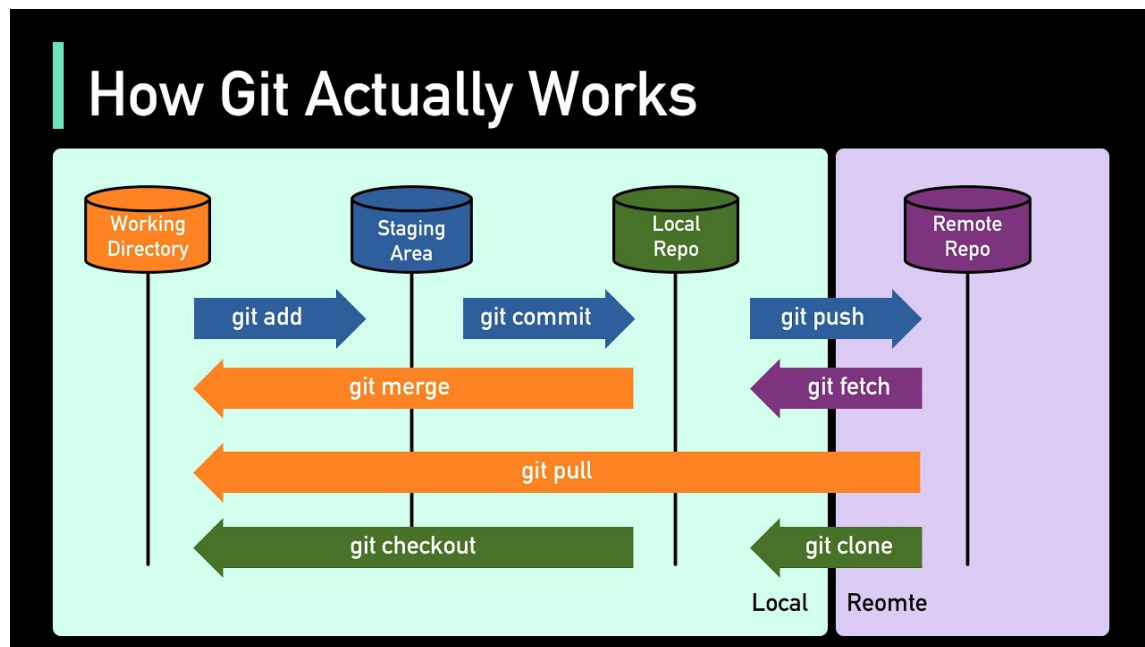


Figure A.1.: This is an overview of how git works along with the commands that make it tick. See [this video](#)

A.7.2. Step 2: Your First Commit

Now, let's add our `data_analysis.R` script to Git's tracking and make our first "commit."

1. Add the file to the staging area:

```
git add data_analysis.R
```

2. Commit the staged file with a message:

```
git commit -m "Initial commit: Add basic data script"
```

The `-m` flag lets you write your commit message directly in the command. Good commit messages are short but descriptive!

A.7.3. Step 3: Making and Undoing a Change

Let's modify our R script. Add a plotting section to the end of `data_analysis.R`.

```
# ... (keep the previous code)

# Create a plot
ggplot(data, aes(x = x, y = y)) +
  geom_point() +
  ggtitle("A Simple Scatter Plot")
```

Now, what if we decided we didn't want this change after all? We can use `git restore` to go back to our last committed version.

```
git restore data_analysis.R
```

If you look at your `data_analysis.R` file now, the plotting code will be gone!

A.7.4. Step 4: Branching Out

Branches are a powerful feature. Let's create a new branch to add our plot without messing up our `main` branch.

1. Create a new branch and switch to it:

A. Git and GitHub

```
git checkout -b add-plot
```

This is a shortcut for `git branch add-plot` and `git checkout add-plot`.

Now, re-add the plotting code to `data_analysis.R`.

```
# ... (keep the previous code)

# Create a plot
ggplot(data, aes(x = x, y = y)) +
  geom_point() +
  ggtitle("A Simple Scatter Plot")
```

Let's commit this change on our new `add-plot` branch.

```
git add data_analysis.R
git commit -m "feat: Add scatter plot"
```

A.7.5. Step 5: Seeing Branches in Action

Now for the magic of branches. Let's switch back to our `main` branch.

```
git checkout main
```

Now, open your `data_analysis.R` script in the RStudio editor. **The plotting code is gone!** That's because the change only exists on the `add-plot` branch. The `main` branch is exactly as we last left it.

Let's switch back to our feature branch.

```
git checkout add-plot
```

Check the `data_analysis.R` script again. **The plotting code is back!** This demonstrates how branches allow you to work on different versions of your project in isolation.

A.7.6. Step 6: Merging Your Work

Our plot is complete and we're happy with it. It's time to merge it back into our `main` branch to incorporate the new feature.

1. **Switch back to the main branch, which is our target for the merge:**

```
git checkout main
```

2. **Merge the `add-plot` branch into `main`:**

```
git merge add-plot
```

You'll see a message indicating that the merge happened. Now, your `main` branch has the updated `data_analysis.R` script with the plotting code!

B. Further reading, cheat sheets, and datasets

No single book can cover everything, and this one does not try to. This appendix collects the resources I reach for most often: quick-reference cheat sheets, free online books that go deeper than we can here, documentation for the key packages used throughout the chapters, and the datasets and interactive tools that support the material. Treat it as a launch pad for the self-directed learning the [preface](#) argues for.

B.1. Cheat sheets

Cheat sheets are dense, one- or two-page summaries you keep open beside your code. Posit (formerly RStudio) maintains a gallery of polished cheat sheets for the tidyverse and the RStudio IDE, and the Bioconductor project publishes one for its core classes.

- [Posit cheat sheet gallery](#) — the hub for all of the sheets below (and the [source repository](#))
- [Base R cheat sheet](#)
- [RStudio IDE cheat sheet](#) — see the [RStudio chapter](#)
- [Data transformation with dplyr](#) — see the [dplyr chapter](#)
- [Data visualization with ggplot2](#) — see the [visualization chapters](#)
- [Tidy data with tidyr](#)
- [Reading data with readr](#) — see [Reading and writing data](#)
- [Factors with forcats](#) — see the [factors chapter](#)

B.2. Free online books

- [R for Data Science \(2e\)](#) — the standard tidyverse introduction
- [Hands-on Programming with R](#) — gentle, project-driven; the source of several figures in this book
- [Modern Data Visualization with R](#)
- [ggplot2: Elegant Graphics for Data Analysis](#) — the definitive `ggplot2` reference

B. Further reading, cheat sheets, and datasets

- [Advanced R](#) — for when you want to understand *how* R works
- [Bioconductor: an introduction](#) — Carpentries-style Bioconductor onboarding
- [Orchestrating Single-Cell Analysis with Bioconductor \(OSCA\)](#) — companion to the single-cell chapter
- [Computational Genomics with R](#)

B.3. Package and project documentation

- [Bioconductor](#) — the project home; every package has a *vignette* worth reading
- [The tidyverse](#) — `dplyr`, `ggplot2`, `tidyr`, and friends
- [mlr3 book](#) — the framework used in the [machine learning chapters](#)
- [ComplexHeatmap reference](#) — production heatmaps beyond `pheatmap`
- [R Graph Gallery](#) and [From Data to Viz](#) — pick the right chart and copy the code

B.4. Interactive practice: swirl

[swirl](#) teaches you R *inside* R: it runs lessons at the console and checks your answers as you go. To get started, install and launch it:

```
install.packages("swirl")
library("swirl")
swirl()
```

Then follow the prompts and pick a course.

B.5. Datasets used in this book

- [BRFSS subset](#) — the Behavioral Risk Factor Surveillance System sample analyzed in the [EDA case study](#)

Most other datasets are pulled directly from packages or public repositories (NCBI GEO, ENCODE, Bioconductor data packages) by the chapters that use them, so there is nothing to download by hand.

B.6. AI assistants

Large language models are a genuinely useful study partner for R and statistics. Rather than list them here, this book devotes a full appendix to using them well — see [AI Tools for Enhanced Learning and Productivity](#).

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

The rise of sophisticated Artificial Intelligence (AI) tools, particularly Large Language Models (LLMs), has drastically altered day-to-day tasks for many. This chapter aims to guide you, as biologists venturing into these computational fields, on how to effectively and responsibly harness these AI tools. The focus will be on using them as powerful assistants to deepen your understanding of statistical concepts, enhance your ability to explore complex biological data with R, and ultimately boost your overall productivity and learning experience. Think of these tools not as a shortcut around learning, but as a versatile companion on your journey.

C.1. Introduction: AI as Your Bioinformatics Learning Companion

You’ve likely heard of tools like ChatGPT, Gemini, Claude, or GitHub Copilot. These are examples of LLMs and other AI-driven assistants that can process and generate human-like text, code, and even help with complex reasoning tasks. Their capabilities are particularly relevant when grappling with new programming languages like R, intricate statistical methods, or the challenge of deriving meaning from large biological datasets.

Table C.1.: Table AI Tools for Bioinformatics Learning

AI Tool	Description
ChatGPT	A conversational AI that can answer questions, explain concepts, and generate code snippets.
Gemini	Google’s AI that can assist with coding, data analysis, and more.
Claude	An AI assistant that can help with writing, coding, and problem-solving.

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

AI Tool	Description
GitHub Copilot	An AI-powered code completion tool that suggests code as you type, based on the context of your project.
Perplexity	An AI that can answer questions and provide explanations across various domains, including programming and data science.

Why should you consider integrating these AI tools into your learning workflow? The benefits are manifold. They can help demystify complex topics by offering alternative explanations, assist in drafting R code for specific analyses, aid in the often-frustrating process of debugging, summarize lengthy documents, and even help you brainstorm approaches for data exploration.

i A Supplement, Not a Replacement

It's crucial to remember that AI is here to *supplement* your learning and critical thinking, not replace it. Your growing expertise in biology, combined with a solid understanding of statistical principles, remains paramount. AI can help you get there faster and overcome hurdles, but the destination is your own deep understanding.

However, it's also important to approach these tools with realistic expectations. LLMs can sometimes provide inaccurate information—a phenomenon often called “hallucination”—or exhibit biases present in their training data.¹ Therefore, your domain knowledge and developing statistical intuition are your best guides in sifting the digital wheat from the chaff.

C.2. AI for Understanding Statistical Concepts and R Functions

One of the most immediate benefits of LLMs is their ability to act as a patient, tireless tutor for statistical concepts. If you find a particular statistical idea challenging, you can ask an LLM to explain it in various ways. For instance, you might prompt: “Explain the concept of a p-value as if you were talking to a biologist who is new to statistics,” or “Describe the key differences between Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) when analyzing gene expression data.” The ability to request analogies or examples directly relevant to biological datasets can be incredibly helpful.

¹Always be prepared to critically evaluate and verify information from AI tools against trusted sources.

Similarly, when you're wrestling with the R language, AI tools can be invaluable. Perhaps you're unsure about the syntax for a `dplyr` verb or the arguments for a `ggplot2` function. You could ask, "How do I use the `dplyr::filter()` function in R to select rows from my dataframe where the 'gene_expression' column is greater than 100 and the 'treatment_group' column is 'Condition_A'?" or "Can you explain the main arguments of the `stats::t.test()` function in R and what its output signifies?" These tools can also help you discover R packages specifically designed for statistical tasks you have in mind.

When it comes to applying these concepts, AI can generate example R code. For common statistical tests like t-tests, ANOVAs, or chi-squared tests, you can request simple, reproducible examples, perhaps even with dummy data that you can then adapt.

 Understand Before You Use

A word of caution is essential here: while AI can generate code, *you* are the scientist. Always strive to understand the code provided. Ensure it aligns with your research question, your data structure, and the assumptions of the statistical test you're performing. Blindly copying and pasting code without comprehension can lead to erroneous conclusions.

C.3. AI for Data Exploration and Visualization Strategies

Exploring a new biological dataset can sometimes feel like navigating uncharted territory. AI can act as a brainstorming partner in this process. You could describe your dataset—its variables, the type of data (e.g., RNA-seq counts, proteomics measurements, microbial abundances), and the underlying biological context—and then ask for suggestions on initial exploratory analyses. For example: "I have a dataset with normalized protein abundance levels for samples from cancerous and healthy tissues, along with patient metadata like age and disease stage. What are some initial plots I could make in R with `ggplot2` to explore potential patterns or differences?"

Speaking of `ggplot2` and other R visualization libraries, LLMs can be particularly adept at helping you craft the precise code for the plots you envision. You can request code for specific plot types, ask for help customizing aesthetics like colors, labels, and themes, or even troubleshoot why your plot isn't rendering as expected. "How can I add a title to my `ggplot`, change the x-axis label to 'Gene Length (bp)', and use a different color palette for my groups?" is a typical query an LLM can handle.

AI can also offer tentative interpretations of statistical outputs or visual patterns. However, this is where your scientific judgment is most critical.

AI Suggestions are Starting Points

While an LLM might suggest that a particular cluster in your PCA plot *could* represent a distinct biological condition, this is merely a hypothesis generated by the model. It is your responsibility to correlate this with your biological knowledge, experimental design, and further statistical validation. AI interpretations are starting points for your own deeper investigation, not final conclusions.

C.4. AI for Boosting Productivity and Troubleshooting

Beyond conceptual understanding and data exploration, AI tools can significantly enhance your day-to-day productivity. For instance, code generation and autocompletion features, whether in standalone LLMs or integrated into development environments (like GitHub Copilot, if available to you), can speed up the writing of boilerplate R code or complete lines of code you’ve started. This is especially useful for repetitive tasks, freeing you up to focus on the more analytical aspects of your work. Remember, the goal is to accelerate your coding, not to have entire complex analyses generated without your deep involvement and understanding.

One of the most common frustrations in programming is debugging. AI can be a powerful ally here. You can paste R error messages directly into an LLM and ask for explanations and potential fixes. If your code isn’t behaving as expected, describe its intended function and the problematic output, and the AI may be able to pinpoint the issue.²

Other productivity boosts include asking AI to summarize lengthy R package documentation or even sections of research articles (always being mindful of access permissions and copyright). Furthermore, if you have a piece of R code that works but feels clunky or inefficient, you can ask an AI for suggestions on how to refactor it to make it more readable, efficient, or aligned with common R programming idioms.

C.5. Best Practices for Prompt Engineering

The quality of the output you receive from an LLM is heavily dependent on the quality of your input—your “prompt.” Effective prompt engineering is a skill that improves with practice.

²Providing a minimal reproducible example (a small piece of code with sample data that replicates the error) greatly improves the chances of getting useful debugging help from an AI.

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

The cornerstone of a good prompt is **specificity and context**. The more information you provide about your goal, your data, the biological question at hand, the specific R package you're using, or the exact error message you're encountering, the more tailored and useful the AI's response will be. Consider the difference:

Vague Prompt	Specific & Contextualized Prompt
"How to plot data in R?"	"How can I create a scatter plot in R using <code>ggplot2</code> to show the relationship between 'gene_expression_log2fc' and 'mirna_expression_log2fc' from my <code>results_df</code> dataframe, with points colored by 'cancer_subtype'?"
"My R code isn't working."	"I'm getting the error 'Error: object 'x' not found' in R when I run this code: <code>plot(x, y)</code> . I defined <code>x</code> in a previous chunk. Why might this be happening and how can I fix it?"

Don't expect the perfect answer on your first attempt. **Iterate and refine your prompts**. If the initial response isn't quite what you need, try rephrasing your question, adding more details, or breaking down a complex request into smaller parts. You can also **specify the desired output format**, such as "Provide the R code for this," "Explain this concept in three simple bullet points," or "List three potential statistical approaches for this problem." Sometimes, asking for multiple perspectives or alternative solutions can also be enlightening. Finally, don't hesitate to experiment with different LLMs if you have access to more than one, as their strengths can vary.

C.6. Ethical Considerations and Limitations

While AI tools offer tremendous potential, it's imperative to use them responsibly and be acutely aware of their limitations.

Accuracy and "Hallucinations": This is perhaps the most critical point. LLMs are designed to generate plausible text, but this text is not always factually correct or logically sound. They can "hallucinate" answers, code, or citations that seem convincing but are entirely fabricated.

You **must** critically evaluate all information and code generated by AI. Cross-reference with trusted sources like official R documentation, textbooks, peer-reviewed literature, and your own developing expertise. Never blindly trust AI-generated content, especially for scientific analysis.

Bias in AI Models: AI models learn from the vast corpus of text and code they are trained on. This training data can contain societal biases, which may then be reflected in the AI's

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

responses. Be mindful of how this could subtly influence suggestions, interpretations, or even the tone of explanations.

Data Privacy and Confidentiality: This is a paramount concern in biomedical research, where the data you work with may be protected by law and by your duty to the people it describes. Two categories deserve special vigilance:

- **PII** (personally identifiable information) — names, dates of birth, addresses, email addresses, identifying numbers, or any combination of fields that could single out an individual.
- **PHI** (protected health information) — the subset of PII tied to health, diagnoses, treatments, or genetic data, regulated under frameworks such as HIPAA in the United States and GDPR in Europe.

Pasting either kind of data into a public chatbot can constitute an unauthorized disclosure, and genomic data is especially hard to truly de-identify — a genome is itself an identifier (Murdoch 2021). The privacy and governance challenges that large language models add on top of this are still being worked out, and formal oversight lags well behind adoption (Meskó and Topol 2023), so the safe default falls to you.

Protect Your Sensitive Data

Under no circumstances should you paste sensitive, unpublished, or personally identifiable data — including PHI/PII and individual-level genomic or clinical data — into public LLM interfaces. Always assume that anything you input could be stored, logged, or used to train future models. Before using any AI tool on research data, confirm what its data-usage policy permits and whether your institution has approved it; many universities and hospitals offer vetted, private-instance deployments with contractual data protections for exactly this reason. For coursework and everyday learning, keep AI to conceptual questions, code on non-sensitive example datasets, and publicly available data.

Over-Reliance and Skill Atrophy: The convenience of AI can be seductive. However, relying on it too heavily can hinder the development of your own foundational R programming and statistical reasoning skills, and there is now empirical evidence that heavy reliance both erodes the skills of experts and reduces the critical thinking people invest in a task (Budzyń et al. 2025; Lee et al. 2025). Use AI as a tool to *aid* and *accelerate* your learning, not to *replace* the hard work of understanding. The ultimate goal is for *you* to become proficient, not for the AI to perform tasks for you. This risk is important enough that the next section treats it on its own.

Reproducibility: If you incorporate AI-generated code or insights into your analyses, ensure you thoroughly understand them, can justify their use, and document them metic-

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

ulously for reproducibility. It can be good practice to note the prompts you used if the interaction significantly shaped your approach, treating it almost like a consultation.

The table below summarizes some key limitations and suggested mitigations:

Limitation	Description	Mitigation Strategies
Accuracy Issues (Hallucinations)	Generates plausible but incorrect/fabricated information or code.	Verify with trusted sources; test code thoroughly; use critical thinking.
Bias	Reflects biases present in training data.	Be aware of potential biases; seek diverse perspectives; critically evaluate fairness of suggestions.
Data Privacy	Risk of exposing sensitive data if input into public tools.	Avoid inputting sensitive/unpublished data; use institutional/private instances if available and vetted; understand terms of service.
Lack of True Understanding	Manipulates patterns in data, doesn't "understand" concepts like humans do.	Don't attribute human-like reasoning; use it for specific tasks, not overarching scientific judgment.
Over-Reliance	Can hinder development of personal skills if used as a crutch.	Focus on learning principles; use AI to assist, not replace, your own effort; regularly practice without AI.
Reproducibility Challenges	AI responses can vary; difficult to document exact "reasoning" of the AI.	Document prompts used for significant code/ideas; thoroughly understand and vet any AI-generated code you use.

C.7. De-skilling, never-skilling, and how to stay sharp

The most subtle risk of AI is not that it gives a wrong answer — it is that it gives a *right* one, so smoothly that you never build the skill yourself. It is worth separating two versions of this:

- **De-skilling** is the *erosion* of a competence you already have. Skills decay when they go unused, and outsourcing them to a machine is a fast way to stop using them. In one striking study, experienced endoscopists who had been working alongside an AI detection tool became measurably *worse* at finding polyps on their own once the AI

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

was taken away — their unaided detection rate dropped after routine exposure to the assistant (Budzyń et al. 2025). The skill atrophied precisely because the machine had been carrying it.

- **Never-skilling** is the failure to *acquire* a competence in the first place. For a learner, this is the bigger danger. If every exercise is answered by an LLM before you have struggled with it, you collect finished code without ever building the mental model that lets you write or debug it yourself. A large survey of knowledge workers found that the more people trusted generative AI for a task, the *less* critical thinking they reported applying to it — they shifted from doing the work to merely checking the machine’s output, and sometimes not even that (Lee et al. 2025).

Both are versions of a long-known phenomenon. **Automation bias** — the tendency to over-trust an automated system and stop scrutinizing it — has been documented in clinical decision support for decades, and it causes the most harm exactly where the stakes are highest (Goddard et al. 2012). A related trap in research is the *illusion of understanding*: AI tools can make a field feel mastered and a result feel explained when neither is actually true (Messerli and Crockett 2024).

💡 Protect the struggle

The productive struggle *is* the learning. This book’s [preface](#) argues that the “Read–Do” loop — reading a little, then immediately trying it yourself — is what moves knowledge into long-term memory. AI is most corrosive when it lets you skip the “Do.” So protect it deliberately: attempt the problem before you ask the AI, and treat the AI’s answer as something to *check against*, not to copy.

A few habits keep AI on the right side of this line:

- **Try first, ask second.** Write your own attempt — even a broken one — before prompting. You will understand the AI’s answer far better, and you will notice when it is wrong.
- **Make the AI explain, not just produce.** Ask *why* a piece of code works, or ask it to walk you through the logic, rather than only requesting the finished snippet.
- **Practice unplugged.** Periodically solve problems with no AI at all, the way a musician practices scales — this is how you find out which skills are actually yours.
- **Verify by hand on small cases.** Run AI-suggested code on a tiny example whose answer you can work out yourself.

C.8. Using AI constructively: when to lean in, and when to step back

AI is not equally appropriate for every task or every moment. A useful mental model is to ask two questions before reaching for it: *How well do I understand this already?* and *How costly is a mistake?*

Lean **in** when the task is low-stakes and reversible, when you are exploring or brainstorming, when you are learning a concept you can then verify, or when you are stuck and need a starting point. These are situations where an occasional wrong answer costs you a little time and nothing more, and where the act of checking the AI's output is itself good practice.

Step **back** — or skip AI entirely — in two situations in particular:

- **Under time pressure.** Deadlines are exactly when you are least able to carefully verify an answer and most tempted to accept it on faith. Automation bias is strongest when attention and time are scarce, so the moment you feel rushed is the moment an unchecked AI answer is most likely to slip through (Goddard et al. 2012). If you cannot afford to verify it, you cannot afford to trust it.
- **In high-stakes environments.** Where an error could harm a patient, corrupt a published result, or expose protected data, the cost of a confident-sounding hallucination is far too high to absorb. In these settings AI should at most be a *second opinion* that a qualified human checks — never the final word — and its use should be governed by your institution's policies rather than left to individual judgment (Meskó and Topol 2023).

One well-studied way to keep yourself honest is a **cognitive forcing function**: a small deliberate friction that makes you engage before you accept. Asking the AI to justify its reasoning, committing to your own answer first, or requiring a manual check before you act all measurably reduce over-reliance and the errors it produces (Bućinca et al. 2021). The goal is not to use AI less for its own sake, but to make sure a human mind — yours — is still in the loop where it matters.

C.9. The Future: AI in Bioinformatics and Data Science

The field of AI is evolving at a breathtaking pace, and its integration into bioinformatics and data science is only set to deepen. We may see more sophisticated AI-powered tools for hypothesis generation, experimental design, drug discovery, and personalized medicine. As future biologists and data scientists, cultivating an adaptable mindset and committing

to continuous learning will be key to navigating and leveraging these advancements effectively. Staying curious about new tools, while maintaining a strong foundation in scientific principles and ethics, will serve you well.

C.10. Practical Exercises and Activities

To help you get comfortable and skilled in using AI tools, we encourage you to try the following:

1. **Conceptual Clarification:** Choose a statistical concept from this course that you find somewhat challenging (e.g., “false discovery rate,” “cross-validation,” or “the assumptions of linear regression”). Use an LLM to ask for an explanation tailored to a biologist. Note down your prompt and the AI’s response. Then, try to re-explain the concept in your own words to a classmate, drawing from (but not merely repeating) the AI’s explanation.
2. **R Code Generation & Adaptation:** Imagine you have a data frame in R named `rna_seq_data` with columns `gene_id`, `sample_name`, `normalized_count`, and `treatment_group` (with levels “control” and “treated”). Ask an LLM to generate R code using `dplyr` and `tidyr` to calculate the average `normalized_count` for each `gene_id` within each `treatment_group`, and then spread the data so that ‘control’ and ‘treated’ averages are in separate columns for each gene. Once you have the code, test it with some sample data (you can ask the AI to generate that too!) and ensure you understand each step.
3. **Debugging Practice:** Take a piece of R code. First, try to understand and debug it yourself. Then, paste the code and the error message into an LLM and ask for help. Compare the AI’s suggestions to your own troubleshooting. Did it help you find the solution faster? Did it explain the error well?
4. **Visualization Brainstorming:** Provide a clear description of a biological dataset, ideally one of your own (e.g., “A dataset containing measurements of 5 different cytokine levels in blood samples taken from patients at 3 time points post-vaccination, with patients also categorized by age group (young, middle, old).”). Use an LLM to brainstorm 3-4 different types of visualizations you could create using `ggplot2` to explore this data. For each visualization, describe what insights it might provide.
5. **Solve it twice.** Pick a small exercise from an earlier chapter and solve it *without* any AI help. Then solve the same problem by prompting an LLM and accepting its answer. Compare: which version do you understand better, and which could you reproduce next week without help? Use this to calibrate where AI genuinely helps your learning versus where it quietly does the learning for you.

C. AI Tools for Enhanced Learning and Productivity in Bioinformatics

As you work through these exercises, make it a habit to document the prompts you use and critically assess the AI's responses. Reflect on when the AI was most helpful, when its advice was off-base, and how you can refine your prompting strategy for better results in the future. This reflective practice is key to becoming a discerning and effective user of AI in your scientific endeavors.

D. Matrix Exercises

D.1. Data preparation

For this set of exercises, we are going to rely on a dataset that comes with R. It gives the number of sunspots per month from 1749-1983. The dataset comes as a `ts` or time series data type which I convert to a matrix using the following code.

Just run the code as is and focus on the rest of the exercises.

```
data(sunspots)
sunspot_mat <- matrix(as.vector(sunspots),ncol=12,byrow = TRUE)
colnames(sunspot_mat) <- as.character(1:12)
rownames(sunspot_mat) <- as.character(1749:1983)
```

D.2. Exercises

- After the conversion above, what does `sunspot_mat` look like? Use functions to find the number of rows, the number of columns, the class, and some basic summary statistics.

```
ncol(sunspot_mat)
nrow(sunspot_mat)
dim(sunspot_mat)
summary(sunspot_mat)
head(sunspot_mat)
tail(sunspot_mat)
```

- Practice subsetting the matrix a bit by selecting:
 - The first 10 years (rows)
 - The month of July (7th column)
 - The value for July, 1979 using the rowname to do the selection.

D. Matrix Exercises

```
sunspot_mat[1:10,]  
sunspot_mat[,7]  
sunspot_mat['1979',7]
```

These next few exercises take advantage of the fact that calling a univariate statistical function (one that expects a vector) works for matrices by just making a vector of all the values in the matrix.

- What is the highest (max) number of sunspots recorded in these data?

```
max(sunspot_mat)
```

- And the minimum?

```
min(sunspot_mat)
```

- And the overall mean and median?

```
mean(sunspot_mat)  
median(sunspot_mat)
```

- Use the `hist()` function to look at the distribution of all the monthly sunspot data.

```
hist(sunspot_mat)
```

- Read about the `breaks` argument to `hist()` to try to increase the number of breaks in the histogram to increase the resolution slightly. Adjust your `hist()` and `breaks` to your liking.

```
hist(sunspot_mat, breaks=40)
```

Now, let's move on to summarizing the data a bit to learn about the pattern of sunspots varies by month or by year.

- Examine the dataset again. What do the columns represent? And the rows?

```
# just a quick glimpse of the data will give us a sense  
head(sunspot_mat)
```

- We'd like to look at the distribution of sunspots by month. How can we do that?

D. Matrix Exercises

```
# the mean of the columns is the mean number of sunspots per month.  
colMeans(sunspot_mat)  
  
# Another way to write the same thing:  
apply(sunspot_mat, 2, mean)
```

- Assign the month summary above to a variable and summarize it to get a sense of the spread over months.

```
monthmeans = colMeans(sunspot_mat)  
summary(monthmeans)
```

- Play the same game for years to get the per-year mean?

```
ymeans = rowMeans(sunspot_mat)  
summary(ymeans)
```

- Make a plot of the yearly means. Do you see a pattern?

```
plot(ymeans)  
# or make it clearer  
plot(ymeans, type='l')
```